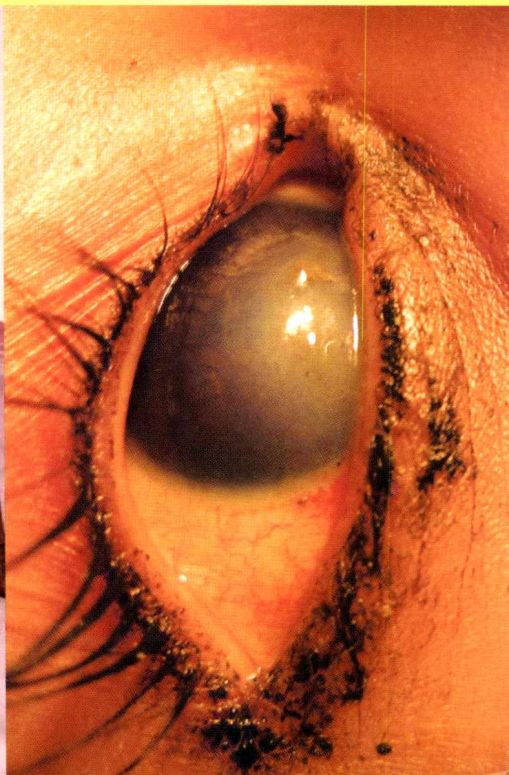
4th Edition^

PARAS

Nutrition and Child Development

KE Elizabeth







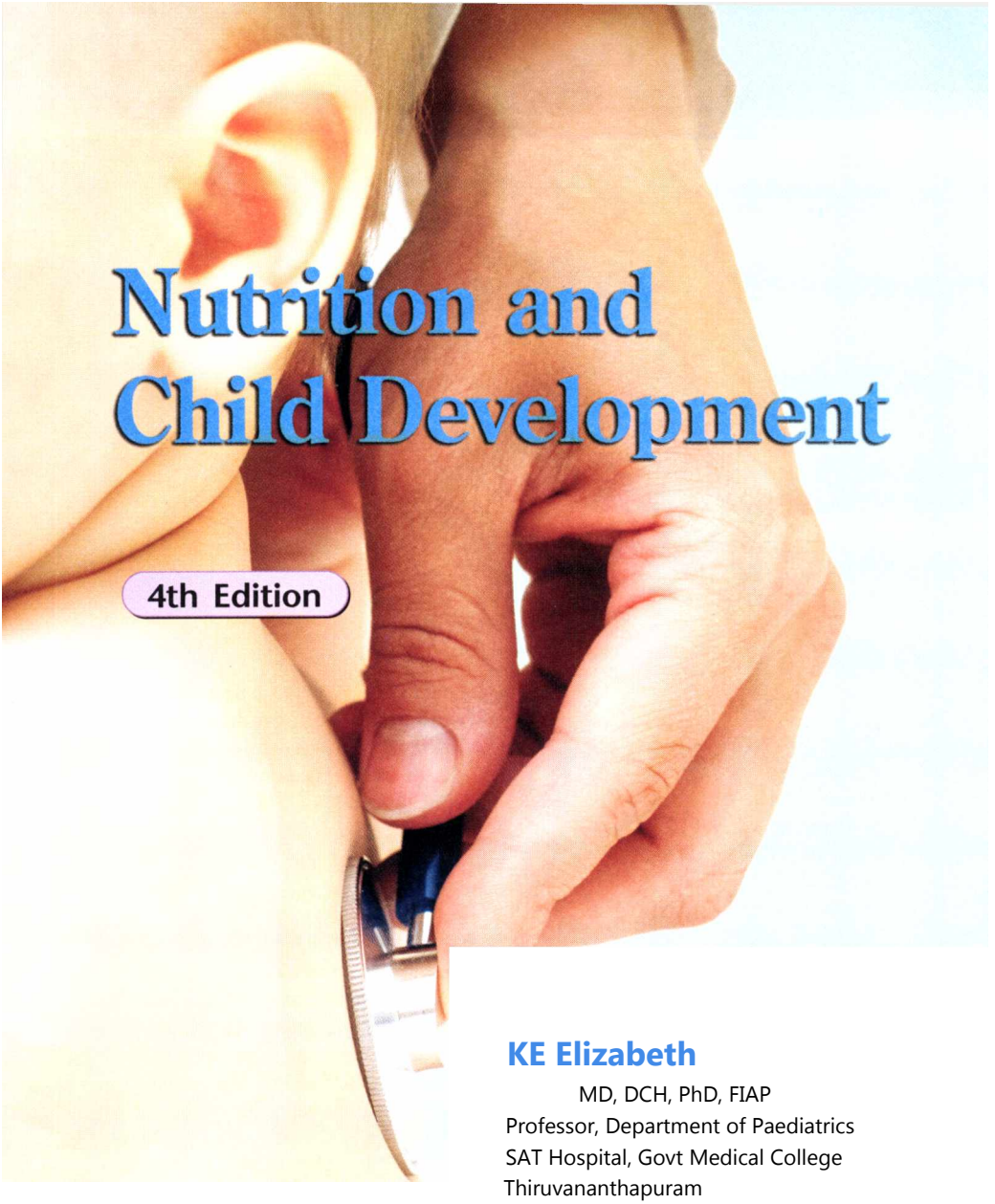
About the author

Dr KE Elizabeth is a talented clinician, academician and researcher in the field of paediatrics and nutrition. She had a brilliant academic career graduating from Medical College, Thiruvananthapuram with a first class and ranked first in her batch in the University of Kerala with gold medals in Medicine and Community Medicine. Subsequently, she obtained DCH and MD (Paediatrics) and PhD (Nutrition and Child Development) from the University of Kerala.

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She has won several honours and distinctions—Dr CO Karunakaran award twice for her original scientific contributions; Indian Academy of Paediatrics Award thrice for the best research paper from teaching institutions of the South Zone; Nana Miniscreen Award for the best TV Programme for Women and Children; Dr TN Krishnan award for her work among the underprivileged children in the coastal area; Senior Award in Community Nutrition, Nutrition Society of India, Hyderabad; and International Ambulatory Paediatric Research Award, Virginia, for her innovations in comprehensive rehabilitation of children with malnutrition.

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Nutrition and Child Development

4th Edition

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To
my husband



and
my daughters



the support, blessings and courage
imparted to me

FOREWORD

In children, nutrition, growth and development are intricately inter-linked-aberrations of one aspect tend to significantly influence the others. Thus, for example, for optimal growth and development of an infant, appropriate nutritional practices play a pivotal role. Medical knowledge on the subject of Paediatric Nutrition, Growth and Development is increasing rapidly, particularly in relation to its epidemiologic, biochemical and clinical aspects. Several of these advances have a direct bearing and practical implications for the practitioners of health care in the developing world. However, there is limited literature on this subject which is properly oriented towards the current needs of the country.

"Nutrition and Child Development" intends to fulfil this felt need and has been conceived with the objective of providing a comprehensive outline of the various aspects of paediatric nutrition and child development, particularly in relation to the developing countries. It is primarily intended for medical and paramedical students and practitioners dealing with the subject.

The author's extensive experience in paediatric practice, teaching and nutrition research has been translated into a relevant volume. This monograph has been logically divided into 10 sections with emphasis on practical aspects. The sections dealing with protein-energy malnutrition, diet in various diseases and child development deserve special mention as they are based on research and practical experience in the Indian context. It is sincerely hoped that this book fulfils the objective of updating the target audience on the subject of nutrition and child development.

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PREFACE TO THE FOURTH EDITION

The changing profile in malnutrition and the expanding horizons in nutrtion and child development have warranted a thorough revision and reorganization of the book. I have taken special care to address these issues in this new edition of the book.

This book is written based on experience and research among children with malnutrition relevant to the Indian context. Throughout the book there is emphasis on interaction between nutrition, growth and development. The various research works undertaken among children with malnutrition are also included. It is hoped that the search will continue for a better understanding of the interaction between host, nutritional and environmental factors. There is no doubt that nutrition related research will be a platform for exploring the various facets of growth and development. It is expected that this book will provide insight into the complexities of the subject and the challenges in front of the clinicians and researchers.

I have revised the chapters, reorganised them and added new topics as per the suggestions of my friends and students. I sincerely acknowledge the suggestions given by Dr Shanti Ghosh, New Delhi, the kind inspiration given by Dr N Edwin, Madurai, and the help rendered by Dr Regi R Chandran. Dr Gibby Koshy, Dr Roy George Jacob, Dr Manu Muraleedharan, Trivandrum, and Ms. Gayathri Thiyagarajan, Chennai, in organising and highlighting the later editions of this handy and succinct book.

The help and inspiration provided by my family, professors, colleagues and students are gratefully acknowledged. I place on record my sincere thanks to Dr YM Fazil Marickar, Professor of Surgery, Former HOD. Medical College, Trivandrum, for the help and inspiration; Dr S Srinivasan, Professor and HOD, Department of Paediatrics, JIPMER, Pondicherry, for the

encouragement; Dr Carol Chu and Dr Angus Dobbie, Department of Clinical Genetics, Leeds, UK, for the inspiration; Dr S Sindhu for the support and the drawings; and Mr Sarath for the typographic assistance. I am indebted to Mr Divyesh Kothari of M/s Paras Medical Publisher, Hyderabad, for his sincere attempts and perseverance in bringing out this book.

KE Elizabeth

PREFACE TO THE FIRST EDITION

Malnutrition is a *"man made disease 'which often' starts in the womb and ends in the tomb"*. Malnutrition and the associated retarding influences cause a lot of morbidity, growth faltering, developmental retardation and significant mortality. There is a wide range of medical and paramedical professionals interested in the subject of nutrition and child development. The examples of the former are general paediatricians, developmental paediatricians, neurologists, endocrinologists, psychiatrists and physicians; and of the latter are nutritionists, dietitians, home science experts, clinical psychologists and special educators. It is important to link them for prevention and management of nutritional disorders. The developmental perspective which is crucial in infants and young children is a very important dimension of this subject.

Malnutrition and mental development is a hot topic as there is lot of controversy as to what is the effect of malnutrition per se on mental development and what is the effect of environmental deprivation on development. There is doubt as to how far the animal studies on malnutrition can be extrapolated to the human settings as the period of gestation as well as brain growth, vary widely in animals and in humans. The ultimate expression of the endowed potential for growth and intelligence is the net effect of the interplay of genetic factors, nutrition and environment.

There is a need for a comprehensive book that addresses the issue of nutrition and child development for use by those who are interested in nutrition and child development including medical and paramedical students and practitioners. As growth and development go hand in hand, these two aspects are combined in this book in relation to nutrition. This book will aid interdisciplinary understanding in an area where a lot of specialists and scientists are involved in clinical management and research.

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ABBREVIATIONS

ARI	—	Acute Respiratory Infection
BFHI	—	Baby Friendly Hospital Initiative
BMR	—	Basal Metabolic Rate
BSID	—	Bayley Scales of Infant Development
CARE	—	Cooperative for American Relief Everywhere
CSSM	—	Child Survival and Safe Motherhood
DDST	—	Denver Developmental Screening Test
DQ	—	Developmental Quotient
FAO	—	Food and Agricultural Organization
GNP	—	Gross Net Production
GOBIFFF	—	Growth monitoring, Oral rehydration therapy, Breast feeding, Immunization, Food supplementation, Female education and Family planning
IAP	—	Indian Academy of Paediatrics
ICDS	—	Integrated Child Development Services
IMR	—	Infant Mortality Rate
LBW	—	Low Birth Weight
MAC	—	Mid Arm Circumference
NCHS	—	National Center for Health Statistics
NNMB	—	National Nutrition Monitoring Bureau
NUT	—	Nutritional Management
OFC	—	Occipito Frontal Circumference
PCM	—	Protein Calorie Malnutrition
PEM	—	Protein Energy Malnutrition
PPE	—	Poverty, Population Growth & Environmental Stress

RCV— Resident Community Volunteers
RDA— Recommended Dietary Allowances
SAT— Sree Avittam Thirunal
STIM— Composite Stimulation Package
SQ— Somatic Quotient
UBSP— Urban Basic Services Programme for the Poor
UIP— Universal Immunization Programme
UNICEF— United Nations International Children Emergency Fund
USAID— United States Agency for International Development
WHO— World Health Organization
WISC— Wechsler Intelligence Scale for Children



FIG. 1 A five-year-old girl with severe stunting in comparison with a normal child of the same age

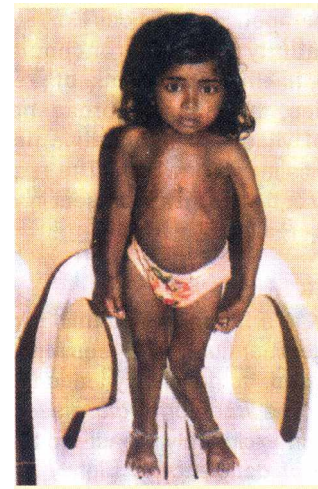


FIG. 2 A six-year-old girl with refractory rickets

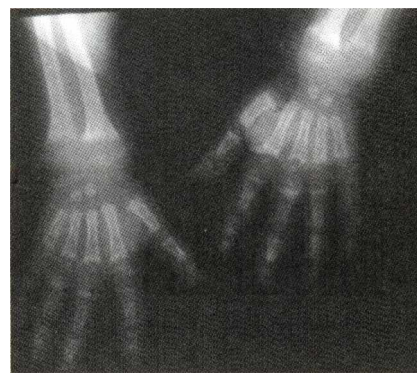


FIG. 3 X-ray of rickets (hands)

Plate I



FIG. 4 An infant with FTT, marasmus and cleft lip and palate

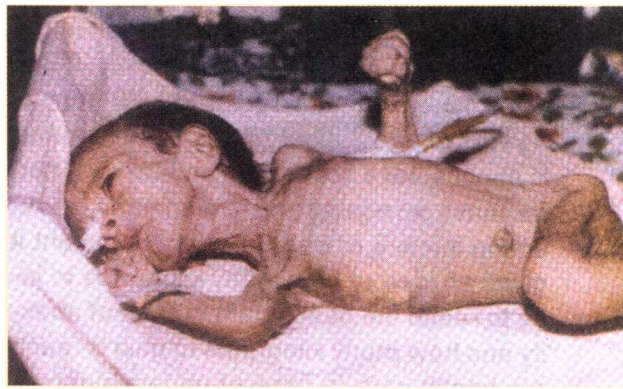


FIG. 5 An infant with marasmus

Plate II

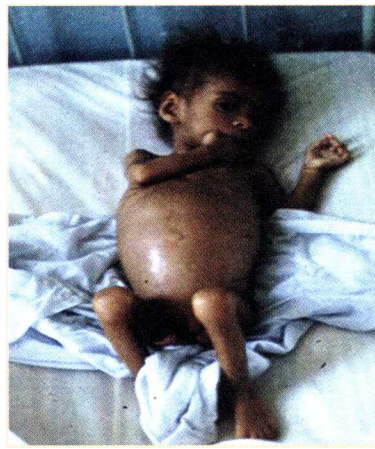


FIG. 6 A child wirh kwashiorkor



FIG. 7 Preterm triplets 600 grams

Plate III



FIG. 8 Scurvy with subperiosteal bleed



FIG. 9 Blount's disease—bow legs

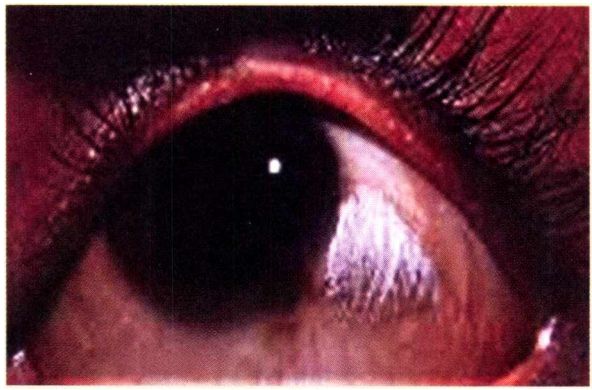


FIG. 10 Bitot's spots—vitamin A deficiency



FIG. 11 Child with marasmus grade IV

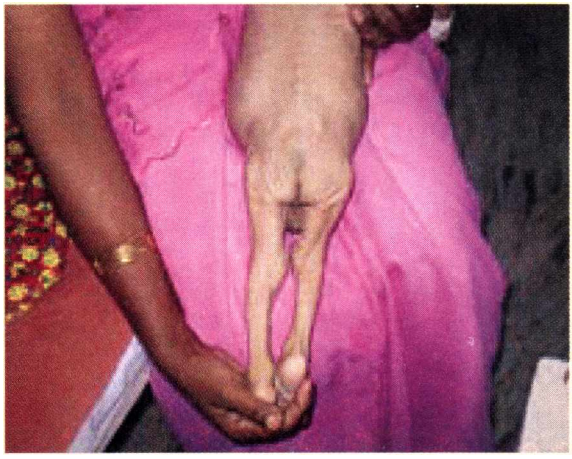


FIG 12 Child with marasmus grade IV

Plate V



FIG. 13 Child with kwashiorkor grade IV

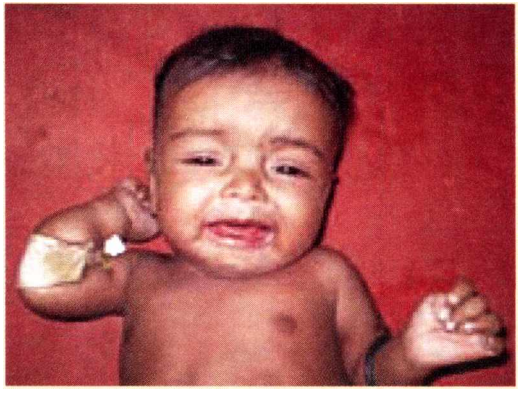


FIG 14 Child with kwashiorkor grade IV

Plate VI

SECTION 1

Infant and Young Child Feeding (IYCF)

"If ever I get a chance, I should love to be reborn just to have the ecstasy of being re-fed by the kindly mother."

— W. Oscar

Introduction

Optimum nutrition is essential for child survival and quality of survival. The word 'nutrition' is derived from 'nutricus' which means 'to suckle at the breast'. Breast milk is the natural food for the infant and it is 'species specific'. Successful breastfeeding is an important child rearing skill to be learnt and practiced.

1.1 Breastfeeding and BFHI

After the introduction of the **'Baby Friendly Hospital Initiative' (BFHI)** in 1992, exclusive demand feeding is accepted as the only mode of early infant feeding. Babies are well known to thrive on breast milk alone during the first 4-6 months of age. The World Alliance for Breastfeeding Action (**WABA) is the global agency for promotion of breastfeeding.** The Breastfeeding Promotion Network of India (BPNI) is the national agency for breastfeeding. Every year, the **'World Breastfeeding Week' (WBW) is celebrated from 1st to 7st of August.** The year 2001 marks the decade of celebration of WBW. Breastfeeding is now accepted as a human right, a right of the baby as well as the mother.

The **BFHI** is a global programme organised by UNICEF. It was **launched in 1992** and **adopted by India in 1993.** BFHI certification is done by the national and state BFHI task forces. By December 1993,38 hospitals were certified and by now, thousands of hospitals have been certified. **Cochin city, Kerala, became the first baby friendly city and Ernakulam, the first babv friendly district.** All the hospi-

tals in this area have been accredited as 'baby friendly'. The recent concept is to change the baby friendly hospitals into 'mother and child friendly hospitals'. The **BFHI plus programme** incorporates other child survival and safe motherhood components like immunization, antenatal care, oral rehydration therapy, acute respiratory infection control programme etc.

1. The Ten Steps in BFHI

In order to actively protect, promote and support breastfeeding, every facility providing maternity services and care for newborn infants should practice the following ten steps:

a) Have a written breastfeeding policy that is routinely communicated to all health care staff.
b) Train all health care staff in skills necessary to implement this policy.
c) Inform all pregnant women about the benefits and management of breastfeeding.
d) Help mothers initiate breastfeeding within an hour of birth.
e) Show mothers how to breastfeed and how to maintain lactation even if they should be separated from their infants.
f) Give newborn infants no food or drink other than breast milk, unless medically indicated.
g) Practise rooming-in and allow mothers and infants to remain together 24 hours a day.
h) Encourage breastfeeding on demand.
i) Give no artificial teats or pacifiers (also called dummies or soothers) to breastfeeding infants.
j) Foster the establishment of breastfeeding support groups and refer mothers to them on discharge from the hospital or clinic.

2. The Ten Policies of BFHI

Based on the above ten steps, the hospital policies are formulated and exhibited. The ten policies of BFHI are the following:

a) Our hospital has an official policy to protect, promote and support breastfeeding.
b) All maternity and child care health staff in the hospital receive training in the skills to promote breastfeeding.
c) All mothers, both antenatal and postnatal, are informed about the benefits of breastfeeding.
d) We assist mothers in the early initiation of breastfeeding, within half hour of birth for a normal delivery, within 4 hours of birth of a caesarean section.
e) All mothers are shown how to breastfeed and how to maintain lactation, even if they should be temporarily separated from their infants.

f) We give newborns no food or drink other than breast milk. Infant foods and breast milk substitutes are prohibited in this institution.

g) We practice, 'rooming-in' by allowing the mothers and babies to remain together 24 hours a day.

h) We encourage all mothers to breastfeed on demand.

i) We strictly prohibit the use of artificial teats, pacifiers, soothers and feeding bottles.

j) We provide follow-up support to mothers for exclusive breastfeeding up to four to six months after birth and continued breastfeeding up to two years of age. We enlist the cooperation of visiting family members to support breastfeeding mothers. Mothers are also advised on whom to contact for assistance in overcoming any problems in breastfeeding.

The operational guidelines for promotional activities are summarized in Table l. 1.

Table 1.1 Operational guidelines for breastfeeding

No.	Contact point	Activity
1.	**Antenatal check-up**	Motivate to exclusively breastfeed. Ensure rest and extra meal, include leafy vegetables and fruits. Undertake physical examination of breast and nipples and correct defects, if any
	Delivery room	Initiate breastfeeding soon after delivery. Discourage prelacteal feeds. Practice rooming-in and bedding-in. Practice exclusive demand feeding
3.	**Primary immunization sessions**	Confirm exclusive breastfeeding. Sort out practical problems. During 3rd dose, advise regarding weaning and hygienic preparation of food
	Measles immunization	Confirm continuation of breastfeeding and weaning foods. Give stress on family pot feeding
	Booster immunization/Pulse polio/Any illness	Ensure breastfeeding and adequate food intake. Stress on hygiene. Advise one extra meal for 2 weeks after an illness

Source: IAP's Policy on Infant Feeding

3. Preparing the Mother for Breastfeeding

The antenatal mother has to be motivated and prepared for breastfeeding. In the last trimester of pregnancy, breast and nipples should be examined for retracted or cracked nipples. Oiling, massaging and applying suction using 'inverted syringe technique' are useful. Primigravidas and those who have experienced difficulty with lactation previously need more care and motivation. **Antenatal mother should take more food, extra 300 kcal and 15 g protein and lactating mother should take extra 400-500 kcal and 25 g protein.** This can be achieved by one to two extra helpings of family food. She should also take plenty of green leafy vegetables, seasonal fruits and fluids.

4. Initiation of Breastfeeding

Baby must be put to breast within **half an hour after normal delivery and within four hours after caesarean sections.** Prelacteal feeds like gold rubbed in water, honey, distilled water, glucose etc., should not be given. These items will satisfy the thirst and will reduce the vigour to suck and may lead to diarrhoea and helminthic infestation. Soon after birth, the baby is awake, alert and **'biologically ready'** to breastfeed and initiation of breastfeeding is very easy. Later on, the baby goes to prolonged sleep and thereafter initiation may be difficult. Breastfeeding can be initiated even when mother is sedated or on IV fluids. In the first *2-* days, small **quantity of colostrum (10-40 ml)** that is secreted is all what the baby needs.

Colostrum is rich in protein and immunoglobulins. The mother and baby should be relaxed and comfortably postured during breastfeeding. Initially they may need some help. The baby's head may be resting on the elbow of the mother and she should support the baby with the same hand. She should also support the breast between the index finger and middle finger of the opposite hand during feeding. **'Rooming-in'** is keeping the mother and the baby in the same room, **'bedding-in'** is keeping the mother and baby in the same bed and **'mothering-in'** is keeping the baby on the abdomen of the mother. These measures ensure **mother-infant bonding** and skin-to-skin contact. Skin-to-skin contact, eye-to-eye contact and mother—infant bonding lead to successful breastfeeding and emotional adjustment. Sucking should be continued as long as the baby desires to suck. This will satisfy the sucking instinct of the baby and will express the **'hind-milk'** which is more nutritious. When sucking takes place only for a few minutes, the baby will get only the **'foremilk'.** This will satisfy only the thirst of the baby and 'hindmilk' has to be fed to satisfy the nutritional demands and to ensure more milk production. It is better to suckle from both the breasts and generally babies finish feeding by twenty minutes.

In case of twin babies, exclusive breastfeeding should be the choice. It is advisable to simultaneously feed them from both the sides or they can be put to

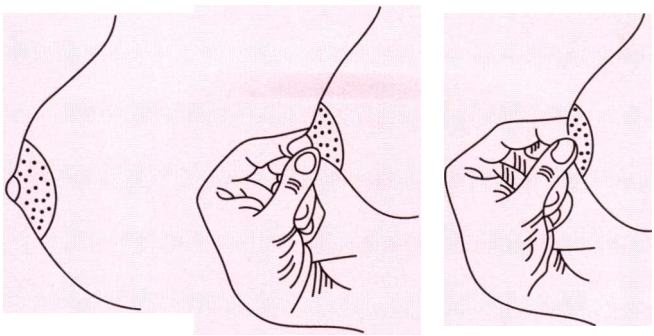
breast alternately one after the other reserving one side for each baby. If weight gain is not satisfactory, they may need extra calories and protein (refer section **1.2).**

5. Reflexes that Help in Breastfeeding

Three reflexes, namely rooting, sucking and swallowing, help the baby in breastfeeding. When the breast nipple is allowed to touch the cheek of the baby, the baby will open the mouth and initiate sucking. This is called rooting reflex. **Sucking and swallowing become coordinated by 34 weeks of gestation.** Sucking by the baby, prolactin **(milk production)** reflex and oxytocin **(milk ejection)** reflex initiate and maintain lactation in the mother. Sucking acts as the afferent stimulus for prolactin and oxytocin reflexes. Oxytocin reflex is also called **'let down reflex'.** Let down reflex will be efficient only when the mother is relaxed and comfortable. Trickling of a few drops of milk from the opposite breast while initiating feeding (let down reflex) gives a positive clue about milk production and ejection. Colostrum is replaced by 'transition milk' in a few days and later on by 'mature milk'. It gradually increases till 6 months after delivery and later plateaus off. Average quantity of milk is 500-800 ml/day.

6. Common Problems during Breastfeeding

a) **Flat or inverted nipples:** The size of the resting nipple is not important. It is just a guide to show where the baby has to take the breast. The areola and the breast tissue beneath should be capable of being pulled out to form the teat. Occasionally, on attempting to pull out the nipple, it goes deeper into the breast; this is true inverted nipple.



a) A short nipple. Is it protractile or not?

b) If you can pull it out like this, then it protracts well

c) If it goes in like this when you try to pull it out, then it is not protractile

Fig. 1.1 Testing a nipple for protractility

Nipple protractility test should be done during pregnancy if there is any doubt (Fig. 1.1). The nipple usually becomes more protractile (capable of being pulled out) as pregnancy progresses and mother should be reassured that she should be able to breastfeed.

Normally, the nipple corrects itself as the child suckles. But in a few cases, the problem persists even after that. In such cases, following inverted syringe technique should be tried (Fig. 1.2):

- Cut the nozzle end of a disposable syringe (10-20 ml).
- Introduce the piston from the ragged cut end side.
- Ask the mother to apply the smooth side of the syringe on the nipple and gently pull out the piston and let her wait for a minute.
- Nipple would then protrude into the syringe. Ask the mother to slowly release the suction and put the baby to breast; at this time it helps the nipple to erect out and baby is able to suckle in the proper position.
- After feeding, the nipple may retract back, but doing it each time before feeding over a peroid of few days will help to solve the problem.

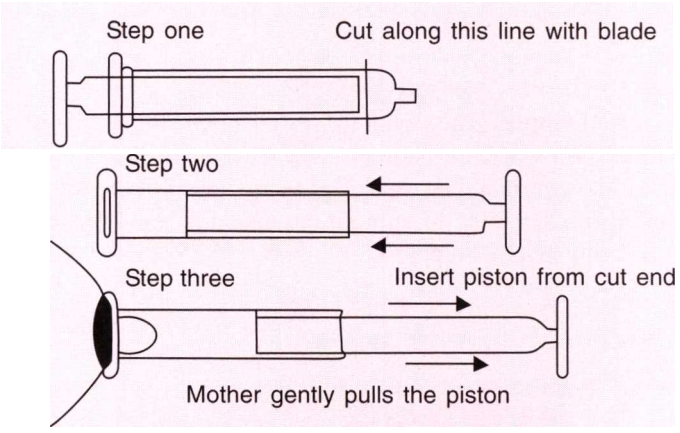
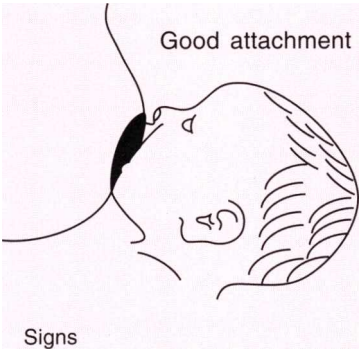


Fig. 1.2 Inverted syringe technique

b) **Fullness and engorgement of the breast:** Fullness of the breast is a frequent problem. However, milk flow continues and the baby can feed normally. If enough milk is not removed, engorgement of breast may result.

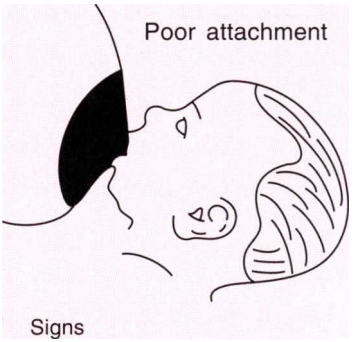
Breast engorgement is an accumulation in the breast of increased amounts of blood and other body fluids, as well as milk. The engorged breast becomes very full, tender and lumpy. The common causes of engorged breasts are: giving prelacteal feeds, delayed initiation of breastfeeds, early removal of the baby from the breast, bottle-feeding and any restriction on breastfeeding.

Engorgement may cause the nipple to flatten, making it difficult for the baby to suckle effectively. The mother too avoids feeding because of a tight and painful breast. This leads to inadequate emptying, decreased production of milk and sometimes infection. Engorgement of the breast can be prevented by avoiding prelacteal feeds, keeping the baby on mother's milk both in hospital and home, unrestricted and exclusive breastfeeding on demand, and feeding in the correct position.



Signs

- Baby's chin is close to the breast
- Baby's tongue is under the lactiferous sinuses and nipple against the palate
- Baby's mouth is wide open and the lower lip turned outwards
- More areola is visible above the baby's mouth than below it
- No pain while breastfeeding
- Baby's cheeks are full, not hollow
- Regular, slow, deep sucks



Signs

- Baby sucks only at the nipple
- Mouth is not wide open and much of the areola and thus lactiferous sinuses are outside the mouth
- Baby's tongue is also inside the mouth and does not cup over the breast tissue
- Chin is away from the breast
- It is painful while breastfeeding

Fig. 1.3 Feeding position

Once engorgement occurs, the baby should be breastfed frequently followed by expression of breast milk. The following measures will help relieve the problem usually within 24 to 48 hours:

- Applying moist heat to the breast 3 to 5 minutes before a feed, followed by gentle massage and stroking the breast towards the nipple
- Expressing enough milk to soften the areola, enabling proper attachment
- Feeding frequently, every 2-2.5 hours or sooner at least for 15-20 minutes each side after milk let down has occurred
- Feeding the baby in a quiet, relaxing place
- Paracetamol may be needed to relieve the pain in the breast

c) **Sore nipple and cracked nipples:** If a baby is not well attached to the breast (Fig. 1.3) he or she sucks only the nipple (poor attachment). It is the most common cause of sore nipples in the first few days. If feeding continues in a poor position, it may lead to a cracked nipple because of physical trauma to this area and later to mastitis and breast abscess. Oral thrush in the baby is another important cause of sore/cracked nipples, but it usually develops after a few weeks of birth. To prevent soreness and cracking of the nipples, attention should be paid to teaching correct feeding positions and techniques to the mother (Fig. 1.3).

If there is pain in the nipple area during breastfeeding, mother should wait until the baby releases the breast, or insert her finger gently into the baby's mouth to break the suction first, so as to avoid injury to the nipple. Then the mother should be helped with attachment and repositioning the baby, so that it will not cause pain. This is the test of correct attachment.

Breastfeeding should be continued on the affected breast as it usually heals after correcting the sucking position. Medicated creams are best avoided as they may worsen the soreness and draw away the attention from the crucial issue.

If the infant has oral thrush, 1 % gentian violet should be applied over the nipple as well as inside the baby's mouth. If the oral thrush in the baby leads to maternal fungal infections and causes an itching in mother's breast, then give systemic antifungal drugs to the mother (miconazole or fluconazole tablets 250 mg QID for 10 days).

For cracked nipples, treatment consists of feeding in correct position, washing the nipple once daily only with water, and exposure of nipple to air and sun as much as possible. Application of hindmilk drop on the nipple after each feed may also help. If mother is not able to feed because of pain she should express milk frequently.

d) **Blocked duct:** If the baby does not suckle well on a particular segment of the breast, the thick milk blocks the lactiferous duct leading to a painful hard swelling. This 'blocked duct' is not associated with fever.

Treatment requires improved removal of milk, and avoiding any obstruction to milk flow. Ensure that the infant is sucking in good position. Some authors recommend holding the infant with the chin towards the affected part of the breast, to facilitate milk removal from that section, while others consider that generally improved attachment is adequate.

Explain the need to avoid anything that could obstruct the flow of milk, such as tight clothes and pinching or scissoring the breast too near the nipple. Encourage the mother to breastfeed as often and as long as her infant is willing, with no restrictions, including night feeds. Suggest that she apply wet heat (e.g., warm compresses or a warm shower) over the breast.

Occasionally, these techniques do not relieve the mother's symptoms. This may be because there is particulate matter obstructing the duct. Massage of the breast, using a firm movement of the thumb over the lump towards the nipple may be helpful. However, this should be done gently, because when breast tissue is inflamed, massage can sometimes make the situation worse.

Unfortunately, blocked ducts tend to recur, but once mother knows what they are due to, and how to treat them herself, she can start treatment early and avoid progression to mastitis,

e) **Mastitis and abscess:** If the blockage of the duct or engorgement persists, infection may supervene. The breast becomes red, hot, tender and swollen. Mastitis must be treated promptly and adequately. If treatment is delayed or incomplete, recovery is less satisfactory. There is an increased risk of developing breast abscess and relapse. A breast abscess may occur sometimes without mastitis.

The main principles of treatment are:

- Supportive counselling
- Effective milk removal
- Antibiotic therapy
- Symptomatic treatment

Mastitis is a painful and frustrating condition, and it makes many mothers feel very ill. In addition to effective treatment and control of pain, she needs emotional support. She may have been given conflicting advice from health professionals. She may have been advised to stop breastfeeding, or given no guidance either way. She may be confused and anxious, and unwilling to continue breastfeeding.

She needs reassurance about the value of breastfeeding; that it is safe to continue; that milk from the affected breast will not harm her infant; and that her breast will recover both its shape and function subsequently. She needs encouragement and effort to overcome her current difficulties.

She needs clear information and guidance about all measures needed for treatment, how to continue breastfeeding or expressing milk from the affected breast. She needs follow-up to give continuing support and guidance until she has recovered fully.

Effective milk removal is the most essential part of treatment. Antibiotics and symptomatic treatment may make a woman feel better temporarily, but unless milk removal is improved, the condition may become worse or relapse despite the antibiotics.

Help the mother to improve her infant's attachment at the breast. Encourage frequent breastfeeding, as often and as long as the infant is willing, without restrictions. If necessary, express breast milk by hand or with a pump until breastfeeding can be resumed.

Antibiotic treatment is indicated if either:

- Cell and bacterial colony counts and cultures are available and indicate infection, or
- Symptoms are severe from the beginning, or
- A nipple fissure is visible, or
- Symptoms do not improve after 12-24 hours of improved milk removal

If possible, milk from the affected breast should be cultured and the antibiotic sensitivity of the bacteria determined. To be effective against *Staph, aureus,* a beta-lactmase resistant antibiotic is needed. For Gram-negative organisms, cephalexin or amoxycillin may be the most appropriate. The antibiotic must be given for an adequate length of time (10-14 days). Shorter courses are associated with a higher incidence of relapse.

Pain should be treated with an analgesic. Ibuprofen is considered the most ffective, and it may help to reduce inflammation as well as pain. Paracetamol is an appropriate alternative.

Rest is considered essential and should be in bed if possible. Helping the woman to rest in bed with the infant is a useful way to increase the frequency of breastfeeds, and thus improve milk removal.

Other measures recommended are the application of warm packs to the breast, which both relieve pain and help the milk to flow. Also ensure that the woman drinks sufficient fluids. Incision and drainage should be done if abscess forms. Breastfeeding should be restarted from the infected breast as soon as possible.

7. How often to breastfeed?

Exclusive demand feeding is the ideal schedule to follow. There is no 'tailor made schedule', as milk production, sucking habits, stomach capacity etc., vary from baby to baby. Practise frequent breastfeeding initially and allow 'self-regulation' by the baby. The mother can soon find out the average time interval the baby will rest after a feed. She can adjust her rest period in between. Almost all mothers can be relied upon to practise demand feeding. She will know why her baby is crying; e.g., is it to sleep? Is it due to illness? etc. There is no need to give boiled and cooled water or fruit juice in between while the baby is on exclusive demand feeding. A full-term appropriate for gestational age (AGA) baby who is thriving well does not need multivitamin drops as well.

8. Burping after feeding

Babies tend to take in a lot of air during feeding. This will lead to abdominal distension, colics, regurgitation etc. To get rid of this, the mother has to do 'winding' or burping. The baby can be put on the left shoulder, the head has to be supported with mother's left hand and then with the right arm support the buttocks and gently pat on the baby's back with the right hand. Slowly air will escape

and the baby will become comfortable. Burping can also be done in other positions, e.g., place the baby prone in the mother's lap and gently pat on the back.

9. Positioning of babies after feeding

Babies can be put in the right lateral position after feeding. This will prevent aspiration. **Prone position is not currently recommended as it is found to be associated with higher incidence of sudden infant death syndrome** (SIDS) in some studies.

10. How to know whether breast milk is sufficient or not?

Most of the mothers and grandmothers are worried whether breast milk is sufficient or not? They may put pressure on the doctor to prescribe an infant milk substitute (IMS). When mothers come complaining that breast milk is not sufficient, a patient listening is required. The following points will help in decision making. Is the baby frequently passing plenty of pale-coloured urine? Is the baby passing 1-6 liquid stools per day ? Is the baby gaining weight? If yes, the baby is getting enough milk.

Next ask whether the mother is offering other feeds or feeding bottle in between. This preload will decrease the vigour of sucking and will lead to incomplete emptying of breast and suppression of lactation. Feeding bottles cause **'nipple confusion'.** Sucking from the bottle is a totally different and at the same time a more easier art compared to breastfeeding. When both are offered, babies who generally tend to be lazy, resort to the more easier technique of bottle feeding. If baby requires mother's milk and bottle feeding, the complete emptying of the breast is very essential before the bottle feed is started.

Watching the baby feeding is the next step. Wrong posturing and wrong techniques must be corrected. The baby should suck on the areola and not on the nipple. Make use of the rooting reflex and ensure optimum attachment to the breast. Breast engorgement, sore nipples, retracted nipples etc., may need treatment. Examine the baby for local problems like cleft palate, prematurity, oro-motor dysfunction and see if the baby can suck on the areola. In babies with cleft palate, expressed breast milk (EBM) should be given using a palada (gokarnam), long spoon or long dropper. A feeding plate that covers the defect can also be used during feeding. The dental surgeon can easily make a feeding plate for the baby. Also look for the 'let down reflex'. When present, it ensures optimum milk production and ejection. The mother must be reassured and motivated. The services of the 'support group' comprising doctors, nurses, voluntary agencies, satisfied mothers etc., as per BFHI guidelines, may be utilized for this. Alleviate stress, anxiety and embarrassment in the mother. Call them again for follow-up and watch the progress. It will be gratifying to see the improvement.

11. Weight Gain in Exclusively Breastfed Babies

The previous growth charts were prepared on babies who had dual feeding. They steadily gain weight and attain normal development. The slogan **"breast milk for brain growth and cow's milk for body growth"** is worth stressing. And now WHO has issued the recent growth charts which are based on studies done on exclusively breastfed babies. These growth charts show how children should grow.

12. How long to breastfeed?

Breastfeeding should be continued well into the second year of life. It is better to breastfeed till two years of life; the period of maximum brain growth and myelination. After 4-6 months of age, weaning foods should be offered in addition to breastfeeding.

13. Whether to breastfeed when the baby or the mother is ill?

Breastfeeding should be continued when the baby is ill. It should be discontinued only if there are gastrointestinal contraindications to oral feeding. It can be given during infections like rhinitis, viral fever, diarrhoeal diseases, respiratory infections, asthma etc. It is the most easily digestible food for the ill baby. It will be the best pacifier to the sick baby and it often acts as a life saviour to many babies. It will satisfy the nutritional and fluid demands and will offer anti-infective and immunological factors. Babies may suck with less vigour and so they may be offered more frequent feeds. **Expressed breast milk (EBM)** should be given if the baby cannot suck. This will prevent suppression of lactation. Babies with congestive heart failure do very well on EBM as it has a very low sodium content. A few drops or small quantities of EBM given to sick babies on IV fluids has been shown to paint the gut with immunological factors, to promote gut function and to reduce the incidence of necrotising enterocolitis (NEC).

Breastfeeding can be continued during most of the maternal illnesses including viral fever, mastitis, breast abscess, UTI, TB, hepatitis B etc. If the mother is an open case of TB, she should be initiated on chemotherapy and the baby should be put on chemoprophylaxis. In India, where TB is rampant and the chance for drug resistance is high, it is better to give INH and rifampicin instead of INH alone. After 3 months, ensure that mother is sputum negative and do Mantoux (Mx) test on the baby. If Mx test is negative, stop drugs and give BCG. If Mx test is positive, continue treatment for a total of 6-9 months. In hepatitis B, the baby can be given hepatitis B specific immunoglobulin, followed by hepatitis B vaccination. In AIDS, as long as there is no caretaker or agency to take up the feeding and care of the baby, breastfeeding may be continued. This is the only possible option in many cases even though there is a chance of HIV transmission through breast milk. **The chance for perinatal transmission of AIDS is almost 30%**. In

mastitis and breast abscess, temporary stoppage and expression of breast milk from the affected side may be required. In postpartum psychosis, breastfeeding can be allowed under supervision. In sore nipples, ensure proper attachment of the baby to the areola, apply milk or oil on the nipple, expose the nipple to air and treat oral thrush in the baby by clotrimazole mouth paints. It can also be applied on the nipple.

14. Contraindications to Breastfeeding

Even though there are a few temporary contraindications to breastfeeding, permanent contraindications are very rare. **Congenital lactose intolerance and galactosaemia are contraindications.** These are extremely rare conditions and such babies cannot be given animal milk also. In acquired lactose intolerance which is temporary, breastfeeding can be continued. Similarly, breastfeeding can be continued in the so-called 'breast milk jaundice' thought to be due to 3-alpha 20-beta pregnanediol which may inhibit bilirubin conjugation. The baby will improve with phototherapy. **Intake of antimalignant drugs, antithyroid drugs and antipsychotic drugs (lithium) are considered contraindications to breastfeeding.**

15. Medications to the Lactating Mother

All drugs taken by the mother will be excreted in breast milk, most of them in low concentrations up to less than 1%. Anticancer drugs cause immunosuppression and affect neonatal growth. Antithyroid drugs and radioactive iodine appear in higher concentrations than in plasma and cause damage to thyroid gland in the infant. **Propylthiouracil is found safe.** Dicumarol can cause bleeding in the infant. **Warfarin is safe.** Cimetidine appears in higher concentrations and can cause suppression of gastric acidity and stimulation of CNS. Ergot therapy can cause ergotism in the baby that manifests as vomiting, diarrhoea, collapse and convulsions. **Oral pill, thiazides, pyridoxine, nicotine and bromocriptine suppress lactation.** Laxatives taken by the mother can cause diarrhoea in the baby. Milk of magnesia, liquid paraffin and glycerine suppositories are safe. Antibiotics are secreted in breast milk and can cause GI upset and diarrhoea in the baby.

16. The Options to a Working Mother

Employment is a bottleneck in exclusive breastfeeding. A working mother can have several options and can select any as per the **'cafetaria approach'.**

a) Exclusively breastfeed as long as possible until a few days prior to resuming work.
b) Extend maternity leave till 4-6 months or avail half pay or loss of pay leave if possible.
c) Take the baby to the day care centre or creche at the work place. Take initiative to start one if there is no such facility.

d) Change the work place nearer to the house or change the residence nearer to the work place.

e) Express and keep EBM to be given while the mother is away, keeping in mind the hygiene factors, refrigeration and pasteurization techniques are prudently followed.

f) Feed before leaving for work, on returning from work, during nights and during holidays.

g) Around 4^{lh} month onwards, start giving complementary foods a few days before joining work.

17. The Advantages of Breastfeeding

The numerous advantages of breast milk are beyond description and understanding.

a) The **physical** benefits are optimum fluidity and warmth.

b) It is very **economical.** The approximate cost to artificially feed a baby less than 6 months of age is estimated to be more than one-third of the average family income, i.e., almost more than the per capita income. We are unable to afford this at national level, community level or at family level.

c) It is very **convenient.** There is no need to carry or sterilize utensils. It can be made available anywhere at any time.

d) It is very **physiological.** It is the sweetest milk with high lactose content. The protein is easily digestible. The lipids are rich in essentia] fatty acids, **long chain polyunsaturated fats (LCP),** phospholipids and prostaglandin precursors. It supplies enzymes like amylase, lipoprotein lipase, bile salt stimulated lipases (BSSL), oxidases, lactoperoxidases, leucocyte myeloperoxidase etc. These enzymes increase digestibility and also act as defence against microbes. It also contains growth regulating factors, growth promoting factors and growth modulators. LCPs promote brain growth and reduce dyslexia and hyperactivity.

e) **Biochemically** it is superior. **The protein is mostly whey protein (80%) rich in a-lactalbumin and lactoferrin** and **the rest is casein (20%) Lactablumin is rich in tryptophan which is the precursor of serotonin which plays an important role as neurotransmitter. Lactoferrin ensures absorption of iron and zinc and it is bacteriostatic as well.** It binds iron and makes it unavailable to the bacteria. Alpha-casein and lactoglobulin, which are allergens, are absent in human milk. Even though protein is lower in breast milk, non-protein nitrogens are high The non-protein nitrogen in breast milk plays a significant role in the growth and development of the infant. It is also rich in binding proteins that bind thyroxin, B_p, vitamin D etc. **The calcium-phosphorous ratio is more than 2** and it ensures calcium absorption. **Lactose promotes calcium and magnesium absorption.**

The solute load is low due to low level of protein, and certain minerals. It ensures gentle load on immature infant's kidney. At the same time there is provision of optimum vitamins and mineral essential for healthy growth.

f) **Microbiologically** it is sterile with least chance of contamination. **Lactoferrin is bacteriostatic and inhibits *E coir,* it binds iron and makes it unavailable to *E coli.*** Peroxidases and lipases kill bacteria. **Bile salt stimulated lipase** (BSSL) **kills amoeba and Giardia. Para amino benzoic acid (PABA) is important in protection against malaria.** The relative deficiency of PABA in human milk leads to supression of parasites to subclinical levels and sufficient antigenic stimulus for immune response.

Transfer of maternal antibodies and T lymphocytes may also offer some protection against malaria. The bifidus factor and acidic pH associated with human milk leads to colonisation by *Lactobacillus bifidus*. Breastfeeding also facilitates the exchange of microbes between mother and infant via skin contact and exposure to microbiota in the immediate environment. In breast-fed infants bifidobacteria constitute from 60 to 90% of the total faecal microbiota, while lactobacilli comprise less than 1 %.

g) **Immunologically,** it is extremely safe and is non-allergenic. It supplies passive immunity. Macrophages, lysozymes and complements offer immunity to the baby. It also supplies acute phase reactants. Nutritional composition of breast mik supports the gut microflora which plays essential role in enhancing the immunity of the infants. It contains immunoglobulins, secretory components and secretory IgA (SIgA). SIgA offers surface protection to the respiratory and GI tracts. Immunoglobulins other than SIgA are generally split up in the gut. **SIgA are produced in the mammary gland by plasma cells that originate from immunocompetent lymphoid tissue, namely, gut associated lymphoid tissue (GALT) and bronchus associated lymphoid tissue (BALT) by virtue of enteromammary and bronchomammary axes.** IgG and IgM levels become undetectable in the second month of lactation. Secretory IgA may resist proteolytic degradation in the neonatal gut and may offer some protection. Breast milk supplies **T** and B lymphocytes. **T** lymphocytes are responsible for transfer of immunological memory.

The **'bioactive factors'** in milk are proteins like lactoferrin, non-protein nitrogen like nucleotides, enzymes, hormones, growth factors, factors for host defence, oligosaccharides, mucins, probiotic substances and polyamines. The **bifidus factor** promotes the growth of lactobacilli. **Polya-mines like spermine, spermidine and putrescine promote cell growth and differentiation. Putrescine is a precursor of gamma amino butyric acid** (GAB A). GAB A is an inhibitory neurotransmitter.

h) **Psychologically,** it ensures emotional stability and personality development due to close contact with the mother and **mother-infant bonding.**

i) It also ensures a lot of **maternal benefits.** It helps to decrease postpartum bleeding and also helps in the involution of uterus by virtue of oxytocin. It helps to burn off extra fat that has accumulated during pregnancy under the effect of various hormones. It may also decrease the incidence of breast and ovarian cancers.

j) **Epidemiologically** it decreases morbidity and mortality. It is estimated that a **breastfed baby is 14 times less likely to die from diarrhoea, 4 times less likely to die from respiratory diseases** and **2.5 times less likely to die from other infections than a non-breastfed infant.**

18. The Factors in Breast Milk that Promote Growth and Mental Development

The current slogan 'breast milk for brain growth and cow's milk for body growth' stresses the importance of breastfeeding in mental development. Breast milk plays a role in various stages of cell division to infant behaviour. It contains amino acids specific for brain development. It is **rich in sulphur-containing amino acids. Cysteine:methionine ratio is high** and this compensates for low cysteine-methionine conversion which is essential for CNS development. **It is rich in taurine which is an important neurotransmitter and neuromodulator for brain and retina.** It contains low amounts of aromatic amino acids like tyrosine and phenylalanine that are less utilized by preterm infants. It offers **a high tryptophan to neutral amino acid ratio** which controls brain serotonin synthesis. The **amino acids are mostly in 'trans' form** whereas in microwaved formula they change to 'cis' form which are neurotoxic.

Breast milk contains essential fatty acids (EFA) and Long Chain Poly unsaturated Fatty Acids (LCPUFAs) in a different ratio which depends on the diet of lactating mother. **Brain lipids are mostly long chain polyunsaturated fatty acids** (LCPs) which are the result of metabolic conversion of essential fatty acids (linoleic and linolenic acids).Linoleic acid (of)) is a precursor of arachidonic acid and linolenic acids for DHA (a8). LCPUFAs are playing very important biological role in infancy. Thus arachidonic acid and docosa hexaenoic acid (DHA) are important in neural and visual development. Arachidonic acid is the precursor of prostaglandin playing a crucial role in immunity and inflammatory modulation. The optimal balance of LCPUFAs (omega 3 & 6 series) are influencing the right immune response maturation in infants. Antenatally placenta is the source of these fatty acids, whereas breast milk is the source after delivery. DHA levels tend to be very low in formula-fed infants due to low conversion of linolenic acid into DHA in infancy. Thus the additional supplementation of balanced LCPUFAs is recommended for formula-fed infants. The EFA are also important for myelination of brain. Palmitic acid in beta position ensures adequate fat absorption from the gut.

Presence of choline, acetylcholine, phospholipid precursors and carnitine ensures optimum metabolism and brain development. **Carnitine levels are found to be low in preterms** and supplementation is required.

Breast milk is a rich source of hormones and **growth factors** like thyroid stimulating hormone (TSH), thyroxine, growth hormone releasing factor (GHRF), insulin, somatostatin, epidermal growth factor, prolactin, neurotensin, nerve growth factor (NGF), trophic factors and beta casomorphin. Human beta casomorphin is a CNS growth factor and it also mediates high concentrations of hormones in breast milk than in maternal serum. NGF leads to dendritic arborization. Enzymes like lysozyme, peroxidase and xanthine oxidase that promote cell maturation are found to be more in colostrum. Breast milk ensures better oxygen saturation and increases the bioavailability of trace elements like copper, magnesium, cobalt, selenium, iron, zinc etc. It contains less poisonous residues than cow's milk which are neurotoxic like chromium, aluminium, manganese etc. Exclusively breastfed preterms have shown higher IQ scores and lesser neurological sequelae. They are better adjusted and have better cognitive abilities.

19. Comparison between Human Milk and Cow's Milk

Milk is species specific. Cow's milk with its high protein and solute load is suitable for the calf which is ambulant and self-feeding within a few hours after delivery. Human milk with its low protein and solute load is suitable for the slower somatic growth in the baby and for rapid brain growth in the first two years of life. Both of them contain equal calories and hence it is not advisable to dilute cow's milk, but at the same time more water is needed to excrete the high solute load in cow's milk. American Academy of Pediatrics (AAP) and European Society for Paediatric Gastroenterology and Nutrition (ESPGAN) are not recommending unmodified cow's milk for infant feeding. The major differences between human and cow's milk are given in Table 1.2.

a) **Protein:** Protein content in cow's milk is three times more than that in human milk. However, it is biochemically different and less digestible. It forms thick curds. Casein content is four times more in cow's milk which requires more HC1 for digestion.
   i) **Casein:** Alpha casein is maximum in cow's milk, whereas beta casein is more in human milk. The former may act as an allergen.
   ii) **Whey protein:** It is four times more in human milk than in cow's milk. In human milk, it is mainly **lactalbumin** and **lactoferrin** (80%). In cow's milk, it is mainly lactoglobulin which is negligible in human milk. Hence it is one of the causes of intolerance to cow's milk. Lactoferrin is bacteriostatic and it increases iron, zinc and magnesium absorption.

      It binds iron and makes it unavailable to *E.coli.* Among the amino acids, **glutamic acid is maximum and glycine is absent in human milk.** Glycine is a non-essential amino acid.

Table 1.2a Comparison—human and cow's milk (100 ml)

Item	Human milk	Cow's milk
Non-protein nitrogen	0.2 g	0.03 g
Protein	1.1 g	3.0 g
Casein: Whey	40:60	80:20
	(lactalbumin & lactoferrin)	(lactoglobulin)
Lactose	**7 g**	**4.5 g**
Fat	3.8 g	3.7 g
EFA	13%	2%
P/S ratio	1.2:1	1:2
Ash/minerals	0.25 g	0.75 g
Ca:P ratio	> 2	< 2
Sodium	0.7 mEq	2.2 mEq
Potassium	1.4 mEq	
Vit. K	15 ng	60 ng
Vit. E	2 mg	0.4 mg
Osmolarity	7.9 mOsm	22.1 mOsm
Energy: Protein ratio	70:1	25:1
Calories	67 cal	67 cal

iii) **Other components in human milk:** Albumin, essential amino acids, lysozymes, immunoglobulins like IgG, IgM, SIgA, acute phase reactants like alpha-1 antitrypsin, alpha-1 antichymotrypsin, binding proteins of thyroxine, corticosterol, vitamin D, folate and B_{12}, secretory components, growth modulators, growth factors, digestive enzymes like milk lipases, amylase etc., are present in human milk. The peroxidase activity is due to the leucocyte myelo-peroxidase and lactoperoxidase. **In frozen banked breast milk, oxidation can lead to cholesterol oxides that are angiotoxic.**

Milk lipases kill amoeba and giardia: Milk lipases are of two types: (a) Lipoprotein lipase (LL) and (b) Bile salt stimulated lipase (BSSL). They facilitate fat absorption and hydrolyse bacterial lipids. Lingual lipase that increases on sucking is not active in gavage feeding and hence milk lipase is very important. **Bifidus growth-promoting factor promotes lactobacilli. Lactobacilli and lactic acid that help in digestion are called probiotic substances.**

SIgA is a molecule of IgA bound to two molecules of secretory component and it is resistant to proteolysis in gut and offers surface protection to GI and respiratory tracts. Growth factors are of two types: **(1) Growth regulating factors**—somatostatins and (2) **Growth mediating factors**—somatomedins A, C and insulin like growth factors—IGF1, IGF11. (IGF1 and somatomedin C are identical.)

T ble 1 2b Composition of nutrients in human and cow's milk (per 100 ml)
3 6

Fatty acids	Human milk	Cow's milk
Fat (g)	3.8	3.7
Linoleate (g)	0.51	0.07
Protein (g)	1.2	3.3
Carbohydrate (g)	7.0	4.8
Minerals (g)	0.21	0.7
Vitamin A (ng)	53	34
Vitamin D (IU)	0.4-10	0.3-4
Vitamin E (mg)	0.2 0.1	
Vitamin K, (ng)	0.3	0.7
Vitamin C (mg)	4.3	1.8
Thiamine (BJ (ng)	16	42
Riboflavin (B_2) (ng)	43	157
Niacin (PP) (|xg)	172	85
Vitamin B_6 (ng)	11	48
Folic acid (ng)	0.18	0.23
Pantothenic acid (mg)	0.25	0.34
Vitamin B_{12} (ng)	0.18	0.4
Biotin (ng)	2.0	22
Choline (mg)	1.3	1.2
Inositol (mg)	45	8
Taurine (mg)	5	0.5
Carnitine (mg)	0.8	1
Sodium (mg)	16	58
Potassium (mg)	55	137
Chloride (mg)	43	103
Calcium (mg)	33	125
Phosphorus (mg)	15	96
Magnesium (mg)	4	12
Iron (mg)	0.05-0.15	0.1
Iodine (|ig)	7	21
Copper (mg)	0.04	0.03
Zinc (mg)	0.53	0.38
Manganese (ng)	traces	traces

Table 1.2c Comparison of fatty acid profile of human and cow's milk

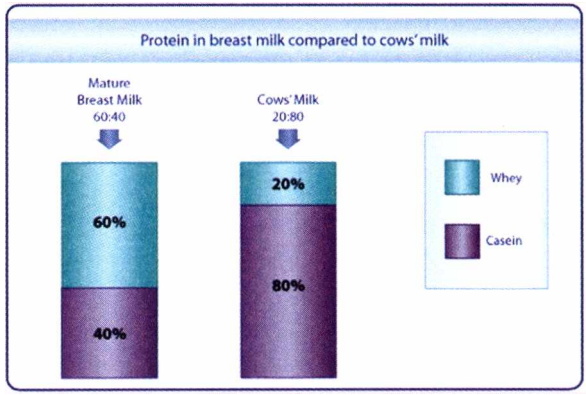
Fatty acids	Human milk	Cow's milk
Saturated		
4:0 butyric		3.0
6:0 caproic		1.0
8:0 caprylic	0.19	1.0
10:0 capric	1.1	3.0
12:0 lauric	4.8	2.0
14:0 myristic	7.2	10.7
16:0 palmitic	23.4	26.3
18:0 stearic	8.0	12.1
Monounsaturated		
16:1 palmitoleic	3.4	4.5
n-9 18:1 oleic	35.3	33.3
Polyunsaturated		
n-6 18:2 linoleic	13.4	2.0
18:3 y-linoleic	0.17	
20:4 arachidonic	0.45	0.1
n-3 18:3 a-linolenic	0.94	1.0
20:5 EPA	0.18	
22:5 DPA	0.17	
22:6 DHA (cervonic)	0.3	

Non-protein nitrogen: The non-protein nitrogens in human milk are urea, amino acids, peptides, nucleic acids, choline, creatinine, creatine, uric acid, ammonia, polyamines, nucleotides, N-acetyl glutamine, N-acetyl neuraminic acid etc. These are called **bioactive factors,** which are lacking in cow's milk.

b) **Lipids:** Lipids present in human milk are unsaturated fat, essential fatty acids, prostaglandin precursors, fat-soluble vitamins, steroids, LCPs and phospho-lipids. Milk fat depends upon maternal fat intake. Vegetarian diet increases polyunsaturated fatty acids (PUFA) and sea fish intake increases the levels of eicosa pentaenoic acid (EPA) and docosa hexanoic acid (DHA). **Human milk also has high carnitine** content, which increases energy metabolism by mitochondrial oxidation and transport of EFA. **Cow's milk contains mostly saturated fat.** The polyunsaturated to saturated fat (P/S) ratio is **1.2:1** in breast milk compared to 1:2 in cow's milk.

c) **Carbohydrate: Human milk is the sweetest milk** closely followed by ass milk, due to lactose content. Lactose is the main carbohydrate in breast milk. Breast milk which plays a role in growth of healthy intestinal flora, enhanced

Table 1.2d Ratio of whey and casein in breast milk and cows' milk



- Breast milk is whey-protein predominant. The average whey/casein ratio in breast milk is 60:40 (i.e., whey proteins represent 60% of the total protein and casein only 40%). The major whey proteins in human milk are:
  - o-lactalbumin (about 40%)
  - Lactoferrin (30%)
  - Immunoglobulins (IgA) (15 to 20%)

  Serum albumin and lysozyme are also present in whey proteins
- Cows' milk is casein-predominant. The average whey/casein ratio in cows' milk is in the range of 20:80 (i.e., whey proteins represent only 20% of the total protein and casein 80%). The major whey proteins in cows' milk are:
  - p-lactoglobulin (60%; not present at all in human milk)
  - a-lactalbumin (20%).
  - Immunoglobulins are of the IgG type, and lactoferrin and lysozyme are present in only trace amounts.

calcium absorption. Only mammalian breast tissue can synthesize lactose and hence **milk is the only natural source of lactose.** It is a disaccharide made of glucose and galactose, and thus it is the only source of galactose for optimal brain development of growing infants. Apart from lactose, approximately 15% of carbohydrates are presented by galacto-oligosaccharides (GOS), which contributes to development of microflora and digestion. Though cow's milk also has lactose as the main carbohydrate, the content is only half of human milk and % GOS is negligible. Thus the usage of cow's milk in infancy does not support proper development of healthy intestinal flora,

d) **Vitamins: Human milk is a good source of vitamins except vitamin K and D,** especially in vitamin D - deficient mothers. Vitamin K is synthesized by

intestinal flora and vitamin D is synthesized in the skin from cholesterol with the help of UV light. Cow's milk is deficient in more number of vitamins,

e) **Minerals:** In breast milk though minerals like iron, zinc etc., are present only in small quantities, the bioavailability is much better than cow's milk due to the presence of carrier proteins like lactoferrin. Cow's milk has excess of sodium, potassium and chloride and thus increasing the solute load.

20. Artificial Feeding

When the mother is unavailable, critically ill or no more, and in any case breast feeding is not possible, the baby may have to be fed artificially (infant formula or unmodified bovine milk). The decision of choosing the formula feeding can be done only by the healthcare professional, taking into consideration the socio-economical status of the family. This advice has to be supported by detailed information regarding the hygienic preparation in right proportion. The caretaker should be reliable and should be informed of the consequences of this decision. Full-strength formula (1:1) is prepared by adding one level measure of powder to one ounce (30 ml) of water. 150-165 ml/kg/day milk can be given in 6-8 feeds. However the exact information on the label and pack should be carefully read before the usage of the content. Ensure the usage of clean cup and spoon or gokarnum (palada) instead of feeding bottles.

There are different options of formula available. Most of the formulae are made of bovine milk which is specially modified to suit the infant's physiological requirements. Usually the starter formula (No. 1) is designed especially for young infants from 0-6 months according to their requirements. After six months when the infant is introduced to complementary foods, follow up formula (No. 2, 3) is advised as a main fluid part of the infant's diet.

There is a range of formulae for infants with special requirements. Hypoallergenic formula is based on cow's milk hydrolysed protein which depends on the level of hydrolysation and can prevent or treat the allergic/atopic diseases. Formula based on the fermented bovine milk ensures better digestibility and prevention of gastrointestinal infections. For infants with lactose intolerance (primary/secondary) special low lactose, lactose-free formula have been designed. For premature, low birth weight infants there are special infant formulae with medium chain triglycerides, LC PUFAs and the nutritional composition as per the recommendations. Soya-based formulae due to nutritional incompleteness and high risk of cross-allergy (30%) are not advisable for usage in feeding premature infants, term infants below 5-6 months of age, and for allergy prevention and treatment cases. The enormous cost, excess, deficiency and omissions in formulation, the wide variations in preparation and dilution, and chances of bacterial contamination during mixing make formula feed undesirable. In instances where formula feeding is required, one has to look for a formula more closer to breast milk, fortified with all essential vitamins and minerals. Also the addition of

functional benefits of infant formula might be of importance in special conditions. When formula feeding is not feasible, cow's milk could only be the last alternative. This is because of the excess of few minerals like sodium, potassium and less of the important nutrients like iron, vitamins C etc. **Also it is not advisable to dilute cow's milk as it will lead to overdilution and very low calories.** In that case, boiled and cooled water should be offered in between to reduce solute load and to prevent constipation. Thirst is the best guide to know how much water is to be given. In diluted cow's milk, the net nitrogen and calories will be very low and child is more likely to develop malnutrition. However, in the first week of life, 1:1 diluted, in the second week 2:1 diluted, in the third week 3:1 diluted and in the fourth week onwards undiluted cow's milk can be given **if and only if absolutely indicated under lower socioeconomic conditions.** Baby must be given boiled and cooled water in between when cow's milk is given. Formula milk is more digestible than any form of unmodified bovine milk as the formula composition is modified to suit the infant's needs.

21. The Adverse Effects of Artificial Feeding

Malnutrition due to dilution and infection due to contamination are the most important side effects of all types of artificial feeding. Untreated bovine milk feeding has more potential risk for allergy development. In addition to allergies, the high sodium content in cow's milk may lead to salt-sensitive hypertension in susceptible individuals. High saturated fat content may be the forerunner of hypercholesterolaemia. High levels of low density lipoproteins (LDL) and saturated fat may lead to coronary artery disease and cerebrovascular disease. Iron deficiency is more common in the unmodified bovine milk feeding. Iron deficiency that occurs in the developing age may decrease D_2 (dopaminergic) receptors and may produce irreversible behavioural changes and dyslexia. Cow's milk protein is found to have dopaminergic effects. Higher incidence of multiple sclerosis and schizophrenia have been reported in unmodified bovine milk feeding than in breastfed individuals.

Higher incidence of diabetes mellitus due to beta cell destruction, dyslexia and lowered IQ due to deficiency of LCPs also have been described. In small infants as the gut is not mature enough, there is chance for unsplit protein to escape into the circulation and cause sensitization. Cow's milk protein intolerance (CMPI) is usually due to lactoglobulin or alpha-casein and may cause diarrhoea, respiratory allergy and eczema. The **osmolarity** of 221 mOsm/L in cow's milk as against **79 mOsm/L in human milk** will increase additional load on the immature kidney. Due to the increased demand for water to excrete this solute load, there is chance for dehydration and constipation. Iron deficiency is the rule in those on cow's milk. This is due to poor availability and absorption of iron and due to enteric loss of blood. Low vitamin C and lactoferrin and high phosphate also lead to decreased iron absorption.

Lack of breast milk and subsequent weaning on to starch-based food with high phytate also lead to mineral deficiency. Low lactose leads to reduced calcium and magnesium absorption. High phosphate in cow's milk reduces calcium absorption and there is increased chance for hypocalcaemic tetany and convulsion. Saturated fat is more in cow's milk and essential fat is only 2% compared to 13% in human milk.

High levels of tyrosine and aromatic amino acids that are not utilized can lead to azotaemia and acidosis. It will also increase energy wastage in the form of very high specific dynamic action (SDA) up to 30% in comparison to 5% with breast milk. **Energy to protein ratio is 70 in human milk compared to 25 in cow's milk.** Pesticide residues up to 4-fold and toxic metal residues up to 8-fold than acceptable levels may be present in cow's milk, e.g., cadmium which is neurotoxic, arsenic which is respiratory, CNS and skin toxic and lead which is neurotoxic and haematotoxic. Manganese is toxic to basal ganglia and can lead to dyslexia. Fungal residues and carcinogens may also be present in cow's milk. Polychlorinated biphenyl (PCB) residues are also more in cow's milk. **The chances of necrotising enterocolitis (NEC) is also more among those on artificial feed.** The flora in such infants is unfavourable and is mainly coliforms.

Usage of cow's milk should not be advised except in situations when the mother is away, no more or it is unavoidable. If animal milk is given, the solute load increases. In buffalo milk, the excess saturated fat needs to be removed by separating the cream. Commercial infant milk formula is generally not advised due to high cost and the usual tendency to give it very dilute. However, infant milk substitutes may be prescribed in severely malnourished babies or conditions of lactation failure. This is to exhibit 'baby friendliness' when the fight is between life and death. This should be prescribed only when medically indicated.

22. Infant Milk Substitute (IMS) Act

The IMS, Feeding Bottles and Infant Foods (Regulation and Production, Supply and Distribution) Act, 1992, was passed to protect, promote and support breastfeeding. The five important points from the act are:

a) No person shall use any health care system or the display of placards or posters relating to, or for the distribution of, materials for the purpose of promoting the use or sale of infant's milk substitutes or feeding bottles or infant foods.

b) No booklets, leaflets, brochures, posters, feeding bottles, cot tags, stickers, clinic cards, prescription pads and similar materials which advertise infant foods or formula should be permitted.

c) There should be no display of artificial infant feeding products in health care facilities. **No samples of infant milk formula or infant food can be distributed.**

d) Company marketing personnel, no matter what they are called, **should not be permitted to have direct contact with mothers.**

e) Only parents who need to artificially feed their infants should be instructed. Instruction should be given only by healthcare professional, which includes a doctor, paramedical staff or community workers. **The instructions should include details on the superiority of breast milk and breastfeeding with a clear warning about the health hazards of artificial feeding.**

The Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act, 1992 allows the dissemination of information **only to the health workers about the scientific and factual matters relating to the use of infant milk substitutes or infant foods.**

Practitioners should discourage every unethical practice. Careful instructions should be given whenever IMS is prescribed, (see Appendix 4)

23. Animal Milk

As mentioned earlier, unmodified cow's milk is unsuitable for feeding during infancy especially in the first half of infancy. Buffalo milk has higher fat content (7 g), mostly saturated fat. Goat's milk is less allergenic and contains more EFA and potassium when compared to cow's milk. Sodium is lower than in cow's milk. However, **goat's milk may predispose to folate deficiency and brucellosis.** Hence it is scientifically proved that any form unmodified bovine milk is unsuitable for infant feeding.

1.2 Feeding of Low Birth Weight (LBW) and Preterm Babies

The incidence of LBW is about 30% (NFHS 2005). Majority of them continue to be small and add to the pool of malnutrition. Sucking and swallowing become coordinated only around 34 weeks of gestation. The growth velocity is much higher in preterm than in term babies; but their nutrient stores are very little.

Gut maturation is inadequate in preterms. However, they tend to advance their "biological clock" and adapt to extrauterine nutrition. Gut maturation is mediated by gut hormones like enteroglucagon, gastrin, motilin and neurotensin. Gastric inhibitory peptide (GIP) increases insulin release and thereby glucose tolerance. A rise in glucagon induces hepatic enzymes like phosphoenol pyruvate carboxykinase (PPCK), the key enzyme in gluconeogenesis. Total parenteral nutrition (TPN) has been shown to produce mucosal atrophy due to low levels of gut hormones and hence at least "minimal enteral feeding" should be given whenever possible. A few drops of colostrum given to a sick baby on IV fluids can paint the gut with immunoglobulins and can promote gut maturity. Enteral feeding also reduces hyperbilirubinaemia. The recommended dietary allowances (RDA) for preterms recommended by the European Society for Paediatric Gastro-

enterology and Nutrition (ESPGAN) is generally accepted. The relevant items are given in Table 1.3.

Table 1.3 RDA for preterm babies

No.	Item	Requirement
1.	Energy	110-165 kcal/kg/day
2.	Fluids	150-200 ml/kg/day
3.	Protein	3-3.5 g/kg/day
4.	Vit. A	1000 IU/day
5.	Vit. D	400 IU/day
6.	Vit. E	15 IU/day
7.	Folic acid	50 ng/day
8.	Vit. C	10-60 mg/day
9.	Calcium	100-200 mg/kg/day
10.	Phosphorus	50-150 mg/kg/day
11.	Magnesium	6-20 mg/kg/day
12.	Zinc	1-2 mg/kg/day
13.	Iron	2.5 mg/kg/day

Vitamins are advised 2 weeks after birth and iron after 6-8 weeks.
(*Source*: RDA of preterms, ESPGAN.)

1. Protein

Protein intake up to 4 g/kg is recommended; however, higher doses are shown to produce azotaemia, hypoglycaemia, hyperaminoacidaemia especially tyrosinaemia and metabolic acidosis. Enzymes for degradation of tyrosine are found to mature late. Hence whey protein with lower concentration of aromatic amino acids like tyrosine and phenylalanine is preferable to casein. Whey protein is also rich in taurine and cysteine. Synthesis of taurine and synthesis of cysteine from methionine is defective in preterms.

2. Fats

Fat malabsorption and steatorrhoea can occur in preterms due to reduced amounts of pancreatic lipase, carboxylic ester hydrolase, bile acids and lingual lipase. Bile salt stimulated lipase (BSSL) in human milk promotes fat absorption. Human milk contains 8% linoleic acid; but some formulae contain unphysiologically high amounts (>20%). Long chain poly unsaturated fatty acids (LCP), more than 18 carbon atoms, are homologous to EFA. Rapid accumulation of LCP occurs in the brain in the third trimester and postnatally. LCPs include adrenic acid and arachidonic acid (n-6) and EPEA and DHEA (n-3 series).

Human milk contains adequate LCP for brain maturation. Carnitine facilitates transport of long-chain fatty acid across mitochondrial membrane for oxidation. Preterms have defective synthesis of carnitine. Human milk is rich in carnitine. Choline is needed for synthesis of acetylcholine and phospholipid. About half the choline is derived from diet. Medium-chain triglycerides are useful as they do not require hydrolase for digestion and absorption. The ESPGAN recommends not to have more than 40% of MCT in a preterm formula as this may lead to abdominal distension, and increased gastric aspirate. MCT increases calcium, magnesium absorption and tends to spare dietary nitrogen as well.

3. Carbohydrate

Lactose enhances Ca and Mg absorption and ensures favourable bacterial flora. Premature infants have transitional lactose intolerance due to immature infants' system. That's why very high lactose content in formula leads to osmotic diarrhoea. Glucose polymers like maltodextrin which are partially digested can reduce osmolality. And hence they are preferred in preterm and low birth weight formula.

4. Energy and Fluid

110-165 kcal/kg/day is the recommended energy and 150-200 ml/kg/day is the recommended fluid. Fluid is started as 60-80 ml/kg/day and is increased in increments of 10 ml/kg/day. The calories are also slowly increased. IV fluid 10% dextrose is given for 2-3 days and if there are no further problems like respiratory distress (RDS), hypoxic ischaemic encephalopathy (HIE) etc., oral feeding can be started.

5. Macrominerals/Macroelements

The intake of sodium, potassium, chloride, calcium, phosphorus and magnesium should be optimum. Magnesium deficiency may impair calcium homeostasis. Hypernatraemia may occur with some preterm formula. Calcium and phosphorus supplements may be needed to prevent rickets and osteopenia in preterm. Calcium is generally given in a dose of 40 mg/kg/day with aCa:P ratio of 2:1, assuming the rest of the requirement to be met from dietary intake.

6. Micro Minerals/Trace Elements

Iron deficiency can occur by 6-12 weeks and hence 2.5 mg/kg/day of iron starting from 6-8 weeks of age may be given. Zinc supplementation has been shown to increase weight gain. Zinc is found to be low in banked milk. Transient hypothyroidism can occur in preterm due to lack of iodine in iodine-deficient areas. Copper supplement is not generally needed.

7. Vitamins

Due to reduced stores and defective absorption, they tend to benefit from "pharmacological doses" of vitamins. One dose of vitamin K 0.5-1 mg is beneficial in all

LBW babies to prevent haemorrhagic disease of the newborn. Nonetheless a fortified formula could be beneficial.

8. Choice of Milk

Out of the various options, mother's own preterm milk is found superior. Others are banked milk, expressed breast milk, milk fortified with human milk protein by lacto-engineering and ordinary and special formula. The composition of banked term and preterm milk are given in Table 1.4.

a) Preterm milk (PTM): Preterm milk (PTM), the milk of mothers who have delivered preterm, is the major source of nutrients to the preterm babies. Preterm milk is different from term milk with a higher concentration of total nitrogen, protein (up to 2.2 g%), sodium, chloride, magnesium, iron, copper, zinc, IgA etc. Thus milk is not only 'species specific', but also 'baby specific'. The high protein content reduces to 1.3 g% by 6 weeks.

b) Expressed breast milk (EBM): It can be foremilk or hind milk with lower or higher fat and energy respectively, depending upon the time of expression.

c) Drip breast milk (DBM): It is the milk collected from contralateral breast due to let down reflex during feeding. It has lower energy content.

d) Ordinary or special formulae: Ordinary formulae are designed for term infants. As premature babies have very high and special requirements to catch up growth of term infants, special infant formula should be given. The infant formula of preterm babies should contain more proteins, multiple carbohydrates, MCT and LC-PUFAs as source of energy and brain development; and more minerals and vitamins compared with the routine infant formula. In other words, the premature formula should be more nutrient denser to ensure the optimal growth of premature babies without the overloading of the infant's immature organs. A comparison between the various formulae is given in Table 1.5. Preterm babies are born at a critical stage of rapid body and brain growth. They have low body stores of nutrients and have increased demand due to fast growth rate and frequent illnesses after birth.

The best choice for premature baby is preterm mother's milk. However, the fortification of even the preterm breast milk is desirable. The best milk strategy available should always be preferred. Larger volumes of nutritionally poorer milk should be adopted as tolerated. Milk pooled from mother who delivered prematurely offers an option to cut down on volumes. With a smaller budget, breast milk fortification and/or preterm formula can be used for special groups such as very low birth weight infants and those with poor growth on maximal volumes of standard milk. Vitamins and iron should be provided to all infants born weighing less that 1.5 kg. As has been reported, deliberated 1:1 mixing of preterm formula with breast milk reduces the volume required for better growth. Mothers are often forced to resort to artificial feeding. This practice should be condemned at any cost and successful breastfeeding should

Table 1.4 Composition of term milk and preterm milk (PTM)/100 ml

Item	Term	PTM 1st week	PTM 2nd week	PTM 3rd week	PTM 4th week	PTM 5th week	PTM 6th week
Protein (g)	1.1	2.3	1.9	1.6	1.5	1.4	1.3
Sodium (mmol)	0.6	1.7	1.3	1.2	0.9	0.8	0.8
Potassium (mmol)	1.5	1.7	1.5	1.3	1.3	1.2	1.2
Calcium (mmol)	0.8	0.7	0.7	0.7	0.7	0.7	0.7
Phosphorus (mmol)	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Energy (kcal)	67	64	67	67	67	67	67

Table 1.5 Comparison of various ordinary and LBW Infant Formulae***

Composition		Starter (per 100 g)	Follow up 1 (per 100 g)	Follow up 2 (per 100 g)	LBW Formula (per 100 g)
Energy	kcal	489	471	468	504
Total Fat	g	23	19.5	19.0	25.9
Milk Fat	g	12.6	12.0	12.0	13.0
Veg Fat	g	9.95	7.00	6.70	10.30
MCT	g				2.59
Lecithin	g	0.45	0.50	0.30	
Linoleic acid	g	1.76	2.4	2.4	5.9
Alpha linolenic acid	mg	250	236	234	6
Milk Protein	g	12	14.2	14.2	12.5
Carbohydrates	g	58.4	59.3	59.8	55.6
Sugar		NIL	3.6	13.8	NIL
Total Ash	g	3.6	4.0	4.0	3.0
Moisture	g	3	3.0	3.0	3.0
Minerals					
Sodium	mg	190	200.0	180.0	197.0
Potassium	mg	500	500.0	500.0	520.0
Chloride	mg	270	320.0	320.0	314.0
Calcium	mg	440	450.0	480.0	832.0
Phosphorous	mg	240	320.0	330.0	416.0
Magnesium	mg	50	48.0	48.0	61.0
Iron	mg	5.9	6.2	6.5	9.1
Iodine	ng	74	99	98	136
Copper	mg	0.3	0.28	0.28	0.60
Zinc	mg	3	3.0	3.0	5.0
Manganese	ng	55	70	70	310
Selenium	ug	14.5	14.5	14.5	52.0

Vitamins					
Vitamin A	mgRE	390	360	360	1215IU
Vitamin D	H9	5.25	5	5	13
Vitamin E	mgTE	3.3	3	3	24IU
Vitamin K	H9	45	21	21	67
Vitamin C	mg	50	46	46	190
Thiamin	mg	0.34	0.65	0.65	1.00
Riboflavin	mg	0.74	1.0	1.0	0.4
Niacin	mg	3.2	3.00	3.00	25.00
Vitamin B6	mg	0.38	0.35	0.35	1.00
Folic acid	H9	98	100	100	200
Panthothenic acid	mg	2.1	2.2	2.2	10.0
Vitamin B12	H9	1	0.7	0.7	2.9
Biotin	H9	12.5	12.0	12.0	20.0
Choline	mg	49	47	47	81.2
cystine*	mg				183
Tyrosine*	mg				212
Taurine**	mg	34			30.3
Carnitine**	mg	6			10.1
1:1 Dilution /100 ml	kcal	67	67	67	80
Scoop Size	9	4.6	4.7	4.8	4.5

* Cystine and tyrosine are essential amino acids in LBW infants
** Taurine and carnitine are essential for preterm,LBW, term infants till 6 months
*** LBW formulation per 100 ml gives 80 kcal, which is higher than term formula

always be aimed at. Mothers should be involved in the NICU care of babies and in breastfeeding. However, some babies may need artificial feeding. In the 'Kangaroo mother care (KMC) programme', the mother provides warmth, nutrition and nursing care to the baby. KMC is a novel method where mothers are used as incubators and as main source of stimulation and nutrition.

9. Human Milk Fortifiers (HMF) for Preterm Babies

The nutritional composition of expressed breast milk can be insufficient to meet the high nutritional requirements of premature baby. Human milk fortifiers (HMF) are commercially available products that can be added to expressed breast milk (EBM). HMF contains protein or protein hydrolysate, fat, carbohydrate, sodium, calcium, phosphorus, copper, zinc, vitamins etc. The HMF powder should be added in EBM; however, expression of milk is not always easy. The quality of EBM varies depending upon the time and mode of collection; e.g., colostrum is rich in protein, sodium, minerals and immunoglobulin; hindmilk has higher fat and lower protein than foremilk. In drip breast milk (DBM), the milk that drips spontaneously from the contralateral breast during feeding, the energy may be as low as 45 kcal/100 ml. (See Table 1.6)

10. Non-nutritive Sucking

It is important for orofacial development, for maturation of sucking reflex and for establishment of lactation. Hence, allow the baby to suck on the breast as early as possible and as long as possible even when no milk is secreted.

11. Lactobezoars

These are milk residues that accumulate in the stomach. These may develop due to high calorie-dense preterm formula. These may be visible on X-ray after air insufflation of stomach. These are self-limited.

12. Mother-Infant Bonding

When the care of the preterm/LBW baby is undertaken by a third person, e.g., nurse, mother-infant bonding reduces. When the mother is the primary caretaker, mother-infant bonding is established. Her bacteria will colonize on the baby. These bacteria will not generally cause infection in the baby unlike the bacteria of the caretaker. This is due to the transplacental antibodies. Occasionally it is noted that a preterm/LBW baby who is not thriving well is not easily accepted by the family as in the case of a full-term baby who is thriving well. Prolonged separation between the mother and baby will increase the gap further and it will also lead to suppression of lactation. Hence, as far as possible mother should be included in the care of the preterm from the very beginning. In the 'Kangaroo mother care method', mother looks after the baby and gives warmth and breastfeeding to the baby.

Table 1.6 Composition of human milk fortifier (HMF) per 2 g sachet (example)

Item	Quantity
Energy (kcal)	6.5
Protein (g)	0.2
CHO (g)	1.2
Fats (g)	0.1
Sodium (mg)	1.5
Potassium (mg)	3.9
Chloride (mg)	4.4
Calcium (mg)	50
Phosphorous (mg)	25
Magnesium (mg)	4
Vitamin A (IU)	730
Vitamin D (IU)	250
Vitamin E (IU)	1.3
Vitamin K (mg)	1.1
Vitamin C (mg)	5
Thiamine (mg)	12
Riboflavin (mg)	20
Niacin (mg)	230
Pyridoxin (mg)	25
Folic acid (mg)	40
Bn (mg)	0.05
Pantothenic acid (mg)	1
Biotin (mg)	0.5
Zinc (mg)	0.18
Copper (mg)	35
Manganese (mg)	1.7

13. Catch-up Growth

In a full term baby, the catch-up is about 200 g/week after the first 10 days of life. Initially there is slight loss of weight and the birth weight is regained by 10 days. In preterm, the catch-up can be up to 10 times for the age or up to 5 times for the length. The preterm is expected to grow on par with the intrauterine growth or as per the corrected age.

Corrected age = Chronological age - Period of prematurity

14. Warm Chain

This refers to the maintenance of optimum temperature of LBW babies during transport and during procedures and while giving care.

FEEDING OF THE LBW AND PRETERM INFANTS

Exclusive demand feeding is best for preterm babies. Those who are not thriving well may need milk formula or human milk fortifiers (HMF). These are to be prescribed only when absolutely indicated and are to be given under supervision as collection of milk and mixing need extra care. Haphazard addition of low molecular weight substances will increase osmolality and renal solute load and there is chance for bacterial contamination.

Route of Feeding

Babies above 34 weeks gestation and weight above 1.8 kg can be put to breast. In infants less than 34 weeks gestation and less than 1500-1800 g birth weight, start with gavage feeds and slowly switch over to oral feeding. EBM is always preferred. Up to 0.5-1 ml/hour may be given to very immature babies to enhance gut maturation. Gravity assisted feeding in 10-20 min is preferred to bolus feeding from a syringe with piston. Large preterms can be initiated on feeding within two hours of birth. 60, 90, 120, 150 ml/kg/day can be given on the first 4 successive days. Up to 180 ml/kg/day on day 10 and 200 ml/kg/day on day 14 may be achieved.

1-3 hourly feeds can be given in smaller to larger babies and if the aspirate is less than 10% of the previous feed, the same schedule can be continued. An initial feeding schedule is given in Table 1.7.

The initial feed may be distilled water followed by 5% glucose and then colostrum/expressed breast milk (EBM).

Abdominal distension and blood in stool should alert the possibility of NEC. If feed volumes need to be reduced below the total fluid requirement, an IV infusion should be considered to make up the requirement. In very immature or sick babies, when enteral feeding is started using nasogastric tube, it is advisable to use a continuous infusion instead of bolus feeding.

Regulatory norms of this country

Table 1.7 Feeding schedule for LBW babies

Birth weight	Quantity	Frequency	Increments
< 1 kg	1 ml	1-2 hr	1 ml/day
1-1.5 kg	2-3 ml	2-3 hr	1 ml/alt feed
1.5-2 kg	5-6 ml	2-3 hr	1-5 ml/feed
2-2.5 kg	8-10 ml	2-3 hr	5-10 ml/feed

1.3 Complementary Feeding Practices

Weaning or complementary feeding after 6 months is extremely important due to high risk of micronutrient deficiencies and malnutrition. Even though babies may thrive on breast milk alone during the first 6 months of life, they become biologically fit to accept semisolids after 4 months of age.

It is essential to prevent growth faltering. Weaning means 'to accustom to' or 'to free from a habit'. It is the process to accustom the baby to semisolids and solids in order to gradually free the baby from the habit of sucking at the breast. Weaning is defined as 'the systematic process of introduction of suitable food at the right time in addition to mother's milk in order to provide needed nutrients to the baby' (UNICEF, 1984). Weaning is the second step for self-existence. The first step is cutting of the umbilical cord.

The term 'complementary feeding' is now preferred because weaning implies abrupt stoppage of breastfeeding, at least to some mothers.

1. Time of Complementary Feeding or Weaning

Birth weight doubles by 4 months of age and the nutritional demands gradually increase and the calcium and iron stores get depleted. But the breast milk supply increases till 6 months and then it plateaus off. By five months of age, the weight doubles and becomes around 6 kg and the baby needs 600-700 kcal/day and around 600 ml of breast milk can supply only 400 kcal. By four months of age, the baby achieves head control and develops hand mouth coordination and starts enjoying mouthing. Also that the extrusion reflex perishes, intestinal amylase matures and the gut becomes ready to accept cereals and pulses (legumes). Gum hardens prior to tooth eruption and the baby enjoys gumming semisolids. Thus the baby is 'biologically ready' to accept semisolids by 4-6 months of age. Early weaning is often due to ignorance and leads to contamination and infection due to unhygienic preparation. Dilute weaning foods also lead to malnutrition. Late weaning leads to growth faltering and malnutrition.

2. Continuation of Breastfeeding

Breast milk should continue to be the main food of the baby even when weaning is started. To minimize interference with normal breastfeeding, it should be given between breastfeeds. Breastfeeding should continue for as long as feasible, preferably till two years of age. This is important as the first two years is a period of rapid brain growth and breast milk contains factors essential for brain growth and development.

3. Complementary Foods

Complementary foods can be home made or instant foods. In any case, it is better to start from mono cereals, followed by multi cereals and cereals-pulse combination. Cereal like rice is the best choice to start weaning as it is gluten free and easily digestible. The first cereal could be rice, which is gluten free and easily digestible. After that mother can make different combination with wheat, pulse, vegetables. They should be locally available, economical and acceptable. Cereal-pulse combination is better due to fortification of amino acids as cereals generally lack lysine and pulses lack methionine. Tubers, fruits, biscuits and banana powder are also popular weaning foods. Each type of complementary foods (home-made or instant) should be analyzed for the advantages and disadvantages. The advantage of homemade weaning cereals is that they are economical, easily available, culturally accepted, and closer to family food and versatile. However, it is quite difficult to keep the nutritional value of home food as per the high requirements of faster growing baby. Addition of jaggery for calories and minerals, milk for protein and oil for calories can make homemade food more nutrient denser. However the digestibility, presence of micronutrients and vitamins and bioavailability is a big concern due to processing and cooking time. The instant complementary food offers balanced nutrient content as per the recommendations for the older infants. The reasonable combination of homemade and instant foods may get the best result in prevention of micronutrient deficiencies and development of healthy family food habits.

4. Family Pot Feeding

The acceptance of food from the family food should be a part of the mixed feeding regime.

It is essential to switch over gradually to the usual family food. It can be given in a thickened and mashed form from the family pot without adding hot spices. Provide little extra oil or ghee, green leafy vegetables and seasonal fruits to the baby. The infant should grow up, accustomed to the traditional foods. Idli, dosai, soups, payasam etc., are very good for babies. A new food should be introduced in the morning session and only one item should be introduced at a time.

- Around 6 months of age: After 4 months of age, cereal-based porridge (ragi, suji, rice etc.) enriched with jaggery/sugar, oil/ghee and animal milk can be started. Start with 1-2 spoonfuls and gradually increase to 1/2 to 1 cup per day in 1-2 servings in addition to breastfeeding. Fruit juice also can be started.
- 6—9 months of age: After 6 months of age, introduce mashed items from the family pot enriched with jaggery/sugar and oil/ghee. Mashed rice with pulses, mashed tubers and vegetables, soups, mashed fruits, biscuits, egg yolk fol-

lowed by white etc., can be given 4-5 times a day in addition to breast milk. Egg white may be allergenic in some.

- 9-12 months of age: After 9 months, introduce soft food that can be chewed, avoiding hot spices. Chappathi and other hard items can be made soft by adding little milk. A variety of food from family pot can be given 4-6 times a day gradually increasing the quantity. By one year of age, the baby should be taking everything cooked at home. This is called 'family pot feeding'. A one-year-old child should eat half of what the mother eats.

5. Bridging the Calorie and Other Nutrient Gap

The calorie gap can be bridged by using oil/ghee and sugar and selecting 'high density food items' that will not swell much on cooking; e.g., egg, potato etc. Cereal-pulse combinations, roots and tubers, vegetables, especially green leafy vegetables and others, seasonal fruits, milk products, egg, fish, meat etc., given to the baby will bridge the nutrient gap. Predigested instant foods are nutrient dense. Frequent feeding is desirable as it aids in good acceptance by the infant. Soaking and malting of grains will increase digestibility and vitamin content. Sprouting or germination will enhance vitamin content and make it 'amylase rich food' (ARF) and will decrease the bulk on cooking. Fermentation enhances vitamin C and digestibility; e.g., curd/yogurt. It also increases shelf-life. The once a day introduction of instant food could be a way of balancing the nutrient gap and one-step solution to prevent malnutrition. Quality instant foods offer balanced nutrients including macro and micronutrients, with good bioavailability.

6. Developing Readiness for Family Foods through Varied Textures and Tastes

It is very essential to introduce varied textures throughout complementary feeding period. Under normal scenario, the mother tends to give a soft, completely mashed food for a longer period. This might not satisfy the baby's urge to chew with the development of teeth and preparation for textured family diet could be difficult. It is essential to advice the mother to differentiate the texture through the preparation and cooking methods. A soft to coarser to bigger bite texture will be a positive approach towards developing the baby for acceptance of family foods. Introducing new tastes with addition of vegetables, fruits will expose the baby to healthy eating practices. It is essential to practice the child towards good nutrition, and healthy eating, right from the complementary feeding period.

7. Preparation and Storage of Weaning Foods

Careful hygienic preparation and storage of weaning food is important. Hand washing with soap and water should be practiced before cooking and feeding. The food stuffs should be freshly prepared. Precooked ready-to-mix cereal-

pulse combinations can be prepared and stored in airtight containers, e.g., SAT mix which is a combination of roasted and powdered rice, wheat, black gram and powdered sugar in the ratio 1:1:1:2. In case of using instant baby foods, detailed reading of preparation instruction on the pack should be done.

8. Careful Feeding Practices

The feed should be carefully fed. There should be a careful selection of weaning foods and advice should be given to the mother by the health care professional. In thick consistency, the mother should not be adding more water to the feed as it might lead to dilution of the nutrients which would lead again to malnutrition. The caretaker should be informed and trained on the right feeding practices.

9. The weaning or Complementary Bridge and the Safety Net to Prevent Malnutrition

Most of the children fall into the pit of malnutrition during the weaning and post-weaning phase. Some even succumb to it. Jelliffe has suggested a 'three plank protein bridge' to prevent PEM. Mothers are expected to make the 'weaning bridge' or the bridge of complementary feeding to carry the children across the pit of malnutrition during liquid to solid transition. The three planks include (1) Continued breastfeeding, (2) Introducing vegetable protein and (3) Animal protein. Some mothers do not make a bridge at all and some others make a bridge that may collapse into the pit. So a 'safety net' is needed beneath the bridge (Fig. 1.4). This includes utilization of supplementary feeding programmes as in ICDS, which ensures extra 300 kcal/child/day. Those who do not avail this facility should arrange extra feeding either in the play school in the form of group eating or at home using the 'Akshayapatra'. It a special container for the child into which small pieces of food can be added in order to make the child eat during play.

FEEDING OF CHILDREN

1. Toddlers (1-3 years of age)

A toddler needs more than half the food that the mother eats. This should be given in frequent servings. As toddlers are more interested in play and as they have a physiological anorexia and reduced growth rate than infants, they must be coaxed to eat. Eating while playing, group eating and eating from a special vessel 'Akshayapatra', into which pieces of food stuff can be added on, may be adopted. They often enjoy eating from their own special vessel.

2. Preschool Children (3-6 years)

A preschool child should eat half the quantity of food that the father eats. They are interested in group play and in exploring and mastering the environment. They should be coaxed to eat. Group eating and supplementary feeding from the

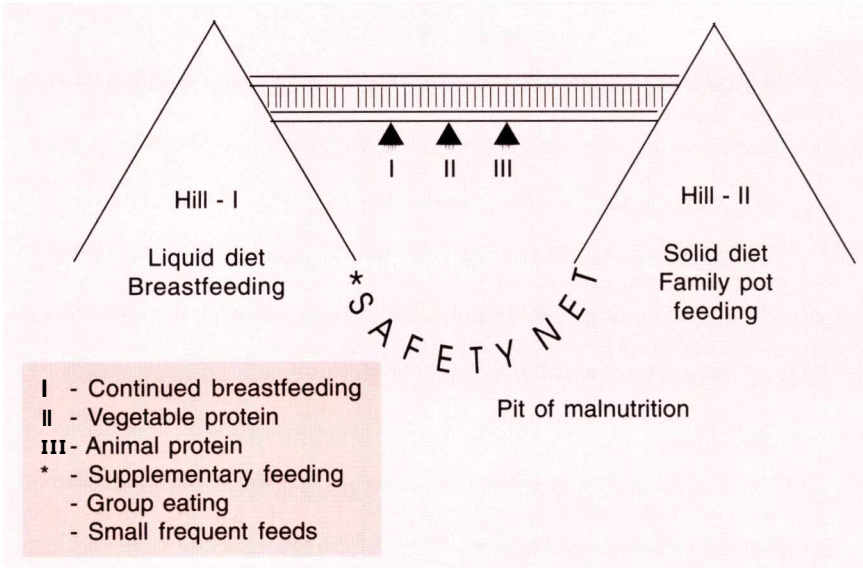


Fig. 1.4 Weaning or complementary bridge & safety net to prevent PEM

ICDS anganwadis should be made available to them in addition to family pot feeding. Vegetables and fruits should be given to them to ensure a good supply of vitamins and minerals. They enjoy variety in food items.

3. School-going Children

They should eat three-fourth of food that the father eats. They should take a balanced diet and should not miss meals especially breakfast which is the brain's food.

4. Feeding During and After Illness

Breastfeeding and feeding of easily digestible soft food items should be continued during illness. Starvation should be avoided unless medically advised. The child should be coaxed to eat small quantities every 2-3 hours. After the illness, give an extra meal for 1-2 weeks to regain the lost weight.

5. Growth and Development Monitoring

Frequent weighing and recording on the growth chart are desirable. A flat curve or a downward curve should be of concern and appropriate intervention should be initiated. Medical check-up, investigations, prompt diagnosis and treatment of intercurrent infections and extra feeding are the interventions. Developmental milestones should also be of concern and early intervention should be undertaken if there is developmental delay.

6. The Following Ten Commandments in Nutrition are Very Useful

a) Be 'baby friendly' and initiate breastfeeding soon after birth, preferably within minutes after delivery.

b) Practice exclusive demand feeding during the first 4-6 months of age.

c) Continue breastfeeding as long as possible, preferably till two years of age, the period of rapid brain growth and myelination.

d) Start building the weaning or complementary bridge at the age of 4-6 months by introducing semisolids that 'the child can eat and not drink'.

e) Slowly switch over to family pot feeding and empower the baby to take everything cooked at home by one year of age.

f) Make a safety net for the young child in the form of supplementary feeding, group eating or small frequent feeds using the 'Akshayapatra concept' to prevent malnutrition.

g) Ensure a balanced diet that includes all the various food items and nutrients.

h) Ensure extra nutrition during special physiological needs like adolescence, pregnancy, lactation and old age. Don't starve the child during illness and offer easily digestible food items including breast milk and give an extra meal for 1-2 weeks after an illness to regain the lost weight.

i) Ensure micronutrients and antioxidants by including green leafy vegetables (GLV), green yellow orange (GYO) vegetables and fruits etc., and also utilize micronutrient supplementation programmes like vitamin A, iron, folic acid, iodine etc.

j) Ensure quality of survival and overall development by non-nutritional interventions like socioeconomic advancement, standards of sanitation, immunization, periodic deworming, and protected water supply, control of alcoholism, family harmony, tender loving care (TLC) and developmental stimulation.

7. Re-lactation

It is the resumption of breastfeeding following cessation or significant decrease in breast milk. This is possible through motivation support, frequent suckling and drop and drip method. Supplementary suckling technique (SST) can be tried.

1.4 Commercial Preparations

Introduction

It is very essential to stress the significance of breastfeeding whenever we discuss anything other than breast milk. There is no other food for infants as good as breast milk and breastfeeding the best way to ensure mutual health of both the

baby and mother. Since the early 2000s, research attention has been focused on the potential long-term benefits of breastfeeding in childhood and beyond.

Breast milk is the best nutrition for infants and is used as the 'gold' standard for good infant nutrition at birth. It provides the right nutrients (protein, fat, carbohydrate, vitamins, minerals, and water) in the right quantities to sustain normal growth and development for the first months of life. In addition to its nutrient content, breast milk contains a host of additional components that benefit infants. Breast milk can also provide the basis for good nutrition even after 6 months of age, until a child is fully weaned.

Infants grow most rapidly during the first 6 months of life, making this period a critical time for nutrition. Although breast milk is the ideal way to feed a baby, there are situations. Keeping this in mind, it is extremely important that a medical practitioner is completely aware of the commercial preparations available.

However, this should be done only with the objective of establishing the right "Baby friendliness", and sustenance of life and good nutrition. The replacement feeding (RF) should be an alternative only when it is acceptable, feasible, affordable, sustainable, and safe (AFASS).

Some of these conditions are the following:

- Breast feeding is contraindicated for infants with galactosaemia, congenital lactose intolerance
- Mothers who have H1V-AIDS
- Mothers who use drugs of abuse
- Mothers who take certain medications like antimetabolites and chemotherapeutic agents and radioactive isotopes

Unsuitable breast-milk alternatives include whole cows' milk, evaporated or sweetened condensed milk, rice gruel or diluted porridge, cassava flour, sugared tea/coffee, which can be mistakenly used as significant sources of fluid and energy.

The main cause of malnutrition and micronutrient deficiency diseases is primarily due to wrong choice of food, improper feeding practices, and incorrect techniques. So it becomes essential to understand the significance of feeding suitable breast-milk substitutes and safe alternative for infants who are not breastfed.

Cow's milk can be adapted for formula feeding (Fig. 1.5), in the form of starter and follow up formulas. Unfortunately we are dependent more on western data than on Indian standards for these commercial preparations The CODEX and ESPGAN standards are the guiding tools as they are constantly upgraded.

INFANT FORMULA

Infant formula is usually produced by adapting the composition of cow's milk to achieve a composition closer to breast milk. The key steps involved are: diluting

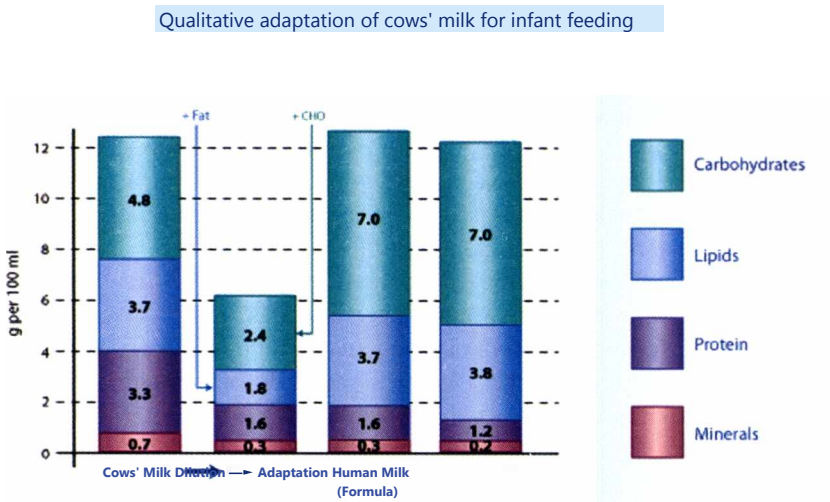


Fig. 1.5 Qualitative Adaptation process of cow's milk for infant feeding

cow's milk with water to reduce the protein and mineral content, adding carbohydrate and fat. and modifying mineral profile to adjust nutrient content.

Types of Starter Formulas

Starter formulas may be whey-adapted, casein-predominant, acidified, hypoallergenic, or therapeutic (specialty).

7. *Whey-Adapted Formula*

Whey-adapted starter formula has whey protein added to cows' milk protein, to achieve a whey/casein ratio usually >1 and an amino acid pattern closer to that found in mature breast milk. Mature breast milk has a whey/casein ratio of 60/40. These formulas generally have mineral concentrations similar to those of breast milk with the use of demineralised whey. This is the most commonly used starter formula (Table 1.8).

2. *Casein-Predominant Formula*

Skimmed cow's milk is the main source of protein in casein-predominant formula. Because cow's milk protein contains more casein than whey protein, these infant formulas are called "casein-predominant" and their whey/casein ratio is < 1. Casein predominance means that it takes longer for this formula to pass through the infant's stomach. Due to the slower gastric passage, they are said satisfy the baby for a longer period of time and are often appreciated for their satiating effects.

Table 1.8 Composition of stages of whey-predominant infant formula

Composition		Starter (Per 100 g)	Follow upl (Per 100 g)	Follow up 2 (Per 100 g)
Energy	kcal	495	476	473
Total Fat	g	23.7	20.0	20.0
Milk Fat	g	12.6	0.0	0.0
Veg Fat	g	10.5	19.35	19.35
Lecithin	g	0.6	0.65	0.65
Linoleic acid	g	3	2.0	2.3
Alpha Linolenic Acid	mg	260	238	238
Milk Protein	g	10.2	14.3	14.3
Carbohydrates	g	60.2	59.1	58.9
Total Ash	g	2.9	3.6	3.8
Moisture	g	3	3.0	3.0
Minerals				
Sodium	mg	120	185.0	185.0
Potassium	mg	460	430.0	430.0
Chloride	mg	350	340.0	300.0
Calcium	mg	260	460.0	480.0
Phosphorous	mg	130	270.0	320.0
Magnesium	mg	45	40.0	48.0
Iron	mg	6	6.2	6.5
Iodine	^g	75	99	99
Copper	mg	0.28	0.28	0.28
Zinc	mg	3.2	3.0	3.0
Manganese	ng	40	73	73
Selenium	ng	14.5	14.5	14.5

contd

Composition		Starter (Per 100 g)
Vitamins		
Vitamin A	mgRE	390
Vitamin D		5.25
Vitamin E	mgTE	3.3
Vitamin K	ng	45
Vitamin C	mg	51
Thiamin	mg	0.35
Riboflavin	mg	0.75
Niacin	mg	3.2
Vitamin B6	mg	0.38
Folic acid		100
Panthothenic acid	mg	2.1
Vitamin B12	ug	1
Biotin	H9	12.5
Choline	mg	50
Taurine	mg	34
Carnitine	mg	6
Ca : Ph Ratio		2
Fe / Zn		1.9
Vit C: Fe		2.66
Vit E: LA		1.1
PRSL*	mOsm/L	89
RSL		68

* Potential renal solute load

Follow upl (Per 100 g)	Follow up 2 (Per 100 g)
360	360
5	5
3	3
21	**21**
46	48
0.65	0.65
1.0	1.0
3.00	3.00
0.35	0.35
100	100
2.2	2.2
0.7	0.7
12.0	12.0
32	32
0	0
0	0
1.70	1.50
2.1	2.2
2.32	2.31
1.5	1.3
119	**120**
86	85

Although the protein source differs between whey-adapted and casein-predominant formulas, the growth effects of these two types of formulas are quite similar. However, the following observations can be made:

- Serum concentrations of the essential amino acid threonine and branched-chain amino acids are greater when infants are fed a whey-adapted formula, but the implications of these findings are unknown.
- Serum concentrations of the amino acids methionine, tyrosine, and phenylalanine are greater when infants are fed casein-predominant formulas. High levels of these amino acids have been associated with pathological conditions, but at concentrations far higher than those observed in routine infant feeding.
- For preterm infants, the greater cysteine content of whey-adapted formulas may be an advantage. Furthermore, lactobezoars may be more common when casein-predominant formulas are used in preterm infants.

3. Acidified Formulas

Acidified formulas have been biologically acidified by a microorganism or they can also be directly acidified by using lactic acid. The type of lactic acid produced is important. For example, only the L-form of lactic acid is metabolized and therefore acceptable in infant feeding. The D-form of lactic acid is not metabolized and can cause metabolic acidosis. An acidified formula provides the benefits of a finer, more digestible curd and a reduced risk of formula contamination. Therefore, these formulas are indicated for infants who present with poor digestion and in situations where hygiene may be poor and the risk of formula contamination is high.

4. Hypoallergenic (HA) Formula

Hypoallergenic formulas are cow's milk based formulas that have been specially treated to break down the protein chains into shorter chains of amino acids. The proteins in hypoallergenic formulas have been moderately hydrolyzed and processed by a protease to reduce their allergenicity. The benefit of such formulas is a reduction in the incidence of symptoms of potentially allergic origin, such as eczema, rhinitis, urticaria, etc. This benefit is more pronounced in infants with allergy or family history of allergy. However, in absolute numbers, more babies who do not have a family history of allergy also benefit from these formulas.

5. Therapeutic or Specialty Formulas

These formulas comprise a broad group of formulas with specific properties that are useful when feeding infants with special dietary needs. They include:

- Formulas designed to meet the special needs of low birth-weight (LBW) infants.
- Lactose-free formulas.

- Formulas designed to provide nutritional support to infants with diarrhoea.
- Formulas hypoallergenic containing extensively hydrolyzed protein for infants who are allergic to cow's milk.
- Thickened infant formulas to reduce regurgitation; these are referred to as anti-regurgitation (AR) formulas.
- Formulas designed for infants who have metabolic problems, such as phenylketonuria.

New Components in Infant Formulas

New components that have been added to infant formulas in recent years include long-chain polyunsaturated fatty acids, probiotics, prebiotics, nucleotides, and antioxidants.

Long-Chain Polyunsaturated Fatty Acids (LC-PUFAs or LCPs)

Breast milk is a rich source of LC-PUFAs, which have been proven to have clinical benefits in the diet of preterm babies. Two such fatty acids—arachidonic acid (AA) and Docosahexaenoic acid (DHA)—are found in cell membranes and are important for the development of a baby's brain, eyes, and nervous system. The sources could be algae oil or fish oil.

Probiotics and Prebiotics

Probiotics are live bacteria that live in harmony with and confer health benefits to their host. In the body, they act mainly in the large intestine where they can provide a barrier that prevents the adherence of pathogenic bacteria (e.g., those that cause food poisoning). They also work in harmony with the immune system to support and enhance its effectiveness.

Prebiotics are non-digestible food ingredients that benefit the host by selectively stimulating the growth and/or activity of probiotics, thus improving the host's health. In other words, they are components that can act as food for probiotics, thus encouraging the growth and colonization of a normal and healthy intestinal flora. Prebiotics are found naturally in breast milk, as well as in certain fruits and vegetables.

Immediately before birth, the intestine of a newborn is sterile. The colonization of bacteria in the gut begins with the process of birth, the infant's exposure to the environment, and milk feeds. The type of bacteria that colonize the gut relates to the type of feed a baby receives, which means breastfed babies have a different microflora than formula-fed babies. Not only are Bifidobacteria and lactobacilli naturally found in breast milk, but breast milk also has such a composition that it favours the growth of specific beneficial bacteria. In general, the predominant flora of a breastfed infant is composed of Bifidobacteria, while formula-fed infants have a much more diverse flora like coliforms.

Probiotics can be added to infant formula and also to infant cereals—the most common strain being *Bifidobacteria lactis,* which has been shown to increase the amount of Bifidobacteria in the large intestine. Scientific research has shown that probiotic-supplemented formulas can reduce the incidence of diarrhoea. A reduction in antibiotic-associated diarrhoea has also been shown.

There are a number of important considerations when adding probiotics to infant formulas and cereals, including:

- The probiotics must be safe and effective for infants and must comply with legislative requirements.
- The benefits of probiotics discussed above are in term infants; their role in preterm infants is investigational.
- Spores of any form are not probiotics by definition and are not considered safe.

Nucleotides

Nucleotides are the building blocks of deoxyribonucleic acid (DNA), which comprises the set of instructions or the code for the auto-reproducing component of every cell in the body. Nucleotides, when present in formula, are important for growth and development, and also serve as important cofactors in cellular signaling and metabolism.

Antioxidants

Infant formulas also contain antioxidants such as beta-carotene. vitamin E or selenium, which protect the body against cell damage from free radicals. Certain other vitamins and minerals that are essential for healthy growth and development also play the role of antioxidant.

WEANING PREPARATIONS AND COMMERCIAL COMPLEMENTARY FOODS

Several paediatric recommendations exist for introducing foods into a complementary feeding diet like the Committee on Nutrition of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition (ESPGHAN) and the WHO. The recommendations are introducing weaning foods around six months of age, and giving follow-on formula in a quantity of not less than 500 ml daily along with complementary foods.

ESPGHAN has recommended that introduction of gluten be avoided in selected families and communities.

Avoiding foods that may contain high amounts of nitrates (e.g., spinach and beets) during the early months.

Delaying the introduction of highly allergenic foods, such as egg white and sea fish.

Homemade Preparation (HMP) versus Commercial Preparation (Table 1.9)

Mothers can choose between feeding home-made prepared foods and commercially-prepared foods. HMP is recommended as it is cheaper and variety can be achieved.

For a complementary feeding diet, cow's milk contains too much protein, sodium, and saturated fats and not enough essential fatty acids, vitamin D, and iron. According to the ESPGHAN Committee on Nutrition, the median protein intake of infants who are fed cow's milk is 20 to 100% higher than that of infants fed infant formulas or follow-on formulas between six and 12 months of age; this is two to three times higher than the calculated "safe level of protein intake." In addition, consumption of cow's milk can lead to increased gastrointestinal blood loss in young infants. As a result, the incidence of iron-deficiency anemia at 12 months of age is substantially higher in infants who are fed cow's milk. Therefore, ESPGHAN, as well as the American Academy of Pediatrics (AAP), have concluded that cow's milk is unsuitable for infants during the first year of life. To ensure adequate nutrients during complementary feeding, along with a variety of home-made preparations, one feed of commercial feed with all enriched nutrients can be given in affordable situations.

Table 1.9 Features of Homemade Preparations and Commercial Weaning Foods

HMP	Commercial Weaning Preparations
■ Variety is unlimited; any food available for home cooking can be prepared for the baby (provided it suits the developmental stage of the infant) ■ Can be fresh and unprocessed ■ Can be ground, pureed, or sieved for proper consistency ■ Can be culturally acceptable and available ■ Shift to family pot is comfortable ■ Are more economical	■ Mostly nutrient dense and prepared as per the standards ■ Standardized for high quality ■ Easy to use ■ Energy density is specified ■ Often include added vitamins (e.g., vitamin A, C) and minerals (e.g., iron) which are generally non bioavailable in HMP ■ Can be used to construct a balanced diet of convenience ■ Overcome problems of seasonal availability of certain foods (i.e., provides variety, regardless of season) ■ Especially helpful for working mothers

Types of Commercial Weaning Preparations

Standard Cereals or Processed Cereal Based Complementary Food

These are commonly called as weaning food or supplementary food based on cereal and /or legume (pulses), millets, nuts and edible oil seeds processed to low moisture content and so fragmented as to permit dilution with water, milk or other suitable medium. Table 1.10 gives the composition of commercial cereals.

Complete Cereals or Milk-Cereal Bbased Complementary Food

These are foods based on milk, cereal and/or legumes (pulses), millets, nuts and edible oil seeds processed to low moisture content and so fragmented as to permit dilution with water.

Pulpy Weaning Foods

These are high-quality pulp of selected single or combination of fruit and vegetable pulps packed in suitable containers. They do not have added colours or preservatives. These are not available in India.

Follow-On/Follow-Up Formulas

Starter infant formulas are not optimal because they contain too much fat and not enough calcium. In some circumstances, they may even be too low in protein or iron (e.g., low-iron starter formulas).

Also called follow-up formula, as "a food intended for use as a liquid part of the weaning diet for the infant from the 6th month onwards till two years of age." The statement also says that these "products are not breast-milk substitutes and should not be presented as such."

In general, follow-on formulas contain larger quantities of certain nutrients when compared to starter formulas. Follow-on formulas should continue to provide about 40% of the infant's energy intake, with about 500 ml to be consumed per day. (refers composition of stages of whey-predominant infant formula)

Protein

The protein content will be sufficient to support growth. The protein used is from good-quality sources in the form of milk, eggs, and fish.

Calcium

According to Codex Standard, follow-on formulas should contain at least 90 mg of calcium per 100 kcal; between 50 and 140 mg/100 kcal.

Iron

The amount of iron in follow-on formulas is higher than that in starter formulas, due to increased demands and less supply from milk.

Table 1.10 Composition of nutrients of cereal preparations

Composition		Complete cereal (with milk) (100 g)	Standard cereal - To be mixed with milk for complete nutrition* (100 g)
Energy	kcal	411	370
Fat	g	9	0.5
Linoleate	9	1.5	
Protein	g	15	6.0
Carbohydrates	g	67.5	85.4
Sugar		11.8	10.4
Dietary fibre	g	1	0.5
Total ash	g	4.5	3.6
Moisture	g	3	4.0
Minerals			
Sodium	mg	100	
Potassium	mg	335	200.0
Calcium	mg	400	190.0
Phosphorous	mg	330	140.0
Iron	mg	7	12.0
Iodine	ng	50	
Copper	mg	0.3	0.30
Zinc	mg	2.5	2.5
Vitamins			
Vitamin A	mgRE	360	400
Vitamin D	ng	5	6
Vitamin E	mgTE	2	3
Vitamin C	mg	50	65
Thiamin	mg	0.5	0.55
Riboflavin	mg	0.6	0.4
Niacin	mg	3	5.00
Vitamin B_6	mg	0.66	
Folic acid	R9	25	21
Panthothenic acid	mg	1.5	
Vitamin B_{12}	ng	0.75	
Biotin	ng	10	37.0

* Standard Cereal can be made versatile by addition of curd, fruit juices or home made preparations

Fat

The fat content of most weaning foods is low, as is the percentage of energy provided by fat (12% at four to seven months of age, 25% at eight to 12 months). Therefore, the fat content of a follow-on formula should be higher than 3.3 g/100 kcal, and it should provide at least 35% of the total energy in the formula (about 4.0 g/100 kcal). The most recent EU Directive on the composition of follow-on formula recommends a fat content between 4 and 6 g/100 kcal. As with infant formula, the linoleic acid content should be between 300 mg and 1200 mg/100 kcal.

TODDLER'S GROWTH & NUTRITION

Toddlers fit the "junior" stage of development in which the child is ready for a strong nutritional foundation that involves more adventurous foods in a complete and balanced daily diet. Healthy eating is important for toddlers in order to:

- Provide the energy and nutrients they need for growth and development.
- Encourage development of skills through touch, smell, and taste.
- Establish eating behaviours and practices that can form the basis for lifelong, health-promoting eating habits.

These are some of the important physical and psychomotor developments that occur at this stage in life:

- After having tripled their weight in infancy, the growth that occurs during the one-to-three years as a toddler includes 50% gain in body weight, 6-10 cm gain in length per year, doubling the size of the brain.
- Reaching, grasping, and releasing are nearly fully mature at this age. Better coordination and more complicated gestures (e.g., wrist rotation and flexion, elbow lifting) develop. This is the "learning" age—the age for all kinds of experiments in the area of feeding, such as holding spoons, self-feeding, and drinking from a straw.

As growth rate declines, children's appetite declines (physiological anorexia) and they may eat less. With the acquisition of language and motor skills, young children may seem distracted at mealtime. This is part of normal development.

Nutritional Requirements

They should have small, regular, nutritious and energy-dense meals that include a variety of foods. The addition of snacks is important and should contribute significantly to their daily intake of nutrients (Table 1.11).

Commercially-Prepared Junior Foods

Junior foods are the natural follow-up to infant cereals, jarred baby foods, and

Table 1.11 Important Nutritional Requirements for Toddlers

Requirements	Description
Energy	Toddlers need 100-110 kcal/kg/day. Compared to adults, who require about 40 kcal/kg/day.
Proteins	Toddlers need proteins for growth, muscle formation, and synthesis of antibodies to resist infection. However, too much protein can overload their kidney functions.
Fat	Fat is an energy-dense nutrient; a lack of fat in a toddler's diet will induce abnormal fatigue and is associated with slow weight gain.
Essential fatty acids	They play in metabolic processes. It is important to protect the heart by avoiding saturated fatty acids and giving more mono- and poly-unsaturated fats.
Calcium	It is needed for strong bones and teeth.
Iron	Iron deficiency is relatively common and is responsible for excessive fatigue, loss of concentration, and apathetic behaviour. It is also associated with a reduced capacity to learn and an increased sensitivity to infections.
Others	Zinc is necessary for growth and healthy immune function. Deficiency is associated with impaired ability to learn, skin problems, and recurrent infections. Others are iodine, FA, vitamin A, D etc.

follow-on formulas. They also provide a natural link to general family foods. A wide variety of junior foods (mainly in the area of breakfast cereals, snacks, finger foods, cookies and biscuits, and even complete meals) is now available specifically for toddlers. However, commercially-prepared foods for one-to-three-year olds must provide the right balance and range of nutrients that toddlers require. Studies show that the diet of many toddlers is nutritionally inadequate.

Foods for toddlers should not contain excess salt. The most common problem areas are lack of essential fatty acids, iron, and vitamin E, marginal intakes of calcium, too much protein, too much rapidly-absorbed carbohydrates. The foods that serve toddlers best are those that contain prime-quality, well-balanced nutri-

ents, provide the right balance of vitamins, minerals, and dietary fibers, avoid excess salt and sugar and avoid the use of food additives such as flavours, colours, or preservatives.

Milk Products for Toddlers

Similar to follow-on formulas, growing-up milks (GUMs) have been designed as a substitute for regular cow's milk in the toddler's diet to provide a more balanced source of nutrition. They supply the valuable nutrients contained in cow's milk (calcium, vitamin B_2, vitamin B_6) and those missing in cow's milk and in the toddler's diet (essential fatty acids, iron, vitamins A and D, trace minerals).

Unsuitable Milks for Toddlers

Skimmed milk and low-fat milks (1 to 2% fat) are not recommended during the first 2 years of life because of their high protein and electrolyte content and low caloric density and low in the EFA, linoleic acid.

Conclusion

Breastfeeding is the best way to feed infants during their first months of life. However, when it is not possible to feed breast milk, alternatives to provide the best of the nutrients possible should be advised. Nonetheless these alternatives can never replace the goodness of breast milk and natural foods available.

1.5 Feeding Related Problems and Picky Eating

Good nutrition is essential from birth for healthy growth and development in children. Feeding and eating experiences early in life also shape the quality of nutrition and dietary preferences throughout childhood.

The most common queries in paediatric clinics is in the area of feeding. And the entire family of the child poses feeding related issues as the major reason for weight loss.

The remarkable role of a paediatrician in eliminating the doubts about feeding related problems cannot be undermined.

Among infants, the usual feeding problems are underfeeding, overfeeding, aerophobia, gas colics etc. Burping after each feeding is essential to prevent aerophobia. Posseting is regurgitation of small amounts of food. This is often due to overfeeding or lack of burping. Rumination (merycism) is a psychological disorder in which the child brings out small quantities of food back into the mouth and again chews it and swallows it. This is similar to 'chewing the cud' by certain animals. This needs psychological evaluation and counseling. Constipation and loose stools also may occur. High solute formula and cow's milk may lead to constipation. Food intolerance and bacterial contamination can lead to loose stools.

Anorexia and decreased food intake are usual problems among toddlers. They want to select the food and to self-feed and the mothers tend to disagree with this. 'Meal times' are often converted to 'war times' and some children may even go on 'hunger strike'. Another peculiarity of this period is that the growth rate reduces and they are more interested in play and in exploring the environment. So the mother will have to coax the child to eat during play. In general, we have to respect the child and set good eating habits.

Escalating obesity rates among young children across the globe has prompted interest in investigating the role of children's eating styles in the maintenance of healthy weights and dietary adequacy. It has been estimated that, 20-45% of infants and children are affected by feeding problems that have been brought to the attention of a health-care professional at some point of their development. These feeding and growth-related problems can place excessive stress on the family, and can negatively impact the physical, intellectual, social and academic development of the child.There is increasing recognition that problematic eating behaviours in childhood may be precursors to eating behaviours later in life. Eating behaviours can vary on a continuum from picky eating to disinhibit overeating or binge eating.

Eating disorders are an important diagnosis in children as they have significant medical and psychiatric morbidity and mortality. The incidence of these eating and weight related disorders are occurring in increasingly younger aged children, and evidence shows that this "dysfunctional" eating leads to an increased risk for eating disorders. Children's food acceptance patterns develop and change dramatically during the first few years of life, presenting parents with the difficult challenge of providing nutritionally complete diets for their young children. Dietary intake of infants begins with a liquid diet, a transition to complementary foods occurs in the latter six months, and, by 24 months, most children are on adult food pattern in the form of family pot feeding.

Children with feeding disorders are characterized by an inability or refusal to eat or drink sufficient quantities or types of food to sustain weight and to meet nutritional requirements for growth. Eating behaviours in childhood may vary on a continuum ranging from picky eating, irregular eating, overeating, to uninhibited or binge eating. The childhood eating behaviours may be influenced also by factors like mother's exposure to and acceptance of a new food, family characteristics, number of exposures to a new food, perceived opportunities to taste a new food, verbal praise given in a social context and early feeding patterns.

Definitions

There are various eating disorders in children like picky eating, infantile anorexia and anorexia nervosa. Picky eating is at one extreme of the continuum. It is also known as 'neophobic', 'fussy eater', 'choosy', and 'problem eaters' across studies.

Picky eating is a common problem for many children. Picky eating may cause concern for parents about adequacy of the child's diet and they are also more likely to pressure a child to eat if they perceive the child to be underweight.

Research based on picky eating in children has shown that these children tend to: eat small meals, eat slowly, be less interested in food. They have acceptance of a limited number of foods, unwillingness to try new type of foods, limited intake of vegetables and other foods, and exhibit strong food preferences.

Children with infantile anorexia develop their illness because of difficulty with the transition to self-feeding. Criteria for diagnosis include: toddler food refusal for at least one month; acute and/or chronic malnutrition; and intense parent-toddler conflict and/or distraction and talking during feeding. Infantile anorexia, unlike so-called "picky eating," is a subtype of infantile feeding disorder and is characterized by failure to gain weight or weight loss over at least 1 month, rare interest in food or expression of hunger, age of onset before the child is 3 years old, and the exclusion of trauma to the oropharyngeal area or other medical conditions. This is defined as a feeding disorder of separation and is characterized by food refusal by the infant with intense conflict in the mother-infant relationship over issues of autonomy, dependency, and control. Occurs after the infant has learned to regulate himself and has established an attachment to the primary caregiver.

Developmental Stage of Feeding Disorders

Problematic eating behaviour is common in toddler years as young children express their independence and become more engaged in social interactions. However, as young children transit to the preschool phase, problematic eating behaviours such as picky eating typically subside. Preschool children fall into 'Erikson's early childhood developmental stage', which encompasses two- to six-year-olds, who are learning to balance a greater sense of responsibility with a need to control sudden impulses. It is well established that as children make the transition from toddler to preschool years, picky eating plays an integral part of their development. Picky eating is an extremely common phase of development that does not usually cause significant health, developmental or social problems in future. Reasons for picky-eating behaviours in children are complex and can include poor food quality, inappropriate eating behaviours of parents or caregivers, lack of food variety, and the family social climate.

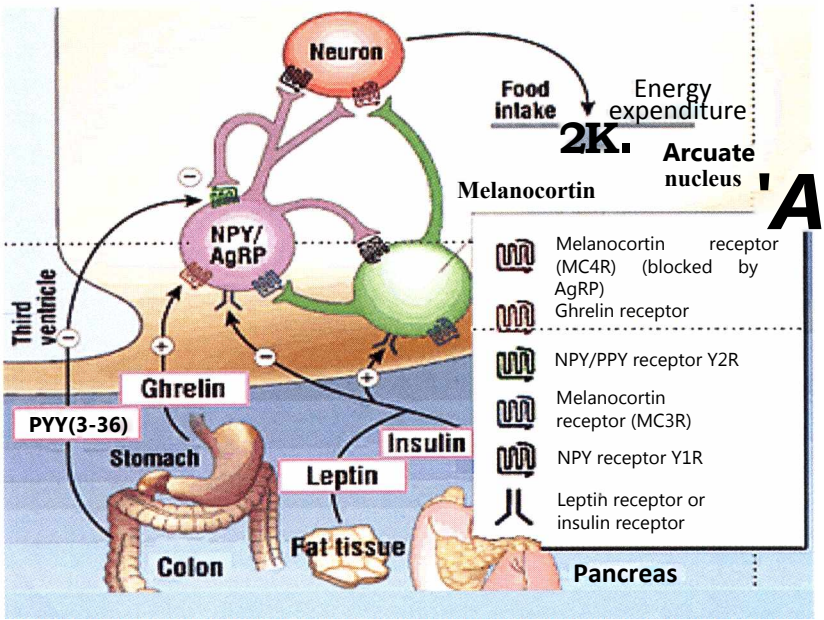
Anorexia nervosa is rare in children and is characterized by the severity of eating restraint, denial and deception, hyperactivity and compulsive exercising, rapid rate and extent of weight loss, depressed mood and obsessive-compulsive behaviour.

Leptin and Ghrelin - The Satiety and Hunger Signals

Two of the major neuroendocrine signals are thought to have a crucial role in the regulation of body weight and appetite: leptin and ghrelin. Accumulating evidence supports the model that stimulation of leptin and ghrelin-responsive pathways, including the central melanocortin system, in the hypothalamus, contributes to the maintenance of body weight and control of appetite. A reciprocal rhythmic pattern of two afferent hormonal signals, anorexigenic leptin and orexigenic ghrelin, imparts rhythmicity to the neuropeptide Y (NPY) system, the final common pathway for appetite expression in the hypothalamus.

Leptin is a 16 kD hormone belonging to the cytokine family of peptides. It is synthesized and secreted from fat cells and is the product of the obesity gene (ob). Leptin is produced by adipose tissue and acts at several receptors in several hypothalamic nuclei, importantly one called the ventral medial nucleus known as "satiety center." Leptin stimulates a feeling of fullness or satiety while ghrelin stimulates appetite. Leptin has been shown to affect appetite, metabolism. An increase in levels of leptin in the blood causes a decrease in feeding behaviour. There are various leptin receptors throughout many tissues in the body including the brain. In particular, neurons in the hypothalamus have been found to have the receptors for leptin. Neuropeptide Y (NPY) containing neurons and a-melanocyte stimulating hormone (a-MSH) neurons in the hypothalamus, both contain the leptin receptor. NPY neurons in the hypothalamus are known to stimulate feeding behavior. Leptin causes a decrease in the activity of NPY neurons, a-MSH neurons in the hypothalamus inhibit feeding behaviour. Leptin stimulates a-MSH containing neurons. Considering only these two actions of leptin in the hypothalamus, it is clear that one of the main central actions of leptin is to reduce appetite. Interactions between ghrelin, leptin, and pancreatic polypeptide control appetite and gastro intestinal motility. It is particularly interesting that, in the framework of the hunger-wake link, the effects of ghrelin and those of leptin are constantly in opposition. Both ghrelin and leptin carry out their effects on feeding behaviour through neurons in the arcuate nucleus. This nucleus and the peptides it synthesises are vitally important for the control of appetite and control interaction between the arcuate nucleus and the melanocortin system.(Fig. 1.6). At least part of ghrelin and leptin signalling is mediated by an ascending neural network through the vagus nerve and brainstem nuclei that ultimately reaches the hypothalamus. Gastrointestinal peptides produce appetite and satiety through endocrine and/or neural pathways (Fig. 1.6).

Hormones that control eating such as, leptin and insulin (lower part of the figure) circulate in the blood at concentrations proportional to body-fat mass. This leads to decrease in appetite by inhibiting neurons that produce the molecules NPY and AGRP, while stimulating melanocortin-producing neurons in the arcuate-nucleus region of the hypothalamus, near the third ventricle of the brain. NPY and AGRP stimulate eating, and melanocortins inhibit eating, via other neu-



Endocrine and neuronal pathways of appetite and satiety
®' ' *(Source:* Michael, 2002)

rons (top). Activation of NPY/AGRP-expressing neurons inhibits melanocortin-producing neurons. The gastric hormone ghrelin stimulates appetite by activating the NPY/AGRP-expressing neurons.

Ghrelin is a 28 amino-acid gastric peptide that acts to increase energy stores in the body and to stimulate the release of growth hormone (GH) from the pituitary gland. It has got appetite-stimulating properties. Ghrelin, which is negatively regulated by leptin and IL-1 beta, is secreted by the stomach and increases arcuate NPY expression, which in turn acts through Y(l) receptors to increase food intake and decrease energy expenditure. Gastric peptide ghrelin may thus function as part of the orexigenic pathway sending 'hunger signal'. Leptin on the other hand, sends the 'satiety signal'.

Epidemiology

Little is known about the prevalence of problematic eating behaviours in healthy, normally developed children and the extent to which these behaviours change over time in different social contexts and also their effect on body weight. It is very difficult to establish the prevalence of problematic disorders like picky eating and childhood anorexia as there are many differences in the populations and

also the definitions for eating behaviours in children across the globe. The peak time for picky eating is the toddler or preschool years. At age 2 or 3, up to 20 percent of children are picky eaters. Carruth (2004) conducted a large survey of families with young children ranging from 4 months of age to 24 months of age to determine the prevalence of picky eating, based on parental perceptions. It was reported that, by 24 months of age, 47% of males and 54% of females were picky eaters and interestingly, the prevalence of picky eaters did not differ with ethnicity, or socioeconomic status.

Pathogenesis

7. *Supertaster Theory*

Picky eaters may be born that way as genetic supertasters.This may be explained by a physiological mechanism based on the supertaster theory. The ability to taste sweetness and bitterness may be genetically related to the number of taste buds on a person's tongue. According to this theory, about 25% of the population has many more taste buds than rest of the general population. The so-called genetic supertasters, for example, may have as many as 1,100 taste buds per square centimetre of tongue, while a more accepting eater may have as few as 11 taste buds in the same-size area. Supertasters find certain fruits and vegetables like grapes, broccoli, cabbage and cauliflower intolerable, bitter and avoid them at all costs. Children are more likely than adults to be supertasters, suggesting that the sensitivity to bitterness diminishes over time.

2. *Slower Growth Rate*

Children belonging to the toddler age group tend to grow more slowly compared to the other stages of development. There could be variation in appetite as well. The quantity of food needed in this age group is also very small comparatively. 'Physiological anorexia' and slow growth rate may be an important reason for picky eating and among toddlers.

3. *Emotional State and Mood Swings*

Toddlers are very much influenced by change in mood and emotions. This could affect their eating habits also, as they do not make decisions like adults based on reasoning.

4. *Parental Influence*

Early infant eating patterns and behaviours are influenced by parental food choices and control over the feeding process. Children develop picky eating habits by modelling after their parent's fussy eating habits and it was more likely to develop, when children were punished, bribed or rewarded for their eating habits. It is well established that the family environment is a key influence on children's

eating behaviours. Parents play a vital role in the consumption patterns of their children because they control what is purchased for the child to eat. Previous food experiences may be a predictor for picky eating. Exposure to less variety of foods may predispose a child to develop picky eating. Parent-child attachment related to feeding is another area that needs further exploration, and may have implications on the extent of breastfeeding and a reduced risk for disordered eating patterns.

5. *Neophobia and Slow Adaptation*

Some children are naturally more sensitive to taste, smell and texture and just need more time to get used to the taste of new foods. Study shows that most children will try a new food only after being offered it 10-15 times. It is normal for most young children to dislike foods due to a bitter or spicy taste or because of their colour or due to chewing difficulty. Neophobia is fear of new items.

6. Genetic and Environmental Influences

Picky eating as well as feeding disorders, like clinical eating disorders, are genetically based, with traits and propensities carried in the DNA. The rather modest contribution of genetics to the occurrence of eating disorders suggests that the family environment might well have a significant role. The parenting of infants of mothers with bulimia nervosa is disturbed, especially in relation to child feeding. There is a strong relationship between feeding problems in childhood and eating disorder in the mother. The mechanisms responsible for this relationship are not understood. Two aspects of the family environment were strongly associated with the presence of child feeding problems: mealtime disorganization, and a mother-child relationship characterized by strong maternal control and dishar-

mony. Problematic family background and relationships increase the risk for development of eating disorders.

PICKY EATING

Children who are labelled as "picky eaters" demonstrate food avoidance and usually eat only a limited number of foods. Picky eaters comprise children who "always" ate a different meal from that eaten by other members of the family, those who "often" refused to eat the right food, and those who "often" refused to eat. Picky eaters eat only a selected group of foods and refuse to try new foods. Picky eating is usually a stage that children outgrow, a normal part of childhood development seen in toddlers who are learning to be independent. Picky eating is associated with eating small meals, eating slowly and accepting a limited number of foods.

Picky Eating and Associated Conditions

7. Low Birth Weight and BMI

Studies reveal associations between low birth weight and underweight, low birth weight and picky eating, and picky eating with low BMI. It is possible that for some low-birth-weight children, there is an underlying metabolic alteration that occurred in uterus that subsequently programs infants to becoming picky eaters, and that picky eating, in turn, subsequently keeps some children in the lower weight trajectory.

2. Future Eating Disorders

Picky eating habits may also be precursors or warning signs of the development of an eating disorder later in life. There is increasing recognition that problematic eating behaviours that manifest in early childhood may be a precursor to maladaptive eating later in life.

Marchi and Cohen (1990) reported that picky eating and pica were associated with anorexia nervosa and bulimic symptoms respectively in adolescence. There is a dramatic increase in the risk of developing an eating disorder when an individual had experienced eating problems earlier in life.

3. Other Associations

Picky eating can be a manifestation of obsessive/compulsive disorder and autistic spectrum disorders. Pickiness in children may be associated with more negative parent affection, more negative food interactions. Picky eaters exhibited decreased sucking patterns during neonatal assessment and this association is more significant with female babies.

Complications/Consequences

It is possible for extreme picky eating to have negative significant health consequences later in life. Picky eating may result in growth failure, susceptibility to chronic illness, and even death if not properly treated. If the poor feeding behavior is severe enough to cause growth faltering, long-term growth and development can be negatively affected.

Nutritional Deficiencies

Parents of picky eaters may be concerned about the nutritional adequacies of their child's diet. Poor eating habits among children create nutritional imbalances, which further reduce appetite or increase carbohydrates cravings. Deficiencies in zinc and vitamin Bj contribute to anorexia along with general malnutrition. "Picky eaters" are also at risk for nutritional deficiencies that can further contribute to growth faltering, and to increased susceptibility to infection. Picky eaters consume fewer total fats, less energy and less protein than children never reporting

picky eating behaviours. Picky eaters are more likely to consume less than dietary recommendations for fruit and vegetables, and meat and alternatives. Many picky eaters choose high-calorie, low-nutrient foods. In some cases food restriction can lead to permanently stunted growth. Studies have reported that carbohydrate, vitamin C, thiamine, riboflavin, niacin, vitamin E, and iron were significantly lower for the picky eater groups in the age group seven to eight months, and energy, total fat, folate, vitamin B thiamine, riboflavin, vitamin B_6, calcium, magnesium, and zinc were lower for the picky eater group at nine to 11 months.

Infantile Anorexia

Infantile anorexia is a severe feeding disorder that typically occurs during the toddler years. Infantile anorexia is the most common serious eating disorder in this age group.

A common finding with childhood anorexia is the frequent high prevalence among boys who have this disorder. In children, boys have been reported to account for between 20-25% of the cases. In Childhood anorexia the core psychopathology is phobic avoidance of normal body weight. The level of serotonin activity in brains of anorexic children was found to be abnormally high. Although normal levels of serotonin are believed to be associated with feelings of well-being, these pumped-up levels of hormones may be linked to feelings of anxiety and obsessional thinking, classic traits of anorexia.

Anorexia Nervosa

This is a disorder of adolescents and young children. The diagnosis of anorexia nervosa in children has been quite rare, controversial and frequently delayed. Depressive symptoms appear earlier or more commonly in childhood anorexia nervosa, possibly as a result of the faster rate of physical deterioration and anorexic symptoms escalate with weight loss creating a vicious cycle.

As a consequence, these children tend to fall off the growth chart; they appear to have normal head circumference with low weight and height percentiles, regardless of parental growth parameters. Table 1 summarizes the complications of anorexia nervosa.

Differential Diagnosis

The differential diagnosis for picky eating in children is limited and includes food allergies or intolerance, lactose intolerance, celiac sprue, gastroesophageal reflux disease (GERD), food refusal in children with GERD, oral hypersensitivity or post-traumatic feeding disorder of infancy.

The differential diagnoses for childhood anorexia are food avoidance, emotional disorder, pervasive refusal syndrome, selective eating disorder, food refusal and functional dysphagia.

Management

Comprehensive assessment of child feeding problems should include observation of a family meal, as well as wider assessment of the quality of the mother-child relationship. Parents and other child caregivers can provide opportunities for children to learn to like a variety of nutritious foods by repeatedly exposing them to these foods, overcoming their tendency to reject unfamiliar foods. Based on Erikson's model, it is anticipated that encouraging children to try new foods will lead to greater initiative in trying different foods in a variety of settings. By pinpointing the factors associated with picky eating and infantile anorexia, researchers have identified new areas of treatment focus, namely that parents can improve their toddlers' eating habits by better understanding their children's temperament and how to cope with their toddlers' behaviours during mealtime. Structured behavioural interventions, systematic desensitization, and cognitive behavioural therapies all prove useful in these situations. Nutrition education and food tasting experiences provide preschool children with a greater sense of initiative in making healthy food choices and tasting new foods. Rewarding and reinforcing children for trying new foods on their own is a positive approach. Gains in weight and height were higher and incidence of infection, mainly upper respiratory infection, was significantly lower in children who received nutrition supplement and counseling.

Children with this disorder require care by a nutritionist and a child psychiatrist skilled in the treatment of infantile anorexia. The main goal of treatment is to remove the conflict and battle of wills from the mealtime. Maternal characteristics and perceptions of their toddlers' temperament characteristics should be addressed in treatment for infantile anorexia. The best way to address infantile anorexia and restore normal growth is by helping the parents reduce stress and control issues around mealtimes. A treatment that focuses on helping toddlers with internal regulation of eating can decrease mother-toddler conflict and struggle for control during feeding. This will improve weight gain in such children.

The treatment of anorexia nervosa in children is a lot more complex than weight restoration. Dietary treatment is obviously important as a major goal in treatment of children with anorexia nervosa. New research shows that irregular or difficult toddler temperament, parental insecurity and parental pursuit of thinness contribute to food refusal and picky eating in toddlers, and are related to infantile anorexia. Developmental delays and picky eating frequently occur together. So better to determine the cause and correct the problem from the inside out.

Although picky eating can be considered as a behavioural issue and treated with behavioural modification, sometimes it may need nutritional interventions as well. Given the high rate of eating disorder among mothers of children with feeding problems, it is essential when assessing children with such disturbance to investigate the mother's eating history.

NEAT Programme

Nutritional Education aimed at Toddlers (NEAT) Program was developed by Horodynski (2004) to improve the knowledge, attitudes, mealtime practices and dietary intake of rural, low income caregivers and toddlers in the home. It was believed that negative attitude towards nutrition led to poor parental feeding practices, which in turn led to picky eaters. There was improvement in the healthier habits and mealtime practices among toddlers after the implementation of this program.

Prognosis

Problematic eating behaviours to some extent are to be expected in the toddler years as young children express their independence; however, as young children transition to the preschool phase, problematic eating behaviours such as picky eating typically subside. Neophobic behaviours will disappear as children grow older. Studies have shown that it usually peaks around preschool age and then declines until about age 10 years and after that food habits remain almost fairly steady. For the most part, picky eating is a childhood phase. The majority of picky eaters resume normal eating behaviour by their tenth birthday, though some may take relatively minor picky eating habits on into adulthood. It is possible that childhood anorexia may represent a more biological or genetic form of the disorder with comparatively poor prognosis.

Conclusion

Good nutrition and healthy food choices are an important component of child health and development. The nutrition choices that young children learn to make affect them throughout their lifetimes. The optimal time to teach nutrition is in the preschool years before unhealthy habits are established and while children are eager to learn. It is important to realize that parents play a huge role in children's self-perceptions of themselves. Understanding the leptin and ghrelin mediated signals and the leptin ghrelin hypothalamic axis will throw more insight into eating disorders in children.

Early recognition and differentiation of infantile anorexia, anorexia nervosa and picky eating in children and timely intervention is essential to prevent future nutritional and psychological disorders. Further research into the investigation on behavioural validation and multidimensional development of eating disorders needs to be performed in order to gauge the true extent of the relationship between childhood eating disorders including picky eating, anorexia and other nutritional problems. Locally available and culturally acceptable programmes *like* NEAT may help a long way in health promotion, prevention and early intervention of eating disorders and their consequences.

SECTION 2

Normal Growth and Growth Assessment

"You are what you eat."

—*Brillat Savarin*

2.1 Normal Growth of Children

A child is not a 'miniature adult'. Childhood is a totally different physiological state compared to adulthood. Growth and development go hand in hand in a child.

FACTORS INFLUENCING GROWTH

The embryo, formed by the fusion of the sperm and the ovum, grows and matures into an adult. A human adult has 10^{14} cells. *Growth is 'increase in size of the body or quantitative growth ' leading to physical maturation. Development is 'maturation of function or qualitative growth' leading to mental maturation.* Growth and development are unique in each person within the limits of normalcy. A wide variety of factors affect growth and development.

The factors affecting growth and development are the following:

1. *Host or genetic factors* like genetic makeup (genotype)
2. *Physical expression (phenotype)*
3. *Demographic factors* like race and sex
4. *Nutritional factors* like maternal nutritional status, breastfeeding and weaning practices, diet during illness, supplementary feeding
5. *Environmental factors* like socioeconomic status
6. *Physical factors* like climate and pollution
7. *Biological factors* like infections, exercise, drugs and chemicals
8. *Emotional factors* like mother and infant bonding, mother and child interaction, family harmony, stimulation and tender loving care (TLC)

Antenatal, natal and postnatal factors like infections, irradiation, drugs and placental insufficiency affect the potential for growth and development to a great extent. '*Ecosensitivity*' is the sensitivity of the organism to the environment. It varies from person to person and is probably genetically determined.

NORMAL GROWTH

1. Stages of Growth

The different stages of growth are:

a) ***Ovum (0-14 days)***
b) ***Embryo (2-9 weeks)***
c) ***Foetus* (9** weeks-birth)
d) *Newborn* (first 28 days of life)
e) *Infant* (first year of life)
f) *Toddler* (1-3 years)
g) *Pre-school child* (3-5 years)
h) *School child* (5-9 years)
i) *Adolescence* (10-19 years). Adolescence is divided into prepubertal, pubertal and postpubertal stages.

2. Growth of Different Tissues

Different tissues grow at different rates. *Somatic growth* of the body is very fast in the first 2-3 years of life and then it slows down. The second spurt is seen in adolescence and then it practically stops. ***Brain growth*** is maximum in the first two years of life and then it slows down. **It** becomes **almost 80% by two years of age.** *Lymphoid growth*, especially of the tonsils and lymph nodes, picks up slowly and peaks before puberty and then it slows down. *Gonadal growth* is noted around puberty only.

Important fetal growth events

Heart beat	**4 weeks**
Circulation	**8 weeks**
External genitalia	10-12 weeks
Bile secretion	12 weeks
Foetal movement	14 weeks
Early swallowing	14 weeks
Meconium	**16 weeks**
Respiration	18 weeks
Surfactant	**20 weeks**
Phonation	22 weeks
Early sucking	28 weeks
Coordinated sucking	34 weeks and swallowing

3. Foetal Growth

By 3 weeks of gestation, the menstrual period is missed. The tri-laminar embryo has ***ectoderm, endoderm and mesoderm.*** **By 10-12 weeks, the external genitalia is distinguishable to identify the sex of the foetus.** Heart beats by 4 weeks, circulation starts by 8 weeks. Respiratory movements occur by 18 weeks. Early swallowing occurs by 14 weeks and sucking by 28 weeks. But, **sucking and swallowing become coordinated only by 34 weeks.** Foetal movements occur by 14 weeks and phonation by 22 weeks. Surfactant is detectable by 20 weeks, bile by 12 weeks and meconium by 16 weeks of gestation. *Meconium is desquamated intestinal cells and intestinal juice.* Almost all the neurons appear by mid-gestation and later they migrate and acquire their various connections.

4. Newborn

The newborn has around 3 kg weight (2.5-4 kg) and 50 cm length (45-55 cm). The head circumference is 35 cm (33-37 cm). **The mid-point of the body is umbilicus unlike pubic symphysis in the adult.** The upper segment to the lower segment ratio from the vertex to pubic symphysis and the pubic symphysis to the heels respectively is 1.7:1. The respiratory rate is 40/minute and the heart rate is 140/minute. Most newborns lose up to 10 per cent weight initially and regain birth weight by 10 days and thereafter the **weight gain is around 200 g/week in the first 3 months,** 150 g/week in the next 3 months and 100 g/week in the next 6 months.

5. Underfives

Children under 5 years of age are a vulnerable group with high morbidity and mortality. Their growth is a direct reflection of their nutritional status.

a) **Weight:** The birth weight doubles by 4 months, triples by one year, quadruples by two years. Thereafter, 2 kg is added on every year till six years and thereafter 3 kg is added on every year till puberty. (Table 2.1)

b) **Length:** The birth length is 50 cm, it becomes 66 cm by 6 months, 75 cm by 1 year and 87 cm by 2 years. It doubles by 4 years and thereafter 6 cm is added on every year till puberty. **Birth length triples by 12 years.** (Table 2.1)

c) **Head circumference:** At birth, the head circumference is 35 cm, it increases to 40 cm by 3 months, 43 cm by 6 months, 45 cm by 9 months, 47 cm by one year, 49 cm by two years and 50 cm by 3 years. The approximate increase is 2.0 cm/month in the first 3 months, 1 cm/month in the next 3 months and 0.5 cm/month in the next 6 months. (Table 2.1)

d) **Chest circumference:** At birth, the head circumference is more than the chest circumference and **it equalises by 1 year.** Thereafter, the chest circumference is more than the head circumference. The chest circumference is measured at the nipple midway between inspiration and expiration. In malnutrition, chest circumference will remain less than head circumference.

Table 2.1 Bedside calculation for weight*, height**, head circumference

Age (years)	Weight (kg)	Height (cm)	Head circumference (cm)
Birth	3	50	33-35
3/12	5	60	39-40
6/12	7	66	42-44
9/12	9	71	44-45
1	10	75	45-47
2	12	87	47-49
3	14	94	49-50
4	16	100	50-51
5	18	106	50-52
6	20	112	51-52
7	23	118	
8	26	124	
9	29	130	
10	32	136	
11	35	142	
12	38	150	

*Add 2 kg/year in 1-6 years of age and add 3 kg/year thereafter till puberty.
**Add 6 cm/year after 2 years of age till puberty

Table 2.2 Formula for growth parameters

Weight (kg) (Weech's)	Infant	Age in months + 9
	1-6 yr	(Age in years x 2) + 8
	7-12 yr	$\frac{\text{(Age in years x 7 - 5)}}{2}$
Height (cm) (Weech's)	2-12 yr	(Age in years x 6) + 77
Head circum-ference (cm) (Dine's)	Infant	Length

The bedside calculation, Weechs 'formula or the National Center for Health Statistics (NCHS) reference standards and WHO growth charts are used to derive the expected weight, height, head circumference etc. The formulae for growth parameters are given in tables 2.1 and 2.2. The WHO accepted NCHS reference standards are given in tables 2.3 (a) & (b).

The anterior fontanel (AF) **closes by 9-18 months** and the ***posterior fontanel (PF)*** **closes by 2—4 months.** *Wide AF and PF* are seen in hypothyroidism, hydrocephalus and in rickets. Preterm babies generally have higher increments of growth, on par with intrauterine growth and they may catch up with others by two years of age. The catch-up growth may be up to 10 times that for the age or 5 times that for the height of the child. It is less in those with congenital anomalies and in full-term small for gestational age (SGA) babies.

6. Teeth Development

The teething appears generally between 3 and 7 months. ***The temporary, deciduous or milk set*** has 20 teeth (8 incisors, 4 canine and 8 molars). These **appear by two and a half years of age.** (No. of teeth = Age in months - 6.) The first tooth appears by 5-9 months. By 1 year of age, 6-8 teeth are present. The infants tend to be irritable, drool throughout the process of teething. In the permanent set, there are 32 teeth (8 incisors, 4 canine, 8 premolar and 12 molars) and the first **molars appear by 6 years.** Eruption of the second molar marks puberty. The eruption of the third molar (wisdom tooth) is variable and occurs after 18 years of age (Fig. 2.1).

7. Skeletal Maturation

In full-term newborn babies, 5 ***ossification centres*** **are present,** namely, lower end of the femur and the upper end of the tibia in the knee and 3 tarsal bones, namely, *talus, calcaneus and cuboid* in the ankle. The head of the humerus is present by 2 months of age and the head of femur by 4-6 months. **By 5-6 months, 2** ***carpal bones,*** **capitate and hamate, appear.** The lower end of radius appears by 2-3 years and lower end of ulna by 7 years. The third carpal, triquetrum, appears by 2-4 years and thereafter 1 carpal bone appears every year. The 8th carpal bone, pisiform, appears by 12 years of age. Except for the first two, there is high variability in the appearance of the other carpal bones. ***The ossification centres appear first on the left side of the body and then on the right side.***

These ossification centres are useful in assessing the bone age of the child. The *bone age is delayed in hypopituitarism, hypothyroidism, severe malnutrition and constitutional delay. It is advanced in precocious puberty.* **Fusion of capitulum with the shaft at elbow predicts puberty within a year.**

Table 2.3a Growth chart percentiles: Weight (kg), height (cm), head circumference (cm): 97th, 50th and 3rd centiles, agewise up to 18 years

	Boys			Girls		
	kg	cm	cm	kg	cm	cm
Birth	4.2	54.8	37.2	3.9	53.9	35.9
	3.3	50.5	34.8	3.2	49.9	34.3
	2.5	46.2	32.6	2.3	45.8	32.1
3 mo	7.6	66.1	43.1	6.9	64.2	41.7
	6.0	61.1	40.6	5.4	59.5	39.5
	4.2	56.1	38.4	4.0	54.9	37.3
6 mo	9.7	72.9	46.2	8.9	70.9	44.6
	7.8	67.8	43.8	7.2	65.9	42.4
	6.0	62.8	41.5	5.6	61.0	40.3
9 mo	11.1	77.3	48.1	10.4	75.6	46.4
	9.2	72.3	45.8	8.6	70.4	44.3
	7.4	67.4	43.5	6.7	65.3	42.3
1	12.2	81.2	49.3	11.5	79.6	47.6
	10.2	76.1	47.0	9.5	74.3	45.6
	8.2	71.0	44.8	7.6	69.0	43.5
1/2	13.8	88.1	50.6	13.0	85.7	49.1
	11.5	82.4	48.4	10.8	80.9	47.1
	9.3	76.7	46.3	8.6	75.1	45.0
2	15.0	94.0	51.4	14.3	92.6	50.1
	12.6	87.6	49.2	11.9	86.5	48.1
	10.1	81.3	47.3	9.6	80.3	46.1
2½	16.2	98.7	52.2	15.7	97.7	50.8
	13.7	92.3	49.9	12.9	91.3	48.8
	10.9	85.8	48.0	10.5	84.9	47.0
3	17.5	103.2	52.8	17.0	102.3	51.4
	14.7	96.5	50.5	13.9	95.6	49.3
	11.8	89.9	48.6	11.3	88.8	47.6
3½	19.3	106.7	53.	19.1	105.3	52.2
	15.7	99.1	50.8	15.1	97.9	49.8
	12.4	91.5	48.2	12.1	90.6	47.4
4	20.5	111.0	53.8	20.4	109.2	52.6
	16.7	102.9	51.2	16.0	101.6	50.2
	13.1	94.9	48.6	12.8	94.0	47.8
4½	21.8	114.9	53.9	21.6	113.0	52.9
	17.7	106.6	51.4	16.8	105.1	50.5
	13.9	98.2	48.9	13.4	97.2	48.1

contd.

	23.2	118.6	54.1	22.9	116.7	53.2
5	18.7	109.9	51.6	17.7	108.4	50.8
	14.7	101.3	49.1	14.0	100.1	48.4
	26.2	125.2	54.4	25.8	123.9	53.6
6	20.7	116.1	51.9	19.5	114.6	51.2
	16.3	107.0	49.4	15.3	105.4	48.8
	29.8	131.3	54.6	29.7	130.9	53.9
7	22.9	121.7	52.1	21.8	120.6	51.5
	17.9	112.1	49.6	16.7	110.3	49.3
	34.1	137.0	54.8	35.0	137.7	54.1
8	25.3	127.0	52.3	24.8	126.4	51.7
	19.5	116.9	49.8	18.3	115.0	49.3
	39.2	142.8	55.0	41.3	144.5	54.3
9	31.4	137.5	52.8	28.5	132.2	51.9
	22.7	126.0	50.3	20.2	120.0	49.5
	45.2	149.0	55.3	48.2	151.2	54.7
10	31.4	137.5	52.8	32.5	138.3	52.2
	24.8	130.6	50.4	22.5	125.4	49.7
	51.7	155.9	55.8	55.3	157.8	55.2
11	35.3	143.3	53.1	37.0	144.8	52.7
	24.8	130.6	50.4	25.2	131.7	50.2
	58.7	163.8	56.4	62.0	164.4	55.6
12	39.8	149.7	53.6	41.5	151.5	53.2
	27.6	135.5	50.8	28.3	138.7	50.8
	65.9	172.0	57.0	68.0	169.7	56.0
13	45.0	156.5	54.1	46.1	157.1	53.6
	31.2	140.9	51.2	31.7	144.6	51.2
	73.2	179.2	57.5	73.0	172.9	56.2
14	50.8	163.1	54.6	50.3	160.4	54.0
	35.9	147.0	51.7	35.2	147.8	51.8
	80.1	184.2	57.6	76.8	174.5	56.3
15	56.7	169.0	54.8	53.7	161.8	54.2
	40.9	153.8	52.0	38.3	149.1	52.1
	86.4	187.1	57.7	79.1	175.0	56.4
16	62.1	173.5	55.0	55.9	162.4	54.3
	45.7	160.0	52.3	40.8	149.9	52.2
	91.6	188.6		80.8	175.0	
17	66.3	176.2		56.7	163.1	
	49.6	163.9		42.3	151.1	
	95.3	189.2		79.9	174.9	
18	68.9	176.8		56.6	163.7	
	52.0	164.4		42.9	152.5	

US NCHS Reference Standards recommended by WHO

Growth parameters recommended by WHO— NCHS reference standards: 50th centile, agewise

Male				Female		
wt	**Length**	**OFC**	**Age**	**Wt**	**Length**	**OFC**
kg	**cm**	**cm**	**mo**	**kg**	**cm**	**cm**
3.3	50.5	34.8	**0**	3.2	49.2	34.3
4.3	54.6	37.2	**1**	4.0	53.5	36.4
5.2	58.1	40.3	**2**	4.7	56.8	38.0
6.0	61.1	40.6	**3**	5.4	59.5	39.5
6.7	63.7	42.0	**4**	6.0	62.0	41.0
7.3	65.9	43.0	**5**	6.7	64.1	42.0
7.8	67.8	43.8	**6**	7.2	65.9	42.4
8.3	69.5	44.5	7	7.7	67.6	43.5
8.8	71.0	45.0	**8**	8.2	69.1	44.0
9.2	72.3	45.8	**9**	8.6	70.4	44.3
9.5	73.6	46.0	**10**	8.9	71.8	45.0
9.9	74.9	46.5	**11**	9.2	73.1	45.4
10.2	76.1	47.2	**12**	9.5	74.3	45.6
10.4	77.2	47.5	13	9.8	75.5	45.9
10.7	78.3	47.8	**14**	10.0	76.7	46.2
10.9	79.4	48.0	**15**	10.2	77.8	46.5
11.1	80.4	48.2	**16**	10.4	78.9	46.8
11.3	81.4	48.3	**17**	10.6	79.9	47.0
11.5	82.4	48.4	**18**	10.8	80.9	47.1
11.7	83.3	48.6	**19**	11.0	81.9	47.2
11.8	84.2	48.8	**20**	11.2	82.9	47.4
12.0	85.1	49.0	**21**	11.4	83.8	47.5
12.2	86.0	49.1	**22**	11.5	84.7	47.7
12.4	86.8	49.2	**23**	11.7	85.6	47.9
12.6	87.6	49.3	**24**	11.9	86.5	48.1
13.6	92.3	49.9	**30**	12.9	91.3	48.8
14.6	96.5	50.5	**36**	13.9	95.6	49.3
15.5	99.0	50.7	**42**	15.0	98.0	49.5
16.0	103.0	51.0	**48**	15.9	102.0	50.0
17.7	106.6	51.4	**54**	16.8	105.1	50.5
18.7	109.9	51.9	**60**	17.7	108.4	50.8

OFC—occipitofrontal circumference

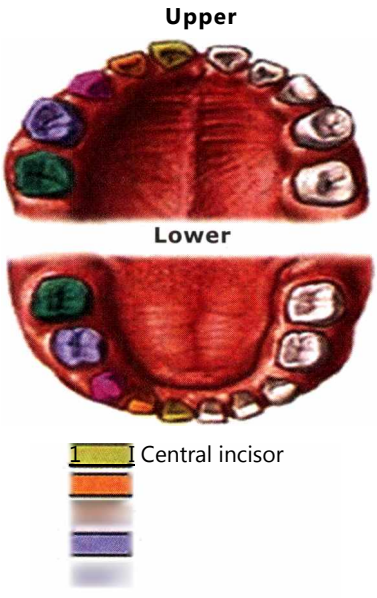


Fig. 2.1 Teeth development

2.2 Growth Pattern of Low Birth Weight (LBW) Babies

Normal birth weight is the first wealth of a baby. Low birth weight (LBW) is a **'hard social indicator'** associated with infant mortality, morbidity, physical and developmental retardation and reduced survival and quality of survival. LBW poses a great challenge due to the economic burden on the family, the society and the nation, apart from the physical and mental sequelae in the individual. The ultimate growth and intelligence of an individual is the expression of the endowed genetic potential. Nutrition and environment are the two important factors that influence this. The **growth of a baby is genetically programmed.** LBW is a consequence of growth failure in the early stage and is likely to result in failure to reach the endowed genetic potential.

Birth weight is the sum total of the intrauterine environment. LBW is < 2.5 kg, irrespective of gestational age. **Very low birth weight (VLBW) is < 1.5 kg and extremely low birth weight (ELBW) is < 1 kg.** LBW is a heterogeneous group consisting of preterm babies and full-term small-for-gestational-age (SGA) babies. The SGA babies have intrauterine growth retardation. In fact, their malnutri-

tion starts in the womb. Depending upon the gestational age, babies are classified into **preterm babies with < 37 weeks of gestation and full-term babies with > 37 weeks of gestation. Depending upon the weight and gestation, babies are classified into appropriate-for-gestational-age (AGA), small-for-gestational-age (SGA) and larger-for-gestational-age (LGA) babies.** Some of the preterm LBWs may be AGA or SGA, but all full-term LBWs are SGA babies. SGA babies are again classified into malnourished, hypoplastic and mixed groups. **Hypoplastic babies** have reduced growth potential and congenital anomalies. These are the products of pre- or periconceptional and early insults like chromosomal and genetic disorders, teratogens etc. They generally do not catch up with the peer group in growth and development. Those with normal organogenesis, but reduced nutrition and placental insufficiency especially in the later period of gestation become **malnourished babies.** They generally do not have congenital anomalies and they grow and catch up with the peer group, if appropriate intervention is given early. Those with early as well as late problems constitute the **mixed group** with guarded prognosis. Similarly, preterm babies without congenital and other problems do extremely well and catch up with the peer group in growth and development by 2 years of age. Hypoplastic babies have reduced weight, height and head circumference and are **symmetric in appearance.** Malnourished babies have reduced weight but length and head circumference are almost normal and are **'asymmetric' in appearance. The Ponderal index is used to categorise these babies.**

$$\text{Ponderal index (PI)} = \frac{\text{Weight (g)}}{\text{Length}^3\text{ (cm)}} \times 100$$

PI > 2.5 is normal. **In malnourished babies, it is < 2** and in hypoplastic babies, it is between 2-2.5. The author has reported that BMI of 13 and MAC of 9 cm will pick up normal babies and 8 cm will pick up babies > 2 kg.

The union of the sperm and ovum results in a monocellular zygote with tremendous growth potential. **At conception, the weight is only 0.005 mg.** The weight increases 65 million times to achieve a birth weight of 3 kg. **At 2 weeks, the embryo weighs 1 gand measures 2.5 cni.** At 12 weeks it is 14 g and 7.5 cm in size and at 28-weeks, it is 1 kg and 35 cm. At 40 weeks or full-term gestation, the weight is around 3 kg and the length around 50 cm. The preterm babies have less metabolic and nutritional reserve. Fat stores are less because **fat is mostly deposited in the last 6 weeks of gestation, i.e, after 34 weeks gestation.** In a 28 weeks gestation preterm baby weighing I kg, the fat is only 1% of body weight; it is around 3.5% at 32 weeks and 16% in a full-term baby. **Glycogen is deposited in the liver only in the last 4 weeks of gestation.** At around 32 weeks gestation, glycogen per unit weight of liver tissue is only one-fourth and total body carbohydrate is only 9 g compared to 34 g in a full-term baby. Similarly, protein, calcium, iron

and other stores are also very little in them. Thus the body composition of preterms differ from that of term babies in many ways. In full-term SGA babies also, the metabolic and nutritional reserve will be reduced.

In early foetal life, growth is due to increase in the number of cells. In mid-foetal life, there is increase in both cell number and size. **By mid-gestation, the total number of neurons in the brain is achieved.** In later foetal life, there is increase in cell size, migration and organization. Thus it is evident that growth failure in early part of gestation is associated with permanent retardation of growth potential. Those with growth failure in later part of gestation can be effectively rehabilitated by providing early and optimum nutrition and stimulating environment before the critical period of growth is over, i.e., first 2 years of age. They generally catch up with the peer group by two years of age. **By two years of age, the body growth and weight approximates 20% of the adult, whereas brain growth becomes almost 80%.**

Soon after birth, there is about 10% weight loss in most babies. Then the birth weight is regained by 10-14 days. In a term baby, the initial weight gain may be 10-40 g/day. The weight gain during infancy is as follows:

Weight gain in the first 3 months (after regaining birth weight)	=	200 g/week
Weight gain in the next 3 months	=	150 g/week
Weight gain in the next 6 months	=	100 g/week

A normal baby doubles the birth weight by 5 months, triples by 1 year and quadruples by 2 years of age. But, in LBW babies, especially preterm babies and those without congenital anomalies, the growth is even faster. It may approximate the intrauterine growth. The intrauterine growth curves for length, weight, and head circumference according to the gestational age are depicted in Fig. 2.2-2.4; generally it is 10 times for the age or 5 times for the length of the baby. A 1 kg preterm baby may increase the weight by 10 times to achieve 10 kg at two years. Large cohort studies of survivors of LBW babies from various countries have revealed that there is a rapid growth in the first 6 months followed by a deceleration. The reflection of the general trend of slow growth noted among normal babies may be observed in them as well. In Indian studies one-third of the LBWs were found to be in the normal range for weight and one-fourth in the normal range for height and head circumference by 4 years of age. Multiple regression analysis have shown that determinants of catch up at 4 years of age were weight and height at one year of age. Weight at one year of age is a reflection of post-natal environment. This slow trend of growth noted in the Indian studies may be due to nutritional and environmental handicaps. Many children lag behind the *controls* at adolescence as well with an ultimate reduced height and weight. The

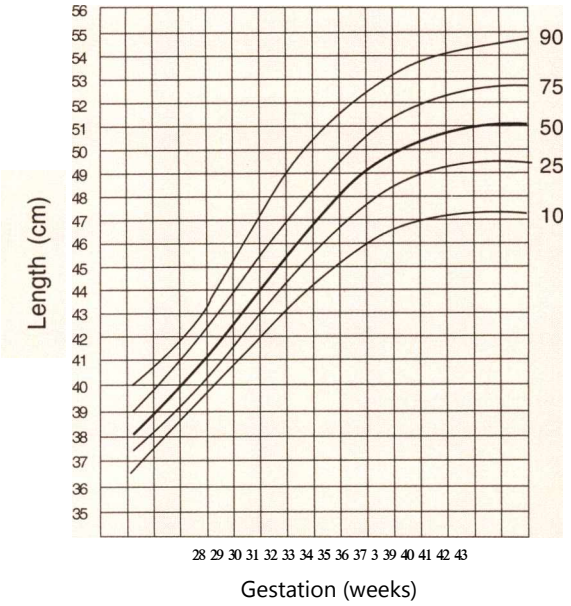


Fig. 2.2 Intrauterine growth: Length, 28-43 weeks (both sexes)

full-term IUGR SGAs remain significantly handicapped in physical growth compared to preterms without congenital anomalies. Two-third of the LBWs achieve normal development, 20% tend to show mild to moderate impairments and 10-20% end up with cerebral palsy and mental retardation. **30% of all cerebral palsies occur in LBW babies.**

Expected weight and weight gain for a preterm baby can be calculated based on the intrauterine growth curves till 42 weeks of gestation or expected date of childbirth. Afterwards, the corrected age should be taken into account for the calculation till one year of age; e.g., a low birth weight baby of 1.5 kg born 2 months preterm is now 8 months old:

Corrected age = 8-2 = 6 months

$$\text{Expected weight} = \frac{\text{Corrected age} + 9}{2} = 7.5 \text{ kg}$$

$$\text{Expected weight of a full-term baby at 8 months} = \frac{\text{Chronological age} + 9}{2} = 8.5 \text{ kg}$$

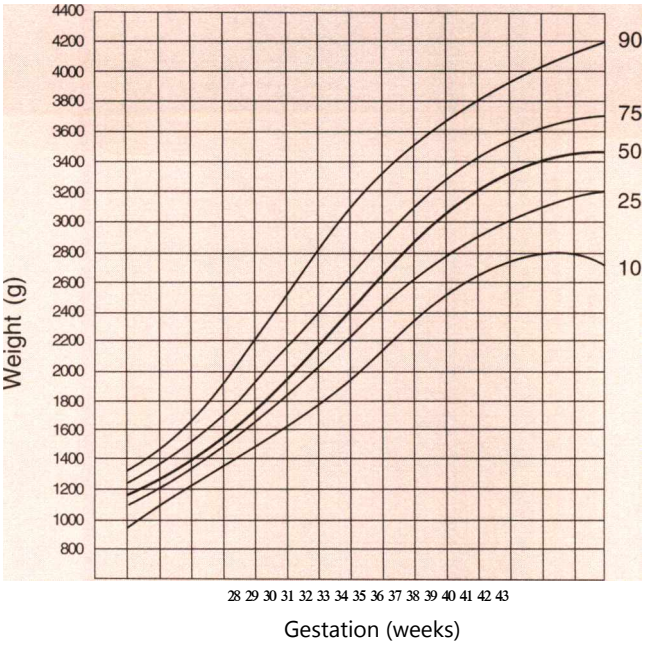


Fig. 2.3 Intrauterine growth: Weight, 28-43 weeks (both sexes)

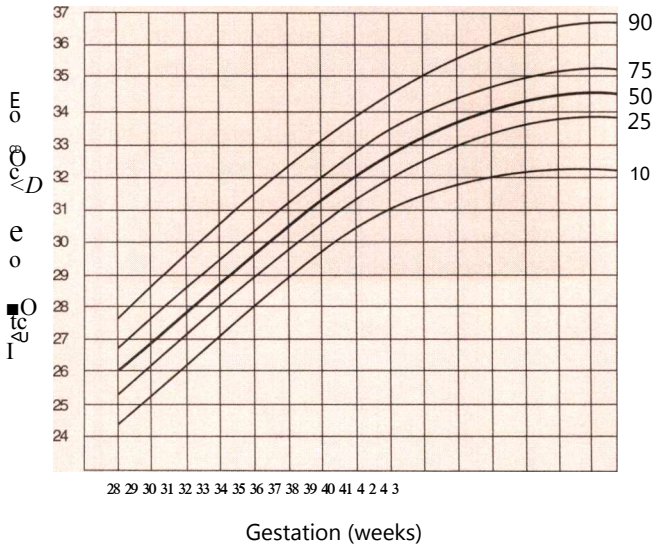


Fig. **2.4** Intrauterine growth: Head circumference, 28-43 weeks (both sexes)

Other long-term sequelae of LBW are also coming to limelight. They seem to be programmed for early degenerative and other adulthood diseases. According to the **Barker hypothesis,** they are more prone to develop diseases like insulin resistance, diabetes mellitus, coronary artery disease, renal diseases and so on. Barker and his colleagues of the British Medical Research Council traced a cohort of men from Hertforshire whose birth weight and weight at one year were available. Those with LBW and low weight at one year were shown to have increased incidence of impaired glucose tolerance, type II diabetes and so on in the middle age. Thus, LBW is associated with physical, developmental, intellectual, behavioural, economic and health problems. This advanced and altered foetal programme is under investigation.

Due to the diverse sequelae, LBW can be perceived as the greatest public health problem facing the globe. But as the aetiology is diverse, prevention is difficult. Considering the diverse aetiology, it can be considered as the **'biological indicator of social deprivation'.** Good quality antenatal care has failed to reduce the incidence of LBW significantly. This warrants effective new strategies aiming at the girl child, adolescent girl, the prospective mother and the antenatal mother. This is called the **life cycle approach.** A **pre-pregnancy weight of around 45 kg and a height of around 145 cm is considered ideal**; at least 42 kg and 142 cm should be achieved. Antenatal care is important to ensure postconceptional early care and for risk assessment. Overall improvement in socioeconomic status, environmental hygiene, adolescent nutrition, micronutrient supplementation and control and treatment of reproductive tract infections (RTI) and sexually transmitted infections (STI) also deserve special mention. Women empowerment and social and financial stability are also important in reducing LBW. Empowerment of women in child rearing skills and ensuring support system in child rearing are also important for child survival and quality of survival. These are now collectively called as **parenting skills.**

Parenting is an art to be learned through experience based on our own upbringing, opportunities to look after sibs, and rekindling and reshaping innate parenting instincts during adolescence and early adulthood through sharing and counseling. In the life cycle approach to reduce LBW and improve birth weight of babies, it is better to start with adolescent care. There are already a number of programmes for infants and young children and pregnant and lactating mothers.

Nutrition and micronutrient supply to adolescents, rubella vaccination to prevent congenital rubella and hearing defect in offspring's, use of reproductive and child health (RCH) services, use of family life education and adolescent clinics can initiate interventions among adolescents.

2.3 The ICP Model of Growth

Childhood is the period of prime growth. Child's growth generally refers to skeletal or linear growth as it is reflected in their height. But every organ system in a child is growing, including their reproductive system and their central nervous system. The growth of all these is dependent on adequate nutrition. So nutrition and malnutrition in infancy and childhood can affect a child's development, but not so in adults.

A useful overall framework for considering linear growth in childhood is provided by the infancy-childhood-puberty (ICP) model (Karlberg, 1987; 1989). This considers a child's growth as made up of three components.

The **infancy component** is continuous with foetal growth (growth before birth), lasts through the first year. This is a period of very rapid growth.

Then follows a period of steady but slower growth, **the childhood component,** which lasts through to the beginning of puberty.

In the **pubertal component** there is a growth spurt, together with a rapid maturation of secondary sexual characteristics, and skeletal growth then stops.

ICP stands for infancy, childhood and puberty stages of growth. The important determinants of growth vary in each of these stages. The determinants are given below:

- Infancy - Nutrition, birth weight and illnesses
- Childhood - Growth hormone, thyroxin, nutrition
- Puberty - Sex steroids, genetic factors and nutrition

Intrauterine growth in turn is influenced by genetic factors (38%), maternal uteroplacental environment (62%). Maternal nutrition, hormones like insulin, IGF, HPL (human placental lactogen), other growth factors, micronutrients, thyroid hormone are also important in intrauterine growth.

In childhood, GH, GH-mediated IGF and thyroid-mediated GH are important determinants of growth. Up crossing and down crossing of centiles to settle to mid parental height (target height) is also observed during this period.

In puberty, the factors are sex steroids, GH-mediated sex steroids and nutrition, micronutrients. Timing of puberty is also important. In maturational growth delay (constitutional delay), the late blooming children remain small till late. But ultimate prognosis and growth are good. There will usually be a similar history in one of the parents, usually the father. Whereas in early blooming children, growth spurt occurs early and epiphysis fuse early resulting in ultimate short adults. During puberty, gynaecomastia may occur in some boys. Source of estradiol in pubertal boys is due to aromatisation of androgen.

During puberty, girls grow faster and earlier. Menarche occurs by 12.5-14.5 yr. Precocity/precocious puberty is diagnosed when second sexual characters occur before 8 yr in girls and 9 yr in boys. Generation by generation, people are thought to grow taller.

Hormonal Regulation of Linear Growth

The growth process is under the control of the endocrine system. However, not only hormones are involved; hormone-binding proteins, growth factors and their binding proteins, as well as the stage of maturity and quantity of the hormone and growth factor receptors on the target cells may play a critical role too. Furthermore, the secretion of hormones, such as growth hormone (GH), follows a pulsatile pattern with higher peaks during the nights as well as during puberty. Such growth regulatory mechanisms interact and change in character over the ages. A further confusing element is that some growth can take place without involvement of central steering mechanisms, as illustrated by an animal study showing that catch-up growth was regulated locally, at the tissue level, and not necessarily by the influence of circulating serum growth factors. For these reasons, the measurement of a hormone or growth factor in a single serum, urine or tissue sample will shed light on only a small part of this very complicated puzzle. It is generally agreed that we have at least three distinct endocrine phases of linear growth. The pattern of postnatal growth is well documented; a high growth rate is observed from foetal life, with a rapid deceleration up to about 3 years of age. This is followed by a period with lower, slowly decelerating velocity up to puberty. Puberty starts with an increased rate of growth, and after the age of peak height velocity has been reached a deceleration is noted until growth ceases.

How foetal linear growth is regulated is not precisely defined and no key circulating hormone has so far been identified. Uterus size, nutritional support and oxygen level in conjunction with insulin-like growth factors and insulin are believed to be involved in regulating foetal growth.

During fetal life the serum GH level is high, and GH receptors have also been detected. Fetal linear growth, however, is known to be almost independent of GH. A lack of growth response to GH during fetal life may be due to immature GH-specific receptors in the growth plate, as noted in the rabbit. GH-deficient children are on average 1-2 cm, or 2-4% shorter than normal infants at birth. Whether this minor deviation is a secondary effect due to the lack of the influential metabolic action of GH or to the lack of a direct effect of GH on the cartilage, is still a matter of debate.

It is more generally accepted that GH is responsible for growth during childhood provided that thyroid hormone secretion is normal. The exact age at which GH begins to control linear growth in humans is still uncertain. The majority of children with isolated GH deficiency grow more or less normally during the

first 6 months of life, but not thereafter. Growth during adolescence is related both to GH and sex steroids - testosterone in males and oestrogens in females. Both GH and sex hormones are needed for normal pubertal growth, although the presence of only one of them is associated with some growth during this period. It is not clear whether GH and sex steroids interact or act independently of each other. It is thus reasonable to conclude that linear growth from birth to maturity is regulated by at least three different growth-promoting systems. Two simultaneously active, superimposed, systems are known to be involved in the adolescent growth spurt. Similarly, a postnatal continuation of the nutritionally driven foetal growth in conjunction with the GH-dependent phase of childhood growth characterizes the growth in the first year of life.

Environmental factors are a more likely cause of the stunting process than the ethnic or genetic background. Maternal illiteracy, poor hygiene, overcrowding, a high disease load and improper and/or contaminated food are all interacting in such environments. Whatever the causative factors are for the faltering process, they will remain if the general living conditions and educational level are not improved. It will take generations before the stunting problem has been eliminated, even in communities with fine financial resources and a well developed health care system. This is called 'Secular trend' which is not uniform in a country like India. In Delhi, per decade. 2.1-2.7 cm but, in other states, only 1.5-2.1 cm is the average height gain/month.

2.4 Growth Disorders and Failure to Thrive (FTT)

FTT refers to a condition when the physical growth of a child is less than expected, usually below the third or fifth centile, or when a child has significant loss of weight in a short time. The extent of malnutrition and the degree of stunting and wasting should be assessed. FTT is divided into three categories:

1. **Organic FTT** (30%): with a known medical condition. This is also called biologic FTT.
2. **Non-organic/psyehosocial FTT** (70%): without any known medical condition. A majority is due to psychosocial neglect. In some, poverty and accidental errors in feeding are noted. This is also called environmental FTT.
3. **Mixed type**

FTT and malnutrition are closely related. *FTT is a medical problem or a label for investigation, whereas PEM is a diagnosis.* The clincal features of FTT are growth retardation, developmental delay, mental changes, behavioural problems and soft neurological signs. Rumination, anorexia nervosa and bulimia may be noted in a few. Neglect of hygiene may be evidenced by diaper rash, dirty

fingers and nails, intertrigo, dirty skin and dress etc. Alopecia on the occiput may suggest that the baby was lying unattended for prolonged periods. Tear in the frenulum and angle of the mouth may indicate force feeding by a rejecting mother. The child may lack eye contact and fails to interact with the mother and the environment. Physical abuse also may be evident in psychosocial FTT.

In preterm low birth weight babies, corrected age can be used to compare physical growth and development till 2 years of age. Organic causes must be looked for in all cases. Table 2.4 summarises the causes of organic FTT. Another approach is to categorise them into a high intake FTT, who take more food, and low intake FTT, who take less food.

Table 2.3 Causes of Organic Failure to Thrive (FTT)

	Organ	Causes
1.	GIT	Cleft lip and palate, GE reflux, pyloric stenosis, lactose intolerance, CMPI, malabsorption, liver and biliary diseases, pancreatic diseases, IBD, coeliac disease, Hirschsprung's
2.	Renal	RTA, UTI, CRF
3.	Neurologic	Mental retardation, cerebral palsy
4.	Cardiovascular	CCF, pulmonary arterial hypertension (PAH)
	Respiratory	Tracheo-oesophageal fistula, cystic fibrosis, bronchiolitis obliterans, asthma, bronchopulmonary dysplasia
6.	Endocrine	Hypopituitarism, hypothyroidism, adrenal insufficiency
7.	Metabolic	Inborn errors of metabolism
8.	Infections	Intrauterine infections, TB, AIDS
9.	Miscellaneous	Syndromic and nonsyndromic malformations, IUGR, chromosomal anomalies, heavy metal poisonings, congenital immunodeficiency

CMPI - Cow's milk protein intolerance
IBD - Inflammatory bowel disease
RTA - Renal tubular acidosis
IUGR - Intrauterine growth retardation

Management includes hospitalization, dietary support and counseling. These are important especially when organic causes are not obvious. Most of the investigation can be undertaken after a week of observation and dietary management. The failure of the parent to give food, the failure of the child to take food and the failure of the child to retain food are the usual causes, that need remedial measures.

In organic FTT, the treatment of the medical condition should be given priority along with dietary management. Allow the infant to feed for 20-30 minutes, offer solids before liquids and avoid distractions during feeding. High calorie formulas that offer more than 20 Cal/ounce and high calorie supplements like oil are useful.

High energy milk that supplies FI00- 100 Cal/100 ml (milk 100 ml, sugar 1 tsp, oil *Vi* tsp), cereal milk and thickened feeds (milk 100 ml + 2 tsp cereal flour or SAT Mix) are beneficial. SAT Mix is a precooked ready-to-mix powdered cereal pulse mixture prepared from rice: wheat:blackgram:sugar in the ratio 1:1:1:2. Family counseling is also important. Weight gain in response to feeding establishes psychosocial FTT. A weight gain of *⁄2 kg/week or 70 g/kg/week is expected.

Mortality rates tend to be high and the ultimate prognosis for physical growth and intelligence is guarded. The prognosis depends upon the age of onset and the duration of FTT. Growth retardation, mental subnormality, behavioural problems and delinquency are sequelae of FTT.

Low birth weight is an important medical and social problem requiring urgent attention with respect to prevention, feeding and special care.

2.5 Growth Charts

1. Growth Charts

Growth charts were **popularized by David Morley.** These are used for growth monitoring. Well baby clinics, primary health centres and ICDS programme utilize growth charts. The weight measurements of a child over a period of time are plotted on the growth chart and any deviation from the normal pattern can be visualized and interpreted. An upward curve in the *'Road to Health '* is ideal (Fig. 2.5). In a child with normal nutritional status, the curve is within the 'road to health'. In a coloured chart, this is the green zone. The curve of those with severe malnutrition will fall in the lower red zone and that for those with mild and moderate malnutrition will fall in the blue and yellow zones. Aflat curve and a downward curve are not desirable. Such children should be investigated and followed up. They must be also be given food supplementation.

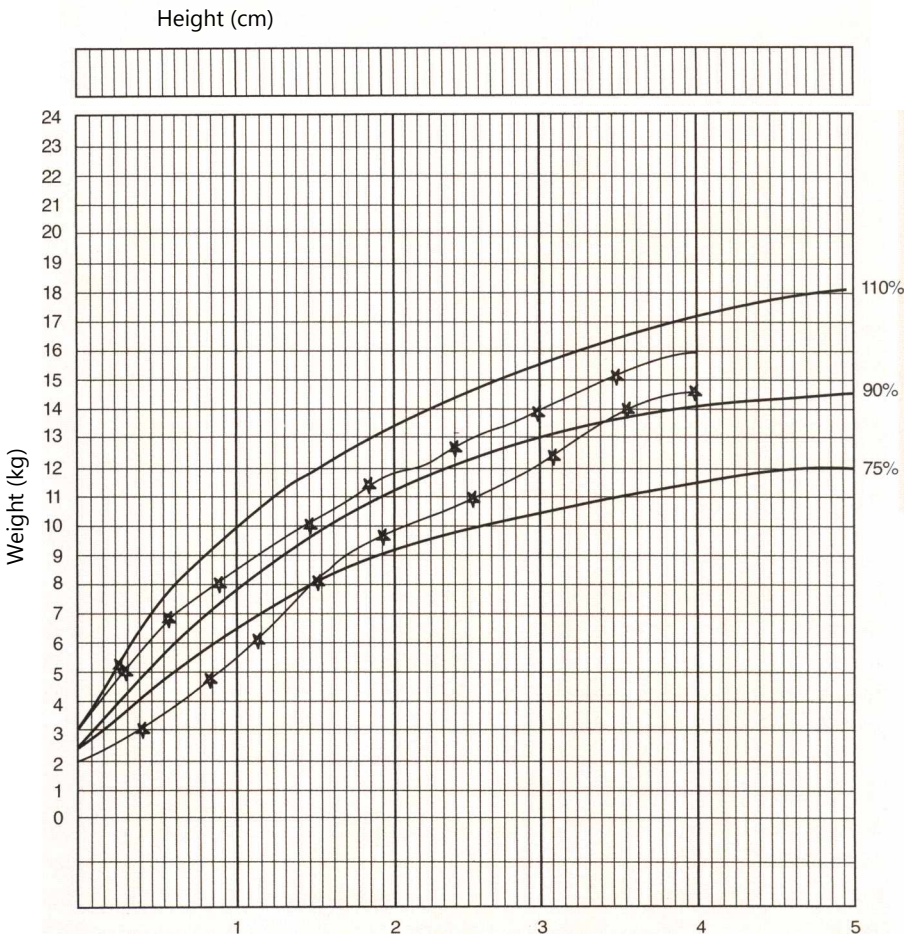


Fig. 2.5 Growth chart/road to health

2. Percentile charts

Anthropometric and other measurements from a large number of normal children when arranged in ascending order will give a bell shaped curve. The curve will be symmetrical and most of the observations will be falling around the centre of the curve. This is called ***'Gaussian distribution***' (Fig. 2.6). *The mean, median and the mode tend to be the same. Mean* is the arithmetic average, *median* is the middle value and *mode* is the most frequently occurring measurement. Most of the observations are towards the middle of the curve and there will be only a few observations at the tail ends.

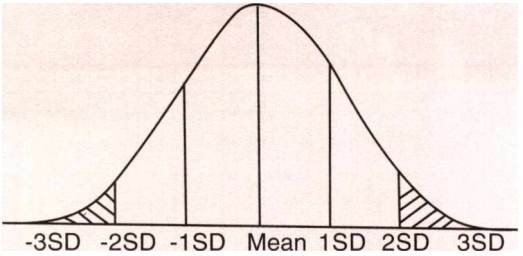


Fig 2 6 Gaussian distribution or bell-shaped curve with mean and standard deviation (SD)

Standard deviation (SD) is the degree of dispersion of the observations away from the mean. *Two-third of the observations fall within one SD above or below the mean. 95 per cent of the observations fall within 2 SD and 99.7 per cent fall within 3 SD.* Values beyond 2 SD are rare and beyond 3 SD are grossly abnormal.

Percentiles are used to represent the position of a particular measurement in the bell-shaped curve. For example, an observation at the 25th percentile denotes that 75 per cent of the measurements tend to be above this and 24 per cent tend to be below this. ***One SD includes observations within 16th and 84th centile and two SD includes those within 3rd and 95th centile.***

3. Growth Velocity

Weight gain or height gain over a unit period of time is velocity and it is a better indicator of growth. It reflects the effectiveness of any intervention, namely, nutritional supplementation, stimulation, growth hormone therapy etc. Weight velocity is 6 kg in the first year. In pre-school child, it is 2 kg/year and in a school child, it is 3 kg/year till puberty. Height velocity is 25 cm in the first year, it is 12.5 cm in the 2nd year and thereafter it is 6 cm/year till puberty.

4. Predicted Height in a Child Based on Mid-Parental Height (MPH)

In short stature, the height of the parents is measured and the midparental height (MPH) is considered. The predicted height is MPH plus or minus 2.5 cm. MPH is computed as follows:

$$\text{MPH (cm) for boys} = \frac{\text{(Paternal height + Maternal height)}}{2} - 1\text{-}6.5 \text{ cm}$$

$$\text{MPH (cm) for girls} = \frac{(\text{Paternal height} + \text{Maternal height})}{2} - 6.5 \text{ cm}$$

5. Mid Parental Height (MPH) Centile

The mid parental height can be compared with the NCHS growth percentile chart at 18 years of age for the respective sex of the child and the mid parental height centile can be noted as 50^, 25^ or 5^ centile etc. The centile at which the child is growing at present is then compared to the *mid paren tal height centile.* If both are in the same centile, *genetic short stature* is confirmed. If the child's height centile is less than the mid parental height centile, the child needs investigation for short stature like endocrine and metabolic work up. If the child's height centile is more than the mid-parental height centile, the ultimate height is expected to be more than that of the parents.

6. WHO Charts Based on Multicentre Growth Reference Study (MGRS)

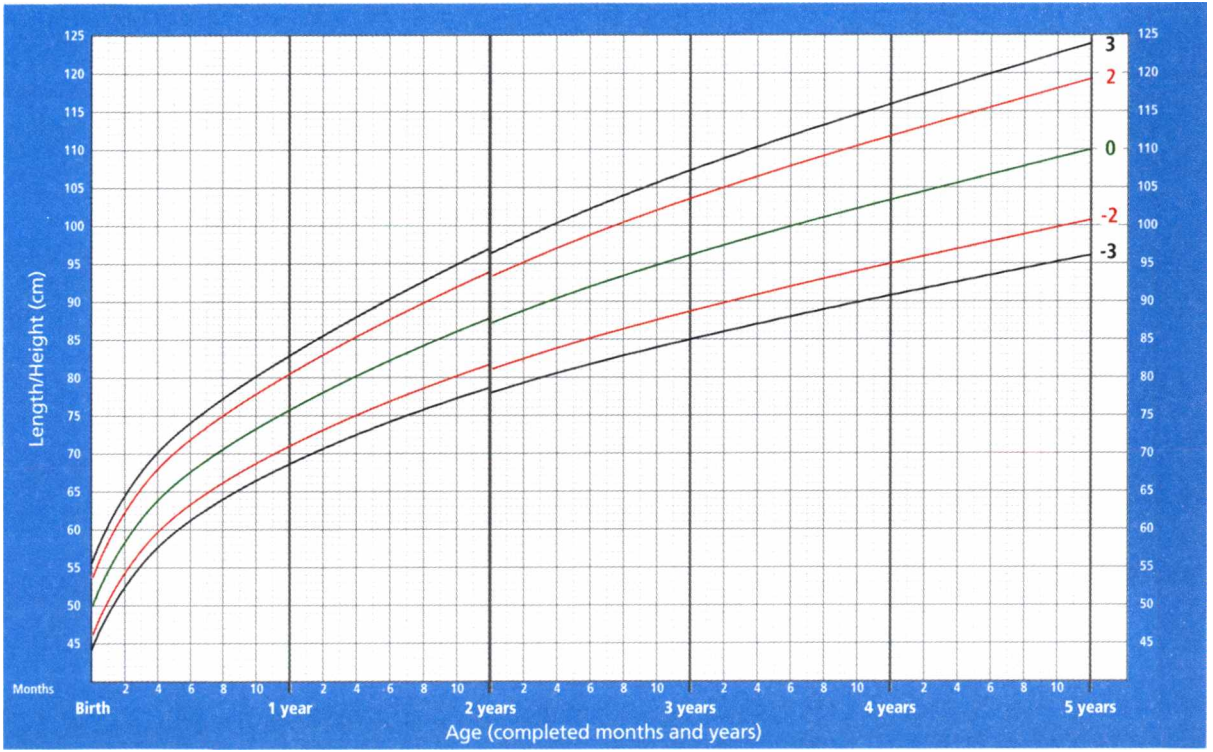
The participating countries include Brazil, Ghana, India, Norway, Oman, and United States. The data was collected by trained staff using a common protocol. The study was designed to combine a longitudinal follow up of children from birth to 24 months and a cross sectional study of children aged 18 to 71 months. Children were selected from communities where there were no environmental constraints to growth. They were healthy term infants who had no known illness or conditions that might affect their growth, and were breast fed as per the international feeding guidelines. The new growth reference is based on breastfeeding as the biological norm. The measurements include weight/age, height/age and weight/height. Data on BMI was generated for children under 5 yr for the first time. Other measurements include head circumference, mid-arm circumference, triceps and sub-scapular skin folds. Current references do not provide values for these parameters. In addition, key motor milestones like sitting, standing and walking were measured, linking motor development to physical growth.

In WHO growth charts (Z scores) < 2 SD indicates low weight and < 3 SD indicates very low weight.

Length/height-for-age BOYS

World Health Organization

Birth to 5 years (z-scores)

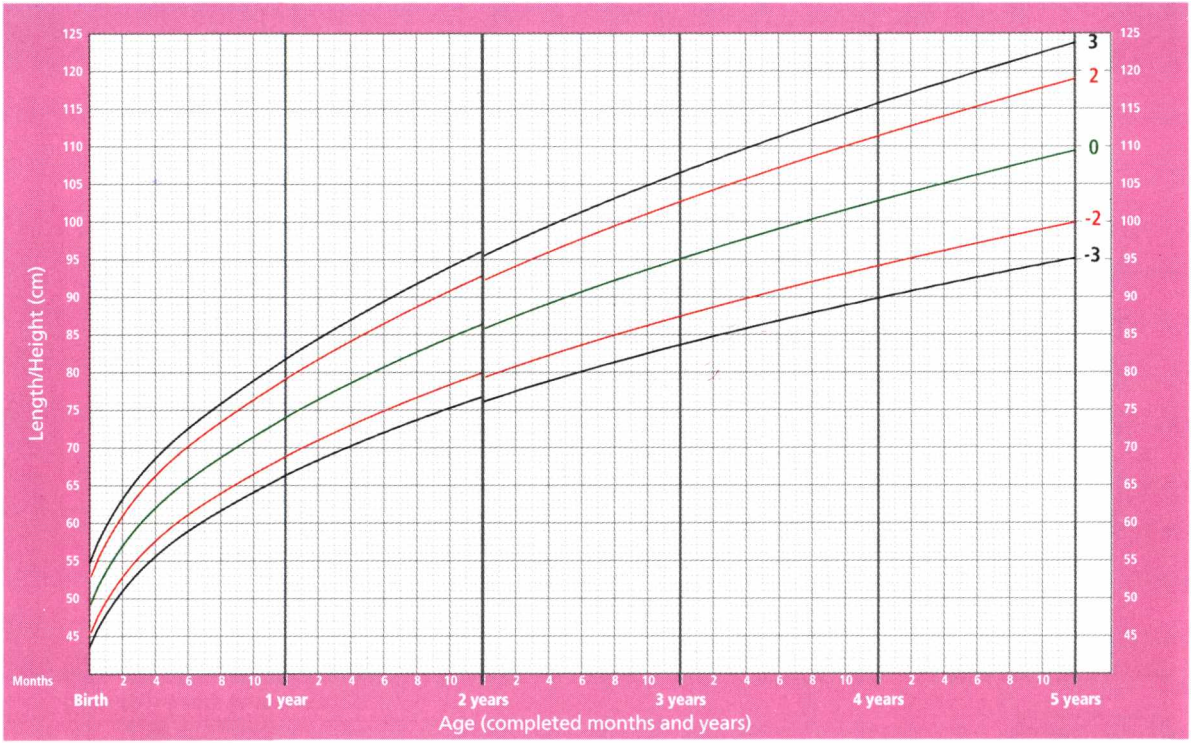


WHO Child Growth Standards

Length/height-for-age GIRLS

World Health Organization

Birth to 5 years (z-scores)

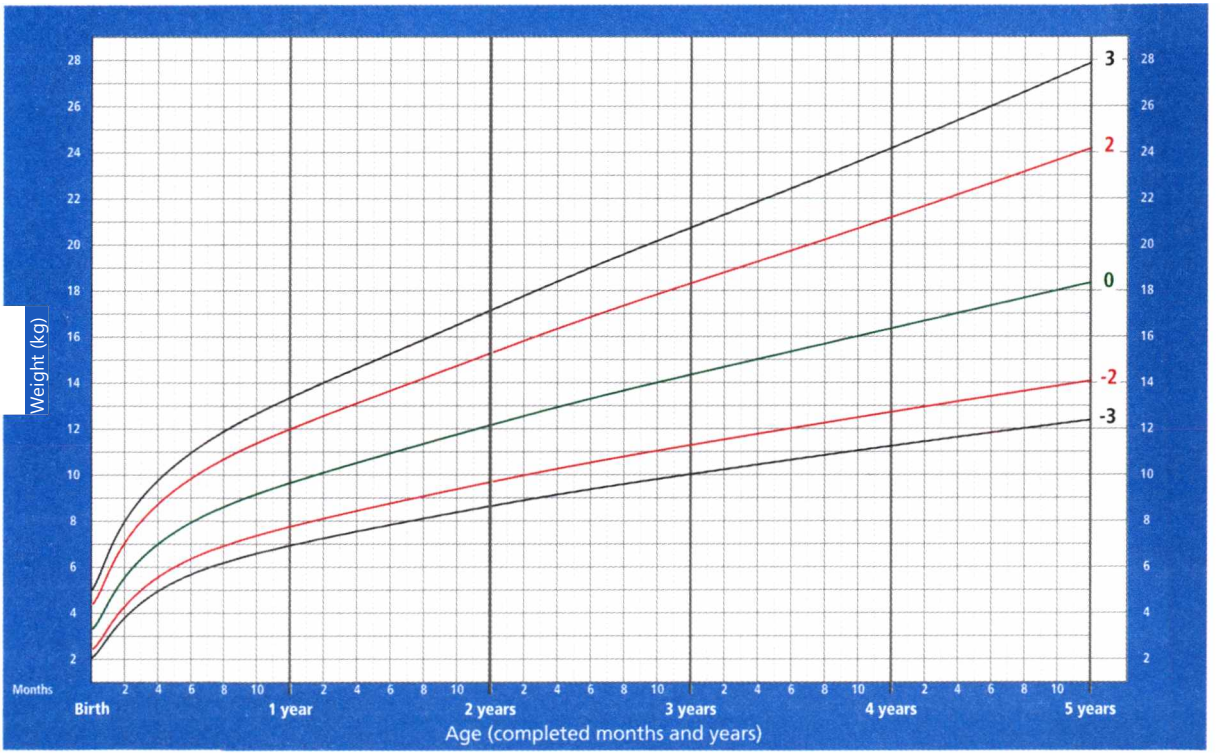


WHO Child Growth Standards

Weight-for-age BOYS

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Birth to 5 years (z-scores)

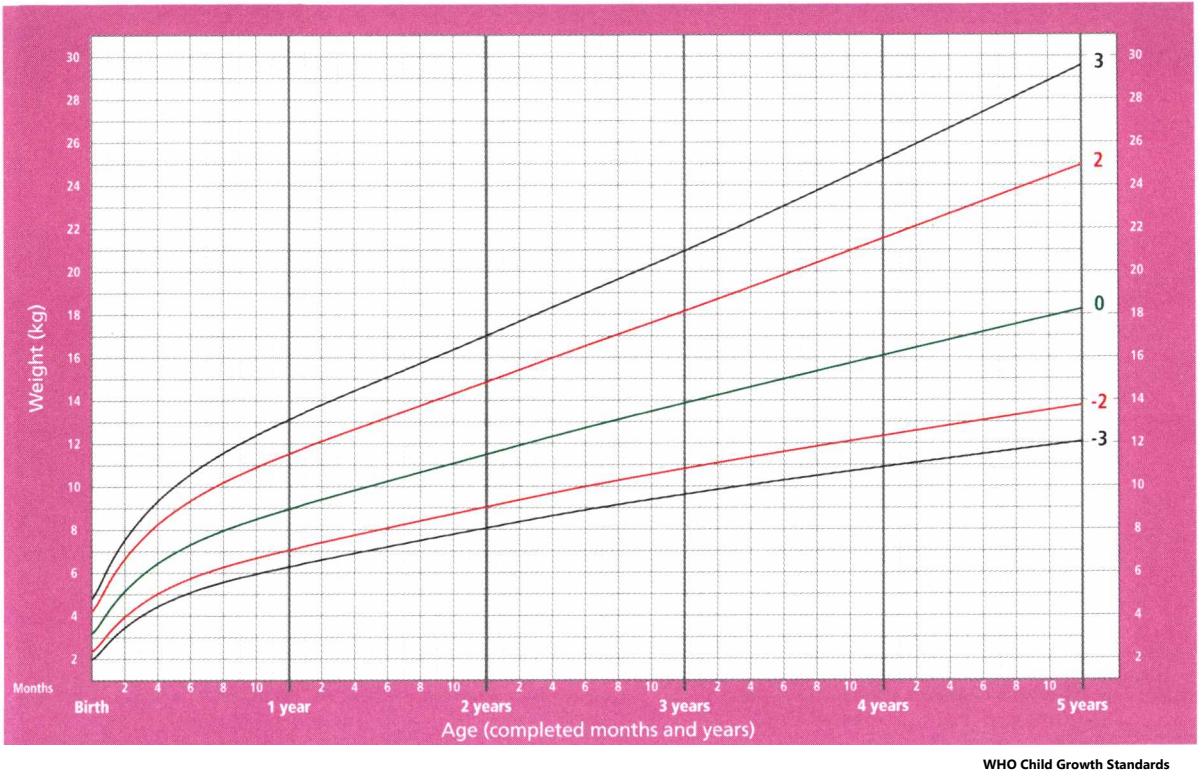


WHO Child Growth Standards

Weight-for-age GIRLS

Birth to 5 years (z-scores)

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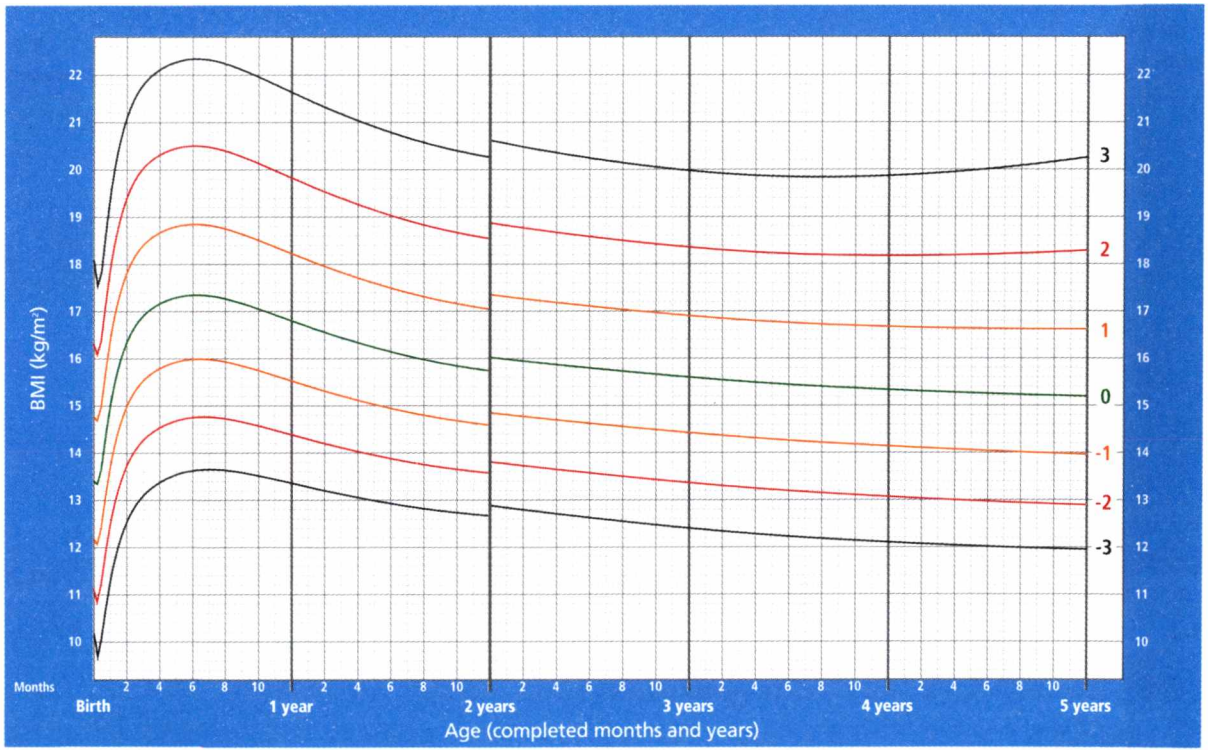


WHO Child Growth Standards

BMI-for-age BOYS

Birth to 5 years (z-scores)

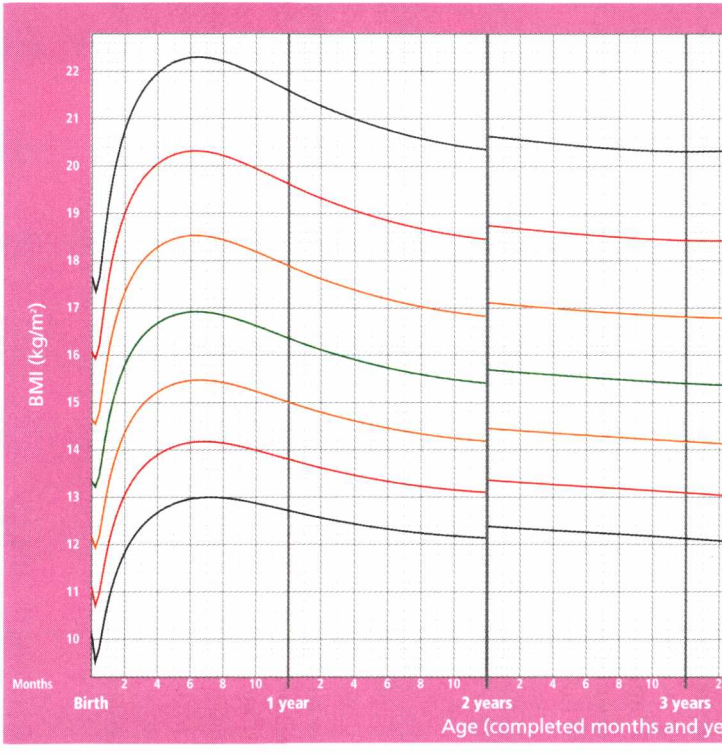
World Health Organization



WHO Child Growth Standards

BMI-for-age GIRLS

Birth to 5 years (z-scores)



WHO Child Growth Standards

SECTION 3

Applied Nutrition

"Let food be thy medicine and medicine thy food."

—Hippocrates (431 BC)

Application is the best output of any research. And Applied Nutrition is putting to use general principles of the science of human nourishment to address or solve specific problems. This also enhances the research base and forms the foundation of research. Identification of changes through right application of the existing nutrition principles helps to correct and contribute to the wellness of the human society.

3.1 Proximate Principles

Food items are divided into energy-yielding foods (carbohydrates and fats), body-building foods (protein) and protective foods (vitamins and minerals). The major nutrients, namely, carbohydrate, protein and fats are called **proximate principles or major nutrients.** Fibre is the undigestible complex carbohydrate that contributes to bulk and helps in gastrointestinal (GI) function. Water is also essential for life. According to Atwater calorie conversion factors, carbohydrate and protein yield 4 Cal per gram and fat yields 9 Cal per gram. However, short chain fats supply lesser calories than 9 Cal/g. Carbohydrate, fats, protein are the major nutrients; sodium, potassium, calcium, phosphorus etc., are the macronutrients and vitamins and minerals are the **micronutrients.**

1. Carbohydrates

Starches, sugars, milk, cereals, roots, tubers etc., yield carbohydrates. Starch is a complex carbohydrate made of several glucose units. Lactose, the milk sugar, is a disaccharide made of glucose and galactose. **Maltodextrin** is a glucose polymer

obtained by partial enzymatic hydrolysis of corn starch. By virtue of its molecular weight being 5 times more than glucose, maltodextrin reduces osmolarity by 5 times. Oils, fats and eggs do not contain carbohydrate. 50-55% of the total energy is derived from carbohydrates.

2. Fibre

Fibre is unabsorbable complex carbohydrate—cellulose, hemicellulose, gums, pectins, lignins, mucilages etc. They increase bulk of the food and prevent constipation and colon cancers. They swell and hold a lot of water and increase GI transit time. They bind bile salts and decrease absorption of cholesterol. Pectin and gums reduce blood sugar levels. Fenugreek seeds (methi/uluva) contain up to 40% gum. Adequate fibre can lower serum cholesterol in hypercholesterolaemic patients and blood glucose in diabetics. Very high fibre content may bind trace elements. Indian diet supplies up to 40 g fibre per day. **The suggested intake in adult is 20-40 g/day or 200-300 mg/kg/day in children.**

3. Protein

The word 'protein' means 'of prime importance'. They are important components of the tissues in the body and help in body building and tissue repair. They form enzymes, hormones and antibodies. When diet is deficient in energy, protein will be broken down and will be wasted as energy and will not be utilized for body building. The **specific dynamic action (SDA)** or the **thermic effect of food (TEF)** for protein (29%) is much higher than that of carbohydrate and fat (5%). This is the energy used during digestion. Amino acids are the building blocks of protein. There are 24 amino acids, 8 of them are **essential amino acids** which cannot be synthesized by the human body. They are **valine, leucine, isoleucine, lysine, tryptophan, methionine, threonine and phenylalanine.** Histidine is essential in infants. In low birth weight babies, arginine, cysteine and taurine are also essential amino acids. **Valine, leucine and isoleucine are the branched chain amino acids.** All others are non-essential amino acids.

Among the amino acids, glutamine is present in largest amount in plasma. It is termed **'antistress nutrient'**. It is a neuroregulator and a precursor for GAB A. Tryptophan gets converted to serotonin and niacin. As it is the precursor of serotonin, it is called **'nature's sleeping pill'** and as it is the precursor of niacin (B_3), it is also called **'provitamin B_3'.** 60 mg tryptophan is equivalent to 1 mg niacin. Glycine is the simplest amino acid.

The best quality protein provides amino acid pattern close to that of tissue protein. Breast milk and egg protein satisfy these criteria. Due to easy availability and storage convenience, egg is considered as the **'reference protein'.** It is a complete protein that contains all the essential amino acids.

The quality of a dietary protein is based on the extent to which it deviates from the reference protein. The chemical score of a food item is the percentage of

the limiting amino acids in the food item compared to the level of the same amino acid in the reference proteins, e.g., **lysine is the limiting amino acid in cereals whereas methionine is the limiting amino acid in legumes (pulses).** They are incomplete proteins. In cereal-pulse combination, the chemical score improves. In growing children, at least 10-15% of the calories should come from protein. Table 3.1 gives the protein quality of common food items based on biological value, net protein utilization and protein efficiency ratio.

Table 3.1 Protein quality of food items

Item	BV	NPU	PER
Egg	96	96	3.8
Cow's milk	90	85	2.8
Meat	74	76	3.2
Fish	80	74	3.5
Rice	80	77	1.7
Wheat	66	61	1.3
Bengal gram	74	61	1.1

a) **Digestibility coefficient (DC)** refers to the amount of absorbed nitrogen compared to the total nitrogen present in the food item. Cooking improves digestibility. Trypsin inhibitor present in soyabean and egg white, that decreases digestibility, gets destroyed on cooking.

$$DC = \frac{\text{Absorbed nitrogen}}{\text{Food nitrogen}} \times 100$$

b) **Biological value (BV)** refers to the amount of retained nitrogen compared to absorbed nitrogen.

$$BV = \frac{\text{Retained nitrogen}}{\text{Absorbed nitrogen}} \times 100$$

c) **Net protein utilization (NPU)** refers to the amount of retained nitrogen compared to the nitrogen present in the food item.

$$NPU = \frac{\text{Retained nitrogen}}{\text{Food nitrogen}} \times 100$$

d) **Protein efficiency ratio (PER)** refers to the gain in weight of experimental animal per unit weight of protein consumed.

4. Fats (lipids)

Fats are concentrates of energy and they provide essential fatty acids (EFA). EFAs are polyunsaturated fatty acids (PUFA). EFA used to be termed the **"Queen of Vitamins'** and then it was designated as **vitamin F.** EFA are linoleic acid and linolenic acid. Both linoleic acid (C18:2/n-6/PUFA) and alpha linolenic acid (Cl 8:3/ n-3/PUFA) must be present in the ratio 5:1 to 10:1. Among PUFA, omega-6 to omega-3 (n6-n3) ratio of 5:1 is desirable. High omega-6 and low omega-3 content as in groundnut, sunflower and safflower oils (Table 3.2) can predispose to free radical disease, angiotoxicity, impaired immune function, reduced glucose tolerance, increased platelet aggregation and albuminuria. In coconut oil, PUFA is only 2%, but the ratio is less than 5:1. In ghee it is 3:1 and in mustard oil it is 1.5:1. Those who consume n-6:n-3 PUFA in a ratio more than 10:1 should consume n-3 rich food items like greens, legumes and fish. Fats also contain fat- soluble vitamins.

Triglycerides are neutral fats with 3 fatty acids and glycerol. The EFA are converted into **long chain polyunsaturated (LCP)** fats by a series of chemical reactions called elongation and desaturation. LCPs of n-6 series derived from linoleic acid (LA) are arachidonic acid (ARA) and adrenic acid (ADA). LCPs of n-3 series derived from alpha linolenic acid (ALA) are eicosa pentaenoic acid (EPA) and docosa hexaenoic acid (DHA). LCPs are the building blocks of brain lipids. **Deficiency of LCPs may lead to co-morbid conditions like dyslexia, dyspraxia and hyperactivity.** DHA is important in brain function and EPA is cardioprotective.

There are two types of dietary fats, visible and invisible fat. Invisible fat is present in the food items and visible fat includes oil. ghee, butter etc. Average Indian diets supply 25-30% of calories as fats. Maximum permitted is up to 45%. Hence in the diet of children and in nutritional therapy of malnourished children 10-15% of the desired calories can be supplemented as visible fats. 1-3% of the calories should come from EFA. Polyunsaturated/saturated (P/S) fat ratio of 1.2:1 present in breast milk is accepted. Phrynoderma (toad skin) responds to EFA along with fat soluble vitamins like vitamin A. Animal fats generally provide unsaturated fat except cod liver oil and sardine that contain EFA. Vegetable oils are rich in EFA. Coconut oil contains less EFA, but more **medium chain triglyceride (MCT) that can be absorbed directly into the portal vein even during fat malabsorption.** MCT are C8-C10 fatty acids. Monounsaturated fatty acid (MUFA) raises **HDLor good cholesterol.** Table 3.2 gives fatty acid composition in various oils. EFA is essential for normal foetal and infant growth, for brain growth and visual acuity.

Table 3.2 Fatty acid composition in oils

Item	SFA (%)	MUFA (%)	PUFA (%) n6	n3
Coconut oil	92	6	1.6	0.4
Corn oil	13	25	61	1
Groundnut oil	18	49	33	0
Olive oil	14	77	8	1
Sunflower oil	11	20	69	0
Safflower oil	10	13	77	0
Palmolein	40	48	11	1
HVO	76	19	3	2

SFA - Saturated fatty acid, MUFA - Monounsaturated fatty acid, PUFA - Polyunsaturated fatty acid, HVO - Hydrogenated vegetable oil (Dalda)

Omega-3 or n-3 fatty acids, eicosa pentaenoic acid (EPA-C20:5/n-3) and docosa hexaenoic acid (DHA-C22:6/n-3) are derived from EFA and are cardioprotective. In Eskimos, even though 60% of the calories come from fats, they have very little mortality from cardiovascular diseases due to high intake of EPA and DHA. Omega-3 fatty acids, especially EPA, compete with arachidonic acid in the cyclo-oxygenase pathway. Normally, arachidonic acid is converted to prostacyclin and thromboxane A,. Prostacyclin is a potent vasodilator and platelet deaggregator, whereas thromboxane A_2 is a potent vasoconstrictor and platelet aggregator. When large doses of EPA was given to volunteers as in the Eskimos' diet, a decrease in thromboxane A_2 was noted. **High arachidonic acid/EPA ratio is postulated to be an important risk factor for coronary artery disease.** Fish oil rich in EPA is commercially available as Promega, Omega 500, Maxepa etc. Other n-3 rich foods are green leafy vegetables, legumes and fish.

Not more than 10% of energy should come from saturated fatty acid (SFA), 10% should come from MUFA and 10% from PUFA. SFA includes lauric acid (C12), myristic acid (C14), palmitic acid (C16) and stearic acid (C18). In coconut oil, 50% is lauric acid and in palm oil, 45% is palmitic acid. Myristic acid is most hypercholesterolaemic. **LDL cholesterol is a risk factor for atherosclerosis and is called 'bad cholesterol' and HDL cholesterol is called 'good cholesterol'.** SFA increases LDL cholesterol. SFA in butter fat is 65%, in mutton fat is 54%, in pork fat is 39% and in chicken fat is 34%.

Cholesterol is synthesized from 2 carbon fragments like acetates, acetic acid, acetyl-CoA etc. This is called **endogenous cholesterol** and that derived from

food is called exogenous cholesterol. It is an essential component of body tissues and lipoproteins. It forms nearly 0.2% of adult body weight. It has no calorific value. Up to 2.0 g cholesterol may be synthesized by an adult and the biosynthesis is inversely proportional to dietary cholesterol. Cholesterol biosynthesis reduces during fasting and physical exercises like yoga. Excess cholesterol is converted into bile acids and excreted into intestine and eliminated as coprosterol. Dietary fibre reduces cholesterol level. Cholesterol is present as free or bound forms in the ratio 1:3. It is bound to esters of fatty acids or lipoproteins. Cholesterol bound to beta-lipoprotein is low density lipoprotein (LDL) cholesterol (70%) and that bound to alpha-lipoprotein is high density lipoprotein (HDL) cholesterol (30%). Small amounts are bound to very low density lipoproteins (VLDL) and the rest is bound to intermediate low density lipoproteins (ILDL) and very high density lipoproteins (VHDL). Hypercholesterolaemia, especially LDL, predisposes to coronary artery disease. Heredity, smoking, sedentary habits, emotional stress, hormonal imbalance, diet, diabetes etc., predispose to hypercholesterolaemia.

Cholesterol Ratios

The total cholesterol to HDL cholesterol ratio is a number that is helpful in predicting an individual's risk of developing atherosclerosis. The number is obtained by dividing the total cholesterol value by the value of the HDL cholesterol. (High ratios indicate higher risks of heart attacks, low ratios indicate lower risk). High total cholesterol and low HDL cholesterol increase the ratio, and is undesirable. Conversely, high HDL cholesterol and low total cholesterol lower the ratio, and is desirable. An average ratio would be about 4.5. Ideally we want to be better than average if we can. Thus the best ratio would be 2 or 3, or less than 4. If a person has a total cholesterol of 200 mg/dl and an HDL cholesterol of 40 mg/dl, the total/HDL cholesterol ratio is 5:1. The goal is to keep this ratio below 5:1, with the ideal being below 3.5:1.

LDL/HDL Ratio

The LDL/HDL ratio is actually a purer ratio than total cholesterol/HDL, because LDL is a measure of "bad' cholesterol and HDL is a measure of "good" cholesterol, whereas the total cholesterol is the sum of HDL, LDL, and the VLDL. Adding up the values for the HDL, LDL and VLDL makes up the total cholesterol measurement. If a person has HDL cholesterol of 60 mg/dl and LDL cholesterol of 180 mg/dl, the LDL/HDL ratio turns out to be 3:1. If a person has an LDL/HDL ratio lesser than 3.5:1, he is said to have a healthy level. However, the ideal HDL/LDL ratio is 2.5:1. Therefore, the doctors advise the patients to maintain at least an LDL/H DL ratio less than 3.5:1.

HDL/LDL Ratio

When comparing "good cholesterol" (HDL) to "bad cholesterol" (LDL), there is a ratio that may be used. When using it, the goal is to keep the ratio of HDL/LDL above 0.3, with the ideal being above 0.4. If a person has HDL cholesterol of 60 mg/dl and LDL cholesterol of 180 mg/dl, the HDL ratio turns out to be 1:0.3. If a person has an HDL/LDL ratio more than 0.3, it is said to be a healthy level. However, the ideal HDL/LDL ratio is 0.4. Therefore, the advise is to maintain at least an HDL/LDL ratio above 0.3.

Even though the total cholesterol/HDL ratio is not as accurate or pure as the LDL/HDL ratio, the former is more commonly obtained because the total cholesterol is easier and cheaper to obtain than the LDL cholesterol level.

5. Energy

The traditional unit of energy is 1 **kilocalorie** (kcal/Cal). It is **the amount of heat necessary to raise the temperature of 1 kg of water by 1°C from 14.5°C to** 15.5°C. According to the International System, the unit of energy is joule. 1 calorie = 4.184 joule. The requirement of an adult sedentary male, i.e., 2400 kcal, is termed as one unit. This is roughly the requirement of an adolescent boy. The requirement of an adolescent girl is slightly less, i.e., 2100 kcal.

3.2 Vitamins, Minerals and Micronutrients

Vitamins and minerals **form the protective foods and are also called functional foods.** Those with RDA < 100 mg/day were traditionally called **micronutrients.** Currently those with requirement in micrograms or milligrams are included as micronutrients. Nutrients called major nutrients are carbohydrate, protein and fats; others are called minor nutrients. It is essential to remember that these micronutrients are certainly of major significance in child health.

VITAMINS

Fat-soluble Vitamins

1. *Vitamin A*

Vitamin A (retinol) is a fat-soluble vitamin. Carotene is converted to vitamin A. It has an important role in vision, immunity and integrity of skin and mucous membrane. Vitamin A is bound to retinol binding protein (RBP) and pre-albumin. **Vitamin** A **deficiency is the commonest vitamin deficiency found in India.** Deficiency leads to **xerophthalmia, night blindness** (nyctalopia), **total blindness, hydrocephalus and increased bacterial binding to the mucous membrane.** The eye manifestations are the following:

a) **XI**A Xerosis conjunctiva

b) XI B Bitot's spots
c) **X2** Xerosis cornea
d) **X3** A Corneal ulceration
e) **X3 B** Keratomalacia
f) **XN** Night blindness
g) **XS** Corneal scarring
h) **XF** Fundoscopic changes

As vitamin A deficiency is a preventable cause of blindness, pulse doses of vitamin A are advised at 6 months interval in children. It is needed in areas where the prevalence is > 0.5%. **Serum retinol is the main indicator apart from clinical eye signs of deficiency. Dark adaptometry is a simple tool to look for deficiency.** In early deficiency, the ability of the pupil to constrict under illumination is impaired. By flashing a light at one pupil and covering the other, the degree of impairment of the pupillary reflex can be estimated

According to the **Vitamin** A **Prophylaxis Programme,** children between 9 months and 5 years are given 9 megadoses of vitamin A concentrate at 6 months interval. The first 2 doses are integrated with measles vaccination and DPT Ist booster. For infants, the dose is 1 ml. equivalent to 1 lakh IU and in children it is 2 mL. In **Costa Rica, vitamin** A **fortified sugar is used.** In case of deficiency, extra doses are advised in addition to the prophylaxis (Table 3.3). In a community setting, it can be given one month apart as per the recommendations of the Child Survival and Safe Motherhood (CSSM) programme. In case of keratomalacia, atropinization and ophthalmology consultation are needed. Vitamin A is given 5000 IU/kg/day for five days followed by 25,000 IU/day till recovery (Nelson, *Textbook of Pediatrics,* 15th edition). When water-soluble injectable or oral preparations are given (Aquasol A, Arovit), vitamin A concentrate should also be given to replenish the stores.

Table 3.3 Treatment of vitamin A deficiency

Oil miscible Vitamin A	**< 1 year (IU)**	**> 1 year (IU)**
Immediately	1 lakh	2 lakhs
Next day	1 lakh	2 lakhs
After 2-4 weeks	1 lakh	2 lakhs
Severe PEM (repeat monthly till PEM resolves)	1 lakh	2 lakhs

Source: WHO/UNICEF/IVACG task force, 1988

Staining of the eye with 1% Rose Bengal and low serum retinol are diagnostic of deficiency. Hypervitaminosis A can lead to GI upset, pseudotumour cerebri, skin desquamation, dry hair and hyperostosis of tibia. Children are attracted to vitamin A and D capsules and parents tend to encourage them to take these capsules in large numbers. This can result in hypervitaminosis. **Intake of 25,000 IU or more during early pregnancy may cause congenital malformation in the baby.** Dietary sources do not cause hypervitaminosis. **Hypercarotenaemia** colours plasma and skin.

The natural sources are green, yellow and orange (GYO) vegetables and fruits, milk and milk products, egg yolk, red palm oil, fish, fish liver oil and lemon grass oil. Beta-carotene, the provitamin A, is important for its antioxidant properties. It is abundant in the coloured vegetables and fruits. **The daily requirement is 1500 IU (500 mg). 1** (xg = 3 IU. One molecule of beta-carotene yields 2 molecules of vitamin A. As this conversion is not very effective, in practice, **6 pg of carotene is considered equivalent to 1 pg of retinol.**

Preparations: Aquasol A—50,000 IU/capsule, Inj. 50,000 IU/ml
Arovit—50,000 IU/tab, Drops 50,000 IU/ml
Adexolin/A & D—Vit. A 5000 IU, Vit. D 400 IU

2. *Vitamin D*

Vitamin D is a fat-soluble vitamin essential for bone growth and calcium absorption. **It is synthesized in the body from 7-dehydrocholesterol with the help of around 288 nm** UV **light.** It is hydroxylated to the active form 1,25-dihydroxy cholecalciferol in kidney and liver respectively. In kidney and liver disorders deficiency can occur. Cholecalciferol (D_3) is the animal source and ergosterol (D_2) is the plant source. The active form is given the status of a hormone. The active form (calcitriol) promotes bone resorption and mineralisation and intestinal calcium and phosphorus absorption. When serum calcium is high, 24,25-dihydroxy cholecalciferol is produced. It is thought to be the inactive form. Preterm babies with rapid catch-up growth, those not exposed to sunlight, babies of mothers with severe Vitamin D deficiency and those with fat malabsorption are prone to develop deficiency. In preterms, deficiency may manifest as early as 8 weeks of age. Vitamin D deficiency leads to rickets in growing children and osteomalacia in adults. It may also lead to hypocalcaemic tetany. In rickets, serum phosphorus level is below 4 mg%, serum alkaline phosphatase (SAP) is raised to thousands, serum calcium is low or normal and there is aminoaciduria. **In rickets,** failure of mineralisation leads to bony deformities, large head, wide open anterior fontanel (AF), frontal, parietal and occipital bossing **(hot cross bun appearance** or **caput quadratum), craniotabes, pigeon chest, rachitic rosaries, pectus carinatum, Harrison's sulcus, knock knee, lateral bowing of tibia, widening of wrist, double malleoli etc.** X-ray of long bones shows cupping, lipping and flaring of metaphysis and widening of growth plate (physis). The epiphysis is

seen far below due to the uncalcified osteoid. On healing, a line of calcification will appear at a point it should normally occur at that age and later the gap between the line and the flared metaphysis will get mineralised, i.e., bone will grow towards the shaft.

Deficiency is treated with 6 lakhs IU of vitamin D oral or IM. Lower doses are also found to be effective. If radiological healing does not occur in 2-3 weeks, another dose can be repeated. If healing does not occur in 4-6 weeks with 2 doses, it is branded as refractory rickets. Oral vitamin D, may also be given 2000-6000 IU/day or 0.5-2 meg 1,25-dihydroxy D, may be given daily till recovery. An account of various types of rickets is given in Table 3.4. Infantile tetany due to reduced vitamin D and calcium absorption can occur. In latent tetany, serum calcium is less than 7-7.5 mg/dl and tetany becomes manifest by **Chvostek, Trousseau** and **Erb signs.** The traditional practice of exposing the babies to sunlight after oiling is good for vitamin D synthesis. Rickets/osteopenia is also seen in preterm and LBW babies. With the popularisation of flat system of housing, wearing of purda, and industrialization and air pollution sunlight exposure is becoming less. As a result, breastfed babies and young children are developing nutritional rickets. The prevalence of osteomalocia is also more among adults.

Blount's Disease: It is a condition with bowing of legs. It is due to arrest of posteromedial aspect of upper tibial epiphysis leading to overgrowth of lateral epiphysis. Predisposing factors are obesity, early walking and black race.

Hypervitaminosis D leads to GI upset, hypotonia, polyuria, polydypsia, hypercalcaemia, hypercalciuria, metastatic calcification, clouding of cornea and conjunctiva, retinopathy and renal damage. Aluminium hydroxide is useful in treatment.

The daily requirement is 400 IU (10 mg). **I** mg = 40 IU. Egg yolk, fish and tropical plants are rich sources of Vitamin D. Summer milk produced by cows grazing on green plants is found rich in vitamin D and A.

Preparations: Calcirol sachets—(vitamin D_3) 60,000 IU/sachet oral, Inj. Arachitol—(Vitamin D_3) 3 lakhs and 6 lakhs IU/ml; Alpha D_3—1 Alpha hydroxylated D_3 tab 0.25 mg, 1 mg (Alpha D_3 - Alphacalcidol is used in kidney diseases). Alphadol tab 0.25 mg; One Alpha (calcitriol) tab 0.25 mg; 1 mg, Rocaltrol (calcitriol) tab 0.25 mg; Adexolin—A& DCap: A5000IU, D400 IU.

3. *Vitamin E*

Vitamin E (tocopherol) is a fat soluble vitamin. Due to its **antioxidant** property, it is used for various therapeutic indications like **prevention and treatment of retinopathy of prematurity (ROP), bronchopulmonary dysplasia (BPD), haemolytic anaemia of prematurity, myopathies, neuromuscular diseases, thrombosis, fibroadenosis etc.** Alpha, beta, gamma and delta tocopherol and tocotrienols are the compounds with vitamin E activity. **The alpha compounds are most bio-potent.**

Table 3.4 Causes and salient features of various types of rickets

Type	S. Ca	S. P	S. AP	Amino-aciduria
Hypophosphataemic rickets				
1. Deficiency	N/D	D	I	+
2. Malabsorption	N/D	D	I	+
3. Liver disease	N/D	D	I	+
4. Anticonvulsants	N/D	D	I	+
5. Phosphate deficiency	N	D	I	-
6. Vit. D resistant X-linked primary hypophosphataemia	N	D	I	-
7. Fanconi syndrome	N	D	I	+
8. Renal tubular acidosis	N	D	I	-
9. Oncogenic hypophosphataemia	N	D	I	-
Hypocalcaemic rickets				
1. Vit. D dependent (Type 1)	D	N/D	I	+
2. Vit D dependent (Type 2)	D	N/D	I	+
Hyperphosphataemic rickets				
Renal osteodystrophy	N/D	I	I	+
Related conditions				
1. Hypophosphatasia	N	N	D	+ PEA
2. Metaphyseal dysostosis				
Jansen's	I	N	I	-
Schmidt's	N	N	N	-

N - Normal, **D** - Decreased, I - Increased, **PEA -** Phospho 1 Amine

The requirement is 5-15 IU (5-15 mg). It is present in cereals and vegetable oils. As more applications of Vit E are yet to be known, it is called the **'shady lady of nutrition".** Vit E supplementation is needed in fat malabsorption and cholestasis and in premature infants. 15-25 IU/day may be given in such cases. Large doses may prevent neurological abnormalities in those with cholestasis and abetalipoproteinaemia. **Vitamin E in excess may lead to necrotising enterocolitis in the newborn.** It is not easily transported across the placenta.

Preparations: Evion, Tocofer, E Cod—100 mg, 200 mg, 400 mg, 600 mg pearls; Evion Drops 50 mg/ml.

4. *Vitamin K*

Vitamin K is a fat-soluble vitamin synthesized by the intestinal flora. It participates in **oxidative phosphorylation.** It increases concentrations of prothrombin (factor II), proconvertin (factor VII), plasma thromboplastin component (factor IX) and Stuwart-Prower factor (factor X). Other **vitamin K dependent factors are factors S and C that are anticoagulants, factors Z and M that stimulate platelet activity and calcium binding protein osteocalcin that promotes coagulation and calcium metabolism.** Naturally occurring vitamin K is K, (phytomenadione) and IC (menaquinones) is of bacterial origin. Synthetic vitamin K (Synkavit) in large doses may produce hyperbilirubinaemia. The requirement is not characterised. Deficiency leads to prolongation of prothrombin time and bleeding tendency. **Haemorrhagic disease of the newborn** (HDN) is due to vitamin K deficiency. The incidence of HDN is more in breast-fed babies due to delayed colonisation of gut and due to **lower concentrations of vitamin K in breast milk** (15 mcg/dl) compared to cow's milk (60 mcg/dl). Hence vitamin K 0.5-1 mg is given to all LBW and high-risk babies. **In HDN, vitamin K 2-5 mg is given.** In severe cases, blood transfusion is also needed. In HDN, estimation of 'protein induced in vitamin K absence' **(PIVKA)** is found useful. Unlike prothrombin time, it tends to remain positive for longer periods and even after blood transfusion. Vitamin K is also given in liver diseases, persistent diarrhoea, malabsorption etc., usually once in 2-4 weeks. In liver diseases, vitamin K is given daily for 3 days to correct prothrombin time.

Dicoumarol and salicylates reduce utilisation of vitamin K by the liver and thereby lead to hypoprothrombinaemia. Vitamin K can be given orally or parenterally.

Preparation: Menadione sodium (water soluble) 10 mg/ampoule.

Water-soluble Vitamins

1. *Vitamin B complex*

B complex factors are water soluble and are lost during washing, cooking, milling etc. A summary of the relevant details on B complex factors and their sources are given in Table 3.5.

a) **Thiamine (B,):** Thiamine plays an important role in the metabolism of carbohydrates, alcohol and branched chain amino acids. The main deficiency diseases are **beriberi and Wernicke-Korsakoff syndrome** (WKS).

Beriberi is a rare disease now; wet beriberi manifests as high-output cardiac failure, dry beriberi presents with neuritis and infantile beriberi presents with aphonia and combined features of dry and wet beriberi. It promptly responds to thiamine, 10-100 times the requirement.

WKS is usually seen in alcoholics, those who fast such as in hunger strike and those who have persistent vomiting such as hypereniesis gravidarum. In 1880, Wernicke described an encephalopathy with ophthal-

Table 3.5 Vitamin B complex factors

Vitamin	Requirement	Sources	Deficiency
Bj (thiamine)	0.5-1.5 mg	Yeast, outer layer of cereals, pulses & nuts	Beriberi, CCF, neuritis, encephalopathy
B_2 (riboflavin)	0.5-1.5 mg	Milk, milk products, egg, liver, green leafy vegetables, wheat	Cheilosis, stomatitis, glossitis, neovascularisation of cornea, nasolabial dyssebacea
B_6 (pyridoxine)	0.5-1.5 mg	Meat, liver, vegetables, cereals	Neuritis, convulsion, anaemia
B_3 (niacin)	5-15 mg	Cereals, nuts, meat	Pellagra, diarrhoea, dermatitis, dementia
Folic acid	50-150 ng	Vegetables, pulses, liver	Megaloblastic anaemia, knuckle pigmentation
	0.5-1 ng	Animal food	Megaloblastic anaemia, CNS degeneration

moplegia, nystagmus, apathy, ataxia and stupor followed by a memory disorder. In 1887, Korsakoff described a psychosis and a memory disorder with inability to retain new memories. The encephalopathy responds to thiamine, but the psychosis responds slowly or not at all.

Anaphylaxis may occur with thiamine injection. The body contains about 30 mg of thiamine only.

b) **Riboflavin (B_2):** Riboflavin has a vital role in cellular oxidation. Body stores are neglibible and deficiency may manifest in about 3 months. The manifestations are angular stomatitis, cheilosis, atrophic papillae on tongue, nasolabial dyssebacea and **neovascularisation of cornea.** Angular stomatitis can also occur in deficiencies of niacin, pyridoxine or iron. It is rare in milk drinking countries. It is a common deficiency in South India, where the staple diet is polished rice. It was also called vitamin G.

c) **Niacin (B_3):** Niacin (nicotinic acid and nicotinamide) is part of NAD and NADP co-enzymes. Tryptophan is converted to niacin and 70 g of protein intake provides about 12 mg of niacin. **Pellagra** caused by niacin deficiency is now rare. Pellagra leads to photosensitive dermatitis, diarrhoea and dementia. **Casal's necklace and glove and stocking type of dermatitis can occur in the exposed parts.**

It used to be common in maize eaters, in which tryptophan is the limiting amino acid and the bound form of niacin in maize is not bioavailable. Excess leucine in maize is antagonistic to tryptophan and niacin. Secondary pellagra may occur in those on low protein diet, and **Hartnup disease, an inborn error of tryptophan absorption.** N iacin causes flushing. As it lowers plasma cholesterol and triglycerides, it is useful in combined hyperlipidaemias. High doses may cause gastric irritation, hyperuricaemia, impaired glucose tolerance and jaundice.

d) **Pyridoxine (B_6):** Pyridoxal, pyridoxamine and their phosphates and pyridoxine have B_h activity. Deficiency produces neuritis, anaemia and convulsions. Drugs like hydralazine, penicillamine, oestrogens, INH etc., interfere with B_6. It keeps up the GAB A level, which is an inhibitory neurotransmitter. It is also beneficial in homocystinuria, hyperoxaluria, sideroblastic anaemia and radiation sickness. Excess may causes sensory neuritis.

e) **Cyanoeobalamine ($B_{|2}$):** It is called the 'red vitamin' and is present only in animal foods, fish and milk. Even though colonic bacteria synthesize it, it is not bioavailable. It has a large molecular weight of 1350 and the dietary B is absorbed from terminal ileum with the help of the intrinsic factor secreted from the stomach. **Deficiency occurs in pernicious anaemia.** B_p and folic acid take part in DNA synthesis and deficiency of both leads to megaloblastic anaemia. B_p helps in synthesis of fatty acids in myelin and **deficiency leads to subacute combined degeneration of spinal cord and knuckle pigmentation.** In infants and young children, deficiency may lead to neuroregression.

f) **Folic acid (B_n):** Folic acid (folacin) or pteroyl monoglutamic acid is the primary vitamin. But. pteroyl polyglutamate is less absorbed. Most folates are in the reduced form (tetrahydrofolate) which is the active form. It is important in DNA synthesis and deficiency leads to **megaloblastic anaemia, diarrhoea,** and **knuckle** and **periungual pigmentation.**

Folic acid given periconceptionally can prevent neural tube defects in the baby. It should also be supplemented in those with haemolytic anaemias. It was also called vitamin M. It is also useful in chronic diarrhoea.

Riboflavin deficiency is the commonest deficiency in South India where intake of milk is low and polished rice is the staple diet. Pyridoxine deficiency becomes overt with drugs like INH, hydralazine, oral pill etc. Pyridoxine excess also can lead to sensory neuropathy. Pyridoxine increases GABA level and thereby control convulsion. Biotin, choline and niacin have relaxing effect on anxiety and **niacin is called 'Nature's Valium'.** Tryptophan is the provitamin of niacin. It is called **'Nature's sleeping pill'.**

g) **Others: Pantothenic acid is also referred to as vitamin B_.** It is beneficial in **'burning feet' syndrome.** This was described by Dr Gopalan. **Biotin is also called vitamin H.** Its deficiency produces hyperaesthesia, hallucination and

dermatitis. **Avidin** present in raw egg is the antagonist of biotin. Choline is part of phospholipids and acetylcholine. It can be synthesized from methionine.

Preparations: Neurobion, Polybion, Becosules, BerocinC, Visyneral Inj., Neurobion, Macraberin, Polybion, MVI. Thiamine—Beneuron 5 mg; riboflavin—Lpabol 20 mg. Riboflavin 10 mg; Pyridoxine—Benadon 40 mg, B-Long 100 mg, folic acid—Folik, Folinal, Folvite, Megafit 5 mg; pantothenic acid—Sigma Pantothenate Inj. 50 mg/ml, 100 mg/tab as Calcium Pantothenate, 16.5 mg/tab in Supradyn; biotin—0.25 mg/tab in Supradyn.

h) **Pyrroloquinoline quinone (PQQ):** It is the new vitamin to be discovered since 1948, when vitamin B_{12} was found. PQQ was discovered in 1979 from a bacterium, and afterwards it was reported to be present in common foods. PQQ, which must be supplied by the diet, acts as a cofactor in enzyme-catalyzed reduction-oxidation (redox) reactions, much as nicotinamides and flavins. As a mammalian redox cofactor necessary for the degradation of the amino acid lysine, PQQ qualifies as a member of the B vitamin family. Researchers noted decreased conception and fertility in PQQ-deficient mice, and the babies born to PQQ-deprived mothers grew at slower rates than offspring from mice given supplemental PQQ.

2. *Vitamin C*

Vitamin C converts proline to hydroxyproline, which is a constituent of collagen. Vitamin C (ascorbic acid) is a water soluble vitamin that is lost during washing and cooking. It is involved in collagen synthesis and bone and teeth formation. It also increases iron absorption. It is also important as an antioxidant due to its reducing property. **The deficiency leads** to **scurvy,** defective bone growth, bleeding gums and delayed wound healing. There will be subperiosteal bleeding and calcification. This may manifest with severe pain and **pseudoparalysis. Scorbutic rosaries** are tender and angulated due to subluxation. X-ray shows **ringed epiphysis** and a zone of destruction at the metaphysis. A dense white line called the **"white line of Fraenkel'** is seen at the metaphysis due to excessive calcification at the metaphyseal end. This is because the osteoid is not moving due to defective collagen synthesis and mineralisation occurs over and over at the same place. Calcification may occur on the sides beyond the metaphysis and broken chips of bone may be seen at the corner due to weight bearing. This is called **'corner sign'.** Excess can cause increase in urinary oxalate and urate excretion with a tendency for renal stones, dyspepsia and diarrhoea. The requirement is 40-50 mg/day. Amla, citrus fruits, sprouting cereals, potatoes and pulses are rich sources of vitamin C. Bioflavanoids are water-soluble compounds that absorb vitamin C.

Preparations: Celin— 100 mg, 500 mg
Suckcee, Limcee—500 mg chewable tab.
Cecon drops—100 mg/ml
Eldervit inj.

MACROELEMENTS/MACROMINERALS

These are elements present in higher concentrations in the body—more than 0.01% body weight. The clinical aspects of macrominerals are given in Table 3.6.

1. Calcium

Calcium is essential for bone and teeth formation, functions of calcium channels, normal muscle contraction, nervous activity and coagulation. Milk and milk products, oysters, crab, fish, mutton, leafy vegetables, roots, gingelly seeds and millets like ragi are rich sources of calcium. 99% of body calcium is present in the bone. A **difference in pH of 0.1 produces a difference of 10% in ionized calcium.** Thus alkalosis and administration of alkalie can produce tetany. Phosphate in milk and oxalates and phytates in grains and vegetables decrease the absorption of calcium. The Ca:P ratio of less than 2 as in cow's milk reduces calcium absorption. Thus artificially fed babies are prone to develop hypocalcaemia. LBW babies, preterms (PT), infants of diabetic mothers (IDM) and those with protein-energy malnutrition (PEM) are more prone to have hypocalcaemia. Hypocalcaemia produces tetany and convulsion. In deficiency, 100-200 mg/kg/day may be given.

Excess can produce hypercalacaemia and is noted with vitamin D intoxication, immobilization, milk-alkali syndrome, hyperparathyroidism etc. It may lead to calciuria and nephropathy. **Idiopathic hypercalcaemia is associated with 'elfin facies' and supravalvular aortic stenosis (William syndrome).** The requirement is 500-1000 mg/day and normal serum calcium is 8-11 mg/dl. The product of Ca and P (Cadman's product) = 40 is ideal for mineralisation.

Preparations: Ostocalcium$B_{]2}$—82 mg elemental Ca/5 mlas0.21gCa$P0_4$. (5 ml/kg of Ostocalcium supply 82 mg/kg of elemental calcium and hence large volumes may be needed.) Shelcal, Trical D, Calrich—Ca Carbonate 250 mg elemental Ca/5 ml and 250 mg/tab.

2. Phosphorus

Phosphorus is closely linked to calcium and most of the calcium is deposited as calcium phosphate in bone and teeth. It is also a component of nucleic acid, phosphate esters, ATP, 2,3-diphosphoglycerate (2,3-DPG) and buffer systems in the body. It is important in cellular metabolism, oxygen transport and acid-base balance. Cereals, pulses, nuts and oil seeds are rich sources. 80% of phosphorus in plant source is present as phytate. Phosphorus present as component of phytate is not available for absorption. The level of phytates tends to be low in polished rice and germinated seeds. Deficiency can occur in LBW, PT and PEM. Those **getting parenteral nutrition and those on ventilators may develop deficiency** and it may result in shift of oxygen dissociation curve to the left, hypoxia.

Table ■1.6 Macroelements/macro minerals in clinical nutrition

Element	Sources	Functions	Deficiency	Clinical features	Requirement	Remarks & toxicity
Calcium	Milk group, millets, greens, fish etc., oysters, crabs	Constituent of bone and teeth, muscle contraction, blood clotting, nervous action in calcium channel	LBW, preterms, PEM, intake of excess cow's milk, infant of diabetic mother (IDM)	Hypocalcemic tetany/ convulsion. **Inv.** S.Ca (7-11 mg/dl).	500-1000 mg/day Treatment Oral: 100-200 mg/kg/ day Inj: 0.5-1 mL/ kg/dose 10% Ca gluconate IV	Oxalates & phytates inhibit absorption Ca/P ratio > 2 is ideal **Toxicity:** Hypercalcaemia
Phosphorus	Cereals, legumes, nuts, oil seeds, milk group	Constituent of bone and teeth, ATP, nucleic acid, 2-3-DPG, buffer system and phosphate esters	LBW, PEM	Deficiency is rare, but leads to hypoxia in TPN & those on ventilators. Hypophosphataemia occurs in rickets. **Inv.** S.P (3-5 mg/dl)	500-1000 mg/day	40-80% of P in cereals is present as phytates and is not bioavailable
Sodium	All foods, common salt	Constituent of body fluids, maintains homeostasis, extracellular cation	Excessive sweating, diarrhoea, diuretics, water intoxication, SIADH	Hyponatraemia, Hypotension, dehydration, shock, lethargy, seizure. **Inv.** S.Na (135-145 mEq/L)	Common salt 10-15 g/day, 2-3 mEq/kg/day 5 g = 85 mEq NaCI	**Toxicity:** Hypernatraemic dehydration, intracranial haemorrhage

contd.

Potassium	All foods, fruits, greens	Constituent of body fluids, intracellular cation	PEM, diarrhoea, diuretics, diabetic ketoacidosis
Magnesium	Cereals, nuts, legumes, greens	Constituent of bone, PEM, IDM cardioprotective role, tetany cellular metabolism, shares some properties of Ca	

Hypokalaemia, hypotonia, ileus, flaccid paralysis. **Inv.** S.K (3.5-4.5 mEq/L) **ECG:** Flat/inverted T waves	2-3 mEq/kg/day **Treatment** 3-4 mEq/kg/day	**Toxicity** Hyperkalaemia, ventricular fibrillation, death. **ECG** Tall tented T waves > 10 divisions
Hypomagnesaemia, seizures, apathy. **Inv** S.Mg (1.5-3 mg%)	200-300 mg/day **Treatment** 0.1 to 0.2 ml/kg 50% MgSOJM 12 hrly for 2-3 days	**Toxicity** Hypermagnesaemia, drowsiness, resp depression, absent DTR, conduction defects and coma

and RBC, WBC and platelet dysfunction. The requirement is 600-1000 mg/day. Protein-rich food stuffs are rich in phosphate. Hyperphosphataemia is noted in chronic renal failure. Dietary restriction is needed in them. Hypophosphataemia occurs in rickets. Normal serum phosphorus is 3-5 mg/dl.

Preparations: Ostocalcium B_{12}5 ml = Ca 82 mg, D_3 200 IU, B_{12} 2.5 mg and Ca_3 ($P0_4$)t 0.21 g; Aquamin 54 mg/tab & 15 ml syrup; Multivite FM; Supradyn tab.

3. Sodium

It is the most important constituent of body fluids and cells. It maintains osmotic balance and keeps cells in proper shape. Sodium is lost in urine and sweat and addition of NaCl to food is essential to give taste and to maintain osmotic balance. The usual intake of salt is 10-15 g/day. Deficiency can occur in diarrhoea, excessive sweating, with diuretics, water intoxication and syndrome of inappropriate secretion of antidiuretic hormone (SIADH). Hyponatraemia produces hypotension, dehydration, vomiting, lethargy, shock and seizures.

Excess occurs in states of oedema, salt poisoning and hypernatraemia. In salt restricted diet, intake is restricted to 4-5 g/day. The requirement of sodium is 2-3 mEq/kg/day and normal serum sodium is 135-145 mEq/L. **Hypernatraemic dehydration produces cellular dehydration, irritability and intracranial haemorrhage.**

Preparations: 0.9% N.saline/DNS-154 mEq/L.

4. Potassium

It is an important component of cell and body fluids. It is the main intracellular cation. It is rich in fruits and greens. Deficiency is noted in diarrhoea, diuretic therapy, PEM, diabetic ketoacidosis (DKA) and with frequent salbutamol nebulization. Hypokalaemia produces abdominal distension, ileus and acute flaccid paralysis (AFP). ECG may show flat T waves and U waves. Excess is noted in acute renal failure and tissue injury. It produces tall tented T waves in ECG and may progress to ventricular fibrillation and death. The requirement is 2-3 mEq/kg/day and normal serum potassium is 3.5-4.5 mEq/L.

Preparations: Potklor, Potasol Oral—1.5 ml = 2 mEq

Inj. KC1—1 ml = 2 mEq; ELIZ solution (Appendix)

5. Magnesium

Magnesium is important for cellular metabolism, is cardioprotective and it shares many of the properties of calcium. It also acts as an antagonist to calcium and this property is made use of in the treatment of bronchial asthma. Calcium produces bronchospasm by activating contractility of smooth muscle, increasing the secretions and by releasing acetylcholine (Ach). Mg decreases calcium uptake by the cells, inhibits smooth muscle contractility, inhibits histamine and Ach release and depresses excitability of smooth muscle fibre. Thus **it has bronchodilator**

and anti-inflammatory properties. In asthma which is refractory even after 3 salbutamol nebulizations, Mg can be given as a slow drip; 0.2 ml/kg 25% $MgS0_4$ in 30 ml glucose in 30 minutes. It is not recommended in high fever and hypotension. Deficiency is noted in IDM and PEM. Hypomagnesaemia may cause convulsion, apathy and cardiovascular diseases. In hypomagnesaemia, Inj. 50% $MgS0_4$0.1-0.2 ml/kg IM 12 hourly can be given for 1-3 days; up to 0.8 mEq/kg/ dose can be given. Cereals, pulses, nuts, green leafy vegetables etc., are rich sources. Excess can produce hyporeflexia, respiratory depression, drowsiness, coma and death. **The antidote is calcium.** Serum magnesium is 1.5-3 mg/dl. Above 5 mg/dl produces symptoms and above 15 mg/dl is fatal. The requirement is around 200-300 mg/day or 3-6 mg/kg/day. 12 mg of magnesium is equivalent to 1 mEq.

Preparations: 25% $MgS0_4$ = 2 mEq/ml, 50% $MgS0_4$ = 4 mEq/ml Calcimax—calcium 150 mg. Mg 25 mg, Zn 1.5 mg, and Vit D 200 IU; Aquamin—35 mg/tab and 15 ml syrup; Multivite FM; Supradyn tab ELIZ solution (Appendix)

TRACE ELEMENTS/MICROMINERALS

Out of the 109 well-characterised elements. 14 trace elements are considered essential for human growth and function. They are iron, iodine, zinc, copper, chromium, selenium, manganese, cobalt, molybdenum, nickel, vanadium, silicon, arsenic and fluorine. Tin is also now included in the list.

Table 3.7 summarises the clinical aspects of trace elements. Trace element deficiencies noted with total parenteral nutrition are given in Table 3.8. Deficiency and excess can be detected from hair with accuracy. It is important to take a balanced diet in order to get all the trace elements. Excess dietary fibre, phytates and oxalates reduce trace element absorption.

By definition, **trace elements** are **those present in concentration less than 0.01% body weight,** i.e., 0.1 mg/g or 100 mg/g.

1. Iron

Iron is the mineral essential for synthesis of haemoglobin and for oxygen transport. It is also important in oxidation-reduction reaction and enzyme function. It is the **most abundant mineral on earth** and **it produces the commonest deficiency as well.** Iron deficiency leads to microcytic hypochromic anaemia, lack of concentration and koilonychia. **It produces geophagia (eating mud), amylophagia (eating raw rice)** and **pagophagia (eating ice cubes).** Deficiency is due to reduced intake, increased demand due to physiological states, worm infestations, blood loss and malabsorption. LBW babies are more prone for deficiency.

Excess intake of cow's milk leads to deficiency. **Milk is a poor source of iron** and very little iron present in cow's milk is not bioavailable. Cow's milk also produces blood loss and milk protein sensitive enteropathy in some. Iron present in breast milk though less is more bioavailable and is absorbed due to the presence of lactoferrin. So it is essential to advice breastfeeding over unmodified

Table 3.7 Trace elements/microminerals in clinical nutrition

Element	Sources	Functions	Deficiency	Clinical features	Requirement	Remarks & toxicity
Iron	Fish, meat group, cereals, legumes, greens, jaggery, cooking in iron vessels, asafoetida, turmeric, dates, watermelon etc.	Constituent of haemoglobin and enzymes, role in oxygen transport	LBW, blood loss, hook worm, whip worm, mal-absorption, poor intake	Pallor, dyspnoea, CCF, irritability, lack of concentration, pica, koilonychia **Inv** S.iron (50-150 ng/dL) Iron binding capacity (100-400 ng/dL), S.ferritin (50-250 mg/mL), blood smear hypochromic microcytic anaemia	Prophylaxis: 2 mg/kg/day. Older children 10-20 mg/day. **Treatment** Oral: 6 mg/kg/day for 3-4 months. **Inj** Weight in kg x Deficit in g% x 2.5 + 25% for stores in div. doses. Packed cell transfusion: 5-10 ml/kg. Always treat the cause	Oxalates, phytates & Zn inhibit absorption. Vit. C, cobalt and acid medium increase absorption. **Toxicity** GI upset, quiescent phase, metabolic derangement, acidosis, hypoglycaemia, hepatic failure and strictures. Gastric lavage with soda bicarbonate and desferrioxamine. Desferal 50 mg/kg followed by 10 mg/kg/hr x 24 hours IV. S. iron > 1800 jig% is fatal.
Iodine	Sea foods, Constituent of drinking water, thyroxine iodized salt		Low content in water, excessive intake of cabbage, cauliflower	Goitre, hypothyroidism, still birth, CNS defect **Inv** Urinary iodine, PBI, T_3, T_4 TSH, iodine uptake study	50-150 ug/day	Iodized salt to contain 15 ng/g (15 ppm); up to 30 ppm is added to tackle loss. Excess can cause goitre
Copper	Liver, fish, meat, oyster,	Constituent of enzymes, TPN, PEM,	LBW, preterm, nep-	Hypochromic anaemia, neutropaenia, hypo-	1-2 mg/day	**Toxicity** Hepatitis, cirrhosis, haemolytic

	legumes, ceruloplasmin, hrotic syndrome pigmentation of hair,					anaemia, Zn deficiency
	competes with Zn & Molybdenum for absorption	role in haemo-poesis and bone meta-bolism		bony defects **Inv.** S. Cu (75-150 ng/dL) S. ceruloplasm (10-50 |ig/dL)		
Zinc	Liver, beef, oyster, cere-als, nuts, grapes; phy-tates reduce absorption	Constituent of enzymes, role in protein and nucleic acid synthesis	PEM, hepatitis, TPN, nephrotic enteropathica	Growth retardation, anorexia, gonadal atr-syn, acrodematitis ophy, alopecia, derm-atitis, diarrhoea, redu-ced taste sensation. **Inv** S. Zn (60-150 |ug/dL), Zn in hair	5-15 mg/day **Treatment** 1-2 mg/kg/day	Used as adjuvant in Wilson's disease. **Toxicity** Gl upset, Cu deficiency
Chro-mium	Yeast, liver, cereals, nuts, cocoa, pepper	Facilitates in-sulin action	PEM, TPN	Hyperglycaemia, encephalopathy. **Treatment** **Inv** S.Cr (0.02 fig/dL)	10|xg/day 180 ^g single dose	Renal failure, dermatitis
Fluo-rine	Drinking water, sea water	Constituent of bone and	Poor content	Dental caries	1-5 mg/day. Drinking water	Dental fluorosis, genu valgum esp. with

	foods, tea, cheese	teeth	
Selenium	Meat group, greens, garlic	Antioxidant, co-factor, maintains liver integrity	PEM, TPN, poor soil content
Manganese	Cereals, legumes, greens, tea	Component of superoxide dismutase, role in oxidative phosphorylation	TPN
Nickel	Chocolate	Component of urease, nickel, plasmin; stabilizes membranes	TPN
Silicon		Cross-linkage of collagen	TPN
Vanadium	Protein rich food		TPN PEM

	up to 1 ppm	sorghum intake. >2-3 ppm in drinking water needs defluoridation by alum and lime
Keshan cardio-myopathy, arthritis, myalgia, growth retardation, liver necrosis, risk of cancer. **Inv** S.Se (13 ng/dL)	100 ng/day	Dental caries, alopecia, garlic odour in breath
Growth retardation, reddening of hair, increased prothro-mbin time **Inv** S. Mn (0.06 ng/dL)	1-5 mg/day	Encephalitis, goitre, cardiomyopathy, chole-stasis. Iron decreases Mn absorption
Inv. S. nickel (0.02 ng/dL)	Not known	Dermatitis, liver necrosis, nasal and lung cancers
Growth retardation, defective bone growth	Not known	Granuloma and fibrosis of lung
Associated with nutritional oedema	Not known	

Table 3.8 Trace element deficiencies noted with TPN

Element	Deficiency
Zinc	**Periorificial crusting dermatitis,** bullae in hands and feet, alopecia, diarrhoea
Copper	**Refractory hypochromic anaemia, neutropenia,** subperiosteal haematoma, soft tissue calcification, **osteoporosis**
Selenium	**Cardiomyopathy,** myopathy, myalgia
Chromium	**Hyperglycaemia,** glycosuria, peripheral neuropathy, encephalopathy

Manganese Reddening of hair, weight loss, hypocholesterolaemia

Molybdenum Tachycardia, irritability, coma, central scotoma

bovine milk feeding to prevent iron deficiency at least for the first two years of life. Phytates and oxalates in cereals, high phosphate in cow's milk and excess of zinc also reduce absorption. Thus, children who take only unmodified bovine milk and rice tend to have iron deficiency. Acid medium, vitamin C and cobalt increase iron absorption. Consumption of tea, coffee etc., with food will reduce iron absorption and **consumption of lemon juice, fruit juice and curd with food will increase absorption** due to the presence of vitamin C. Haeme iron from animal source is better absorbed. For anaemia prophylaxis, Folifer tablets with 20 mg elemental iron and 100 mg folic acid are given to children for 3 months. For treatment, up to 6 mg/kg elemental iron should be given for 3 months. The requirement of iron is 10-20 mg/day. In severe deficiency, packed cell transfusion is given. For parenteral iron therapy, the dose is calculated as follows: **Deficit in g% x weight in kg x 2.5 + 25% of the calculated dose to replenish stores (mg).** It is present in green leafy vegetables, cereals, pulses, (3 Gs—grams, grains and greens), molasses, egg, meat etc. Cooking in iron pots is also beneficial. Blood loss as in hookworm and whipworm infestation, haemorrhoids, menorrhagia etc., leads to severe iron deficiency. Excess of iron can occur in haemolytic anaemia with frequent blood transfusions. Haemosiderosis is the result. A number of cases of iron poisoning have occurred as children are attracted to the colourful Folifer tablets supplied in bulk from the primary health care centres to the mother and the child. Desferrioxamine is the antidote. Serum iron is 50-150 |ig/dl and serum iron binding capacity is 100—400 (ag/dl. It is increased in iron deficiency. Serum ferritin is the indicator of iron stores and is 50-250 (ig/ml. Serum iron

> 1800 (ig/dl is fatal. **Weekly iron prophylaxis may be beneficial in children and adolescents.** Normal Hb (g/dl) levels in various age groups (WHO 1968): Newborn, 13.0; 2-6 months, 9.5; 6 months to 6 years, ll;6-12years, 12; adult male, 13; female, 12; pregnancy, 11.

Preparations: Folifer (paed)—iron 20 mg, folic acid 100 mg/tab; Folifer (adult)—iron 100 mg, folic acid 500 mg/tab; Inj. Imferon—iron 50 mg/ml; Tonoferron—iron 25 mg/ml in drops, 250 mg/5 ml in syrup; Lysiron—iron 5 mg/ml drops; Mumfer—50 mg/ml drops; 50 mg/5 ml syrup; Pink—100 mg elemental iron, 50 mg folic acid.

2. Iodine

It is essential for the synthesis of thyroid hormones (thyroxin). Deficiency leads to hypothyroidism, endemic goitre and growth retardation. Intrauterine deficiency may lead to mental retardation. **The requirement is 100-150 mg/day.** About 2/3 of this is generally derived from the drinking water and 1/3 from diet. Goitrogens in vegetables of brassica species (cabbage) interfere with utilization. Iodine fortified salt that contains potassium iodate 15 ppm (15 mg/g) is useful to meet requirements especially in mountainous areas. In commercial iodized salt, up to 30-50 ppm iodine is added to cover losses. Double fortified salt contains potassium iodate and ferrous sulphate. Excess of iodine will also lead to goitre.

Serum iodine is 50-150 |ig/dl. Urinary iodine excretion is reduced in deficiency. Urinary iodine is a very good indicator of deficiency, which can be tested using dipstick. Improved kits for testing iodized salt are also now available. Excess iodine may cause iodism and dermatitis.

Preparations: Aquamin—iodine 15 mg/tab, and 15 ml syrup

3. Zinc

Zinc is a cofactor in various enzymes and is important in protein and nucleic acid synthesis. **Zinc deficiency leads to growth retardation, hypogonadism, skin changes and diarrhoea. Acrodermatitis enteropathica responds to zinc therapy.**

It produces crusting dermatitis in periorificial regions, bullae in palms and soles and diarrhoea. Deficiency leads to decreased taste sensation and alopecia. Deficiency occurs in PEM, TPN, hepatitis and nephrotic syndrome. Deficiency produces thymic atrophy and **serum thymulin levels can be used to detect early deficiency.** The requirement is 5-15 mg/day. Meat, cereals, pulses, vegetables, nuts and fruits supply zinc. Phytates inhibit absorption of zinc. In deficiency, 50-150 mg zinc sulphate is given, i.e., 1-2 mg/kg elemental zinc/day. **Excess can produce iron and copper deficiency.** So, **it is useful in the treatment of Wilson's disease.** Excess can also produce GI upset. Normal serum level is 60-150 (ig/dl. In malnourished children values as low as 20-25 (ig/dl have been noted which imprroved marginally after food supplementation. It appears that they need zinc supplements in addition to food. The sources are liver, beef, oyster, nuts, cereals and

grapes. Zinc supplementation has been shown to result in better catch up in height in those with LBW and malnutrition.

Zinc sulphate 20 mg tablets are now available in RCH kits supplied to health workers. 10 mg elemental Zn/day x 14 days is given to infants, 6 mon with diarrhea and 20 mg for those > 6 mon. Zn is found to improve epithelial repair, brush border enzymes, T-cell immunity, permeability of intestine and regulation of water and electrolytes. Thus, Zn will decrease diarrhea duration, stool output and also prevent subsequent episodes for next 2-3 months.

Preparations: Ulceel—$ZnS0_4$ 220 mg/tab; Zevit—elemental Zn 10 mg/5 ml syrup, 22.5 mg/cap, 2.5 mg/ml drops; Zincosules—5 mg/5 ml; Aquamin, Multivite FM, Supradyn, Visyneral Zn—Zn 20; Rezimune—20 mg elemental zinc

4. Copper

Copper is involved in cross-linkage of connective tissues, haemopoiesis, neurotransmission, lipid metabolism, iron absorption and oxidative enzymes. The requirement is 1-3 mg/day. **Copper deficiency may produce neutropenia, refractory hypochromic anaemia, hypopigmented hair, osteoporosis, soft tissue calcification, subperiosteal haematoma** and **impaired myelination.** Deficiency is noted in preterms, LBW, PEM, TPN and nephrotic syndrome. Oversupplementation of zinc may lead to copper deficiency. Cereals, pulses, nuts, vegetables, fruits and fish contain copper. **Excess copper intake** and cooking in copper vessels are identified as **causes of Indian childhood cirrhosis (ICC).** Excess can also lead to haemolytic anaemia and zinc deficiency. Normal serum copper is 75-150 (ig/dl. Levels as low as 10-20 |ig/dl has been noted in PEM which only mildly increase after food supplementation for 3 months. Copper competes with zinc and molybdenum for absorption.

Preparations: Aquamin—copper 300 mg/tab, 300 mg/15 ml syrup;
Multivite FM—copper sulphate 100 mg/cap; Supradyn

5. Chromium

Chromium is important in glucose tolerance and **facilitates insulin action.** Deficiency is noted in PEM and TPN. 180-250 mg single dose of chromium is found to increase glucose tolerance. Deficiency produces glycosuria, hyperglycaemia, encephalopathy and peripheral neuritis. The requirement is 10-70 |ig/day. Chromium is present in yeast, liver, cereals, pulses, nuts, vegetables and fruits. **Excess can lead to dermatitis and renal failure.** Normal serum chromium is 0.02 |ig/dl.
Preparations: Aquamin—chromium 20 mg/tab, 20 mg/15 ml syrup

6. Manganese

It is important as an enzyme cofactor especially in superoxide dismutase (SOD), oxidative phosphorylation and in bone mineralization. Deficiency occurs in PEM and TPN. Iron decreases manganese absorption. Deficiency produces growth

retardation, weight loss, red hair, **hypocholesterolaemia** and increased prothrombin time. The requirement is 1-5 mg/day. Cereals, pulses, nuts and vegetables are sources of manganese. Excess can lead to cholestasis, encephalopathy, basal ganglia disorder, goitre and cardiomyopathy. Normal serum manganese is 0.06 mg/dl.

Preparations: Aquamin—Mn 500 mg/tab and 15 ml syrup;
Multivite FM, Supradyn

7. Fluorine

It is a component of bone and teeth and is important for prevention of dental caries. It is found in drinking water, sea foods, tea and cheese. Up to 1 ppm in drinking water is desirable. Excess fluoride > 2-3 ppm in water leads to fluorosis of bone. This requires defluoridation using alum and lime. Excessive intake of sorghum may produce **fluorosis.** The requirement is 1-5 mg/day.

8. Molybdenum

Molybdenum is a component of xanthine oxidase and is important in uric acid metabolism. It is rich in legumes, greens and meat group. Deficiency can occur in total parenteral nutrition (TPN) and due to poor soil content. **Deficiency produces** tachycardia, central scotoma, irritability, coma and probably **increased incidence of mouth and oesophageal cancers.** Excess may unmask hyperuricaemia, gout and genu valgum. Excess molybdenum competes with copper for absorption. The requirement is 250-500 mg/day. Normal serum molybdenum is 0.07 jag/dl.

Preparations: Aquamin—Mo 50 mg/tab and 15 ml syrup

9. Selenium

Selenium is cardioprotective, is an important antioxidant cofactor and it maintains liver integrity. It is rich in grains, meat group and garlic. Selenium deficiency may lead to growth retardation, liver necrosis, arthritis, myalgia, myopathy and **Keshan cardiomyopathy.** Deficiency also may increase the incidence of cancers. Deficiency occurs in PEM, TPN and poor soil content. **Selenium excess may lead to dental caries, alopecia and garlic odour in breath.** The requirement is 100 mg/day and normal serum selenium is 10-15 mg/dl.

Preparations: Aquamin—Se 20 mg/tab and 15 ml syrup

10. Cobalt

Cobalt is part of vitamin B_{12} and is also required for iodine utilization. It is rich in meat group. Deficiency occurs in PEM and TPN. Deficiency produces anaemia and goitre. Cobalt increases iron absorption. **Excess can lead to goitre and cardiomyopathy.** Normal S. cobalt is 0.007 (Xg/dl.

11. Nickel

It is a component of urease and nickelplasmin and it stabilizes membrane. Deficiency is noted in TPN. **Excess can produce dermatitis, liver necrosis and nasal/lung cancers. Chocolate contains a lot of nickel.** Serum nickel is 0.02 jag/dl.

12. Vanadium

It is associated with nutritional oedema. Deficiency occurs in PEM and TPN. All protein-rich foods are good sources of vanadium.

13. Silicon

It is important in cross-linkage of collagen. Deficiency is seen in TPN. It leads to growth retardation and defective bone growth. **Excess can lead to granuloma and fibrosis of lung.**

14. Arsenic

It is important in nail and hair growth. Excess arsenic as found in cow's milk is toxic to skin, CNS and respiratory system. For infants, multivitamin and mineral preparations are available as drops. The contents of a standard preparation are given in Table 3.9.

Preparations: Dexvita drops, Visyneral drops.

Table 3.9 Vitamin and mineral content in commercial preparations/ml

Item	Unit	Quantity
Vitamin A	(IU)	1250
Vitamin D	(IU)	330
Vitamin E	(IU)	8
Vitamin C	(mg)	40
Thiamine	(mg)	1
Riboflavin	(mg)	1.37
Niacin	(mg)	10
Pyridoxine	(mg)	1
Folic acid	(MS)	100
Pantothenic acid	(mg)	3
Biotin	(mg)	50
Zinc sulphate	(mg)	11
Choline	(mg)	50
Inositol	(mg)	15

15. Vitamins

Vitamins are substances found in small amounts in several food items that are needed for growth, normal metabolism and health. The fat-soluble vitamins are A, D, E and K and water-soluble vitamins are B complex factors and C. (3 (Beta) carotene, the precursor of vitamin A, vitamin E and vitamin C have antioxidant properties. The non-glyceride components in the diet, namely carotenoids and tocols (tocopherol and tocotrienols), are **precursors of vitamin** A **and vitamin** E **respectively and play an important role in arresting free radical diseases.** They play a crucial role in antiaging, antitumour, lipid peroxidation and immunomodulation processes. They protect against coronary artery disease, stroke and regulate **serum lipid profile. 0.6 mg of beta-carotene is equivalent to 1 unit of vitamin** A. (Refer sections 5.1 and 5.2)

16. Minerals

Minerals like calcium, phosphorous, magnesium etc., are important in bone and teeth development and in maintaining homeostasis. Trace elements like iron, iodine, zinc, copper etc., are also crucial for life, (refer Section 4.4)

ANTINUTRIENTS IN FOOD ITEMS

There are several antinutrients in food items, like trypsin inhibitors, phytates, oxalates, tannins, goitrogens etc.

a) **Trypsin inhibitors:** Trypsin inhibitors are present in legumes, soyabeans, white of egg etc. They are heat labile and can be got rid of by cooking. They inhibit trypsin in the gut and lead to indigestion of protein. **Ovomucoid is the trypsin inhibitor in duck's egg white.**

b) **Phytates:** Phytates are hexa phosphates of inositol and are present in cereals. **They bind iron, zinc, calcium, magnesium and can lead to deficiencies.** Refined grains like polished rice and germinated grains contain less phytates. During germination, phytates reduce due to enzymatic breakdown. The phosphate in phytate is not bioavailable.

c) **Tannins and caffeine:** Tannins are condensed polyphenolic compounds present in legumes, millets, fruits, tea etc. They bind iron and interfere with iron absorption. They also bind proteins. Removal of seed coat and addition of milk to tea reduce tannin. Caffeine in coffee is reduced by roasting. It is a stimulant. Theobromine in cocoa is a stimulant.

d) **Oxalates:** Oxalic acid and calcium oxalates are plenty in legumes, vegetables like spinach, drumstick and curry leaves, amla nuts, coffee and tea. They interfere with calcium absorption and predispose to oxaluria and urinary stones. Pyridoxine is useful to reduce oxalates.

e) **Goitrogens**: They are **thiocyanates, isothiocyanates, glucoinolates etc., present in the vegetables belonging to Brassica family and Crucifera family,** like

cabbage, cauliflower, radish, soya-beans, peanuts etc. They block iodine uptake and cause goitre.

0 **Excess fibre:** Even though dietary fibre is undigestible, it is an essential component of food. Excess fibre can bind trace elements and can cause trace element deficiency.

g) **Beta oxalyl amino alanine (BOAA)** present in **kesari dhal** leads to lathyrism. ***Lathyrus sativa*** seeds (kesari dhal) can be **parboiled** to get rid of the toxic amino acid.

h) **Cyanogenic glucosides (linimarin):** It is present in some varieties of tapioca (Cassava). This is converted to gaseous cyanide. Cyanate to thiocyanate conversion can occur if there is excess protein consumption. Cyanogenic glucosides in tapioca can be removed by leaching out with water. This is called **double cooking,** by draining the initial water and recooking using fresh water.

i) **Avidin** in duck egg is antagonistic to biotin which can be eliminated by cooking.

j) **Xenobiotics** are compounds that are added to impart flavour or taste. Some of them cannot be metabolized and in large and frequent doses may cause toxicity, e.g.. **monosodium glutamate** (aginomoto). Some of them may be carcinogens.

k) **Aflatoxins:** The **fungus** *Aspergillus flavus,* which affects groundnut, produces aflatoxins and this can cause liver damage.

3.3 Food Groups and Recommended Dietary Allowances

1. Food Items

a) **Cereals:** Cereals form the staple diet in India, e.g., rice, wheat, maize. Smaller grains are called millets. Cereals generally lack lysine. **Rice is richer in lysine** and NPU is better. **Parboiling** leads to retention of vitamins, increases shelf-life and ability to resist insects. Off flavour due to parboiling can be reduced by sodium chromate. In parboiling, rice is first boiled and then steamed after draining the water. **Polished rice has low levels of vitamin B complex. Wheat lacks lysine and threonine.** Ragi is rich in minerals like calcium, iodine etc. **Cereals do not contain vitamin A and vitamin C except yellow maize which contains a-carotene.** Phytates and tannins in cereals interfere with mineral bioavailability. Cereals contain 7-11 g% protein and 2-5 g% fat. Wheat, ragi, oats and barley contain gluten. Gliadine in gluten causes coeliac sprue due to certain amino acid sequences (motifs) that cause sensitization.

b) **Legumes (pulses):** Pulses are rich sources of protein (up to 22 g%). Pulses meet the protein requirement of vegetarians. **Pulses lack vitamin** A **and vita-**

min C; but germinated legumes contain vitamin C. **Pulses are deficient in methionine.** Cereal-pulse combinations lead to supplementary effect of protein. A 4:1 or 3:1 proportion is enough for supplementary effect. Soyabean contains 43 g protein and 430 kcal per 100 grams. Oligosaccharides in pulses cause flatulence.

c) **Roots and tubers:** They are rich in carbohydrate and are good sources of energy and calcium. Carrots contain P-carotene and potatoes contain vitamin C. Cyanogenic glucosides in tapioca can be eliminated by leaching out with water.

d) **Vegetables:** Vegetables add variety to diet, provide minerals, vitamins and fibre. **Green, yellow, orange and red (GYOR) vegetables** are sources of (3-carotene.

Green leafy vegetables (GLV, greens): They are rich sources of calcium, iron, beta-carotene, vitamin C and B complex. They are cheap and at least 50 g should be consumed every day. They can be easily grown in the backyard of the house.

e) **Fruits:** They are rich sources of vitamins and fibre. Green, yellow and orange (GYO) fruits contain beta-carotene. Amla, citrus fruits and guava are rich in vitamin C. Dried fruits like dates supply iron. Plantains, jack fruit etc., supply energy. Seasonal fruits should be encouraged. Papaya can be grown in the backyard to supply fruits in all seasons.

f) **Milk and milk products (milk group):** Milk is a good source of protein, calcium and vitamins. **It is deficient in iron and vitamin** C. Iron and calcium absorption is interfered by high phosphate content in cow's milk. Human milk has high lactose content (7 g%). Buffalo milk has high fat content (7 g%). It is mostly saturated fat. Milk and milk products should be included in the diet of growing children.

g) **Non-vegetarian foods (meat group):** Egg, fish, meat etc., are included in this group. Eggs supply good-quality protein, vitamins and fat. It is deficient in carbohydrate and vitamin C. Fish, meat and chicken are good sources of protein and vitamins. Fish is a good source of calcium. Omega 3-PUFA in fish is protective against cardiovascular diseases.

h) **Fats, oils, nuts and oil seeds:** Nuts and oil seeds are good sources of protein, vitamins and fat. Visible fat includes oils, butter, ghee, hydrogenated oils (Dalda) etc. Fats are rich concentrates of energy.

They improve palatability and supply EFA and fat-soluble vitamins. Fats increase gastric emptying time. Total calories from visible fat should not be more than 10-15%; the maximum permitted limit is 20%. This is based on the fact that average Indian diet supplies 20-25% calories as fat and this can be enhanced to 30-4-5%. Red palm oil (RPO) is a rich source of (3-carotene up to 800 mg/g. However, after processing of oil to RBD oil (refined, bleached and deodorized), the levels reduce. Very high PUFA content as in safflower oil is

undesirable due to peroxidation. Coconut oil is deficient in EFA, but it contains medium chain triglycerides (MCT). Hydrogenated vegetable oils (HVO) are solid fats with good shelf-life, but they contain more trans isomers which elevate LDL cholesterol and contain less EFA. Unless fortified, they are deficient in EFA.

i) **Sugars:** Sugar, honey, jaggery etc, are energy concentrates. **Jaggery is rich in iron.** Excessive intake of sugars is not desirable due to wide fluctuations in blood sugar and high incidence of caries among children,

j) **Condiments and spices:** These include asafoetida, cardamom, chillies, cloves, coriander, cumin seeds, fenugreek seeds, garlic, ginger, nutmeg, pepper, poppy seeds, tamarind, turmeric etc. They increase palatability. Some of them supply a-carotene, vitamins and minerals. **Green chillies supply P-carotene. Garlic is a rich source of selenium.** Turmeric contains high levels of iron, but also contains tannin that interferes with iron absorption,

k) **Salt:** Normally, 10-15 g/day of common salt is consumed. In salt-restricted diet, 0.5-1 g/day is recommended. In iodized salt, potassium iodate is fortified. In **double fortified salt,** iron is also added as ferrous sulphate with sodium bisulphate or as ferric ortho-phosphate without causing colour change.

1) **Water:** Water intake should be optimum, 5-6 glasses/day in toddlers and 8-10 glasses/day in older children. The Holliday and Segar formula is used for fluid calculation (Table 3.19). The present weight of the child may be used for fluid calculation in children unlike for calorie calculation. For calorie calculation, the expected weight is used, as the recommended dietary allowance (RDA) is for the age of the child.

2. Food Guide Triangle

The food guide triangle gives a guideline to the choice of food groups in day-to-day practice (Fig. 3.1). Cereals and pulses should be consumed in 6-10 servings a day. Vegetables should be consumed in 3-5 servings; fruits, milk and milk products and non-vegetarian items like fish, poultry, meat in 2-3 servings; and fats, oils and sugars sparingly. The standard serving size of food groups is given in Table 3.10.

Each servings can have **exchange items.** The exchange items are selected to supply around 100 kcal per exchange. The usual cereal exchanges are 1 katori cooked rice, 2 chapati, 2 idli, 2 slices bread etc. About 1/2 to 3/4 katori of any pulses can be exchanged. About 100 g of any vegetable can also be exchanged. Among snacks, 2 vadai, 1 sandwich, *Vi* katori upma etc., can be exchanged. Fats, oils and sugars should be used in moderation. The nutritive value of raw food items are given in Table 3.11 and the nutritive value of cooked items according to household measures are given in Table 3.12.

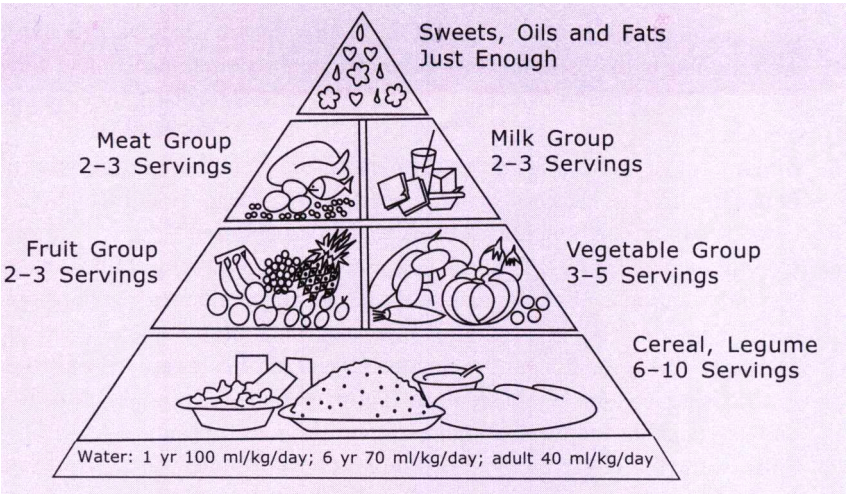


Fig. 3.1 Food guide triangle for day-to-day choice of food

Table 3.10 Serving size of various food groups

Food group	Serving size	Servings/day
Cereal, pulse, bread	1 slice or 1 oz	6-10
Vegetables	*Vi* cup	3-5
Fruit	1	2-3
Milk, milk products	1 cup	2-3
Meat group	2 oz	2-3
Fats, oil	2-3 tsp	Sparingly
Sugars	To taste	Sparingly

cup = 240 ml, 1 glass = 200 ml, 1 katori = 150 ml, 1 ladle = 30 ml, 1 oz = 30 ml, 1tablespoon = 15 ml, 1 teaspoon = 5 ml

Vegetarianism is good, but the degree counts. Vegetarians have low risk of obesity, cardiovascular diseases and colon cancer. Strict **vegans** are at risk of calcium, iron, vitamin B_p deficiency. Bran of grains and germinated seeds are useful in them. **Lacto-ovo-vegetarians** take milk and egg and have very little nutritional risk except haeme iron. **Fruitarians** are at risk of protein, sodium and other deficiencies.

Table 3.11 Nutritive value of common foods/100 g

Item Protein	(g)	Fat (g)	Fibre (g)	CHO (g)	Energy (kcal)	Iron (mg)
Cereals/Grains						
Rice	7	0.5	0.2	78	350	0.7
Ragi	7	1.3	3.6	72	330	3.9
Wheat	11	1.5	1.2	71	350	5.3
Maize	11	3.6	2.7	66	340	2.3
Pulses/Legumes						
Bengal gram	17	5.3	3.9	60	360	4.6
Black gram	24	1.4	0.9	60	350	3.8
Green gram	24	1.3	4.1	57	340	4.4
Red gram	22	1.7	1.5	58	340	2.7
Soya bean	43	19.5	3.7	20	430	10.4
. Leafy vegetables						
Agathi 8		1.4	2.2	12	93	3.9
Amaranth	4	0.5	1.0	6	45	3.5
Cabbage	2	0.1	1.0	5	27	0.8
Cauliflower green	6	1.3	2.0	8	66	40.0
Chekkurmanis	7	3.2	1.4	12	100	28.0
Coriander leaves	3	0.6	1.2	6	44	1.4
Curry leaves	6	1.0	6.4	18	100	0.9
Drumstick	6	1.7	0.9	12	92	0.9
Spinach	2	0.7	0.6	3	26	1.1
IV. **Roots & tubers**						
Arrow root Flour	0.2	0.1	-	83	340	1.0
Beet root	1.7	0.1	0.9	9	43	1.2
Carrot	0.9	0.2	1.2	10	50	1.0
Colocasia	3.0	0.1	1.0	21	97	0.4
Onion (big)	1.2	0.1	0.6	11	50	0.6
Onion (small)	1.8	0.1	0.6	12	60	1.2
Potato	1.6	0.1	0.4	22	100	0.5
Radish	0.7	0.1	0.8	3.4	17	0.4
Sweet potato	1.2	0.3	0.8	28	120	0.2
Tapioca	0.7	0.2	0.6	38	160	0.9
Yam	1.2	0.1	0.8	18	80	0.6
Other vegetables						
Ash gourd	0.4	0.1	0.8	2.0	10	0.8
Bitter gourd	1.6	0.2	0.8	4.2	25	0.6

contd.

Brinjal	1.4	0.3	1.3	4.0	25	0.4
Cauliflower	2.6	0.4	1.2	4.0	30	1.2
Cucumber	0.4	0.1	0.4	2.5	13	0.6
Drumstick	2.5	0.1	4.8	4.0	25	0.2
Kovai	1.2	0.1	1.6	3.0	20	0.4
Ladies finger	1.9	0.2	1.2	6.4	35	0.4
Pumpkin	1.4	0.1	0.7	4.6	25	0.4
Snake gourd	0.5	0.3	0.8	3.3	18	1.5
VI. **Nuts and oils**						
Almond	20	58	1.7	11	655	5.0
Cashew nut	21	47	1.3	22	600	5.8
Coconut (fresh)	4.5	41	1.0	13	444	1.7
Coconut (dry)	6.8	62	6.6	18	660	7.8
Groundnut	25	40	3.1	26	560	2.5
VII. **Condiments & spices**						
Chillies (dry)	16	6.2	30	31	250	2.3
Coriander	14	16.0	32	21	288	7.1
Fenugreek	26	5.8	7.2	44	330	6.5
Garlic	6.3	0.1	0.8	29	145	1.2
Ginger	2.3	0.9	2.4	12	67	3.5
Pepper	11.5	6.8	15	50	300	12.4
Tamarind	3.1	0.1	5.6	67	280	17.0
Turmeric	6.3	5.1	2.6	69	350	67.8
VIII. **Fruits**						
Amla	0.5	0.1	3.4	14	58	1.2
Apple	0.2	0.5	1.0	13	59	0.6
Banana	1.2	0.3	0.4	27	116	0.4
Dates (dry)	2.5	0.4	3.9	75	317	7.3
Grapes	0.6	0.4	2.8	13	58	0.5
Guava	0.9	0.3	5.2	11	50	0.3
Jack fruit	1.9	0.1	1.1	20	88	0.6
Lemon	1.0	0.9	1.7	11	57	0.3
Musambi	0.8	0.3	0.5	9	43	0.7
Mango	0.6	0.4	0.7	17	75	1.3
Water melon	0.2	0.2	0.2	3	16	7.9
Orange	0.7	0.2	0.3	10	48	0.3
Papaya	0.6	0.1	0.8	7.2	32	0.5
Pineapple	0.4	0.1	0.5	10.8	46	2.4
Tomato	0.9	0.2	0.8	3.6	20	0.6
IX. **Meat group**						
Fish	20-60	1-10	-	0-5	100-300	1-50
Beef		8.0	10	0.5	0.2 400	18

contd.

Egg	13	13	-	-	173	2
Chicken	26	0.6	-	-	109	-
Mutton	20	13.0	-	-	194	2.5
Pork	18	4.4			114	2.2
Milk and milk	**products**					
Cow's milk	3.2	4.1	-	4.4	67	0.2
Buffalo's milk	4.3	6.5	-	5	117	0.2
Human milk	1.1	3.4	-	7.4	65	0.3
Cheese	24	25	-	6.3	348	2.1
Skimmed milk powder	38	0.1	""	51	357	1.4

Table 3.12 Food values in household measures

Items	Protein (g)	Energy (kcal)
Cow's milk 1 glass (200 ml)	6	120
Cooked rice 1 cup	4	175
Ragi flour 6 tsp	2	100
SAT mix 6 tsp	2.5	125
Cooked dhal 1 tsp	0.5	10
Egg 1	6	80
Fish 1 oz (10 cm piece)	6	80
Mutton 1 oz (8 bits)	6	50
Bread 1 oz (1 slice)	2	70
Dosai 1	2	70
Idli 1	2	50
Chappathi 1	2	70
Puri 1	1	35
Vada/bonda 1	1	50
Upma 1 cup	6	250
Sugar 1 tsp	-	20
Jaggery 1 tsp	-	20
Ghee/Butter 1 tsp	-	36
Mashed potato 1 tsp	-	40
Plantain 1	0.5	50
Groundnut 10	1	20
Pappadam 1	0.5	20

contd.

Biscuit 1	0.5	20
Coffee 1 cup	1.8	80
Tea 1 cup	1.0	60

3. Balanced Diet

A balanced diet is one which supplies all the nutrients in the right quantity and proportion. It is essential for growth, to maintain good health and to prevent deficiencies. Carbohydrate should yield 55-60 per cent of the calories, protein should yield 10-15 percent of the calories and fat should yield 30-35 percent of the calories.

Table 3.13 Balanced diet for an adult/adolescent boy (2400 kcal)

Food item	Quantity	Kilocalories
Cereal	400 g	1400
Legumes	60 g	230
Roots & tubers	50 g	50
Vegetables	50 g	50
Green leafy vegetables	50 g	50
Fruits	50 g	50
Milk/Curd	250 ml	150
Oil/Fats	30 g	270
Sugar	30 g	120
Total		2370

The diet should also contain enough vitamins, minerals, fibre, water etc. It should preferably include all the food items listed in the food guide triangle (Fig. 3.1). The composition of a balanced diet for an adolescent boy which is equal to that of an adult sedentary male, is given in Table 3.13. **The requirement of an adult sedentary male is 2400 kcal, i.e., 1 unit of energy** (also refer Table 3.14)

Exchanges for Non-vegetarians

1. 50% legume + 1 egg/30 g meat or fish
2. No legume, 1 egg + 30 g meat or fish
3. No legume, no egg, 60 g meat or fish

Table 3.14 The approximate RDA of vitamins and minerals

1. Vit. A	1500 IU/day (500 jxg)
2. Vit. D	400 IU/day (10 pg)
3. Vit. E	5-15 IU/day (5-15 mg)
Vitamin B Complex	
4. B, (Thiamine)	0.5-1.5 mg/day (1 mg/1000 cal)
5. B_2 (Riboflavin)	0.5-1.5 mg/day
6. B_e (Pyridoxine)	0.5-1.5 mg/day
7. B_3 (Niacin)	5-15 mg/day
8. B^ (Folic acid)	50-150 pg/day
9. B_{12} (Cyanocobalamine)	0.5-1.5 peg/day
10. Vit. C	40 mg/day
Macro elements	
11. Calcium	500-1000 mg/day
12. Phosphorus	800-1000 mg/day
13. Magnesium	200-300 mg/day
Trace elements	
14. Iron	10-20 mg/day
15. Iodine	50-150 pg/day
16. Copper	1-2 mg/day
17. Zinc	5-15 mg/day
18. Fluoride	1-5 mg/day
19. Manganese	1-5 mg/day
20. Selenium	100 pg/day
21. Molybdenum	200-500 pg/day
22. Chromium	10 pg/day

4. Nutritional Supplementation During Pregnancy and Lactation

During pregnancy, 300 kilocalories extra and during lactation 500 kilocalories extra should be supplemented. This can be achieved by taking **I** extra meal

Pregnancy (+ 300 kilocalories)

Item	Quantity	Kilocalories
Cereal	30 g	115
Legume	15 g	60
Milk	200 ml	120
Sugar	5 g	20
Total		315

Lactation (+ 500 calories)

Item	Quantity	Kilocalories
Cereal	60 g	230
Legume	30 g	120
Milk	200 ml	120
Sugar	5 g	20
Oil	2.5 g	20
Total		510

during pregnancy and 2 extra meals during lactation utilizing the food items available in the house. Legumes, milk products, vegetables and fruits should be ensured.

5. Food Processing and Storage

A. **Cooking:** Cooking improves digestibility, taste, flavour and appearance. Over-cooking is harmful as it leads to loss of nutrients. The various methods of cooking are:

a) Boiling: Do not boil for too long and use only enough water. The left over water should be used in soups or other liquid preparations.

b) Steaming: This does not cause much loss of nutrients.

c) Pressure cooking: This retains nutrients as left over water is very little.

d) Frying: Use only small amount of oil and it is not advisable to reuse the oil. See that the oil is at the right temperature before food is put in. Deep frying at high temperatures produce volatiles and chemicals like perioxides, aldehydes, ketones, hydroperoxidases, polymers, cyclic monomers etc. They are toxic to lung, kidney, CVS, CNS and immune systems.

Better options are:

a) Use of oil at room temperature as add on, e.g., salad oil

b) Use of small amounts of oil with fried mustard

c) Sauting, use of tomato as a base can reduce the oil requirement

d) Shallow or stir frying

e) Deep frying only when absolutely necessary; start with small amount of oil and add small amounts of fresh oil to replace oil used up in frying.

B. **Fermentation:** It enchances vitamin C and increases digestibility.

C. **Precautions while using vegetables and fruits**

1. Use fresh vegetables and fruits only.
2. Consume raw whenever possible.
3. Wash them before cutting and do not wash after cutting.
4. Cut them only at the time of cooking or eating.
5. Do not soak them in water for prolonged time. This will lead to loss of

vitamins and minerals.

6. Peel them carefully and preferably after cooking to retain more vitamins and minerals.
7. The coloured leaves are more nutritive. The green leafy tops of vegetables and coloured leaves should not be discarded.
8. Avoid overcooking and use only small amounts of water for cooking.
9. Cereals like rice should be cooked with only *2-2^{x}h* times of water.

D. **Washing:** Wash food items in clean water to remove dirt, bacteria, helminthic ova etc. Avoid prolonged soaking in water to retain water-soluble vitamins. Washing after cutting leads to nutrient loss.

E **Storage:** Cereals, pulses etc., should be stored clean in tightly covered containers. Vegetables should be stored in cool, dry place or in refrigerator after removing damaged or spoiled parts. Store onions, potatoes etc., in dry open baskets.

F. **Pickling:** It increases shelf-life.

G **Parboiling:** This is an ancient Indian method of hot soaking followed by steaming of paddy. This enables the vitamins to percolate inside. The off flavour can be eliminated by adding small quantities of sodium chromate. Parboiling improves shelf-life and offers resistance to destruction by insects. Parboiling also helps to remove the toxic amino acid (BOAA) in kesari dhal.

6. Improving Nutritive Value

Soaking, malting and sprouting improve nutritive value.

a) **Soaking:** Soaking in water for sometime initiates enzyme action and germination. It increases vitamins, digestibility and also makes cooking easier. Soaking of rice and blackgram before grinding for idli is a good practice. All legumes may be soaked before cooking.

b) **Malting:** Soaking of some cereals like ragi (Ragimalt) etc., overnight improves digestibility and enhances vitamins. This is called malting.

c) **Sprouting/germination:** Germination of cereals and legumes augments digestibility, increases vitamins, reduces bulk on cooking and decreases phytate levels. Digestibility increases due to production of amylase.

d) **Amylase rich food (ARF):** Wheat, bajra, jowar, moong dhal (green-gram) etc., can thus be converted to **amylase rich food** (ARF). Wheat yields maximum amount of amylase. ARF is particularly useful in persistent diarrhoea and malnutrition. Amylase powder can be prepared and stored. Small quantities can be added to the child's diet.

ARF can be prepared as follows:

i) Soak the item with 3 times water for twelve hours.

ii) Drain the water and wrap in moist muslin cloth and keep in a cool dry place for 48 hours to allow germination. Sprinkle water in between.

iii) Dry the germinated grain in sunlight for 8 hours.

iv) Devegetate and remove the sprouts
v) Roast the dried grain in low flame (80°C) for 10 minutes till they are brittle and malt aroma develops.
vi) Grind the roasted grain into fine powder.

The powder can be stored in airtight container for a month. Small quantities of ARF powder can be added to rice-dhal gruel or other porridge to reduce viscosity and thereby increase calorie density. This is easily digestible due to the presence of amylase. This will not swell much on cooking. Due to the low levels of phytates, the bioavailability of minerals increases and ARF is rich in vitamins.

7. Swelling of Food Items and High-density Foods

Items that do not swell on cooking are called high density food items, e.g., egg, potato, banana etc. It has to be borne in mind that pulses increase two times on cooking, rice increases three times and wheat increase four times on cooking. Rice, dhal etc., will also acquire a lot of moisture on cooking. Theoretically, 80 g of raw rice on cooking should increase three times to fill a cup, but in practice it will hold only rice cooked from 50 g. This is due to the extra moisture it acquires on cooking. So the calorie and protein content of 50 g raw rice is assigned to that of one cup of cooked rice, i.e., 4 g protein and 175 cal.

8. Spirulina

Spirulina, an algae, is now recommended as a safe food. It is a rich **biological source** of protein (65-70%), vitamins and minerals. It contains chlorophyll, xanthophyll and phytocyanin pigments and traces of cholesterol. It is a rich source of PUFA. It is now used as a food supplement or as a low volume, high value food. As high temperature cooking reduces its value, it may be taken with snacks or cold beverages. As it is a single-celled algae with no cellulose in the cell wall, it is digested and assimilated fast. Spirulina was used in ancient Africa and Mexico and was called Douhe ar tecuitlatl. It is used as a slimming agent in US. The presence of phenylalanine in it is said to signal the brain to stop hunger pangs leading to reduction in food intake. Spirulina is also used as a balm or anti-wrinkle cream. It contains a lactiferous galactogogue that increases breast milk. It reduces blood sugar and cures hepatitis and glaucoma. The phytocyanin, a blue pigment in spirulina, is also used in Japan as a safe bio-lipstick and eyeliner. The usual coal tar based cosmetics are carcinogenic. Spirulina also helps in skin metabolism, cell regeneration and skin secretion. It may also serve as a bio-pesticide and bio-fertilizer.

Even though it is used as a weight reducer and cosmetic in developed countries, it has been used to beat malnutrition in India. The protein conversion efficiency ratio is very high and 1 gram of it can supply double the vitamin A requirement of a child. Children like the taste of spirulina. It can be given mixed with food.

It can be grown in farms or even in a pot at home. *Spirulina platensis* or *fusiform* can be added to about 20 litres of the fertilizer, suphala, which contains urea, magnesium sulphate and sodium carbonate in a wide mouthed pot of about 35 litres size. The pot is then kept outdoors in the sun and is stirred occasionally. The algae multiplies using sunlight and the nutrients. It is harvested periodically after a week by running the medium through fine nylon cloth. The green paste is scraped and allowed to dry in the shade till it forms flakes. Spirulina is being grown in many villages in Tamil Nadu. The greatest advantage is that this biological molecule can replace several chemical supplements given to a growing child or a child with malnutrition.

9. Non-nutrient Components of Diet or Phytochemicals

The non-nutrient components of diet are now coming to limelight. These are certain **phytochemicals** that have antimutagenic properties. **Functional food** is that which promotes health and prevents diseases, over and above its nutritive value. These items contain phytonutrients or phytochemicals. These include herbs, spices, vegetables and fruits. Change in dietary patterns and lifestyle diseases like cardiovascular diseases, diabetes and cancer have caused a renewed interest in the medicinal properties of food. Consequently, **new food sciences have emerged, namely, nutraceuticals, medicinal foods, functional foods phytonutrients etc.**

Antioxidants like beta-carotene, vitamins E and C selenium etc., offer protection against cardiovascular diseases. Cancers are due to a series of mutational events and **antimutagenic agents** could be beneficial in cancer protection and treatment, (also refer Section 3.6)

Garlic, onion etc., that belong to the allium family contain the phytochemicals allium and allyl sulphide, that are antimutagenic. Boiling and frying do not alter these phytochemicals. **Curcumin** present in turmeric provides yellow colour, flavour and digestive properties. It is anticarcinogenic and also has anti-inflammatory, antifungal, antioxidant and wound healing properties. Cooking at high temperatures does not alter the properties. Antioxidant properties of fresh green vegetables, fruits and grains are decreased on cooking. The effect of food processing on antioxidant property is also being evaluated. Antioxidants scavenge oxidants and free radicals and protect against coronary artery diseases, stroke, cancer, diabetes, asthma etc. **Flavanoids, carotenoids and anthocyanins are antioxidants.** The other phytochemicals include detoxifying agents like indoles, isothiocyanates, phytosterols, non-starch polysaccharides (NSP) or dietary fibre like cellulose, hemicellulose, gums, mucilages, pectins and lignins and also alkaloids like caffeine, non-protein amino acids and plant sterols. The various phytochemicals and the sources are given below (Table 3.15).

Table 3.15 Sources of phytochemicals

Phytochemical	Source
Thiocyanates, indoles, lutein, xeaxanthin, sulforaphae, isothiocyanates	Cabbage, beet greens, arugula, broccoli, brussels sprouts, kale, mustard greens
Carotenes	Mangoes, carrots, pumpkin, apricots
Allium, allylsulphide	Onions, garlic leeks
Limonene	Lemons and citrus fruits
Lycopene	Tomatoes, water melons
Resveratrol, ellagic acid, cyanidin, quercetin	Strawberries, grapes, raspberries
Genistein, phytosterols, saponins, protease inhibitors	Soyabeans, mung beans, peanuts, dried beans

10. Digestion and Absorption

All the three major nutrients, carbohydrates, fats and proteins are split by hydrolysis using enzymes.

a. *Enzymes*

Salivary glands secrete saliva which contains ptyalin (alpha amylase), mucus and lingual lipase. Salivary secretion is regulated by nervous system.

In the stomach, gastric (oxyntic/parietal) glands secrete HC1, pepsinogen, intrinsic factor and mucus. Pyloric glands secrete mucus, gastrin and small quantities of pepsinogen. Pepsinogen is converted to the active form pepsin by HC1 and the already formed pepsin. **Keanin is the enzyme that acts upon milk in infants.** Gastric juice also contains small quantities of gastric lipase and gastric amylase. Gastric secretion is regulated by nervous and hormonal mechanisms. Gastrin is the major hormone that regulates secretion. Secretin and cholecystokinin (CCK) are the other gut hormones that are secreted by the small intestinal mucosa. Gastrin, secretin, and CCK regulate pancreatic secretion. Secretin regulates bicarbonate secretion and gastrin and CCK regulate the exocrine enzyme secretions. Pancreatic juice contains amylase, lipase and proteolytic enzymes like trypsin, chymotrypsin, carboxypolypeptidases, ribo- and deoxyribonucleases etc.

The proteolytic enzymes are secreted as proenzymes to prevent autodestruction and are later on converted to the active forms. Trypsinogen and chymotrypsinogen are activated to trypsin and chymotrypsin by enterokinase

which is secreted by intestinal mucosa. Liver, the largest gland in the body, secretes bile and CCK mediates contraction of gall bladder and release of bile. Bile acids, cholic acid and chenodeoxycholic acid are synthesized from cholesterol and they are conjugated with glycine and taurine into glyco- and tauro-conjugated bile acids. The salts of these acids secreted into the intestine have emulsifying or detergent function and help in micelle formation and fat absorption. Secretin increases bile secretion also. Small intestinal enzymes are intestinal peptidases, lipases and four disaccharidases, namely, lactase, sucrase, maltase and isomaltase (alpha dextrinase). Enterokinase secreted from intestinal mucosa activates trypsinogen to trypsin. The large intestine secretes a lot of mucus and absorbs water. It also plays a role in absorption of short chain fatty acids that are produced by fermentation of unabsorbed carbohydrate. These fatty acids are important source of energy in short bowel syndrome where absorption from the small intestine is reduced.

b. *Carbohydrate (CHO)*

CHO are mostly polysaccharides (starch) and disaccharides. Ptyalin (alpha amylase) in saliva, gastric amylase and pancreatic amylase hydrolyse starch into maltose and glucose polymers like maltotriose and dextrins. Disaccharides are split by the four disaccharidases, namely, lactase, sucrase, maltase and isomaltase (Fig. 3.2). Amylase from amylase rich food (ARF) also can play a role in digestion. 20% of the ingested carbohydrate passes to the large intestine. It is fermented to produce short chain fatty acids. Some of these fatty acids are absorbed and act as sources of energy.

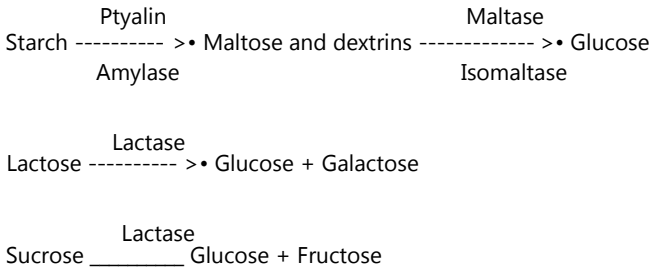


Fig. 3.2 Digestion of carbohydrate

The monosaccharides like glucose, galactose and fructose are easily absorbed. Arabinose, mannose, xylose etc., are the other monosaccharides.

c. *Fats (lipids)*

Neutral fats are triglycerides, three fatty acids with one glycerol. Lingual lipase.

human milk lipase and gastric lipase initiate hydrolysis. Emulsification of fat is mediated by bile salts. This is followed by complete hydrolysis by pancreatic lipases into fatty acids and monoglycerides (Fig. 3.3). After absorption into the epithelial cells, they are reconstituted into triglycerides and are incorporated into chylomicrons along with beta lipoproteins. Cholesterol and phospholipids are also incorporated into chylomicrons and are transported through lymph into thoracic duct. **Short chain fatty acids and medium chain triglycerides (MCT) are directly absorbed into the portal blood and hence can be absorbed even during fat malabsorption.**

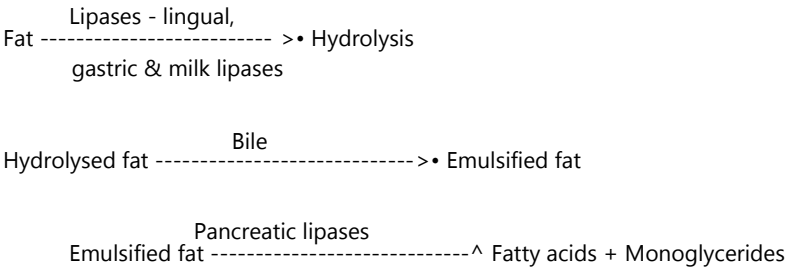


Fig. 3.3 Digestion of fat

d. *Proteins*

Proteins are long chains of amino acids bound together by peptide bonds. Proteins are hydrolysed into proteoses, peptones and polypeptides by pepsin and further hydrolysed by pancreatic proteolytic enzymes (trypsin, chymotrypsin, carboxypolypeptidase) and intestinal peptidases and are absorbed as amino acids (Fig. 3.4).

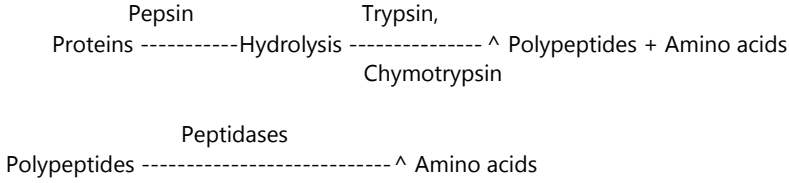


Fig. 3.4 Digestion of protein

Most of the nutrients are absorbed from jejunum and duodenum. Vitamin B_{12} is absorbed from ileum. Absorption occurs by diffusion and active transport.

RECOMMENDED DIETARY ALLOWANCES (RDA)

The RDA varies according to the reference standards and it is often a source of confusion to many. Different authors advise different amounts as the ideal intake.

1. Definition of RDA

RDA is not the same as the average requirement or the average minimum requirement. According to the American Food and Nutrition Board, **RDAs** are **levels of intake of essential nutrients that on the basis of scientific knowledge, are adequate to meet the known nutrient needs of all healthy persons.** Thus RDAs take into account a margin of safety, equal to two standard deviations (SD) for individual minimum requirement. Thus RDA covers the needs of over 99% of all individuals in a group and is prescribed at plus two standard deviations.

2. ICMR Recommendations

The RDAs as per the ICMR recommendation (1998) is given in Table 3.16. and 3.17. This is almost on par with the American recommendation (American National Academy of Sciences) and it appears to be too much for the average sized Indians. This is because the ICMR recommendation is fixed at +2 SD and not at the 50th centile. This is the ideal or maximum requirement and it is meant to cover the well nourished from high socioeconomic status including those who are little obese. This is widely accepted because we should not be guilty of keeping low recommendations for our children. Many clinicians still feel that the average sized child cannot consume so much with our present diet which is not calorie dense. Moreover, the people of Kerala who consume the lowest per capita calories among Indians are shown to have better nutritional status than the others. Hence, a search can be made for a more realistic RDA for the average child or at least for the minimum RDA. The ICMR RDA is considered the maximum or the ideal, but there should always be a range rather than a fixed number. For this, various other calculations are used. The latest view is to reduce the energy requirement and to increase the calcium requirement.

3. Coefficient of Calorie Rrequirement

The coefficient of calorie requirement (NIN, Hyderabad) is another simple method to calculate RDA. Assuming 2400 kcal as 1 unit of energy, the RDA is expressed as a proportion of this (Table 3.18). This is lower than the ICMR recommendations. 1 unit is the requirement of an adult sedentary male. It is also considered the requirement of an adolescent boy.

4. Holliday and Segar Formula

Another time tested method to calculate the RDA of calories is the Holliday and Segar formula. RDA is expressed **as that for the age of the child** and not for the present weight of the child. Hence, the expected weight for the age and not the

Table 3. **16** Recommended Dietary Allowances for some Micronutrients, 2009 (proposed)

Group	Activity/ age	Body wt. kg		Thiamine mg	Riboflavin mg	Niacin mg	Pyridoxine mg	Vitamin A ng Retinol	p-Carotene		Folic acid mg		B_{12} mg	Vitamin C Ascorbic acid mg	
		1989	2009	1989/	2009	1989/	2009	1989/2009	1989	2009	1989	2009	1989/2009	1989	2009
Adult reference man	Sedentary Moderate Heavy	60	60	1.2 1.4 1.6	1.4 1.6 1.9	16 18 21	2.0	600	2400	4800	100	200	1.0	40	50
Adult reference woman	Sedentary Moderate Heavy	50	50	0.9 1.1 1.2	1.1 1.3 1.5	12 14 16	2.0	600	2400	4800	100	200	1.0	40	50
Woman	Pregnant	50	50	+ 0.2	+ 0.2	+ 2	2.5	750	2400	6400	400	500	1.0	40	50
	Lact. (0-6 M)	50	50	+ 0.3	+ 0.3	+ 4	2.5				150	300	1.5	80	80
	Lact. (6-12 M)			+ 0.2	+ 0.2	+ 3	2.5	950	3800	7600	150	300		80	80

Infant	0-6 M	5.66	5.5	55 fig/kg	65 Hg/kg	0.71	0.1
	6-12 M	8.6	8.4	50	60 ng/kg	0.65 ng/kg	0.4
Children	1-3 years	12.2	12.4	0.6	0.7	8	0.9
	4-6 years	19.0	18.1	0.9	1.0	11	
	7-9 years	26.9	25.2	1.0	1.2	13	1.6
Boys	10-12 years	35.4	34.3	1.1	1.3	15	1.6
Girls	10-12 years	31.5	35.0	1.0	1.2	13	1.6
Boys	13-15 years	47.8	47.6	1.2	1.5	16	1.6
Girls	13-15 years	46.7	46.6	1.0	1.2	14	2.0
Boys	16-18 years	57.1	55.4	1.3	1.6	17	2.0
Girls	16-18 years	49.9	52.1	1.0	1.2	14	2.0

350	1400		25	25	0.2	25	25
	1400		25	25		25	25
400	1600	3200	30	80	0.2	40	40
400	1600	3200	40	100	1.0	40	40
600	2400	4800	60	120		40	40
600	2400	4800	70	140	1.0	40	50
600	2400	4800	70	140	1.0	40	50
600	2400	4800	100	160	1.0	40	50
600	2400	4800	100	160	1.0	40	50
600	2400	4800	100	200	1.0	40	50
600	2400	4800	100	200	1.0	40	50

Table 3.17 Recommended Dietary Allowances of Energy Protein and Minerals for Indians, 2009 (proposed)

Group	Activity age	Body wt. kg		Energy, Kcal		Protein, g		Fat, g		Calcium, mg		Fe, mg		zn
		1989	2009	1989	2009	1989	2009	1989	2009	1989	2009	1989	2009	20(
Adult Refe-rence man	Sedentary Moderate Heavy	60	60	2425 2875 3800	2318 2727 3475	60	60	20	15-20	400	600	28	17	12
Adult Reference Woman	Sedentary Moderate Heavy	50	50	1875 2225 2925	1899 2234 2854	50	55	20	15-20	400	600	30	21	10
Woman	Pregnant	50	50	+ 300	+ 365	+ 15	+ 23 82.2	30	12.5	1000	1200	38	38	12
	Lact. 0-6 M	50	50	+ 550	+ 600	+ 25	+ 16.5 77.9	45	17.5	1000	1200	30	16	12
	Lact. 6-12 M			+440	+ 520	+ 18	+ 11 70.2	45	17.5	1000	1200	30	16	12
Infant	0-6 M	5.4	5.5	108/ kg	563	2.05/ kg	1.16/kg	25.0				-	-	–
	6-12 M	8.6	8.4	98/kg	625	1.65/ kg	1.69/kg	25.0	25.0	500	500			-

contd.

Children	1-3 years	12.2	12.4	1240	1036	22
	4-6 years	19.0	18.1	1690	1350	30
	7-9 years	26.9	25.2	1950	1691	41
Boys	10-12 years	35.4	34.3	2190	2189	54
Girls	10-12 years	31.5	35.0	1970	2008	57
Boys	13-15 years	47.8	47.6	2450	2748	70
Girls	13-15 years	46.7	46.6	2060	2328	65
Boys	16-18 years	57.1	55.4	2640	3017	78
Girls	16-18 years	49.9	52.1	2060	2070	63

15.7	25	25.0	400	600	12	7	5
20.3			400	600	18	13	7
29.6		20.0	400	600	26	15	8
39.9	22	20.0	600	600	34	21	9
40.4	22	20.0	600	700	19	27	9
54.2		20.0	600	800	41	32	11
51.9		20.0	600	700	28	27	11
61.5	22	15.0	500	600	50	27	12
52.1	22	15.0	500	600	30	26	12

Table 3.18 Coefficient of Calorie Requirement

Age (years)	Co-Unit	Energy (kcal)
1-3	0.4	960
3-5	0.5	1200
5-7	0.6	1440
7-9	0.7	1680
9-12	0.8	1920
12-21	1.0	2400
Adult male		
Sedentary	1.0	2400
Moderate	1.2	2880
Heavy	1.6	3840
Adult female		
Sedentary	0.8	1920
Moderate	0.9	2160
Heavy	1.2	2880

Source: National Institute of Nutrition, Hyderabad

present weight is used for calculation of required calories. The same formula is often used for fluid calculation. For fluid requirement calculation, the present weight is often used instead of the expected weight. The formula and the requirement as per the expected weight for age are given in Table 3.19.

5. Bedside Calculation of Calorie Requirement

The 'bedside calculation of calorie requirement' is often followed by clinicians (Table 3.20). For a one-year-old child, the requirement is 1000 kcal, and for each completed year, 100 kcal, is added on till puberty. This is lower than the ICMR recommendations and can be regarded as the minimum requirement. Many of our children are not getting even the minimum requirement, leave alone the ideal or maximum requirement. It has to be borne in mind that the malnourished child requires more calories and protein for catch-up growth and to replenish the stores.

6. Bedside Approximation of Expected Weight and Height

The expected weight can be calculated using the Weech's formulas (refer Section 2.1). A bedside approximation is very helpful to derive the expected weight. **Birth weight doubles by 5 months and triples by 1 year** (i.e., 10 kg) **and quadruples by 2 years** (i.e., 12 kg). Thereafter, add 2 kg/each year among children up to 6 years of age and thereafter add 3 kg/each year till puberty. This is included in Table 2.1. Similarly expected height also can be easily derived. **At birth, length is 50 cm, it becomes 75 cm at one year and 87.5 cm at two years. Birth length**

Table 3.19 Holliday and Segar formula for calculation of calories

Holliday and Segar Formula

Up to 10 kg - 100 kcal/kg
10-20 kg - 1000 + 50 kcal for each kg above 10 kg
Above 20 kg - 1500 + 20 kcal/each kg in excess above 20 kg

Age (years)	Expected weight (kg)	Energy (kcal)
1	10	1000
2	12	1100
3	14	1200
4	16	1300
5	18	1400
6	21	1520
7	24	1580
8	27	1640
9	30	1700
10	33	1760
11	36	1820
12	39	1880

Table 3.20 Bedside calculation of calories

Age (years)	Energy (kcal)
1	1000
2	1100
3	1200
4	1300
5	1400
6	1500
7	1600
8	1700
9	1800
10	1900
11	2000
12	2100
Adolescent boy	2400
Adolescent girl	2100

doubles by 4 years and thereafter 6 cm/year is added on till puberty. **Birth length triples by 12 years.**

7. RDA of Vitamins and Minerals

The approximate bedside calculation of RDA for the various vitamins and minerals are given in Table 3.21 and the ICMR recommendations in Table 3.18. (Refer section 5.1 to 5.4)

Table 3.21 The approximate RDA of vitamins and minerals

1. Vit. A	1500 IU/day (500 |ag)
2. Vit. D	400 IU/day (10 ng)
3. Vit. E	5-15 IU/day (5-15 mg)
Vitamin B Complex	
4. B, (Thiamine)	0.5-1.5 mg/day (1 mg/1000 cal)
5. B_2 (Riboflavin)	0.5-1.5 mg/day
6. B_6 (Pyridoxine)	0.5-1.5 mg/day
7. B_3 (Niacin)	5-15 mg/day
8. B_n (Folic acid)	50-150 |ig/day
9. B_{12} (Cyanocobalamine)	0.5-1.5 |icg/day
10. Vit. C	40 mg/day
Macro elements	
11. Calcium	500-1000 mg/day
12. Phosphorus	800-1000 mg/day
13. Magnesium	200-300 mg/day
Trace elements	
14. Iron	10-20 mg/day
15. Iodine	50-150 |ig/day
16. Copper	1-2 mg/day
17. Zinc	5-15 mg/day
18. Fluoride	1-5 mg/day
19. Manganese	1-5 mg/day
20. Selenium	100 |ig/day
21. Molybdenum	200-500 ng/day
22. Chromium	10 ng/day

8. Balanced diet

The food items to be included in the balanced diet is given in Table 3.22 and Figure 3.1. Also refer Table 3.23.

Table 3.22 Balanced Diet for Infants, Children and Adolescents - ICMR 1998 (Number of portions)

Food groups	g/Portion	Infants	Years						
		6-12 months	1-3	4-6	7-9	10-12		13--18	
						Girls	Boys	Girls	Boys
Cereals & millets	30	1.5	4	7	9	9	11	10	14
Pulses	30	0.5	1	1.5	2	2	2	2	2
Milk (mL)	100	5[a]	5	5	5	5	5	5	5
Roots & tubers	100	0.5	0.5	1	1	1	1	1	1
Green leafy vegetables	100	0.25	0.5	0.5	1	1	1	1	1
Other vegetables	100	0.25	0.5	0.5	1	1	1	1	1
Fruits	100	1	1	1	1	1	1	1	1
Sugar	5	5	5	6	6	6	7	6	7
Fats/oils (visible)	5	2	4	5	5	5	5	5	5

[a] Quantity indicates top milk. For breastfed infants, 200 mL top milk is required.
One portion of pulse may be exchanged with one portion of egg/meat/chicken/fish.
For infants, introduce egg/meat/chicken/fish around 9 months
Specific recommendations as compared to a sedentary woman
Children:

1-6 years	1/2 to 3/4 the amount of cereals, pulses and vegetables and extra cup of milk.
7-12 years	Extra cup of milk.
Adolscent girls	Extra cup of milk.
Adolescent boys	Diet of sedentary man with extra cup of milk.

Table 3.13 Balanced diet for an adult/adolescent boy (2400 kcal)

Food item	Quantity	Kilocalories
Cereal	400 g	1400
Legumes	60 g	230
Roots & tubers	50 g	50
Vegetables	50 g	50
Green leafy vegetables	50 g	50
Fruits	50 g	50
Milk/Curd	250 mL	150
Oil/Fats	30 g	270
Sugar	30 g	120
Total		2370

3.4 Recent Concepts: Probiotics, Antioxidants and Rainbow Revolution

With the evolving nutrition concepts across the world, it is critical to understand certain terminologies which have gained momentum in health and disease over a period of time.

1. Probiotics, Prebiotics and Synbiotics

The idea that bacteria can affect health is not new. Much effort has been directed at eliminating bacteria, through antibiotics, to improve the health of children. A different approach to bacteria is gaining popularity roughly 100 years later. This approach uses the idea of "probiotics", a general term for nutritional supplements containing one or more cultures of living organisms (typically bacteria or yeast) that when introduced to a human or animal, have beneficial impact on the host by improving the endogenous microflora. Because of the difficulty in changing the intestinal or colonic microbiota permanently, successful colonisation with a probiotic is usually transient as the gastrointestinal tract has many defences that inhibit colonisation like the gastric acid, duodenal bile, mucin and the gut immune system. Successful probiotics are capable of resisting the insults and transiently take up residence in the gut. By definition, they also must be of human origin and have some health-promoting benefits proven by clinical trials.

Efforts are being directed at developing foods that promote the growth of probiotic strains of bacteria that can be administered alone, "prebiotics", or in combination with a probiotic bacteria as one concoction, "synbiotics".

a. The Gastrointestinal Ecosystem

At birth, the gastrointestinal tract is sterile; within hours, bacteria ingested during the birthing process rapidly colonise the gut. The GI tract soon contains about 10 times as many bacteria as there are cells in the body. Many of these species remain unidentified. It is this microflora which is responsible for priming the GI immune system. After this initial colonisation, a person's individual gut flora remains remarkably constant throughout life. The gut immune system learns to recognize and tolerate these bacteria acquired during early infancy. Consequently, it is very difficult to permanently change this flora after this time. Constant interaction occurs between this endogenous flora and potentially pathogenic microorganisms. Even under optimal conditions, bacteria, viruses and toxins penetrate the intestinal barrier and invade the adjacent parts of the body. The integrity of the barrier can be altered dramatically by changes in the secreted antibodies, the mucous layer and the immune systems and through direct and indirect interaction of the other organisms. The endogenous flora plays a major role in the control of the each of these aspects of the mucosal barrier. In addition, changes in this microflora may affect the distal immune system and therefore can have effects beyond the GI tract.

b. Probiotics in Health

Health and well-being depend on complex and dynamic interplay between factors that control vital processes such as appetite, energy balance, metabolic rate and stress response. Lifestyle and eating habits are in part responsible for each person's overall health status.

Perhaps an equally important factor that has been overlooked is the intake of probiotic bacteria. Traditionally, fermentation was used as a method of preserving foods. Ingestion of these foods would expose the host to prebiotic organisms; the same or similar to these are being used today. However, the western diet contains dramatically decreased number of fermented foods exposing the host to as few as one millionth of the organisms to which our ancestors were exposed. It is not surprising that astronauts, who receive a diet low in fibre and antioxidants, return to earth with significantly decreased count of endogenous probiotic bacteria. It may not be a coincidence that increases in inflammatory conditions, allergic disorders, obesity, heart disease and cancers have paralleled the decreasing content of probiotics in the western diet.

c. Probiotics in Intestinal Disease

It would seem logical that most health-promoting benefits of probiotic organisms

have been studied in the context of various GI disorders. Most clinical studies have involved either the treatment or prevention of GI disorders, especially acute diarrhoea in children, antibiotics associated diarrhoea and *Clostridium difficile* associated diarrhoea.

Infectious diarrhoea: Probiotics have been shown to be successful in the treatment or prevention of various types of diarrhoea including rotavirus, *Clostridium difficile* and traveler's diarrhoea.

Saavedra et al published a landmark study in the prevention of infectious diarrhoea using children < 2 years admitted to a chronic care facility; a standard infant formula was supplemented with 2 strains of probiotic bacteria *(Bifidobacterium bifidus* and *Streptococcus thermophilus*). Children were randomized to receive this or the standard formula and were followed up for the development of diarrhoea and rotavirus shedding. Not only did the probiotic group develop lower rate of diarrhoea (7% versus 31%) but also rotavirus shedding was decreased substantially in the probiotic group (10% versus 39%). A Peruvian study evaluated probiotics (lactobacillus GG) in a high-risk population (undernourished children < 6 years) and showed statically fewer episodes of diarrhoea. Breast-fed children did not seem to benefit at all, raising the possibility that probiotics provide a similar action as breast milk in the prevention of infection. However, the effect of probiotic use on shedding of a specific pathogen is most interesting. This effect of bacterial supplementation to prevent infection with a viral agent seems to show an immune modulatory effect of probiotics, beyond simple barrier protection. This theory is further supported in a study by Isolauri et al., who demonstrated that the duration of hospitalization from rotavirus diarrhoea was significantly shorter in children who were given oral rehydration and LGG (as fermented milk product or freeze dried powder) than in a similar group who received ORS and placebo. Another similar study from Majamaa et al., showed children receiving LGG had a decreased duration of diarrhoea and also significant increase in the number of cells secreting IgA against rotavirus (immune modulating effect on the host by probiotics).

Research may reveal that not only LGG but other organisms such as *Lactobacillus plantarum* may have higher efficacy for certain diseases.

Clostridium difficile **diarrhoea:** Infection and symptomatic diarrhoea with C. *difficile* is often precipitated by antibiotic use, which disrupts the endogeneous flora that suppress the growth of C. *difficile* under normal conditions. Even though infection with C. *difficile* usually responds well to treatment with oral antibiotics such as metronidazole and vancomycin, relapses are common and difficult to treat. Another biotherapeutic agent *Saccharomyces boulardii* so-called because a live organism though not human derived, is an yeast that has been used in prevention of relapsing *C.difficile* diarrhoea. Colonization was not reduced whereas toxin production was. Lactobacillus GG has also been shown to

reduce the risk of antibiotic associative diarrhoea by nearly 75% in children in studies in USA and Finland.

Traveller's diarrhoea: As evident in some of the studies of infectious diarrhoea, probiotics seem to prevent viral-induced diarrhoea better than bacterial diarrhoea. This may be the reason that probiotics have yet to be shown conclusively to decrease the incidence of traveller's diarrhoea. *S. boulardii* seems to have stronger effects on bacterial diarrhoea. Whereas LGG has been shown to be more effective against viral and idiopathic diarrhoea.

Inflammatory bowel disease: An increasing body of clinical data supports a role for probiotics in the treatment of IBD. These diseases are multifactorial in causation with many of the proposed factors in the development of IBD related to the endogenous microflora.

A probiotic consisting of a non-pathogenic strain of *Escherichia coli* and mezalazine has shown effectiveness in induction and maintenance of remission in ulcerative colitis. Similarly, a probiotic preparation consisting of 3 strains of bifidobacteria, 4 strains of lactobacilli and 1 strain of *Streptococcus* has been used to maintain remission in patients with ulcerative colitis. The same combination was shown to have efficacy in preventing pouchitis in ulcerative colitis after colectomy.

Regarding Crohn's disease, oral therapy with LGG resulted in increased number of IgA-secreting cells in children providing a means for reducing antigenic stimulation—all these have some beneficial effect on reducing intestinal permeability and disease activity. Especially interesting is the recent development of so-called "turbo-probiotics", which are genetically engineered to deliver active compounds. An example is the probiotic bacteria *Lactococcus lactis* that was engineered to secrete IL-10, an anti-inflammatory cytokine currently undergoing study in its activity to treat acute Crohn's disease.

Helicobacter pylori: Probiotic bacteria may be antagonistic to *Helicobacter pylori. Lactobacillus salivarius* has been shown to inhibit the attachment of *H. pylori* in vitro. In clinical practice, an open labeled, randomized trial of triple therapy with or without the addition of *L. acidophilus* was conducted in 120 patients with *H. Pylori.* Eradication rates were higher (87% versus 70%) in the group supplemented with the probiotic.

d. Allergy

A striking example of probiotic effects outside the intestine is their ability to reduce the symptoms of atopic dermatitis. An RCT among exclusively breast-fed children with eczema showed that the group receiving probiotic supplementation had significant to complete resolution of eczema as compared to controls. Explanation as to why this difference occurred may be increased production of the anti-inflammatory cytokine IL-10.

Another example of the potential effects of probiotics on atopic disease comes from the study by Kalliomaki et al. In their study, pregnant women with a family history of atopic disease (allergic rhinitis, eczema or asthma) were randomized to receive LGG or placebo beginning a month before their expected date of delivery and for 6 months after delivery. The probiotic was supplied to the infant in two ways:

1. Indirectly through the breastfeeding mother who took the probiotic orally
2. Directly from a spoon

At 24 months, the rate of atopic eczema was reduced by 50% in children who received the supplementation with LGG indirectly or directly.

The timing of administration might be the crucial aspect of probiotics. It has been shown that a lower ratio of "good" to "bad" bacteria early in life precedes the development of atopic disease.

Recent studies from Finland have shown a reduced incidence of milk allergy in toddler given LGG during early infancy. Likewise, treatment of milk allergic toddler with LGG seems to ameliorate both the extent and severity of allergic eczema.

e. Immune Regulation

It appears that at least some probiotics may be capable of both down regulating the allergic response and in enhancing immune response against potential pathogens. LGG has been shown to increase antibody response to rotavirus infection and to its vaccine. It has also been shown to enhance antibody response in adults given typhoid vaccine. A potential beneficial effect of the immune enhancement has been seen in a study that children with cystic fibrosis treated with LGG had a reduced incidence of severe respiratory infections when matched with a group of placebo-treated controls.

A Finnish study also suggests that probiotics may be useful in preventing respiratory infection that is at a mucosal site not in direct contact with the site of colonisation by the probiotic.

f. Probiotics in Critical Care

The incidence of multiorgan failure in the ICU is high, with infection playing a prominent role. These patients are often treated with broad-spectrum antibiotics to help reduce acquired infections. It is ironic that although the overall incidence of infections has reduced, the overall mortality rate has not.

An underlying explanation for this phenomenon may be related to the type of nutrition that critically ill patients receive and its effects on the intestinal ecosystem.

Recent studies have suggested that enteral nutrition may be comparable with parenteral nutrition in terms of delivery of nutrients but it is far better at mediating the acute phase response. Enteral nutrition has been shown to en-

hance natural killer cell activity, lymphocyte number and response to antigenic stimulus.

This aspect of nutrition may be in part caused by "prebiotic effect"—the delivery of nutrients in the form of non-digestable compounds to the endogenous flora. They serve as substrates for the above flora providing important nutrients such as short chain fatty acids, polyunsaturated fatty acids, amino acids, polyamines, vitamins, antioxidants, growth factors and coagulable factors to the ecosystem. Parenteral nutrition circumvents this process, may cause atrophy of the mucosa and clonic microflora. As a result, patients may be more susceptible to infection because of microbial translocation and systemic response that follows.

g. Commonly Studied Probiotics

Lactobacillus acidophilus
Lactobacillus GG
Lactobacillus reuteri
L plantarum 299 V
Bifidobacteria bifidus and *longum*
Streptococcus thermophilus
Saccharomyces boulardii (yeast)

Side Effects

Probiotics are by and large considered safe. Disseminated fungaemia has been seen with the administration of *S. boulardii* which was treated successfully with antifungal therapy.

h. Prebiotics and Synbiotics

Prebiotics refers to foods that promote the growth of probiotic strains of bacteria. These compounds are neither hydrolysed nor absorbed from the GI tract and usually are fermented by the beneficial bacteria in the colon.

Food ingredients most likely to meet these criteria are fructans (inulin and oligofructans); soya bean oligosaccharides, galacto oligosaccharides (present in breast milk), lactulose and lactitol. These low molecular carbohydrates occur naturally in onion, chicory, garlic, tomatoes, wheat and bananas. They are fermented by the *Bifidobacteria* in the colon resulting in production of short chain fatty acids that are subsequently absorbed. Short chain fatty acids are the primary fuel for the colon and promote the absorption of sodium and water. By stimulating *Bifidobacteria* they may have protective effects against colorectal cancer and infectious colitis, improve lipid metabolism, exhibit beneficial effects of fibre and increase the bioavailability of minerals, improve host defences and have a low carcinogenic potential. It has been suggested that oligosaccharides may be beneficial in preventing NEC in preterm babies.

Another potential use of prebiotics is in combination with probiotic bacteria. This "synbiotic" may offer advantages in the survival of the probiotic bacteria. Some potential synbiotic combinations include the following: Bifidobactrium with fructo oligosaccharides (FOS), Lactobacilli with lactitol.

Conclusion

Probiotics are truly a timeless concept. Initially standard components of the human diet, potentially beneficial bacteria and yeast have been eliminated through modem methods of preparing food. Although the concept is not new, the science of probiotics and prebiotics is in its infancy. When used appropriately, they are a beneficial adjunct to proven therapies, have the benefit of providing a stabilizing influence on the delicate balance in the ecosystem that consists of man and his flora.

2. Type I & Type II Nutrients

The nutrients in a food item may be regulated and highlighted by a claim by the manufacturer or may be naturally occurring.

a. *Type I Nutrients*

When a nutrient is subject to a regulatory minimum (e.g., vitamin C added to a flavoured drink) or a regulatory maximum (e.g., low fat), the total amount highlighted by claim including any amount due to natural occurrence is called 'Type I Nutrient'. Minimum 5 samples, 250 g each should be analysed and the mean result should be reported. Less than 90% of the declared value or less than 30% in any one sample with respect to minimum and more than 110% of the declared value or more than 170% in any one sample with respect to maximum claimed will be deemed as out of compliance.

b. *Type II Nutrients*

A nutrient which is present in a food and for which there is no regulated minimum or maximum is called 'Type II Nutrient' (e.g., vitamin C in fresh orange juice). At least 12 samples should be analysed and the test result should be at least equal to 80% of the value declared on the label.

3. Rainbow Revolution

After green revolution for cereals, white revolution for milk, blue revolution for fish, rainbow revolution is now in limelight to cultivate and consume coloured (violet, indigo, blue, green, yellow, orange and red, especiallly 'GYOR') vegetables and fruits for micronutrients and antioxidants. Providing variety of fruits and vegetables with variation in colour surely ensures the provision of adequate micronutrients and antioxidants as well. The liberal utilization of coloured vegetables and fruits in the diet ensures the adequacy of essential vitamins, minerals and antioxidants required to prevent the hidden deficiencies. Green farming is yet another intervention with reduced use of chemical fertilizers and pesticides.

4. Free Radicals and Antioxidants

In the current day scenario, the most sold concept is the antioxidants and forms an integral part of the anti-ageing therapies. Nonetheless, antioxidants play a very critical role in the nutritional status of a child.

Free oxygen radicals and reactive oxygen species (ROS) play a role in aging and various disease process and antioxidants act as protective scavengers.

a. Types of Free Radicals

We are exposed to environmental electromagnetic radiation, both natural (cosmic radiation) as well as from man-made sources. If this radiation is of the low wavelength types, like gamma rays, it can split water in the human body to generate hydroxyl radicals.

H_2O=OH-+H+

This hydroxyl radical is designated as the **free oxygen radical.** Gomberg first discovered free radicals in 1900. In the early part of the 20th century, biochemists attributed the rancidity of stored and packed foods like pork, fish, butter, etc., to free oxygen radicals liberated by contaminating microorganisms or alternatively by gamma radiation used for sterilization. It was only as late as 1968 when an antioxidant enzyme was discovered in the body, superoxide dismutase, that the role of free radicals was speculated in the human body. Free oxygen radicals are implicated in over 100 human diseases. They are particularly causative for carcinogenesis, arterial ischaemia, inflammatory, degenerative and neoplastic diseases and aging itself.

In health, over 98% of the oxygen consumed by a cell is converted to form water molecules. This is accomplished by the enzyme cyctochrome oxidase system located in the mitochondria. The remaining 1-2% of the unutilized oxygen escapes this pathway to form free oxygen radicals.

A free radical simply joins on by addition to another non-radical to render the latter to a free radical. This process proceeds serially to form more and more free radicals leading to a progressive increase in tissue damage. Alternatively, when two free radicals meet, the above chain reaction gets terminated.

Oxygen-derived free radicals have a life span of only a few micro seconds. Oxidation of carbohydrates and fats commonly occurs for releasing energy. During oxidation, there is loss of electron(s) from an atom or a molecule. This generates free oxygen radicals. Oxygen is usually involved in the process of oxidation. Hence, most of the radicals freely formed are free oxygen radicals.

As long as oxygen remains in the body as dioxygen (0_2) its behaviour is easily controllable. However, once free oxygen radicals are formed, cell membranes are attacked and weakened to result in cell damage or death. This can occur to any cell of the body.

Superoxide radical is relatively inactive, but may interact with metals such as iron to produce the highly reactive and damaging hydroxyl radical. **Hydroxyl radical** is the most reactive oxygen-free radical. It primarily attacks lipids in the cell membrane and this is called lipid peroxidiation.

Reactive oxygen species (ROS): Include superoxide free radical, hydrogen peroxide and hydroxyl free radical. Of these, hydroxyl free radical is toxic and short-lived. A **free radical is defined as any species capable of independent existence and contains one or more unpaired electrons.** An unpaired electron is one that remains alone in an orbital. If an atom consists of such an unpaired electron, it is called a free radical. Its consequent tendency to acquire an electron from other substances makes it highly reactive.

When oxygen is reduced to water in the mitochondria, catalyzed by cytochrome oxidase, four electrons are acquired. Electrons may be gained one at a time, by reduction. The individual molecules in reduction are highly reactive and potentially damaging to tissues.

Other sources of reactive species are xanthine oxidase, which produces superoxide; cyclo-oxygenase, which produces hydroxyl and peroxyl radicals and nitric oxide radical produced in vascular endothelium and the macrophages.

ROS are partially reduced. Examples are: Hydrogen peroxide (H_2O_z), hydroperoxy radical (HOO), hypochlorous acid radical (HOCI). Under certain conditions, reactive oxygen species have potential to enter free radical reactions to form the more toxic free radicals. Another reactive oxygen species, which is not a free radical, is **singlet oxygen (O').** In this, a rearrangment of electrons has occurred, which allows it to react faster with biological molecules as compared to normal oxygen.

b. Source of Free Radicals

Free radicals are generated exogenously by drugs, chemicals, pesticides, industrial pollutants, tobacco smoke, sun light and ionizing radiation. Endogenously free radicals are being constantly formed in lysosomes, peroxisomes, nuclear endoplasmic reticulum, plasma membrane and cytosol.

c. *Role of Free Radicals*

1) Biological role

i) Modulation of inflammatory process by regulating prostaglandin synthesis

ii) Checking infection by killing phagocytosed bacteria

iii) Maintaining vascular tone by controlling production of endothelium-derived relaxing factors

iv) Assisting in the detoxification process

2) **Oxidative stress:** In the body, generation of oxygen-derived species and the level of antioxidant defence systems are approximately balanced; hence, it is

easy to shift the balance in favour of the oxygen-derived species and upset cell biochemistry. Any imbalance is called oxidative stress. Most cells can tolerate a mild degree of oxidative stress because they have repair systems, which recognize and remove oxidatively damaged molecules, which are then replaced. In addition, cells may increase the antioxidant defences in response to the stress.

3) **Free radical-induced diseases:** During phagocytosis, which is an important and primary defence mechanism of the body, there is liberation of superoxide radicals. This is normally kept in check by the body's natural enzyme defences. However, if hydrogen peroxide is concomitantly present, the superoxide radical separates to form the hydroxyl radical. The latter can stay only for a few microseconds but can damage cell membranes as well as deoxyribonucleic acid (DNA).

   i) **Diseases by attacking cell membrane**
   Lipid peroxidation occurs and by a chain reaction, plenty of peroxyl and lipid radicals are produced. This results in loss of fluidity and structural integrity without consequent loss of secretory, enzymatic cellular recognition and other membrane functions. Cell lysis ultimately occurs and the degradation products cause increased vascular permeability, oedema and inflammation. Peroxidation is implicated in diseases like retinopathy of prematurity (ROP), adult respiratory distress syndrome (ARDS) and rheumatoid arthritis.

   ii) **Diseases by attacking DNA:** Hydroxyl radical causes chemical adulteration of DNA. Imperfect repair of such damage is activation of oncogenes and thereby carcinogenesis. It is presumed that the DNA in each human cell receives about 10,000 oxidative 'hits'per day. Much of this damage is repaired, but not all.

   At moderately high concentration, ROS such as hydrogen peroxide act as signal transduction messengers. The mechanism can be understood by knowing the regulation of gene expression by antioxidants. Two well-defined transcription factors, nuclear factor (NFkB) and activator protein (AP)1, have been identified to be regulated by intracellular redox state. The regulation of gene expression by oxidants, antioxidants and redox state has emerged as a novel sub-discipline in molecular biology that has promising therapeutic implications. Binding sites of redox regulated transcription factors are located in the promoter region of a large variety of genes that are directly involved in the pathogenesis of diseases like AIDS, cancer, atherosclerosis and diabetic complications.

   iii) **Diseases** by **the attack of proteins:** Free radicals can fragment cross-links or aggregate proteins. The consequences include interference with ion channels, failure of cell receptors and failure of oxidative phosphorylation.

d. Antioxidants

Antioxidants can be defined as the substances whose presence in relatively low concentrations significantly inhibits the rate of oxidation of the targets. Over the years, the human body has developed strategies to protect it from the uncontrolled side effects of free radicals with antioxidative substances and enzyme systems.

Nutricines are components of food which are considered for their direct contribution to nutrition. Examples include antioxidants, non-digestible carbohydrates, natural acids, enzymes and lecithins. Nutricines provide the crucial link between health and nutrition.

The normal cell controls or prevents the adverse effects of free radicals by:

- Physically separating the free oxygen radicals from the susceptible molecules of the human body
- Providing molecules that effectively compete for oxygen
- Rapidly repair the damage caused by free radicals
- Lysing and inactivating free radicals by removing damaged molecules

Classification of Antioxidants

These are classified in various ways (Table 3.24)

Table 3.24 Classification of antioxidants

Intracellular antioxidants	Superoxide dismutase, catalase, glutathione peroxidase
Extracellular antioxidants	Transferrin, haptoglobin, albumin, extracellular superoxide dismutase and catalase, bilirubin, mucus, glucose, vitamin C, urate
Lipoprotein antioxidants	Vitamin E, beta-carotene, retinyl stearate, lycopene
Membrane antioxidants	Vitamin E, beta-carotene, coenyzme Q
Nutritional antioxidants	Vitamin E (tocopherols and tocotrienols), beta carotene, vitamin C, phytochemicals like flavanoids, flavones, flavanols, cinnamic acid, cumarin derivatives, phytoalexane derivatives, selenium as a co-factor for glutathione peroxidase, cysteine, taurine
Synthetic antioxidants	Selenium compound like glutathione peroxidase, food additives like propyl gallate, butylated hydroxyanisole, antioxidant drugs like allopurinol, desferrioxamine, N-acetyl cysteine

1. Depending upon their category
   a) Enzymes
      i) Natural: Physiological
         - Superoxide dismutase
         - Glutathione peroxidase
         - Catalase
      ii) Synthetic
         - A selenium compound, which mimics glutathoine peroxidase
         - Food additives such as propyl gallate, butylated hydroxy anisole (BHA) and butylated hydroxytoluene (BHT)
   b) Vitamins
      - Beta-carotene/vitamin A
      - Vitamin E
      - Vitamin C
2. Depending on mechanism or stage at which they act
   a) Preventive antioxidants
      i) Natural: Physiological—a variety of enzymes function as antioxidants, catalase, glutathione peroxidase. These prevent the hydroperoxides from getting free radicals.
      ii) Synthetic: Metal ions like iron contain unpaired electrons. Desferrioxamine is a powerful synthetic iron chelator that strongly inhibits iron-dependent lipid peroxidation.
   b) Chain breaking antioxidants
      i) Natural: Physiological—these have the ability to trap the peroxy radicals already liberated, e.g.: superoxide dismutase, vitamin A, vitamin C.
      ii) Synthetic: Many drugs designed for other purposes have antioxidant activity, e.g.: Probucol can circulate in the plasma lipoproteins and reduce their susceptibility to oxygen.
   c) Miscellaneous antioxidants. These act by different mechanisms.

Beta-carotene: Vitamin A is a unique antioxidant. It protects by trapping free radicals. Haptoglobins, haemopexin, transferrin, caeruloplasmin, urate, coenzyme Q etc., are some of the biologically important antioxidants. **Three antioxidant enzymes,** superoxide dismutase, catalases, glutathione peroxidase and **three antioxidant vitamins,** vitamin A/beta-carotene, vitamin E, vitamin C, constitute the antioxidant system in the body.

Superoxide dismutase has a relatively short life, 6-10 minutes, which limits its application as a therapeutic agent.

Glutathione peroxidase is of major importance in the defense system of cells. Glutathione is present intracellularly in the cells and takes part in the antioxidant functions. **Glutathione is a selenium-dependent enzyme** and levels of this element below 100 mg/L in the body results in the sub-optimal levels of this natural antioxidant.

Vitamin A and E function in lipid phase and vitamin C acts in the aqueous phase. These vitamins provide major antioxidant benefits.

5. Smart Nutrients, Super Nutrients

It has long been suspected that the relative abundance of specific nutrients can affect cognitive processes and emotions. Newly described influences of dietary factors on neuronal function and synaptic plasticity have revealed some of the vital mechanisms that are responsible for the action of diet on brain health and mental function. Several gut hormones that can enter the brain, or that are produced in the brain itself, influence cognitive ability. The brain friendly nutrients are referred to as **'smart nutrients'.** Out of the several smart nutrients four are currently called **'super nutrients'**. These are vitamin A, iron, folic acid and omega-3 fatty acids (Fig 4.15). Dietary supplementation is being promoted. Some work is also going on with respect to mega vitamin and mega mineral therapies.

6. Superfoods & Super Antioxidant Foods

When it comes to superfoods, the most basic ones are the super greens or green foods. These include spirulina, chlorella and blue-green algae. They are the best superfoods to start with if you are just beginning to add these health wonders to your world.

What are Super Greens?

The green foods are most well known for their superior concentrations of chlorophyll. Super greens are sources of algae. They grow in the water. They are the sole foods of many marine species. The three green foods that are considered to be superfoods are chlorella, spirulina and blue-green algae.

Spirulina Pacifica: This super green food is incredibly rich in iron, beta carotene, calcium, protein, phytonutrients, enzymes and antioxidants. This is such a potently nutritious food that it is listed as one of the top ten superfoods in the world by highly acclaimed nutritionist and superfood guru, David Wolfe. According to Wolfe, spirulina is so nutrient rich that some say a person could live on spirulina alone for quite some time.

This form of algae is close to 70% protein. No other food contains this much protein. It is a complete protein as it contains all essential amino acids and amino acids. Because of its high concentration of chlorophyll it is a potent blood detoxifier and builder.

Klamath Lake Blue-Green Algae: This particular type of algae is known as the superfood for the brain. It is a potent antiinflamatory agent as well as an immune enhancer. It also has strong antiviral components. Blue-green algae is high in chlorophyll as well which makes it an important blood purifier. It is high in beta carotene, B complex vitamins, fatty acids, enzymes, essential amino acids

and nucleic acids which are essential for growth and repair. The essential amino acid make-up of blue-green algae is almost identical to that of human blood.

Chlorella: According to David Wolfe, chlorella supports the function of the brain and liver, improves digestion and elimination, helps regenerate the body, detoxifies the blood, protects against radiation, relieves inflammation, supports healthy weight loss, enhances immur.c function and accelerates the healing process. If that is not enough to convince you to use this super food, you might consider taking it to improve your decision making abilities.

7. Truly Zero Calorie Foods

Water is the only true zero calorie food that occurs naturally. Water is a great addition to your diet in a variety of ways. Eight 12-ounce servings of water every day are suggested as a healthy part of any diet. However, it has an added benefit when you're trying to diet to lose weight: it helps fill you up!

Water can be frozen into ice and crushed or shaved in order to add a different consistency to your diet.

In addition to water, diet soda pop is found in zero-calorie versions. It comes in name brands and generics and is typically sweetened with artificial sweeteners. Quite a bit of debate surrounds these beverages and whether they are healthy additions to your diet because of the artificial sweetener aspect. Experts offer conflicting claims that artificial sweeteners cause health problems or contribute to obesity.

8. Negative Calorie Foods

There are several fruits and vegetables that occur in nature that while they have calories when consumed, actually result in negative calories to the body after digested. It is because these foods cause the body to extend more calories to digest them than they have through their nutritional content that they become zero calorie foods.

These foods can basically be consumed in whatever quantity you want without feeling guilty. They are healthy and provide nutritional content your body needs. Keep in mind, however, that these foods only have this negative calorie effect when consumed without extras added to them like butter, sugar or dips.

Table 3.25 Negative calorie foods

Asparagus
Apple
Beet
Cranberries
Broccoli
Grapefruit

Cabbage
Lemon
Carrot
Mango
Cauliflower
Orange
Celery
Pineapple
Chile peppers
Raspberries
Cucumber
Strawberries
Dandelion
Tangerine
Endive
Garden cress
Garlic
Green beans
Lettuce
Onion
Papaya
Radishes
Spinach
Turnip
Zucchini

9. Super Antioxidant Foods

There's more than one way to tackle those damaging free radicals, reports Jacqui Ripley. The seven super antioxidant foods are:

Cooked Tomatoes

"The bright red colour of tomatoes is supplied by a photochemical called lycopene, which is in the same family of carotenoids (natural fat-soluble pigments) as the orange-coloured betacarotene in carrots," says nutrition consultant Lorraine Perretta (myvitality.com). Research shows that lycopene can be absorbed more efficiently by the body if processed into ketchup, juice, sauce and paste.

A proven antioxidant, lycopene appears to be different from other carotenoids because its concentration in body tissue tends to be higher. Lycopene is deposited in the liver, lungs, prostate gland, skin and colon. Research has suggested that frequent consumption of tomato products or lycopene may be associated with a lower risk of prostate cancer. A study of more than 47,000 men, conducted by Harvard Medical School, concluded that those who ate tomato sauce or other types of cooked tomatoes two or more times a week had a 20% less chance of developing prostate cancer.

Turmeric

Curcumin is a compound found in turmeric. The Alzheimer's Society reports that research from the University of California in Los Angeles has suggested the idea that curcumin might play a role in slowing the progression of this disease. Mice were injected with amyloid proteins to create conditions similar to those in patients suffering from dementia. They suggested this key ingredient of curry helped to clear amyloid from the brain and that its relative lack of side effects and combination of anti-inflammatory and antioxidant properties could be beneficial. “It’s too soon to say curcumin, or drugs based on it, could be potential preventive treatments,” says Clive Ballard, director of research for the Alzheimer’s Society, “but if you like spicy Indian food, enjoy it.”

Blueberries

“New research has declared red, purple and blue fruits - blackberries, blueberries, cranberries, black grapes included - to be the anti-ageing food of the 21st century,” says Perretta. "The active ingredient is a plant chemical group called anthocyanidins, which are powerful antioxidants.” Anthocyanidins have been found to prevent collagen from breaking down - the elastic protein in skin, joints, and veins and arteries that carry nutrients to the brain. They are thought to be 50 times more powerful than vitamin E.

“They are robust nutrients and survive various food processes, so when fresh berries are not available, canned and frozen berries are nutritious alternatives,” says Perretta.

Wheatgrass

Chlorophyll, the substance that makes green plants green, is seen as a useful blood tonic. “Foods such as wheatgrass, algae, seaweeds and green vegetables help to ‘build’ the blood,” explains Patrick Holford in his book *The Optimum Nutrition Bible* (Piatkus). “Research has shown that components of chlorophyll found in foods, when taken in very small purified amounts, may stimulate the production of red blood cells in the bone marrow.” Rich in enzymes, vitamins, minerals and trace elements, this highly nutritious substance also contains useful detoxification and cleansing properties.

Broccoli Sprouts

“Glucosinolates are phytochemicals that were once thought to be toxic to humans and act as natural pesticides,” says nutrition expert Judith Wills and author of *The Food Bible* (Quadrille). “They are found mainly in cruciferous and green vegetables - cabbage, brussels sprouts, kale and cauliflower - where the stronger the taste, the higher the potency of the chemicals. Broccoli is a particularly rich source of glucosinolates which breaks down into a substance called

sulphoraphane that appears to have a strong anti-cancer effect by stimulating our natural defences."

Sulphoraphane is a compound that was recognised in broccoli spouts by researchers at the Johns Hopkins University School of Medicine in Baltimore and the French National Scientific Research Centre. They discovered it kills the bacterium *Helicobacter pylori,* the bug widely thought to be responsible for the majority of cases of stomach ulcers and stomach cancer.

Grapefruit

"Bioflavonoids act as potent antioxidants which can bind to toxic metals and escort them out of the body," says Holford. "They have a synergistic effect on vitamin C, stabilising it in human tissue. Furthermore, they have an antibiotic effect which accounts for their anti-infection properties and are also anti-carcinogenic." Bioflavonoids usually appear to be most powerful in fruit, probably because the sugars help the flavonoids to be absorbed. Taxifolin and rutin are two important flavonoids found in citrus fruit, including grapefruit. "Many years ago, bioflavonoids were classed as vitamin P and then more or less dismissed as of no significance," says Wills. "Now we know better."

Onions

"There is a subgroup of flavonoids called flavonols, one of which - the most researched, and probably the most abundant in foods - is quercetin," says Wills. Found not only in the skins of onions but also in apples, black tea and red wine, Wills says a high quercetin intake has been linked with a lower risk of coronary heart disease and may also help to prevent cataracts. This antioxidant also boosts antihistamine properties which may help to relieve allergic and asthma symptoms.

10. Genonutrients

Genonutrients are substances (nutrients) found in foods and plants that have the ability to affect gene "expression." An example could be something as simple as antioxidants, like vitamin C, or polyphenols found in red wine (also refer Section 9.1).

SECTION 4

Triple Burden of Malnutrition

"If you eat wrongly, no doctor can cure; and if you eat rightly, no doctor is needed." —*Victor G Rocine (1930)*

4.1 Undernutrition and Severe Acute Malnutrition (SAM)

Majority of the children in India who live below the poverty line in an environment of multideprivation and starvation have physical and developmental retardation. It has been estimated that in India, **65 percent, i.e., nearly 80 million children under five years of age suffer from varying degrees of malnutrition.**

Sociodemographic factors like neglect of the girl child, large family size and lack of child spacing and family welfare methods (unplanned maternity) have an adverse effect on child survival and child development.

Environmental factors like parental education, socioeconomic status, sanitation, standard of living, parental attitudes and child rearing practices influence the growth and development of children.

Nutritional factors like improper breastfeeding practices, weaning practices and diet during illness influence the growth and development of children. Maternal malnutrition, low birth weight (LBW) and recurrent infections are other important factors that lead to malnutrition. **Malnutrition is often found to start in the womb and end in the tomb.** Severe forms of malnutrition like marasmus and kwashiorkor represent only the tip of the iceberg. Many more suffer form moderate, mild or invisible malnutrition . Malnutrition increases morbidity and mortality.

Total development of children is influenced by genetic, environmental and nutritional factors. During development of the brain, the most important phase of neuronal proliferation and migration occurs in the intrauterine period. Hence antenatal care and health of the girl child who is the prospective mother are of utmost importance. After birth, the first two years of life include a period of rapid brain growth and myelination. **By two years of age, brain attains 80 percent of**

growth and myelination becomes almost complete. Hence any programme aimed at the developing brain should be started before the age of two. Autopsy studies and animal studies have shown that **malnutrition causes** structural changes in the growing brain and **a reduction in the total number of cells, especially the glial cells.** The DNA, RNA, protein, total lipid, cholesterol and phospholipid contents of brain decrease in malnutrition. In various studies conducted in the Department of Paediatrics, Sree Avittam Thriunal (SAT) Hospital, Medical College, Trivandrum, **malnourished children were found to have a statistically significant reduction in serum and CSF proteins and lipids, serum trace elements and other macro elements.** Serum enzymes, developmental quotient (DQ), motor nerve conduction velocity and brain stem auditory evoked potentials (B AEP) were also low when compared with appropriate controls. Such developmental screening tests are of value only if they result in appropriate interventional strategies.

Among the various interventional strategies, single point interventions like nutritional supplementation and primary health care have failed to deliver the desired outcome. There is an interplay of various factors that influence the intellectual development, namely, genetic, nutritional and environmental. But, the contribution of each of them is difficult to separate and evaluate. The effect of malnutrition in reducing the intellectual achievement is very difficult to separate from other associated retarding social and environmental factors. There are some suggestions that stimulation along with nutritional supplementation may be a better choice. Hence, efforts are needed to develop composite intervention packages including several inputs. Regardless of the precise mechanism, it has been established that growth, development and intelligence of malnourished and socially deprived children are at risk. Multideprivation including malnutrition has been thus identified as the most important constraint in the total development of children. Hence, the mode of intervention has to be multidisciplinary including primary heath care, proper care during illness, nutritional supplementation, developmental stimulation, psychosocial support, environmental health and socioeconomic advancement. The package has to be integrated with the existing child welfare programmes.

ASSESSMENT OF ENVIRONMENT

Environment means surroundings, conditions or influences. It is the sum total of everything that influences any individual from inside and outside the body. The external or macro-environment includes all living and non-living things with which he is in constant interaction. It can be divided into **physical, biological and psychosocial** components. These are not watertight compartments, but are closely related. Some epidemiologists have given the term microenvironment to the personal and domestic environment. The internal environment pertains to each and every tissue, organ and system and their harmonious functioning within the body. If the environment is favourable to the individual, he can make full use of

his physical and mental capabilities. **Health** implies *a continuous adaptation and adjustment to the environment in order to ensure optimal function of body and mind.* Improvement in human adaptation to natural environment can lead to longer life and better quality of life. **Health has also been defined as complete physical, mental, social and spiritual well-being.**

1. Physical Environment

Physical environment is the term applied to the non-living things and physical factors like air, water, soil, climate, heat, light, noise, radiation etc., with which man is in constant interaction. Man's victory over the physical environment is responsible for the improvement in health. But in doing so, he has created new health problems such as air pollution, water pollution, noise pollution, radiation hazard and urbanization.

The proportion of population having access to safe water and sanitation facilities is a very useful indicator of health. All households still do not have safe water in the home or within 15 minutes walking distance and adequate sanitary facility in the home or in the immediate vicinity. The National Sanitation Foundation of USA has defined sanitation as **'a way of life'.** WHO has defined *environmental sanitation* as '*the control of all those factors in man's physical environment which exercise a deleterious effect on his physical development, health and survival'.* It is not merely sanitary disposal of human excreta, but the science of controlling the whole environment. Being a way of life, it must come from within the people. The term **environmental health** is now being used instead of environmental sanitation.

2. Biological Environment

Biological environment is the term to denote all the living things which surround man, including human beings. The microbes that are useful and harmful are included in this. In the fight for survival, some of them cause diseases and destruction. A harmonious interrelationship and a peaceful co-existence do not always endure.

3. Psychosocial Environment

Psychosocial environment includes complex factors that affect personal health, health care and community well-being. It includes cultural values, customs, habits, benefits, attitudes, morals, religion, education, lifestyle, and social and political organization. The various socioeconomic conditions that determine the psychosocial environment are economic status or purchasing power, education, occupation, political system and others like dependency ratio, family size, housing condition, per capita calorie intake etc.

4. Microenvironment

The domestic environment and the personal influences are included in microen-

vironment. It includes physical, biological and psychosocial components. The quality of the child's home environment influences the mental abilities. Both human and animal studies have shown that stimulating environment may substantially prevent the unfavourable effects of malnutrition . Thus, home and the social environment of the child need a careful scrutiny in order to find out how and why some children coming from the same economic class are less vulnerable and more capable than others in preventing the adverse effects of malnutrition . The **mother-child unit** is the most important factor influencing this ability. The importance of 'mothering' in child development is to be stressed. Home influences which outweigh the effect of all other environmental impacts combined together determine the fundamental organization of children's behaviour. **Microenvironment** of a child is **the immediate home environment with special emphasis on the mother and her psychosocial functions** (Appendix). Maternal attitude towards the child, family harmony, interrelationship with family members, friends and neighbours are considered in the microenvironment. Malnutrition impairs the mother-child interaction and the ability of the child to interact with the environment. The role of the father should also be taken into account. **The parent child unit is also an important concept** instead of mother and child unit. Child rearing and mother craft are now included in parenting skills.

5. Assessment Tools

Socioeconomic status, sanitary conditions, housing conditions, and microenvironment are considered in the assessment of the environment. Initially, a multiple social index which included income, diet, living space and stability of the family was used to score the environment. **The socioeconomic status is commonly assessed based on education, occupation, and income as in the Kuppuswami scale** (Appendix).The housing condition, sanitary facilities, availability of electricity, recreational facilities like radio, TV and conveyance facilities also are considered in some scales. Microenvironment can be assessed by scoring the maternal attitude of acceptance and rejection and supportive system to the mother. Standards of sanitation can be assessed using a scale which includes the source of drinking water, toilet habits, cleanliness and food habits as in the Briscoe scale (Appendix). Income, occupation, standard of housing, sanitation, nutrition and level of provision of health, educational, recreational and other services collectively indicate an index of the standard living. Families are divided into **above or below the poverty line (APL or BPL). The poverty line is around Rs 20,000 per annum in a five member family** or **Rs 372 per capita per mensem (2000 AD).**

6. Ten Commandments in Environmental Health

a) Safe drinking water
b) Disposal of excreta

c) Proper disposal of all wastes
d) Control and prevention of air pollution
e) Noise reduction and control
0 Proper housing standards
g) Proper ventilation and lighting
h) Prevention of radiation exposure
i) Control of biological hazards including microbes, animals and man himself
j) Legal measures for the above

The various scoring system likes SE status, standards of sanitation and microenvironment are give in Appendix.

ASSESSMENT OF NUTRITIONAL STATUS

Nutrition forms the most predominant influence on the development of the growing child. Human survival has always depended upon food and hence, nutrition has determined his place of living and his way of living. Assessment of nutritional status can be done by evaluating:

1. Dietary factors
2. Clinical features of malnutrition
3. Anthropometric measurements
4. Biochemical parameters
5. Morphological parameters
6. Radiological parameters, and
7. Epidemiological data regarding morbidity and mortality.

1. Dietary assessment

Nutritional status is related to the nutrient intake especially among preschool children. Accurate diet assessment is difficult and time consuming. A good rapport with the mother is essential for correct replies to the questions. Clinicians can have an overall assessment regarding breastfeeding practices, weaning practices and food intake prior to the illness by a **24-hour recall method.** Average of a three days recall during the mid week is recommended by some. **A food frequency table** to record the frequency of intake of each food item is also desirable, i.e., thrice a day, once a day, twice a week etc. The standard serve for each item has to be defined prior to this. Weighting the uncooked as well as the cooked food and then assessing the nutritive value of food eaten are vary good methods, but often not practical. The service of a dietitian is ideal for accurate assessment. The calculated intake should be finally compared with the Recommended Dietary Allowances (RDA) for the age (refer Section 3.3). A rough idea about the adequacy of vitamins and minerals in the diet should also be obtained,

a) *Breastfeeding and weaning practices* are the two most important dietary habits that determine child health as well as morbidity and mortality. Breastfeeding forms the integral part of the well-known **child survival pack-**

age, the components of which are **growth monitoring, oral rehydration therapy, breastfeeding, immunization, food supplementation, female education and family planning (GOBIFFF).** Another comprehensive package is NIMFES (Fig. 4.14). Child nutrition is the priority of this era. Bottle feeding is identified as a cause for increased morbidity and mortality. A filthy medicine bottle with a teat on it, wrapped in a dirty cloth with files on the teat is a common sight and is obviously a source of infection. Moreover, the teat may cause nipple confusion especially in a newborn baby. This leads to inadequate sucking at breast and suppression of lactation. The art of breast sucking and bottle sucking are entirely different. Supplementary feeding can easily be given using an ordinary spoon or traditional spoon called 'palada' (gokarnam).

b) *Diet during illness:* Maternal beliefs regarding diet during common childhood illnesses are often wrong and unscientific. It is not uncommon to starve the child during diarrhoea, measles, respiratory infection etc. Mothers must be taught to continue feeding during illness and to select easily digestible food items during illness. The convalescence should also be given due importance to ensure addition of all essential nutrients.

2. Clinical Assessment

In clinical assessment, features indicating wasting, oedema, vitamin deficiencies and those specific for various diseases should be looked for. The absence of such clinical signs denotes normal nutritional status.

3. Present Diet

While eliciting the dietary history, the average food consumed prior to the illness or hospitalization should be recorded. Establish good rapport with the mother and as far as possible avoid leading questions that evoke only a positive response, e.g., aren't you giving milk? Aren't you giving egg? etc. Instead make the mother narrate like this: "My child wakes up at 5 am, takes 3 biscuits and cup of milk and then sleeps. He gets up at 7 am and takes 1 dosai with chutney. He has a banana and a cup of milk at 10 am", and so on. Then calculate the calories and protein from the items consumed (refer Table 3.12). The calculated calories and protein are then compared with the RDA (Tables 3.16-3.20) and expressed as follows: Compared to the ICMR recommendations there is a calorie gap of 500 kcal and the protein gap is 5 g. It may also be added that the child is getting breast milk or supplementary feeding in addition to the above intake.

4. Anthropometric Assessment/Auxology

Anthropometry is a simple valuable tool and **the gold standard for evaluating the nutritional status,** but it has many limitations. Adequate precautions are to be

taken during measurement and the procedures utilized are to be standardized and checked frequently for accuracy. Intra-observer and inter-observer reliability should be established first, measurements must be taken according to the standardized techniques and the equipments should be checked periodically for accuracy.

a) ***Weight.*** It is measured using a beam scale or **Salter type** scale with pants in which the child is placed. The beam should be properly balanced and should move freely when at rest and the pointer should be on zero. The scale should be set on a flat horizontal surface. The shoes should always be removed and children should be weighed with as little clothing as custom permits. Weight is read either directly or by contact with any other object. Weight is either read directly or by balancing the beam, depending on the type of scale. The result should be read only after the beam reaches its balance point or the pointer becomes motionless. Occasionally, children are so restless that no balance point can be reached. In such cases, **double weighing** is done; first the mother is weighed alone and then the mother is weighed holding her child and the difference is computed. As accuracy is less satisfactory, this is used as a last resort only. It is always preferable to record both the weights before doing the subtraction. For older children, the weight should be accurate to the nearest 500 g and for small children to 100 g. In newborns, 20-50 g also may be important. Electronic scales are expensive. The bathroom scale is not accurate and hence not advisable.

b) *Height:* Below the age of two years, a horizontal measuring rod or infantometer is used. **Height measured in lying down posture is called length.** Length measurement needs two people. Shoes are removed and the child is placed on the back on a flat surface. One person, preferably the mother, maintains the top of the child's head against the fixed vertical head board with the child's eyes directed upwards. The other person firmly presses the knees together and down so that they touch the horizontal surface and then moves the mobile foot board so that it touches the heels when the feet are at right angle. Accuracy must be to the nearest 0.5 cm. Beyond the age of two years, a vertical measuring rod or anthropometer is used. The wall itself may also be graded, provided the zero is located exactly at the angle formed by the ground and the wall and the ground must be perfectly horizontal. The child must stand bare-footed and the heels, buttocks, shoulders and occiput touching the wall and looking straight ahead. The chin should be straight **(Frankfurt plane).** The observer reads the measurement directly after lowering the cursor or placing a horizontally held book or wooden board in order to touch the top of the head. The hair should be completely flattened. Accurancy must be to the nearest 0.5 cm. Stature below 3 SD is abnormal and < 0.4 centile may be diagnostic of endocrine causes.

c) ***Mid arm circumference (MAC):*** Between one to five years of age, the arm circumference remains fairly constant. **Measurement is performed on the left arm, midway between the acromion and the olecranon.** The measuring tape is held gently without pressing the soft tissues. The tape must be flexible and non-stretchable and unaffected by temperatures. The reading should be accurate to the nearest 0.1 cm. **Reading below 12.5 cm indicates severe PEM,** 12.5-13.5 cm moderate PEM and above 13.5 cm is normal. MAC is a good test to identify children with risk of dying. But it is not suitable for continued growth monitoring as it increases only very slowly during the one to five year period. Recently 12 cm and 13 cm have been suggested as cut-off in some studies conducted in Trivandrum instead of 12.5 and 13.5 cm. It is 27-30 cm in women and 30-33 cm in men.

d) ***Head circumference:*** While measuring the head circumference, the maximum occipitofrontal circumference (OFC) is measured by placing the flexible, non-stretchable tape firmly over the most prominent region of the occiput and frontal crests. The measurement is taken accurate to the nearest 0.1 cm.

e) ***Chest circumference:*** It is measured at the nipple and is related to OFC. In early infancy, OFC is more than chest circumference and by one year of age both are equal and thereafter the chest circumference is more than OFC. In PEM, chest circumference may continue to be less than OFC, i.e., OFC to chest circumference ratio > 1.

f) ***Skinfold thickness (SFT'):*** The skinfold thickness at triceps is measured to the nearest 0.1 cm by means of the **Harpenden calipers. This gives an indication of the subcutaneous fat** and indriectly the calorie reserve in the body. Subscapular SFT can also be measured under the scapula (Fig. 4.1).

g) ***Somatic quotient (SQ):*** The average of weight, height, OFC and MAC each expressed as a percentage of the expected is termed SQ. This is not very useful as each of the components has varying significance.

h) ***Upper segment-lower segment ratio:*** At birth itis 1.7:1, at 6 months 1.6:1,at 1 year it is 1.5:1, at 2 years it is 1.2:1, and it is 1:1 in the adult. Upper segment is measured from vertex to pubic symphysis. **The ratio becomes 1:1 by 9-10 years of age.**

i) ***Mid parental height (MPH):*** In growth retardation and short stature, mid parental height should be estimated. It is a good predictor of adult height of the child. It is estimated as follows:

$$\text{MPH for male child} = \frac{\text{Paternal height} + \text{Maternal height}}{2} + 6.5 \text{ cm}$$

$$\text{MPH for female child} = \frac{\text{Paternal height} + \text{Maternal height}}{2} - 6.5 \text{ cm}$$

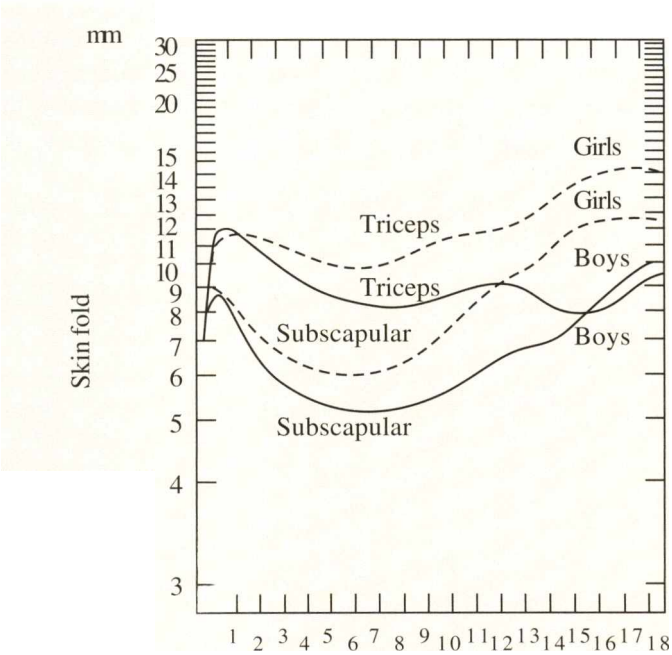


Fig. 4.1 Skin fold thickness according to age and sex (Tanner 1975)

MPH centile is then obtained by comparing it with the reference standard for 18-20 years of age. Normally, the child is expected to grow on par with the MPH centile. If the child's present height is less than MPH centile, the child needs investigation for endocrine or other causes of growth retardation,

j) *Reference standards'.* The anthropometric measurement must be compared to appropriate standards. It is preferable to use the reference standard from the same population taking care that the subjects do not suffer from malnutrition or infection. But the ICMR reference standards give unacceptably low values. Various other Indian standard are also available, collected from well nourished children belonging to high socioeconomic status. However, the National Center for Health Statistics (NCHS) references are recommended for use in India. At present the NCHS references are the best available for use irrespective of ethnic or social considerations as they comply most satisfactorily with WHO criteria. This is also called the WHO standards. The fiftieth centile is regarded as 100 per cent and references are available for weight for age, height for age, weight for height and head circumference separately for boys and girls. **The Weech's formula is used by some professionals to find out the expected weight forage** (Table 4.**1**). But, there are some difficulties in knowing the exact age of the child. Some mothers do not often remember the

correct age of the child especially the date of birth. Therefore, weight of the child cannot always be compared to the expected weight for age. This difficulty can be overcome by asking the mother to relate the date of birth to some important event like village festival. Age can then be estimated referring to the indigenous calendar.

Table 4.1 Formula for average weight, height and head circumference in children

Weight	*kg*
Birth	3
3-12 months	Age (month) + 9 / 2
1-6 years	Age (year) x 2 + 8
7-12 years	Age (year) x 7 - 5 / 2
Height	*cm*
Birth	50
3 months	60
6 months	66
1 year	75
2-12 years	Age (year) x 6 + 77
Head circumference	*cm*
Birth	35
Infant	Length (cm) + 9.5 ± 2.5 / 2
3 months	40
6 *months*	43
1 year	47
2 years	49
3 years	50
4 years	50.4
5 years	50.8

k) ***Age-independent anthropometric indicators'.*** These indications do not require consideration of the age. The labile tissues like subcutaneous fat and muscle are more reduced in malnutrition than the static tissues like skeleton. Hence the ratio between the labile and static tissues can be calculated and compared with the normal.

i) ***The bangle test:*** This is done by slipping a bangle with an inner diameter of 4 cm up the forearm. If it crosses the elbow, the child is malnourished. It is a simple test, but is less sensitive as the elbow circumference represents the thickness of the bone.

ii) ***The Shakir's tape:*** This is a plastic tape with coloured zones, green, yellow and red representing more than 13.5 cm, 12.5-13.5 cm and less than 12.5 cm respectively to measure MAC.

iii) ***The Quae stick:*** It is the short name for Quacker arm circumference stick. It is a rod with two sets of markings, one indicating the height and the other, the MAC for the corresponding height. The arm circumference is measured and the stick is placed behind the standing child. If the height is more than the expected height for the measured arm circumference, the child is considered malnourished.

iv) ***The modified Quae stick:*** This utilizes a rod that is coloured green, yellow and red that represent normal nutritional status, borderline and severe malnutrition respectively. The upper zone is red.

v) ***The Nabarrow's thinness chart:*** A graphic chart that represents the expected weight for height has been prepared by Save the Children Fund. In the severely malnourished child, the head touches the upper red zone when the child is made to stand against the column on the chart for the recorded weight of the child.

vi) ***The MAC to head circumference (MH) ratio (Kanawati):*** A ratio of 0.28-0.314 indicates mild malnutrition; 0.25-0.279, moderate malnutrition ; and less than 0.249, severe malnutrition.

vii) ***The head circumference to chest circumference ratio:*** A ratio of more than one is normal and less than one indicates malnutrition in children above 9 months of age.

viii)77ie ***mid arm circumference to height ratio:*** A ratio less than 0.29 indicates severe malnutrition and 0.32-0.33 indicates normal nutrition.

ix) ***Rao and Singh weight/height^2 ratio:*** Ratio above 0.0015 is normal, 0.0013-0.0015 indicates moderate malnutrition and less than 0.0013 indicates severe malnutrition.

x) ***Ponderal index of weight/height^3 ratio*** *:* This ratio above 2.5 is normal, 2-2.5 indicates symmetric IUGR and less than 2 indicates asymmetric IUGR (malnourished).

xi) ***Dughadale weight of weight/height$^{1.5}$ ratio:*** A ratio of above 0.79 indicates normal nutrition and below 0.79 indicates malnutrition.

xii) *Body mass index and Quetlet index:* This is expressed as weight in kg/ height' expressed in m. It is a very good index of body's reserve or loss of fat. The extent of wasting, tendency for obesity and obesity can be assessed using reference curves. In adults, BMI 18.5-25 is considered normal. Recent studies suggest that less than 15 may be considered as moderate malnutrition and < 13 as severe underweight (UW) in growing children. In adults, BMI > 25 indicates overweight (OW) and > 30 indicates obesity (OB). The corresponding figures in growing children are 22 and 25 respectively (Also refer section 12.4).

$$\text{Quetlet index} = \frac{\text{Weight (kg)}}{\text{Height}^2\text{ (cm)}} \times 100$$

Normal is > 0.15

$$\text{Body Mass Index (BMI)} = \frac{\text{Weight (kg)}}{\text{Height}^2\text{ (m)}}$$

UW = BMI < 5th centile, OW = BMI > 85th centile, OB = BMI > 95th centile

xiii) ***Mid arm muscle circumference (MAC):*** This is calculated by the following formula: Mid arm muscle circumference = MAC-(3.14 x SFT) cm.

5. Classification of Malnutrition

Malnutrition is generally classified according to weight for age. Chronic malnutrition is classified according to height for age and acute malnutrition according to weight for height.

a) ***Classification according to weight for age:*** Weight for age is the most commonly used parameter to classify nutritional status.

i) ***Gomez's classification:*** **Gomez and his associates are credited with the first classification of malnutrition which came in 1956.** It has three degrees. The details are given in Table 4.2. All cases with oedema are included in third degree malnutrition irrespective of weight for age as suggested by Bengoa in 1977.

ii) ***Jelliffe's classification:*** It has four degrees of malnutrition and it was proposed in 1965. It is detailed in Table 4.3.

iii) ***Wellcome Trust or International classification:*** It is a clinical classification suggested by the Wellcome Trust in 1970. It is based on weight for age and the presence or absence of oedema. The details are given in Table 4.4

vi) ***Indian Academy of Pediatrics (IAP) calssification:*** It is the most popular classification in India proposed by IAP in 1972. It has four grades of malnutrition, as detailed in Table 4.5. The scientific reason why severe malnutrition (weight below 60%) is further classified into grade III and IV is not clearly explained; It may be to highlight that very severe malnutrition exists. If the patient had oedema of nutritional origin, the letter 'K' is placed along with grade of malnutrition in order to denote kwashiorkor.

b) ***Classification according to height for age***: The calculation based on weight for age does not help to exclude other obvious syndromes with short stature. Moreover, it does not imply whether the malnutrition is of recent or past onset. Amost simultaneously two workers. Waterlow from London and McLaren from Beirut, independently came out with the height for age and weight for height concept to indicate stunting and wasting respectively in 1972. Height for age is used to grade stunting. It indicates past or chronic malnutrition. Classification based on height for age is given in Table 4.6

c) ***Classification according to weight for height***: It is used to grade wasting. Wasting indicates recent or acute malnutrition. Classification based on weight for height is detailed in Table 4.7.

Table 4.2 Gomez's classification according to weight for age

Nutritional status	**Weight for age (Harvard) (% of expected)**
Normal	- >90
First degree PEM	- 75-90
Second degree PEM	- 60-75
Third degree PEM*	- <60

**AII cases with oedema to be included in third degree PEM irrespective of weight for age. (PEM - Protein-Energy malnutrition)*

Table 4.3 Jelliffe's classification according to weight for age

Nutritional status	**Weight for age (Harvard) (% of expected)**
Normal	- >90
First degree PEM	- 80-90
Second degree PEM	- 70-80
Third degree PEM	- 60-70
Fourth degree PEM	- <60

Table 4.4 Wellcome Trust classification of malnutrition

Weight for age (Boston) (% of expected)	Oedema	Clinical type of PEM
60-80	+	Kwashiorkor
60-80	-	Underweight
<60	-	Marasmus
<60	+	Marasmic Kwashiorkor

Table 4.5 The IAP classification of malnutrition

Nutritional status*	Weight for age (°/o of expected)
Normal	>80
Grade I PEM	71-80
Grade II PEM	61-70
Grade III PEM	51-60
Grade IV PEM	< 50

* *If the patient has oedema of nutritional origin, the letter K is placed along with the grade of PEM in order to denote kwashiorkor*

Table 4.6 Calssification according to height for age

Height for age (% of expected)	Waterlow's classification	McLaren's classification	Visweshwara Rao's classification
Normal	> 95	> 93	> 90
First degree stunting/short*	90-95	80-93	80-90
Second degree stunting	85-90	-	-
Third degree stunting/dwarf*	< 85	< 80	< 80

**Terminology used in McLaren's classification*

Table 4.7 Calssification according to weight for hight

Weight for height (% of expected)	Waterlow's classification	McLaren's classification
Normal	> 90	> 90
First degree wasting/ mild wasting*	80-90	85-90
Second degree wasting/ moderate wasting	70-80	75-85
Third degree wasting/ severe wasting*	< 70	< 75

**Terminology used in McLaren's classification*

d) ***WHO cut-off for assessment of malnutrition in community studies.*** The WHO cut-off to estimate malnutrition in population analysis is the mean value minus two standard deviations (SD). As adopted from Waterlow's classification, the combined position of two indicators, i.e., weight for height and height for age distinguishes between wasting caused by acute malnutrition and stunting caused by chronic malnutrition (Table 4.8).

Table 4.8 WHO-cut-off for assessment of PEM in the community

Cut-Off	H/A	W/H	H/A & W/H
> Mean - 2 SD	Normal	Normal	Normal
< Mean - 2 SD	Stunted	Wasted	Stunted & wasted

H/A - Height for age, W/H – Weight for height

e) ***Standard deviation (SD) score/Z score****:* The SD score is used in population studies. Percentage of the median is claculated first to interpret data at population level and Z score is then calculated.

$$\text{Perccentage of the median} = \frac{\text{Measured individual value} \times 100}{\text{Reference median}}$$

Practically, 80% of the reference median for weight for age and weight for height and 90% for height for age correspond to 2 SD below the median. Third centile corresponds to median munus 2 SD. Each indiviudal value can be expressed as -1.8 SD. +1.9 SD etc.

$$\text{SD/Z score} = \frac{\text{Measured individual value - Reference median}}{\text{SD of the reference median}}$$

f) *WHO classification of malnutrition*

i. Acute and chronic malnutrition

W/A	H/A	W/H	Interpretation
Normal	Normal	Normal	Normal
Decreased	Normal	Decreased	Acute malnutrition
Decreased	Decreased	Normal	Chronic malnutrition
Decreased	Decreased	Decreased	Acute-on-chronic malnutrition

ii. Moderate and severe undemutrition

Features	Moderate	Severe
Oedema	No	Yes
Weight-for-height (wasting)	70-79%	< 70%
Height-for-age (stunting)	85-89%	< 85%

6. Biochemical Indicators of Malnutrition

The striking biochemical changes include lowering of serum protein, especially the albumin fraction, enzymes like esterase, amylase, lipase, cholinesterase, alkaline phosphatase and lactic dehydrogenase, carrier proteins like transferrin, caerulopasmin and beta-lipoprotein, essential amino acids, essential fatty acids, serum calcium, phosphorus, sodium, potassium, iron, magnesium etc. The hydroxyproline index and careatinine height index are also low. **Urinary creatinine gives an indirect evidence of muscle mass.** Serum protein electrophoresis shows low albumin band and low alpha-2 beta-globulin bands. Alpha-2 and beta globulin bands represent the carrier proteins. **The alpha-1 globulin band that represents acute phase reactants and gamma globulin band that represents antibodies produced against infection are found to be raised.** Reduction in carrier proteins is an early indicator of malnutrition .

Reduction in serum albumin is slow. Low serum albumin is well known to predict mortality. Low serum total lipids and phospholipid also have been suggested as possible predictors of mortality in a study. The synthesis of acute phase reactants is given more priority in protein deficiency states than the synthesis of carrier proteins.

7. Morphological Indicators of Malnutrition

It is centered around the changes that occur in the mucosa and hair shaft. **In the buccal smear, above 70 per cent of cells may be seen mutilated** as aganist less than 10 per cent in normal children. The hair shaft size is reduced and most of the cells in the growing end are noted to be in the resting phase of telogen and only very few in the growing phase of anagen. The mineral content of the hair root may also be reduced. Curly hair may straighten up in malnutrition . **Difference in texture of hair is an early sign.**

8. Radiological Indicators of Malnutrition

There may be some retardation of bone age, osteoporosis and rarely evidence of rickets or scurvy. Rickets usually manifests in the rehabilitation phase when the child starts growing. Transverse lines that represent periods of arrested growth at the growing end of long bones may be noted prior to the onset of frank malnutrition. The bone age usually corresponds to the height age rather than the chronological age.

9. Epidemiological Assessment

Vital statistics like infant mortality, neonatal mortality, perinatal mortality, still birth and one to four year mortality are the usual indicators selected to evaluate the nutritional status of a community. When the nutritional status improves, the mortality comes down. **Under five mortality rate (U5MR) is used to rank the nations of the world in the descending order. India ranked 49 with a U5MR of 72. The country with highest U5MR is ranked No.l to ensure highest priority.**

ECOLOGY AND SPECTRUM OF PROTEIN-ENERGY MALNUTRITION (PEM)

1. Definition of PEM

WHO (1973) has defined **PEM as a** ***range of pathological conditions arising from coincident lack, in varying proportions, of protein and calories, occurring most frequently in infants and young children and commonly associated with infections.***

2. Terminologies Related to PEM

There are various terminologies related to PEM. Man has understood that very little can be done about something that has no name. In the fifteenth century, the term atrophy was used to denote a serious nutritional disorder of mankind. The term went on changing several times before the present name of PEM came into existence. Malnutrition in the broad sense can mean overnutrition or undernutrition, but PEM is restricted to undernutrition.

a) ***Protein-calorie malnutrition (PCM):*** **Jelliffe** coined the term PCM to in-

Table 4.9 National Health Indices

Item	India	Goal 2000	Kerala
IMR/1000 live births	54	60	13
PMR/1000 live births	46	35	
U5MR/1000 live births	72	10% annual reduction	32
MMR/1000 live births	4	< 2	0.8
Low birth weight (LBW%)	33	10	20
Crude birth rate-CBR/1000	26	21	17
Infants(%) 85		85	90
Immunization			
Antenata!(°/o) 80		100	100
Antenatal care-ANC%	80	100	100
Deliveries(%) by trained persons	34	100	100
Vit. A deficiency blindness(%)		0.3	
Total fertility rate	2.8		1.8
GNP $340			
Life expectancy	64		74
Adult literacy(%)	66		90
Goitre rate% (6-11 years)	9		
Moderate & severe PEM(%) (under fives)	46		29
Moderate & severe stunting(%) 38			27
Moderate & severe wasting(%)	19		**12**
Crude death rate (CDR)/1000	8		**6**
Age at marriage	18		22

IMR - Infant mortality rate, PMR - Perinatal mortality rate
U5MR - Underfive mortality rate, MMR - Maternal mortality rate.

Source: The State of the World's Children, UNICEF, 2009 Sample Registration Survey; National Family Health Survey, II

elude all clinical types of malnutrition, in 1959. Even through there was an effort to introduce the term 'Protein Calorie Deficiency Disease', it was abandoned by the Nutrition Expert Committee in favour of PCM. The International System of units proposed the replacement of the term 'calorie' by 'joule' (1 cal = 4.184 J) as a unit and the term 'energy' for general use. The has resulted in the term PEM instead of PCM.

b) *Energy protein malnutrition* . To emphasize the energy gap rather than protein deficiency and also to stress the overall energy crisis of mankind, the term 'Energy Protein Malnutrition' has been suggested by some. But pending general agreement on improved nomenclature, the term PEM is retained.

3. Specturm of PEM

The spectrum of PEM includes severe as well as mild forms of PEM. The severe forms of PEM are kwashiorkor, marasmus and marasmic kwashiorkor.

a) *Kwashiorkor*. **This was first recognized by Prof Cicely Williams in 1933** from Gold Coast. She attributed it to protein deficiency. She observed that this was the disease of the first child when the second was on the way displacing the first child from the breast. She named it kwashiorkor. **The word was taken from the Ga language of Ghana, which means 'kwa-ni-oshi korkor' implying 'pretend not to mind the korkor (second one), the disease of the first child; 'red boy',** due to the characteristic pigmentary changes, was yet another term for kwashiorkor. Later on, the term was interpreted as the disease of the **'deposed child'.** Workers from West Indies identified a syndrome similar to kwashiorkor with prominent cheeks and oedema and suggested the term **'sugar baby'** in order to stress the dietary orign of the disease. A classic case of kwashiorkor is apathetic, miserable, stunted in growth and has oedema, hepatomegaly, anaemia and hair and skin changes. Grading of kwashiorkor is as follows:

Grade I—pedal oedema; grade II—I + facial oedema; grade III—II + paraspinal and chest oedema; grade IV—III + ascites. The triad of kwashiorkor is growth retardation, odema and mental changes.

b) *Marasmus '*. This was recongnized hundreds of years ago as a major contributor to infant mortality. The word marasmus is derived from the Greek word marasmos, which means **'wasting'.** Affected children exhibit extreme wasting and have an old man appearance with just skin and bones. Grading of marasmus is as follows:

The wasting often starts in the axilla and groin (grade I marasums), then in the thigh and buttocks (grade II), followed by chest and abdomen (grade III) and lastly the buccal pad of fat (grade IV) which is metabolically less active.

The wasting of the brown fat occurs first because it is metabolically more active and is important in thermogenesis. Children with marasmus are generally alert and have good appetite. Later on they become irritable.

c) ***Marasmic kwashiorkor:*** When marasmic children develop oedema, it is termed marasmic kwashiorkor.

d) ***Prekwashiorkor:*** Affected children have poor nutritional status and certain features of kwashiorkor like hair changes, moon face and hepatomegaly but do not have oedema.

e) ***Nutritional dwarfing:*** Prolonged PEM starting fairly early in life and going on over a number of years without developing kwashiorkor or marasmus results in nutritional dwarfing. The terms **bonsai children** or **pocket editions** are now popularly used for this condition. These children have evidence of stunting, but no wasting. Micronutrient deficiency is now thought to be a cause for stunting.

f) ***Underweight:*** The child is malnourished, but does not have any features of marasmus or kwashiorkor. The weight for age is 60-80% of the expected.

g) ***Invisible PEM:*** This is not very evident. Toddlers who show breast addiction must be suspected to have invisible PEM. It has been reported that 'the average moderately malnourished child in the 6-24 months age range looks entirely normal, but is too small for age, has lowered resistance to infection and therefore easily succumbs to illness. The child receiving only 60 per cent o f the calorie requirement may give no outward sign of hunger beyond a frequent desire to breastfeed'. Use of growth charts is perhaps the best way to monitor nutritional status and to identify children with invisible PEM. The invisibility renders prevention and cure difficult. A flat curve or a downward curve in the growth chart is of concern even when it is within the 'road to health.'

h) ***Early lactation failure (ELF) syndrome:*** Early lactation failure, abrupt stoppage of breast-feeding and early introduction of dilute starch-based liquid diet without any good-quality protein is found to result in severe undernutrition, with skin changes, apathy, hypoalbuminaemia, anaemia, oedema and micronutrient deficiency disorders. In lactation failure, many mothers do not start cow's milk protein; instead wean the babies less than 4-6 months on to dulute starch-based preparations. A different type of kwashiorkor with even keratomalacia has been observed in very young infants as shown in Fig. 4.2. This kwashiorkor is different from what was earlier observed in toddlers when the second baby is born.

"Kwashiorkor - In fact, a name means very little except to classify a certain conception. Until pathologists and biochemists can give us more precise information about the defects, we may well accept the word kwashiorkor in all its cacophony". —Cicely Williams, 1953

Since kwashiorkor was described in 1933, many children were diagnosed with this form of malnutrition. The distinguishing feature was the presence of



A



B

Fig. 4.2 ELFS. *A,* Kwashiorkor; *B,* Keratomalacia.

oedema, and the rapid demise of these children. Oedematous malnutrition had been described in Europe since the eighteenth century and associated with monotonous diets. It was Williams who gave it the African name kwashiorkor after the notion in the Ga language 'kwa ni oshi korkor', meaning "pretend not to mind korkor (= the second one)". Williams carefully suggested that "as maize was the only source of the supplementary food, some amino acid or protein deficiency cannot be excluded as cause".

In spite of Baby Friendly Hospital Initiative which spread in Kerala since 1990s we are seeing clusters of very young infants with severe PEM, similar to kwashiorkor. Classically described during famine, poverty and illiteracy this form of PEM is reported from Kerala where none of the risk factors are present. In fact, the mothers had good education status and had consulted various doctors. The abrupt stoppage of breastfeeding often followed an episode of diarrhoea.

Common finding was early stopping of breast milk and weaning to ragi by

bottle. Although weight recording had shown PEM, absence of road to health charts couldn't identify the growth failure in these infants.

This stresses the need for strengthening breastfeeding and optimum complementary feeding practices

4. Magnitude of the Problem of PEM

PEM is global health problem, more prevalent in the developing countries. **'It often starts in the womb and ends in the tomb'.** PEM is a disease of multideprivation and proverty, affecting nearly 150 million children under the age of five years in the world. Out of the 120 million children in India, over 75 million are estimated to suffer from visible PEM. Severe malnutrition represents only the tip of the iceberg. Many suffer from invisible PEM which is difficult to monitor and makes prevention and cure difficult. It has been estimated that for every diagnosed case of PEM, there are 10 others with borderline malnutrition, undetected in the community. During the decade 1981-90, as reported in the 'State of the World's Children', in India, 63 per cent of children under five years of age suffered from moderate or severe PEM. It is much higher than the 38 per cent reported during 1975-81 period. This is to be viewed with serious concern, and necessary steps are to be taken to analyze the situation and arrive at feasible solutions. The high prevalence of PEM in India focuses attention on the fact that after independence **a second India** has been added, whom she is not able to feed properly. Moreover, India consists of a dual society of a small group of well-fed and a large group of ill-fed people. India is supporting over 15 per cent of the world's population with only 2.4 per cent of global land area. It is interesting to note that every sixth person in the world is an Indian.

5. Prevalence of PEM

In India, household nutrition surveys done periodically by the National Nutrition Monitoring Bureau (NNMB) provide data on the nutritional status of children as well as the prevalence of PEM. According to the 1980 and 1990 reports, **Kerala had the highest percentage of children with normal nutritional status.** A dip in the nutritional status noted in the national level during the last decade is seen reflected in the Kerala scene too. Table 4.10 shows the percentage distribution of underfive children according to the nutritional status in India as a whole and in Kerala. The infant mortality rate (IMR), the percentage of low birth weight (LBW) babies, the prevalence of PEM, the prevalence of wasting indicating acute PEM and the prevalence of stunting indicating chronic PEM are some of the indicators selected to evaluate the efficacy of health care system in a nation (Table 4.9). Kerala is ranked 1 as per nutritional status, 6 as per energy consumption (Table 4.11). The effects of PEM are multidimensional. The most imporant among them are: reduced activity, reduced growth, increased susceptibility to infection, reduced intellectual capability and performance, reduced work efficiency, and increased mortality.

Table 4.10 Percentage distribution of children according to nutritional status—Gomez's classification

Year	Place	Normal	1° PEM	2° PEM	3° PEM
1970	India	3	14	65	18
1980	India	15	48	32	5
1980	Kerala	22	55	18	5
1990	India	10	38	44	6
1990	Kerala	18	47	33	2
1993	India	47	33	18	12
1993	Kerala	71	11	12	6

Source: NNMB Annual Reports, Family Health Survey, 1993 and The State of the World's Children

Table 4.11 Ranking of states according to nutritional status of underfive children and energy consumption

State	Ranking* nutritional status	Ranking** energy consumption	Average consumption Energy (kcal)	Protein (9)
Kerala	1	6	2140	53
Tamil nadu	2	7	1871	46
Maharashtra	3	5	2211	62
Andhra Pradesh	4	4	2340	56
Karnataka	5	2	2431	65
Gujarat	6	3	2375	69
Madhya Pradesh	7	1	2614	83

Source: NNNB data (1988- 90)

*Rank of state with lowest malnutrition - 1

**Rank of state with highest energy consumption - 1*

6. Ecology of PEM

PEM is a disease of multifactorial deprivation. According to Jelliffe, the ecological factors leading to PEM are conditioning influences, cultural influences, socio-economic factors, factors related to food production and intake as well as availability and utilization of health and other services.

a) *Conditioning influences:* **Low birth weight and infections are the most important conditioning influences responsible for malnutrition, especially in small children.** Diarrhoea, acute respiratory infection, vaccine prevent-

able diseases (VPD) like measles, whooping cough and TB and helminthiasis are the common infections that initiate malnutrition and aggravate existing malnutrition.

b) *Cultural practices*: Due to various cultural influences like food habits, custmos, beliefs, traditions, religion, food fads, coooking practices, child rearing practices, attitudes and superstitions, people tend to consume poor diet when good ones are available. Thus, lack of food is not the only problem. Often there is starvation in the midst of plenty. The tragedy is that these cultural factors adversely affect the most vulnerable group, i.e., infants, toddlers and expectant/lactating mothers. The presonal likes and dislikes in selection of food are called **food fads.** It may stand in the way of correcting nutritional deficiencies.

The various cooking practices like draining away the water at the end of cooking, prolonged boilling in open pans, peeling the vegetables prior to cooking etc., tend to reduce the nutritive value of foods. Premature curtailment of breastfeeding, adoption of bottle feeding and change over from locally available food to commercially prepared refined foods are some of the child rearing practices that adversely affect the nutritional status of children. In most communities, men eat first and women and girls eat last and least. The health of the woman and the girl child who is the prospective mother is thus adversely affected. Alcoholism is another sociocultural factor that leads to proverty and malnutrition in the society and family. In Kerala, the rice giving ceremony is often delayed and postponed leading to malnutrition among infants. The influence of 'fast food and colas' is another hazard.

c) *Socio-economic factors:* Factors like poverty, ignorance, illiteracy, lack of knowledge regarding the nutritive value of food, unhygienic environment, large family size, lack of spacing, overcrowding and unemployment influence the quality of life and also the health status. Socioeconomic advancement is the final answer to the problem of malnutrition.

d) *Food-production and intake:* Abrupt withdrawal of breast milk, delayed and inadequate weaning, lack of food supplementation for the target group like young children and women are the direct determinants of malnutrition. Scarcity of food is a factor responsible for malnutrition at the family level, but it is not true at the national or global level. It is often a problem of uneven and inefficient distribution. It is said that there will be very little malnutrition in India if all the available food can be equally distributed according to the physiological needs. However, production of good quality food has to improve to alleviate malnutrition and poverty. India is a dual community with a well-fed minority and an ill-fed majority. Food is also wasted due to ineffective storage.

e) *Availability and utilization of health and other services:* Inadequacies in nutrition and health education, nutritional surveillance, nutritional rehabilitation, primary health care, immunization, early diagnosis and prompt treatment, referral services etc.. contribute to and also perpetuate malnutrition. In Kerala, all the services are available at door step; but utilization is below optimum. In some other states, availability of services is poor.

7. Changing Profile of PEM

Recently, a change in the profile of PEM has been noted. There is a decrease in the hospital incidence of kwashiorkor and marasmic kwashiorkor. The more common types now are underweight, nutritional dwarfism and marasmus.

CLINICAL FEATURES OF PEM

1. Clinical signs of PEM

Clinical examination provides clues for assessing the nutritional status as well as the severity of the disease. Jelliffe has classified the clinical signs for easy identification and interpretation (Tabic 4.12).

a) *Growth retardation:* **It is the most common feature of PEM,** evidenced by weight loss, wasting and stunting. Growth retardation is present even in a kwashiorkor case.

b) *Hepatomegaly:* This is due to fatty infiltration, which starts in the periphery of the lobule and gradually extends to the centre. Histologic evidence will be present, even when fatty liver is not clinically evident.

c) *Hair changes:* The changes are more evident at the root of the hair in acute PEM. The hair becomes sparse, easily pluckable and hypopigmented—**hypochromotrichia.** In some children, the hair becomes red and this led to the term red boy to denote kwashiorkor. When the nutritional status is regained, the root becomes pigmented and the tip is seen hypopigmented—**flag sign.** There may also be straightening of curly hair and change in texture.

d) *Skin changes:* These indicate severe degree of malnutrition and may occur rapidly in fatal cases. The skin becomes hypopigmented, hyperpigmented, erythematous or jet black in colour. This **'flaky paint dermatosis'** is pathognomonic. It occurs more often in the extremities than trunk and the hyperpigmented patches peel off to expose raw or hypopigmented areas. The cracked lesion in the flexures, groin and buttocks which is infected and ulcerated is called **crazy pavement dermatosis. Deficiencies of tyrosine, niacin, zinc and vitamins are attributed in the pathogenesis of these skin changes.** Secondary infection with fungi and bacteria, especially cutaneous diphtheria, also may occur. The term **'nutritional dermatosis'** is also applicable.

Table 4.12 Clinical signs of malnutrition (Jelliffe)

Organ	Signs
1. Hair	Lack of lustre
	Thinness and sparseness
	Straightness
	Dyspigmentation
	Flag sign
	Easy piuckability
2. **Face**	Diffuse depigmentation
	Naso-labial dyssebacia
	Moon face
3. **Eyes**	Pale conjunctiva
	Bitot's spots
	Conjunctival xerosis
	Corneal xerosis
	Keratomalacia
	Angular palpebritis
4. **Lips**	Angular stomatitis
	Angular scars
	Cheilosis
5. **Tongue**	Oedema
	Scarlet and raw tongue
	Atrophic papillae
6. Teeth	Mottled enamel
7. **Gums**	Spongy, bleeding gums
8. Glands	Thyroid enlargement
	Parotid enlargement
9. **Skin**	Xerosis
	Follicular hyperkeratosis
	Petechiae
	Pellagrous dermatosis
	Flaky-paint dermatosis
	Scrotal and vulval dermatosis
10. **Nails**	Koilonychia
11. **Subcutaneous tissue**	Oedema
	Amount of subcutaneous fat reduced
12. **Muscular and skeletal systems**	Muscle wasting
	Craniotabes
	Frontal and parietal bossing
	Epiphyseal enlargement (tender/non-tender)
	Beading of ribs
	Wide open anterior fontanelle

contd

	Knock-knees or bow legs
	Diffuse or local skeletal deformities
	Deformities of thorax
	Musculo-skeletal haemorrhages
13. **GIT**	Hepatomegaly
14. **Nervous system**	Psychomotor change
	Mental confusion
	Sensory loss
	Motor weakness
	Loss of position sense
	Loss of ankle and knee jerks
	Calf tenderness
15. **Cardiovascular**	Ca rd io meg a l y/**microcardia**
	Tachycardia

e) *Mucosal changes:* These are due to various vitamin deficiencies and secondary infection including moniliasis. Glossitis, stomatitis, cheilosis etc., are common.

f) *Purpura or bleeding:* It may be seen in those with Gram-negative septicaemia, disseminated intravascular coagulation and vitamin C and K deficiency.

g) *Oedema:* It is seen in kwashiorkor and marasmic kwashiorkor. Mooning of face is often noted in kwashiorkor. Effusion into the serous cavities may occur in severe oedema. Isolated ascites may be due to associated liver disease or intestinal TB.

h) *Mental changes:* Irritability and apathy are the common changes noted in PEM. They are multifactorial in origin. Affected children fail to interact with the environment and even with the mother. Social smile also regresses. Brain oedema, electrolyte imbalances, hypokalaemia and hypomagnesaemia are suggested to be the causes. Alteration in neurotransmitter synthesis and release is found to be another major cause. PEM reduces playful exploratory activity, motivation and arousal.

i) *Tremors:* These are characteristically seen during treatment. Deficiency of vitamin B factors due to increased demand, electrolyte imbalance, imbalance in the production of inhibitory substances like gamma-aminobutyric acid (GABA) and dysmyelination are thought to be the causes. Frequent blinking, tremulous cry due to vocal cord tremor and tremors of the body designated **'kwashi shake'** are rarely noted. The tremors generally subside after some time.

PATHOLOGY, PATHOGENESIS AND ADAPTATIONS IN PEM

1. Pathogenesis and Adaptations in PEM

The malnourished child adapts maximum to the unfavourable circumstances as

well as to the calorie and protein gap. This can be compared to the state of **"hibernation"** in animals. Malnourished children reduce their activity, curtail their growth and bring down their **basal metabolic rate (BMR)** in order to save energy for survival. The energy expenditure in a normal child is given in Table 4.13. However, in a malnourished child due to associated diarrhoea and protein-losing enteropathy, the faecal loss will be more than in a normal child and the **specific dynamic action** (SDA) will vary depending on the type of food consumed. Protein rich food advised may suddenly increase the specific dynamic action (30%) whereas carbohydrate rich food tends to keep it low (5%). **The adaptation hypothesis put forward by Gopalan in 1967** helps to explain the pathogenesis of PEM. The adaptation in PEM is basically biochemical and hormonal. The high level of catabolic hormones including cortisol causes muscle and fat breakdown. The anabolic hormones like insulin and insulin-like growth factors maintain near normal anabolism to prevent oedema and fatty liver by enabling the synthesis of albumin and beta-lipoproteins from the available pool of amino acids. But in kwashiorkor, due to the stress of malnutrition or due to the sudden stress of an added infection or other catastrophe, the adaptive mechanisms tend to fail. However, the reduced activity in a malnourished child as part of the adaptation reduces the mother-child interaction and the ability of the child to explore and master the environment. The reduction in BMR and lack of insulating fat lead to hypothermia which may prove fatal.

Table 4.13 Energy expenditure in a normal child

Item	Energy expenditure (%)
BMR	50
Activity	25
Growth	12
Faecal loss	8
Specific dynamic action	(SDA) 5

2. Why do some children develop marasmus and some others develop kwashiorkor?

a) *Variable calorie and protein intake:* Predominant protein deficiency was thought to be the cause for kwashiorkor and predominant calorie deficiency, the cause for marasmus. Protein deficiency in experimental animals has demonstrated a syndrome similar to kwashiorkor. But later, many worker,s including Gopalan from India, have pointed out that both the syndromes occur in children taking similar diet and sometimes both occur in the same family among siblings.

b) *Time bound development of kwashiorkor and marasmus:* Viteri in 1964 suggested that time is the determining factor. If the malnutrition is chronic, children may take the long marasmic route and adapt. If acute, they may take the kwashiorkor route and fail to adapt to the situation.

c) *Theory of adaptation:* Gopalan in 1967 postulated the theory of adaptation. If the child adapts to malnutrition, marasmus results and dysadaptation leads to kwashiorkor. Thus the capacity to adapt physically and biochemically determines the outcome.

d) **Golden's theory of free radicals:** Postulates that free radicals play a role in oedema, skin changes and fatty liver.

3. Physio-chemical and Hormonal Mechanisms in Adaptation and Dysadaptation

I. *Cortisol:* **Cortisol is the major catabolic hormone in the body.** It also has anabolic effects especially in carbohydrate metabolism. The stress of PEM and infection leads to increased cortisol (hydrocortisone) secretion. *Cortisol helps to maintain blood glucose, amino acid pool and fatty acid pool* (Fig. 4.3).

a) ***Effect on carbohydrate metabolism:*** It increases gluconeogenesis from amino acids, increases glycogenesis and also decreases glucose transport and utilization (antagonistic to insulin and synergistic to growth hormone) and keeps up blood glucose level.

b) ***Effect on protein metabolism:*** It mediates proteolysis and decreases protein stores. It decreases amino acid uptake and protein synthesis in extrahepatic cells and keeps up blood amino acid pool. It also mediates conversion of amino acid to glucose and amino acid uptake and protein synthesis in the liver alone. In the liver, DNA transcription to RNA is promoted (anabolic effect).

c) ***Effect on lipid metabolism:*** It mediates lipolysis from subcutaneous fat and keeps up fatty acid pool. However, it decreases utilization of fatty acid for energy.

d) ***Regulation of cortisol:*** Any type of stress, trauma or infection leads to cortisol release. ACTH from pituitary mediates release of cortisol. In PEM, it is released in response to stress and infection and it mediates gluconeogenesis. glycogenesis and helps to maintain glucose, amino acid and fatty acid pool by effective catabolism. In **kwashiorkor, cortisol is not high sustained** probably due to adrenal exhaustion. Free cortisol level is high due to lack of carrier protein. This may lead to mooning of face.

n. *Insulin:* Insulin is the main anabolic hormone in the body. Others are somatomedins and insulin like-growth factors. *Insulin promotes utilization of glucose, protein synthesis and prevent lipolysis* (Fig. 4.4).

a) ***Effect on carbohydrate metabolism:*** It mediates glucose transport and

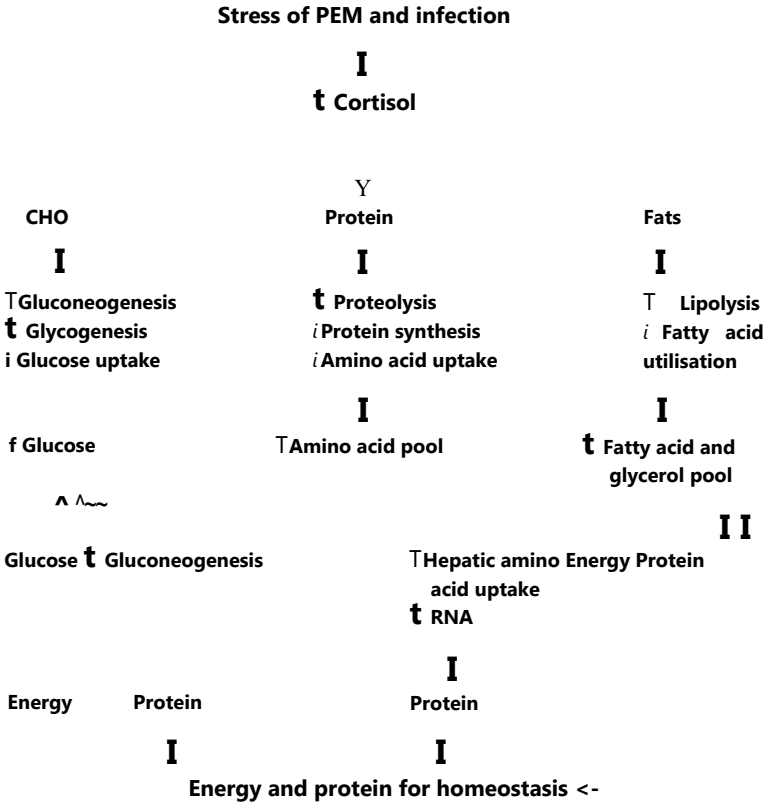


Fig. 4.3 Role of cortisol in PEM (maintain glucose, amino acid & fatty acid)

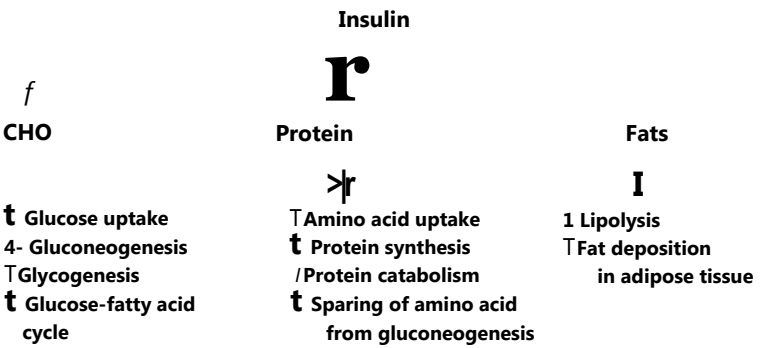


Fig. 4.4 Role of insulin in PEM
(glucose utilization, protein synthesis and fat sparing)

utilization. Only brain cells are permeable to glucose without insulin mediation. It decreases gluconeogenesis and spares amino acids for protein synthesis. It converts glucose to fatty acids and mediates deposition in subcutaneous fat (glucose-fatty acid cycle). It increases glycogenesis.

b) *Effect on protein metabolism:* Along with growth hormone it promotes growth (synergistic to GH), increases amino acid uptake and protein synthesis. It reduces protein and albumin catabolism. It decreases gluconeogenesis and leads to sparing of amino acid pool. Insulin mediates switching on and off of protein synthesis.

c) *Effect on fat metabolism:* It prevents lipolysis. It increases fatty acid pool by conversion of glucose and promotes fat deposition (fat-sparing effect). Lack of insulin leads to reduced glucose utilization, protein synthesis and growth. Lack of insulin will increase lipid utilization and fat deposition in liver as fatty liver.

Thus it also switches on and off carbohydrate and fat utilization depending upon the blood glucose level. Insulin levels increase with high blood sugar and leads to carbohydrate utilization. With low blood sugar, insulin levels reduce and lead to lipid utilization.

d) *Regulation of insulin:* Blood sugar plays a key role in the regulation. Hyperglycaemia stimulates and hypoglycaemia decreases insulin release. Food intake and insulinotropie hormones like gastrin, gastric inhibitory peptide, secretin, cholecystokinin etc., increase insulin levels. **Amino acids like arginine, lysine and hormones like growth hormone and cortisol stimulate insulin release.** Severe protein depletion, lack of food intake, persistent hypoglycaemia, alteration in amino acid pool, hypokalaemia, (3-cell destruction and (3-cell exhaustion due to over stimulation by growth hormone and cortisol play a role in reducing insulin. In severe PEM, insulin mediates protein synthesis for homeostasis and keeps up the anabolic limb. In kwashiorkor and marasmic kwashiorkor, lack of insulin leads to failure of anabolic limb leading to lack of synthesis of albumin and (3-lipoprotein resulting in oedema and fatty liver.

III. ***Growth hormone (GH):*** When starvation and protein depletion are at a peak, GH secretion increases. Low somatomedin is another stimulus for GH secretion. ***It conserves carbohydrate increases, protein synthesis and increases fat mobilization*** (Fig. 4.5).

a) *Effect on carbohydrate metabolism:* It conserves glucose by reduced tissue uptake of glucose and utilization (antagonistic to insulin and synergistic to cortisol). It also increases glycogenesis.

b) *Effect on protein metabolism:* It increases protein synthesis (synergistic to insulin). It increases tissue amino acid transport and DNA transcription to RNA. It also reduces protein catabolism. It has a protein-sparing effect.

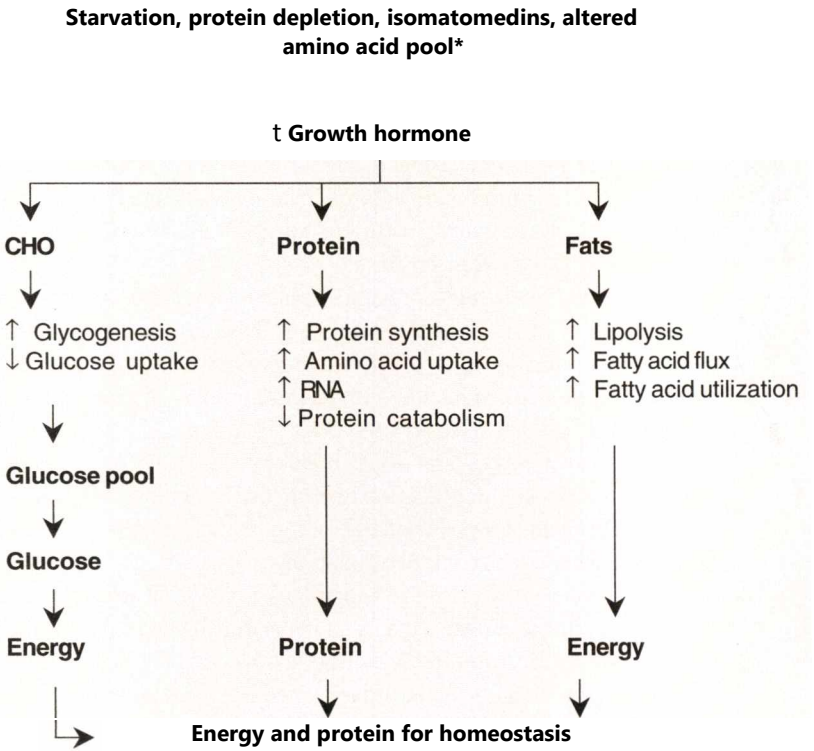


Fig. 4.5 Role of growth hormone in PEM (glucose conservation, protein sparing and fat utilization)

c) *Effect on lipid metabolism:* It leads to lipolysis and fatty acid flux and increases fatty acid utilization.

d) *Regulation:* It is regulated by somatostatin and somatomedins. Starvation, hypoglycaemia, reduced fatty acid pool, protein depletion (S. albumin < 3 g%) and altered amino acid pool (reduced branched chain amino acid and alanine) mediate increased GH levels in prekwashiorkor and kwashiorkor. In short, growth hormone conserves glucose, spares protein and increases fat utilization.

IV *Somatomedins:* Somatomedins A and C and insulin-like growth factors (IGF) are more reduced in kwashiorkor than in marasmus. IGF-I and somatomedin C are now found to be identical.

V *Glucagon:* It is generally normal or may be variable depending on the number of alpha cells in the pancreas.

VI. *Thyroxin:* It is usually normal, but may be low if there is associated iodine malabsorption. However, conversion of T_4 to T in the tissue is decreased and may be one of the causes for low BMR.

Selective cell damage, cell exhaustion, hypokalaemia, low gut insulinotropic factors, lack of food intake, hypoglycaemia and protein depletion decrease insulin level in kwashiorkor. **Chromium deficiency is probably another factor that reduces the action of insulin.** 250 mg chromium chloride administration was found to improve glucose tolerance. Cortisol levels are found to be more in marasmus than in kwashiorkor; but free cortisol is more in kwashiorkor probably due to lack of binding proteins.

The various hormonal changes are summarized in Table 4.14.

Serum albumin level above 3.5 g/dl is considered normal and below 2.5 g/dl is considered pathological. Serum albumin level of 3 g/dl is considered to be the critical value below which there is low serum (3-lipoprotein concentration leading to fatty liver and low branched chain amino acid and alanine levels leading to increase in GH levels. Cortisol and insulin levels tend to reduce when serum albumin falls below the critical level.

Glycogen in liver can be readily converted to glucose; but muscle glycogen cannot be released into blood due to lack of glucose phosphatase enzyme. Muscle glycogen is utilized for energy production in the muscle only.

4. Dysadaptation in Kwashiorkor

In kwashiorkor, catabolism is not to the same extent as in marasmus and **anabolism is very low.** Thus, homeostasis is not maintained and this results in fatty liver and oedema (Fig. 4.6 and 4.7). Essential amino acid pool, the anabolic hor-

Table 4.14 Hormones in PEM

Hormone	Marasmus	Kwashiorkor	Remarks
GH	N/H	VH	Low IGF & somatomedins
Glucocorticoids	VH	H	Stress, infections
Insulin & IGF	N	L	(3-cel 1 dysfunction
Somatomedins	L	VL	Altered amino acid pool
Glucagon	N/V	N/V	a/P cell ratio
Thyroxin	N/V	N/V	Low T_4 to T_3 conversion, Iodine malabsorption

H - high, **VH** - very high, **L** - low, **VL** - very low,
N - normal, **V** - variable

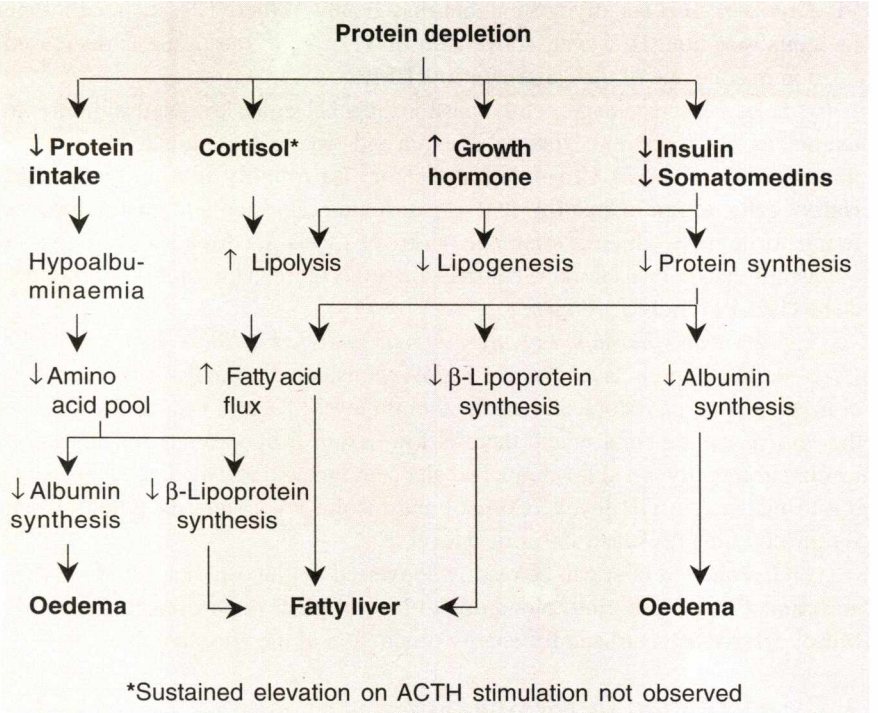


Fig. 4.6 Role of hormones in kwashiorkor

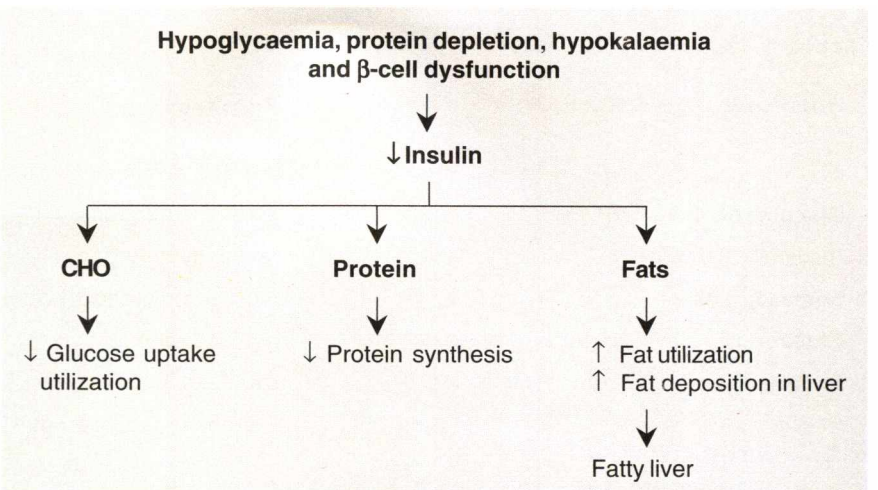


Fig. 4.7 Role of insulin deficiency in kwashiorkor

mones including insulin, insulin-like growth factors and somatomedins are low. Hence protein synthesis, especially that of albumin and b-lipoprotein, is very low and gross hypoalbuminaemia produces oedema. Mobilization of fatty acid from subcutaneous tissue mediated by cortisol and growth hormone produces fatty acid flux into the liver via the blood stream, but they fail to be transported out as b-lipoproteins are low. Moreover, lipogenesis is low as the anabolic hormones are low. Thus, fatty liver results and it starts in the periphery and proceeds to the center. Vanadium deficiency, free radicals and potassium depletion also add to the pathology.

5. Pathogenesis of Oedema in Kwashiorkor

Hypoalbuminaemia decreases plasma volume, cardiac output and glomerular filtration rate and thus decreases filtration of sodium and water. Decreased arterial BP and low peritubular hydrostatic pressure mediates increased tubular reabsorption of sodium and water. Increased reabsorption of sodium and water is also facilitated by the renin-angiotensin-aldosterone cycle triggered by the decreased renal blood flow.

Increased ADH and its decreased inactivation by the liver also produce water retention. **Ferritin is also said to have ADH-like action leading to water retention. Vanadium deficiency and free radicals are other causes.** Thus the pathogenesis of oedema in kwashiorkor is multifactorial (Fig. 4.8) and hypoalbuminaemia is only one of the causes for oedema.

6. Pathology

The pathological changes are summarised in Table 4.15.

Table 4.15 Pathological changes in PEM

Item	Remarks
Proteins	- Reversal of albumin/globulin ratio - Increased NE/E amino acid ratio; > 3.5 in kwashiorkor
Carbohydrate	- Low glycogen, hypoglycaemia—often asymptomatic
Lipids	- Increased NE/E fatty acid ratio; > 3 in Kwashiorkor
Electrolytes	- Normal/high sodium, low potassium
Water	- Increased TBW; high ECF/ICF ratio
Minerals	- Low Ca, P, Mg, K

NE - Non-essential, E - Essential, TBW - Total body water

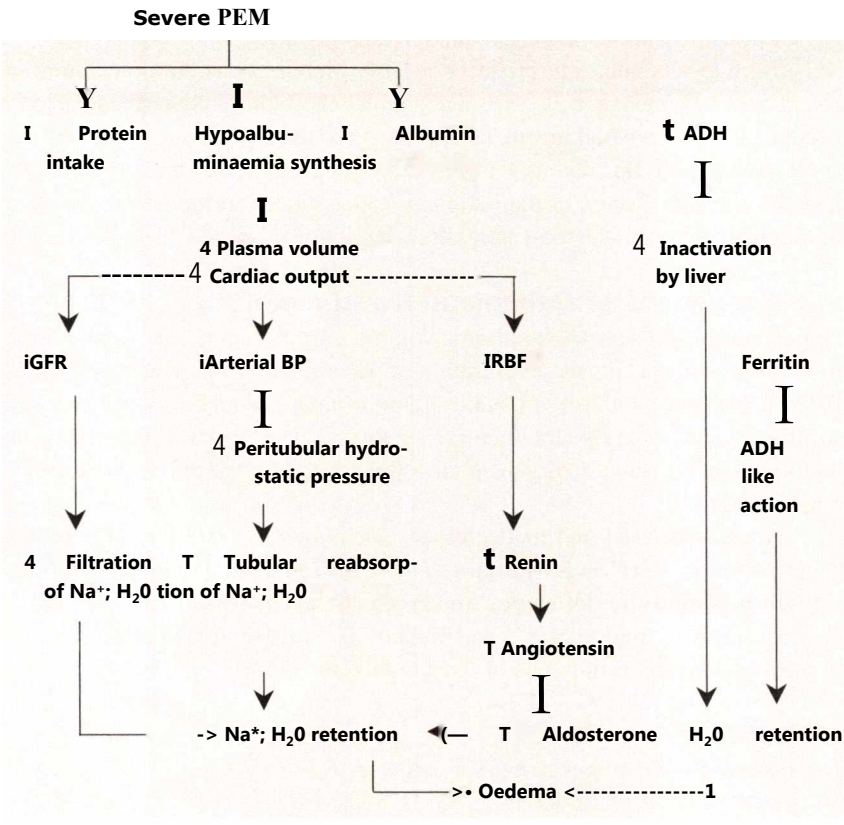


Fig. 4.8 Pathogenesis of oedema

A. Proteins

Serum proteins are low. Serum albumin is low and serum globulin is almost normal. Serum albumin is often maintained in marasmus. In kwashiorkor, the albumin/globulin ratio is reversed.

a) *Albumin:* The fall in serum albumin is seen only about 3 weeks later due to some adaptations. Albumin pool may decrease by 50%. The adaptations to keep up serum albumin are:
   i) Shift from extravascular pool to intravascular pool.
   ii) Albumin catabolism decreases by 50%.
   iii) Half-life of albumin increases.
   iv) Urinary nitrogen decreases.
   v) Protein synthesis first decreases in the muscles and later in the liver.

   When serum albumin level falls below 3 g%, oncotic pressure starts falling and mooning of face and fatty change in the liver start. **It is reported that**

when serum albumin falls below 2 *g%,* **mortality increases from 4% to** 63%. In one of our studies in those with serum albumin above 3 g%, the mortality was 2%; in the 2-3 g% range, the mortality was about 25%; and below 2 g%, the mortality was 50%. But nowadays since marasmus is more common with low normal S. albumin level, other predictors of mortality have to be looked into. In serum protein electrophoresis, low gamma globulin fraction has been identified as a predictor of mortality.

b) *Globulin:* Total globulin may be normal, but in serum protein electrophoresis, the various fractions may show alteration.

   i) **a,-globulins, a,-globulin band may be increased in PEM.** They are the acute phase reactants and are given priority in synthesis when there is infection, a,-antitrypsin, a-fetoprotein, o^-acid glucoprotein, trypsin, plasmin, thrombin etc., are some of the a,-globulins.

   ii) **a,-globulins may be low.** These are the binding proteins like caeruloplasmin, haptoglobin, a,-antithrombin, zinc, a,-glucoprotein etc.

   iii) **P-globulin:** (3-globulins are generally low. They are the (3-lipoprotein, transferrins, hemopexin, etc. Serum transferrin below 0.45 mg% indicates severe PEM.

   iv) **y-globulin** are usually increased. Rarely, some may have severe immunological derangement especially in terminal cases and in them low y-globulin may be seen.

c) *Amino acid pool:* It comes down by 50%. In kwashiorkor, the non essential/ essential amino acid ratio increases. Normally, it is 1.5 and in kwashiorkor it is above 3.5. Branched chain amino acids (BCAA), threonine and tyrosine also decrease. BCAA are valine, leucine and isoleucine.

Plasma aminogram in PEM: Plasma amino acid pool may decrease up to 50% in severe PEM. In kwashiorkor, the essential amino acids especially branched chain amino acids and threonine reduce and there is a fall in essential to non-essential ratio. Insulin decreases amino acid pool by transporting them into the muscle for protein synthesis. Cortisol leads to catabolism of protein and increases amino acid pool. Cortisol and growth hormones (GH) are complementary in maintaining blood glucose; but are antagonistic in their effect on protein metabolism. GH stimulates the passage of amino acid into the tissue as part of its anabolic effect. The level of alanine is also found to be low in kwashiorkor. In prekwashiorkor states, it is high as it is neither utilized for gluconeogenesis nor metabolized to urea. But when anorexia and kwashiorkor set in, it falls as it is utilized as a glycogenic substrate. **Lysine, phenylalanine, tyrosine and histidine levels are maintained in kwashiorkor. Alanine levels reflect a balance between the opposing effects of protein and energy deficiencies.** The low branched chain amino acid and alanine levels are found to have high correlation with GH levels.

d) *Enzymes:* All enzymes are low normal or low. Muscle enzymes fall much before serum enzymes. Protein deficiency leads to decreased urea production and decreased hydroxyproline excretion. Creatinine excretion is also decreased as the muscle mass is low. Creatinine is generally related to the height of the child and a ratio is calculated between creatinine excreted by the child and a normal child with the same height **(creatinine height index).**

B. Carbohydrates

Glycogen reserves in liver and muscle get depleted and there is chance for hypoglycaemia. Blood sugar below 40 mg% is regarded as hypoglycaemia, but it may be often asymptomatic. Below 30 mg% it is often symptomatic. 70% of glycogen is derived from pyruvate and lactate, 20% from glycerol and below 10% from gluconeogenic amino acids.

C. *Lipids*

In serum fatty acid pool, the essential fatty acids decrease in kwashiorkor. The non-essential/essential fatty acid ratio increases above 3.

Fatty liver: 40% of body fat gets deposited in liver and 40% of the wet weight of liver may be due to fat in kwashiorkor.

Serum lipids: Serum lipids may be low normal or high and is inversely proportional to the extent of fatty liver. Defatting stage may show high values. Lipogenesis makes the levels gradually normal. However, low serum phospholipids and total lipids have been observed as probable predictors of mortality.

D. Electrolytes

Total body sodium increases and potassium decreases. However, hyponatraemia may be seen in some children with water retention **(dilutional hyponatraemia).** Very low serum sodium is said to be a bad prognostic sign.

E. Water

Total body water increases and there is an increase in ECF/ICF ratio. The extracellular fluid increases and the water balance becomes similar to that of the newborn baby.

F. Minerals

Minerals like iron, zinc, magnesium, copper, calcium, phosphorous and trace elements decrease.

C. *Endocrine Glands*

Most of the endocrine glands show general atrophy. This was the initial observation during autopsy among severely malnourished children. Later on when Radio Immune Assay and ELISA were introduced, most of the endocrine functions were

found to be fairly normal in spite of the organ atrophy. Selective beta-cell destruction of pancreas may occur.

H. Heart

Heart usually shows microcardia. But cardiomegaly is seen in some especially among those with CCF, anaemia and probably thiamine and selenium deficiency **(nutritional cardiomyopathy).** ECG will reflect the changes of microcardia, cardiomegaly and electrolyte changes like hypokalaemia.

I. Liver

Liver shows fatty change starting from the periphery to the centre. Coagulation factors may be low and liver function may be altered especially when there is ascending cholangitis. Raised serum bilirubin has been reported as a bad prognostic sign.

). GIT

There may be mucosal atrophy, delayed mucosal repair and disaccharidase deficiency. Cholangitis and ascending pancreatitis are seen in a few cases due to bacterial overgrowth in the upper gut and duodenum.

K. Immune System

Malnourished children have deficient immunological defences. It is comparable to that in AIDS.

a) *Non-specific immunity:* Non-specific mechanisms like mucosal and skin barrier, chemotaxis, bacterial killing, lysozymes, complements, interferon, interleukins II etc., are defective.

b) *Humoral immunity:* It is more or less maintained and hence, immunizations are effective. **But antibody production to diphtheria toxoid may be slightly low.** The low secretory IgA level in malnourished children predisposes to respiratory and gastrointestinal infections.

c) *Cell-mediated immunity:* **It is low in malnourished children.** Thymic atrophy, lymphopenia and reduced T helper to suppressor cell ratio have been documented. Mantoux and other skin tests may show false-negative result in malnourished children. There may also be delayed positivity. Some recent evidences suggest that infection may be the initial problem causing deterioration in host nutrition and immunity. This is more so in those with normal birth weight.

L. PEM, Morbidity and Mortality

PEM is the most important and basic hurdle in the **'triple-M complex'** of **malnutrition, morbidity and mortality. PEM** and infection are the two major public health problems. **PEM** is particularly serious and rampant during the post weaning period. **Malnutrition begets infection and infection begets malnutrition.**

Both are more common among poor children. The vicious cycle of malnutrition and infection often sets in following diarrhoea and measles. More than three-fourth of children requiring hospital admission are found to be malnourished. Very high mortality has been reported in severe PEM. Some studies have in fact brought out a **'threshold effect' in PEM,** which is responsible for the high mortality in severe PEM. Weight below 70% of the expected is considered high risk for morbidity and mortality. PEM is found to account for about four million deaths in children. **It is still the first killer disease (54%) followed by acute respiratory infection (20%) and diarrhoea (18%) in the global perspective.**

M. Immunity and Cytokine Response in Malnutrition

The World Health Organization defines malnutrition as "the cellular imbalance between supply of nutrients and energy and the body's demand for them to ensure growth, maintenance, and specific functions." It has been recognized that malnutrition is the most common cause of immunodeficiency worldwide. Actually, malnutrition and infection interact in a vicious cycle and the presence of one easily leads to the development of the other. There are several mechanisms involved in this relationship. PEM impairs cell-mediated immunity, phagocytic function, and the complement system. It also diminishes immunoglobulin (IgA, IgM, and IgG) concentrations and cytokine production. Micronutrient deficiencies associated with PEM also adversely affect the immune response.

In response to infection, the immune system first executes innate and then subsequently acquired host defense functions of high diversity. Both processes involve activation and propagation of immune cells and synthesis of an array of molecules requiring DNA replication, RNA expression and protein synthesis and secretion, and therefore consume additional anabolic energy. Mediators of inflammation further increase the catabolic response. Nutritional status of the host critically determines the outcome of infection

PEM is a common cause of secondary immune deficiency and susceptibility to infection in humans. Apart from deficiencies in single nutrients, such as vitamins, essential fatty acids, amino acids, iron, and trace elements, undernutrition greatly increases susceptibility to infectious diseases, especially children belonging to the lower socioeconomic strata. Malnutrition is responsible, directly or indirectly, for 54% of the 10.8 million deaths per year in under five children and contributes to every second death (53%) associated with infectious diseases in developing countries. Infection causes energy loss on the part of the individual, which reduces productivity on the community level and perpetuates the alarming spiral of malnutrition, infection, disease, and poverty.

Severe malnutrition during childhood affects thymic development, which compromises immunity in children by a long-term reduction of peripheral lymphocyte counts. This immunodeficiency represents a key factor in susceptibility to infections and has therefore been termed nutritionally acquired immunodefi-

ciency syndrome. In severely malnourished patients, both acquired immunity like lymphocyte functions as well as innate host defense mechanisms like macrophages and granulocytes are affected. Diminished immune functions render undernourished patients more susceptible to infections. In addition to promoting acute and chronic infections, PEM impairs the linear growth of children, leading to a further reduction in food intake, nutrient absorption, direct and catabolic nutrient losses and increased metabolic requirements. It has been suggested that acute phase response and proinflammatory cytokines directly affect the bone remodelling required for longitudinal growth. Stimulation of an immune response by infection increases the demand for energy and associated substrates, leading to a vicious cycle of adverse nutritional status and increased susceptibility to infection. Inflammatory conditions such as mediators of sepsis increase the catabolic disease state.

Cytokines are substances that play an important role in coordinating the inflammatory response of the body to various external and internal stimuli. There are two classes of cytokines: Proinflammatory and anti-inflammatory. The proinflammatory cytokines are essential to initiate defense against various pathogens. In certain conditions, there is overproduction of the proinflammatory cytokines resulting in counterproductive effects. The anti-inflammatory cytokines downregulate the inflammatory process by suppressing production of the proinflammatory cytokines and help to balance the inflammatory response. Similarly excess secretion of anti-inflammatory cytokines may have deleterious effects on organ function. The proinflammatory cytokines include IL-1B, IL-6, IL-8, TNF-a, and IL-2, and the anti-inflammatory cytokines include IL-1 receptor antagonist, IL-4, IL-10, and IL-13. Cytokines are also capable of autoregulation; IL-10 and IL-4, produced in response to pro-inflammatory cytokines, suppress the production of other pro-inflammatory cytokines. An excessive pro-inflammatory cytokine response is manifested by SIRS (systemic inflammatory response syndrome) which is balanced by a compensatory anti-inflammatory response syndrome (CARS) reflected in the increased production of IL-4 and IL-10. An uncontrolled inflammatory response due to an imbalance between the pro-inflammatory mediators like free radicals, and anti-inflammatory mediators like free radical scavengers like zinc, selenium etc. plays an important role in the development of the clinical syndrome of malnutrition. Therefore, nutritional modulation that affects both sides of SIRS/CARS equation should hold greater promise in treatment of malnutrition.

Usually children are hospitalized under such circumstances and given milk products as therapeutic food. It has been shown that addition of curd/yogurt and leaf protein concentrate (LPC) can be beneficial in tackling the problem of malnutrition and in improving immunity and achieving immunomodulation. It has also been shown that cytokines (TNFa, IFNy, IL-10, IL-4) may serve as biological markers to assess the effect of functional foods like curd or LPC on immunity in

malnutrition. Curd and LPC help to maintain the balance in cytokine production by increasing the production of pro-inflammatory and anti-inflammatory cytokines.

In the context of what is known as the 10/90 gap areas, i.e., 10% of global health research funding is being targeted to health problems that account for 90% of the global disease burden and hence, research on infection and malnutrition are highly warranted for scientific, economic, and ethical reasons.[1011] To conquer malnutrition, cost-efficient and practical approaches need to be established. Measures to counteract acute malnutrition and immunosuppression are the need of the hour.

EFFECT OF PEM ON MENTAL DEVELOPMENT

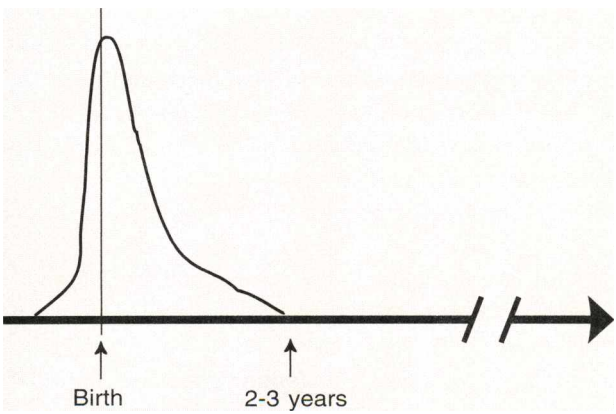
1. Effect of PEM on Nervous System

The effect of PEM on brain is largely restricted to animal models and a few autopsy studies in malnourished children. The slow reduction in brain weight compared to the body weight and an increased brain weight to body weight ratio noted in malnourished children led to the earlier concept of **'brain sparing'** in PEM. Animal studies have shown **reduced brain weight, cell count, synaptic connections, myelin content etc., as an effect of malnutrition during the period of brain growth.** Cerebellum which develops later showed more marked changes. However, the applicability of these studies to the human situation is questionable. The physiology, biochemistry, the period of gestation, the period of brain growth and the timing of birth in relation to maturity vary in humans. In the growth velocity curves, the age axis is measured in days in rats, weeks in pigs and months in humans. It is difficult to malnourish human foetus till mid-gestation unlike in animal models. **The number of neurons becomes complete by mid-gestation.** But neuronal migration, glial proliferation, cerebellar growth and myelination continue during the first two years of life. Evidence so far suggests that the changes are somewhat similar to those reported in animals. **Reduction in glial cells, DNA, RNA, protein, total lipid, cholesterol, phospholipid and myelin content of brain has been reported in PEM.** Adverse effects on tissue growth, developmental differentiation, neuronal migration, myelination, synaptic transmitters, tissue enzymes, dendritic arborization and axonal orientation have also been demonstrated in PEM.

The considerable reduction in dendritic arborization in PEM is depicted in Fig. 4.9. *Maximum insult occurs during the period of maximum brain growth.* The **'critical period' of brain growth** extends from mid-gestation through the early pre-school years. The critical period of brain growth is summarized in Fig. 4.10. It is the period of neuroglial proliferation. **By two years of age, there are billions of cells in the cortex and each cell may have up to 14,000 connections.** The neuronal pattern at birth and the neuronal connections at two years are depicted in Fig. 4.11. The weight of the brain is 350 g at birth, i.e., 25% of the adult



Fig. 4.9 Connections of the neuron in a one-year-old normal child and a child with severe marasmus



Critical period

<--------▶

Neurons divide

----------▶

Cerebellar neurons divide

■*---------- //—*►

Fig. 4.10 Critical period of brain growth

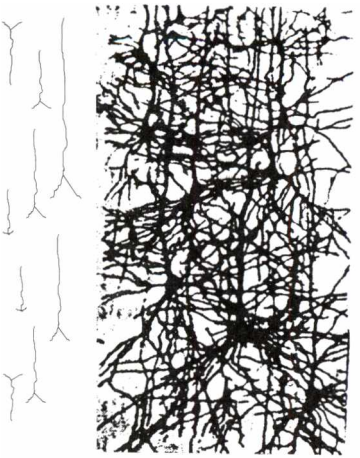


Fig. 4.11 Dendritic arborization at birth and at 2 years of age

size. By 6 months, it becomes 50%, by 2 years it becomes 75%, by 5 years it becomes 90% and by 10 years it becomes 95% of the adult brain. The adult brain weighs about 1234 g. *During the critical period, the brain has biosynthetic abilities that do not persist into later life.* The germinal cell population for neurons becomes inactive after early development, making it impossible to generate new neurons, if substrate becomes available only after the critical period is over.

In the spinal cord, reduction in the number and size of the anterior horn cells and degenerative changes have been demonstrated. This has been termed 'kwashiorkor myelopathy' by Udani. Motor nerve conduction velocity is found to be low in PEM. All the changes noted in the peripheral nerves due to acute malnutrition have been shown to be completely reversible on rehabilitation. This is by virtue of the process of continuing myelination.

The head circumference which correlates with brain size, is found to falter during the period of malnutrition to some extent. It has been shown to be related to nutritional as well as socioeconomic status of the child and maternal head circumference and nutritional status. The recent concept is that the head growth is controlled more by genetic factors than by environmental factors, including nutrition. There is almost 95 percent chance of becoming retarded if head circumference is outside 3rd and 97th centiles. The adverse effects on the brain have also been documented by EEG studies, CT and MRI scans. EEG may show slowing of the waves, and CT and MRI scans show cerebral atrophy.

Reduction in head circumference, developmental quotient, motor nerve conduction velocity, brainstem auditory evoked potentials (BAEP) and CSF cho-

lesterol and phospholipid content have been demonstrated in various studies in our centre. Such screening tests are of value only if they result in appropriate intervention. The old theory of 'brain sparing' in PEM has been discarded and the present concern is the **vulnerability of the central nervous system to malnutrition and the quality and degree of detriment and also its reversibility.** The longer the duration of PEM and more unfavourable the environment, the greater will be the adverse effect on the growing brain.

2. Long-term Effects of PEM on Growth and Intelligence

Catch up growth in weight is almost always satisfactory with intervention but the catch-up in height tends to fall short of well nourished controls. Continuing malnutrition and late intervention are responsible for permanent stunting of growth. The head circumference also continues to remain reduced. In a cohort study from a backward community, nearly 50 per cent of children were found to deteriorate in their nutritional grade and the mortality was found to be three per cent during a one-year follow up without any intervention.

In animal studies by Dobbing, malnutrition during the 'critical period' of brain growth has been shown to cause permanent deficit in the number of cells, DNA and cholesterol content of brain and a permanent reduction in the head circumference. **Cerebellum was found to be most vulnerable.**

On the basis of the concept of **'critical period',** one might expect that younger the child, the greater would be the ultimate effect. But available data from humans provide very little support to this concept. This is due to the reversibility and biosynthetic abilities during this period.

Three points are to be considered while discussing the long-term effects of PEM, *the permanent impairment of the growth potential that may pass on to the subsequent generation, the adaptive and catch up processes within the individual and the interfering variables from outside that may maintain, enhance or reduce the defects.* Positive influences that arise from intervention tend to reduce the effects. Negative influences that arise from persisting malnutrition and deprivation tend to enhance the effects.

It is universally agreed that children with PEM from an environment of deprivation have developmental delay and they perform significantly less well when compared with well-nourished control children from a reasonably good environment. However, the controversy is on the degree of deficit attributable to PEM alone. Malnutrition is one of the important factors in the total adverse environment that influences intelligence, but a direct one-to-one causal relationship with intelligence is yet to be established. **PEM is known to magnify** thu **adverse effects of socioeconomic deprivation.**

Stunting of growth has been correlated with poor performance in some studies. But the genetic endowment and environmental factors that influence both stature and mental performance were not considered in most of the studies.

In a study on aboriginal children in which the socioeconomic factors were matched, *the tall children performed better on intelligence testing.* Several short-term prospective studies have shown that children who suffered from PEM scored less in cognitive tests than the controls. But in most studies, the duration of follow-up was not long enough to exclude the possibility of eventual catch-up. However, some studies have shown that permanent stunting of performance parallels physical stunting. There was no significant difference in the extent of the deficit during the age range 11 to 17 years, indicating no further catch up during this period of follow-up.

Significant deficit in the performance of motor coordination tasks has been demonstrated, that may parallel the clumsiness reported in animal models due to cerebellar involvement. In PEM, general reasoning and perceptual abilities and sensory integration seem to be most severely affected followed by short-term memory and learning skill and lastly language ability. Chronic PEM is thought to have various effects during school age like withdrawal from competitive situations, low exploration, low social interaction, timid and anxious behaviour, poor impulse control, poor frustration tolerance and erratic activity.

In general, PEM leads to decrease in physical and mental growth. It produces functional isolation referred to as **'transient autism''** and reduces the ability to explore and master the environment. It leads to irritability, apathy and decreased ability to adapt. It also increases morbidity and mortality. Almost 50% of the malnourished children are unable to complete school education. Malnourished children show decreased overall academic performances.

Clumsiness, poor motor response, poor planning and directionality and poor sport activities are disadvantages noted in **gross motor development.** Decreased motor skills, decreased organisation of movements, poor recognition of images, decreased ability to explore, to write and draw, decreased differentiation of shapes and letters and decreased ability to transfer information from one sensory form to another are the hurdles in **fine motor and adaptive development.** Decreased ability to communicate and to make sentences, poor spelling and writing skills, decreased perception and ability to understand in spite of repeated telling and decreased ability to follow concepts and instructions are important disadvantages noted in **language development.** Regarding **personal social development,** they tend to have behavioural problems, functional isolation, high dependency, poor social interaction, concentration, memory and discussion abilities. They are withdrawn, anxious and lack self esteem and self-confidence. But in clinical practice, gross motor delay is found to be more in malnourished children, which may catch up with good nutrition.

MANAGEMENTOFPEM

PEM is a complex syndrome of nutritional deficiency often coupled with infection. Over 80% of the underfives are estimated to be malnourished and around

10% suffer from severe malnutrition. Around 25% hospital beds are occupied by patients whose major problem is malnutrition. In more than 50% of deaths in children, malnutrition is the direct or indirect cause. PEM is most rampant in the weaning and post weaning period. PEM is a silent killer in many children. Care of children with PEM includes (1) Ambulatory care of mild and moderate PEM and (2) Hospital care of marasmus and kwashiorkor (severe PEM).

1. Assessment

PEM is a disease of multideprivation and hence history taking should include socioeconomic background, family size, child spacing, emotional deprivation, immunization status, size at birth, customs, cultural practices, feeding practices, superstitions, food fads etc.

Anthropometry should be evaluated with due importance to weight for age, height for age, weight for height, body mass index (BMI), mid arm circumference, chest circumference and head circumference. The measurement should be compared to NCHS reference standards. The Wellcome Trust classification is a simple clinical classification to categorise malnourished children depending on the child's weight. The degree of stunting and wasting can be assessed based on height for age and weight for height respectively. Development should be assessed with due importance to gross motor, fine motor adaptive, language and personal social milestones. Clinical evaluation should include wasting, oedema, apathy, hair changes, skin changes, mucous membrane changes, vitamin and mineral deficiencies, hepatomegaly, infections and complications like dehydration, hypothermia, hypoglycaemia, shock, heart failure etc. (Table 4.12)

Marasmus is the commonest type of severe PEM. Extreme wasting with just skin and bones and old man appearance are the typical features. As the appetite is generally good in marasmus, treatment is more easy and effective. **Marasmic kwashiorkor is more difficult to treat** as the appetite is poor and homeostasis is more altered. Appearance of oedema in a marasmic child leads to marasmic kwashiorkor. **Kwashiorkor is the worst form of severe PEM with gross oedema and florid mental changes.** The appetite is very poor, homeostasis is grossly altered and children with kwashiorkor often present with life-threatening medical emer-gencies. A **comparison between marasmus and kwashiorkor is given in Table 4.16.** In marasmus there will be steady weight gain on treatment, but in kwashiorkor, there will be initial weight loss follwed by weight gain. This is called 'tick' sign.

In marasmus, exclude other causes of failure to thrive (FTT) like inborn errors of metabolism and in kwashiorkor exclude other causes of oedema. In severe stunting, rule out other causes of stunting. **The causes of growth retardation are the following:**

1. Racial/genetic
2. Intrauterine growth retardation (primordial/syndromic malformations)

Table 4.16 Comparison between marasmus and kwashiorkor

Item	Marasmus	Kwashiorkor
Appearance	Old man appearance, skin and bones, generalized wasting	Mooning of dependent upper limb
Age group	Infants	1-5 years
Prevalence	Common	Rare
Weight	< 60%	60-80%
Growth retardation	+ +	+
Oedema	Nil	+ +
Apathy	Nil/mild	+ +
Mood	Usually alert	Irritable
Appetite	Good	Very poor
Hair changes	Nil/mild	+
Skin changes	Nil/mild	+
Fatty liver	Absent/mild	+ +
Infections	+	+ +
Life-threatening medical emergencies	+	+ +
S. protein & albumin	Low normal	Very low
Carrier proteins	Low normal	Very low
Anabolism	+	Very low
Catabolism	+ +	+
Response to treatment	Good	Poor

3. Nutritional
4. Emotional deprivation
5. Skeletal disorders
6. Metabolic disorders
7. Endocrine disorders
8. Chromosomal disorders
9. Constitutional delay
10. Chronic systemic disorders

2. Investigations

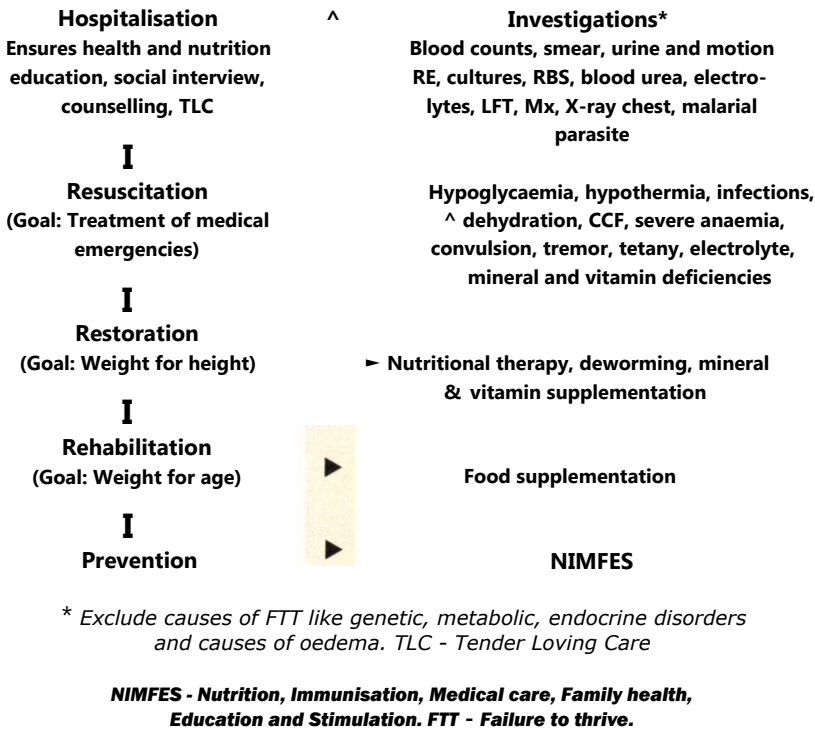
Blood: Hb, counts, peripheral smear, blood sugar, urea, electrolytes, serum protein, albumin, blood culture. Mantoux test, X-ray chest; urine routine examination and culture and motion routine examination, sugar and culture are the investigations that are needed for management. Liver function test, renal function test, lumbar puncture etc., may be done whenever indicated.

3. Management

Care of children with severe PEM is quite a challenging task as they often present with life-threatening medical emergencies. The various steps involved are given in Fig. 4.12.

A. *Resuscitation*: The initial resuscitation management should depend upon the clinical presentation. Life-threatening medical emergencies should be managed immediately. Management of associated complications is summarised in Table 4.17.

a) *Hypothermia:* Hypothermia, hypoglycaemia and Gram-negative sepsis often go together. Keep the child warm, covered up, use a goose neck lamp if needed and practise 'bedding-in' with the mother. Start early feeding and start antibiotics.



* *Exclude causes of FTT like genetic, metabolic, endocrine disorders and causes of oedema. TLC - Tender Loving Care*

NIMFES - Nutrition, Immunisation, Medical care, Family health, Education and Stimulation. FTT - Failure to thrive.

Fig. 4.12 Various steps in the management of PEM

Table 4.17 Management of complications in PEM

Complications	Management
1. Hypoglycaemia	10% glucose 5 ml/kg IV, 10% glucose drip* (2/3 maintenance in oedema). Early and frequent feeding
2. **Hypothermia**	Warm bed, bedding-in with mother, double clothing, treatment of hypoglycaemia & sepsis
3. **Septicaemia/ infection**	Ampiciilin + gentamicin. Add cloxacillin SOS
4. **Dehydration and electrolyte imbalance**	ORS 70-100 ml/kg in 4 hr. In severe dehydration, IVF Ringer Lactate 100 ml/kg in 3-6 hr (in oedema, restrict fluid; half as RL and half as 10% glucose), KCI
5. **CCF**	Frusemide 1-2 mg/kg (avoid digoxin). Add KCI 1-2 mEq/kg & spironolactone 2-3 mg/ day for prolonged therapy
6. **Severe anaemia**	Packed cell 5-10 ml/kg and frusemide, iron 6 mg/kg/day x 3 months
7. **Convulsion, tremor**	IV 10% glucose, 10% Ca. gluconate 0.5-1 ml/kg, 50% $MgS0_4$ 0.2 ml/kg IM ql2h x 2-3 days, vitamin $B_{1(}$ B_6, B_{12}, folic acid
8. **Vitamin & mineral deficiencies**	Vit. A** 1-2 lakhs IU orally and repeat after one day and after 1 month, vit. A prophylaxis prog., calcium, phosphorus, zinc, iron, multivitamins

** Holliday & Segar formula: 100 ml/kg for 1st 10 kg, 1000 + 50 ml/each kg above 10, 1500 + 20 ml/each kg above 20.*

***In keratomalacia, 5000 IU/kg vit. A x 5 days and then 25,000 IU/day till recovery. (Nelson, Textbook of Pediatrics. 15th ed.)*

b) *Hypoglycaemia:* In sick children, give 10% dextrose 4 ml/kg stat followed by 10% dextrose as maintenance fluid with 1 mL KCI for each 100 ml of IV fluid (Table 4.17). Use Holliday and Segar formula for calculation of maintenance fluid. Initiate oral feeding as early as possible.

c) *Septicaemia/infections:* Give broad-spectrum antibiotics, crystalline penicillin or ampicillin with an aminoglycoside. Cloxacillin should be added to the cocktail if there is suspicion of staphylococcal infections, i.e., abscess, boils or carbuncles. Malaria may present as diarrhoea without fever.

d) *Dehydration and electrolyte imbalance:* In children with diarrhoea, if there is no dehydration, select plan A therapy according to DTU protocol and give half to one glass ORS or home available fluid (HAF) to prevent dehydration after each stool. In mild to moderate dehydration, select plan B and give 70-100 ml/kg ORS in 4 hours, sip by sip or as IG drip. In severe dehydration select plan C and give 100 ml/kg N Saline or Ringer Lactate (RL) in 3 to 6 hours. It is given in 2 rations: 30 ml/kg in *Vi* an hour in children and 1 hour in infants and 70 ml/kg in 2Vi hours in children and 5 hours in infants. In oedematous children, skip the first ration of 30 ml/kg and give only 70 ml/kg as a slow ration. Half the calculated fluid can be given as NS/RL and the rest as 10% glucose. This will tackle hypoglycaemia and will reduce sodium intake. They usually have hypokalaemia and so add KCI 1 to 2 ml to each 100 ml of IVF. Potassium supplement, 2-4 mEq/kg/day may be continued for 1-2 weeks in kwashiorkor. 15% IV KCI will give 2 mEq/ml and 1.5 ml oral KCI will give 2 mEq. Fruit juice can be encouraged to supplement potassium. Sodium should be restricted to 2-3 meq/kg/day. If there is evidence of acidosis, give sodium bicarbonate 2 mL/kg IV diluted with equal quantity of distilled water. Resomal is beneficial in PEM (appendix).

e) *Heart failure:* In case of heart failure, restrict fluid intake to two-third of the maintenance and give frusemide 1-2 mg/kg/dose. If diuretic therapy is needed for more than a day, in addition to potassium supplementation, add aldactone 2-3 mg/kg/day. **Oedema of kwashiorkor is not treated by diuretics.**

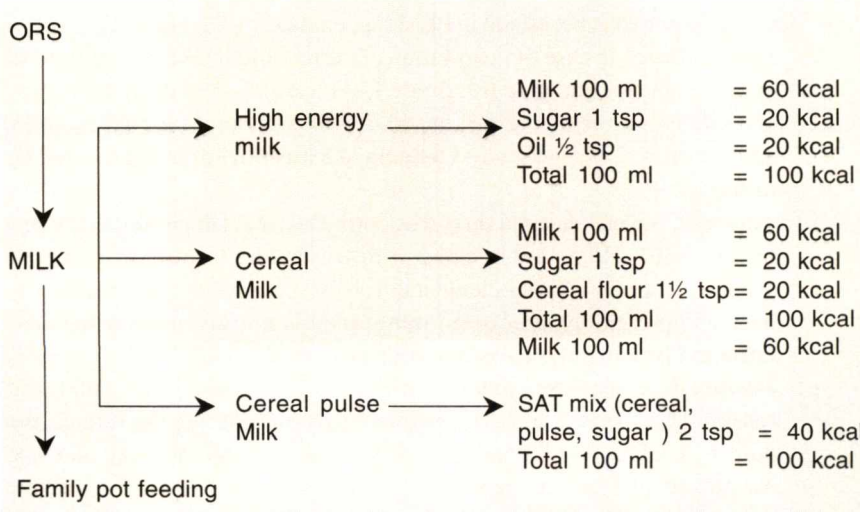
f) *Anaemia:* Severe anaemia should be corrected by small packed cell transfusion, 5-10 ml/kg slowly along with frusemide. In mild to moderate anaemia, start 2-6 mg/kg elemental iron after the patient has been stabilized and preferably after deworming. **Iron is not given early because unbound iron in gut may lead to over growth of** *E. coli.*

g) *Vitamin & mineral deficiencies:* In vitamin A deficiency and in all florid cases of PEM, give 2 lakhs IU vitamin A as a concentrate. For infants, the dose is 1 lakhs IU. The dose may be repeated after one day and one month and also initiate them into the vitamin A prophylaxis programme (also refer Section 5.1). Vitamin K 2-5 mg IM/IV per day may be given for 2-3 days in florid PEM and in those with persistent diarrhoea. All B complex factors, vit. C, vit. E, vit. D, zinc, calcium etc., need to be supplemented. Magnesium is given if there is seizure, tetany or apathy. 50% $MgS0_4$ 0.1-0.2 ml/kg/dose in 2 divided doses is given for 1-3 days. 25%

$MgS0_4$ 1 ml gives 2 mEq, 50% $MgS0_4$ 1 mL gives 4 mEq and oral Mg hydroxide 5 ml gives 2 mEq.

B. *Restoration/nutritional management:* This phase is for restoration of weight for height. This is the second step in the management of severe PEM and often the first step in those with mild to moderate PEM. The various steps involved are depicted in Fig. 4.13.

a) *Calories and protein:* The goal is 150-200 kcal, 3—4 g protein and 100-165 ml fluid intake per kg body weight per day. The calorie requirement is calculated based on the actual weight irrespective of oedema. In practice, it is found to be comparable to the optimum RDA for age as per the ICMR recommendations. Up to 150% of the RDA can be given during nutritional therapy. The RDA for age is advised in a community setting. Accurate weighing and calculation may be difficult in the field setting. The health workers and Anganwadi teachers can easily comprehend the concept of RDA for age both in the well-nourished and malnourished child. Nationwide surveys have shown that the average calorie gap in underfive children is 400 kcal per child. This calorie gap is more among the malnourished. This is bridged through the ICDS programme. Children with severe malnutrition are entitled to get double the ration. The total calorie calculated is to be divided into 6-8 feeds and may be given orally or using a feeding tube. During day time, 2 hourly feeds can be given. A late night



In lactose intolerance, give curd, cereal, pulse, chicken soup and family pot feeding

Fig. 4.13 Nutritional therapy in PEM

and an early morning feed will prevent hypoglycaemia in the night. Tube feeding is necessary in extreme anorexia and apathy,

b) *Feeding methods:* If breast-fed, it should be continued. This will ensure nutritive and non-nutritive sucking and enhance lactation and mother-infant bonding. Both foremilk and hindmilk should be given to the baby. If the baby is below four months and exclusively breast-fed and at the same time malnourished, look for feeding problems like retracted nipple, sore nipple, cleft palate etc. Watch the baby while feeding and look for the let down reflex. Correct local problems and encourage exclusive breast-feeding. Ask the mother to express breast milk after the baby finishes sucking. In some cases, 1-2 gokarnam (palada) milk may be expressed and this should be given to the baby. This will ensure more milk production as well. Weight gain should be monitored every week. If the gain is less than expected, we will have to exhibit **'baby friendliness'** and prescribe extra calories and protein as infant milk substitutes (IMS) like formula milk or cow's milk. Oil supplementation can be tried. Coconut oil with MCT is preferred. **1(>-15% of the total calorie requirement can be given as visible oil** even in newborn babies. This can be added to IMS or semisolids.

In babies above 4 months of the age, in addition to breast milk start giving cereals, legumes (pulses), milk, oil, fruits etc. Milk is usually the best available initial food. 25-30% of the calculated calories may be initiated and slowly increased. ORS followed by dilute milk, whole milk, high energy milk and high energy cereal milk can be given. To make high energy milk, add 1 tsp sugar and *Vi* tsp coconut oil to 100 ml milk in order to give 100 kcal/100 ml. Cereal flour or cereal pulse combination also may be added to 100 ml of high-energy milk to make a thickened feed. A precooked, ready-to-mix cereal-pulse combination can be easily prepared at home by roasting and powdering rice, wheat, blackgram and adding powdered sugar in the ratio 1:1:1:2. This mix can be given as a semisolid, mixed with hot water or hot milk. This mix is a research and development (R & D) product of Nutrition Clinic, Department of Paediatrics, SAT Hospital, Medical College, Trivandrum and is named **SAT mix.** 100 g of this mix supplies 380 kcal and 8 g protein; 6 teaspoon of SAT mix will give 125 kcal and 2.5 g protein. Oil supplementation can be upto 10-15% of the calculated calories. Coconut oil is preferred as it is easily available in small volumes, acceptable and has the advantage of the medium chain triglyceride (MCT). MCT is absorbed even during fat malabsorption. Oil can be given added to the milk, semisolid and solid foods or as such to the child taking care to prevent aspiration of oil. **Isodense formulae that supply 100 kcal/100 ml** are well tolerated (Table 4.18).

Table 4.18 Isodense formula (100 mL = 100 kcal) - F100

Item	Composition	kilocalories
High energy milk	1/2 glass milk, 1 tsp sugar, 1/2 tsp oil	100 kcal/100 ml
Cereal milk	1/2 glass milk, 1 tsp sugar, 1 1/2 tsp cereal flour	100 kcal/100 ml
Cereal pulse (SAT mix)	1/2 glass milk 2 tsp SAT mix	100 kcal/100 ml
Fruit juice	1 orange, 2 tsp sugar, water up to 100 mL	100 kcal/100 ml
Egg flip	One egg, 2 tsp sugar, 3/4 glass milk	200 kcal/200 ml
Curd/Lassee	1/2 glass curd, 2 tsp sugar	100 kcal/100 ml

SAT mix is roasted and powdered rice: wheat: blackgram: powdered sugar in the ratio 1:1:1:2.

In **lactose intolerance,** soya milk may be started in young infants. Continue breastfeeding and give curd, weaning food, chicken soup, oil etc. Family pot feeding can be continued after excluding animal milk temporarily. **Refeeding of the excluded diet may be tried after 1-2 weeks.**

Multivitamins and micronutrients should be started early. Once stable, deworming should be done. Iron therapy is not initiated till carrier proteins are synthesised. Otherwise *E. coli* will thrive on the unbound iron in the gut. As a next step, traditional food items such as rice, idli, upma, tubers, legumes (pulses), payasam, egg, fish, green leafy vegetables, seasonal fruits etc., should be introduced gradually. Ensure **'family pot feeding'** as early as possible. Groundnuts can be given to older children.

A weight gain of 0.5 kg/week in children and 70 g/kg/week in infants is the target. 150-200 g/week is expected in newborns and young infants. But it is difficult to achieve this if intercurrent infections occur frequently. Restoration of weight for height may take about 8-12 weeks. Oedema clears and social smile returns in 1-2 weeks. Return of social smile in severe PEM is an early sign of recovery,

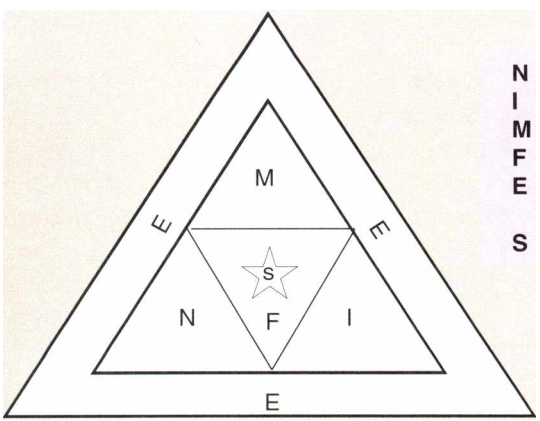
c) *Adjuncts:* Plasma transfusion 10 ml/kg, albumin, Vamin glucose 30 ml/kg, IV Astymin etc., may also be given to sick children. Some may need parenteral nutrition.

C. *Nutrition recovery syndrome:* Apparent worsening with increase in liver size, hypertrichosis, gynaecomastia, parotid swelling, abdominal distension, ascites, splenomegaly and eosinophilia during therapy marks 'nutrition recovery syndrome'. Encephalitis like picture also may occur. It is a self-limited condition and is now attributed to excess hormones secreted during recovery than to protein excess. Self-limited tremors may also occur during treatment **(kwashi shake).** Dysmyelination. vitamin deficiencies, neurotransmitter imbalance and high solute load on the kidneys are the other possible reasons for nutrition recovery syndrome. In practice, the child should be observed and at the same time nutritional therapy should be continued. Refeeding syndrome is another term used with hyperphosphataemia.

D. *Rehabilitation:* The RDA for the age should be given to the child during this phase. Food supplementation from ICDS. immunization, growth monitoring, medical check-up and developmental stimulation should be made available to the children.

4. Prevention

In around l/3rd of children with PEM, it is the sequel of low birth weight. Hence, antenatal care should be emphasized and strengthened. Similarly, the health and nutritional status of adolescent girls should be improved. In others with PEM, it is the sequelae of recurrent infections and lack of feeding. The strategies for prevention can be summarised as 'NIMFES' (Fig. 4.14). N stands for **nutrition** and growth monitoring, I for **immunization,** M for **medical check-up and medical care** during illness, F for **family welfare** (timing, limiting and spacing of births). E for **education** regarding mothercraft, child rearing skills, environmental and personal hygiene and S for **stimulation,** developmental surveillance and tender loving care (TLC).



N - Nutrition
I - Immunization
M - Medical care
F - Family planning
E - Education - health and nutrition
S - Stimulation

Fig. 4.14 NIMFES for comprehensive care of children

4.2 WHO Recommendation for Management of SAM

DIAGNOSIS OF SAM

1. Weight-for-height < 70% of expected.
2. Visible wasting
3. Oedema
4. MAC < 11 cm

Another term for SAM is severe childhood undernutrition (SCU).

WHO GUIDELINES FOR THE IN-PATIENT TREATMENT OF SEVERELY MALNOURISHED CHILDREN (SAM)

A. General principles for routine care (The' 10 steps')
B. Emergency treatment of shock and severe anaemia
C. Treatment of associated conditions
D. Failure to respond to treatment
E Discharge before recovery is complete

General Principles for Routine Care

These steps are accomplished in two phases:

* An initial stabilization phase where the acute medical conditions are managed; and
- A longer rehabilitation phase.
- Note that treatment procedures are similar for marasmus and kwashiorkor.

Step 7. Treat/Prevent Hypoglycaemia

- Blood sugar level <54 mg/dl or 3 mmol/L.
- Assume hypoglycaemia when levels cannot be determined.
- CONSCIOUS CHILD-50 ml bolus of 10% glucose by nasogastric (NG) tube.
- UNCONSCIOUS CHILD, lethargic or convulsing -IVsterile 10%glucose(5 ml/kg), followed by 50 ml of 10% glucose or sucrose by NG tube.
- Start two-hourly feeds, day and night.

Step 2. Treat/Prevent Hypothermia

m If axillary temperature < 35°C, take rectal temperature

- If the rectal temperature is < 35.5°C (< 95.9°F):

Rewarm the child: either clothe cover with warmed blanket and place a heater or lamp nearby or put the child on the mother's bare chest (skin to skin) and cover them - Kangaroo mother care
Feed straightaway

Step 3. Treat/Prevent Dehydration

- Difficult to estimate dehydration using clinical signs alone.
- Assume all children with watery diarrhoea may have dehydration.
- Do not use the IV route for rehydration except in cases of shock.
- Continue feeding.

Assessment of Dehydration in Severely Malnourished Children

Basic format remains the same.
Some signs unreliable:

- Mental state
- Mouth, tongue and tears
- Skin turgor

Oedema and hypovolaemia can coexist.
Diagnosis of dehydration in severely malnourished children

- History of diarrhoea (with large volume of stools)
- Increased thirst
- Recent sunken eyes
- Prolonged CFT, weak/absent radial pulse, decreased or absent urine flow

Difficult using clinical signs alone.
Best to assume that all with watery diarrhoea have some dehydration.
Treat with ORS unless shock is present.

Step 4. Correct Electrolyte Imbalance

- Plasma sodium may be low, though body sodium is usually high. Sodium supplementation may increase mortality.
- Potassium and Magnesium are usually deficient and need supplementation; may take at least two weeks to correct.
- Oedema, if present, is partly due to these imbalances. Do NOT treat oedema with a diuretic.

Step S. Treat/Prevent Infection

m Usual signs of infection, such as fever, are often absent. Give broad-spectrum antibiotics to all.

- Hypoglycemia/hypothermia usually coexistent with infection. Hence if either is present assume infection is present as well
- No complications - Co-trimoxazole
- Severely ill - Ampicillin + Gentamicin
- If the child fails to improve clinically within 48 hours, add: cefotaxime/ ceftrioxone

Status Antibiotics

Infected child or complications present*

. IV AMPICILLIN 50 mg/kg/dose q 6hrly and IV GENTAMICIN 2.5 mg/kg/dose q 8hrly

- Add IV CLOXACILLIN 100 mg/kg/day q 6hrly if staphylococcal infection is suspected.
- Revise therapy based on the culture sensitivity report

For septic shock or No improvement or worsening in initial 48 hours

m Add third generation cephalosporins i.e. IV CEFOTAXIME 100 mg/kg/day q 8hrly

Meningitis

- IV Cefotaxime 200 mg/kg/day IV q 6hrly with IV amikacin 15 mg/kg/day q8hrly

Dysentery

m CIPROFLOXACIN 30 mg/kg/day in 2 divided doses.

- IV ceftriaxone 50 mg/kg/day in od or q 12 hourly if child is sick or has already received nalidixic acid

Step 6. Correct Micronutrient Deficiencies

- All severely malnourished children have vitamin and mineral deficiencies
- Vitamin A orally on day 1
- Give daily:

  Multivitamin supplement
  Folic acid 1 mg/d (give 5 mg on day 1)
  Zinc 2 mg/kg/d
  Iron 3 mg/kg/d after first week

Step 7. Start Cautious Feeding

- Small, frequent feeds.
- Oral or nasogastric (NG) feeds (never parenteral preparations).
- Milk-based formulas such as starter F-75 containing 75 kcal/100 ml and 0.9 g protein/100 ml will be satisfactory for most children.

- 130 ml/kg/d of fluid (100 ml/kg/d if the child has severe oedema).
- If the child is breastfed, encourage to continue breastfeeding.
- A gradual transition is recommended to avoid the risk of HEART FAILURE.
- Monitor during the transition for signs of heart failure.
- If respirations increase by 5 or more breaths/min and pulse by 25 or more beats/min for two successive 4-hourly readings, reduce the volume per feed.

Step 8. Achieve Catch-up Growth

m Readiness to enter the rehabilitation phase is signaled by a RETURN OF APPETITE, usually about one week after admission

- Recommended milk-based F-100contains 100 kcal and 2.9 g protein/100 ml.
- In rehabilitation phase vigorous approach to feeding is required to achieve very high intakes and rapid weight gain of >10 g gain/kg/d.

To Change from Starter to Catch-Up Formula

- Replace starter F-75 with the same amount of catch-up formula F-100 for 48 hours, then,
- Increase each successive feed by 10 ml until some feed remains uneaten.
- The point when some remains unconsumed is likely to occur when intakes reach about 30 ml/kg/feed (200 ml/kg/d)

Step 9. Provide Sensory Stimulation and Emotional Support

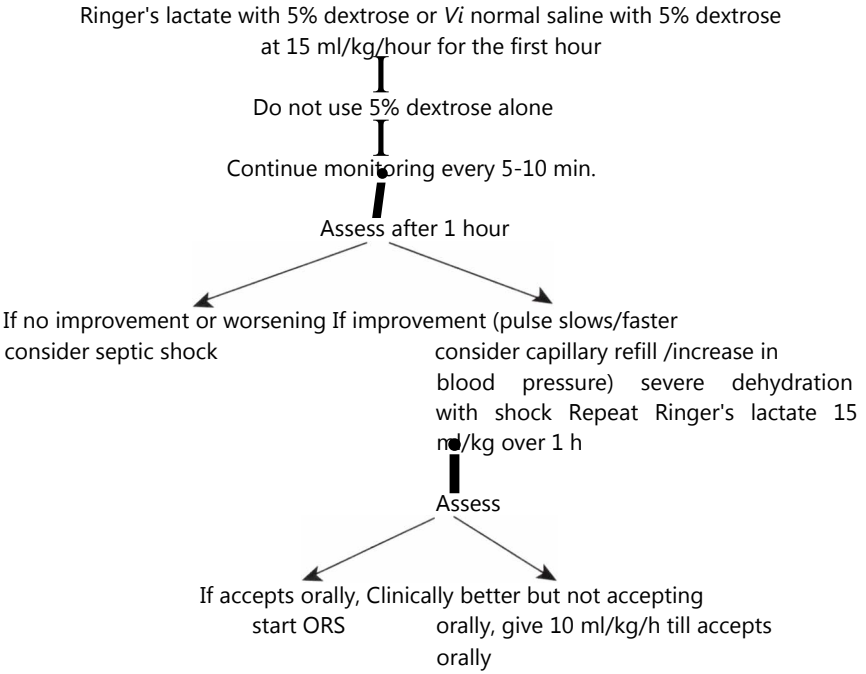
- Delayed mental and behavioral development is present.
- Provide:
  - Tender loving care - return
  - Cheerful, stimulating environment
  - Structured play therapy 15-30 min/d
  - Physical activity as soon as the child is well enough
  - Maternal involvement when possible (e.g., comforting, feeding, bathing, play)

Step 10. Prepare for Follow-up After Recovery

- A child who is 90% weight-for-length (equivalent to -1 SD) can be considered to have recovered.
- Show parent or caregiver how to:
  - Feed frequently with energy- and nutrient-dense foods.
  - Give structured play therapy.
- Advise parent or caregiver to:
  - Bring child back for regular follow-up checks.
  - Ensure booster immunizations are given.
  - Ensure vitamin A is given every six months.

Emergency Treatment of Shock and Severe Anaemia

Fluid Therapy in Severe Dehydration.



Severe Anaemia

- Blood transfusion is required if:

  Hb < 4 g/dl or if there is respiratory distress and Hb 4-6 g/dl

- Give:

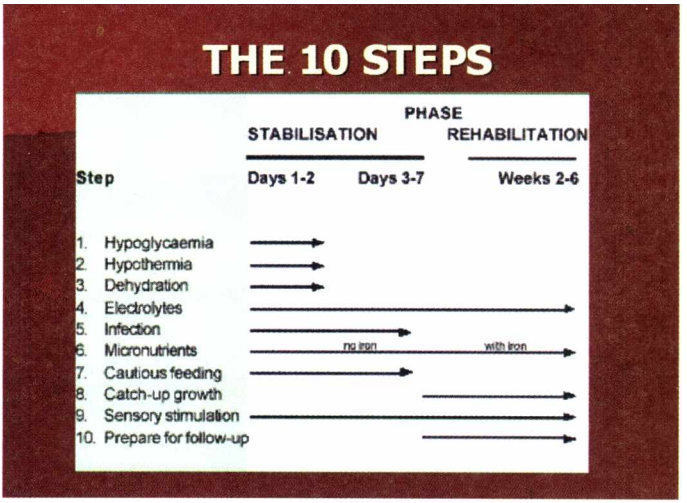
  Whole blood 10 ml/kg slowly over 3 hours
  Frusemide 1 mg/kg IV at start of transfusion

- If CARDIAC FAILURE present, transfuse packed cells (5-7 ml/kg) rather than whole blood

- Monitor RR and HR every 15 minutes. If either of them rises, transfuse more slowly.

Give oral iron for two months to replenish iron stores.

SEVERE ACUTE MALNUTRATION: TREATMENT SUMM ARY



Treatment of Associated Conditions

Vitamin A Deficiency

- If **eye signs** of deficiency, give orally:
  Vitamin A on days 1, 2, 14

>12 months	200,000 IU
6-12 months	100,000 IU
0-5 months	50,000 IU

- If **corneal including/ulceration,** give additional eye care to prevent extrusion of the lens:
  - Instill chloramphenicol or tetracycline eye drops (1%) 2-3 hourly for 7-10 days
  - Instill atropine eye drops (1%), 1 drop three times daily for 3-5 days

Dermatosis

Signs:
- Hypo-or hyperpigmentation
- Desquamation, ulceration, exudative lesions

Zinc deficiency is used in affected children. Skin quickly improves with zinc supplementation

contd

In addition:

- Apply barrier cream (zinc & castor oil ointment, or petroleum jelly or paraffin guaze) to raw areas
- Omit nappies so that the perineum can dry

Continuing Diarrhoea

- Common features but it should subside during the first week of treatment with cautious feeding. In the rehabilitation phase, loose, poorly formed stools are no cause for concern provided weight gain is satisfactory
- Mucosal damage & giardiasis
  - Stool microscopy
  - Give: Metronidazole (7.5 mg/kg 8-hourly for 7 days)
- Lactose intolerance.
  - Only rarely due to lactose intolerance. Treat only if continuing diarrhoea is preventing general improvement
  - Starter F-75 is a low-lactose feed. In exceptional cases:
    - Substitute milk feeds with yogurt or lactose-free infant formula
      - Reintroduce milk feeds gradually in the rehabilitation phase

Osmotic Diarrhoea

- Suspected if diarrhoea worsens substantially with hyperosmolar starter F-75 and
- Ceases when the sugar content is reduced and osmolarity is < 300 mOsmol/L.
- In these cases: use isotonic F-75 or low osmolar cereal-based F-75. Introduce F-100 gradually

Parasitic Worms

- Give mebendazole 100 mg orally, twice daily for 3 days

Tuberculosis (TB)

If strongly suspected (contacts with adult TB patient, poor growth despite good intake, chronic cough, chest infection not responding to antibiotics):

contd

- Mantoux test (false negatives are frequent)
- Chest X-ray if possible
- If test is positive or strong suspicion of TB, treat according to national TB guidelines

Failure to Respond to Treatment

- Good weight gain (> 10 g/kg/day): continue same
- Moderate weight gain (5-10 g/kg/day), check intake and infection
- Poor weight gain (< 5 g/kg/day)
- Inadequate feeding
- Untreated infection
- Specific nutrient deficiencies
- Tuberculosis and HIV/AIDS
- Psychological problems

Discharge

- Recovered/ready for discharge when reaches 90% weight-for-length and no oedema
- Absence of infection
- Eating at least 120-130 cal/kg/day and receiving adequate micronutrients
- Consistent weight gain (of at least 5 g/kg/day for 3 consecutive days) on exclusive oral feeding
- Completed immunization appropriate for age
- Caretakers sensitized to home care
- Return of social smile

Children Discharged Early: What to Do

Recovery complete if 90% W/L;
But can be discharge early for domiciliary if:

- The child: > 1 yr; good appetite and weight gain; no oedema, antibiotic treatment completed. Vit, K, min. given 2 wks
- The mother: available at home, motivated and trained to look after; have resources; reside near hospital.
- Local Health Worker: Can provide support; trained; motivated

Monitoring Feeding at Home Essential:

- Feed frequently at least 5 times a day
- Modify home food to suit F-100
- High energy snacks between meals
- Assistance to complete each meal
- Give electrolyte/mineral solutions
- Breastfeeding should continue

4.3 Obesity & Metabolic Syndrome

Obesity is now a considerable public health problem in most economically advanced countries. The prevalence of obesity in the UK (which is higher than that of most countries in Western Europe), for example, is such that 25% of the adult population is now classified as obese on the basis of a BMI > 30 kg/m^2. The prevalence of the disorder is even higher in the United States, comprising 1 in 3 adults.

The "obesity epidemic" has developed with considerable rapidity over the past 2 decades, with the incidence rising 3-fold in the UK, for example, since the early 1980s[3]. Obesity is associated with a reduction in life expectancy of; 8 y, as well as with an increased risk of several major diseases, including type 2 diabetes, coronary heart disease, and certain cancers (such as breast and colon).

In the case of type 2 diabetes, the risk is particularly marked because being obese increases the likelihood of developing the disease by 10-fold once a BMI of 30 kg/m2 is reached. Furthermore, the greater the degree of obesity, the higher the relative risk.

Although concern with obesity as a health problem has inevitably focused on humans, the disorder and the diseases with which it is associated are also a growing problem in our companion animals. However, in contrast to humans, an objective definition of obesity is lacking in cats and dogs, and the assessment is usually made on the basis of the "body condition score". This, of course, involves a subjective element, unlike BMI, and the cut-off points for normal weight, overweight, and obesity are somewhat arbitrary with no clear reference to the amount of body fat or the threshold for the risk of type 2 diabetes and other associated diseases. As a consequence, estimates of the incidence of obesity in companion animals vary widely; in dogs, for example, these range between 10 and 40% of the population.

We emphasize, however, that the human classification of obesity based on BMI is not without complications. Body builders, for example, have a high BMI in relation to body fat because of their large muscle mass, whereas the threshold for obesity of a BMI of 30 kg/m^2, based primarily on Europeans and North Americans, is now recognized as inappropriate for some other population groups, such as those in South East Asia, where a lower cut-off value is increasingly employed. This is a reflection of the fact that for a given BMI, these populations are likely to have more abdominal fat and are therefore more likely to exhibit the deleterious metabolic consequences that accompany "central obesity", (refer Appendix for lOTF cut-offs of BMI and BMI charts.

There has been a rapid rise in the incidence of obesity, primarily as a result of changes in lifestyle (diet and activity levels). **Obesity has provided considerable impetus for the investigation of the fundamental mechanisms involved in**

the regulation of energy balance. Important developments include the identification of novel factors involved in the control of appetite, such as ghrelin, orexin A, and the endogenous cannabinoids, and the emergence of the concept of "nonexercise activity thermogenesis" (NEAT) provided new perspectives on energy expenditure. Studies on white adipose tissue have led to the recognition that it is an important endocrine organ, communicating with the brain and peripheral tissues through the secretion of leptin and other adipokines.

There is a rapidly expanding list of protein factors released by white adipose tissue, including the key hormone, adiponectin. Of particular note is the range of cytokines, chemokines, and other inflammation-related proteins secreted by white fat as tissue mass rises; indeed, obesity is characterized by chronic mild inflammation. The adipokines provide an extensive network of communication both within adipose tissue and with other organs, and some are implicated directly in the pathologies associated with obesity, particularly the metabolic syndrome. Although the focus remains very much on obesity in humans, the disorder and its sequelae are also a growing concern in companion animals.

Regulation of Energy Balance

The growing concern with obesity has been the main impetus behind much recent research on the regulation of energy balance, reflecting the fact that it has been axiomatic that the disorder is fundamentally a problem of energy balance.

Put simply, **obesity can develop only when energy intake is in excess of energy expenditure, with the differences in input and output buffered primarily by changes in fat stores.** There is an underlying genetic predisposition to obesity, with distinct differences between breeds of dog, for example, in the tendency to become obese.

At a mechanistic level, major developments have occurred recently in the control of energy balance through the identification of novel factors involved in appetite, such as ghrelin, orexin A. and the endogenous cannabinoid system.[9"12] Similarly, important developments in our understanding of energy expenditure have come through the emergence of the concept of "nonexercise activity thermogenesis" (NEAT)5,[13] together with the discovery of new mitochondrial uncoupling proteins (UCP), primarily UCP2 and UCP3[12] However, these novel uncoupling proteins are no longer thought to provide an immediate locus for adaptive thermogenesis in tissues without brown fat.

The primary buffering of energy intake and expenditure is through fatty acid deposition (as triacylglycerols) and release in white fat, an organ that until recently was considered a "poor relation" in energy balance and obesity research. However, this has changed radically over the past few years with the tissue becoming a focus of intense research activity. There are several **reasons** for this change in position: **(1) obesity is defined by the expansion of the tissue, which therefore has to be central in the consideration of the disorder; (2) white**

adipose tissue (WAT) is the primary site of the production of key hormones involved in energy balance, notably leptin; (3) the tissue secretes a number of factors involved in a range of metabolic and physiological processes; some of these factors are implicated in the pathologies associated with obesity, particularly insulin resistance and the metabolic syndrome.

White Adipose Tissue (WAT)

The apparent simplicity of both white adipocytes and of WAT itself, histologically and metabolically, is the key reason why the organ has been relatively ignored until recently. With triacylglycerols constituting up to 85% of tissue weight, it is not surprising that WAT was regarded as essentially limited in function to lipid synthesis and breakdown.

At the cellular level, there is considerable heterogeneity, with mature adipocytes accounting for no more than half of the total cell content of white fat, the tissue containing fibroblasts, endothelial cells, preadipocytes, and macrophages, for example; complexity is also evident at the level of the basic process of glucose transport into white adipocytes; of the 14 members of the facilitative glucose transporter (GLUT) gene family (gene name SLC2A), as many as 8, GLUT1, GLUT3, GLUT4, GLUT5, GLUT8, GLUT10, GLUT12, and HMIT, are expressed in white adipocytes (Yao, Wood, and Trayhum, unpublished observations). Thus, the **process of sugar uptake into white adipocytes** is thought to involve a range of different transport proteins, each with its own distinct kinetic characteristics, and at least one (GLUT4) displaying insulin sensitivity. WAT is a major secretory organ, particularly through the release of fatty acids during fasting.

The tissue also releases other lipid moieties, such as cholesterol, retinol, steroid hormones, and prostaglandins.[21] Cholesterol and retinol are not synthesized byWAT, but rather are taken up and stored within the tissue. Steroid hormone conversions can take place in white adipocytes, such as the activation of 11 -dehydrocorticosterone to corticosterone catalyzed by 11 (3-hydroxysteroid dehydrogenase type I.[22] The enzyme lipoprotein lipase is released from adipocytes for the breakdown of circulating triacylglycerols to fatty acids, which are subsequently stored within fat cells.

In the late 1980s, a further secreted protein from adipocytes was identified, namely, adipsin, a complement-related factor/[23-24]) Adipsin was initially thought to be a direct signal in energy balance, but this was subsequently found not to be the case.

A major step forward in the recognition of the secretory role of WAT occurred in the early 1990s with the discovery that the proinflammatory cytokine tumor necrosis factor-a(TNF-a) is synthesized and released by adipocytes. TNF-a expression increases in obesity, and this cytokine plays an important role in the induction of insulin resistance. TNF-a was shown to have extensive metabolic effects in adipose tissue, including the stimulation of lipolysis and apoptosis.

The pivotal change in perspective on the role of WAT as a secretory organ came with the identification of the hormone leptin in 1994. This followed the search for the Ob gene, a mutation in which is responsible for the obesity of the ob/ob mouse. **Leptin, a 16,000 MW cytokine-like protein,** is a critical hormonal signal from adipocytes in the regulation of appetite and energy balance, interacting with several hypothalamic orexigenic and anorexigenic pathways. Thus, the neuropeptide Y, melanin-concentrating hormone, orexin A, agouti-related peptide, and cannabinoid systems have each been reported to be inhibited by leptin. In contrast, the key anorexigenic systems of pro-opiomelanocortin/melanocortin, cocaine- and amphetamine-regulated transcript, and corticotrophin-releasing hormone are upregulated by the hormone. These multiple effects of leptin result in a powerful suppression of food intake. In addition to inhibiting intake, leptin plays a role in the regulation of energy expenditure; a potent example of this comes from overfeeding studies on normal and ob/ob mice. In one study, lean mice fed a "cafeteria diet" overate by 70% in energy terms with no additional energy deposition; this is a powerful illustration of the much debated phenomenon of diet-induced thermogenesis. Serendipitously, in this particular study, the energy intake of the lean mice fed the cafeteria diet was the same as that of ob/ob mice fed a standard laboratory diet. However, the rate of energy deposition of the obese was 3 times that of the lean. Thus, the ob/ob mutants lacking functional leptin had a greatly reduced capacity for diet-induced thermogenesis.

Adipokines

The identification of leptin led to the recognition that white fat is an important endocrine organ. Indeed, it is now evident that white adipocytes secrete a multiplicity of protein signals and factors termed adipokines. The diversity of the adipokines is considerable, in terms of both protein structure and function. The adipokines encompass classical cytokines (e.g., TNF-cx, IL-6), chemokines (e.g., monocyte chemoattractant protein-1 [MCP-1]), proteins of the alternative complement system (e.g., adipsin), and proteins involved in vascular hemostasis (e.g., plasminogen activator inhibitor-1 [PAI-1]), the regulation of blood pressure (angiotensinogen), lipid metabolism (e.g., cholesteryl ester transfer protein, retinol binding protein), glucose homeostasis (e.g., adiponectin), and angiogenesis (e.g., vascular endothelial growth factor [VEGF]).

From the wide range of adipokines identified over the past few years, it is apparent that white fat is a secretory organ of considerable complexity that is closely integrated into overall physiological and metabolic control.[1516-2141] A corollary to the secretion of such a wide range of protein signals and factors is that WAT communicates extensively with other organs.

Co-culture studies indicated, for example, that adipocytes signal directly to other tissues such as the adrenal cortex, and there is a distinct cross-talk between white adipocytes and the brain through leptin and the sympathetic

nervous system. Indeed, the sympathetic system plays an important role in the regulation of leptin production in white adipocytes, whereas leptin stimulates the sympathetic activity in several organs, including the kidneys and brown adipose tissue.

A number of adipokines are linked to inflammation and the immune response and parallels have been drawn between adipocytes and immune cells. Indeed, preadipocytes are reported to be able to act like macrophages. The inflammation-related adipokines include cytokines, chemokines, and acute phase proteins. Clear evidence for the expression and secretion of the following cytokines and chemokines has been documented: TNF-a, transforming growth factor-(3, IL-lb, IL-6, IL-8, IL-10, MCP-1, and macrophage migration inhibitory factor. Acute phase proteins that have been clearly identified as adipokines are haptoglobin, serum amyloid-A, and plasminogen activator inhibitor-1 (PAI-1). PAI-1 is also, of course, a key agent in vascular hemostasis.

In addition to these factors, several other inflammationrelated adipokines are recognized, including leptin, the angiogenic protein VEGF, and the first of the family of neurotrophins to be discovered, namely, nerve growth factor (NGF).

Importantly, the major adipocyte hormone adiponectin has an anti-inflammatory action in addition to its role in insulin sensitivity and several other metabolic processes.

Introduction

Obesity is a challenging multifactorial problem. It is escalating at an alarming rate across the globe in all age groups, especially among the urban. Various studies have shown that there is up to 5-10% increase in obesity per decade in the latter quarter of last century. Obesity in childhood is an important risk factor for obesity in adulthood and up to 80% of them become obese adults. This phenomenon of tracking warrants prevention and early intervention. Some of the health hazards that are linked to obesity are coronary artery disease, cerebrovascular disease, hypertension, type II diabetes, hyperlipidaemia, orthopaedic disorders, cholelithiasis, hyperuricaemia, early pubertal changes, menstrual irregularities, respiratory infections, obstructive sleep apnoea (OSA) and psychosocial problems.

Obesity is arbitrarily defined as excess adipose tissue in the body. Weight for age, weight for height, body mass index (BMI), skin fold thickness, waist:hip ratio (WHR) (Fig. 4.15) are usually used for evaluation of obesity into apple shaped and pear shaped obesity. Of these, BMI is agreed upon as a reliable indicator that correlates well with body fat estimation. Dual energy X-ray absorptimetry (DEA) is the gold standard in body fat estimation. CT, MRI and USS scans are also useful in assessing fat.

There is a paradox of undernutrition and obesity coexisting in the developing countries like India. It is attributable to urbanization, technology based sedentary life style, high-fat high-sugar junk food, increasing purchasing power, lack of

exercise, excessive TV viewing etc. Another major area of interest in this respect is 'programming' and the Barker hypothesis. Maternal malnutrition begets foetal malnutrition. Intrauterine growth retardation (IUGR) followed by postnatal overfeeding and sudden upward shift in growth curve to higher centiles is now identified as an important risk factor for early onset adulthood diseases. Birth weight reflects the intrauterine environment and is the first wealth of a baby.

A state of energy balance exists in adult humans, since the homeostatic mechanisms can strike a remarkable balance between energy intake and energy expenditure. Because of this balance, body weight and body energy stores are maintained more or less constant in spite of fluctuations in energy intake and expenditure.

Definition

The body mass index (BMI) is well correlated with measurements of body fat. It is defined as the weight (kg) divided by height (m^2). The term obesity and overweight refer to excess in body weight relative to height. Overweight is defined as

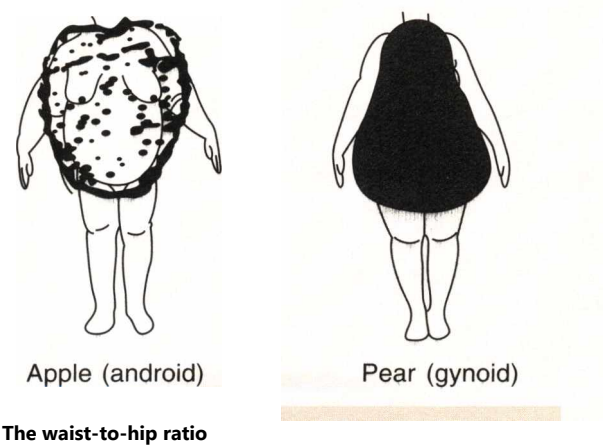


Fig. 4.15 Waist:hip ratio

relative weight up to 20% above normal (25 to 30 kg/m^2) and obesity is relative weight 20% above ideal body weight (> 30 kg/m^2). Even a small excess in energy intake over energy expenditure (25 kcal/day) over a period of time can lead to obesity. Hence, obesity is a mismatch between energy intake and energy expenditure. However, it is still not clear whether obesity is due to excess energy intake or a reduction in energy expenditure.

Biochemical Changes in Obesity

Components of Energy Balance

Energy intake: Calorie or energy content of food. Varies from 4 kcal/g for carbohydrates to 9 kcal/g for fat.

Energy expenditure: Resting metabolic rate + Meal induced thermogenesis + Physical activity energy expenditure

Energy storage: When energy exceeds energy expenditure, a state of positive energy balance occurs. When overfeeding relative to energy needs occurs, the body increases its overall energy stores.

Each of these factors are determined by various factors.

Resting metabolic rate (RMR) is the energy expended by the body to maintain physiologic functions like heartbeat, muscle contraction and respiration. It is the minimum level of energy expended by the body to sustain life. Because resting metabolic rate occurs predominantly in muscle and the major organs of the body, the main source of variability in resting metabolic rate is organ and muscle mass (fat free mass). The relationship between resting metabolic rate and fat free mass decreases with age.

79 kcal/kg—0 to 2.5 years
36 kcal/kg—4 to 7 years
28 kcal/kg—adolescence
21 kcal/kg—adults

The RMR is also not proportionate to the organ size. Skeletal muscle constitutes 43% of the total fat free mass of an adult, but only 22 to 36% of the RMR. The brain constitutes only 2% of the mass but 20% of the RMR.

RMR is also influenced by the fat mass, which contributes 10 to 13 kcal/kg of the RMR. In healthy adults the RMR declines with age. Males have a higher value than females by 50 kcal/d. This difference is independent of the gender difference in the fat free mass and is consistent across the life span.

Meal-induced thermogenesis occurs over an extended period of at least 5 hours. Cumulative energy cost is equivalent to approximately 10% of energy utilized. The thermogenic effect is higher for proteins (30%) and carbohydrates (15%) than for fat (5%). This is because the process of energy storage is efficient

for fat, whereas additional energy is required to convert carbohydrate and proteins to the appropriate storage form.

The physical activity energy expenditure is determined by the amount or duration of activity, type of activity and the intensity with which the activity is performed. The metabolic count of physical activity is expressed as metabolic equivalents or METs which represents multiples of resting metabolic rate (RMR). Sitting quietly after a twelve-hour fast is equivalent of one MET. Physical activity provides the greatest source of flexibility in the energy expenditure system and large changes in energy expenditure can be achieved through physical activity.

Total energy expenditure is similar between lean and obese individuals after considering the fat free mass. Fatness has only a negligible effect on energy expenditure, except for a small effect on RMR. This could be due to the additional cost of weight bearing activity in subjects with greater body fat.

Even though it is often said that obesity is the end result of positive energy balance due to overeating and lack of physical activity, it actually is caused by many complex and interrelated factors. Of these we will consider the role of insulin resistance and elevated free fatty acid (FFA) levels in the genesis of obesity in some detail.

Adipose tissue is the major storage depot of fatty acids and fuel store in humans. The free fatty acids that are released from adipocytes are transported bound to albumin and are removed from circulation within 3 to 4 minutes. Normally FFA concentration is only 5 mmol/L.

Adipocytes take up fatty acids from circulating triglycerides present in chylomicrons and VLDL, by the action of LPL. Inside the adipocytes the fatty acids are reesterifed to TAG for storage. The release of FFA from adipose tissue occurs by the action of hormone-sensitive lipase. Lipolysis is stimulated by catecholamines, cortisol and growth hormone, but inhibited by insulin.

The FFA level in plasma is generally higher in obese than in non-obese individuals. The high FFA concentration has a strong positive correlation with several adverse metabolic consequences of obesity. Elevated FFA expose cells to excess lipid fuel.

The high FFA level may be due to two factors:

1. Increased release of FFA from adipose tissue leading to increased availability of fuel
2. Decreased uptake of FFA by other tissues leading to normal or reduced availability of fuel.

Studies with isotope dilution techniques have shown that excess adipose tissue lipolysis is the major reason for high FFA concentration in upper body obesity. The high level of FFA occurs after a meal when insulin is increased. The

excess FFA contributes to the pattern of metabolic syndrome in obesity. It is also seen that there are differences in the FFA release with the pattern of fat distribution.

Upper body obesity has elevated FFA release in post-absorptive state and impaired insulin-mediated suppression of FFA release. The FFA release is primarily from upper body subcutaneous tissue. This is thought to be due to their greater beta receptor density or sensitivity. The adipocytes in this location are big in size.

Upper body obesity is associated with resistance to the ability of insulin to stimulate glucose uptake, especially in skeletal muscle. The mechanism of FFA-induced insulin resistance in skeletal muscle involves impaired glucose transport. This may be due to an impaired glucose transporter translocation to the cell membrane. The insulin resistance seen in upper body obesity is a possible risk factor for CAD. The decreased insulin sensitivity of peripheral tissues results in decreased oxidative (glycolysis) and nonoxidative (glycogen synthesis) metabolism of glucose. FFA-induced skeletal muscle insulin resistance may be due to a rapid alteration in insulin signaling. The plasma level of insulin is elevated and insulin resistance is established as a significant metabolic abnormality in obesity. The dysregulation of adipose tissue lipolysis in upper body obesity also proves that the metabolic abnormalities in the obese phenotype are mediated by FFA.

Endogenous glucose production is not suppressed by insulin in upper body obesity. Greater availability of insulin exposes peripheral tissues to excessive amounts of insulin. Decrease in FFA has been found to improve insulin mediated suppression of glucose production. The progression of insulin resistance to type II diabetes in obese subjects may be due to beta cell lipotoxicity induced by FFA.

Obese subjects have a dyslipidaemia consisting of hypertriglyceridaemia, low HDL cholesterol and a higher proportion of small dense LDL particles. The increased FFA flux may contribute directly or indirectly to these abnormalities and the resultant increased risk for CAD. Fasting hypertriglyceridaemia is caused by increased hepatic secretion of VLDL. The high FFA delivery to hepatocytes will result in increased triglyceride synthesis. FFA increases the secretion of apoB 100 by a post-translational process. In the presence of higher concentration of FFA, the proportion of secretion of apoBlOO is more than insulin antagonizes VLDL synthesis and increases intracellular degradation the rate of degradation on the other hand of apoB 100. In the prevailing condition of insulin resistance, in obesity the VLDL and apoB 100 synthesis are enhanced.

The decrease in HDL cholesterol is an indirect effect of high FFA. Increased FFA causes a larger VLDL pool which drives the exchange of HDL cholesterol esters for VLDL triglyceride by CETP (cholesterol ester transfer protein). There is an increase in triglyceride-rich HDL particles which are preferred substrate for hepatic lipase.

LDL cholesterol is produced through the metabolism of VLDL particles by LPL. Exchange of triglyceride and cholesterol ester via the CETP occurs between triglyceride rich VLDL particles and LDL. Hepatic lipase acts on these particles to yield dense small LDL particles that are three times as atherogenic as normal LDL particle. Hence the presence of small dense LDL particles in obesity is an indirect effect of high FFA levels. The dyslipidaemia is also contributed by genetic polymorphism of several lipoprotein genes—apoE, LPL and apoB 100 and apoAll.

The cytokine TNF alpha is overexpressed in adipose tissue in obese animals and correlates with fat cell size. The excess TNF alpha production in obesity may play a role in insulin resistance. Resistin is another hormone secreted by adipose tissue, which results in insulin resistance and obesity related type 11 diabetes.

Leptin. the product of Ob gene has no role in the insulin resistance associated with obesity. Obesity does not result from a single factor. Cultural, behavioural and biologic factors control the energy intake and expenditure. Genetic and hormonal factors contribute to individual susceptibility. It has been established beyond doubt that an upper body fat distribution confers a greater metabolic and health risk than a lower body fat distribution. The role of FFA in the genesis of the metabolic syndrome of obesity has also been established beyond doubt.

Adipose tissue is now given the status of an organ. It in fact is having major functions than previously thought. It reflects the reserve food on board and lack of adipose tissue is associated with decreased work efficiency, menstrual and fertility disorders and psychosocial problems. The number and size of adipose tissue increases during gestation and infancy. This continues in puberty at a slow pace. In adulthood, in most individuals, the adipose tissue is usually stable.

It is interesting to note that adipose tissue is also given the status of an endocrine organ. It secretes a 16 kD protein called leptin in proportion to the size and number of adipose cells. The OB gene encodes this protein. It circulates bound to binding proteins and crosses the blood-brain barrier. It attaches to OB receptors in the hypothalamus and choroids plexus and sends a number of signals that result in appetite regulation, feeding behaviour and maintenance of body weight. It also influences gene expression and secretion of neuropeptide Y (NPY). NPY is a potent stimulator of feeding. Glucagons like factor I glucagons, melanocyte stimulating hormone (MSH), urocortin, serotonin, cholecystokinin, enterostatin etc., are the other factors concerned with appetite and regulation and food intake.

Comorbidity

People with a BMI of 25 or above have an increased risk of developing comorbidities, which is further increased with BMI values of 30 or more. Virtually all obese people will have developed physical symptoms by 40 years of age, and the majority will require medical intervention for diseases that develop as a direct result of their obesity by the age of 60 years. For BMI values of 40 or more (severe

or morbid obesity), the risk of a life-threatening disease developing as a direct result of obesity is extremely high.

Obesity not only causes much psychological morbidity, but is also a primary risk factor in the development of hypertension, cardiovascular disease, stroke, diabetes mellitus, hyperlipidaemia, osteoarthritis, and cancer of the breast, ovary, prostate and colon.

Obesity is associated with a considerably increased risk of endometrial cancer (the relative risk is 5.4 for those weighing 40% or more than average), and a greater risk of breast cancer in premenopausal women, and to some extent of bowel cancer in men.

The proportion of common diseases that can be attributed to excess body weight is shown below. Hip fracture is expressed as a negative proportion, as people who are excessively overweight or obese are less likely to experience a hip fracture than those who are underweight.

Proportion of various diseases that are attributable to excess weight (BMI > 27 kg/m²)

Disease	Proportion (%)
Obesity	100.0
Hypertension	24.1
Myocardial infarction	13.9
Angina pectoris	20.5
Stroke	25.8
Venous thrombosis	7.7
Type 2 diabetes	24.1
Hyperlipidaemia	7.7
Gout	20.0
Osteoarthritis	11.8
Gall-bladder disease	14.8
Colorectal cancer	4.7
Breast cancer	3.2
Genitourinary cancer	9.1
Hip fracture	3.5

Obesity leads to premature mortality. A man weighing more than 140% of the average weight is 5.2 times more likely to die of diabetes than a man of ideal weight. Similarly, women who are more than 140% overweight are 7.9 times more likely to die of diabetes than women of ideal weight. After adjustment for age and smoking, the risk of a fatal or non-fatal myocardial infarction among women with a BMI greater than 29 is three times that among lean women.

Osteoarthritis is a common complication of obesity, especially in weight-bearing joints such as the knees and hips. The risk of osteoarthritis is related to the total amount of fat, rather than to the extent of abdominal fat.

People who are obese are more likely to develop gallstones because of their higher output of cholesterol in bile.

Obesity is also associated with reproductive and menstrual disorders. Sleep apnoea is caused by the physical pressure effects of fat on the chest wall and upward pushing on the liver, which compresses the lungs and leads to poor lung ventilation. In addition, fat around the neck of an obese person may compress the trachea.

A newly identified hormone, resistin, links obesity to type 2 diabetes and partly explains how obesity predisposes people to diabetes. Resistin is thought to be secreted by fat cells and then to modify the body's sensitivity to insulin, causing insulin resistance.

'Programming' (?) and Early Onset of Adulthood Diseases

Programming is a relatively new concept that has attracted the attention of many researchers. It is the term given to the idea that there are certain times during early life when the foetus or the infant is susceptible to certain adverse influences that produce life-long effects on organ structure and function. David Barker and his colleagues have observed on cohorts from Hertfordshire and Preston that there is a relationship between birth weight as well as weight at one year of age and adult morbidity and mortality due to coronary artery disease (CAD), cardiovascular diseases (CVD), hypertension, non-insulin dependent diabetes mellitus (NIDDM), renal diseases and so on. Leon and his associates have suggested that the adverse effects are marked among babies with intrauterine growth retardation (IUGR) and are least among premature babies. Thus the weight in relation to the gestational age appears more important. So the question arises, "What is the intrauterine environment of an IUGR baby that leaves the everlasting impact?"

The factors responsible for the relationship between birth weight and adult onset diseases are not very clear and has raised lot of controversies and criticisms. However, there are some suggestions that the endocrine system is a candidate factor in this major issue. The role of maternal glucocorticoids has been highlighted in some animal and human studies. Maternal glucocorticoids are elevated in IUGR. The unrestricted access to cortisol due to adrenal overactivity secondary to activation of pituitary-adrenal axis may explain the relationship between IUGR and subsequent evolution of adult onset disease like hypertension. Thus, the programming of hypertension and adult onset disease may be due to overexposure to maternal glucocorticoids during foetal life. This may be the result of unfavourable and stressful situations during pregnancy. Deficiency of placental 11-beta-hydroxy steroid dehydrogenase may also contribute to high levels of cortisol. In certain periods of nutrient deprivation during pregnancy, placenta may undergo hypertrophy, perhaps to protect the foetus, but the resultant changes in the foeto-placental blood flow may initiate a high systolic blood

pressure (BP). This initiation may have an effect on the developing blood vessels changing their compliance characteristics. These are called 'initiation and amplification'. The protein, 32-33 split proinsulin is now identified as a marker of impaired pancreatic beta cell function. This is a precursor of insulin, but it is not biologically active. This is found to be elevated in IUGR and may have a role in future development of NIDDM.

Amplifiers and Modifiers

A number of confounding factors may intervene during subsequent life of an individual. Obesity, lack of exercise, smoking, stress and strains, high fat diet, micronutrient and antioxidant deficiencies may act as effect modifiers or amplifiers. The 'fast food and cola culture of the west' which has suddenly crept wide and wild into our society is a factor that may result in obesity, degenerative and malignant disorders and micronutrient and antioxidant deficiencies. Early onset of ischaemic heart disease, hypertension and the like even in young women is a paradox that deserves attention.

Weight at one year, which is identified as an important determinant by many workers, is a proxy of some of these modifiers or amplifiers. Socio-economic factors, feeding practices, child rearing skills, environmental and emotional factors etc., influence weight at one year in addition to birth weight. Birth weight is a proxy of intrauterine environment and weight at one year is a proxy of many aspects of early postnatal environment. Normal birth weight babies come from better intrauterine environment and better nourished mothers with lower stress and this itself is a predictor of healthier, lower stress life which in turn would lead to lower adulthood morbidity and mortality.

Thrifty Gene Hypothesis

The 'thrifty gene hypothesis' has thrown insight into some other aspects as well. It suggests that in earlier periods of human development, some individuals store up energy as fat. During periods of malnutrition, those who store up energy as fat overcome successfully, whereas those who lack this capacity become victims of malnutrition. Thus, those with a 'thrifty gene' have a survival advantage. But, in affluent societies with surplus food supply, storing up of energy as fat may lead to problems. In the midst of plenty, the thrifty gene adaptive mechanism may prove to be a disadvantage.

Hereditary Factors

Obesity is the expression of a complex interaction between genetic and environmental factors including food intake. One-third of the variance of obesity in a given population is determined by heredity. Parental obesity, especially when both parents are obese, is the strongest predictor. Resting energy expenditure (REE) and metabolic rate are now identified to be inherited. Studies done on twins

and on adopted children who follow the BMI patterns of their biological parents reconfirm the genetic theory.

Syndromic Obesity

Syndromic obesity is common among children. Most of them are dysmorphic genetic syndromes with typical facial and physical features, hypogonadism and mental retardation. Prader-Willi syndrome is a prototype of this and hypotonia is the hallmark of this syndrome. Obesity usually starts after infancy. It is due to deficiency or microdeletion of paternally derived chromosome 15. Beckwith-Widemann syndrome is a foetal overgrowth syndrome with excess of IGF II. It is an autosomal dominant condition that affects chromosome 11. Some of them have UDP for chromosome 11, derived from paternal side. Lawrence-Moon-Biedl-Bardet syndrome is associated with obesity, retinitis pigmentosa, polysyndactyly, hypogonadism, mental retardation and nephropathy. It is an autosomal recessive condition. Carpenter syndrome is an autosomal recessive condition with obesity, acrocephaly, craniosynostosis, syndactyly and mild mental retardation. Sex-linked recessive obesity is associated with severe mental retardation, microcephaly and large ears.

Among the environmental factors, lifestyle and eating patterns like energy dense foods, high-fat high-sugar junk food and soft drinks, affordable fast foods have contributed to the explosion in the prevalence of obesity especially among the urban. Technology based sedentary lifestyle, excessive TV viewing, advertisements and lack of exercise are other risk factors. TV viewing is reported to increase obesity by 2% per hour of viewing and so is the case with videogames and surfing the internet. TV viewing reduces metabolic rates, increases snacking, reduces exercise and adds on to information about high-fat high-sugar items that flood the market.

Constitutional Obesity

Constitutional obesity in children is due to excessive calorie intake. It is more common in infancy, around 6 years and in puberty. These children are taller and have advanced bone age. Puberty sets in early affecting ultimate height. External genitalia appear disproportionately small and embedded in fat. This is the most common type of obesity.

Neuropsychiatric Obesity

Neuropsychiatric cases like hypothalamic, pituitary and other brain lesions like craniopharyngioma, psychological disorders like bulimia nervosa etc., lead on to obesity. These conditions disregulate appetite and involve signals that culminate in the ventromedial region of the hypothalamus.

Endocrine Obesity

Among the endocrine disorders, Cushing syndrome is the most important condition. It presents with short stature and obesity. Turner syndrome patients tend to have short statue and obesity due to lack of sex hormone. Hypothyroidism and growth hormone (GH) deficiency also may rarely lead on to obesity. In GH deficiency, obesity is due to reduced energy expenditure and disproportionate weight for height. Polycystic ovarian syndrome is associated with late onset obesity, menstrual irregularities, hirsutism and acne.

Evaluation of obesity should include a detailed history including diet and lifestyle, thorough physical examination, psychological profile and laboratory investigations. The ELIZ health path for adults (EPHA) that incorporates weight, height and BMI in the same chart is very useful in identifying, monitoring and preventing obesity (Appendix). Blood pressure recording, blood sugar and lipid profile are mandatory in all cases of obesity. Other special investigations can be planned after clinical evaluation.

The interventions include dietary approach, exercise, behavioural modification, drug therapy and surgical procedures. Since treatment and outcome of therapy are often disappointing, prevention and identification of high-risk cases are of utmost importance. Obesity is an emerging public health problem and tracking of weight, height and BMI using simple charts like EPHA (appendix 7) can go a long way in tackling this problem.

The factors that affect growth and development are interlinked and faced by several confounding and modifying factors. The essence is that providing a better nurturing environment during intrauterine and postnatal life or rather starting from childhood and adolescent life of prospective mothers may act as a real preventive strategy against most adult onset diseases. At the same time it is important to tackle and control confounders, modifiers and amplifiers like feeding practices, nutritional status and lifestyle. It is better to pave the way for positive health, which is the motto of the era, rather than prevention.

Management

J. Diet

Formal calorie-counting diets may be useful for getting someone who is obese or overweight started on a weight-loss programme, but strict diets are difficult to sustain in the longer term. Most people like variety in their diet and they enjoy ‘treats’. One of the most important aims of any programme is to help patients to recognize ‘danger foods’ (particularly those high in fat), and to help them to increase their own control over eating. In practice, a 600-calorie-deficient diet is normally effective. Calorie counting and fat avoidance can be encouraged by asking the patient to keep a food diary, which can also provide insight when weight loss is not proceeding as planned. It is common for obese and overweight individuals to underestimate their food intake by about one-third perhaps be-

cause of genuine forgetfulness, or self-deception due to a lack of understanding of food composition, particularly with regard to hidden fat. In particular, the eating of snacks tends to be under-reported.

Food is an important part of social life. The diet should not be so defined as to prevent the patient enjoying normal social intercourse, or so strict as to preclude 'treats'. 'Negative dieting' is often counterproductive in the long term. The approach that should be taken should emphasise new food opportunities, new methods of food preparation, and the integration of 'treats' into the overall food plan.

Many patients will have stories of very-low-calorie or quirky diets which have helped them to lose vast amounts of weight rapidly in the past. Sadly, most of these patients will have relapsed subsequently. This emphasises one of the key messages that a weight control programme is not just a 'one-off' diet to give someone a rapid period of weight loss, but rather it is a process of re-education which will affect their whole lifestyle. It is relatively easy to lose weight over a short period, but much more difficult to maintain that weight loss over the longer term. Only improved insight, changed dietary habits, behavioural change and exercise will sustain optimal weight.

The long-term aim is to give people control over what they eat, and not to let food control them.

The Atkins diet: This is a high-protein, low-carbohydrate diet consisting of meat, cheese, etc., and avoiding starches, fruit, sugars and processed food.

Formula diet: This is a balanced eating plan. The formula is to eat meals for which 40% of the calories are derived from carbohydrates, 30% are derived from protein and 30% from fats.

The Hay diet: The Hay diet is also known as 'food combining for health'. It involves keeping starch foods separate from protein foods in order to aid digestion.

Weight-Watchers Pure Points: Each person attending the Weight-Watchers weekly club session is privately weighed, and then there is a group discussion with the club leader to share news, hints and tips.

The 'Pure Points' programme allots points rather than calories to a variety of foods. Participants are allowed a predetermined number of points per day depending on how much they weigh and how much weight they need to lose. Most vegetables count as zero points, which means that participants can eat as many as they like. They can save points from their daily allowance to put towards a special food treat.

Weight-Watchers group members are encouraged to exercise and thus 'earn' extra points to spend on food. For instance, if they walk briskly for 30 minutes, they can add three points to their allowance.

Low-fat diets (LFD): Many diets emphasise the reduction in fat intake which automatically reduces caloric intake, as fat is so high in calories. Very-low-calorie

diets of around 800 kcal per day can induce rapid weight loss, but weight is often regained equally quickly once the diet has ceased. They are not generally recommended, but can be useful under medical supervision for specific reasons, such as the need for a patient to lose weight rapidly on medical grounds.

Low-calorie diets (LCD). These are energy-restricted diets in the range 800-1500 kcal per day. They should contain a balance of protein, fat and carbohydrate, usually with reduced fat. Many of the diets described above fall within the category of a low-calorie diet.

High-dietary-fibre diets: Two studies that investigated the effects of dietary fibre found that fibre supplements were more effective than placebo when given with a diet of 1200-1600 kcal per day. However, the weight loss achieved was no different to that obtained with a comparable low-fibre/low-calorie diet.

Very-low-calorie diets: This type of diet consists of 600 to 800 kcal or less per day. It is usually adopted for several days or weeks in order to achieve rapid weight loss. Lean muscle is lost as well as fat stores. Experts recommend a minimum protein intake of 0.8-1.5 g/kg of ideal body weight and daily vitamin and mineral supplements. Very-low-calorie diets should not be continued for more than four weeks.

Decrease alcohol intake: Alcohol contains nearly as much energy as does fat, at 7 kcal/g. It can compromise a weight-reducing diet by providing hidden calories, and it is thought to alter the pattern of fat distribution, encouraging a 'beer belly'. Excessive amounts of alcohol act as a central depressant and decrease initiative and willpower, reducing enthusiasm for physical exercise.

2. Physical Activity

The Health Education Authority recommends that 'adults should try to build up gradually to take half an hour of moderate intensity physical activity on five or more days of the week. Activities like brisk walking, cycling, swimming, dancing and gardening are good options'.

One of the key insights which health professionals can give to patients involves forging a link between the calorific value of the food a person eats and the exercise which is necessary to burn off those calories. This is particularly useful in cases where the patient is prone to 'snacking'. Most people find these comparisons surprising, and gasp with astonishment. Armed with concepts such as this, they will rapidly learn to recognise that if they eat something extra then it must be balanced with an equivalent extra energy output; otherwise, they must expect an increase in body weight. Examples of exercise types and the calorie & food equivalents:

Activity

- Energy expended per hour in kilocalories
- Food equivalent expended per hour

Driving a car
m 80 kcal
■ Slice of bread

Standing relaxed
. 100 kcal
■ Glass of white wine

Standing doing light work
• 180 kcal
■ Bag of crisps

Walking 5 km in an hour
. 260 kcal
■ 1 *Vi* pints of beer

Walking 7 km in an hour
. 420 kcal
■ *2Vi* oz peanuts

Running 9 km in an hour
. 600 kcal
■ Two chocolate bars

Cross-country skiing (competitive)
. 1440 kcal
■ Roast dinner with sponge pudding

The challenge to healthcare professionals is to drive these obvious health care messages home and to trigger action. Activities that are recommended must be realistic for the individual concerned and appropriate to any other problems they might have. There is little point in recommending a two-mile walk to someone with severe airways disease, but they may be able to manage to climb their stairs once an hour or take a gentle walk to the shops. Activities which fit into the individual's lifestyle and which are easily put into action are the ones that are most likely to succeed in the long term.

For fitter individuals, suggest short 'triggers' such as climbing stairs, running for a bus or walking fast. Help patients to recognise that their longer periods of activity are beginning to pay dividends and that they are gradually becoming able to take on more, and to move faster and more easily.

Specific recommendations for physical activity for people who are obese

a) Build up slowly towards 30 minutes of moderate-intensity activity a day. The

30 minutes can be accumulated throughout the day in 10 to 15 minutes bouts. Moderate intensity means breathing slightly harder than normal, but still within the 'comfort zone' whereby the activity can be done whilst talking at the same time.

b) To achieve optimal weight loss, consider extending some sessions to 45 minutes or longer, as this will encourage the use of fat as an energy source.

c) Increase the amount of daily routine activity, such as gardening, shopping, housework, walking, etc.

d) Decrease the amount of time spent in sedentary activities, and try not to sit down for more than 30 minutes at a time.

e) The most effective activities for achieving weight loss are those that involve large muscle groups, which are aerobic in nature, such as walking, swimming or cycling.

f) Consider weight-bearing exercises such as walking and climbing stairs, as these help to conserve muscle mass and maintain strength and resting metabolic rate.

g) Find physical activities which are enjoyable.

3. *Behavioural Modification*

Any behavioural approach should take into account the fact that eating is a highly reinforcing behaviour. It induces feelings of gratification and pleasure which for some people is their main source of pleasure, and such individuals will not forsake their 'eating for pleasure' habit very readily.

We need to avoid medicalising obesity by applying stringent guidelines to weight management, and to look for new ways of tackling obesity as a society.

In order to understand what intervention to use to try to help patients to lose weight, you need to determine whether each individual is ready to change, to conquer their overweight or obesity and to sustain that weight loss. Then you can match your approach or intervention to the stage at which they are at present.

The five stages of change include the following:

a) Pre-contemplation: 'the stage at which there is no intention to change behaviour in the foreseeable future'

b) Contemplation: 'the stage at which people are aware that a problem exists and are seriously thinking about overcoming it, but have not yet made a commitment to take action'

c) Preparation: 'the stage that combines intention and behavioural criteria, individuals at this stage are intending to take action in the next month, and may have unsuccessfully taken action in the past year'

d) Active change: 'the stage at which individuals modify their behaviour, experiences or environment in order to overcome their problems'

e) Maintenance: 'the stage at which people work to prevent relapse and consolidate the gains attained during the action'

Types of Behavioural Therapy

Cognitive behaviour programme: This includes traditional behavioural approaches such as the following:

- Self-monitoring (e.g., keeping a diary of food eaten and the calorie and/or fat content)
- Stimulus control—developing strategies for an individual to reduce exposure to stimuli which may trigger inappropriate eating
- Coping with cravings and high-risk situations
- Stress management, especially for those who report stress-induced eating behaviour
- Relaxation techniques
- Learned self—control breaking the cycle between certain stimuli to eating particular foods and eating inappropriately
- Problem-solving skills

Other approaches with relevance to weight-loss management include the following:

- Healthy eating advice, and modification of disordered eating patterns (e.g., working with someone who is 'binge eating' to encourage a normal eating pattern)
- Weight management—setting behavioural goals that reflect changes in eating behaviour or exercise habits
- Mood management
- Managing work and family
- Relapse prevention—various mechanisms so that the individual accepts that lapses are to be expected and understands how they might avoid a 'relapse'
- Avoiding self-defeating thinking (e.g., 'all-or-nothing' thinking)
- Improving body image—learning to dissociate body image and self esteem

4. Drug Therapy

Choice of drugs: Most of the anti-obesity drugs that have been used in the past have been withdrawn because they are ineffective or have adverse effects.

Drugs should never be used as the sole element of treatment—other components of managed care should continue. Drug treatment should be discontinued if weight loss is less than 5% after the first 12 weeks, or if the patient gains weight at any time while he is receiving drug treatment.

Combination therapy involving more than one anti-obesity drug is contraindicated.

Drugs that act on the Gastrointestinal Tract

Binding agents: Chitosan (Ultrathin) is a non-digestible fibre that can bind fat and cholesterol and reduce the absorption. It is derived from shell of shellfish. 1-2 capsules can be taken with 1-2 glasses of water 30 minutes prior to major meal. It is used in adults only.

Bulking agents: These are agents such as methylcellulose and ispaghula husk, which are used to induce a feeling of fullness of the stomach. However, there is no evidence that they are beneficial in the long-term treatment of over-weight and obesity. Patients should be told to take plenty of water with tablets, as the latter swell when in contact with liquid, and not to take them before going to bed.

Pancreatic lipase inhibitor (orlistat): Orlistat inhibits fat breakdown in the lumen of the stomach and the small intestine. It inhibits pancreatic and gastric lipases and works by decreasing the hydrolysis of ingested triglycerides, thus reducing dietary fat absorption by around one-third. People who take orlistat excrete about 32% of ingested fat in their faeces, compared with 4.4% in controls. This leads to greater and more rapid weight loss.

Criteria for Prescribing Orlistat

- The licensing criteria and the National Institute for Clinical Excellence (NICE) recommendations require potential patients to lose 2.5kg in the month preceding the first prescription for orlistat. This enables treating doctors to ascertain whether a person is able to maintain a suitably low fat intake and a reasonable amount of physical activity.
- Patients should have documented evidence of a BMI of 30 or above (and no significant comorbidity necessarily) or a BMI of 28 or above with significant comorbidity (e.g., diabetes, hypertension or dyslipidaemia).
- Patients taking orlistat should be monitored and weighed on a monthly basis thereafter as part of a supervised weight management plan.
- Patients who are continuing to be prescribed orlistat should show a 5% weight loss three months after the start of drug treatment and at least a 10% cumulative weight loss six months after the start of treatment.
- Orlistat can only be prescribed for adults aged 18 to 75 years.

Orlistat is not absorbed from the gastrointestinal tract, so there are minimal systemic side-effects. As there is reduced absorption of the fat-soluble vitamins, including vitamins A and D, vitamin supplements may be required. There is no evidence of a link between orlistat and breast cancer, which was originally listed as a possible side-effect. The drug is contraindicated in pregnancy and whilst breastfeeding, and for patients with cholestasis and malabsorption syndromes.

There is some evidence that orlistat helps to reduce the risks associated with comorbidities. In one study, those taking orlistat showed significant improve-

merits in the from of reduced levels of total and LDL cholesterol, fasting plasma glucose and blood pressure. Other studies have confirmed these findings, demonstrating improved glycaemia control, with a reduction in HbAlc and dyslipidaemias in those with type 2 diabetes.

Centrally acting drugs: Centrally acting drugs act on serotoninergic or noradrenergic pathways, or both. In recent years many of these drugs have been withdrawn from use because of the incidence of side-effects. Fenfluramine and the combination of phentermine with dexfenfluramine have been withdrawn because of their link with heart valve defects.

Sibutramine: Sibutramine creates a feeling of satiety by acting as a serotonin and noradrenaline reuptake inhibitor in the brain, with the result that patients feel satisfied after eating smaller quantities of food. It may also increase thermogenesis by a stimulant action on the peripheral noradrenergic system.

The dose of 10 mg once daily is well absorbed from the stomach and has a half-life of 14-16 hours.

Common side-effects of sibutramine include headache, dry mouth, constipation, anorexia, insomnia, rhinitis and pharyngitis in 10-30% of patients. A small mean increase in mean blood pressure of 1-2 mm Hg is observed in patients on sibutramine, and an average increase in heart rate of 4-5 beats per minute.

Phentermine: Phentermine is an appetite suppressant with stimulant qualities, which lead to a modest weight loss in the medium term in people who are more than 15% overweight, when associated with a restricted calorie diet. However, there tends to be rapid weight regain on withdrawal of the drug. The severe side-effect of pulmonary hypertension is uncommon. Adverse reactions such as dry mouth and headache are more common, and the drug can lead to dependence.

It is given in a dose of 15-30 mg daily for 12 weeks or less. Phentermine is not recommended for the routine management of obesity, and is categorised in clinical evidence as ‘likely to be ineffective or harmful’.

Fluoxetine: There is limited and conflicting evidence that the selective serotonin reuptake inhibitor fluoxetine has any beneficial effect on obesity. It is neither licensed nor recommended for this purpose.

Unsuitable drugs for the treatment of obesity: Diuretics, purgatives, hormone treatments (including human chorionic gonadotrophin and dehydroepiandrosterone (DHEA), ephedrine, amphetamines and amphetamine-like substances are unsuitable for the treatment of obesity. Certain compounds are an appropriate part of obesity management in the presence of coexisting conditions (e.g,, thyroxine in cases of biochemically proven hypothyroidism, metformin and acarbose in the presence of non-insulin-dependent diabetes).

5. *Modes of Surgical Treatment (Bariatric Surgery)*

Only experienced surgeons in special centers (bariatrists) should perform these obesity related operations (bariatric surgery). The operation most widely used in

Europe is gastroplasty, by gastroplication or gastric stapling. Gastroplasty is being increasingly performed laparoscopically.

Restrictive Procedures

Gastroplasty

The term gastroplasty implies the changing of the shape of the stomach. This is done by partitioning a pouch a 15 to 40 ml at the top of the stomach, which rapidly fills with food and then empties slowly through a narrow channel into the body of the stomach. The pouch restricts the volume of food that a person can eat by reducing the stomach's functional capacity.

The operation is referred to as gastric stapling because of the line of staples that is used to divide the stomach. The most commonly used procedure is the vertical banded gastroplasty, in which the pouch is formed along the line of the lesser curvature of the stomach and empties through a channel of about 11 mm diameter. The channel or stoma is externally wrapped or banded to prevent stretching or more rapid passage of food. Patients consume a liquid-only diet, supplemented by iron and vitamins, for around three months to avoid breakdown of the stapled joints. They then progress to a carefully balanced diet supervised by a dietician and taken as small regular amounts throughout the day.

Laparoscopic Gastric Banding

This is a technique in which an adjustable band is wrapped round the outside of the stomach in order to prevent distension and restrict food intake. The degree of restriction is altered by increasing or decreasing the pressure through an epigastric or abdominal portal. The pressure alterations are made by the surgical team rather than the patient, who might increase the pressure by too much in order to facilitate more rapid weight loss, or by too little in order to be able to eat more.

These restrictive procedures are technically easy, have low morbidity; do not cause malabsorption (as food eventually passes through the gastrointestinal tract in the usual way), and cost less than other surgical procedures. However, the degree of sustained weight loss may not be as great as that achieved with other procedures. Some patients recognise that high-calorie liquids such as milkshakes, ice-cream and alcohol pass rapidly through the stoma without causing fullness, and they change their diet accordingly, thereby regaining weight. The alteration of pressure within the band can be an uncomfortable procedure, partly due to the increased restriction of the stomach, and partly because of the needle used to access the portal.

Gastric Bypass or Roux-en-Y Bypass

Gastric bypass surgery is widely used as a first-line procedure in obesity surgery. A 10 ml segment is isolated from the body of the stomach, but is surgically separated from the remainder of the organ and anastomosed to the proximal

jejunum, bypassing most of the stomach and the entire duodenum. This restricts food intake in the same way as inducing a degree of malabsorption. This double action is what makes gastric bypass surgery so effective in inducing and maintaining long-term weight loss. Gastric bypass surgery is a larger and more technically demanding operation than gastro-plasty, and malabsorption (especially of iron, folate and vitamin B_p) can occur postoperatively, requiring careful monitoring for life.

Jejuno-ileal Bypass

This procedure was abandoned around 1980 because of the high rate of complications, although patients with late side-effects from the procedure may still be encountered in primary care. More than 90% of the small bowel was bypassed by attaching the beginning of the jejunum to the end of the ileum, leaving a total of only 18 functional inches. This caused rapid transit of food through the bowel, and incomplete digestion, leading to malabsorption and severe steatorrhoea. Subjects could eat an unrestricted diet with no change in eating habits and still lose weight. Those undergoing the operation did lose weight—often over half their excess weight—but complications were common and occasionally life-threatening. These complications included acute hepatic failure, cirrhosis, oxalate nephropathy and chronic renal failure, immune-complex arthritis and malabsorption syndromes. Surgical re-anastomosis may be required to limit the associated morbidity.

Surgery for 'Super-Obese' Patients

Specialist forms of surgery have been designed for 'super-obese' individuals who have a BMI greater than 50 kg/nr, are at least 225% overweight or weigh more than 400 lb. with life-threatening obesity-related morbidity. These radical procedures involve 80% distal gastrectomy and gastro-ileostomy with diversion of biliary and pancreatic secretions to the distal ileum. This is said to result in intense weight loss with malabsorption, especially of the fat-soluble vitamins, folate, vitamin B_p, iron and calcium, all of which need to be monitored and, if necessary, supplemented.

Liposuction

This is a cosmetic procedure that involves the suction of fatty material from under the skin by means of a trochar. Liposuction usually results in the removal of approximately 3 liters of fat, but has sometimes involved the loss of up to 10-12 litres in extreme cases. Although the technique has occasionally been used as a treatment for morbid obesity, it does not normally result in the loss of sufficient fat to be considered in this category.

Jaw Wiring Procedures

These procedures are no longer recommended by some authorities. They have not been tested in a randomised controlled trial, but in one study of 17 obese patients, the significant amount of weight that was lost whilst the wires were in place was regained at an unusually rapid rate once the wires had been removed. The jaws are wired in such a way that the patient can drink but not chew. Strong fixation is needed to resist the strains on the wires caused by coughing or sneezing. Some practitioners have fitted waist cords once the jaw wiring has been removed, to limit the amount of weight regained. One study of 35 patients whose jaws were wired described the 14 patients who stayed the course and had waist cords fitted after the wires were removed as achieving an average weight loss of 33 kg over a period of three years.

Apronectomy

This is not a treatment for obesity, but it is helpful for patients who have lost large quantities of weight and have overhanging folds of excess skin as a result. Other common sites for skin contouring operations following weight reduction are the under-arm area (known as a brachioplasty), and the inner and outer aspects of the thighs. Abdominal apronectomy can be circumferential, involving skin removal round the patient's back. Male subjects may undergo gynaecomastia correction.

It can be psychologically damaging do deny patients such surgery on financial or other grounds after they have followed medical advice diligently, but are left feeling uglier with their hanging skin folds, and with lower self-esteem than when they started. Skin contouring is a technically straight forward procedure, and its satisfying results can help to maintain long-term weight loss.

Artificial Bezoar

This procedure involves the insertion of a balloon or object into the stomach in order to decrease its capacity. It has not proved successful as a treatment for obesity.

Obesity is a difficult problem to tackle and it requires a multipronged approach.

Metabolic Syndrome

The metabolic syndrome is characterized by a group of metabolic risk factors in one person. They include:

- Abdominal obesity (excessive fat tissue in and around the abdomen)
- Atherogenic dyslipidemia (blood fat disorders — high triglycerides, low HDL cholesterol and high LDL cholesterol — that foster plaque buildups in artery walls)
- Elevated blood pressure

- Insulin resistance or glucose intolerance (the body can't properly use insulin or blood sugar)
- Prothrombotic state (e.g.. high fibrinogen or plasminogen activator inhibitor-1 in the blood)
- Proinflammatory state (e.g.. elevated C-reactive protein in the blood)

People with the metabolic syndrome are at increased risk of coronary heart disease and other diseases related to plaque buildups in artery walls (e.g., stroke and peripheral vascular disease) and type 2 diabetes.

"Decrease your weight and increase your grace"

— *William Shakespeare*

4.4 Micronutrient Malnutrition

MICRONUTRIENTS AND MICRONUTRIENT DEFICIENCY DISORDERS (MDDs)

Micronutrients are functional foods that fight diseases and also supply antioxidants. There is a saying that a large helping of onion, a dash of garlic, a sprinkling of turmeric, fresh vegetables and fruits can keep cancer and vascular and heart diseases away. **Micronutrients were regarded as those with RDA < 100 mg/day as per the old definition.** These include the vitamins and the trace elements. (Also refer Section 3.3) These are present in non-staple items like milk, egg, meat, fish, vegetables and fruits. Those micronutrients in limelight at present are vitamin A, folic acid, iron, iodine and zinc. **Spirulina** is a biological supplement of most of these micronutrients.

1. Definition

Micronutrient is a fascinating terminology that has come up in the recent past. These refer to substances that are needed in small quantities by the body like vitamins and minerals, that is, in 'milligrams per day' in contrast to the major nutrients which are required in 'grams per day'. Micronutrients are cofactors of enzymes, gene activators and scavengers of free radicals. These are considered as "magic bullets" by many clinicians and patients as well. Those items that promote brain growth and function are also referred to as **"smart nutrients".** Some of the micronutrients are powerful antioxidants and have also been found to be important in stress, mood changes, depression and so on.

The bulk of human body is made up of 11 major elements namely, **H.** C, N, O, Na, Mg, P, S, Cl, K, Ca and the other elements form a minor part of the body.

Trace elements or microminerals are those minerals present in traces or small concentration in the body, < 0.01% of body weight, i.e., < 0.1 mg/g or 100 mg/g. WHO/FAO suggests the following working definition of a recommended nutrient intake: it is the intake level sufficient to meet the daily nutrient requirements of most individuals in a specific life-stage and gender group and is based on an estimated average nutrient requirement (EAR) plus two standard deviations above the mean: RN1 = EAR + 2 SD EAR

It is important to remember that the Green Revolution has made the production of grains to be ahead of the planned targets. The White Revolution led to more production and consumption of milk and milk products and the Blue Revolution led to increase seafood availability, and the current "Rainbow Revolution' aims at the availability and consumption of green, yellow, orange, red (GYOR) vegetables and fruits, and therefore more micronutrients.

2. Importance of Micronutrients in Human Health

Three of them have been singled out in the World Summit for Children 1990, namely, vitamin A, iron and iodine. These are already in limelight and have relevance on a public health scale and there are national/global prophylaxis programs associated with them, for example, the vitamin A prophylaxis program for children, the anaemia prophylaxis programme for children and mothers and the universal iodization of salt. Recently, folic acid has gained momentum. The others with emerging focus are calcium, zinc, vitamin D, vitamin B_p, copper and so on. Certain micronutrients like vitamin A, vitamin D and zinc have well-recognized clinical features in contrast to others that do not have such specific features. Subtle deficiencies are difficult to make out clinically and some deficiencies do not have specific features. Serum level estimation of micronutrient is costly and is not freely available. Moreover, serum levels are subject to certain drawbacks while interpreting the results as, e.g., serum carrier protein levels may alter the free vs bound form of the micronutrient. The pH of the blood may alter the level of ionized calcium. Low serum zinc levels occur in liver diseases, malignancy, infections etc.

3. Micronutrient Malnutrition (hidden hunger)

This compromises both survival and quality of survival. Malnutrition is claiming over 6 million lives of under-5 children globally every year. It is an important cause of growth and developmental retardation. Iron deficiency contributes to 20% of maternal deaths and lowers IQ points in children by 9 points. Iodine deficiency is the most common cause of preventable mental retardation and even mild deficiency can lower child's IQ by 10 points. Vitamin A deficiency is a well-known cause of preventable blindness. Folic acid deficiency may induce neural tube defects and vitamin D and calcium deficiency result in poor bone formation. Micronutrients include antioxidants that protect the body from tissue damage

and degeneration. Malnutrition during foetal growth and in early life can lead to early onset of adulthood disease as well.

Some studies have shown micronutrient deficient states in our setting and benefits of supplementation of micronutrients among newborns and children with malnutrition, diarrhoea, pneumonia etc. However, as per the current level of knowledge, we do not have a 'rule of thumb' to suggest specific micronutrient supple-mentation. Term babies who are thriving well on exclusive breastfeeding do not need any supplements. Among the LBW babies, there appears to be a differential rational among preterm vs term IUGR babies. Preterms are definite to benefit from supplements as most of the micronutrient transfer occurs in the third trimester. Cord blood levels of micronutrients are reported to show a preferential concentration compared to maternal serum. A study was conducted in the De-partment of Paediatrics, SAT Hospital, Trivandrum, on cord blood micronutrient levels among LBW vs normal weight and preterm vs term small for gestational age (SGA). The data have shown that micronutrient levels were the lowest in preterm followed by term SGA babies and in both groups, the levels were lower than term normal babies. This study highlights the importance of judicious supple-mentation of micronutrients in preterm and term SGA babies, irrespective of ma-ternal nutritional status. Exclusively breast-fed babies in the first semester of life generally do not have significant micronutrient deficiencies, but meeting the demands during and after initiation of complementary feeding is an area of con-cern. Micronutrient malnutrition is also called hidden hunger.

Complementary feeding or weaning is now identified as the 'weakest link in child nutrition'. By 5-6 months of age, when complementary feeding is initi-ated, breast milk supply reaches the maximum and plateaus off subsequently. Birth weight doubles by 5 months and by that time, the calcium and iron stores get depleted.

In our part, where complementary feeding is mostly on vegetarian items and cow's milk, there is a risk of developing deficiency of calcium, iron, zinc, vitamin A, folic acid, vitamin C, vitamin B_{12} and so on. Non-availability of haeme iron and intake of cow's milk increases iron deficiency. Non-haeme iron is only 5% bio available as against 35% from haeme iron. Reduced bioavailability of calcium due to increased phosphate in cow's milk may add to calcium deficiency. Lack of green, yellow, orange, red (GYOR) vegetables and fruits in the diet leads to micronutrient deficiency disor-ders especially vitamin A deficiency (VAD) and lack of citrus fruits leads to vitamin C deficiency. Lack of non-vegetarian items pose the risk of vitamin B_{12} and zinc defi-ciency.

4. Micronutrient Deficiency Disorders (MDDs)

Iron: Iron deficiency is the most common micronutrient deficiency across the globe affecting all ages and all socioeconomic status. Up to 30-60% of pregnant mothers and children have varying grades of anaemia. Grading of anaemia as per

WHO criteria is given in Table 4.19. Anaemia during pregnancy and childhood increases the risk of LBW, maternal morbidity and mortality and also child morbidity and mortality. Apart from increasing morbidity and mortality, iron deficiency anaemia (IDA) decreases physical stamina, learning ability, concentration and work efficiency. IDA during the critical period of brain growth may lead to irreversible changes in the brain by decreasing dopaminergic receptors and increasing opiate receptors (Fig. 4.16).

The newer iron preparations are gastrointestine friendly but it is their bioavailability that is questionable. Iron polymaltose complex is ferric iron. The conventional ferrous iron is better absorbed than the ferric iron. Recently, carbonyl iron is replacing iron polymaltose in the market. The addition of folic acid with iron is found to improve the outcome. The habit of drinking fruit juice especially lime juice will enhance iron absorption from a good meal whereas coffee and tea will hinder iron absorption. Cooking in iron pots may improve iron bioavailability. The greens, grams and grains (3 Gs) are rich in iron, but the bioavailability is very low compared to haeme iron.

Table 4.19 Haemoglobin levels (g%) indicative of anaemia

Subject	Non anaemic	Anaemic	Mild anaemic	Moderate anaemic	Seven anaem
Pregnant women	> 11	< 11	> 10-10.9	7-10	< 7
Women	> 12	<12	> 10-11.9	7-10	< 7
Men	> 13.5	< 13.5	> 10-13.4	7-10	< 7
Newborn	> 14.5				
2 mo	> 9				
6 mo-6 yr	> 11	< 11	> 10-10.9	7-10	< 7
> 6 yr	> 12	< 12	> 10-11.9	7-10	< 7
Adolescent girls	> 12	< 12	> 10-11.9	7-10	< 7
Adolescent boys	> 13	< 13	> 10-12.9	7-10	< 7

Ref: WHO technical report series, No 405, 1968; and Nelson Textbook of Pediatrics, 17th edition

Iron is low in cow's milk and cow' milk allergy may lead to intestinal blood loss. Breast milk iron is highly bioavailable due to the presence of lactoferrin. Ferritin is an indicator of iron stores.

In treating anaemia, daily iron is required till the anaemia gets corrected and the stores get replenished, which may take up to 100 days. But in prophy-

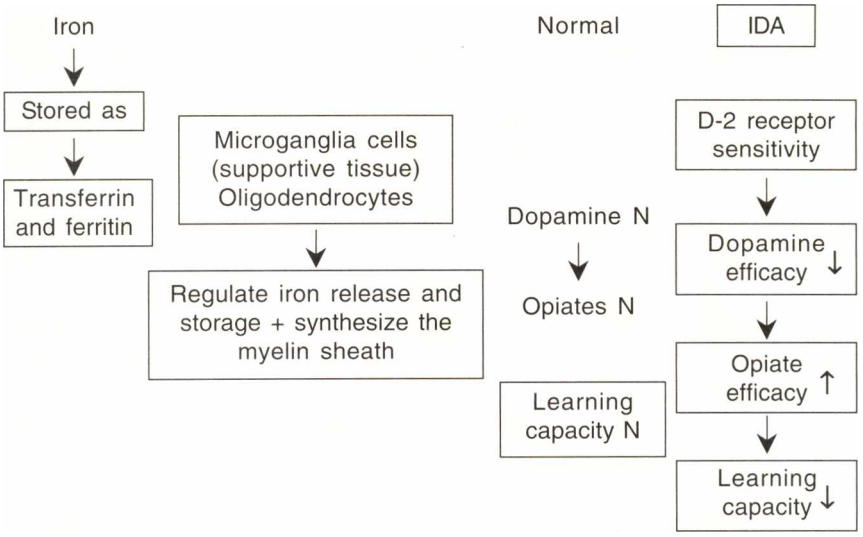


Fig. 4.16 Interaction between iron deficiency (IDA) and learning capacity

laxis, *weekly iron* has revolutionized the situation. The intestinal cells are shed and renewed in 4-5 days. The iron that is taken into the mucosal cell is released into the body only as per the need and the rest gets shed with the mucosal cells. The scope of weekly iron is thus conclusive theoretically. UNICEF has also supported weekly iron in anaemia prophylaxis. It is a wonderful strategy to give weekly iron to adolescent girls, and also boys, to prevent anaemia. It is also expected to reduce the future prevalence of anaemia during pregnancy and improve pregnancy outcome. The economic consequences of iron deficiency anaemia include cognitive delays in children, lower productivity among adults, low birth weight, increased morbidity and mortality among children and mothers. Iron also can correct menstrual irregularities, headache, minor learning problems, lack of concentration etc. This weekly iron program is in full swing in some states of India like Tamil Nadu and is implemented as a directly observed swallow pill on Saturdays. Saturdays are now being earmarked for adolescent care and counseling clinics.

There are some suggestions that iron supplementation may result in increased risk of infectious morbidity. In a malnourished child with diarrhoea and vomiting, iron is often poorly tolerated and may aggravate the GI problem. As long as carrier proteins are not available, unbound iron in the gut may facilitate proliferation of *E coli.* In a well-nourished and stable patient, this risk is less. So iron supplements are advised only after stabilizing the patient and preferably after deworming.

Vitamin A: Vitamin A deficiency is now declining and florid vitamin A deficiency (VAD) leading to blindness is rare. Still, it is prevalent in some backward pockets of the community and among some pregnant mothers who often complain of night blindness. The overall prevalence of VAD in India is up to 6% and in some backward pockets it is up to 12%. Vitamin A prophylaxis programme was started in 1977 and was integrated with Immunization Programme in 1992. It is warranted if the prevalence of VAD is above 0.5%. The indicators of vitamin A deficiency as per WHO that signifies public health importance are given in Table 4.20. Vitamin A deficiency can be made out by serum retinol level, conjunctival impression cytology (CIC), 1% Rose Bengal staining test and dark adaptometry. In this method, the ability of the pupil of the eye to constrict under illumination is tested. By flashing a light at one pupil and covering the other, the degree of impairment can be estimated. Slow pupillary reflex is an early sign of VAD. Hypervitaminosis A and the side effects of vitamin A also require mention in this context. The important ones are its teratogenic effect, hyperostosis of tibia, benign intracranial hypertension (BIH), and the so-called Gulf syndrome. The Gulf syndrome refers to hypervitaminosis A and D due to excessive intake of the attractive A and D/fish oil pearls that are brought from Middle-East countries.

For treatment of VAD, the WHO/UNICEF/IVACG Task Force 1998 recommends vitamin A concentrate 1 lakh units for < 1 year and 2 lakh units for > 1 year of age x 2 doses on 2 consecutive days and repeat doses at 6 months interval. In severe PEM, apart from the initial 2 doses, monthly doses are recommended till recovery. In GI upset, severe deficiency and impending blindness, injectable vitamin A (Aquasol A) may be given. But it should be followed up with vitamin A concentrate to ensure storage. Water-soluble injections do not serve this function of storage.

Table 4.20 Indicators of clinical and functional vitamin A deficiency that signifies a public health problem

Indicator	Minimum prevalence
Night blindness (24-71 months)	> **1%**
Bitot's spot (6-71 months)	> 0.5%
Corneal scars/ulcer (6-71 months)	> 0.05%
Keratomalacia (6-71 months)	> 0.01%

(Ref: WHO indicators for assessing vitamin A deficiency and their application in monitoring and evaluating intervention programmes. WHQ/NUT/1996.10)

Vitamin A maintains the integrity of the skin and the mucus membrane and reduces bacterial binding to the mucosa and protects the surface tracts like GIT and respiratory tract and is also shown to decrease morbidity and mortality due

to ADD and ARI. This is highly effective in post measles cases and hence one dose of vitamin A is recommended in all measles cases.

Iodine: Iodine is a critical nutrient in early nervous system growth. Intrauterine deficiency can cuase mental retardation. Goiter, growth retardation and mental subnormality occur in iodine-deficient areas. Iodine deficiency disorder (IDD) is now identified as a major public health problem all over India, including the coastal districts. The prevalence is reported to be around 16% among preadolescent children. Endemic goiter is due to overt deficiency; simple colloid goiter is thought to be due to subclinical deficency, which is rampant even in coastal districts of Kerala, where it was thought to be not a problem. If the prevalence of goiter is > 5%. endemic goiter is suspected and the urinary iodine will be low in these deficient subjects. A reagent-treated strip or dipstick is being designed to do spot analysis of urine. When urinary iodine is not low in such goitrous area, goitrogens should be suspected as the cause of goiter. Other causes are cobalt deficiency, selenium deficiency and certain infections like *Yersinia.* The WHO criteria for assessing severity of IDD based on urinary iodine is given in Table 4.21. Universal Salt lodisation (USI) was started in 1984 and up to 30-50 ppm potassium iodate is fortified at the manufacturing level to ensure at least 15 ppm to the beneficiary. Excess iodine can lead to iodism, dermatitis, goiter and rarely, thyrotoxicosis. USI programme has had major setbacks due to the disparity in price and quality among the various products available in the market. It is high time that the government ensures the right quality of iodised salt at the right price. Double fortified salt with iodine and iron is another breakthrough in the research field and this is being tried out in Andhra Pradesh. Such research should reach the community.

Table 4.21 Epidemiological criteria for assessing severity of IDD based on median urinary iodine levels

Median value (g/L)	Severity of IDD
< 20	Severe IDD
20-49	Moderate IDD
50-99	Mild IDD
> 100	No deficiency

Ref: WHO/UNICEF/ICCIDD. Indicators for assessing iodine deficiency & their control through salt iodisation. WHO/NUT. 1994.6

Folic acid/folate (vitamin B_u): It is important in cell maturation, especially RBC maturation. Deficiency leads to megaloblastic anaemia and concurrent iron and folic acid deficiency leads to dimorphic anaemia. Folic acid and $B_{,2}$ deficiency may also produce periungual and knuckle pigmentation. Polyglutamates

available in some vegetables are less bioavailable than pteroyl monoglutamate (folic acid). It is the most critical vitamin during the first 4 weeks of pregnancy that regulates neurulation and neural tube closure. Deficiencies lead to nutritional anaemia, premature birth and LBW. Addition of folic acid to iron in pregnancy has led to positive impact. It should be ideally consumed periconceptionally to prevent neural tube defects like meningomyelocele. Dose recommended is 400 mg/ day. It seems to be an ideal gift to a couple who are thinking of the family way. It should also be given to mothers who already had a similiar problem in a previous pregnancy. It is also found to be beneficial in megaloblastic anaemia following anticonvulsant therapy. Some cases of subacute combined degeneration, homocystinuria and stroke syndromes also benefit from folate supplementation.

5. Interaction of Micronutrients and Developmental Biology

Human development proceeds as complex interdependent, yet distinct and precisely integrated biological programmes. Following fertilization, developmental programmes are initiated by selective reading of the genetic code leading to the generation of the various cell types, organs and organ systems that constitute mammals. These programmes are dependent on the expression of the maternal and foetal genomes and regulated by maternally derived supply of nutrients. In fact, most developmental programmes, including organogenesis, limb formation, and central nervous system (CNS) maturation, require specific nutrients for their initiation and progression. Nutrients serve as informational molecules that interact with the genome and trigger or facilitate developmental programmes. Micronutrients act as gene activators.

Mammalian development is dependent critically upon the function of the maternal and foetal genomes and the availability of maternally or placentally synthesized small molecules including hormones, cytokines and growth factors. Mammalian development proceeds as a precisely orchestrated temporal programme that requires coordinated changes in foetal and maternal gene expression as well as a complete and regulated maternally derived nutrient supply (Figure 4.17).

During more advanced stages of foetal development, nutrient deficiencies can impair the progression of morphogenic processes resulting in developmental impairment of varying severity. In many cases, deficiency or excess of particular nutrients at a critical time will impair specific genetically programmed developmental processes. Therefore, the risk of these impairments is associated with specific developmental period or *critical window.* Nuclear receptors, otherwise known as ligand-activated target transcription factors, represent a common mechanism whereby individual nutrients specifically can and do influence maternal andfoetal gene expression and thereby determine cell lineage in the foetus. With the genetic code now solved, nutrition is the safest and most effective means to modify genome structure and function.

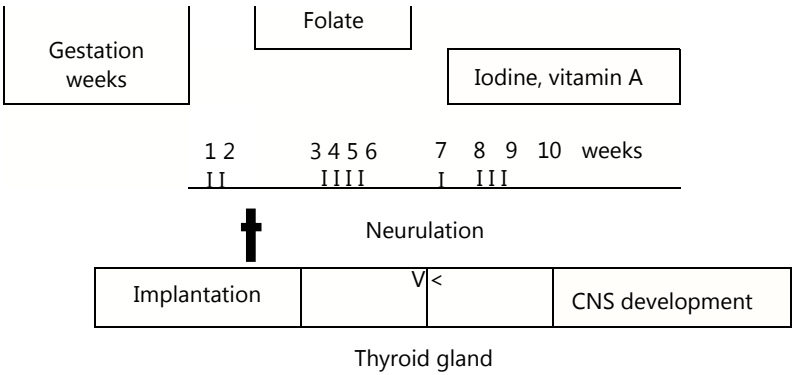


Fig 4 17 **Role 0,:** nutrient:s at a critical time of developmental process, i.e., the *critical window*

Early nutrition affects not only brain development, growth and body composition but also metabolic programming. This influences the occurrence of diet-related adult chronic diseases, immunity, coping up with stress, capacity for physical work, congitive and educational performance as in Figure 4.18.

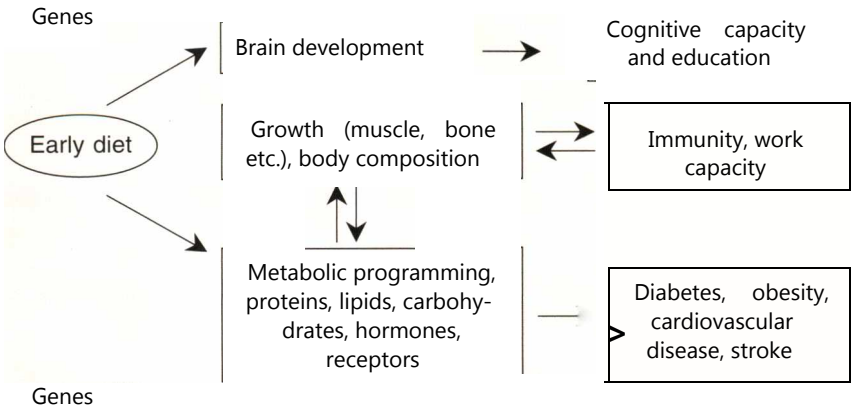


Fig. 4.18 Interaction between early nutrition and genotype

6. Micronutrients interactions—Implications on Preventive and Therapeutic Prescriptions

The interactions among the various micronutrients are well known and have important implications as for example, zinc depletes copper and competes with iron absorption, high phosphorus hinders calcium absorption and may be severe enough to cause hypocalcaemic tetany in artificially fed babies, in spite of high content of calcium in cow's milk. There are some useful interactions also, as for example vitamin A and C enhance iron absorption.

7. Current Recommendation

The methods for micronutrient nutrition are medical supplementation, food fortification and dietary diversification and increased consumption. The former two are expensive and not available to poor and weaker sections of the community. There may also be side effects and drug interactions. So, the best way to improve micronutrient nutrition and to prevent micronutrient deficiency disorder (MDDs) is **food diversification and healthy eating practices.** The 'rainbow revolution' should be intensified to improve supply and consumption of micronutrients. Nutrition action plan should be designed and implemented to achieve this goal. Micronutrient fortification strategies like double fortification of salt with iodine and iron also should reach the needy community. Biological products like spirulina, which is a treasure source of micronutrients, should be tapped instead of pills for each micronutrient, which may act as 'double edged sword' due to toxicity and drug interaction.

SECTION 5

Diet in Critically 111 Patients

"Feeding the sick is the greatest of all virtues."

—*Old Indian Proverb*

5.1 Fluid and Electrolyte Therapy

Fluid and electrolyte therapy is meant to maintain or restore normal volume and composition of body fluids. It is life saving in those with dehydration, blood loss and in postoperative patients.

1. Fluid Compartments

At birth, total body water (TBW) is 78% of the body weight. It gradually decreases and almost reaches the adult value of 55-60% by one year of age. **It is 55% in females and 60% males.** In the foetus, extracellular fluid (ECF) is more than intracellular fluid (ICF) and it reduces to reach the adult pattern by one year of age. Generally, ICF is 30-40% of body weight and ECF is 20-25%. In ECF, 15% is interstitial fluid and 5% is plasma water. The third compartment is transcellular fluid; GI secretions, urine, CSF and intraocular, pleural, peritoneal, pericardial and synovial fluid. The fourth is the slowly exchangeable fluid compartment (5-10% of body weight) present in the bone, cartilage etc. TBW can be calculated using a simple formula. **TBW (L) = Weight (kg) x 0.6 + 0.251**

$$\text{Conversion from mg to mEq is done as} = \frac{\text{mg} \times \text{Valency} \times 10}{\text{Atomic number}}$$

2. Osmolarity

Except for transient changes, the ECF and ICF are in osmotic equilibrium and the total cations and anions are balanced. ECF osmolarity is determined by Na and its accompanying anions Cl and HCO_3 and is roughly double that of serum Na

except when there is hyperglycaemia, hyperlipidaemia etc. If serum Na is 140 mEq/L, S. osmolarity is 280 mOsm/L. Fluid therapy consists of three categories—**maintenance therapy** to replace usual body losses, **deficit therapy** to replace extra losses and **supplemental therapy** to replace ongoing losses. A stable patient may need only maintenance therapy before restoring oral intake, but a dehydrated patient with extra blood/fluid loss may need all the three categories of replacement. Close monitoring of serum electrolytes and central venous pressure (CVP) should be done during prolonged fluid therapy.

3. Maintenance Therapy

This is meant to replace the insensible and renal water losses. Insensible loss is mainly through the lungs and skin and very little quantity through motion. 100 ml fluid/100 kcal will be enough for this. This can be calculated based upon the meter squared system or the Holliday and Segar formula based on weight (Table 5.1).

Table 5.1 Holliday and Segar formula for calculation of calories and fluid

Weight range	Quantity
1st 10 kg	100 kcal or ml/each kg
10-20 kg	1000 + 50 kcal or ml/each kg > 10
> 20 kg	1500 + 20 kcal or ml/each kg > 20

In children, generally calorie requirement is calculated for expected weight (RDA for age) and fluid for present weight. In adult both are calculated for present weight

The average requirement is 1500 ml/M^2/day. i.e., an average adult with 1.73 m^2 of surface area, the requirement is 2595 ml/24 hours (5 pints), i.e., a 65-kg adult will require 2500 ml (5 pints). Up to 30% increase in requirement can occur with excessive physical activity and similar decrease can occur with decreased physical activity in a comatose patient. Up to 12% increase can occur with 1°C rise in body temperature and similar decrease can occur with 1°C fall in temperature in a hypothermic patient. Infants and children require relatively more fluids than adults.

Requirement increases when solute load is high as in diabetes, after infusion of mannitol, radiocontrast dyes etc., and it decreases to two-third in renal failure, cardiac failure and due to **syndrome of inappropriate ADH (SIADH) secretion** as in head injury, pneumonia etc. In SIADH, full maintenance may be given if no oedema or hyponatraemia. Thus the fluid requirement in an adult varies from 1-3 L/m^2/day. In oliguric or polyuric patients, insensible loss plus renal loss should be replaced on a ml to ml basis. Insensible loss is approxi-

mately 400 ml/m²/day, i.e, 700 ml/day in an adult. In children it is proportionately low. Insensible loss is replaced as 10% dextrose and output as 50% normal saline and 50% or 10% dextrose (ref. section 7.3).

a) *Calories:* Calorie requirement is calculated based upon the Holliday and Segar formula. The ideal weight is taken for calculation in children as RDA is for the age of the child and not for the present weight. 100 ml of 5% dextrose/ DNS supply 20 calories. Maintenance fluid with 5% dextrose will supply at least 20% of calories. Even though inadequate, such low calorie intake has some sparing effect on catabolism of proteins. If IV fluid is needed for more than a few days, higher concentrations of dextrose solution with or without amino acids may be needed. **Up to 12% glucose can be given through peripheral vein.**

b) *Glucose:* The glucose intake can vary from 50-300 g/m²/day. 1500 ml/m²/day of 5% dextrose/DNS (i.e., 2500 ml/day in an adult) will supply about 75 g/m²/ day of glucose. In adult, at least 1730 ml of fluid (3'/2 pints) given should be 5% dextrose/DNS in order to supply the minimum glucose, i.e., 50 g/m²/day. Unlike in adults, in children generally 5% dextrose alone is not given; instead paediatric maintenance solutions which supply 5% dextrose, 1/5 NS and electrolytes like Isolyte P are given. In diabetic patients, normal saline is given till RBS reduces to 300. After that, solutions containing glucose can be given.

c) *Sodium:* Sodium requirement varies from 5-250 mEq/m²/day. **Sodium requirement is more in those with increased gastrointestinal losses, fistula, nasogastric drainage etc.** It is less in those with renal, hepatic and cardiac diseases and oedema. Gastric aspirate should be measured and replaced as normal saline. 1 litre of 0.9% N saline or DNS supplies 154 mEq of sodium (Table 5.2). In adults, at least 4 pints should be given in order to supply about 180 mEq/m²/day of sodium. Normal serum sodium is between 135-145 mEq/L. Na deficit is calculated as follows:

Na deficit (mEq) = (135 - S. Na) x Weight (kg) x 0.6 + 0.251

d) *Potassium:* Potassium requirement is 10-250 mEq/m²/day. Potassium losses parallel sodium losses. Due to the high reserve of potassium, it need not be administered if IV fluid is not continued except when hypokalaemia is expected or documented. Serum K^+ level tends to be high in tissue trauma and anoxia due to release of intracellular potassium. In shock and renal dysfunction, hyperkalaemia may be seen. In patientg on diuretics prior to surgery and following gastrectomy, duodenal fistula, ileostomy, ulcerative colitis, rectal tumours etc., potassium loss may be great. Potassium content of GI fistula may be double that of plasma concentration. 15% KCI supplies 2 mEq/mL of potassium. For ordinary maintenance, 1 mL of KCI is added to 100 mL of IV fluid. In hypokalaemia up to 2 mL of KCI can be added to 100 mL of IV fluid (40 mEq/L). Hypokalaemia is often present in PEM. diabetic ketoacidosis, severe asthma on frequent nebulization etc. Hypokalaemia produces hypotonia,

Table 5.2 Composition of common IV fluids (mEq/L)

IVF	Glucose (9/L)	Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	Others (mEq/L)
NS	·	154		154	·
RL	-	131	5	111	$HC0_3$ 29 Ca 4
5% Dextrose	50	-	-		-
3% Saline	-	510		510	-
Maintenance fluid					
Isolyte P* with dextrose	50	27	19	19	$HC0_3$ 27 Mg 3
Isolyte E** with dextrose	50	141	10	103	Citrate 8 Mg 3
Isolyte G*** with dextrose	50	65	17	149	NH_4 70
Sancelyte P*	33	47	5	35	$HC0_3$ 18 Mg 1

Others	mEq/mL
7.5% $HC0_3$	0.9
10% Ca glue.	1.8
15% KCI	2.0
10% KCI	1.3

**For paediatric use **For enteric loss ***For gastric loss*

depressed tendon reflexes, abdominal distension, ileus, slurred speech and cardiac arrhythmia. Cardiac arrhythmias are more common in those with pre-existing myocardial disease. ECG changes in hypokalemia are prolonged QT interval, depression of ST segment and flattening or inversion of T wave. Hyperkalaemia usually occurs with renal dysfunction. Hyperkalaemia produces arrhythmias and cardiac arrest. ECG shows tall tented T waves more than 10 divisions. Hyperkalaemia is treated with 1-3 mEq/kg of sodium bicarbonate IV, 0.5 ml/kg of 10% calcium gluconate and insulin glucose drip. An easy way is to give **10 ml/kg 5% dextrose with 0.1 U/kg of plain insulin as**

a slow drip. In severe cases, exchange resins or dialysis may be needed. Normal serum K^+ is 3.5-5 mEq/L.

e) *Bicarbonate:* Acidosis usually occurs due to renal failure or accumulation of organic acids. Acidosis produces rapid and deep (Kussmaul) respiration. However, overcorrection may lead to tetany. In mild cases, it is corrected only if it is associated with hypokalaemia. The requirement varies from 5-250 mEq/ m^2/day. The usual dose of 0.5 ml/kg of 7.5% soda bicarbonate (1 ml = 0.9 mEq) will raise serum bicarbonate by 1 mEq/L. In metabolic acidosis, solutions that replace chloride by bicarbonate can be given, e.g., Ringer lactate. Bicarbonate deficit is calculated as:

HCO, deficit (mEq) = Desired level (15 mEq/L) - Observed level x Weight (kg) x 0.6+ 0.251

Usually only half-correction is aimed at and so the multiplication factor can be 3 instead of 6.

Alkalosis is rare as kidney has enormous ability to excrete bicarbonate, but alkalosis can occur due to excessive administration of alkali or milk, loss of hydrogen ion as in pyloric stenosis and hypokalaemia. Acidifying agents like ammonium chloride may be rarely needed. Normal serum HCO, is 22-28 mEq/L.

f) *Chloride:* Chloride requirement varies from 0-250 mEq/L. Normal serum chloride is 96-108 mEq/L. Maintenance solutions are commercially available (Table 5.2). The calculated maintenance fluids can be given as readymade solutions or as a combination of normal saline, DNS and 5% glucose with electrolytes as per requirement.

4. Deficit Therapy

Deficit results from blood loss, diarrhoea, vomiting, dehydration, third space loss, starvation etc. Fluid deficit is represented as percentage of body weight lost. **In children, mild, moderate and severe dehydration represent 3-5%, 7-10% and 10-15% loss of body weight as against 3%, 6% and 9% loss in adults.** For 1% deficit, 10 ml/kg fluid is required, i.e., extra 30-150 ml/kg fluid should be given for deficit therapy. But in practice, only 2/3 need be corrected;
i.e., about 20-100 ml/kg is enough. The signs of dehydration are summarised in Table 5.3. The type of dehydration is based upon serum sodium. In isotonic or isonatraemic dehydration, serum sodium is normal; in hypotonic or hyponatraemic dehydration, serum sodium is < 130 mEq/L and in hypertonic or hypematraemic dehydration serum sodium is > 150 mEq/L. It is prudent to know the type of dehydration, but the primary concern should be to detect and correct hypovolaemia irrespective of serum sodium level.

5. Hypovolaemia

It is divided into three categories—covert compensated, overt compensated and decompensated. **Shock is decompensated state.**

Table 5.3 Symptoms and signs of dehydration

Item	Mild	Moderate	Severe
General appearance*	Thirsty, alert restless	Thirsty, drowsy	Drowsy/comatose cold, sweaty, cyanotic
Pulse	Normal	Rapid	Feeble or not palpable
Respiration	Normal	Deep and rapid	Deep and rapid
Systolic BP	Normal	Normal or low postural hypotension	< 90 mm Hg or unrecordable
Skin pinch*	Retracts readily	Retracts slowly	Retracts very slowly > 2-3 seconds
Eyes	Normal	Sunken	Grossly sunken
Tear	Present	Absent	Absent
Mucosa	Moist	Dry	Very dry
Urine output	Normal	Reduced	Not passed for many hours
Thirst*	Normal	Increased	Inability to drink
Body weight loss+	3-5%	7-10%	10-15%
Estimated fluid deficit (ml/kg)	30-50 ml	70-100 ml	100-150 ml
Fluid to be given (ml/kg)	20-30 ml	50-60 ml	60-100 ml

+ Lower range in adult and higher range in children

These are called key or star signs of dehydration

a) *Covert compensated:* This is the commonest type and is often missed. Systemic circulation is maintained at the expense of splanchnic circulation. Symptoms are increased thirst, drowsiness, nausea and hiccoughs. BP is normal. If it is not corrected, it can lead to multi **organ dysfunction syndrome (MODS).** Whenever in doubt, give a bolus of 10-20 ml/kg normal saline or Ringer lactate (RL) and then reassess. If the diagnosis is correct, patient will improve. Otherwise, look for organ dysfunction, serum chemistry and jugular venous pressure (JVP) for overhydration.

b) *Overt compensated:* Patient shows increased sympathetic drive with tachycardia, wide pulse pressure and cold, clammy extremities, but systolic BP is maintained. There may be postural hypotension. There is confusion, drowsiness, tachycardia and tachypnoea. It is better to raise the leg and give one bolus of 10-20 ml/kg NS or RL in half to one hour. In hypoglycaemic patients,

DNS may be given instead of normal saline. If the diagnosis is correct, tachycardia, tachypnoea and wide pulse pressure will resolve. Low dose dopamine 5 mg/kg/minute may be started to ensure renal perfusion.

c) *Decompensated:* This is shock state with decreased perfusion of vital organs in spite of redistribution of blood flow. BP is low and initial tachycardia changes to bradycardia with myocardial hypoxia. 2-3 rations of 10-20 ml/kg IV bolus with reassessment (check BP) may be given till circulation is maintained and urine output is more than 1 ml/kg/hour. Haemacel and blood may be given in bleeding. 5-10 mg/kg/minute dopamine drip is also to be started. When more than 3 bolus rations are needed, blood chemistry, JVP and CVP monitoring should be undertaken; maximum up to 8 rations may be given. Once the serum electrolyte results are available, the type of dehydration can be categorised.

6. Types of Dehydration

a) *Isotonic or isonatraemic dehydration:* **The commonest type of dehydration** is isotonic or isonatraemic, i.e., serum Na 135-145 mEq/L. The deficit therapy is calculated according to the degree of dehydration as mild, moderate or severe deficit requiring 30-150 ml/kg fluid. Due to intracellular shift of water and sodium, only 2/3 of the calculated dose (20-100 mL/kg) need be given in 3-6 hours as normal saline or Ringer lactate. In starving patients, DNS may be given to administer glucose. If 20 ml/kg bolus is given to restore circulation, that much fluid must be adjusted in the rest of the calculation.

b) *Hypotonic or hyponatraemic dehydration:* When serum Na^+ is < 130 mEq/L. hyponatraemic dehydration is diagnosed. This is due to excessive administration of electrolyte-free glucose solution, diarrhoea, diuretic therapy etc. According to the degree of dehydration, fluids, preferably NS/RL, may be administered for deficit therapy. When serum Na is < 120 mEq/L, 12 ml/kg of 3% hypertonic saline can be given at a rate of 1 ml/minute. **In hyponatraemic dehydration, there is lethargy, cellular oedema, cold extremities, cyanosis and shock.** Rarely, convulsions may occur due to cellular oedema.

c) *Hypertonic or hypernatraemic dehydration:* When serum Na is >150 mEq/L, hypertonic dehydration is diagnosed. Cellular dehydration, CNS haemorrhage and sequelae can occur. This should be corrected only slowly at a rate of 10 mEq/L/day. Otherwise, convulsions can occur which may respond to 3-5 ml/kg 3% sodium chloride or 20% mannitol. A suitable regimen to correct hypernatraemic dehydration is to give roughly 2/3 maintenance as 5% dextrose with about 25 mEq/L of sodium as a combination of chloride and bicarbonate. Readymade maintenance solutions are available to provide this (e.g., Isolyte). It can also be prepared by 1/5 N saline in 5% glucose. Each pint should contain saline and glucose. **Electrolyte-free solutions like 5% dextrose should not be given.** In hypernatraemia. **patient is irritable and puffy.**

Skin turgor may be maintained and extremities may be warm. This occurs due to excessive administration of salty formula. Hypertonic dehydration can occur in diabetes, renal failure and following mannitol. Hypernatraemic dehydration is rare.

Fluid therapy is summarized in Table 5.4. In oligo-anuria, hypovolaemia is first corrected by giving 20 ml/kg NS/RL, followed by frusemide 1 -2 mg/kg. If still oliguric, treat as acute renal failure (ARF). Management of ARF is given in Section 8.3.

5.2 Enteral Nutrition

1. Introduction

Twentieth century has the credit of several remarkable inventions and innovations. Total parenteral nutrition (TPN) is an amazing accomplishment of this century. The dusk of this century has witnessed a mushrooming of superspecialities in the field of medicine. In the glamour of performing sophisticated techniques like gamma knife surgery that attracts instant publicity, basic techniques like feeding are relegated to the background. More than 50% of the patients who seek medical care are malnourished and many more develop malnutrition after admission to the hospital. About 54% of deaths occur in malnourished patients. Feeding is as important as breathing to maintain life. Provision of food and water to the sick and needy is considered as the most important of all human virtues.

Nutritional support is necessary when there is reduced intake, inadequate absorption, increased loss and increased demand. It may be difficult to show that nutritional support alters the outcome of many disease processes, but it helps in restoring the nutritional status of the patient and it plays an adjunctive role in the disease and immune process. It improves physical and mental function. It decreases the effects of catabolism. It prevents further weight loss and death from cachexia. It restores normal body tissues during convalescence and reduces the duration of hospital stay.

2. Physiology of Malnourished Patients

Moderate to severe malnutrition affects fat and protein turnover with an enhanced reliance on fat as the energy source. Subsequently there will be erosion of protein stores, fat depletion and expansion of extracellular fluid (ECF) compartment. Cellular function is altered with deficits in membrane potential, alteration in cellular hydration, reduction in key enzymes and deficits in high energy phosphates. These result in functional impairments like respiratory dysfunction, muscle dysfunction, immune dysfunction, impaired wound healing and repair of damaged tissues and psychological dysfunction. Short-term nutritional therapy facili-

Table 5.4 Fluid therapy in various clinical settings

Stable patient	Hypovolaemia	Oliguria/anuria
■ Maintenance fluid/5% dextrose with 1/5 NS ■ Replace gastric aspirate as NS ■ Serum urea, electrolytes after 24-48 hours in prolonged IV fluid therapy ■ KCI in K^+ loss or hypokalemia (1-2 ml_/100 mL of IVF)	***Isonatraemia*** *m* Mild: 20 mL/kg IV bolus NS or RL (1 ration) and reassess ■ Moderate: 2 rations IV bolus and reassess. ■ Severe: 3 rations IV bolus or 2 rations and 1 ration Haemacel or blood if Hb is low or bleeding profusely	■ Correct hypovolaemia ■ Fluid challenge: bolus 20 mL/kg NS or RL followed by frusemide ■ Still oliguric, insensible loss' + last day's output ■ Frequent serum biochemistry ■ Nephrology consultation
Suspected hypovolaemia ■ 20 mL/kg IV bolus NS or RL ■ Blood urea, serum electrolytes after 8-12 hours ■ Reassess and give maintenance	■ Blood urea, serum electrolytes after 8-12 hours ■ Reassess and give maintenance ***Hyponatraemia*** • If dilutional (SIADH), give 2/3 maintenance ■ In Na^+ loss 12 mL/kg 3% saline in 24 hours ■ Daily serum biochemistry ***Hypernatraemia*** ■ 2/3 maintenance as 1/5 N saline with dextrose or readymade solution	

Replace all ongoing losses through drains, fistula, etc. In hypernatraemia, see that each ration contains 1/5 parts NS in glucose. Hypotonic solutions like 5% dextrose alone should not be given. Encourage the patient to drink plain water if oral intake can be allowed in case of hypernatraemia

tates substantial improvement in physiological functions in 3^4- days, but it may not lead to weight gain in the patient. **Weight for height is the goal during short-term nutritional management.**

The metabolic effects of stress response include hypermetabolism, hypercatabolism, increased urinary nitrogen excretion, altered amino acid profile, hyperglycaemia, immunosuppression, altered vascular permeability, acute phase responses like fever and increased acute phase proteins and altered gastric function. The normal metabolic response to starvation is a marked reduction in measured energy expenditure (MEE) and the eneirgy requirements may decrease by 30%.

The state of starvation or severe PEM is equated to that of 'hibernation' among animals. However, during infections or following trauma, the body is unable to economise energy expenditure. **In head injury patients, energy expenditure has been estimated to be as high as 120-250% of the normal.** Even after barbiturate coma or paralysis with pancuronium, the energy expenditure remained 20-30% above the normal. **For every** 1°C **rise in body temperature, 10% calories should be given extra.** At least 10-20% of extra calories have to be provided during illness. In a normal child, the energy expenditure is as follows:

Basal metabolism	50%
Activity	25%
Growth	12%
Specific dynamic action (SDA)	5%
Faecal loss	8%

3. Nutritional Requirements

Any child with malnutrition or any child with inadequate nutrient intake for 7 days or more should receive nutritional support. In a well-nourished child, the calorie and fluid requirements can be calculated based on the Holliday and Segar formula utilizing the observed weight of the child. In a malnourished child also, the fluid requirement can be calculated based on the observed weight. Up to 10 kg body weight, 100 kcal/kg is given. Above 10 kg, 1000 kcal plus 50 kcal/each kg above 10 is given and above 20 kg, 1,500 plus 20 kcal/each kg above 20. However, they need a liberal supply of calories to recoup their weight and to promote growth.

In the community setting, the Recommended Dietary Allowances (RDA) for the age can be prescribed. RDA for the age is estimated using the ICMR recommendations or the bedside calculation. The ICMR recommendations are placed at plus 2 standard deviations in order to cover the requirements of well-nourished children belonging to high socioeconomic status. This is the ideal or

maximum requirement and is almost on par with the American recommendations. It appears to be quite high for a medium or small frame child. The bedside calculation is the minimum requirement and this may be enough for an average frame child. It is calculated as follows: 100 kcal/kg up to 1 year, 1000 kcal at 1 year, thereafter 1000 plus 100 kcal/each year up to puberty. At puberty, the requirement is equated to 1 unit of energy (2,400 kcal).

In the hospital setting, during nutritional management instead of the RDA for the age, a therapeutic calculation can be made; i.e.. 150-200 calories/kg (observed weight)/day in moderate to severe malnutrition respectively.

In a bedridden child. 2/3 of the requirement can be given with at least 10-20% extra calories to meet the stress of fever and illness. Similarly, fluid calculation may be restricted to 2/3 of the requirement in view of syndrome of inappropriate ADH (SIADH) secretion whenever indicated. In oligo-anuria, the fluid should be restricted to insensible loss plus last day's urine output. Insensible loss is calculated roughly 400 ml/m^2. It can be calculated using a bedside calculation.

Newborn	30 ml/kg
Infants	25 ml/kg
1-5	20 ml/kg
Above 5 years	15 ml/kg
Adults	10 ml/kg

Insensible loss is generally given as 10% dextrose. 50%' of the urine output can be given as N. saline and the rest as 10% glucose. However, in severe oedema, fluid and saline can be further restricted. In chronic renal failure and other conditions with growth retardation, RDA (calories) for the height age can be given instead of the chronological age. Oral feeds should be sta-rted early in all sick children on IV fluids. A model menu is given in Table 5.5.

1. **Carbohydrate**
   50-60% of the total calories can be supplied as carbohydrate. In a hospital setting, up to 15-30 g/kg/day of glucose can be given. Maintenance fluid using 5% glucose can supply up to 20% of calories. This is just enough to prevent significant protein catabolism. Oral supplements should be started early.
2. **Proteins**
   10-15% of the calories can be derived from protein. 1.5-2 g/kg/day is usually given. In moderate to severe PEM, up to 3-4 g/kg/day can be given. In practice, when calories are adjusted, protein supply will be much more than what is required. But this may be low-quality protein and not first class protein alone. Total protein intake should not exceed 7 g/kg/day at any cost. In renal failure, it should be restricted to 0.25-0.5 g/kg/day. In hepatic dis-

Table 5.5 Model Menu

5-year-old child with 15 kg weight on maintenance IV fluid and tube feeds

Calories	RDA (bedside)	1,400 kcal
	2/3 RDA	1,000 kcal
	+ 10% extra for illness	1,100 kcal
Fluid	Holliday & Segar	1,250 ml
Protein	1.5-2 g/kg	22-30 g

Item	Fluid (ml)	Energy (kcal)	Protein (g)
Maintenance fluid Isolyte P or 5% glucose with 1/5 saline	500	100	
50% glucose added to drip*	50	100	
High energy milk	400	400	12
Cereal flour 12 tsp or SAT mix 9 tsp		200	4
Isodense orange juice	100	100	
Isodense egg flip	200	200	10
	1250	1100	26

**Isolyte P +50% glucose gives 50 g of glucose in 550 ml making 9.1% glucose solution. 10% glucose can also be made by mixing isolyte P and 25% glucose in the ratio 3:1.*

eases, formulations high in branched chain amino acids (BCAA) are indicated.

3. **Fat**

   25^45% of the calories can be given as fat. In practice, up to 10-15% of the total calories can be given as visible fat. Coconut oil has the advantage of medium chain triglyceride (MCT) whereas others like sunflower oil have the advantage of polyunsaturated fatty acids (PUFA). Coconut oil is deficient in EFA and sunflower oil is deficient in omega-3 fatty acids. So it is ideal to give breast milk to the child whenever possible to prevent EFA deficiency. Up to 3% of the calories should come from EFA. **The polyunsaturated to monounsaturated to saturated fatty acid ratio may be kept at 1:1:1 and the ratio of omega-6 to omega-3 fatty acids should ideally be less than 5:1. MCT is preferred in hepatobiliary diseases and fat malabsorption.**

4. **Mieronutrients**

   The demand for vitamins and minerals increases during illness due to poor

intake, excessive loss and as a coenzyme in metabolic and degradation pathways. This may further increase due to drug intake. 5-10 times the requirement may have to be given during disease states. In hepatobiliary and pancreatic diseases, fat-soluble vitamins may be given up to 5 times the usual requirement and water-soluble vitamins may be given up to twice the usual requirement.

4. Route of Administration

The route of administration can be enteral or parenteral.

The golden rule in nutrition is that "if the but works, use it". Enteral nutrition promotes GI function and stimulates enteric trophic hormones like enterglucagon and gastrin. It prevents and treats malnutrition and ensures early recovery. Other nutrients like glutamine, polyamines, short chain fatty acids and ketones also promote GI function. The role of colostrum as the first immunization to paint the gut with protective factors is extremely useful. Enteral feeding should be tried when there are no GI contraindication to oral feeding, e.g., peritonitis, ischemic enteritis, necrotising enterocolitis, GI bleed etc.

Enteral nutrition is preferred to parenteral nutrition because:

1. It is more physiological.
2. It is simple to administer with very few complications.
3. It is 8 times less costly than parenteral nutrition.

A major limiting factor regarding enteral nutrition is the high osmolarity of the preparations. High osmolar load leads to diarrhoea and nutrient and electrolyte malabsorption. The enteral preparation should be iso-osmolar with plasma (275-295 mOsm/L). Most preparations are at the range of 300^400 mOsm/L. Isodense preparations are those which supply 100 calories/100 ml (Table 4.18). It is possible to make preparations calorie dense by adding fat without increasing the osmolarity. The fear that it may cause diarrhoea and malabsorption is not true. It is well tolerated.

a) *Routes of EN*

i) Oral: Oral is the best route for feeding. Sucking at the breast should always be encouraged in infants. Even when milk supply is minimal, sucking will promote maturation of gut, orofacial development and emotional satisfaction. Both nutritive and non-nutritive sucking are beneficial.

ii) Feeding tubes: Orogastric or nasogastric tubes can be used for feeding the comatose, preterm and kwashiorkor patients. Tube can also be placed beyond the pylorus or into the jejunum (naso-jejunal).

iii) Gastrostomy/jejunostomy: In atresias, stenosis and stricuture, feeding gastrostomy and jejunostomy are useful.

b) *Mode of Administration of EN*

The desired calories can be given as 6-8 feeds/day. 2-hourly feeds during day time and one late night and one early morning feed can be planned.

Tube feed can be given as bolus feeds or as a continuous drip. Bolus feed should be allowed to fall by gravity using the syringe-based system. It mimics normal feeding, but it is less tolerated. Continuous drip can be given using IV drip set or induce peristalsis and will prevent nausea, vomiting, diarrhoea, abdominal pain etc.

5. Complications of EN

i) GIT: Gagging, stricture, oesophagitis, gastritis, nausea, vomiting, abdominal cramps.
ii) Mechanical: Dislodgement, tube migration, inflammation, granulation of skin site, obstruction of tube, aspiration.
iii) Metabolic: Fluid overload, dehydration, **hyper- and hypoglycaemia,** electrolyte imbalance, azotaemia, hypercapnia, hyperphosphataemia, hypercalcaemia, vitamin and mineral deficiencies.
iv) Infection: Aspiration pneumonia

Diarrhoea and abdominal cramps are due to rapidly delivered formula, hypertonic formula, bacterial contamination etc. Nausea and abdominal distension are due to ileus, fat intolerance, aerophagy, large residuals, hyperosmolar formulas and unpleasant odour of formulas. Some of these can be overcome by reducing the osmolarity. Isodense formulas and amylase rich food (ARF) are useful in this context.

Overhydration is due to rapid infusion rate or excess fluids. This can be overcome by thickened feeds, diuretics and by reducing the sodium content. Dehydration is due to reduced intake and overthickened feeds. Azotaemia is due to high proteins or renal insufficiency. Aspiration is due to malpositioned tubes, gastric hypomotility, GE reflux, neurological damage etc. Continuous infusion and infusing the formula beyond the pylorus are beneficial. Prokinetic agents like cisapride also may be indicated in some cases.

Polymeric formula can be used for enteral feeding. Elemental formulas will be needed in those with extremely short bowel. The organic chemical taste and odour render them unpalatable and hence these cannot be taken by mouth. Kitchen-based formulas like isodense formulas can be prepared using an electric blender (Table 4.18). A list of commercial enteral and parenteral preparations are given in Table 5.6.

Table 5.6 Commercial preparations for parenteral and enteral use

I. PN products available in Indian market

A. Amino acid infusions

1. Aminoplasmal — B. Braun
   a) 5% Solution — 500 ml, 250 ml and 100 ml
   b) 10% Solution — 500 ml, 250 ml and 100 ml
   c) Aminoplasmal — 21% branched chain amino acids

2. Proteinsteril Fresenius
   a) Proteinsteril 10%—500 ml, 100 ml
   b) Proteinsteril Hepa—8% and 5%—500 ml
      (42% branched chain amino acids)
   c) Proteinsteril Nephro—500 ml (rich in arginine)
3. Vamin-G (Vamin-9-Glucose) - 500 ml, 100 ml (Pharmacia & Upjohn)
4. Aminocore 5% - 500 ml; 10% - 500 ml (Core Parenterals)
5. Aminoven

B. Lipids (intralipid)
- 10% intravenous lipids are usually given
- MCT/LCT 50:50 combination is preferred

II. **Enteral preparations available in Indian market**

1. Pepti 2000 Elemental diet Renal solute load - 334 mOsm/L 500 ml liquid or 5 sachets Energy 100 kcal	Osmolarity - 40 mOsm/L
2. Bonvit	Renal solute load 330 mOsm/l Energy 100 kcal
3. Recupex 500 g sachet	Low lactose, low residue Energy 220 kcal
4. Tocal 100 g packet	Lactose free, residue free Energy 500 kcal
5. Renocare I	For pre-dialysis state 4 sachets = 1000 kcal
6. Renocare II	4 sachets = 1016 kcal
7. Ten-O-Lip	6 sachets = 1800 kcal
8. Ten-O-Lip-LF	Lactose free 1 sachet = 250 kcal

contd.

9. Glutameal - Plain — 5 sachet = 1400 kcal
   Glutameal - Diabetic 1 sachet = 6.6 g glutamine
   Glutameal - Iron
10. Pramilac — Skimmed milk powder
11. Protal M — 25 g Pack = 86 kcal

III. **Paediatric enteral preparations**

1. Lactodex (Raptakos) Low lactose, low fat for diarrhoea — 100 ml = 55 kcal
2. Lactodex LBW for preterm and LBW — 100 ml = 80 kcal
3. Lactodex follow up (above 6 months of age) — 100 ml = 64 kcal
4. Zerolac (Raptokos) Lactose free — 100 ml = 64 kcal
5. Trophox (Raptakos) Protein supplement — 15 ml = 6 g protein
6. Energex (Indon) Lactose free — 100 g = 500 kcal

5.3 Partial and Total Parenteral Nutrition

1. Introduction

Even though enteral route is the best, some of the hospital patients require enteral with partial parenteral nutrition (PPN) or total parenteral nutrition (TPN). TPN is expensive. The cost is estimated to be as high as Rs. 2,000/day in an adult. It is one-third to half of this in a child. TPN for more than three weeks can produce cholestasis, altered immunity, altered T4/T8 ratio and various nutrient and micronutrient deficiencies. **The American Society for Parenteral and Enteral Nutrition (ASPEN) and the Indian Society for Parenteral and Enteral Nutrition (ISPEN) guidelines** are the following:

a) Use PN only when oral intake is grossly inadequate or when tube feeding is

not possible due to poor tolerance or psychological reasons.

b) Whenever possible, use PPN supplementing the inadequate enteral intake.

c) When PN is indicated and when the patient is not critically ill and it is required for a period less than two weeks, a peripheral venous access is preferable.

d) In critically ill patients, compromised haemodynamically and/or requiring PN for more than 2 weeks, a central venous access is preferable.

e) Peripheral PN should not be considered if the glucose concentration in the infusate exceeds 10-12%. Any concentration of lipids can be infused peripherally.

Early enteral feeding should be the ultimate goal. It is very good to stimulate the gut as soon as possible after a period of starvation.

2. Goals

a) *Calories:* In health, the calories from the RDA are utilized as follows: BMR - 50%, activity - 25%, growth - 12%, specific dynamic action (SDA) - 5% and faecal loss - 8%.

In a bedridden child, activity, growth and SDA reduce and only 2/3 of the RDA need be given. Add 10-15% of the calculated calories towards extra requirement due to fever and illness. For every 1 degree Celsius rise in temperature, give 10% calories more.

b) *Fluids:* The requirement is roughly 1 ml for each calorie. The fluid requirement is usually calculated based on the Holliday and Segar formula and taking into account the actual weight of the child (Table 4.1). In conditions like head injury, pneumonia, meningitis, nephritis etc., where syndrome of inappropriate ADH (SIADH) secretion is probable, reduce to 2/3 requirement. In oligo-anuria, give only insensible loss and last day's output.

c) *Glucose:* 15-30 g/kg/day of glucose can be given. 5% glucose maintenance fluid will supply 20% calories and is enough to decrease protein catabolism. Up to 6-12 mg/kg/minute glucose can be given. 10% dextrose ml/kg/day x 0.07 will give mg/kg/min (Fig. 5.1)

d) *Protein:* The RDA for age or roughly 1.5-2 g/kg/day is given. In renal failure, restrict to 0.25-0.5 g/kg body weight.

e) *Fat:* Up to 30% of total calories can be given as fat. 1-3 g/kg/day fat can be given.

f) *Vitamins and minerals:* The demand for multivitamins and minerals like iron, zinc etc., increases during illness. This is due to poor intake, excessive loss or due to utilization as co-enzyme in metabolic and degradation pathways, e.g., drug intake.

In total parenteral nutrition (TPN), all the nutrition for homeostasis and growth are given through parenteral route. In hyperalimentation, at least 150% of RDA is given to achieve positive nitrogen balance and weight gain. In partial

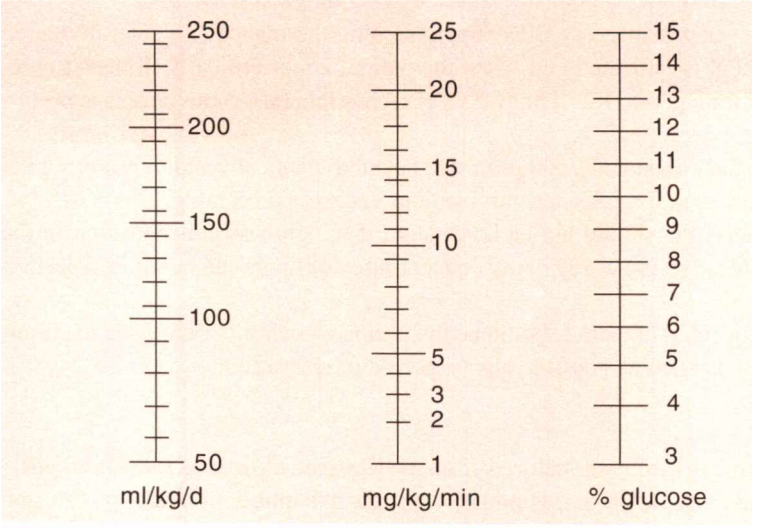


Fig. 5.1 Glucose rate calculator

parenteral nutrition (PPN), 30-50% is given parenterally and the rest through enteral route.

3. Indications for TPN (Table 5.7)

According to Rudrick and Ruberg, TPN is needed for all patients who cannot eat, should not eat or will not eat. The clinical conditions are:

a) Congenital anomalies like tracheo-oesophageal fistula, oesophageal atresia etc.

b) GI diseases like NEC, meconium ileus, intestinal fistulas, short bowel syndrome, peritonitis, inflammatory bowel diseases (IBD)

c) Others like extreme prematurity, severe bums, severe systemic diseases, anorexia nervosa etc.

The absolute indications are when bowel rest is needed for more than 2 weeks and when nutritional requirements exceed the capacity of partial PN. Nutrients to be given are glucose, emulsified fat, amino acids, vitamins, mineral mixture, electrolytes and water. **Oral feeding should be started as early as possible and TPN should be stopped when 70% of the kilocalories is taken orally.**

Table 5.7 Indications for TPN

A. **Neonates**

a) Absolute indications : Intestinal failure (short gut, functional immaturity, pseudo-obstruction)
Necrotising enterocolitis

b) Relative indications : Hyaline membrane disease, promotion of growth in preterm infants, possible prevention of necrotising enterocolitis

B. **Infants and children**

a)	Intestinal failure	Short gut Protracted diarrhoea Chronic intestinal pseudo-obstruction Post-operative abdominal or cardio-thoracic surgery Radiation/cytotoxic therapy
b)	Exclusion of luminal nuti	Crohn's disease
c)	Organ failure	Acute renal failure Acute liver failure
d)	Hypercatabolism	Extensive burns Severe trauma

4. Nutrients (Table 5.8)

a) *Calories:* Newborn 110-125 kcal/kg/day
Children 100-110 kcal/kg/day

The sources of energy are hypertonic glucose, isotonic fat emulsion, amino acid solution and others like fructose, xylose, sorbitol. Up to 0.7-0.8 kcal/ml can be given through peripheral vein.

b) *Glucose:* Dose—10-30 g/kg/day. Rate of infusion is 6 mg/kg/minute. In hypoglycaemia up to 12 mg/kg/minute can be given. Glucose is given as 10%, 20%, 25%, or 50% glucose. **Up to 10—12% glucose can be given through the peripheral vein.** Add 4 ml of 50% glucose or 8 ml of 25% glucose to each 100 ml of 10% glucose to make 12% glucose for PN. 1 g glucose supplies 4 calories.

c) *Isotonic fat emulsion:* As it is iso-osmolar, peripheral vein also can be used (280-330 mOsm/L). The preparations available are Liposyn with 10% saf-flower oil, egg yolk lecithin and glycerol, and **Intralipid** with 10% soyabean oil instead of safflower oil. It is expensive and needs separate IV line. It is tolerated by extreme preterms, those with respiratory distress syndrome and liver

dysfunction. 0.5 g/kg can meet EFA requirements. MCT/LCT ratio 50:50 is preferable.

Table 5.8 Practical guidelines for paediatric PN

A. Newborns (term) (per 24 hr)

Day	CHO (g/kg)	Fat (g/kg)	Amino acid (g/kg)	Fluid (ml/kg)
1	10	1.0	0.75	60
2	12	2.0	1.50	70
3	12	3.0	2.00	80
4	14	3.5	2.50	90

B. > one month, < 10 kg (per 24 hr)

Day	CHO (g/kg)	Fat(g/kg)	Amino acid (g/kg)	Fluid (ml/kg)
1	10	1	1.0	120
2	12	2	1.5	130
3	13	2	2.0	140
4	14	3	2.5	150

C. Children Between 10 -30 kg (per 24 hr)

Day	CHO (g/kg)	Fat(g/kg)	Amino acid (g/kg)	Fluid (ml/kg)
1	4.5	1.5	1.5	100
2	7.5	2.0	2.0	no

D. Children over 30 kg (per 24 hr)

Day	CHO (g/kg)	Fat(g/kg)	Amino acid (g/kg)	Fluid (ml/kg)
1	2	1	1	70
2	5	2	1.5	80

Dose: 1-3 g/kg/day. 10% lipids give 10 g fat/100 mL and supply around 90 calories. 20 calories is derived from glycerol and phospholipid, i.e,. I 10 kcal/ 100 ml. **EFA deficiency can occur in parenteral nutrition without lipids.** It should be given at least once in 3-4 days,

d) *Protein:* Protein hydrolysates are not popular now. Crystalline amino acids like Aminosyn - Abbott, Freamin III - McGaw, Travasol 10% - Baxter, Aminodrip, Vamin (7%), Astymin 9% and Proteinsteril 6% are currently used.

Dose: 2-5 g/kg/day. 9% Astymin provides 9 g/i 00 ml. These can be mixed with glucose-based maintenance fluids. Excess can cause hyperchloraemic acidosis, cholestasis, giant cell hepatitis, amino aciduria etc. Enough calories as glucose and fat should be given to spare amino acids from being used up for energy.

e) *Electrolytes:* Paediatric maintenance solutions can provide most of the electrolytes for normal maintenance.

f) *Vitamins and minerals:* Standard multivitamins (MV1 Paediatric - Armour) should be given at least 2-3 cc/day. Folic acid 100 mg, vitamin K 1-2 mg and B_{12}10 mg are given biweekly IM. Standard mineral mixtures are also added at least 2 ml/day. The composition of a standard preparation is given in Table 5.9. Oral preparation like Aquamin 1-3 tsp/day can also be given. **Excess copper and manganese can aggravate cholestasis.** Some fluids also contain fluoride, iodide etc.

Table 5.9 Composition of a standard mineral mixture (100 ml)

Item	Quantity
Zinc	200 mg
Copper	20 mg
Chromium	0.2 mg
Manganese	5 mg
Selenium	1.5 mg

g) *Fluids:* 70-150 ml/kg fluids are given. Start with lower dose and increase by 10 ml/kg/day. LBW babies require more fluid (Table 5.10). Also refer tables 5.8 and 5.1.

Table 5.10 Fluid requirement of LBW and preterm babies (ml/kg/day)*

Day	Weight < 1 kg	1-1.5 kg	1.5-2.5 kg	> 2.5 kg
1	120	100	80	60
2	130	110	90	70
3	140	120	100	80
4	150	120	110	90
5	160	140	120	100
6	170	150	130	110
7	180	160	140	120
8	190	170	150	130
9	200	180	160	140
10	200	180	160	150

* Dai ly increment 10 ml/kg/day
Use electrolyte free 10% glucose in 1-3 days
After 3 days, Isolyte P: 25% glucose in the ratio 3:1 may be given,
making a 10% glucose solution

5. Guidelines for Calculation of TPN

The guidelines for calculation of TPN are summarised in Table 5.11.

6. Model Calculation

TPN for a 10-month-old baby with 5 kg weight

a) *Goal:*

Calories 5 x 110 = 550 kcal

Fluid 5x 150 =750 ml

Table 5.11 Calculation of TPN: A ready reckoner

Age group	Calories (kcal/kg)	Fluids (ml/kg)	Glucose (9/kg)	Amino acid (g/kg)	Lipids (g/kg)
Newborn*	125	70-150	7-15	2-5	1-3
Infant	110	120-200	15-30	2.5-4	1-3
Child	100	70-150	7-15	2-3	1-3

**In preterm babies, 110-160 kcal/kg & 80-200 ml/kg fluid may be given*

Decide the protein and lipid fraction first and then make up the rest of fluids and calories by adjusting the strength of glucose. In hyperglycaemia, insulin is given in a dose of 0.05 U/kg/hour.

b) *Lipid*

3 g/kg 5x3=15

10% intralipid 15 g= 150 ml = 150 kcal

Start with 0.5 g/kg/day and increase 0.5 g/kg/day up to 2-3 g/kg/day. Intralipids can be given daily or once in 4 days. In some centres, heparin (1 U/mL) is added. **Heparin is inactivated by vitamin** C.

c) *Protein*

4 g/kg 5 x 4 = 20 g

9% Astymin 18 g = 200 ml = 72 kcal

Start with 0.5 g/kg/day and increase 0.5 g/kg/day up to 2-3 g/kg/day. In acidosis, buffer with sodium bicarbonate.

d) *Glucose*

16 g/kg 5 x 16 = 80 g

20% glucose 80 g = 400 ml = 320kcal

Start with 6 mg/kg/minute and increase 2 mg/kg/minute every day up to 12 mg/kg/minute. Start 10% glucose and increase the strength by 2%/day. Unutilized glucose may lead to fatty liver.

e) *Fluid:* 750 ml total (150 + 200 + 400) = 542 kcal (150 + 72 + 320)

f) *Others:* Add sodium, potassium, calcium, phosphorus, magnesium, multivitamin and other minerals. **Phosphate is added first followed by calcium. Sodabicarh should not be mixed with calcium and insulin.**

MVI Paediatric 1-2 ml and mineral mixture 1-2 ml are given. Also give vitamin K 0.5-2 mg/every 2 weeks, vitamin B 0.5 cc IM/every week and iron (Imferon) 1 cc IM/every week.

Ensure that folate 100 mg/day is present in the MVI. The requirement of IV sodium and potassium is 2-3 mEq/kg/day and IV calcium and phosphorus is 20-40 mg/kg/day.

Oral feeding should be started as early as possible and TPN should be weaned in 3^4- days. The quantity of nutrients supplied by commercially available TPN solution at an infusion rate of 135 ml/kg/day is given in Table 5.12. **Glutamine, short chain fatty acid and ketone bodies are usually not available in solutions.** The commercial preparations are given in Table 5.6

Table 5.11 Quantity of nutrients in TPN solution (135 ml/kg/day)

TPN solution	Dose/kg/day
Amino acid	2.2 g
Glucose	27.0 g
Sodium	4.3 mEq
Potassium	4.1 mEq
Chloride	4.1 mEq
Acetate	4.1 mEq
Calcium	1.2 mEq
Phosphorus	1.4 mEq
Magnesium	1.1 mEq
Sulphate	1.1 mEq
Calories	117 kcal
Fluid	135 ml

7. Procedure

Practise strict asepsis and change infusion bottle and sets every day. The solutions should be prepared under laminar flow hoods. Jugular or subclavian cannulation or cut down is done and 22-24 catheter is introduced to enter SVC, curl the

catheter in the skin tunnel and connect through millipore filter to an infusion pump. Use 1.9-2.7 F silastic catheter in subclavian vein. Lipid and 12% glucose can be given through the peripheral vein. It is ideal to have a TPN team consisting of physician, staff nurse, pharmacist, technical assistant and nutritionist. Generally, dextrose and amino acid are given in one hand and intralipid in another line or intralipid is connected with the infusion through a connector beyond the filter.

Currently, mixed system is tried. First amino acid and glucose are mixed and lastly fat is added, slowly shaking the bottle.

8. Monitoring

Daily weight and intake output (I/O) chart. Daily biochemical parameters like Dextrostix (4 times), urine sugar (4 times), urine acetone, electrolytes, acid-base balance. Maintain plasma osmolarity at 285-300 mOsm/L. Weekly LFT, total protein. albumin, creatinine, blood culture and biweekly haemocrit, urea, ammonia, lipid profiles, calcium, phosphorus etc. Serum triglyceride should not exceed 150 mg%. Check serum for turbidity due to fat emulsion.

9. Complications

a) *Catheter related* like perforation, kinking, thrombosis, air embolism, pneumothorax, cardiac arrhythmias.

b) *Metabolic* like hyperglycaemia, electrolyte and acid-base imbalance, azotaemia, hyperammonaemia, aminoaciduria, fatty liver, hyperbilirubinaemia, cholestasis etc.

c) *Nutritional* like weight loss, vitamin, mineral, EFA, and phosphate deficiency, hypervitaminosis, hypermanganesaemia, metabolic bone disease etc.

d) *Sepsis:* is a major complication that should be prevented. The complications are high in untrained hands.

10. Partial Parenteral Nutrition (PPN)

It is more often used as it can be given into a peripheral vein. The composition of a 12% PPN solution is given in Table 5.13.

11. Contraindications for TPN

Hyperbilirubinaemia, azotaemia, shock, thrombocytopenia, extreme prematurity and RDS **are contraindications.** Intralipid is contraindicated in jaundice, bleeding due to hypercoagulability and thrombocytopenia.

12. Percutaneous Insertion of Central Line

Insertion at the bedside

a) *Selection of vein:* Select any big peripheral vein or antecubital vein or temporal vein in front of the ear.

Table 5.13 Composition of a 12% PPN solution

Item	Quantity
50% glucose	40 ml
N.saline	100 ml
$NaHC0_3$	15 ml
KCI	5 ml
Ca gluconate	5 ml
MVI	2 cc
10% glucose	400 ml

Amino acid 30 g/L can be added to this, e.g., Astymin

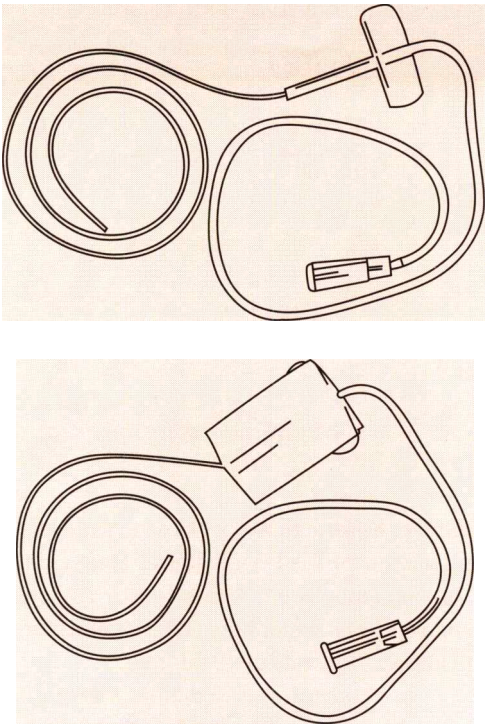
b) *Selection of catheter:* Silastic tubes or Broviac catheters may be used. Silicon catheter with dacron cuff is least irritable. Dow Corning Medical Grade Tubing can be cut and autoclaved. A fine silastic catheter (OD 0.64 mm) is used in babies up to 6 kg and wider bore (OD 0.9 mm) for older children. The fine catheter will allow a flow rate of 40 ml/hour

c) *Procedure:* Except in neonates, oral sedation is given half an hour before. Measure the distance from the vein along its course to the right nipple to reach the right atrium. The length of the catheter should be 15 cm more than this measured length. Follow strict asepsis as in a surgical procedure. Chlorhexidine is used for cleaning the area. **When Broviae catheters are not available, 2 butterfly sets may be used.**

A 19G butterfly with the plastic tubing cut away may be used to thread 0.64 mm catheter and a 17G Venflon cannula for 0.9 mm catheter. The butterfly needle or cannula is inserted into the vein. Blood flow through the needle is controlled by gentle pressure on the needle tip.

Using fine forceps, the silastic catheter is threaded several mm at a time into the needle along the vein till 15 cm remains outside. Change the position of the patient if there is obstruction to catheter advancement. The needle used to enter the vein is now discarded. Fix the catheter to the skin at the site. The end of the catheter is now threaded through the cylindrical plastic needle guard of a 25G butterfly needle in 0.64 mm catheter and 23G in 0.9 mm catheter. This second butterfly needle with the tubing and hub is inserted into the catheter (Fig. 5.2).

The guard is then pulled over the needle. The butterfly needle and its plastic guard are sandwiched between two pieces of elastoplast. The needle may be blunted prior to insertion by rubbing the serrated face of an artery forceps several times up and down along the bevel of the needle. The butterfly needle



Pjg ,j **2** Fine silastic catheter with 25G butterfly suitable for parenteral nutrition in infants

and tubing are flushed with heparinised saline (1 U/ml) prior to insertion. The remaining catheter is coiled on the skin and is covered by a piece of gauze dressing. Secure the whole assembly by elastoplast tape or dynaplast. Flush the assembly with heparinised saline. X-ray can be taken to see that the catheter is in the right atrium after injecting 1 ml of 45% Hypaque. If there is free venous return on suction, parenteral alimentation can be started. Block in the catheter can be corrected by flushing or by injecting 2500 U urokinase and reflushing 3 hours after clamping. The TPN solution is then discarded and new solution set up in view of contamination.

IB. Central Vein Cannulation

Double-lumen **Broviac** catheter is used for central venous catheterisation. It is inserted by surgical cut down cannulation into the internal jugular or subclavian under sedation. The second lumen can be used for blood and blood products and for blood sampling. Braun Cavafix is used in some centres. For young infants,

Cavafix 134-20G outer cannula with inner 22G needle is used and in older children Cavafix 257-16G outer cannula with inner 18G needle is used. The catheter is threaded through the cannula and then the cannula is slipped back and fixed near the hub. The quality of the catheter is very important to prevent breakage within the vein and thereby catheter embolism.

14. Practical Difficulties in the Indian Context

a) Lack of PN units even in major hospitals.

b) High incidence of metabolic, mechanical and infectious complications due to lack of expertise and team work.

c) Non-availability of suitable infusates. All in one PN bags of 1000 ml capacity designed for adults is not cost effective in children. Separate preparation of amino acids, lipids and dextrose are available in 500 ml bags rather than 100 ml bags. Standard vitamin and mineral mixtures are not readily available.

d) Mixing of PN solutions needs extra care and precautions. It should preferably be done in a laminar flow workstation or aseptic isolator cabinet. Amino acids and dextrose can be infused in one bag and lipid in another bag separately using two burette administration systems and a Y-connector. Lipid can also be administered separately. Lipids can also be mixed to the same slowly and in the end.

The concentration of the calories should be kept at a level enough to maintain life, but not high enough to cause metabolic disturbances. Glucose intolerance is uncommon and in infants, it should be regarded as an early sign of sepsis.

15. Setting up of a PN Unit

An ideal major hospital should set up a PN unit with the following infrastructure and personnel and should have an exhibited protocol for reference.

Clinical nutritionist or paediatrician or anaesthetist	Paediatric surgeon	Nursing staff	Pharmacy
Prescribes PN Clinical assessment and monitoring solution during trouble shooting	Insertion of venous lines, central and peripheral	Supervision Asepsis Ensure supply	Ensure supply Actual mixing

PN solution and dosage: The dosage and the selection of PN products depends on the age and weight of the child (Tables 5.5-5.12).

Up to 6-12 mg/kg/minute glucose can be given. 10% dextrose ml/kg/day x 0.07 will give the rate of glucose mg/kg/min. Glucose rate calculator can be used

for this calculation. Intravenous lipids can be given daily or once in 3^4- days. Vitamin K 0.5-2 mg is given every 2 weeks and vitamin B_{12} 0.5 ml and iron (Imferon) I cc IM every week. Ensure folate 100 mg/day in the infusate and calcium and phosphorus 20-40 mg/kg/day. In the commercially available infusates, 135 ml supplies 27 g glucose, 2.2 g amino acid and 117 calories.

Complications and troubleshooting: The complications are:

a) Catheter related like perforation, kinking, thrombosis, air embolism, pneumothorax, cardiac arrhythmias.
b) Metabolic like hyperglycaemia, electrolyte and acid-base imbalance, azotaemia, hyperammonaemia, aminoaciduria, fatty liver, hyperbilirubinaemia, cholestasis etc.
c) Nutritional like weight loss, mineral, EFA and phosphate deficiency, hypervitaminosis, hypermanganesaemia, metabolic bone disease etc.
d) Sepsis: is a major complication that should be prevented.

Complications are high in untrained hands. Trace element deficiencies are summarized in Table 3.8. The risk of catheter infection increases with the number of times the continuity of the infusion system is interrupted. Heparin is suitable for fine silastic catheter that block more easily. Asepsis should be observed while flushing a blocked catheter. After flushing, it is better to discard the infusate in view of contamination. It can be replaced with 10% dextrose solution until a new infusate is set up. If blocked catheter cannot be pushed with heparin (1 U/ml), it is worth injecting 2,500 units urokinase and clamping off for 3 hours and then attempt flushing.

16. Home Parenteral Nutrition (HPN)

Recently, HPN is also under trial. HPN is infused overnight and many patients can return to relatively normal lifestyle.

Oral stimulation, mobility, tender loving care (TLC), developmental stimulation and psychosocial support are essential during PN. Monitoring during PN includes frequent bacteriological cultures, urine and plasma chemistry and haematological evaluation.

In practice, the clinician will have to give both IV fluids and enteral nutrition. This may be essential to administer the required calories and protein. Model calculations are given in Table 5.5 and Table 5.14.

17. Conclusion

Nutritional support is a powerful therapeutic tool that modulates the stress response, provide specific fuels and growth factors. **Nutrition buys time to save patients threatened by diseases.** The ultimate aim of nutritional support should be to ensure survival and quality of survival.

5.4 Diet in Various Diseases

DIET IN A CRITICALLY ILL CHILD

1. Goals

a) *Calories:* **BMR is roughly 22 kcal/kg/ideal weight.** In a bedridden child, activity, growth and SDA reduces and only 2/3 of the RDA need be given. Add 10-15% of the calculated calorie towards extra requirement due to fever and illness. **For every 1 degree Celsius rise in temperature, give 10% calories more.**

Conversion of Fahrenheit to Celsius: $\frac{F-32}{1.8}$

Conversion of Celsius to Fahrenheit: (C x 1.8) + 32

b) *Fluids:* The requirement is roughly 1 ml for each calorie. The fluid requirement is usually calculated based on the Holliday and Segar formula, taking into account the actual weight of the child (Table 14.10). When syndrome of **inappropriate ADH (SIADH) secretion is probable, restrict to 2/3 requirement.** In oligo-anuria, give only insensible loss and last day's output. (Refer Section 8.3)

c) *Carbohydrate:* 15-30 g/kg/day of glucose is needed.

d) *Protein:* The RDA for age or roughly 1.5-2 g/kg can be given. In renal failure, restrict to 0.25-0.5 g/kg body weight.

e) *Fat:* In malabsorption states and hepatobiliary and pancreatic diseases, medium chain triglycerides (MCT) is preferred as coconut oil or cotton seed oil. Up to 10-15% of total calories can be given as visible oil/fat.

f) *Vitamins and minerals:* The demand for multivitamins and minerals like iron, zinc etc., increases during illness due to poor intake, excessive loss and to serve as coenzyme in metabolic and degradation pathways. This may further increase due to drug intake.

2. Routes of Administration

a) *Oral:* It is the preferred route in most cases. Enteral nutrition is tolerated by most patients unless there are GI contraindications to oral feeding, e.g., peritonitis, ischaemic enteritis, GI bleed etc. Enteral nutrition is essential to support life and to sustain GI function. Nutritive and nonnutritive sucking are essential in newborns and infants for maturation of gut, orofacial development and emotional satisfaction.

b) ***Nasogastric (NG) feeding:*** This is ideal in the comatose and in some preterm and kwashiorkor patients. Orogastric, nasojejunal and gastrostomy feeds are also given in certain situations.

c) ***Parenteral:*** Often parenteral/IV nutrition is given through peripheral route. In specific indications partial/total parenteral nutrition (PPN/TPN) are given.

3. Types of Food for Enteral Nutrition

a) ***Family pot feeding:*** Household items like milk, fruit juice, idli, rice, curd etc., can be given in less sick children.

b) ***Liquid diet:*** Soups, kanji water, fruit juice, egg flip, milk, butter milk, high energy milk, cereal milk, milk + SAT Mix (cereal pulse) etc., can be selected. **Isodense (isotonic) liquids that supply 100 kcal/100 ml** are preferred to supply calories and to prevent nausea, vomiting, diarrhoea etc. (Table 4.18) e.g.. **high energy milk** - *Vi* glass milk with 1 tsp sugar and *Vi* tsp oil; **high energy cereal milk** - *Vi* glass milk with 1 tsp sugar and 1 *Vt* tsp cereal flour **high energy cereal-pulse milk** - *Yi* glass milk with 2 tsp SAT Mix; **fruit juice** - 1 orange with 2 tsp sugar made up to 100 ml. **Egg flip - one egg** in 3/4 glass milk with 2 tsp sugar (200 cal in 200 ml). **Butter milk** or lassee - *Vi* glass yogurt with 2 tsp sugar.

c) ***Polymeric & non-elemental diet:*** Also refer Table 5.6. Carbohydrate, protein, fat etc., are processed and then given, e.g., Casec (Calcium caseinate - Mead Johnson), Promod (Whey - Ross), maltidextrin, MCT or coconut oil, com oil to supply long chain triglycerides (LCT), Polycose (glucose polymer-Ross), Ten-o-Lip, Ten-o-Lip LF (lactose free), Ten-o-Tube (Recon Ltd).

d) ***Elemental diet:*** e.g., amino acids, mineral mixtures, glucose, fructose.

e) ***Combination*** of the above also can be given.

4. Method of Administration

The desired calories can be given as 6-8 feeds/day. NG feed can be given as bolus feed allowed to fall by gravity or as a continuous drip. Continuous drip will not induce peristalsis and will prevent nausea, abdominal pain, diarrhoea etc., and allows maximum absorption. Bolus mimics normal feeding; but is less tolerated.

5. Complications

Complications of enteral feeding are:

a) Hypoglycaemia
b) Dehydration
c) Overhydration
d) Hypernatraemia
e) Hypercalcaemia (excess milk)
f) Vitamin and mineral deficiencies

g) Aspiration
h) Azotaemia
i) GI upset etc.

Diarrhoea and abdominal cramps are due to rapidly delivered formula, hypertonic formula, bacterial contamination etc. This can be overcome by decreasing the delivery rate, by reducing osmolarity, lactose content and by switching over to elemental formula/TPN. Vomiting, nausea and abdominal distension are due to ileus, fat intolerance, aerophagy, large residuals, hyperosmolar formula or unpleasant odour of formula. Overhydration is due to rapid infusion rate or excess fluids. This can be overcome by thickened feeds, diuretics and by reducing sodium content.

Azotaemia is due to high proteins or renal/hepatic disease. Dehydration is due to reduced fluid intake and overthickened feeds. Aspiration is due to malpositioned tubes, gastric hypomotility, gastrooesophageal (GE) reflux, neurological damage etc. In such situations, try continuous infusion or infuse the formula beyond pylorus.

6. Model Diet

5-year-old child, 15 kg, with pneumothorax and respiratory distress and refusal to take feeds. A model diet is given in Table 5.14.

Table 5.14 Model diet in a critically ill child

	Fluid	Energy Protein
Maintenance fluid-Isolyte P 500 ml with 25% glucose in the ratio 3:1		200 kcal
High energy milk	200 ml	200 kcal 6.0 g
SAT Mix (cereal pulse) 12 tsp		240 kcal 6.0 g
Coconut oil 2 tsp		80 kcal
Fruit juice (1 orange with 2 tsp sugar)	100 ml	100 kcal
Egg flip (3/4 glass milk + 1 egg + 2 tsp sugar)	200 ml	200 kcal 10.0 g
	lOOO ml	**1020 kcal 22 g**

a) Calories - RDA (minimum)-1400kcal (Bedsidecalculation)
2/3 of the RDA-1000 kcal
+ 10% extra for illness -1100 kcal
b) Fluid - 1250 ml (Holliday and Segar formula for present weight). Fluid restriction due to SIADH = 1000 ml
c) Protein - 1.5-2 g/kg = 22-30 g
d) Oil 10-15% of total colories = 110-165 kcal = 2'/2-4 tsp/day
e) Route of administration - IV and oral/IG

While stopping IV fluids, make up calories and fluid by *Kanji,* gruels, butter milk, sugar and coconut oil. Oil should be restricted to 10-15% of total calories; it can be given added to milk or cereals.

7. Energy Expenditure in Critically Ill Children

The nutritional requirements and the nature of fuel utilization in critically ill children have not yet been defined. Overfeeding can lead to many side effects, including diet-induced thermogenesis, increased carbon dioxide production, and fatty deposition in the liver. Underfeeding, on the other hand, may result in depletion of fat and protein stores and malnutrition. 16 to 20% of critically ill hospitalized children have been reported to develop significant actue protein-energy malnutrition (APEM), some within 48 hr of admission to a paediatric intensive care unit. Nutrient store deficiencies and APEM have been reported to be common early in the course of critical illnesses in children, especially in those < 2 years of age. Many children also may develop acute fluid and electrolyte malnutrition (AFEM).

Numerous equations have been used to estimate the caloric needs of the critically ill child. They calculate the predicted basal metabolic rate (PBMR) and then add the correction factor for the illness to calculate the predicted energy expenditure (PEE). There are many formulas that estimate PBMR in children, the most common being the Harris-Benedict, the Talbot, the Schofield, and the Food and and Agriculture Organization/World Health Organization. The accuracy of the stress-related correction to PBMR in critically ill children is yet to be determined. There are various methods to calculate energy expenditure.

Indirect calorimetry (IC): It is a clinically available tool that can give real-time accurate information of resting energy expenditure (REE) and substrate utilization. This is done by measuring the consumption of oxygen and production of carbon dioxide when a metabolic substrate is burned.

Bio-electrical impedance; Electrical impedance is a newer method used for body composition assessment. A major advantage of the bioelectrical impedance analysis (BIA) is its simplicity.

This procedure involves sending a very small current through the body which is unable to be felt and measuring its resistance. The underlying theory to

this procedure is based on the subjects' height and his/her resistance to a current. Lean tissue offers less resistance to a current as it contains more water and electrolytes than adipose tissue. Another advantage of this procedure is that no special training or skill is required.

While the accuracy of BIA is reported to be similar to skinfold measurements, research is continuing in order to improve the accuracy of the equations. This procedure holds promise for making accurate measurements easily and quickly.

Table 5.15 Equations for calculation of predicted basal metabolic rate (PBMR)

Harris-Benedict equation (kcal/day)
Males: 66.473 + (13.7516 x Wt) + (5.0033 x Ht) - (6.755 x age)
Females: 66.50955 + (9.5634 x Wt) + (1.8496 x Ht) - (4.6756 x age)

WHO equation	**(kcal/day)**
3 yr	Boy: (60.9 x Wt) - 54
	Girl: (61 x Wt) - 51
3-10 yr	Boy: (22.7 x Wt) + 495
	Girl: (22.5 x Wt) + 499
10-18 yr	Boy: (17.5 x Wt) + 651
	Girl: (12.2 x Wt) + 746
Correction factors (% of PBMR added to PBMR)	
Elevated temperature	+ 12% per °C above 37°C (98.6°F)
ARDS	+ 20%
Sepsis	+ 10% to 30% depending on severity
Trauma	+ 10% to 30% depending on severity
Surgery	+ 10% to 30% depending on severity

Present studies show that the measured energy expenditure (MEE) is much lower than the PBMR. The MEE did not differ significantly among disease groups or between medical and surgical patients. MEE was lower in the presence of multiple organ system failure (MOSF). The incidence of hypermetabolic state in critically ill child seems much less than in adults. There are probably several reasons for this low energy expenditure in critically ill infants and children. Mechanical ventilation decreases the work of breathing as well as heat loss through the respiratory tract and the energy expenditure for thermal regulation. Sedation decreases overall energy expenditure by reducing muscle activity. An additional factor is a reduction in nonessential or facultative metabolism such as growth, neurotransmitter synthesis and catecholamine-stimulated thermogenesis. It is

interesting to speculate that critically ill infants and children may be able to forego the energy expenditure required for growth during the acute phase of their illness. Therefore, direct measurement of energy expenditure is the only means currently available for accurate determination of caloric requirements. It would be better to give the caloric intake based on the predicted basal metabolic rate (BMR) than to give the caloric intake based on the predicted basal metabolic rate (BMR) than to give the RDA which leads to significant overfeeding.

Another bedside method is to give 2/3rd of the RDA for the patient with 10% extra for the stress and the illness; the advantage is that it is easy and is not technology dependent.

DIET IN DIARRHOEAL DISEASES

Oral rehydration therapy (ORT) has revolutionised the management of diarrhoea and it also has reduced morbidity and mortality. The control of diarrhoeal disease (CDD) and ORT programme was started in full swing in 1984-85. ORS trials were started in 1971. ***Diarrhoea can be defined as increase in frequency, fluidity and volume or bulk of stool compared to the normal bowel habit of that individual.*** Out of the various terminologies used, the preferred terms are acute diarrhoeal disease (ADD), acute dysentery (invasive diarrhoea), persistent diarrhoea (PD) of more than 14 days duration and chronic diarrhoea with features of malabsorption or failure to thrive.

1. Pathogenesis

Diarrhoeal diseases lead on to fluid and electrolyte malnutrition (FEM) and protein-energy malnutrition (PEM). Osmotic, secretory and invasive processes are the three main pathogenic pathways. Osmotic and secretory diarrhoeas are watery diarrhoeas with small bowel involvement. Villus cell damage leads to reduced digestion and absorption leading to osmotic diarrhoea. Enterotoxins lead to secretory diarrhoea (e.g., cholera, enterotoxigenic *E. coli* - ETEC). In rotavirus diarrhoea, absorption decreases and secretion exceeds absorption. In chronic diarrhoea, crypt cell hyperplasia leads to secretory diarrhoea and villus cell damage leads to disaccharidase deficiency and osmotic diarrhoea. Villus cells are absorptive and crypt cells are secretory. **Villus cell damage may stimulate erypt cell hyperplasia and secretory disorder may crop up on an osmotic diarrhoea.** Invasive diarrhoea produces blood and mucus due to cytotoxin-mediated inflammation in large bowel.

2. Investigations

In ADD, apart from history, investigations are not generally needed because most episodes are self limited to 4-7 days. This is the time needed to replenish mucosal cells from the basal layer. A search for non-GI infections like ARI and

UTI is often worth undertaking. In severe dehydration, renal function tests may be done. In persistent and chronic diarrhoea, stool microscopy, pH, reducing substances, culture, malabsorption studies like stool fat, chromatography for reducing sugar and biopsy may be needed. Stool sample mixed with a few drops of water with or without Sudan Red stain showing more than 6-8 droplets per low power field is abnormal and suggests steatorrhoea. The fat cells tend to move towards the periphery of the slide.

3. Management

Assessment of dehydration and categorisation into plan A, B and C for ORT is the cornerstone in the management of ADD (Table 5.16). ORS contains the following: sodium 90, potassium 20, chloride 80, citrate 10 or bicarbonate 30 and glucose 111 mmol/L. The electrolyte content of cholera stool is as follows; sodium 101, potassium 27, chloride 92; and non-cholera stool is: sodium 56, potassium 25, chloride 55 and bicarbonate 14 mmol/L. Ensure user-friendly ORS. ORS with increased sugar leads to osmotic diarrhoea. These are called oral dehydration solution/ODS. Continued breastfeeding and early feeding from the family pot reduce the duration and severity of diarrhoea and also prevent malnutrition. Convalescing children need an extra meal/day for two weeks. Acute dysentery requires drug therapy for 5 days (e.g., nalidixic acid, erythromycin, tetracycline, furazolidone, or metronidazole). Infestations like whip worm (WW) also need appropriate therapy (mebendazole). Mebandazole is given as retention enema in resistant cases, 2 tablets per day in 100 ml N. saline for 3 days.

Table 5.16 Management of ADD

Plan	Hydration	Quality	Fluid
Plan A	No dehydration	Vi-1 glass/purge	ORS/HAF
Plan B	Some dehydration	70-100 ml/kg in 4 hr	ORS
Plan C	Severe dehydration	100 ml/kg in 3-6 hr*	IVF-RL/NS

HAF - Home Available Fluids, RL - Ringer Lactate, NS - Normal Saline
This is given in 2 rations: 30 ml/kg first ration in 1 hour in infants and V_2 hour in others; and 70 ml/kg 2nd ration in 4 hours in infants and 2 56 *hours in others*

4. Approach to Persistent Diarrhoea

Persistent and chronic diarrhoeas pose diagnostic and therapeutic dilemma. It is important to understand the cause before initiating therapy.

Perianal excoriation suggests osmotic diarrhoea. Steatorrhoea suggests fat malabsorption. Blood and mucus are seen in dysentery, inflammatory bowel

disease (IBD), WW colitis, polyposis etc. Prolapse rectum is seen in WW colitis. Formed, watery and mucus stool with abdominal pain relieved by defaecation associated with normal growth often suggests irritable bowel/non-specific diarrhoea. Osmotic diarrhoea abruptly stops on fasting trial, but secretory diarrhoea due to crypt cell hyperplasia tends to respond slowly. In mucosal damage, crypt cell hyperplasia may occur and this in turn leads to secretory diarrhoea. Anti secretory agents like rececadoril may be beneficial. The causes of chronic diarrhoea are summarised in Table 5.17.

Table 5.17 Causes of chronic diarrhoea

1.	PEM & micronutrient deficiencies	Acrodermatitis enteropathica, lactose intolerance
2.	Post enteritis syndromes -	Sugar intolerance, protein-losing enteropathy, iatrogenic
3.	Ongoing infections	WW, giardia, GB Salmonella, TB, Cryptosporidium, mixed - tropical sprue like/bacterial overgrowth
4.	Allergic/food sensitive enteropathies	Gluten, cow's milk protein, soya protein, eosinophilic gastroenteropathy
5.	Malabsorptions	Cong, di- and monosaccharide intolerance, pancreatic insufficiency, enterokinase deficiency, microvillus atrophy
6.	Immunodeficiency syndromes	Congenital/acquired
7.	Hormonal	Vasoactive intestinal peptide (VIP) syndrome, thyrotoxicosis
8.	Surgical	Peritonitis, pelvic abscess, malrotation, short bowel syndrome
9.	Inflammatory bowel disease (IBD)	Crohn's, ulcerative colitis
10.	Miscellaneous	Non-specific/irritable bowel syndrome

WW - Whip worm, CB - Croup B

Apart from ORT and management of the cause, hypo-osmolar ORS, fasting trial and dietary manipulation by trial and error are often needed. The various steps are given in Fig. 5.3. In severe and prolonged diarrhoea, **hypo-osmolar super ORS** (or rice based) should be tried. Hypo-osmolar ORS is very useful in non-cholera stool (appendix 7). There are two types of hypo-osmolar ORS; one with sodium 75 and glucose 75 mmol/L and the other with sodium 60 and glucose 24 mmol/L. Super ORS is starch-based ORS, starch 40 g instead of glucose 20 g in *one* packet.

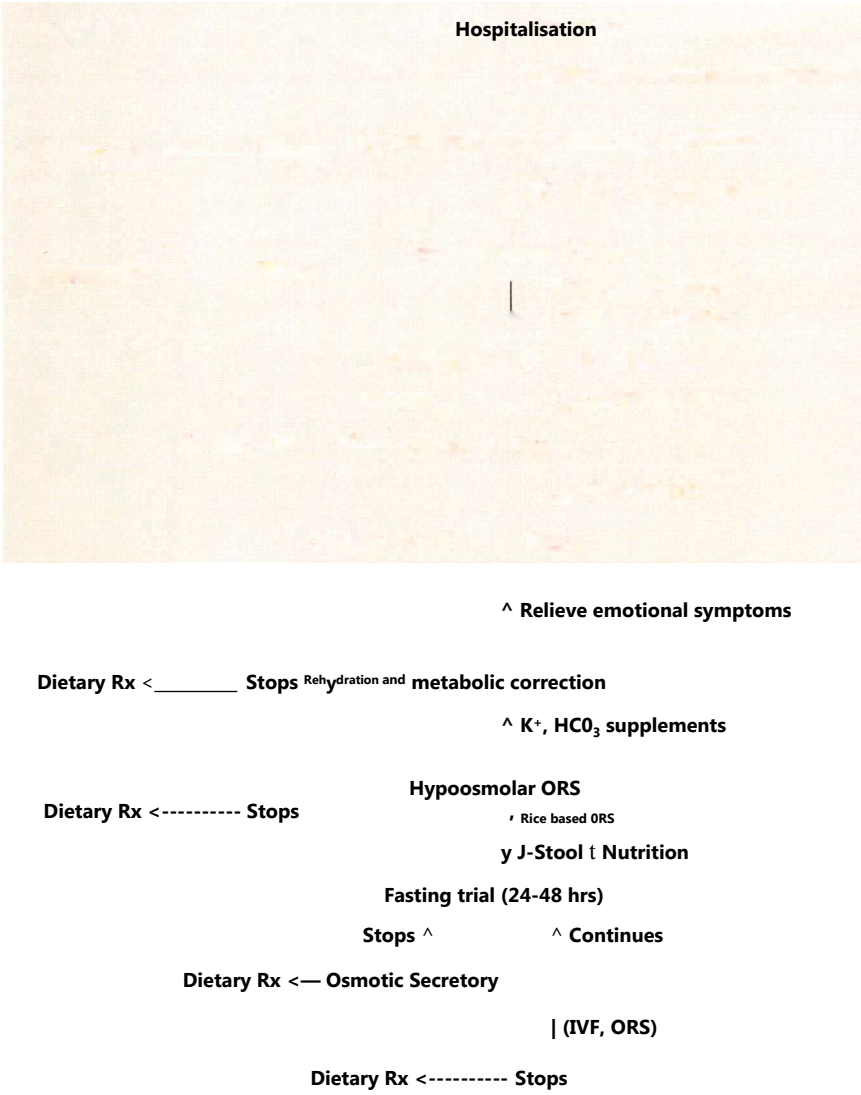


Fig. 5.3 Management of persistent diarrhoea

Hypo-osmolar or rice-based ORS is better accepted and it decreases stool output by 25% and also improves nutrition of the child. Dietary management is most important. Various stepwise diets are given in Table 5.18. Low milk, then milk-free and then starch-free diet can be tried in succession. In osmotic diarrhoea, disaccharide **(lactose, sucrose)** free diets are rewarding (e.g., Prosoyal). Commercial lactose-free preparations are given in Table 5.18. In **cow's milk protein intolerance (CMPI),** milk protein also should be avoided and soya protein can be tried (e.g., Nusobee). In **soya protein intolerance (SPI),** this should

Amylase breaks down starch into maltidextrins and reduces the viscosity and bulk of the porridge. It becomes hypoosmolar and calorie dense and does not swell as much as whole grain on cooking. Germination also enhances the vitamin content of the grain.

Soya milk, casein hydrolysate, whey hydrolysate and comminuted chicken may be tried. **Ten-O-Lip LF and Ten-O-Tube LF** (Recon Ltd.) are polymeric lactose-free diet for oral and NG feeding (Table 5.6). In resistant cases, oral elemental diet may be tried, e.g., glucose, fructose, amino acids etc.

Table 5.18 Dietary management of chronic diarrhoea

1.	Diet A/ Level I diet	Low milk diet (50 ml/kg/day) Curd, rice/sooji gruel with milk
2.	Diet B/ Level II diet	Milk-free diet - cereal-pulse mix/ gruel, amylase rich food
3.	Diet C/ Level III diet	Lactose-Sucrose/starch-free diet soya based, chicken based, egg

A. Low lactose diet (level I) cereal-based diet with little milk

Ingredients	Amount	k cal	Prot (g)
Milk	75 ml	52	2.6
Rice	10 g	34	0.8
Sugar	2.5 g	10	-
Water q.s	100 ml	-	-
Total	100 ml	96	3.4

Liquid consistency 1 kcal/ml, volume 100 ml; lactose 3.4 g.
Suji, or broken wheat (dalia), jaggery and green leafy vegetable can also be added.
Calorie density of the feed can be increased by adding more sugar (but more than 6% may cause osmotic diarrhoea) and oil (up to 3% is well accepted by children).

Lactose-free diet (level 2) Milk-free cereal-based diet

Ingredients	Amount	kcal	Prot (g)
Rice	15 g	48	1.2
Cooked moong	dal 5.0 g	17	1.1
Sugar	2.5 g	10	-
Coconut oil	2.5 g	22	-
Water q.s	100 mL	-	-
Total	**100 mL**	**97**	**2.3**

contd.

C. Lactose and sucrose free diet (level 3) Chicken glucose pure (For infants and children with severe persistent diarrhoea and malabsorption of disaccharides)

Ingredients	Amount	kcal	Prot
Chicken	100 g	110	26
Glucose	50 g	200	-
Coconut oil	50 g	440	-
15% KCI	7.5 ml	-	-
7.5% $NaHC0_3$	20-30 ml	-	-
Total	1000 ml	750 ml	26

The quantity of chicken can be increased up to 200 g according to the protein requirement of the child.

D. Commercial formula (lactose free)

Proprietary name (manufacturer)	Composition per 100 g powder (fortified with vitamins and minerals)				
	Protein (g%)	Carbohydrate (g%)	Fat (g%)	Cal (kcal)	Remarks
Isomil* (Abbot)	Soya protein isolate	50.3	25.46		Lactose free Sucrose (40%)
Soyal (FDC Ltd.)	Soya solid 18.0	Sucrose 5.0	Vegetable oil 18.0	466	Lactose free
Prosoyal (FDC Ltd.)	Soya protein isolate	Maltodextrin 15.6	Vegetable oil 24.0	506	Lactose free Sucrose free
Zerolac (Raptakos Bret)	Soya protein isolate 15.0	Maltodextrin 58.0	Vegetable oil 20.0	472	Lactose free Sucrose free
Nusobee-18 (Wockhardt)	Soya + Casein 18.0	Maltodextrin + Sucrose 56.9	Vegetable oil 20.0	480	Lactose free
Nusobee (Wockhardt)	Soyaprotein 15.6	Maltodextrin 50.0	Vegetable oil 20.0	412	Lactose free Sucrose free
Simyl MCT** (FDC Ltd)	Casein 14.0	Maltodextrin 61.4	Vegetable oil 18.0	412	Lactose free Sucrose free

contd.

Pregestimil***(Mead Johnson)	casein hydrolysed 19.0	Tapioca starch 90.0	Corn oil 27.4	453 Lactose free Sucrose free

** With methionine, saffiower oil 42%, coconut oil 30%, and soy oil 28%*
***With medium chain triglyceride, carnitine, taurine*
****With tyrosine, tryptophan, lecithin and medium chain triglyceride.*

Coconut oil can be given by virtue of its medium chain triglycerides (MCT). Intractable cases may need partial or total parenteral nutrition. It requires patience on the part of the physician and the parents to try the various diets in succession (Table 5.18)—Diet A-C. Some physicians often go to Diet C first rather than from Diet A to C. After exclusion diet, it is important to restart feeding in 2-6 weeks. Lactobacilli, IV amino acids, plasma may also be given as adjuncts. Short chain fatty acids and amylase-resistant starch that can be absorbed from colon are under trial. Small intestine absorbs 22 L/day and colon absorbs 1.5 L flu id/day. Colon can increase absorption four-fold up to 6 L/day.

Table 5.19 Chicken-based diet*

Comminuted chicken	30 g	35 kcal
Glucose/glucose polymer	10 g	40 kcal
Vegetable oil	3.0 g	25 kcal
Mineral mixture	0.8 g	
Water up to	100 ml	
Total	**100 ml**	**100 kc**

** 777/s may be diluted with 5 or 10% glucose*

5. Prevention

Prevention of persistent and chronic diarrhoea is most practical and easy. Usage of ORT, continued feeding and avoiding of inappropriate antibiotics are all that are needed to prevent chronic diarrhoea. Vitamin A, folic acid and zinc supplementation are also found beneficial.

DIET IN RENAL DISEASES

A. Acute Glomerulonephritis (AGN)

Oliguria, haematuria, oedema and hypertension mark acute nephritis. The complications that can occur are hypertensive encephalopathy, congestive cardiac failure, electrolyte imbalance and acute renal failure.

1. Nutrition

a) *Fluid:* Restrict fluid to insensible loss plus last day's output when oliguria is present. Oliguria is defined as **urine output less than 1 ml/kg/hour or 25 ml/kg/day.** Insensible loss of water is 400 ml/m^2/day. Normal glomerular filtration rate (GFR) is 25 ml/nr/minute or 10.20 ml/m^2/min.

i) Mosteller's formula for calculation of surface area:

$$BSA = \sqrt{\frac{\text{Height (cm)} \times \text{Weight (kg)}}{3600}}$$

ii) Bedside calculation of insensible loss:

Infant = 25 ml/kg Adult = lOml/kg NB = 30ml/kg >5yr = 15 ml/kg

1-5 yr = 20 ml/kg

iii) Formula to calculate GFR:

i) Preterm = 0.35 x length + S creatinine

ii) Infant = 0.45 x length + S creatinine

iii) Child = 0.55 x length + S creatinine

If there is oedema and oliguria, give frusemide 1-2 mg/kg. If there is vomiting, give IV fluid. Insensible loss is replaced as 10% dextrose and urine output, 50%- as N. saline and 50% as 10% dextrose,

b) *Calories:* Give liberal calories, RDA for age plus 10% extra for infection/ illness.

c) *Protein:* Give RDA for age if blood urea is normal. **In renal failure, restrict to 0.5-1.25 g/kg body weight.**

d) *Sodium:* Restrict sodium during oliguria and gradually add 1-2 g/day during diuretic phase and slowly increase to 10 g/day.

e) *Potassium:* Potassium is avoided and fruits should not be given during oliguria. If complications are not subsiding with medical management, start peritoneal dialysis. During peritoneal dialysis, fluid and diet restriction are not strictly essential.

2. **Peritoneal dialysis (PD):** Indications—symptomatic uraemia, circulatory overload, hyperkalaemia and hypertension not responding to medical treatment. During PD, input of 20-50 ml/fluid/kg/cycle x 20 cycles are initially planned. 1.5% solution (type I) is generally preferred. Type II or higher strength solutions are used in overhydrated patients. **4.5% solution can be made up by adding 5 ampoules of 25% glucose to 1 Lof type I fluid.** Type II fluid is 8.7%. From 4th cycle onwards, add KCI in a dose of 2 ml/L of PD fluid. Monitor blood urea and serum electrolytes daily.

3. **Diuretic phase:** During diuretic phase, slowly release fluid restriction, protein restriction and salt restriction.

4. **Model diet in acute nephritis** (Table 5.20): Four-year-old child with 15 kg weight, oedema, hypertension and oliguria (urine output 300 ml) with normal B. urea.
   a) Calories: RDA 1300 kcal (1000+ 100 kcal for each completed year; bedside calculation); 10% extra— 130 kcal. Total 1430 kcal
   b) Protein: 1.75 g/kg—27 g
   c) Fluid: Urine output—300 ml
      Insensible loss—15 x 20 = 300 ml
      Total—600 ml
   d) Sodium: *Vi* to 1 g/day

Table 5.20 A model diet in acute nephritis

Item	Qty	Fluid (ml)	Energy (kcal)	Protein (g)	Sodium (mg)
Id 1 is	2	–	100	4	90
Sugar	3 tsp	-	60	-	-
Milk (100 ml, diluted)	1 glass	200	60	3	20
Sugar	2 tsp	-	40	-	-
Sweet bread	2	-	140	4	300
Butter (unsalted)	3 tsp	-	120	-	-
Sugar	3 tsp	-	60	-	-
Butter milk	½ glass	100	30	1.5	10
Sugar	2 tsp	-	40	-	-
Rice	1 cup	-	175	4	5
Ghee/oil/butter	3 tsp	-	120	-	-
Vegetables	50 g	-	-	1	-
Kanji water/Water	½ glass	100	-	-	-
Glucose biscuit	2	-	40	1	-
Butter milk	½ glass	100	30	1.5	5
Sugar	2 tsp	-	40	-	-
Rice	1 cup	-	175	4	5
Ghee/oil/butter	3 tsp	-	120	-	-
Vegetables	50 g	-	-	1	-
Kanji water/Water	*Vi* glass	100	-	-	-
SAT Mix (cereal-pulse)	5 tsp		100	2.5	
Total		**600**	**1450**	**27.5**	**435**

e) What to give? The fluids that can be given are kanji water, butter milk and dilute milk (50-100 ml of milk made up to 1 glass).

The food items that can be given are salt-restricted items like rice, *kanji,* idli, dosai, rice flakes, sugar, jaggery, honey, glucose, oil/ghee, unsalted butter and vegetables. Avoid high proteins, salt and fruits. SAT Mix (cereal-pulse) can be given (100 g = 380 kcal and 8 g protein, 1 tsp = 20 cal and 0.5 g protein).

Other items that can be substituted are rice flakes, ragi, nestum, custard powder, honey, jaggery etc. Avoid high protein, extra sodium and fruits.

B. Diet in Acute Renal Failure (ARF)

1. Causes of ARF
   a) Prerenal—e.g., dehydration, shock, burns
   b) Renal—e.g.. AGN. systemic lupus erythematosus (SLE), acute tubular necrosis, dysplastic/contracted kidneys, haemolytic uraemic syndrome (HUS).
   c) Postrenal—e.g., obstructive uropathy.

   In prerenal, serum creatinine is normal and urinary sodium is low, whereas in renal, both are high. Blood urea is raised in all the three types.
2. Management
   a) *Fluid challenge:* 20 ml/kg NS or RL or more till central venous pressure (CVP) is normal.
   b) *Diuretic challenge:* 1-2 mg/kg frusemide if well hydrated or 2 ml/kg 20% mannitol if underhydrated. If oligo-anuria persists, treat as ARF
   c) *Fluids in ARF:* IVF if oral is not tolerated. Quantity—insensible loss + last day's output. Type of fluid—insensible loss as 10% dextrose and output, 50% as N.saline and 50% as 10% dextrose.
   d) *Sodium:* No extra sodium when there is oligo-anuria and hypertension.
   e) *Sodium bicarbonate:* Acidosis is corrected by 1-2 ml/kg 7.5% Soda bicarb. This may be given 12 hourly.
   f) *Potassium:* To tackle hyperkalaemia, correct acidosis by giving soda bicarb, administer calcium gluconate 0.5-1 ml/kg IV 12 hourly and give insulin glucose in severe hyperkalaemia. 10 ml/kg of 5% glucose with 0.1 U/kg insulin may be given as a slow drip. Cation (K) exchange resins and dialysis may be needed in severe cases.
   g) *Phosphate:* To control hyperphosphataemia, prevent absorption by giving aluminium hydroxide and supplement calcium. Avoid calcium phosphate (e.g., Ostocalcium) and give calcium carbonate (e.g., Shelcal). Avoid protein-rich food which tends to have high phosphate.
   h) *Protein:* Restrict protein intake to 0.5-1.25 g/kg/day. Provide essential amino acids. Nonessential pool tends to be high due to contribution by

urea and other nitrogen wastes,

i) *Others:* Treat anaemia (packed cells and erythropoietin in chronic cases), hypertension (Nicardia), infection (antibiotics) and coagulopathy (FFP). Antibiotics other than penicillins need dosage modification.

3. Recovery phase: During recovery phase, slowly increase fluids, protein and sodium.
4. Model diet in ARF (Table 5.21): Four-year-old with 15 kg weight, hypertension and ARF. Output 100 ml.

Table 5.21 A model diet in acute renal failure

Item	Qty Fluid (ml)	Energy (kcal)	Protein (g)	Sodium (mg)
Milk (100 ml, diluted)	1 glass 200	60	3	20
Sugar	2 tsp	40	-	-
Unsalted idli	2	100	4	10
Sugar	3 tsp	60	-	-
Butter (unsalted)	75 g	600	-	-
Sugar	10 tsp	200	-	-
SAT Mix	8 tsp	160	4	-
Nestum rice	5 tsp	100	1	-
Sugar	5 tsp	100	-	-
Milk (100 ml, diluted)	1 glass 200	60	3	20
Sugar	2 tsp	40	-	-
Total	400	1520	15	50

Note: Up to 45% of the calories can be derived from fat. Other items that can be substituted are rice flakes, ragi, custard powder, honey, jaggery, sweet bread, butter, bun etc.

a) ***Calories:***	RDA	- 1300 kcal	
	15% extra	- 195 kcal	
	Total	- 1495 kcal	
b) ***Protein:***	1 g/kg/day	- 15 g	
c) ***Sodium:***	No added salt		
d) ***Fluid:***	Insensible loss	- 15 x 20 =	300 cc
	Urine output	- 100 cc	
	Total	- 400 cc	

C. Diet in Chronic Renal Failure (CRF)

CRF is usually due to end-stage renal disease. Patients present with growth retardation, anaemia, uraemia, acidosis, hypertension, oedema, hyperphosphataemia, rickets etc. The treatment modalities include dietary management with or without dialysis and ultimately renal transplantation whenever possible.

1. **Dietary management:** The goals are:
   a) to reduce nitrogen intake
   b) to maintain nitrogen balance
   c) to cover essential amino acid requirement
   d) to supply enough calories
      i) *Energy:* Infant 100-120 kcal/kg/day
         Children 80-100 kcal/kg/day
      In stunted children, RDA for height age is given rather than for chronological age.
      ii) *Protein:* High protein will aggravate acidosis, hyperkalaemia and hyperphosphataemia, whereas low protein will reduce BUN, improve renal function and reduce GI and neurological symptoms like nausea, vomiting, muscle cramps, convulsion, neuropathy etc. Milk is rich in phosphate and meat is rich in potassium. Protein intake is based upon the extent of CRF, mild/moderate or severe (Table 5.22).
      iii) *Fluids:* Thirst controls fluid intake and fluids should be given without producing water retention. If fluid retention occurs, give diuretic and restrict sodium and consider dialysis if water retention and weight gain increase in spite of diuretics.
      iv) *Sodium:* Sodium excretion is almost constant in CRF. Excess intake will lead to hypertension and fluid retention. Restrict salt intake to 300-600 mg/day in infants and to 1-2 g/day in older children.
      v) *Potassium:* Restrict potassium intake and give soda bicarb, calcium gluconate and potassium exchange resin (1 g/kg/day), if there is hyperkalaemia. Hypokalaemia can occur at any time. Give small dose of potassium or fruit juice if serum K is low normal or low.

Table 5.22 Protein allowance in CRF according to GFR (g/kg)

	Protein (g/kg)		
Type of CRF	**0-1 yr**	**1-5 yr**	**5-10 yr**
Mild (GFR 20-40)	1.8	1.4	**1**
Moderate (GFR 5-20)	1.4	1	0.8
Severe (GFR < 5)	**1**	0.8	0.6

2. Acidosis: Give 0.5-2 ml/kg soda bicarb in divided doses. Always correct hypocalcaemia to prevent tetany before giving soda bicarb.
3. **Hyperphosphataemia and renal rickets: Hyperphosphataemia and rickets may develop in three months period after onset of CRF.** Hence X-ray wrist should be taken after three months. Restrict phosphate intake (high protein diet). Give aluminium hydroxide (1 ml/kg/day), calcium (1 g/day) and alpha D 0.25 mg/day or up to 0.05 mg/kg/day. Alpha D is available as 0.15 mg and 1 mg capsules. Serum Ca, phosphorus and serum alk. $P0_4$ase should be monitored every 2-4 weeks and maintain serum phosphorus level at 4-6 mg%. Cow's milk is rich in phosphate and hence phosphate-low formula may be used, e.g., Similac.
4. **Anaemia:** Decreased erythropoietin and reduced RBC life span lead to anaemia. Packed cell transfusion and administration of erythropoietin are beneficial. Erythropoietin is given in a dose of 25-100 units/kg IV/SC thrice weekly. Higher doses are needed initially.
5. **Diet in peritoneal dialysis:** Strict dietary restriction is not required during PD; except in severe oliguria, oedema or hypertension.
6. **Classification of CRF:** CRF is classified according to GFR or symptoms. Glomerular filtration rate (GFR) does not approximate adult values till the third year of life. GFR is standardized to the surface area (1.73 m^2) of a 70-kg adult and is expressed as ml/min/1.73 m^2.
7. **Model diet in mild-moderate CRF** (Table 5.23)
   5-year child with 15 kg, normal BP. Height 98 cm. Height age 4 years. Urine output 400 ml (> 1 ml/kg/hour) S. K normal.
   a) *Fluid:* According to thirst or insensible loss + last day's output 15 x 20 - 300 ml + 400 ml = 700 ml or up to two-third maintenance 800 ml
   b) *Calories:* RDA for height age 1300 kcal + 20% extra = 1560 kcal.
   c) *Protein:* 1.4 x 15 = 21 g.
   d) *Sodium:* Restrict to 500 mg/day (no added salt).
   e) *Potassium:* Fruit juice can be given if S. K^+ is normal.
8. **Model diet in severe** CRF (Table 5.24)
   5-year child with 15 kg weight, oedema and hypertension. Height age 4 years. Urine output 600 ml (> 1 ml/kg/hour) S. K normal.
   a) *Fluid:* According to thirst or insensible loss + last day's output 15 x 20 = 300 ml + 600 ml = 900 ml or two-third maintenance.
   b) *Calories:* RDA for height age 1300 kcal + 20% extra = 1560 kcal.
   c) *Protein:* 0.8 x 15 = 12 g of high biological value.
   d) *Sodium:* No added salt.
   e) *Potassium:* Fruit juice can be given if S. K is normal.

Table 5.23 A model diet in mild to moderate CRF with normal potassium

Item	Qty	Fluid (ml)	Energy (kcal)	Protein (g)	Sodium (mg)
Fruit juice (orange/ lime/tomato)	1	100	100		20
Sugar	2 tsp		40		
Sweet bread	2	-	140	4	300
Butter (unsalted)	75 g	-	600	-	-
Sugar 5 tsp	-	100	-	-	
Milk (100 ml diluted)	1 glass	200	60	3	20
Sugar 2 tsp	-	40	-	-	
SAT Mix 4 tsp	-	80	2	-	
Unsalted idli	2	-	100	4	10
Sugar 3 tsp	-	60	-	-	
Butter milk	1 glass	200	60	3	20
Sugar 2 tsp	-	40	-	-	
Kanji water/Water	1 glass	200	-	-	-
Cooked rice	1 cup	-	175	4	5
Vegetable	50 g	-	-	1	-
Total	700	1595	21	375	

DIET IN HEART DISEASE WITH CONGESTIVE CARDIAC FAILURE (CCF)

Cyanotic congenital heart diseases (CCHD) may lead to marked growth retardation than acyanotic CHDs. However, PDA, VSD and pulmonary hypertension are common conditions that lead to growth faltering. This is attributable to CCF, hypoxia, acidosis, respiratory infection, poor coordination of sucking and swallowing and tiredness during feeding. 10-30% extra calories may be needed due to infection and the hypermetabolic state. Weight gain is essential to control infection and to plan surgery as well. Fluid and salt restriction are required in CCF.

Table 5.24 A model diet in severe CRF

Item	Qty	Fluid (ml)	Energy (kcal)	Protein (g)	Sodium (mg)
Milk (100 ml diluted)	1 glass	200	60	3	20
Sugar	2 tsp	-	40	-	-
Unsalted butter	75 g	-	600	-	-
Sugar	5 tsp	-	100	-	-
Butter milk	1 glass	200	60	3	40
Sugar	2 tsp	-	40	-	-
Cooked rice*	1 cup	-	175	3	5
Vegetable	50 g	-	-	1	-
SAT Mix (cereal-pulse)	4 tsp	-	80	2	-
Fruit juice	1	100	100	-	20
(orange/lime/tomato)	10 tsp	-	200	-	-
Sugar					
Kanji water/Water	2 glass	400	-	-	-
Glucose	5 tsp	-	100	-	-
Total		900	1555	12	85

* custard can be given

1. Goals
   a) *Fluid:* In oedema, restrict to insensible loss + last day's output or two-third maintenance.
   b) *Calories:* RDA for age + 10-30% extra.
   c) *Protein:* RDA for age or up to 10-15% of total calories as protein of high biological value.
   d) *Sodium:* Restrict to *V*2-*I* g/day.
2. Model diet in CCF (Table 5.25)
   2-year child with 8 kg weight and oedema, output 300 cc.
   a) *Fluid:* 20 x 8 = 160 ml + 300 ml = 460 ml or 2/3 maintenance—500 ml
   b) *Calories:* RDA 1100 kcal + 20% extra = 1320 kcal
   c) *Protein:* RDA or up to 10% of calories = 120 kcal = 30 g
   d) *Sodium:* Restricted to *Vi* g/day.

Table 5.25 A model diet in CCF

Item	Qty	Fluid (mL)	Energy (kcal)	Protein (g)	Sodium (mg)
Unsalted idli	2	–	100	4	10
Sugar	3 tsp	-	60	-	-
Milk	1 glass	200	120	6	30
Rice	1 cup	-	175	4	5
Dhal	4 tsp	-	60	2	5
Vegetable	50 g	-	-	1	
SAT Mix (cereal-pulse)	12 tsp	-	240	6	-
Kanji water	½ glass	100	-	-	-
Glucose	2 tsp	-	40	-	-
Bread	2	-	140	*4*	300
Butter (unsalted)	3 tsp	-	120	-	-
Sugar	3 tsp	-	60	-	-
Plantain	1	-	100	0.5	30
Butter milk	1 glass	200	60	3	15
Sugar	3 tsp	-	60	-	-
Total		500	1335	30.5	395

DIET IN RESPIRATORY DISEASES

Respiratory diseases contribute to highest morbidity and mortality among infants and children. These lead to hypermetabolic state and at the same time lead to anorexia, vomiting and semistarvation. Warm fluids help in expectoration and have soothing effect on the throat. Encourage small frequent feeds from the family pot. Solid sugars like sweets, chocolates etc., that stick to the teeth and tonsillar crypts may act as nidus for infection. Vitamin A supplementation is known to decrease respiratory and diarrhoeal diseases.

Vitamin A deficiency leads to increased bacterial binding to the mucosa. Vitamin E supplementation may decrease reactive oxygen species (ROS) and oxygen free radical disease in preterm babies on ventilators.

In asthma, **exclusion diet may be needed. In hypoallergenic diet, citrus fruits, groundnuts, chocolates, cow's milk, egg, meat, fish etc., are avoided step**

by step. Exclusion diet is tried only if they fail to respond to round-the-clock bronchodilators or if food allergy is evident.

Goals

Calories: 10-20% extra calories are needed in acute and chronic respiratory diseases. Convalescing children should get 1-2 extra meals/day for at least 2 weeks to restore weight and to prevent malnutrition.

DIET IN HEPATIC DISEASES

The goal is to provide adequate calories and electrolytes and to prevent hypoglycaemia, hypoalbuminaemia, hypokalaemia etc. Liberal carbohydrates and fruits, adequate protein and fat according to tolerance are given in mild diseases. High fat decreases gastric emptying and may aggravate nausea. MCT is better tolerated when there is decreased bile flow. Phospholipid extracts from soyabeans (Essentiale) is found to help in liver regeneration and to improve appetite. L-ornithine-L-aspartate (Hepamerz) orally or IV is beneficial in liver disorders. Ursodeoxycholic acid (UDLIV) is effective in cholestatic jaundice.

Silymarin in a dose of 10-20 mg/kg/day in 3 divided doses for 1 month is also beneficial. It contains flavanoligans namely silybinin, silycristin and silydianin. It is a strong antioxidant and prostaglandin synthetase inhibitor.

1. **Hepatic encephalopathy**

   The aim is to reduce ammonia level and to support the liver.

   a) Avoid protein by mouth.
   b) Sterilize the gut by oral ampicillin or neomycin.
   c) Lactulose 1-2 ml/kg/day in divided doses or till there is diarrhoea (up to 30 ml/dose).
   d) Lactisyn or lactobacilli may be given orally.
   e) Ryle's tube aspiration and bowel wash.
   f) Calorie requirement is RDA for age plus 10-20% extra calories. As much calories as possible should be given as 10% glucose enriched with 25% dextrose. Up to 12% glucose can be given through the peripheral vein.
   g) Blood transfusion and salt-free albumin.
   h) Supplement vitamin K and fresh frozen plasma (FFP).
   i) Give hepatic drip to supply fluid and calories (Table 5.26)
   j) Glucagon 0.03 mg/kg/day up to 1 mg/dose for 3 days helps in liver regeneration and to prevent hypoglycaemia.
   i) Supplement branched chain amino acids valine, leucine, isoleucine which help in liver regeneration (proteinsteril hepa)

2. **Recovering hepatic encephalopathy**

   Add protein 10 g/day and gradually increase up to 1 — 1.5 g/kg/day. Branched chain amino acids like valine, leucine, isoleucine are metabolised in the muscle and kidney unlike non-branched chain amino acids that are metabolised in the liver. Food items which give branched chain amino acids alone cannot be

Table 5.26 Composition of hepatic drip

Item		Quantity
N. saline	-	100 ml
10% dextrose	-	400 ml
KCl	-	5 ml
Ca gluconate	-	5 ml
MVI	-	2 ml

selected and hence give food items with low protein, e.g., protein of vegetable origin rather than animal origin. Medium chain triglyceride (MCT) can be given in children with decreased bile flow and fat malabsorption.

3. **Chronic liver disease**
   In chronic liver disease, ensure RDA plus 10-20% extra calories for malabsorption and altered liver function. Protein enough to meet RDA can be given unless in hepatic coma. Supply MCT and fat-soluble vitamins in view of decreased bile flow. Restrict fat if there is cholestasis. Prolonged cholestasis associated with fat malabsorption leads to deficiency of fat-soluble vitamins and calcium. Vitamin K injections should be given twice a month. High dose of vitamin A, D and E also should be supplemented. Vitamin E and D deficiency tend to be troublesome. Vitamin E deficiency is associated with neurological symptoms like ataxia, neuropathy etc. Water-soluble preparation of vitamin E up to 15-25 IU/kg/day and vitamin D up to 1000 IU/kg/day may be needed in some. Water-soluble vitamin A (Aquasol A) preparations are available as oral and injectable preparations. But, only fat-soluble preparations lead to vitamin A storage. (Oral Vitamin A 1 cap = 50,000 IU and inj 1 mL = 50,000 IU.) Table 5.27 summarises the special nutritional deficiencies and their management in chronic liver disease.
4. **Liver disease with ascites and oedema**
   Ascites is due to hypoalbuminaemia, secondary aldosteronism and/or portal hypertension. Salt and fluid should be restricted and N. saline may be avoided in hepatic drip. Aldactone (aldosterone antagonist) can be given 3-5 mg/kg/day in 4 divided doses. Plasma and albumin infusion are beneficial.

DIET IN MALABSORPTION AND OTHER GI DISORDERS

1. **Malabsorption**
   Among carbohydrate malabsorption, **disaccharidase deficiency is the most common, e.g., lactose intolerance.** This is diagnosed by watery diarrhoea and perianal excoriation due to acidic stool, presence of more than 2% reduc-

ing substance in motion at 2 separate testings and a pH less than 5.6. Continue breastfeeding, stop artificial feeding and give curd or yogurt. Soya milk may be started if necessary in very young infants. In older children, cereals, pulses etc., can be continued along with family pot feeding. (Table 5.18)

Table 5.27 Special nutritional demands in chronic liver disease

Deficiency	Management
Fat malabsorption	MCT as coconut oil/MCT formula
Fat-soluble vitamins	
Vitamin A	Water-soluble Vitamin A - 10,000-15,000 IU/day
Vitamin E	50-400 IU/day as tocopherol or TPGS (Tocopherol polyethylene glycol - 1000 succinate) Maintain Vit. E:Serum lipid ratio of 6 mg/g in children and 8 mg/g in adults
Vitamin D	5000-8000 IU/day of D_3 or 3-5 mg/kg/day of 25- OH cholecalciferol or 1-2 mg/day of 1,25-Di-OH cholecalciferol (Calcitriol)
Vitamin K	2.5-5 mg on alt. day as water soluble menadione orally or inj. vit. K 5 mg twice a month
Water-soluble vitamins	Twice the requirement daily
Others	Retention of cholesterol and bile acids (itching, xanthomas); choleretics/UDCA (ursodeoxycholic acid)15-20 mg/kg/day or cholestyramine 8-16 g/day

Protein malabsorption is called **creatorrhoea** and chicken when given is found to be passed out as such. These patients need supplementation of protease enzyme (Festal N, Pankreoflat, Serutan).

Fat malabsorption is called **steatorrhoea.** MCT can be given and supplement lipase enzyme (Pankreoflat, Serutan, Festal N). Fat-soluble vitamins should also be supplemented (refer Table 5.27).

2. **Diet in pancreatic insufficiency**

In cystic fibrosis or pancreatic insufficiency, **steatorrhoea** occurs. Fat as MCT can be given. They also need pancreatic enzyme supplementation. The dose of pancreatic enzymes is calculated based on the dose of lipase; up to 1500 IU/kg/meal may be given along with meals and snacks (Pankreoflat/ Festal N). Festal N = Lipase 6000 IU, amylase 4500 IU, protease 300 IU. If

steatorrhoea does not improve, give antacids and ranitidine to prevent digestion of enzyme in the drug by HC1 and also give omeprazole, proton pump inhibitor to normalize digestion. The drug should be given mixed with a meal.

3. **Diet in bowel resection or short bowel syndrome**
   Short bowel syndrome results when more than 20% of the intestine is resected. Parenteral nutrition may be needed at this stage. **In the first stage of 3-4 weeks, there is profuse watery diarrhoea. In the second stage up to 6 months, there is reduction in diarrhoea and adaptation; in the third stage after 6 months, patient becomes stable and tolerates many food items.** In short bowel syndrome, the carbohydrate that is passed unabsorbed into the colon gets fermented and gives rise to **short chain fatty acids** like acetate, propionate, butyrate etc. Colon plays an important role in absorbing these fatty acids that serve as a source of energy. Most of the nutrients are absorbed in upper small intestine. Terminal small intestine is the site of B absorption. **In ileal atresia or resection, B,, deficiency can occur.** However, the intrauterine reserve is sufficient for 1-10 years. In suspected or proved deficiency, give inj. $B_{|2}$1 mg every month. High oral doses may lead to mucosal diffusion.

DIET IN DIABETES MELLITUS

In children **insulin-dependent diabetes** (type I) is common. **Non-insulin-dependent diabetes** (type II) is extremely rare. **Malnutrition related diabetes (secondary diabetes)** is also reported from developing countries. **In syndrome** X, **there is hyperinsulinism and insulin resistance and is more common in those with central obesity.** In children, initial control of diabetes is difficult as they tend to have 'brittle diabetes'. After some months, the insulin requirement may come down and it is called the 'honeymoon phase'.

Maintain adequate carbohydrate, fat and protein ratio: carbohydrate 50-60%, protein 10-15% and fat 20-30%. Avoid fasting and feasting. High fibre, low fat diet with adequate carbohydrate and protein is ideal for diabetic children.

1. Meal Planning

The goal is to ensure normal growth and to keep FBS <115 mg/dl, PPBS < 126-140 mg/dl, S. cholesterol < 200 mg/dl, S. LDL cholesterol < 130 mg/dl, HDL > 50 mg/dl, S. triglyceride < 160 mg/dl and glycated Hb (6-8 g) within normal limits. There should not be wide fluctuations in blood sugar and so timing of meals and composition of diet should be relatively fixed and at the same time without monotony. **Fasting and feasting may be avoided as far as possible. Meal times must be regular and quantity should be consistent.** Sodium should be restricted to 3-5 g/day if there is hypertension and cholesterol should be restricted to 300 mg/day. BMR is roughly 22 kcal/kg ideal weight. Bitter things like bitter gourd may stimulate beta cells of the pancreas.

a) *Carbohydrate:* Avoid rapidly absorbed mono- and disaccharides and refined sugars like glucose, sugar, honey, sweets, sweet drinks etc., and encourage complex carbohydrates. Tubers should be restricted. Whole wheat is considered better than rice by some as it contains **'Acarbose' which allows slow carbohydrate absorption.** Rice is generally consumed in larger quantities than wheat and hence wheat is recommended by some.

b) *Fibre:* Fibre is unabsorbed plant polysaccharide. It delays carbohydrate absorption and decreases hyperglycaemia. The suggested intake is 20-35 g/day. It increases insulin receptors and decreases insulin requirement. Pectin, gum etc., present in fibre bind bile salts and increase their excretion. Since bile salts are derived from cholesterol, this will reduce serum cholesterol. Fibre relieves constipation. Whole wheat, coriander, carrot, brinjal, cauliflower, ladies finger, mango etc., contain 1-3% fibre; and ragi, pulses, ground nut, peas, guava etc., contain 3-5% fibre. In ripe mango, the fibre content reduces.

c) *Low fat:* Low fat increases insulin binding and reduces LDL and VLDL cholesterol and thereby decreases the incidence of atherosclerosis in diabetes. **The polyunsaturated/saturated (P/S) ratio 1.2:1 is usually recommended.** The ratio of polyunsaturated to monounsaturated to saturated fat may be equal for practical purpose. It may be better to give vegetable fat that contains PUFA. Avoid animal fat, hydrogenated oil (Dalda) etc. Fish and chicken are preferred than beef and egg. Turmeric, Bengal gram, onion and garlic reduce cholesterol.

d) *Fruits:* When the blood sugar is well controlled, half to one fruit can be allowed at the expense of a snack or after exercise. The fruit can be selected based upon the carbohydrate content of the fruit (Table 5.28).

Table 5.28 Carbohydrate content of vegetables and fruit

Less than 5% carbohydrate	Cabbage, spinach, snake gourd, brinjal, bitter gourd, cucumber, drumstick, radish, plantain flower, watermelon, tomato
5-20% carbohydrate	Agathi, cauliflower, beet root, plantain stem, peas, ladies finger, plum, papaya, peaches, musambi, lime, grape, strawberry, mango, orange, apple
Greater than 20% carbohydrate	Cereals, pulses, potato, jam, jack fruit, banana/plantain

The carbohydrate, protein, fat ratios of common food items are given in Table 5.29 and this has to be taken into account in meal planning.

Table 5.29 Proportion of carbohydrate, protein and fat in common food items

Item	Carbohydrate (%)	Protein (%)	Fat (%)
Rice	78	7	0.5
Whole wheat	70	11	1.7
Ragi	72	7	1.0
Pulses	60	22	0.7
Bread	52	8	6.7
Vermicelli	78	8	0.5
Milk	4.5	3	4
Pappad	0.4	18	0.3
Egg	-	13	13
Mutton	-	18	13
Fish	-	20	1.9
Chicken	-	26	0.6
Plantain/Banana	27	1.2	0.3
Orange	11	0.7	0.2

2. Model diet (Table 5.30)
   11-year-old child with 25 kg with diabetes
   Calories - 2000/day (RDA)
   Carbohydrate - 50-60%
   Protein - 10-15%
   Fat - 20-30%

(At least 100 g carbohydrate is needed to prevent ketosis) Up to 300 g/day can be given to supply 1200 kcal from carbohydrate (55%), 300 kcal from protein (15%). 600 kcal from fat (30%).

Ensure appropriate timings to prevent ups and downs in blood sugar values. The calories will have to be spread over as breakfast 20%, lunch 20%, dinner 30% and midmorning, midafternoon and evening snacks 10% each. One snack may be omitted and 10% may be added to the lunch (e.g., midafternoon) if three snacks cannot be taken.

Table 5.30 A model diet for a child with diabetes

6 AM	Bed coffee or green tea	1 cup
8 AM	Puttu/iddli/bread	2 pieces
	Green gram/Bengal gram*	2 table spoons
	Milk without sugar	1 cup
10.30 AM	Butter milk/lime juice without sugar	1glass
	Toast/biscuit	
1 PM	Rice	*IV₂* cups
	Dhal/pulse	2 table spoon
	Vegetables	50 g
	Fish/chicken	2 pieces
	Onion-tomato-cucumber salad	V₂ cup
4 PM	Coffee/tea without sugar	1 cup
	Toast/biscuit/vada/baji	2
8 PM	Chapati	2
	Dhal/pulse	2 table spoon
	Vegetables	50 g
	Fish/chicken	2 pieces
	Salad	*V₂* cup
9 PM	Milk without sugar	1 glass
	Fruit	Vi—1 (only if RBS is normal and exercise is adequate)

*Germinated cereals and pulses are better.

Avoid sugar, sweetened drinks and sweet bakery items.

3. **Exchange system**

   Exchange system is useful to ensure fixed energy intake and to avoid monotony. 1 exchange will supply 10 g carbohydrate (Table 5.31).

4. **Glycaemic index of foods**

   It is currently under study, i.e., the effect of the food on the blood glucose concentration. It is defined as the increase in RBS following ingestion of a food as percentage of increase following ingestion of a standard food (glucose). The glycaemic index of glucose is 100%; rice, 72%; wheat, 65%; ice cream, 42%; potato, 90%; soyabean and peanut, 20%; apple, 40% etc, This may vary from time to time depending upon whether it is taken on empty stomach or full stomach etc., and hence it is not of much practical significance.

5. **Exercise**

   Exercise will reduce insulin requirement, reduce LDL cholesterol, increase HDL cholesterol and avoid obesity. Extra calories for exercise may have to be provided (Table 5.32).

Table 5.31 Carbohydrate exchanges

1 exchange = 10 g carbohydrate

* ½ slice bread
* 2 cream crackers
* 2/3 cup unsweetened orange juice
* 1 medium apple or orange
* 2/3 cup cornflakes
* 1 small potato
* 1 plantain (small)
* 1 glass milk
* 1 cup porridge
* 1 cup thin soup
* 1 glass yogurt (curd)
* ½ banana

Avoid exercise until RBS is< 300 mg%; if RBS is 180-300 mg%, there is no need to increase carbohydrate intake; and if RBS < 180 mg%, give extra allowances.

Table 5.32 Carbohydrate allowances for exercise

Exercise	Allowances	Food items
Mild	10-15 g/hour	1 fruit or 1 starch exchange
Moderate	25 g before exercise and 10-15 g/hr	1 sandwich + 1 glass milk and 1 fruit or starch exchange
Severe	50 g before exercise and 10-15 g/hr	1 sandwich + 1 glass milk and 1 fruit or starch exchange

6. Insulin

Usually 1 unit/kg/day is the requirement. It can be given as 2 rations; 2/3 in the morning and 1/3 in the evening, 20 minutes prior to food. Each ration should be 2/3 intermediate acting and 1/3 plain insulin. This can be given by mixing the two or using commercial combinations, e.g., Mixtard (Bovine/ Porcine). **When the requirement is >2 U/kg/day, it indicates insulin antibodies and then switch over to Human Mixtard.** Bovine insulin varies from human insulin in three amino acids and porcine in one amino acid; both induce antibodies. **Human insulin is the best,** but very expensive. It does not evoke insulin antibodies.

7. **Sweeteners**

Non-nutritive sweeteners (saccharin, aspartame, sucramate and acesulphame K) are used in diabetic diet for improving palatability without increasing energy intake. Nutritive sweeteners (sorbitol and fructose) are useful in baking. They increase energy intake and tend to have GI side effects when > 50 g/day is consumed. Fructose in excess may be channelled into glucose pathway. Saccharin in very high doses is thought to produce hepatic malignancy.

DIET IN HYPERLIPIDAEMIAS

Hyperlipidaemias are of two types: (1) primary (familial hyperlipidaemia syndromes), (2) secondary. Those with hyperlipidaemias are at high risk for heart disease; but many do not develop clinical heart disease. It is true that adult cardiovascular disease may have its roots in childhood and adolescence. Cholesterol and triglycerides circulate as lipoproteins. The protein component of lipoproteins is called apolipoproteins. The dietary lipoproteins are chylomicrons secreted by the intestine. **Low density and very low density lipoproteins (LDL and VLDL)** are synthesized by the liver. They are also called **'bad cholesterol'. High density lipoproteins (HDL)** are synthesized by liver and small intestine and are called **'good cholesterol'.** HDL contain, phospholipids and proteins and they accept cholesterol and esterify it. Serum lipids should be estimated after 8 hours standard fast as food intake leads to large variation.

In children less than 2 years, the diagnosis is usually made when serum drawn for some other purpose is found lipaemic. Lipaemia retinalis and xanthomas also may lead to the diagnosis.

A. Primary Hyperlipidaemias/Genetic Dyslipidaemias

They were originally classified as Type I, II, III, IV and V depending upon whether chylomicrons, LDL, HDL etc., were raised alone or in combinations; but recently they are classified according to the genetic defect and are termed **'Genetic Dyslipidaemias'.** They include familial combined hyperlipidaemias (FCHL), familial hypertriglyceridaemias (FHTG) and lipoprotein lipase (LPL) deficiency.

B. Secondary Hyperlipidaemias

They are due to obesity, hypothyroidism, nephrotic syndrome, glycogen storage disease, lipid storage disease, diabetes mellitus, congenital biliary atresia, cholestasis, hepatitis, anorexia nervosa, SLE etc.

1. *Screening*

Screening for hyperlipidaemias should be undertaken in high-risk children and offspring of parents with hyperlipidaemias. If the serum cholesterol is <170 mg/dl, no treatment is required except revaluation at least after 5 years. If serum cholesterol is between 170-200 mg/dl. reestimate and if the average is > 170 mg/dl, do

lipid profile. If it is more than 200 mg/dl, lipid profile is advisable. Serum cholesterol < 170 mg/dl, TG < 160 mg/dl, **HDL** > 50 mg/dl and **LDL** <110 mg/dl are acceptable.

2. *Dietary Management*

The golden rule is that for better compliance, all the family members should switch over to the low-fat diet.

a) *Step I diet:* When **LDL** cholesterol is > 110 mg/dl, give step I diet with 30% of total calories from fat with equal proportion of saturated to monounsaturated to polyunsaturated fat. Restrict cholesterol to 300 mg/day (1 egg = 250 mg cholesterol) or to 100 mg cholesterol/1000 kcal intake.

b) *Step II diet:* When **LDL** cholesterol is >130 mg/dl, give step II diet with only less than 7% of calories from fat. Restrict cholesterol to 200 mg/day or to 66 mg/1000 kcal intake. Skimmed milk powder can be used instead of ordinary milk formulas in infants and breast milk may be continued. Skimmed milk powder alone may lead to essential fatty acid (EFA) deficiency. In non breast fed babies fed on skimmed milk powder, vegetable oil may be given to supplement 1-3% of calories in order to prevent EFA deficiency. In lipoprotein lipase **(LPL)** deficiency, MCTs that are directly absorbed into the portal vein can be given. **Cholesterol content of food items is given in Table 5.33.**

Table 5.33 Cholesterol content of various foods

Item	Quantity	mg
Egg	1	250
Meat	100 g	135
Cheese	100 g	56
Liver	100 g	400
Milk	200 g	45
Skim milk	200 g	5
Ice cream	100 g	54

3. *Other precautions*

Ensure high-fibre diet which has cholesterol-lowering properties. Avoid hydrogenated vegetable oils which contain trans fatty acids (Dalda, Margarine). Antioxidants like vitamin C. **beta carotene, vitamin** E, **have protective effect against heart disease.** Green tea is a good source of antioxidants. Encourage fruits, vegetables, fish oils etc. Mushrooms provide a totally fat-free item.

4. *Pritikin approach (diet and exercise)*

This was first introduced by Nathan Pritikin in 1974 in order to 'live longer'. It is

sometimes prescribed for calorie conservation, coronary artery diseases, hypertension, non-insulin-dependent diabetes etc., along with exercise.

Pritikin Lifetime Eating plan I allows 10% kcal from fats, 10-15% kcal from protein, > 80% kcal from carbohydrate, 30—40 g dietary fibre/1000 kcal, 4 g NaCl and 100 mg/day cholesterol. Whole grain, beans, pea, vegetables, fresh fruit, non-fat meat, non-fat cheese etc., are allowed. Meat group is restricted to 1 oz/week. This is different from the western refined diet and it simulates Indian diet.

In Pritikin Therapeutic Diet II, in addition to the above, alcohol and cholesterol are further restricted. Cholesterol is restricted to 250 mg/day.

5. *Drug Therapy*

Drug therapy is needed if LDL cholesterol is > 190 mg/dl or if LDL > 160 mg/dl with additional risk factors like obesity, diabetes, hypertension and family history of coronary artery disease. Dietary management should be continued while on drug therapy.

a) *Cholestyramine:* 2-12 g twice daily may be used and titrated to keep S. LDL cholesterol <110 mg%. Supplement extra fat-soluble vitamins as their absorption will be affected. Colestipol may also be tried.

b) *Lovastatin:* Lovastatin and other HMG-CoA reductase inhibitors may be used if absolutely essential. Dose is 20 mg/day. Long-term side effects of these drugs are under study.

c) *Nicotinic acid:* Nicotinic acid which is used in adults may be tried in children as well. It is found to decrease VLDL.

d) *Gemfibrozil:* It reduces VLDL formation.

e) *Fish oil capsule/syrup:* Fish oil rich in omega-3 fats like eicosa penta enoic acid (EPEA) is found to reduce thromboxane A_2 which is a potent vasoconstrictor and platelet aggregator. Tablets like Promega, 0mega-500, Maxepa etc., may be beneficial in high-risk patients. High arachidonic acid (omega 6) to EPEA (omega 3) ratio is an important risk factor for coronary artery disease. The preferred ratio is 5:1. Hence EPEA is beneficial along with diets and oils like sunflower oil rich in arachidonic acid.

DIET IN OBESITY

Obesity is increasing in epidemic proportions. Obesity in children may have effects in adulthood and may be the forerunner of increased mortality, cardiovascular diseases, diabetes and atherosclerosis. 10-30% of obese adults have had childhood obesity. Family obesity and onset of obesity in puberty are additional risk factors. Obesity is associated with increased plasma insulin, lipids, lipoproteins and blood pressure. Weight for age is not enough for calculating obesity. Body Mass Index (BMI) is a more useful index. Obesity is due to excess adipose tissue that imparts health risk. Adipose tissue is found metabolically

active. **Leptin are peptides synthesized bv adipose cells that act upon hypothalamus and affect food intake.** Obesity is graded according to BMI. **Obesity can be present early (starting obesity) or it may start around puberty (creeping obesity).** The causes of obesity are hereditary, syndromic, constitutional, neuropsychiatric and endocrine.

1. Classification

The ideal body weight in an adult can easily be calculated using the **'Broca's index'. Ideal weight (kg) = height in cm -100.** The range for male is height -100-105 and that for female is height - 100-107. Weight for height greater than 120% is considered obesity. Body Mass Index (BMI) is the best as it correlates with subcutaneous and total body fat. **BMI is defined as weight/height^2 (kg/m^2).** BMI is slightly less in females than in males. BMI correlates with BP and blood lipids. BMI value indicates risk for obesity or overweight in pubertal children. BMI value above 25 indicates risk for obesity and above 30 indicates overt obesity (85th and 95th centile, respectively) in young adults. As per International

Obesity Task Force (IOTF), **BMI 18.5-25 is considered normal, 25-30 is grade I obesity, 30-40 is grade II obesity and 40 plus is grade III obesity.** BMI < 18.5 is considered chronic energy deficiency (CED) or underweight in adults. But in adolescents < 15 is considered underweight, > 22 is considered overweight and > 25 is considered obesity.

The incidence of childhood obesity in developed countries is 10-15%. When food intake exceeds expenditure, body fat stores increase and when positive energy balance continues over a period of time, obesity sets in. The number and size of adipocytes increase. Fat cells are laid in the 3rd trimester of pregnancy and it triples by one year and slowly increases till adolescence. The number increases during infancy, childhood and puberty. **Maximum number of adipose cells are laid around 10th month of life.** Obese children have hyperinsulinism and insulin resistance. Meals high in refined sugars and protein cause more secretion of insulin and insulin inhibits lipolysis and utilization of free fatty acid and increases fat synthesis. Offering a bottle to satisfy a fretful child and early introduction of high-calorie solids can lead to obesity. Children with less activity and outdoor play and those with increased television viewing are at risk of developing obesity. The sedentary nature of the leisure time and TV advertisement of food items add to the risk. 100 calories above RDA can result in 5 kg fat deposition per year. There are two peaks of incidence; during infancy and adolescence. Those with obesity are taller with advanced bone age and genitalia appear small and embedded in fat. Puberty is slightly earlier and hence ultimate height may be less. They tend to have more emotional problems due to cultural prejudice and stigmatization. Those with BMI above 95th centile for age and sex or above 30 should have assessment of BP and fasting lipid profile.

2. Complications

Pickwickian syndrome is an extreme form of obesity with cardiorespiratory diseases, alveolar hypoventilation, polycythaemia, cyanosis, CCF and somnolence. Higher concentration of oxygen can be deleterious in them due to washing out of $C0_2$ which is the only respiratory drive. Repeated infections, sleep apnoea, hypoventilation, slipped epiphysis, hypertension, diabetes mellitus and gall bladder disease are also seen in those with obesity. Central obesity or apple shaped obesity is more harmful than pear shaped obesity. Waist hip ratio (WHR) > 0.9 in males and > 0.8 in females is undesirable.

3. Assessment

Those with BMI > 85-95th centile or > 25-30, skinfold thickness (SFT) > 85th centile and weight for height >120% should be taken for further assessment. SFT may be increased at triceps, subscapularis, biceps and supra-iliac regions. Syndrome like Prader-Willi, Lawrence-Moon-Biedl, Cushing's and pseudohypoparathyroidism should be looked for. These conditions contribute to 1% of childhood obesity. Evaluate bone age and look for increased intracranial tension (ICT). In those with short stature, perform full endocrine work up. BP and lipid profile should be included in the work up. In pituitary and adrenal disorders, bone age is reduced.

4. Treatment

Diet, exercise and behavioural modification

Behavioural modification is the most important intervention in weight reduction. Strict eating schedule, distractions, increase in activity, avoidance of tempting circumstances and stimulus, diet records and rewards are the means to achieve the goal. During infancy, food should be given only at sign of hunger and avoid tempting by showing attractive food. **Drastic reduction in food and strenuous exercise are not good for a growing child.** Family involvement and change in lifestyle of the family are often rewarding. Constant monitoring is needed to prevent relapses.

Diet should contain all essential nutrients and exercise management should be optimum. Exercises that increase fitness like walking, cycling, swimming etc., are good. Brisk walking for 30 minutes, running for 15 minutes or volleyball play for 45 minutes a day is found very beneficial. Walking 3 miles/hour will expend 300 kcal/hour and can lead to loss of 3 kg in 3 months. **Exercise with modified diet is beneficial than exercise alone. 1**200-1600 kcal diet is given to the adolescent child instead of the RDA of 2400 calories, which is equivalent to I unit of kcal, i.e., restrict to 2/3 of the requirement. In severe obesity, more than 50% reduction is aimed at, i.e., 1000 kcal diet.

5. Model Diets

a) *Diet I:* By avoiding ghee, butter, oil, gravy, milk cream, sugar, jaggery, chocolates, bakery items and choosing low-calorie dense items like idli instead of high-calorie dense items like upuma (e.g., 2 idlies = 100 Cal instead of 1 cup upuma = 250 Cal) can induce a deficit of approximately one-third of calorie intake. This is most practicable and acceptable to the family. Avoid eating in between meals.

b) *Diet II:* This is a highly restrictive diet rich in protein, called **'Protein Modified Fast' (PMF)** with or without added carbohydrate. PMF is associated with ketonuria within 2 days and requires close monitoring. It is also associated with nitrogen loss, lymphopenia, and low transferrin levels. Hypocaloric diet enhances lipolysis.

Addition of little carbohydrate leads to increase in insulin, reduction in lipolysis and fat utilization and leads to protein catabolism and negative nitrogen balance. But, addition of carbohydrate will prevent ketosis and monitoring will become difficult. PMF leads to weight loss of 3 kg in first week followed by 1 kg/week thereafter. Renal, hepatic and cardiovascular functions should be monitored. Hypotension, cardiac arrhythmias and alopecia have been noted.

i) **Precautions**

- PMF should supply 1.5-2.5 g/kg protein/day according to ideal weight.
- Protein should be derived from lean meat, fish, eggs etc.
- Supplement KC1 (25 mEq), calcium (800 mg), multivitamins and minerals.
- Include salad and low calorie vegetables without oil.

A model diet for a 10-yr-old obese child with 55 kg is given in Table 5.34.
Ideal weight 30 kg, Protein 1.5-2.5 g/kg = 30 x 1.5 = 45 g. Energy < 50% of RDA = 1000 kcal.

ii) *PMF with added carbohydrate:* To the above PMF diet, add two plantains (100g)that supplies 100 calories and 1.2 g protein extra or 100 g potatoes that supplies 100 calories and 1.6 g protein.

iii) *PMF diet for adults:* This can be given only under strict supervision. The principles are: no oil is used on salads, tea and coffee always black, meat never fried. No eating between meals, use little salt, don't drink too much liquid. Avoid candy. The diet must be followed rigidly. Eat every food listed at the correct time for results. If for some reason the diet could not be followed, start again from the beginning (Table 5.35). **This is not generally given in children.** This is done for two weeks only. A loss of 10 kg (20 lb) is expected in these two weeks. After this time, there will be diminished appetite.

Table 5.34 Diet II - Protein modified fast (PMF) - 1000 kcal diet for $_g$ io-year-old obese child

Food items	Raw (9)	Cooked quantity	Energy (kcal)	Protein (g)
Rice	100	2 cups	375	7
Dhal	25	10 tsp	95	5
Milk	300	l½ cups	200	9
Egg	80	2	160	12
Soya	25	10 tsp	110	11
Greens	50	5 tsp	-	1
Id 1 i	-	1	50	1
Total			990	46

Table 5.35 A protein-modified fast (PMF) for adults

Time	Items
Breakfast	- Grapes, 1 or 2 eggs, coffee or tea. This is the same every day except the 1st and 8th days, when 3 eggs can be taken
1st & 8th days	- Breakfast as above, take 3 eggs Lunch : 3 eggs, tomatoes, coffee and tea Dinner : 3 eggs, combination salad, 1 piece dry toast, grapes, coffee and tea
2nd & 9th days	- Breakfast as above Lunch : 1 or 2 eggs, tomatoes, spinach, coffee/tea Dinner : Steak, tomatoes, lettuce, celery, olives, cucumbers, coffee and/or tea
3rd & 10th days	- Breakfast as above Lunch : 1 or 2 eggs, tomatoes, spinach, coffee/tea Dinner : 2 lamb chops, celery, tomatoes, cucumbers, coffee and tea
4th & 11th days	- Breakfast as above Lunch : Combination salad, grapes, coffee/tea Dinner : Eggs, cheese, spinach, dry toast, coffee and/or tea

5th & 12th days - Breakfast as above
Lunch : Fruit salad (fresh, unsweetened), coffee/tea
Dinner : Steak, celery, tomatoes, cucumbers, coffee and tea

6th & 13th days - Same as the 5th and 12th days

7th & 14th days - Breakfast as above
Lunch : Chicken (baked or boiled), tomatoes, grapes, coffee/tea
Dinner : Canned vegetable soup, chicken, tomatoes, cucumbers, coffee and/or tea

Steak—mutton rib. Lamb chops—meat of lamb without fat

Any item listed can be taken at the proper time. A loss of 5 kg (10 lb) each week is not expected, but a loss of total 10 kg (20 lb) in two weeks is expected.

6. Drugs

Drugs have little role in children because the appetite suppressing effects wane off very fast. Amphetamines are best avoided due to psychomotor stimulation and addiction. Fenfluramine, phenmetrazine and diethylpropion are tried by some to help them adhere strictly to their diet. Dexphenfluramine is said to be more safe. Sibutramine is new, 5-12 mg/day as a single dose.

7. Surgery

Surgery has no role in childhood obesity. In adults with weight for height more than 200%, liposuction, gastric bypass (bariatric surgery) etc., are tried.

8. Adipose Tissue as an Organ

Recently, adipose tissue is upgraded as an organ like the liver which produces a hormone called leptin, which can 'speak to the brain'. The OB gene encodes for this protein. **Leptin is the 'information minister' of the body that updates the brain about the need to eat and the energy on board.** The discovery of leptin has shown fat as much cleverer than it was thought to be before. It plays a crucial role in protecting bone and organs, immune system and reproduction. Leptins belong to cytokines which are hormonal signals that regulate immune system and fight against infections. Fat is essential for reproduction and health of the foetus. Anorexics tend to have irregular periods and variable fertility. Fat ensures enough energy on board for the growing foetus during pregnancy.

DIET THERAPY IN INBORN ERRORS OF METABOLISM (IEM)

The diagnosis is suspected when infants normal at birth, present with failure to

thrive (FTT), acidosis, persistent vomiting, developmental abnormalities, organomegaly, convulsions and peculiar smell of urine. A history of similar deterioration in a previous sibling normal at birth should suggest the diagnosis of IEM. Early recognition and elimination diets can make these infants survive. Treatment includes restricted or exclusion diet and cofactor therapy.

A. Amino Acids

/. *Phenylketonuria (PKU)*

This is the most common IEM. Phenylalanine (PA) is an essential amino acid degraded by the tyrosine pathway. Deficiency of phenylalanine hydroxylase or its cofactor tetrahydro biopterin (THBP) leads to accumulation of PA, with a mousy or musty odour of phenylacetic acid in urine. Infants present as **'blue eyed blondes'** with seborrhoea/eczema, hypertonia and seizures. **Mother with PKU taking high phenylalanine diet can have abortions, babies with mental retardation, microcephaly and congenital heart disease (CHD).** Mental retardation is due to elevated PA. Blue eyes and blondness are due to decreased melanin as a result of decreased tyrosine. **Ferric chloride test yields green colour.** PA is an essential amino acid and so it cannot be totally eliminated, 25-50 mg/kg body weight is the requirement.

Administration of tetrahydrobiopterin (THBP) and neurotransmitter precursors like 5-OH tryptophan and L-dopa may be beneficial.

Low phenylalanine formula (LPF): LPF like Lofenalac (Mead Johnson) is the diet of choice. Tyrosine should be provided liberally as it is synthesized from PA. Overtreatment can lead to PA deficiency. The goal is to maintain serum PA level between 3-15 mg%. Rigid diet can be released after 6 years of age. Dietary restriction should be reintroduced during pregnancy. PA content of food items is given Table 5.36.

Model diet in PKU: Principle: Avoid high-protein items like fish, egg, meat, cheese etc. Limit medium-protein items like cereals, pulses, bread etc., and liberally use low PA items like tubers, vegetables and fruit, e.g., tapioca, cucumber, apple, grapes, guava, mango, papaya etc.

A model diet is given in Table 5.36. 3-year-old child with PKU, weight **1**0 kg. Calories RDA (minimum) = 1200 kcal. PA requirement (up to 50 mg/kg) 50 x 10 = 500 mg, Protein 1.5 x 10-15 g

2. *Tyrosine*

Tyrosine is synthesized from PA. It is the precursor of dopamine, adrenaline, noradrenaline, melanin and thyroxine. In tyrosinaemia, tyrosine levels are elevated and urine smells rancid, fishy or rotten cabbage like. **Ferric chloride test yields transient green colour.** The clinical presentations vary according to the subtypes. Diet low in PA, tyrosine and methionine is beneficial. **Vitamin** C **enhances the optimal functioning of dioxygenase enzyme. Liver transplantation is effective.**

3. *Alkaptonuria*

It is due to deficiency of homogentisic acid reductase. Homogentisic acid is a degradation product of tyrosine. It results in **ochronosis** and **arthritis** and **black colour of urine** on standing due to oxidation of homogentisic acid. There is no effective treatment.

4. *Homocystinuria*

This is the second common type of IEM. Homocystine is a degradation product of methionine. Conversion of homocystine to cytathionine is blocked due to cystathionine synthetase deficiency. It leads to **Marfan phenotype, mental retardation** and **subluxation of lens** (ectopia lentis). Thrombosis of arteries and veins can occur. Dietary restriction of methionine and large doses of B_{12} (1-2 mg/day), B_6 (200-1000 mg/day) and folic acid (1-5 mg/day) are beneficial. Legumes that lack methionine can be given.

Table 5.36 Phenylalanine content of food items

Item	Qty	Energy (kcal)	Protein (9)	PA (mg)
Ragi/sago porridge	30 g	100	2	100
Jaggery	20 g	160	-	-
Milk	50 ml	30	1.5	75
Papaya/guava	50 g	25	0.3	-
Rice	1 cup	175	3.5	130
Greens	50 g	-	1.0	25
Cucumber salad	50 g	35	1.5	5
Carrot halva (carrot 30 *g*, sugar 20 g, ghee 10 g)	175	0.5	10	
Vegetable soup	1 cup	100	2	-
Vegetable	100 g	-	1.0	-
Oil	1 tsp	40	-	-
Tapioca/potato	50 g	75	0.5	15
Arrowroot porridge	30 g	100	0.1	5
Milk	50 ml	30	1.5	75
Sugar	3 tsp	120	-	-
Grapes	50 g	25	0.3	5
Total		1190	15.7	445

Include items low in protein. In homocystinuria, methionine level increases to > 30 mg%. Cysteine and betaine (trimethyl glycerine), a methyl group donor (6-9 g/day) is found to lower homocystine levels. Low methionine formulas are available, e.g., Methionacid—protein hydrolysate without methionine (methionine 0.2g%). Restrict methionine to 20-25 mg/kg. Ensure adequate calories as carbohydrate, fats and also vitamins and minerals. Avoid food items rich in protein like egg, flesh food, milk etc.

a) Model diet in homocystinuria (Table 5.37)
   A 3-year-old child with homocysteinuria, weight 10 kg
   i) Energy: RDA—1200 kcal
   ii) Methionine: 25x10—250mg
   iii) Protein: 10x1.5—15g

Table 5.37 A model diet for homocystinuria

Item	Qty	Energy (kcal)	Protein (g)	Methionine (mg)
Cooked rice	1 cup	175	4	75
Greens	50 g	-	1	100
Bread	2 slice	140	4	-
Oil	3 tsp	120	-	-
Vegetable salad/soup (beet root *Vi* cup, carrot *Vi* cup, green 50 g)	1 cup	180	2	10
Banana	1	50	1	10
Apple	1	30	1	2.5
Cucumber	1	50	1.5	7.0
Ragi/sago porridge	30 g	100	2	25
Jaggery	20 g	160	-	-
Arrow root porridge	30 g	100	0.1	10
Sugar	3 tsp	120	-	-
Total		1225	16.6	239.

5. *Cysteine/cystine*

Cysteine is synthesized from methionine and two cysteine molecules oxidize to form cystine. In cystinuria and cystinosis there is no effective treatment. Due to low solubility of cystine, there may be familial renal calculi. Cysteamine and phosphocysteamine may be useful in the treatment.

6. *Tryptophan*

a) *Indicanuria:* In tryptophan malabsorption, tryptophan is converted to indole in the gut by bacterial action. Indole is converted to indican after absorption and gets oxidized to indigo blue leading to blue diaper syndrome. It can also occur in blind loop syndrome. **In blue diaper syndrome, there is hyperealciuria and nephrocalcinosis.** Treatment includes low-protein diet.

b) *Hartnup disorder.* Tryptophan is the precursor of niacin and serotonin. Neutral amino acids (alaine, serine, threonine, valine, leucine, isoleucine, phenylalanine, thyroxine, tryptophan and histidine) transport defect leads to Hartnup disorder which was first described in Hartnup family. **Photosensitive dermatitis is the essential feature with episodic psychological disorder and ataxia.** It responds to high doses of niacin 50-300 g/day and high-protein diet.

7. *Branched chain amino acids*

Disorders of valine, leucine, isoleucine and organic acidaemias are included in this. The common features are vomiting, acidosis, dehydration and refusal of food. Ketosis and skin manifestations are seen in some.

a) *Maple syrup urine disease (MSUD):* The defect in branched chain 'keto acid dehydrogenase' leads to sweet odour of maple syrup (burnt sugar) in body fluids and urine and there is elevation of the branched chain amino acids in blood. **Ferric chloride test leads to navy blue colour.** They present with feeding difficulty, shrill cry, alternate hyperactivity and flaccidity, semi coma and convulsion. MSUD **(Mead Johnson) formula** low in branched chain amino acid is available. Small amounts of branched chain amino acids should be provided as they cannot be synthesized by the body. This diet should be continued at least till myelination is complete. Large doses of **thiamine** can be added as it is the cofactor for the enzyme. It benefits patients with partial deficiency (thiamine-responsive MSUD). Peritoneal dialysis is also tried. The acceptable levels of valine is 2.6 mg%, of isoleucine is 0.9 mg% and of leucine is 2.6 mg% in the plasma.

b) *Isovaleric acidaemia:* It is a defect in leucine metabolism. **Sweaty feet odour in urine** is characteristic. A low-protein diet 1-1.5 g/kg/day with glycine 250 mg/kg/day and carnitine 100 mg/kg/day are beneficial. **Glutaric acidaemia also produces sweaty feet odour.**

c) *Multiple carboxylase deficiency:* This is an enzyme in the branched chain amino acid metabolism. This disorder is associated with dietary biotin deficiency or biotin utilization defect. Biotin deficiency leads to hyperaesthesia, dermatitis, hallucination etc. **Tom cat urine smell is characteristic. Biotin** 10 mg/day is found beneficial.

d) *Propionic and methyl malonic acidaemias:* These produce ketosis, hyperammonaemia, hyperglycinaemia, neutropenia and thrombocytopenia.

Ferric chloride test yields purple colour. They were termed **ketotic hyperglycinaemias.** Ketosis yields brown red colour with ferric chloride test. They respond to low-protein diet with L-carnitine (50-100 mg/kg/day) and peritoneal dialysis. Propionate precursors (isoleucine, valine, methionine and threonine) deficient diet is available as Milupa OSI (Milupa Corporation). Hyperglycinaemia is due to inhibition of glycine cleavage enzyme by the organic acids. Propionic acid is a catabolite of isoleucine, valine, methionine, threonine, odd-chain fatty acids and cholesterol. Methyl malonic acid is a catabolite of propionic acid. In propionic acidaemia, large doses of biotin and in methyl malonic acidaemia, large doses of vitamin $B_{|2}$ (1 mg/day) are found beneficial. Vitamin $B_{|2}$ metabolites act as coenzymes.

8. *Glycine*

It is a non-essential amino acid synthesized from serine and threonine. Glycine is absent in breast milk. Glycine cleavage system consists of 4 proteins, P, T, H and L

a) *Hyperglycinaemias:* These occur as a spectrum of conditions that produce ketosis, e.g., **propionic acidaemia, methyl melonic acidaemia, isovaleric acidaemia** etc. These were called ketotic hyperglycinaemias. Nonketotic hyperglycaemias are due to glycine cleavage system disorders especially P protein (80%) and T protein.

b) *Nonketotic hyperglycinemias:* Mild varieties present as mental retardation, microcephaly, myoclonic seizures, but severe forms rapidly progress to coma and death. Low glycine diet (breast milk), low protein diet, sodium benzoate, folate and exchange transfusion are tried. Strychnine and diazepam counter CNS effects of glycine.

9. *Oxalosis and Hyperoxaluria*

Oxalic acid is derived from glyoxylic acid and ascorbic acid and by absorption from the gut. They present with oxaluria, oxalate stones and nephrocalcinosis. Normal oxalate excretion is 10-50 mg/day. Large doses of B_6 (100-120 mg/day) can inhibit oxalate excretion. Oxalate-rich food like horsegram, kesari dhal, amaranth, curry leaves, drumstick leaves, spinach, plantain flower, almond, cashew nut, gingelly seeds, amla, tea, coffee etc., should be avoided.

10. *Urea Cycle Disorder with Hyperammonaemia*

Disorders of arginine, citruline, ornithine metabolism are included in this. Hyperammonaemia produces vomiting, refusal to feed, tachypnoea and coma. Older children may present with agitation, confusion and ataxia.

Endogenous breakdown of protein should be avoided by adequate calories. Essential amino acid can be supplied intravenously (0.25 g/kg/day). IV lipids (1 g/kg/day) is found beneficial to supply calories. For long-term management,

restrict protein to 0.5-1 g/kg/day. Sodium benzoate 250 mg/kg, sodium phenylacetate 250 mg/kg, arginine hydrochloride 200-300 mg/kg as 10% solution and peritoneal dialysis are found beneficial. Arginine should be avoided in arginase deficiency and in hyperammonaemia due to organic acidaemias. Carnitine is also added as benzoate and phenyl acetate cause carnitine depletion.

11. *Histidine*

This is due to histidase deficiency that converts histidine to urocanic acid. **Histidinaemia produces green brown colour with ferric chloride test.** They present with growth retardation, mental retardation and speech defects. Histidine-deficient diet is the treatment (low-protein diet). Unlike PKU, maternal histidinaemia does not produce ill effects in the offspring.

The essential amino acid content of various food items are given in Table 5.38. The recommended dietary allowances of the essential amino acids are given in Table 5.39.

B. Lipid Metabolism

1. **Refsum disease and Refsum syndrome:** Refsum disease is a peroxisomal disease. Peroxisome is a subcellular organelle concerned with fat and amino acid metabolism. In Refsum disease, **very long chain fatty acid (VLFA) metabolism is defective.** In Refsum disease, children present with **broad based gait, ataxia, sensorineural hearing loss** and **atypical retinitis pigmentosa.** Treatment consists of administration of cholic and deoxycholic acid (100-250 mg/day) to reduce toxic bile acid intermediates and ethyl ester of docosahexa enoic acid (DHEA) 200-250 mg/kg. The intake of phytanic acid, a long chain fatty acid should be reduced.

   Refsum syndrome manifests in 2nd and 3rd decade with icthyosis, chronic polyneuritis, progressive paralysis, ataxia, atypical retinitis pigmentosa, deafness and ECG changes. Phytanic acid containing diet, like green vegetables, spinach, nuts, coffee and dairy products should be avoided.
2. **Lipid storage disorders:** Lipid storage disorders like Gaucher's. Niemann-Pick etc., do not have any effective dietary management.
3. **Lipoprotein metabolism and transport disorders:** They manifest as hyperlipidaemias (refer Section 8.9).

C. Carbohydrate Metabolism

Defects in metabolism of galactose and fructose and glycogen storage disorders are the usual types.

I. Fructose: In fructose metabolism, defects like benign fructosuria and hereditary fructose intolerance, honey and sugar (sucrose) should be avoided. Sucrose contains glucose and fructose. Symptoms mimic galactosaemia as

Table 5.38 Essential amino acid content in common foods/100 g

Item	Prot. (9)	Argi. (mg)	Hist. (mg)	Lysi. (mg)	Tryp. (mg)	PA (mg)	Tyre (mg)	Meth. (mg)	Cyst. (mg)	Thre. (mg)	Levc. (mg)	Isol. (mg)	Vali. (mg)
Cereal group													
Maize	11	516	285	356	71	516	427	214	178	498	1281	427	534
Ragi	7	351	152	257	117	363	257	246	164	281	807	468	562
Rice	7	523	142	251	87	305	316	164	98	251	545	327	412
Wheat	11	548	246	321	132	529	340	170	265	340	775	416	529
Pulses/legume group													
Bengal gram	17	1562	438	1206	137	986	493	219	219	601	1589	877	849
Black gram	23	1997	653	1536	269	1190	538	346	307	845	1920	1306	1190
Green gram	23	1920	653	1766	230	1344	384	307	230	768	1958	1344	1229
Horse gram	21	1866	669	1830	246	1338	-	246	458	810	1901	1302	1373
Kesari	27	2210	722	2120	226	1173	-	135	316	631	1849	1849	1128
Peas	19	1796	410	1386	189	882	536	158	221	756	1355	882	945
Red gram	22	1285	893	1714	143	1642	464	214	214	714	1607	893	928
Soya bean	42	3110	1037	2764	553	2073	1451	553	691	1658	3317	2211	2211
Greens													
Amaranth	3.9	154	83	160	45	115	122	45	26	90	267	186	179
Cabbage	1.8	130	38	70	20	58	35	17	20	64	99	66	75
Drumstick leaves	6.5	407	150	342	107	310	-	118	139	268	492	300	375
Spinach	1.9	112	45	128	32	106	99	35	26	93	170	96	112
Tuber group													
Beetroot	1.6	86	27	111	16	57	46	19	32	68	89	54	62
Carrot	0.8	35	13	32	6	29	20	10	8	29	43	32	43
Onion	1.2	32	13	55	17	34	-	13	-	17	32	17	27
Potato	1.6	86	26	83	26	70	44	23	13	57	99	70	81
Radish	0.7	77	18	30	2	30	-	6	-	25	46	33	43
Sweet potato	1.2	53	17	49	21	51	29	19	6	53	68	55	72
Tapioca	0.7	70	13	35	10	22	12	6	11	24	36	30	29
Vegetable group													
Bitter gourd	1.6	70	21	55	10	62		39		65	109	96	96
Cauliflower	2.5	122	50	151	38	97	-	42	-	109	185	126	147
Cucumber	0.4	28	5	16	3	8	-	4	-	10	16	11	13
Ladies finger	1.8	69	33	63	12	42	81	24	18	42	72	45	57
Nuts group													
Almond	20	2198	466	533	167	999	599	333	166	566	1499	833	1032
Cashew nut	21	2165	441	983	373	915	-	605	-	678	1229	1085	1220
Gingelly	18	2198	498	498	234	1084	674	527	352	674	1465	733	850
Ground nut	25	2795	567	932	243	1256	972	243	324	689	1620	972	1134

Fruits group													
Apple	**0.2**	**5**	**4**	**11**	**2**	**5**	**3**	**2**	**2**	**7**	**12**	**7**	**8**
Banana	**1.2**	**68**	**70**	**51**	**13**	**49**	**30**	**15**	**32**	**36**	**61**	**48**	**49**
Dates	**2.4**	**60**	**20**	**76**	**“**	**68**	**20**	**20**	**48**	**72**	**104**	**60**	**88**
Grapes	**0.5**	**37**	**18**	**11**	**2**	**10**	**9**	**17**	**8**	**14**	**10**	**4**	**14**
Grava	**0.8**	**-**	**.**	**27**	**8**	**-**	**-**	**8**	**-**	**-**	**-**	**-**	**-**
Mango	**0.6**	**-**	**-**	**83**	**13**	**-**	**-**	**7**	**-**	**-**	**-**	**-**	**-**
Papaya	**0.6**	**-**	**-**	**40**	**13**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**
Meat group													
Beef	**21**	**1476**	**720**	**1944**	**252**	**936**	**792**	**576**	**288**	**1008**	**1836**	**1152**	**1188**
Egg	**13**	**852**	**320**	**937**	**192**	**767**	**533**	**447**	**298**	**682**	**1108**	**873**	**969**
Chicken	**25**	**1449**	**662**	**2070**	**248**	**1035**	**869**	**662**	**331**	**1035**	**1904**	**1366**	**1325**
Mutton	**18**	**1273**	**503**	**1510**	**237**	**740**	**622**	**444**	**237**	**858**	**1421**	**918**	**947**
Pork	**18**	**1226**	**568**	**1704**	**269**	**837**	**688**	**538**	**239**	**927**	**1555**	**1017**	**1076**
Milk group													
Cow’s milk	**301**	**112**	**87**	**255**	**46**	**163**	**153**	**82**	**26**	**143**	**306**	**173**	**204**
Human	**1.1**	**45**	**31**	**76**	**20**	**41**	**41**	**18**	**20**	**523**	**94**	**59**	**56**
Curd	**301**	**100**	**80**	**240**	**40**	**165**	**185**	**85**	**30**	**155**	**340**	**160**	**235**
Cheese	**23**	**926**	**77**	**2007**	**309**	**1351**	**1312**	**694**	**154**	**1003**	**2470**	**1390**	**1853**

Table 5.39 RDA of essential amino acids among children

Item	RDA (mg/kg)
Valine	30-90
Leucine	45-160
Isoleucine	30-70
Lysine	60-100
Phenylalanine	25-125
Methionine	25-50
Cysteine	25-50
Tyrosine	25-125
Threonine	35-80
Tryptophan	4-10
Histidine (infants)	25

sucrose is added as a sweetener to milk and baby foods. Progressive liver disease may occur.

2. **Galactosaemia:** This is characterised by galactosaemia, galactosuria and cataracts. By the time the diagnosis is made, cataracts set in. At least the younger siblings should be helped by avoiding galactose and lactose in the diet. Lactose is the milk disaccharide that contains glucose and galactose.
   a) *Types:* In galactokinase deficiency, phosphorylation of galactose to galactose-1-phosphate is prevented leading to accumulation of galactose. In galactose-1-phosphate uridyl transferase deficiency (transferase deficiency), the formation of uridine diphosphate galactose (UDP gal) is prevented leading to accumulation of galactose and galactose 1-phosphate. The clinical manifestations are persistent **vomiting, FTT, jaundice, hepatosplenomegaly, cirrhosis, cataract, mental retardation** etc. Cataract may be only sign in galactokinase deficiency. **Urine Benedict's test is positive; but glucose oxidase test is negative.** Guthrie's microbiological assay is positive due to the presence of galactose.
   b) *Dietary management:* Milk sugar (lactose) contains 50% galactose. So avoid all milk and milk products. **'Non dairy creams'** prepared from soya protein may be used. Non-fat milk solids and lactose are added to baked foods to improve texture and nutritive value and for its 'browning properties'. So avoid all baked items. Galactose is present in some complex starches (stachyose, raffinose) in peas, soyabeans, vegetables etc. Hence formula prepared from soya protein isolate (Nusobee, Zerolac) can be given whereas whole soya flour preparations are avoided (Soyal, Prosoyal). Among fleshy foods, avoid organ meat like liver. Strict dietary restriction is necessary in early years which can be somewhat relaxed during school age.

      The RDA for calories, protein etc., are calculated as for a normal child. Table 5.40 gives the items to be included and excluded in galactosaemia.

3. **Glycogen storage disorders (GSD):** These enzyme disorders produce abnormal concentration or structure of glycogen. Some of them manifest with early morning hypoglycaemia. They benefit by frequent day and night time feedings or continuous night time NG feeding. They also benefit from taking uncooked corn starch with slow absorption. Cornflakes meal for dinner is beneficial. With such dietary regimen, growth will improve, hepatomegaly will regress and hypoglycaemia and lactic acidosis will become manageable.

D. Mucopolysaccharide Metabolism

Mucopolysaccharides are glycosaminoglycans that contain alternating carbohydrate residues of N-acetyl hexosamine and uronic acid. Mucopolysaccharidoses (MPS) result from enzyme defect that leads to accumulation of mucopolysac-

charides. There is no effective dietary management. Mucolipidosis exhibit features of both MPS and lipidosis.

Table 5.40 Food items to be included and excluded in galactosaemia

Item	Include	Exclude
Milk group	Soya protein milk, non dairy creams of soya protein (avoid whole soya flour)	All milk and milk products including breast milk and sodium caseinate
Cereals - Pulse group	Cereals, pulses sparingly	All pulses if erythrocyte enzyme is low
Fats	Oil, nut, margarine without milk	Butter, cream, cheese, ghee, margarine with milk
Fruits and vegetables	Fresh fruits and vegetables	Peas, vegetables canned or processed with lactose
Meat group	Muscle, egg, fish	Liver

E. Purine and Pyrimidine Metabolism

Purine and pyrimidines along with ribose and deoxyribose and phosphate are the essential components of RNA and DNA.

1. **Gout and hyperuricaemias:** Elevation of uric acids (purine) leads to gouty arthritis and gouty tophi and hyperuricaemia. **Hyperuricaemia** may produce uric acid stones. Allopurinol reduces uric acid production, reduces excretion of uric acid and increases excretion of oxypurines. Probenecid is also effective in increasing uric acid clearance. Excess intake of molybdenum increases the risk of gout as it is an essential constituent of xanthine oxidase in uric acid metabolism.

   Xanthine is the precursor of uric acid. Food items that are rich in uric acid like beef, chicken, pork, liver, sardine, spinach, cauliflower, chocolate, cocoa etc., should be avoided.
2. **Lesch-Nyhan syndrome:** This is due to the total deficiency of **hypoxanthine-guanine phosphoribosyl transferase enzyme.** Children present with **motor delay, choreoathetosis, hyperreflexia, spasticity** and **self-destructive behaviour.** Treatment includes avoidance of purine-rich food like sweet breads.

F. Porphyrias

These are disorders of haeme biosynthesis. Haeme is composed of ferrous iron and protoporphyrins. Dietary management includes provision of liberal calories, carbohydrates, p-carotene (120-180 mg/day) is found beneficial for photosensitive skin lesions due to its antioxidant property that tackles oxygen radicals.

KETOGENIC DIET FOR MYOCLONIC SEIZURES

Ketogenic diet is found beneficial in infants and children with myoclonic seizures. Myoclonic epilepsies of childhood include benign myoclonus of infancy (BMI) with normal EEG, typical myoclonic epilepsy of early childhood (TME), complex myoclonic epilepsies (CME) like Lennox-Gestaut syndrome with myoclonic and tonic seizures, West syndrome with infantile myoclonus, hypsarrhythmia and mental retardation, juvenile myoclonic epilepsy (JME) and progressive myoclonic epilepsies (PME) with CNS degeneration. Ketogenic diet produces its effect by increasing the inhibitory neurotransmitter GABA.

80% of the calories is supplied by fat and rest as carbohydrates. Protein is restricted. 60% of the fat can be as MCT. Adequate vitamins and minerals especially calcium should be supplied. Ketogenic diet is unpalatable and not liked by children above 2-3 years. MCT is present in coconut oil and cotton seed oil. The child is fed on fat and oil and urine is tested for ketone bodies. The child should be under strict observation.

SECTION 6

Food Poisoning and Food Allergy

"More people have died of food than of famine."

—Old American Proverb

6.1 Food Poisoning

Food poisoning often manifests as diarrhoea and vomiting. It is characterised by the following: (1) Onset of vomiting, diarrhoea, abdominal cramps or other constitutional symptoms within 2-24 hours of ingestion of the contaminated food, (2) Occurrence of illness in at least 2 or more persons partaking the same food with a similar clinical pattern in the affected persons, (3) Usually caused by bacteria or its toxins in the affected persons. This definition excludes symptoms produced by chemical contaminants in the foods, diseases caused by infected food but with longer incubation periods like hepatitis, food-borne diseases with systemic symptoms like enteric fever and non-infective food intolerances and allergies. The 6 Fs that lead to food-borne diseases are: flies, food, fluid, fingers, faeces and fomites. The common bacterial food poisonings and the clinical features are given in Table 6.1.

Table 6.1 Features of bacterial food poisoning

Clinical features	Incubation period	Causes	Common food items
Nausea & vomiting	< 1-6 h	*Staph, aureus* (preformed toxins A, B, C, D, E)	Ham, poultry, cream-filled pastries, potato and egg salad, mushrooms
		Bacillus cereus (emetic toxin)	Fried rice, pork

	Heavy metals (copper, tin, cadmium, zinc)	Acidic beverages
Histamine < 1 h response and GIT symptoms	Histamine (scombroid)	Fish (bluefish, bonito, mackerel, mahi-mahi, tuna)
Neurologic 0-6 h including paraesthesia and GIT symptoms	Tetrodotoxin, ciguatera	Puffer fish (amberjack, barracuda, grouper, snapper)
	Paralytic compounds	Shellfish (clams, mussels, oysters, scallops, other mollucks)
	Neurotoxic compounds	Shellfish, mussels
	Domonic acid, monosodium glutamate	Chinese food
Neurologic 0-2 h and GIT symptoms	Mushroom toxins of an early onset	Mushrooms
Moderate- to-severe abdominal cramps & watery diarrhoea 8-16 h	*B cereus enterotoxin*	Beef, pork, chicken, vanilla sauce
	Clostridium perfringens enterotoxin	Beef, poultry, gravy
16-48 h	*Caliciviruses*	Shellfish, salads, ice
	Enterotoxigenic *E. coli, Vibrio cholerae* 01 & 0139	Fruits, vegetables Shellfish
	V cholerae non-01	Shellfish
Diarrhoea, 16-72 h fever, abdominal cramps & watery diarrhoea	*Salmonella*	Poultry, pork, eggs, dairy products, including ice cream, vegetables, fruit
	Shigella	Egg salad,
	Campylobacter jejuni	Poultry, raw milk
	Invasive *E.coli*	Vegetables
	Yersinia enterocolitica	Pork chitterlings, tofu, raw milk
	Vibrio parahaemolyticus	Fish, shellfish

Bloody diarrhoea, abdominal cramps	72-120 h	Enterohaemorrhagic *E. coli*	Beef (hamburger), raw milk, roast beef, salami, salad dressings
Methaemoglobin poisoning	6-12 h	Mushroom toxins of late onset	Mushrooms
Hepatorenal failure	6-24 h	Mushroom toxins of late onset	Mushrooms
GIT symptoms then blurred vision, dry mouth, dysarthria, diplopia, descending paralysis	18-36 h	*Clostridium botulinum*	Canned vegetables, fruits and fish, salted fish, bottled garlic
Extraintestinal manifestations	Varies	Brainerd disease	Unpasteurized milk
		Brucella	Cheese, raw milk
		Group A *Streptococcus*	Egg and potato salad
		Listeria monocytogenes	Cheese, raw milk, hot dogs, cole slaw
		Trichinella spiralis	Pork
		Vibrio vulnificus	Shellfish

1. Staphylococcal Food Poisoning

Preformed enterotoxins A, B, C, D and E are usually involved in staphylococal food poisoning. These are heat-resistant polypeptides.

S. aureus is the culprit. The common sources are ham, pork, canned beef and cream-filled pastry. The incubation period is l-6 hours, usually 3 hours. It causes nausea, vomiting, diarrhoea and collapse. Fatality is rare and is due to dehydration. Diagnosis can be established from vomitus, stool, food and from purulent lesion on the body of the food handler or nasal swab. Isolation and phage typing and enterotoxin testing are helpful in confirmation. Treatment is by proper hydration. Antibiotics are not helpful.

2. Bacillus cereus

It is an aerobic spore-bearing Gram-positive bacillus. The common sources are

fried rice, meat balls, boiled beef, barbecued chicken, cream and vanilla sauce. There are two clinical types. One-third of the cases present as the 'emetic type' with vomiting, collapse. Diarrhoea may be a late symptom. The incubation period is 1-6 hours. The 'diarrhoeal type' has a longer incubation period and presents with diarrhoea, collapse and vomiting. The incubation period of the diarrhoeal type is 6-14 hours, usually 9 hours. Diagnosis can be obtained from vomitus, stool or food. Special peptone-beef extract, egg yolk, agar is required for culture. Serotyping is also important. Treatment is only symptomatic. It is usually self-limiting.

3. Clostridium perfringens

Clostridia are Gram-positive spore-bearing obligate anaerobes that are present in the human intestinal flora. Toxins are elaborated only during sporulation. Food poisoning is caused by heat labile enterotoxin, which is the component of spore coat. It causes epithelial damage and secretion from the ileal epithelium. Type A produces acute watery diarrhoea and type C produces bloody diarrhoea with perforation and peritonitis. Type C disease is also called 'enteritis necroticans' or 'pigbel' and is spread by poorly cooked pork. It may be fatal. The other sources are beef, chicken and turkey. Clostridial food poisoning usually occurs by ingestion of an inadequately cooked meat, especially when large quantities are cooked during parties or festivals. Cooking may not kill organisms in large chunks of the meat. Sporulation may occur when there is a significant time lag between cooking and eating. The incubation period is 8-22 hours, usually 12 hours. Diagnosis can be obtained from stool, rectal swab or food. Special culture and typing are required for confirmation. Treatment is symptomatic.

4. Vibrio parahaemolyticus

It is a rare type of food poisoning. The incubation period is 2—48 hours, usually 12 hours. The manifestations are diarrhoea, collapse, nausea and vomiting. Fatality is rare. The sources are sea food and rarely salted vegetables and salt water. Sea water may also be a source. Diagnosis can be confirmed by culture and serotyping from stool, rectal swab or food. Treatment is symptomatic. Tetracyclines or trimethroprim-sulfamethoxazole may be beneficial.

5. Yersinia enterocolitica

It produces collapse, diarrhoea and vomiting. Pharyngitis, arthritis, adenitis and rashes may also occur. The sources are chocolate milk, raw milk or pork. The incubation period is 2 hours to several days, usually 3 days. Diagnosis can be obtained by culture or serology. Stool from the food preparer is the best diagnositc material. Treatment includes symptomatic management. Streptomycin, tetracycline, chloramphenicol and cotrimoxazole are effective against the *Yersinia.*

6. Listeria monocytogenes

It causes diarrhoea, nausea, collapse and rarely fatality. Blood may be present in the stool. The sources are dairy products, milk, raw vegetable, cole slaw, poultry and beef. The incubation period is variable. Stool and rectal swab are the sources of the diagnostic material. Special cultures are required for confirmation. Symptomatic treatment is effective. Ampicillin is also useful.

7. Campylobacter jejuni

It produces diarrhoea, collapse and sometimes blood in stool. The sources are milk, chicken, and beef. Pet animals also may act as sources. The incubation period is 1-4 days or may be longer. Stool and rectal swabs can yield *Campylobacter* in a special culture. Erythromycin and macrolides are effective, apart from symptomatic treatment.

8. E. coli

Toxin producing *E. coli* produce diarrhoea, nausea, vomiting, collapse and rarely fatality. Haemolytic uraemic syndrome also may occur. The incubation period is 1-5 days. Salads and beef are the sources. Stool and rectal swabs may yield the organisms. Serotyping and special tests for toxin production are helpful. Treatment is symptomatic.

9. Salmonella

This is the most common type of food-borne disease leading to diarrhoea. However, due to the long incubation period and sporadic involvement, it is often not identified as a food poisoning. The incubation period is 5-72 hours, usually 24 hours. The non-typhoidal salmonella associated with food poisoning are the following:

S. enteritides
S. typhimurium
S. heidelberg
S. newport; and
S. hadar

Salmonella colonise on all domestic animals and egg, and egg preparations are the most common sources. Meat and poultry are also sources. The presentation includes diarrhoea, collapse, nausea, vomiting and rarely blood in stool. Fatality also may occur. Food, stool or rectal swab from the patient or the food handler can yield the organism. Special culture and phage typing for *S. typhimurium* are required for confirmation. A special Salmonella-Shigella culture medium is available.

Salmonella may cause ileo-caecal mucosal ulceration and a systemic invasion. It may lead to seeding in bones, joints, meninges, gall bladder, heart, etc.

Toxic megacolon and haemolytic uraemic syndrome may also occur. Antibiotics like nalidixic acid, co-trimoxazole or the 3rd generation cephalosporins are indicated in the 'dysenteric type' of disease. In mild cases, antibiotics may lead to development of drug resistance and overgrowth by other resistant organisms. However, infants and elderly patients may be given antibiotics even in mild cases.

10. Shigella

It produces collapse, fever, diarrhoea, blood in stool, nausea and vomiting. The sources are milk, salads, potato, tuna and turkey. The incubation period is 1-7 days. Stool or a rectal swab from patients and food handlers and food may yield the organism. Salmonella-Shigella culture medium and typing are useful. Cotrimoxazole, nalidixic acid and 3rd generation cephalosporins are effective, apart from symptomatic treatment.

11. Clostridium botulinum

Botulinum toxin is the most lethal toxin and the lethal dose is l(h7 mg/kg. A-G are the antigenic types. A, B, E and F cause human botulism. It causes a neuromuscular blockage and death by a respiratory paralysis. *Clostridium botulinum* is a Gram-positive, spore-bearing obligate anaerobe that is present in soil, dust and agricultural products. Spores are heat resistant, whereas toxin is heat labile and is destroyed by heating at 80°C for 5-10 minutes. Food-borne botulism is due to preformed toxin in preserved food. Infantile botulism is due to ingestion of spores that elaborate toxin in the gut. Wound botulism is due to deposition of spores in the wound.

Canned or preserved foods, especially low-acid home canned food lead to botulism. The symptoms start within 18-36 hrs after the ingestion of contaminated food. The manifestations are nausea, vomiting, diarrhoea followed by blurred vision, dry mouth, dysarthria, diplopia and descending paralysis. It blocks acetylcholine release at the neuromuscular junction.

Differential diagnosis includes myasthenia gravis and Guillain-Barre syndrome. Infantile botulism may lead to sudden infant death syndrome (SIDS). Constipation or prolonged intestinal transit time is noted in infants who develop botulism. Treatment is only supportive. Antibiotics like cotrimoxazole, nalidixic acid or cephalosporins may be used for secondary infection. Aminoglycosides may potentiate neuromuscular blocking action of toxin. Prevention is by adopting safe methods for canning and discarding suspicious food and reheating of home canned food to 80°C for at least 5 minutes.

12. Mushrooms

The poisonous mushrooms are *Amanita phalloides, A muscaria, A. bresa* and *Galerina venenata.* Toxicity includes GI upset, haemolysis and multi-organ failure. *A. muscaria* produces parasympathomimetic symptoms due to muscarinic

alkaloids. Atropine is the antidote for *A. muscaria* poisoning. Gastric lavage is indicated in mushroom poisoning. *A. phalloides* poisoning is managed symptomatically with IV fluids and glucose. Dialysis and penicillin are also useful. Penicillin inhibits the uptake of the toxin into the liver. Cytochrome and thioctic acid are also under trial.

13. Shellfish and Other Sea Foods

During 'red tides' or 'dinoflagellate booms', the dinoflagellate, *Ptychodiscus brevis* elaborates several neurotoxins like 'saxitoxin' that inhibit the sodium-potassium pump and nerve conduction. Filter feeding molluscs like black mussel and sea scallop concentrate these toxins and ingestion of these shellfish may lead to poisoning. Toxicity occurs in 1/2-2 hours after ingestion. The manifestations are GI upset, glove and stocking type of paraesthesia, oral paraesthesia, vertigo, ataxia, a sensation of floating, hot-cold reversal in temperature sensation and diaphragm paralysis. Treatment includes supportive care and mechanical ventilation. Some of the symptoms may persist for several weeks.

'Ciguatoxin' produced by dinoflagellate and concentrated by ciguatera fish may also present with symptoms similar to saxitoxin. IV calcium and mannitol are found beneficial.

'Scombrotoxin' and histamines encountered in Scombroid fish may produce allergic reactions. Treatment includes gastric lavage, antihistamines and ranitidine orcimetidine.

6.2 Food Allergy

Food allergy is a troublesome affair for the patient, the family and the doctor. Food intolerance and aversion are often mistaken as food allergy.

1. Definitions

a) *Food intolerance:* Any form of unpleasant, reproducible adverse reaction to a food, which is neither psychologically nor immunologically based.

b) *Food allergy:* Clinical reactions to food components which are caused by pathological immune reactions; manifesting in any organ but principally in the GI tract, skin and the respiratory tract.

c) *Food aversion:* Psychological difficulties in relation to ingestion of particular foods, but which does not occur when the same food is given in a disguised form.

2. Pathophysiology

The intestinal mucosa is continually exposed to a variety of antigens from vari-

ous sources like food, microbes, etc. The intestine identifies antigens in the lumen by permitting small quantities to cross the endothelium and interact with the mucosal and systemic immune system. A breakdown in the gut mucosal barrier or an early excessive exposure to some antigens, can lead to a pathological state and sensitization.

The following are the components of the GI mucosal barrier:

a) Non-specific
   - Proteolysis in the stomach/intestine that breaks down the protein structure
   - Mucus in the gut
   - Gut motility
   - Microvillus membrane

b) Specific
   - Gut associated lymphoid tissue (GALT)
   - Secretory IgA
   - Macrophages
   - T lymphocytes and B lymphocytes.

3. Common Food Allergens

A variety of food items are identified as food allergens.

a) *Cow's milk protein:* Among cow's milk protein, beta lactoglobulin is most allergenic. About 25 other proteins are also known. Usually more than one protein are involved. Food processing may uncover antigenic proteins and therefore processed cow's milk may be more allergenic than raw milk. Human breast milk is known to contain foreign food proteins like cow's milk, egg, wheat, peanut etc. These proteins are able to elicit an allergic response even in breast-fed infants.

b) *Soya protein:* Soya protein allergy often develops as a cross-reacting response in infants with a proven cow's milk allergy. There is no proof that soya is less allergenic than cow's milk. Cross reactivity also exists between legumes, soyabeans, peas and peanuts. Soyaprotein intolerance is usually induced when infants with persistent diarrhoea are treated with soya milk. When the gut is immature or damaged, the whole proteins tend to get absorbed leading to sensitization.

c) *Peanuts, cocoa, peas:* These are often the causes of a hyperreactive airway disease or asthma.

d) *Egg white:* It is a potent allergen in some. Yolk is not an allergen. Vaccines prepared on chick embryo like MMR, mumps, measles, etc., may evoke a severe reaction. Currently human diploid cell (HDC) culture vaccine is available for measles and rubella.

e) *Citrus fruits:* These may aggravate a hyperreactive airway disease.

f) *Shrimp, crustaceans and fish:* These are rare causes of food allergy.

g) *Spices, yeast:* These too, are in the line of food allergens.
h) *Ingredients of Chinese cooking:* Aginomoto or monosodium glutamate is often the culprit.

4. Pathogenesis

The three important factors involved are the following:
a) Genetic predisposition
b) Early exposure, and
c) Defects in gut mucosal barrier.

Food antigens cause a mucosal damage. It also leads to systemic immune reactions and manifestations in the skin, respiratory tract and other organs. In children with two atopic parents, the risk of developing food allergy is close to 60%. The genetic basis is, however, not clear. A high IgE concentration in the cord blood correlates well with later development of food allergy. Antigen exposure early in life and duration of breastfeeding also determine the development of food allergy. Antigens or protein get absorbed from an immature gut or when there is a mucosal injury like diarrhoeal episodes. Otherwise, proteins are split up into amino acids before absorption.

Small amounts of ingested antigens eventually lead to sensitisation, while larger amounts induce gut tolerance. Intrauterine sensitisation is also possible, either by transplacental transfer of food allergens or by the action of maternal anti-idiotypic antibodies.

Food allergy can manifest in several ways. **Anaphylaxis or type I hypersensitivity reaction** mediated by specific IgE antibodies is manily responsible for immediate food allergies. These occur within an hour of ingestion and manifest manily in the skin and the GIT. These children usually have a family history of atopy, elevated total serum IgE, positive skin prick test as well as elevated specific IgE food antibodies.

Intermediate type of food allergies are mediated by type III hypersensitivity reactions (Arthus type) based on IgG antibodies and immune complexes. They occur from I -24 hr after ingestion and clinical symptoms are mainly gastrointestinal. Skin prick tests against offending antigen are usually negative.

The role of cell-mediated immune reactions in food allergy is unclear but it is believied that they play a part in late manifestations like enteropathy and malabsorption.

5. Clinical Manifestations

The gut has a dual role in food allergy. It functions as a target organ and as a vehicle for delivery of antigens to cause reactions in other organs. The usual clinical manifestations are the following:
a) General

- Anaphylaxis, shock

b) Gastrointestinal
- Nausea, vomiting
- Diarrhoea, malabsorption, failure to thrive
- Intestinal loss of blood, protein
- Abdominal pain, bloating

c) Respiratory
- Sneezing, rhinorrhoea
- Wheezing, cough
- Bronchospasm, dyspnoea

d) Skin
- Lip swelling, angioedema
- Itching, rash, urticaria
- Eczema

e) Others
- Joint swelling, arthralgia
- Headache, apathy
- Irritability, hyperkinesis

Gastrointestinal reactions are most common, occurring in about 50-80% and the manifestations may be immediate, intermediate or late. Early onset symptoms produce vomiting, diarrhoea and collapse. Late gastrointestinal manifestations include enteropathy, colitis and malabsorption usually induced by cow's milk.

6. Diagnosis

The diagnosis of gastrointestinal food allergy is often difficult due to practical problems. A good history and a reproducible reaction to elimination followed by challenge are most important. Laboratory tests may be of some supportive help. The various tests in gastrointestinal allergy include:

a) *Elimination and challenge:* This is the gold standard. Elimination leads to improvement and challenge reinduces the symptoms.

b) *Skin prick tests:* These are reliable in immediate food sensitivity. However, these are negative in Type III reactions.

c) *Immunological tests:* These are only of supportive use. The following tests are usually done:
- Total S. IgE
- Serum IgE food antibodies (RAST)
- Intestinal biopsy
- Serum IgG food antibodies (RIFT, ELISA)
- Malabsorption studies.

7. Treatment

Elimination diet and drugs are the mainstay in the treatment of food allergies.

a) *Elimination diet*: Indiscriminate use of elimination diets without a firm diagnosis, often causes malnutrition. This should be discouraged. Once the offending food allergen is identified, that alone should be excluded from the diet. Cow's milk protein allergy is the commonest food allergy in infants. Breast milk is the diet of choice if there is no significant lactose intolerance. Since some infants with cow's milk allergy also have a secondary soya protein allergy, soya-based formula should not be considered as the first choice of therapy. For infants who are not breast fed, hypoallergenic protein hydrolysate formula is recommended. Casein and soya hydrolysates are less allergenic than whey formula.

   Food intolerance and aversion to a particular food may be differentiated during elimination diet. Elimination of a single food is easy in older children. Most food allergies disappear eventually since tolerance develops with increasing age. Elimination diets should therefore be discontinued after a few years, generally by 3 years.

b) *Drug therapy*. The role of drugs in the therapy of food allergy is limited. Oral disodium cromoglycate may be useful in non-gastrointestinal symptoms like asthma caused by food allergy. Its effectiveness in gastrointestinal symptoms is not proven. Antihistaminics are also tried. Oral ketotifen is still under trial. Prostaglandin synthetase inhibitors like aspirin and indomethacin also have no proven efficacy. Steroids are indicated only in eosinophilic gastroenteritis. Hydrocortisone and adrenaline are required in anaphylaxis or angioedema to tide over the acute crisis. Ranitidine also may be beneficial.

8. Prognosis

By 3 years of age, most children are able to tolerate previously allergic foods. This transience of food allergy in the majority of children is due to maturation of the gastrointestinal mucosal barrier system and tolerance. Hence the prognosis is generally good.

SECTION 7

Life Cycle Approach in Nutrition

"Catch them young or better catch them before they hatch out."

—Elizabeth KE

7.1 Foetal Programming and Foetal Origin of Adulthood Diseases (FOAD)

Forsdahl (1972) suggested that growing up in poverty caused 'permanent damage' perhaps due to a 'nutritional deficit', which resulted in 'life-long vulnerability' to an affluent adult lifestyle and high fat intakes. Barker (1986) proposed that the roots of cardiovascular disease lay in the undernutrition in foetal life and early infancy after studying Hertfordshire birth cohort (1911-1930). There is some evidence that body proportions at birth show stronger associtation with cardio vascular disease (CVD). For example, low Ponderal index (weight/length) predicted coronary heart disease (CHD) better than birth weight.

Blood pressure, type 2 diabetes, insulin resistance, and combinations of these, the insulin resistance syndrome (IRS), are consistently related to low birthweight. Barker and Hales have suggested that it should be renamed the **'small baby syndrome'.** Although lipids show some associations with size at birth, these are weaker and less consistent. Arterial intima media thickness and carotid stenosis, examined using ultrasound, are increased in lower birthweight men and women and flow-mediated dilation, a measure of endothelial function, is reduced in such young adults and children of lower birthweight. People who were heavier at birth tend to become 'fatter' adults as measured by body mass index. However, this may reflect increased lean mass rather than adiposity but central obesity has been linked to small size at birth.

Accelerated childhood weight again (upward crossing of centiles) was associated with higher blood pressure in young adults in the UK. Early adiposity rebound was also associated with increased adult diabetes. Whet determines the

age of adiposity rebounds is unknown, but in Finland, lower weight at one year predicted an earlier rebound.

Adult obesity also adds to the effects of low birthweight. The effects of adult BMI on high blood pressure, type 2 diabetes, and insulin resistance are greater in individuals of low birthweight. The intriguing corollary of this is that high birthweight may protect against the adverse effects of adult obesity.

THE 'FOETAL ORIGINS OF ADULT DISEASE (FOAD)' HYPOTHESIS

Barker hypothesised that the associations between small size at birth or during infancy and later CVD reflect permanent effects of foetal undemutrition (Fig. 7.1). The fetus is dependent on the transfer of nutrients from the mother and adapts to an inadequate nutrient supply in a number of ways: prioritization of brain growth at the expense of other tissues such as the abdominal viscera, reduced secretion of or sensitivity to the foetal growth hormones, insulin and IGF-1 and up-regulation of the hypothalamo-pituitary-adernal (HPA) axis. The FOAD hypothesis proposes that although occurring in response to a transient phenomenon (foetal undernutrition) these adaptations become permanent or 'programmed' because they occur during critical periods of early development.

Programmed changes may include reduced insulin sensitivity; low muscle mass, pancreatic beta cell mass and nephron numbers; altered arterial structure, and up-regulation of the HPA axis and sympathetic nervous system (Fig. 7.1). The FOAD hypothesis proposes that these changes also render the individual more susceptible to the effects of environmental stressors such as obesity arising in later life. Nutritional effects on fetal growth are also shown by the drop in birthweight observed during famine.

Foetal growth depends on the uptake of nutrients at the end of a complex materno-fetal supply line. This includes not only the mother's food intake but also her intermediary metabolism, endocrine status, cardiovascular adaptations to pregnancy (e.g., plasma volume expansion which influence uterine blood flow), and placental structure and function.

In India, the mean full-term birthweight is 2.6-2.7 kg, almost 1 kg lower than in Western Europe. A world map of intrauterine growth retardation highlights South Asia as the worst affected region.

South Asian men and women have a higher fat mass, more centrally distributed fat, and a higher risk of obesity-associated disease than white Caucasians.

According to the FOAD hypothesis, increasing child and adult obesity in combination with persistently poor fetal growth creates a high risk for adult CVD and diabetes. There is certainly an epidemic . In Pune Children's Study, CVD risk factors were measured in 201 children bom in the KEM Hospital in Pune.

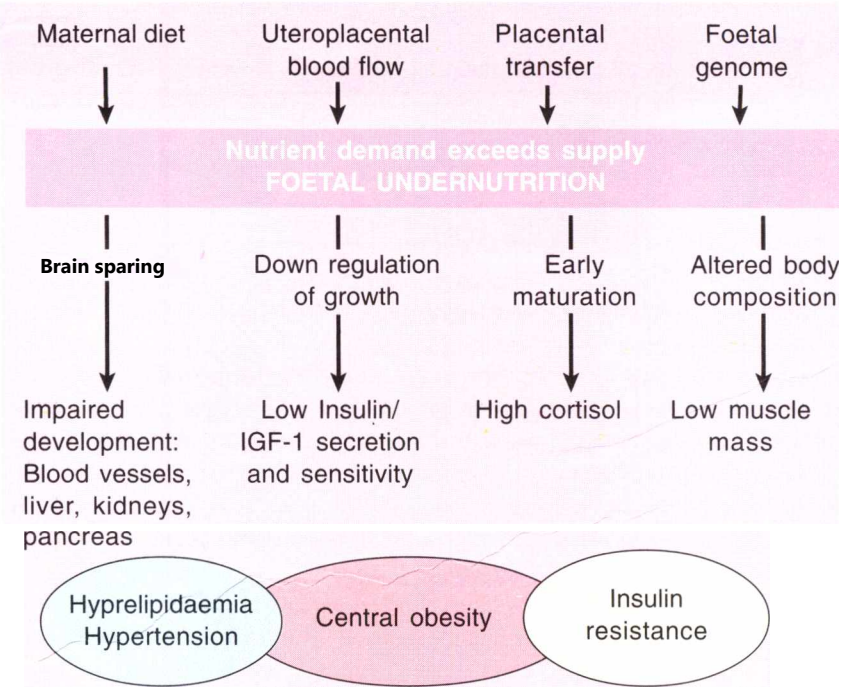


Fig. 7.1 The Foetal origins hypothesis

Birth weights were obtained from labour ward records. When they were initially studied, at the age of 4 years, lower birthweight children had higher plasma insulin and glucose load and higher IGF-1 concentrations, but not other evidence of an adverse CVD risk profile.

Further evidence came from studies carried out at the Holdsworth Memorial Hospital (HMH) in Mysore, South India. Among men and women aged over 45 years, the prevalence of coronary heart disease (CHD), defined using ECG changes and an angina questionnaire was 11% overall, and as in the Western studies, higher in those of lower birthweight. Lower birthweight was associated with higher two-hour glucose concentrations, higher rates of type 2 diabetes and impaired glucose tolerance, and higher serum triglyceride concentrations.

Two prospective studies based on large birth cohorts are currently under way in New Delhi and Vellore.

Thrifty Genotype and Thrifty Phenotype

The foetal origins hypothesis has also been called the 'thrifty genotype'. The thrifty genotype and phenotype are interesting issues. Neel suggested that these genes conveyed a 'fast insulin trigger' and thus the ability to store food rapidly as fat (savings), which become diabetogenic in modern setting of plentiful nutrition. The thrifty phenotype hypothesis, on the other hand, suggests that the undernourished foetus develops insulin resistance and other metabolic changes as a strategy for immediate survival, to down-regulate and prioritise growth (economy), for which it pays a price later in life, generally after the reproductive period. Both thrifty genes and the thrifty phenotype could become detrimental on exposure to plentiful nutrition.

Foetal insulin hypothesis is yet another issue. Hattersely proposed that, since insulin is a major growth hormone in foetal life, genes associated with either insulin resistance or reduced insulin secretion would lead to reduced foetal growth as well as an increased risk of adult diabetes. This is called the 'foetal insulin hypothesis'.

Twin studies have classically been used to distinguish between genetic and environment effects. Twins are smaller at birth than singletons but do not experience greater cardiovascular morbidity or mortality. Studies linking CVD risk to the difference in birthweight within twins pairs have shown inconsistent results.

The time trends in CVD and type 2 diabetes in western countries and in different socioeconomic groups during the 20th century, and the recent rise in developing countries, suggest a susceptibility to environment changes, which could either have genetic basis (thrifty genotype) or arise from foetal programming (thrifty phenotype).

The 'genes versus environment' debate is currently stimulating a great deal of hypotheses-testing research in this field. With increasing understanding of epigenetic effects and gene-environment interaction, it is no longer possible to think of diseases as being 'either genetic or environmental'. Thus the term geno-environmental is coined.

It is clearly possible to permanently alter gene expression by manipulation of intrauterine nutrition (Fig. 4.17). Such epigenetic effects may persist across generations.

Small Baby Syndrome and Foetal Malnutrition

The term 'fetal malnutrition' has been coined to refer to thin babies with loose skin fold who have low weight for length and low Ponderal index, irrespective of whether they are above or below 2.5 kg. A birthweight of 2.5 kg is the cut off chosen to mark low birth weight (LBW) irrespective of the gestational age. The expected weight for height can be obtained from intrauterine growth curves (Fig. 2.2-2.4).

Such babies with foetal malnutrition have long-term effects and are the candidates for the small baby syndrome with early adult diseases.

Table 71 The average weight and length of foetus as per gestational age

Gestational age	Length	Weight
28 wk	32 cm	1.1 kg
32 wk	42 cm	1.7 kg
36 wk	48 cm	2.6 kg
40 wk	51 cm	3.3 kg

The FOAD hypothesis is attractive because it suggests that several common degenerative diseases could be prevented by improving maternal health and foetal development.

There is no evidence that the current paediatric policy of encouraging catch-up growth during infancy in low birthweight babies is harmful. However, prevention and treatment of obesity in childhood should become a public health priority. Current data suggest that the greatest benefit in terms of future risk-reduction will be in low birthweight individuals. Further research is needed into effective interventions to prevent and treat obesity.

Tracking of BMI from newborn to adulthood seems to be logical and an intervention by itself. Therapeutic lifestyle changes (TLC) are recommended to nullify the effects of obesity and environment. Maintenance of weight for height, optimum BMI and waist/hip (W/H) ratio should be the goal for all children, adolescents and adults.

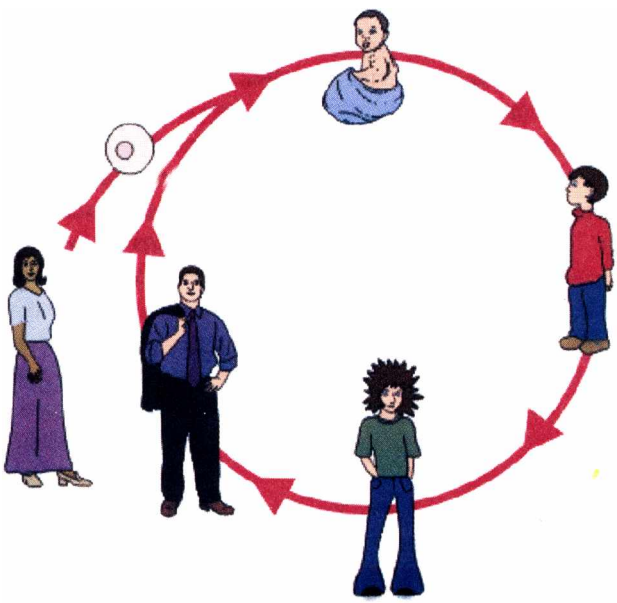
A W/H ratio of 0.95 is desirable in men and 0.8 in women. A BMI cut-off between 18 and 22 is now accepted to be ideal for South Asians. The BMI cutoff values and the ELIZ health path for the various age groups are given in Appendix.

Human placental lactogen (HPL) also called **human chorionic somatomammotropin,** is a polypeptide placental hormone. Its structure and function is similar to that of human growth hormone. It modifies the metabolic state of the mother during pregnancy to facilitate the energy supply of the foetus. HPL is an anti-insulin. In a bioassay HPL mimics the action of prolactin, yet it is unclear if HPL has any role in human lactation. HPL affects the metabolic system of the maternal organism. HPL increases production of insulin and IGF-1 and increases insulin resistance and carbohydrate intolerance. Chronic hypoglycaemia leads to a rise in HPL. HPL induces lipolysis with the release of free fatty acids, increase in insulin secretion and insulin resistance. With fasting and release of HPL, free fatty acids become available for the maternal organism as fuel, so that relatively

more glucose can be utilized by the foetus. Also, ketones formed from free fatty acids can cross the placenta and be used by the foetus. These events support energy supply to the foetus in states of starvation.

7.2 Girl Child In Focus

Life Cycle approach



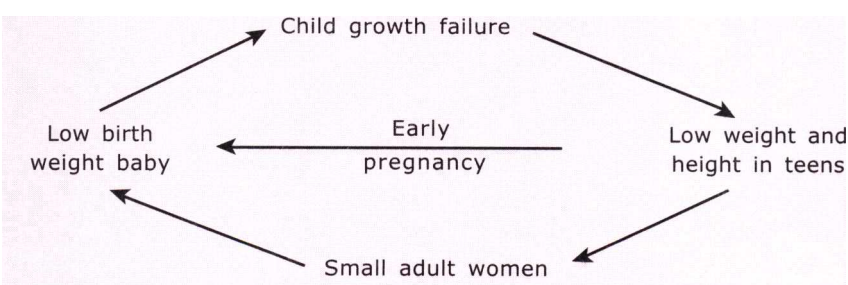


Fig. 7.2 Intergenerational cycle of growth failure

The Girl Child

The girl child is a special child and should be considered as a 'privileged child'. The girl child is the 'prospective mother'. She has to contribute a lot to the future citizens, the future health of the nation and to human existence. Her health, her education and her status in the society are the determining factors for the future of the family, the society and the nation at large.

According to the demographic profile, more number of females per every 1000 male population is considered a favourable ratio. This also points to the need for a "positive attitude towards the girl child" and females. However, in practice, there is "obvious neglect" of the girl child, at least in some communities and in some parts of India. This neglect starts right at the time of birth and continues thereafter. She is subjected to infanticide, child labour, physical and sexual abuse, child marriage, dowry deaths and even to the cruel practice of burning alive in the name of 'sathi'.

In some communities, the general custom of distributing 'sweets' and sharing joy at the birth of a male child is not practiced when a girl child is born. Some husbands and mothers-in-law punish the women for giving birth to girl children. Some husbands even go to the extent of deserting the wives and remarrying. Female infanticide is also a reality. The cost of getting her married is one reason for the negative attitude towards the girl child. Hence the present dowry system should be totally banned.

There is an increased incidence of malnutrition among the girls. In the average families, the girl child and women eat last and least. This leads to chronic malnutrition, micronutrient deficiencies and stunting of growth in the girls. Thus the older girls, the adolescent girl, the prospective mother, the antenatal and the postantal mother, are malnourished and stunted. This may be an important contributing factor to the high prevalence of low birth weight (LB W) babies in India, in spite of improved antenatal care.

During the event of an illness, the male child is given preference over the female child in seeking medical care. The hospital statistics are in accordance with this. It is also a fact that morbidity and mortality rates are more in males than in the females. This is partly due genetic, metabolic, degenerative and immunological disorders, infections and accidents that are more common in males. It appears as if "Mother Nature" has a preference for female sex. The life expectancy is also more among females.

Coming to education, primary school enrollment ratio is 113:91 for boys and girls and secondary school enrollment ratio is 59:38 (1990-95). The male literacy rate in India is 66% compared to 38% in females (1995). There is a saying that "when we educate a boy, we educate a man, and when we educate a girl, we educate a generation and a nation too."

Socially, the girl chlid and women are subjected to multideprivation as well as the physical and sexual abuses. She has to do hard work and undertake multiple

responsibilities without any acceptance or remuneration. Most of the girls are exploited in the form of child labour. The girl child has to assume 'mother figure' to the younger siblings. They have to do household work and also additional labour. Women are given less remuneration for equal work than men in most jobs.

In the state of Kerala which stands apart in vital statistics, there is no willful or obvious neglect of the girl child. The girl child is given education, more than the boys. However, in most families and even among mothers, there is a preference for the male child. They often opt to have one more child hoping that the next will be a male child. Male children are considered as future 'bread winners' and future heads of the families. In olden days, the girl child and women of Kerala had high esteem and social status in the royal families and in most communities. Now this trend is declining. But the women of Kerala are assuming additional responsibilities as earning members for self existence and social security. They have numerous responsibilities like house keeping, child rearing, income generation, providing transport facilities, medical care, etc. Hence, they need a lot of physical, emotional and financial support.

Mass media education utilising the print and the electronics media should be undertaken to change the attitude of the society towards the girl child. The government of Tamil Nadu has started a programme to adopt the so-called 'unwanted and abandoned girl child'. Financial allowance is also being given to mothers who deliver female children.

Information, education and communication (IEC) can go a long way in creating a positive attitude towards the girl child. She should no more be considered a burden, but a "blessing to the family and society".

7.3 Adolescent Nutrition & Growth

Introduction

Adolescence is a period of rapid growth and perhaps **the last chance to grow. It is a transitional stage in human life.** The exceptional rapid growth in this stage is characterized by a lot of individual variations that poses difficulty in defining normality. In our society where priorities have been based on morbidity and mortality rates, adolescents have been overlooked by health planners. But recently their presence, care and counselling are coming to limelight. There is a high prevalence of malnutrition and anaemia among them. Studies have shown that two-third of adolescent girls are malnourished and anaemic.

Adolescence is a period of high stress both physically and psychologically. Puberty sets in a little earlier in tropical countries. **Precocious puberty is diagnosed when signs of secondary sexual characters are seen before the age of eight years in girls and before the age of nine years in boys.** Puberty is a period

of rapid growth. The growth at this period is under the influence of nutrition, genetic factors and hormonal factors. An adolescent boy is expected to take one unit of calories, i.e., 2400 kcal. This is the requirement of an adult sedentary male. The girl child may require a little less amount of calories due to the smaller frame of the body (2100 kcal). Malnutrition is common among adolescents due to various reasons. Many of them consume less food due to lack of time as they are engaged in a period of stressful studies in the school final and college levels. Many of them refuse to carry tiffin boxes and resort to taking 'fast food' and soft drinks. These food items do not supply enough calories, vitamins and minerals. Some of them especially girls, eat less food as they want to remain slim. In many families, the adolescent girls eat last and least due to poverty and ignorance. Some of the adolescents have psychological problems like anorexia nervosa and bulimia nervosa.

Increased sports and other activities in adolescents warrant extra food which is often not obtained. Adolescent pregnancy is another problem faced by some. The adolescent child should take a balanced diet (Tables 3.13-3.22 and Fig 3.1). Health and nutrition education can help in achieving this goal. Nutrition of the adolescent girl should get priority as she is the prospective mother. Improving the nutrition of the girl child will help to reduce the incidence of low birth weight babies.

ADOLESCENT EATING HABITS

1. Eating Behaviour

The ten facets of eating behaviour in adolescents that need attention are the following:

a) Missing meals like breakfast, lunch etc. **Breakfast is brain's food** and missing the breakfast can even affect classroom performance. Many do not cherish carrying tiffin boxes with them.

b) Eating snacks and confectionery. These provide mostly empty calories, lead to micronutrient deficiencies and caries tooth. Some of the adolescents skip major meals like lunch and eat only snacks.

c) Fast food. Fast food provides some nutrition, but it is not a balanced diet. Very little information is available about the nutrient composition of fast food. Moreover, deep frying and refrying in the same oil are potentially injurious to health in view of the liberated oxides, peroxides and other free radicals. Food additives like ajinomoto and certain colouring agents and dyes are antinutrients.

d) Soft drinks and other fun drinks. These supply only empty calories and may also contain antinutrients. Moreover, they kill the appetite and promote skipping of meals.

e) Start of alcohol consumption. This is a dangerous turn and leads to alcohol related accidents and crimes.

f) Distinctive likes and dislikes for food like aversion to roots and tubers, steamed food items etc., and liking for fried food, fast food etc.

g) Variable food consumption. On some days, they tend to take high energy and on some days they tend to take very low energy.

h) Low intake of some nutrients especially micronutrients. Among the nutrients, calcium, iron, iodine and fluorine have attracted attention from olden days, but others like vitamin A, E, and C, zinc, copper, selenium etc., are now coming to limelight. **Zinc supplementation has shown to improve weight and height gain in those with low birth weight and malnutrition.**

i) Unconventional meals. Eating of various uncoventional combinations and permutations that cause anxiety in elders.

j) Adolescent dieters. Special dieting is resorted to, to remain slim. On the contrary, some want to become tall or muscular. Psychological disorders like anorexia nervosa or bulimia nervosa are also encountered in some. Acne vulgaris is another problem that necessitates dieting. Almost all people have acne at some time during adolescence. Avoiding of fatty foods, chocolate, soft drinks and beer and supplementation of polyunsaturated fats, vitamin A and zinc are found to be beneficial. The eating habits of adolescents are important because when they settle down as adults, they have to establish the eating behaviour of a new family.

2. Eating Healthy Campaign

Based on the above observations, the following principles are recommended for the eating-healthy compaign.

a) **"Moderation in all things":** Ensure a balanced diet by including all food groups every day. The intake of fats, oils and sugars should be kept to the minimum. The food guide triangle will help in the choice of a balanced diet (Fig. 3.1). Include cereal-pulse combinations, roots and tubers in adequate quantities, vegetables and fruits liberally, milk and milk products and meat group in moderation and oils, fats and sugars in small quantities.

b) **"Variety is the spice of life":** Ensure variety within each food group, e.g., not always the same vegetable. Food fads should be avoided as it may evolve as a social handicap.

c) **"What is enough":** The adolescent boy should eat as much as the father eats (2400 kcal) and the adolescent girl should eat a little more than what the mother eats (2100 kcal).

d) **"One man's meat is another man's poison":** Vegetarianism/non-vegetarianism. Some claim that vegetarians are healthier than others. This depends on the degree of vegetarianism. **Vegetarians** have low risk of obesity, coronary artery disease, and hypertension and colon cancer. **Vegans,** who eat no animal products, are at risk of calcium, iron and vitamin B_p deficiency. Whole grains and germinated seeds are beneficial in them. **Lacto-ovo-vegetarians**

who take milk and egg have very little nutritional risk except iron deficiency due to lack of haeme-iron. **Fruitarians** eat only fruit and are at risk of protein, sodium and other deficiencies.

e) **"Enough is as good as a feast":** The best or the perfect diet. We can only choose a better diet; there is no best or perfect diet. Humans are omnivorous with enzyme system that can adapt to a variety of foods.

f) **Micronutrients and antioxidants:** Beta-carotene, vitamin A, vitamin C, folic acid, iron, iodine, zinc are essential. Weekly iron during adolescence is a useful intervention.

The concept of eating healthy should be advised with patience, perseverance and a sense of humour. They should be cautioned about drinking while driving, excessive slimming, overeating etc. Parents have more influence than they believe. Parents can choose the foods and drinks they buy, cook and store in the refrigerator.

The adolescent's food habits are laid down in the family and the family appears to be the one and the main influence. The others are the peer group, the evolving independent personality and the society in general. Adolescence is a transitional stage with upset food habits, but soon they settle down and reestablish the eating behaviour of the family and work out a compromise set of food habits with the partner. Thus, they decide the eating habits of the future families.

ADOLESCENT GROWTH

Genetic, hormonal and nutritional factors, childhood food habits and emotional balance account for variations in growth of adolescents.

Growth spurt accounts for sudden increase in height and weight, changes in body proportion, changes in facial features, changes in body structure and fat deposition. The last but not the least is the development of sexual organs and appearance of secondary sexual characteristics. Sexual maturation is marked by 'menarche' in girls and 'nocturnal emission' in boys.

Velocity of growth is different in different peroids of life. It is high during the first years of life, then slows down and again reaches its peak during the adolescent years. During puberty, boys gain about 20-38 cm in height, while girls about 16-25 cm and also 20 kg weight in boys and 16 kg weight in girls. At the cessation of growth, boys are taller than girls, although in the early adolescence the girls are taller. Body segments grow at different rates at different stage of puberty, e.g., legs begin to grow earlier than the trunk, hands and feet grow at a faster rate, the shoulders widen in boys and the hips widen in girls.

- It is the growth hormone and sex hormones which play a major role in growth spurt.
- Pubertal height spurt begins at an average age of 12 years for girls and 14

years for boys. Signs of puberty before 8 years in girls and 9 years in boys is precocious puberty.

- About 25% of adult weight is gained during this period.
- Adolescent growth failure is never noted on time because after early childhood nobody bothers to monitor growth.

Causes for Growth Failure

1. Normal variation needs only reassurance. This is called constitutional delay.
2. Pathologic due to genetic, chronic illness, nutritional problems and endocrine problems. This needs to be detected early and given proper management. Growth failure resulting from reduced nutritional intake is a major community health problem.

Adolescence is the second and perhaps the last chance for growth and rehabilitation and prevention of future risks like growth failure, psychological problems, birth of low birth weight babies and so on.

Pubertal growth spurt starts earlier (nearly 2 years) in girls than in boys. In girls growth spurt is in the following order:

- Selective fat deposition resulting in a feminine contour
- Breast development, enlargement of breasts, areola and nipples
- Development of the pubic hair, growth of hair in the armpits and around external genital organs
- Peak growth velocity—increase in height, weight, widening of the hips
- Menarche—onset of menstruation

In boys, growth spurt is in the following order:

- Body more muscular
- Grows facial hair, moustache begins to appear
- Voice becomes deep
- Grows hair in the axilla, chest and in the groin
- Onset of nocturnal emissions

Sexual development is assessed using Tanner's Sexual Maturity Rating (SMR) scale. SMR scale for boys and girls are given Table 7.2 and 7.3 Menarche usually occurs in SMR stage 2-3.

Adolescent Growth Assessment

Teenage is generally equated with adolescence, but the recent view is to include 10-19 years as adolescent period. Adolescence is a transition period and growth during adolescence is an important determinant of adult body size. Almost 25% of adult status is achieved during this period. Moreover, adolescence is the second and perhaps the last chance for nutritional rehabilitation and prevention of future risks like growth failure, psychosocial problems, birth of low birth weight (LBW) babies and so on. **The three dimensions of growth during adolescence are physical growth, mental and emotional blooming and sexual development**

Most of the Indians, especially the rural and the urban poor fail to achieve the full-endowed genetic potential for growth and intelligence. This is due to the sub-optimum interaction of host factors with nutrition and environment.

Table 7.2 Sexual Maturity Rating (SMR) scale

SMR stage	Age in years	Pubic hair	Penis	Testes
1	10-12	None	Preadolescent	Preadolescent
2	12-14	Sparse, long, slight pigmentation	Slight enlargement	Enlarged scrotum, pink texture altered
3	14-16	Darker, starts to curl	Longer	Longer
4	16-18	Resembles adult like type but less in quantity	Larger, increase in breadth, increase in glans size	Larger, darker scrotum
	18-20	Adult distribution, spread to medial surface of thighs	Adult size	Adult size

Table 7.3 SMR in

SMR stage	Age in years	Pubic hair	Breasts
1	10-12	Preadolescent	Preadolescent
2	12-14	Sparse, lightly pigmented, straight	Breast and papilla elevated, increase
3	14-16	Darker, beginning to curl	Breast and areola enlarged, no contour separation
4	16-18	Coarse, curly, abundant but amount less than in adult	Areola and papilla form secondary mount
5	18-20	Adult feminine triangle, spread to medial surface of thighs	Mature nipple projects

High prevalence of low birth weight (LBW) babies and adolescent growth failure are indicators of social deprivation. Growth failure is the net effect of influences like socioeconomic factors, poverty, illiteracy, ignorance, childhood malnutrition, anaemia, worm infestations and gender disparity. Beauty consciousness, adolescent dieting, alcohol and drug abuse, missing meals, fast food and cola culture, eating snacks and confectionery, undesirable like and dislikes for foods etc., also add to the problem. Lack of nutrition and health education (NHE) and ineffective information, education and communication (IEC) activities that are expected to lead to informed decision making are also contributing factors.

Literature scan on adolescent growth reveals that the 50th centile height among affluent Indians approximate 50th centile of NCHS reference standards till 6 years in boys and 10 years in girls and gradually slip to 10th-20th centile by adolescence. Similar trend occurs with weight also. Even though affluent Indians are shorter and lighter than Westerners, they are comparable to their counterparts of Asian origin. Growth pattern of adolescent Indian boys and girls (10-19 years) as per the National Nutrition Monitoring Bureau (NNMB) data is depicted in Fig 7.3 & 7.4. The secular trend among adolescents for height in a study from Delhi showed a secular increase of 2.1 cm for boys and 2.7 cm for girls per decade. But, it was only 1.5 cm for boys and 2.1 cm for girls in other regions. Thus the secular trend is not consistent and is variable. The age of onset of menarche in girls also varies from 12.5 to 14.5 years. This is noted to be progressively decreasing. The maximum physical growth occurs 1-2 years before menarche. Weight gain is maximum during 12-16 years in both sexes and height gain is maximum in girls during 11-13 years of age and 13-15 years of age in boys. In conclusion, adolescent growth chart in relation to age alone is not the right answer. Sexual development and height also need to be incorporated in the charts. These are, however, exhaustive and complicated.

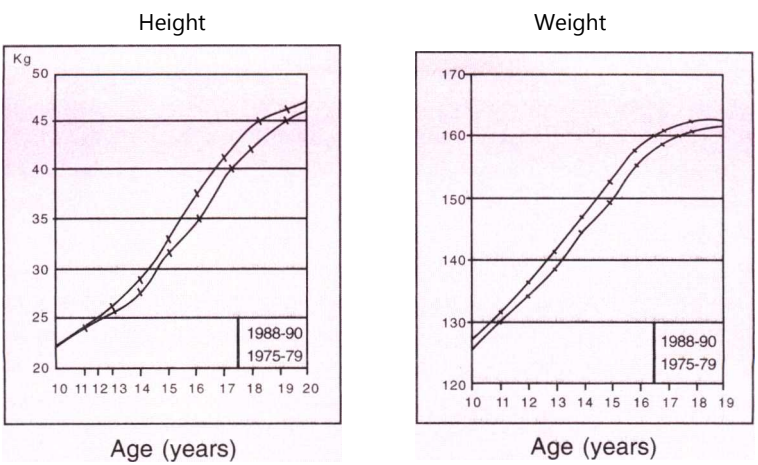
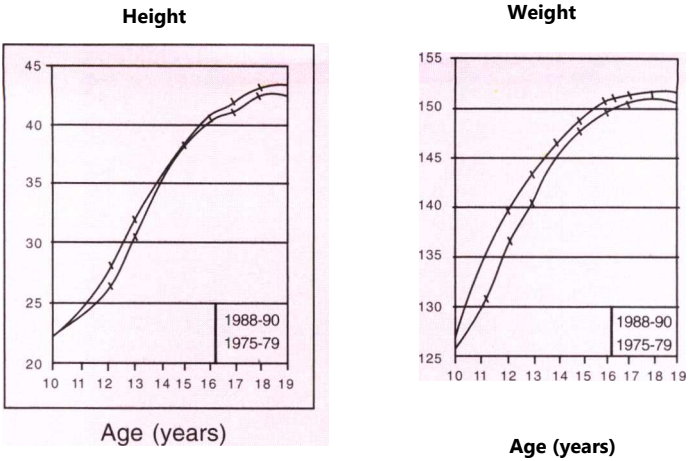


Fig. 7.3 Growth pattern of adolescent boys



Growth pattern of adolescent girls
(Source: NNMB, N1N Hyderabad, 1975-1990)

'9'

Growth assessment is an essential component of health surveillance and the gold standard for growth assessment is anthropometric measurements using standardized equipments and procedures. Weight, height and upper segment to lower segment (U/L) ratio are generally measured. **The 50th centile of the normal bell shaped distribution curve is the median and it is also termed** standard value or 100% of the expected. The WHO recommended standard reference growth chart is the NCHS standard validated in 1963-1975 and updated in 2000 on more than 20,000 well-nourished, healthy US population from birth to 18 years who represent various ethnic groups and immigrants. The future generation is expected to have better nutrition and growth. **Even though the NCHS reference curves are considered ideal for underfives across the globe, these are less applicable to adolescents especially from developing country like India.** Adolescent growth is linked to the onset of puberty and various genetic and hormonal factors. The early growing and the late maturing adolescents included in a particular age group tend to level off the growth peak and camouflage the issue. It appears as though adolescent children grow more gradually and over a long period than they do. The growth curves may indicate poor or excessive growth when the child is growing normally, that is, when growth happens to be later or earlier. Growth is an ongoing process rather than a stable quality. A child in the 5th centile of weight for age may be growing normally, failing to grow or may be recovering from growth failure. Further, there is no universally agreed upon criterion for growth failure; most consider it as below 5th centile, some consider it as below 3rd centile.

The nutritional requirement as per the ICMR (1998) recommendation of adolescent boys is 2450 kcal and adolescent girls is 2060 kcal. According to the bedside calculation, these are 2400 (1 unit) and 2100 respectively. It may be understood that the nutritional requirement of an adolescent boy is equal to that of an adult sedentary male, i.e., equal to that of the father. The requirement of adolescent girls is more than that of an adult sedentary female, i.e., more than that of the mother. (Refer tables 3.13-3.22)

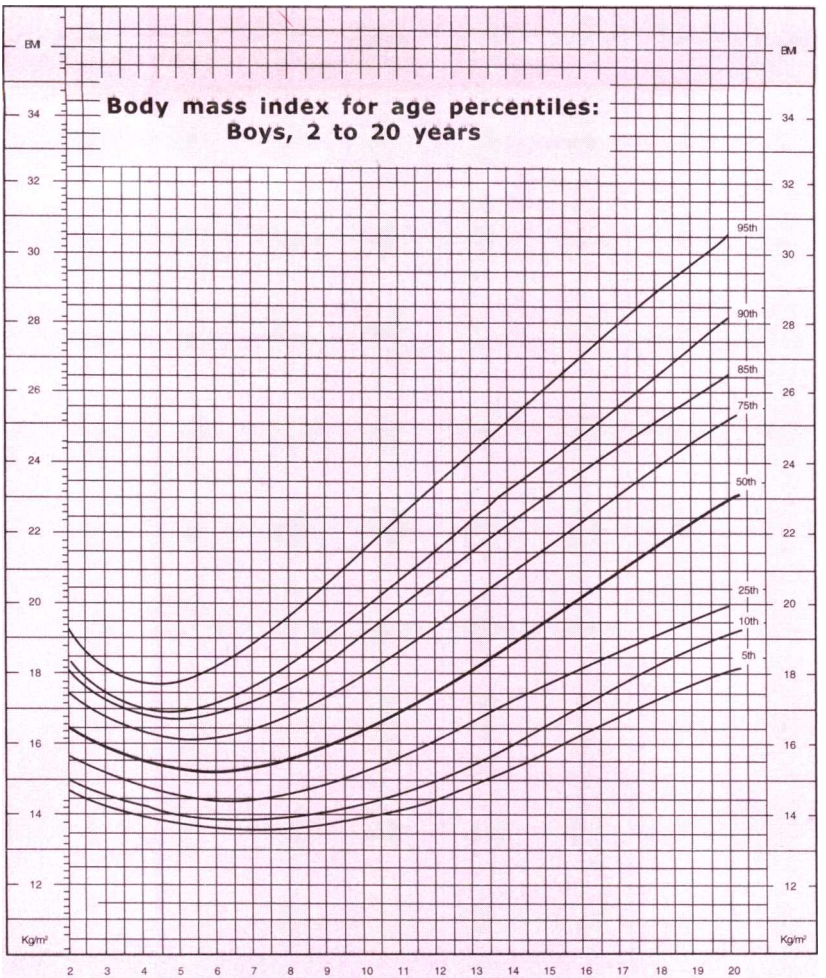
Among adolescents, normal variation in the timing of growth spurt can lead to misdiagnosis of growth disorders. The relationship between growth and sexual maturity is an important confounder. Special growth charts have been developed for early, mid and late maturing adolescents. There is also the risk of diagnosis of growth disorders especially with particularly tall or short parents, if parental height is overlooked. Mid parental height is also a significant variable that is to be considered. Charts have been designed based on mean mid parental height. Mid parental height (MPH) is approximately the average final height expected in a child. It is calculated as follows:

$$\text{MPH for boys} = \frac{\text{Paternal height} + \text{Maternal height}}{2} + 6.5 \text{ cm}$$

$$\text{MPH for girls} = \frac{\text{Paternal height} + \text{Maternal height}}{2} - 6.5 \text{ cm}$$

Acute malnutrition leads to fall in weight for age or more precisely weight for height leading to wasting. Chronic malnutrition leads to fall in height for age and stunting. When growth parameters fall below the 5th centile, it is customary to express it as percentage of the standard value to grade the severity of growth failure. It is important to differentiate growth failure due to undernutrition from genetic, constitutional, endocrine, metabolic, skeletal and congenital causes. Specialized charts are available for US children with various conditions like achondroplasia and Down, Turner and Klinefelter syndromes.

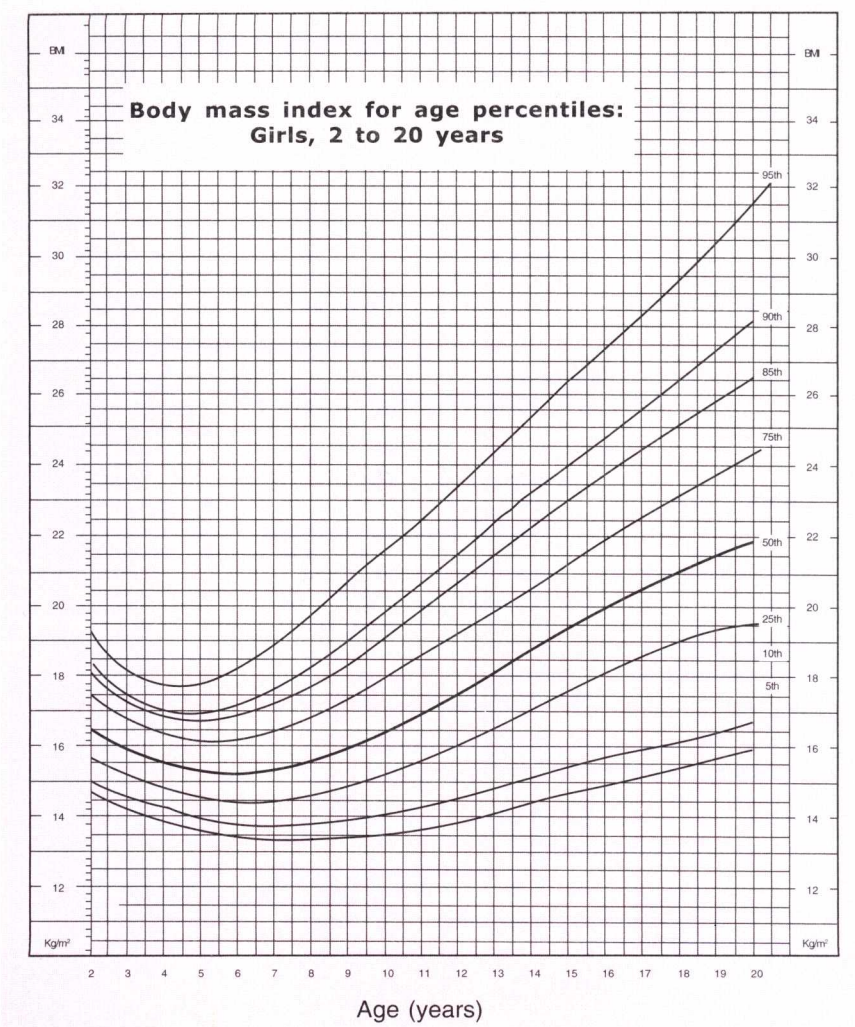
Growth charts for age can also confirm obesity, if weight for age exceeds 120%. The relationship between weight and height is popularly expressed as body mass index. The body mass index (BMI) is calculated as weight (kg) per height^2 (m^2). Standards for BMI have been developed for 1-19 years old population (Fig. 7.5 & 7.6). Raised BMI is an estimate of adiposity. **However, measurement of electrical impedance is the best estimate of body composition.** BMI is predictive for adult obesity-related morbidity and mortality. Obesity is slowly emerging as a problem in our transitional society. Reduced BMI is indicator of chronic energy deficiency (CED), which is a fairly common problem in a develop-



Body mass index for age percentiles: Boys, 2-20 years

(Source: NCHS and CDC [US], 2000)

ing country. Thus, incorporation of BMI seems appropriate and valid. **BMI changes with age; at birth, the median is as low as 13 kg/m², it increases to 17 at one year, decreases to 15 at 6 years and slowly increases to 21 by adulthood.** It is largely age independent in a particular age range and the differences between both sexes is very minimal. The International Obesity Task Force (IOTF) has defined BMI > 18.5 as normal; BMI > 25, corresponding to 90th centile as overweight; and > 30



Fia 7 6 Body mass index for age percentiles: Girls 2-20 years

(Source: NCHS and CDC [US] 2000)

corresponding to 97th centile as obesity. Around 13 years, the corresponding figures are 22 and 25 respectively. BMI 18.5 corresponds to 12th centile in adults. In addition to weight, height and BMI. upper segment to lower segment ratio is also usually calculated. An upper segment to lower segment ratio of 0.9-1 is expected among adolescents. It is 1:1 in adults. Higher U/L ratio is characteristic of short limb dwarfism and bone disorders like rickets. Bone age is found to correlate well with the stage of pubertal development.

The existing growth charts prove to be insufficient due to confounding variables like age of onset of puberty, stature of parents etc. A growth chart based on weight, height and BMI seems appropriate for practical purpose. Even though BMI above 18.5 is considered normal, many studies have shown that most of our adolescents especially below 13 years have a BMI below 18.5. Rao and Singh have also pointed out this and have suggested that values < 15 indicate underweight or Chronic Energy Deficiency (CED) and <13 indicates severe underweight. The upper cut-off of 22 for overweight and 25 for obesity in young adolescents when growth is still continuing seems appropriate and in accordance with IOTF standardisation. In late teens and adults, in whom growth is over, a cut-off value of 18.5 and 25 seems appropriate. BMI above 25 denotes risk for obesity or overweight and above 30 is definite evidence of obesity (Fig. 7.7 and 7.8). A single chart that shows weight, height and BMI is warranted. Thus incorporating weight, height and BMI, in place of age, sex, pubertal growth and parental stature, the **'ELIZ Health Path for Adolescent children'(EHPAC)** was designed for practical purposes (Appendix). This health path is very helpful due to the following reasons:

1. It is very simple to use and demonstrate. The weight and height can be plotted in the same chart and BMI can be directly read from the right margin of the chart. It avoids the tedious calculation of BMI.
2. The same chart can be used for both sexes and is sufficient for preliminary screening of a large number of children. For detalied evaluation of an individual child, age and sex specific data and centile charts may be used.
3. In addition to incorporating weight, height and BMI in the same chart, it also depicts the various curves denoting normal range, undernutrition, overweight (tendency for obesity) and overt obesity.

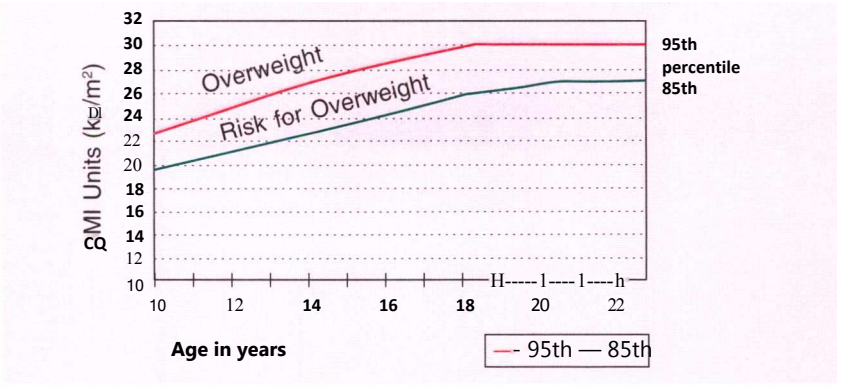


Fig. 7.7 Male body mass index 85th and 95th percentiles.
(Source: Must et al., 1991)

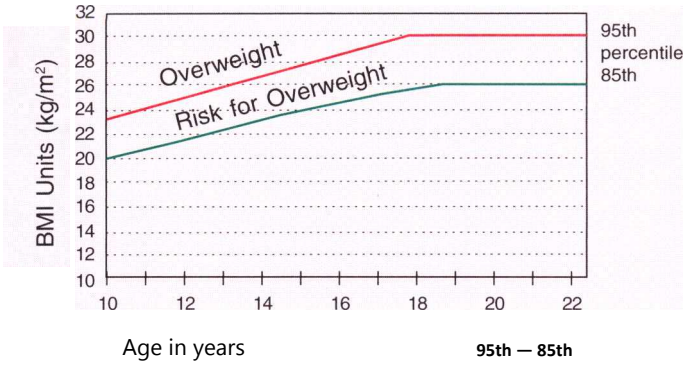


Fig. 7.8 Female body mass index 85th and 95th percentiles *(Source: Must et al., 1991)*

4. It can diagnose both undernutrition and obesity and also shows the desirable weight range for the stature of an individual. This information can help to curtail future obesity as well as purposeful dieting and slimming especially in adolescent girls who consider themselves obese even though they are in the normal range.
5. The purposeful omission of a curve below BMI 15. e.g.. at BMI 13, will avoid false satisfaction among adolescents with undernutrition that they are also above a curve. This is important as adolescence is the last and the final chance for them to grow and maintain normal body proportions.
6. This health path can also be used by adults to maintain optimum body proportions and thus remain fit and keep away from many of the lifestyle disease. ***(Ref. ELIZ health path for children and adults in Appendix)***

ADOLESCENT CARE AND COUNSELLING

Adolescents need special care, support and counselling. Teenage clinics can be organised in health centres and hospitals on Saturdays. Teenage clubs can be organised under NGOs and resident's associations. The Adolescent Friendly Hospital Initiative (AFHI) is another interesting endeavour in this respect.

Adolescent growth reflects the adult size, the productivity of the future nation and even the size and health of future citizens. So, it is important to give ample importance to adolescent nutrition and growth monitoring. UNICEF has recommended **45 kg and 145 cm as the ideal prepregnancy weight and height of girls.** Pre-pregnancy weight and height are now identified as important determinants of birth weight of future babies. Micronutrient deficiencies, especially iron deficiency, is yet another important cause for reduced productivity and LBW

babies. Adolescent growth failure needs prevention and treatment. The following suggestions are put forward for the same:

1. Ensure optimum growth and development in children including adolescent boys and girls.
2. Ensure a balanced diet in all age groups and optimum nutrition.
3. Ensure 45 kg weight and 145 cm height among prospective mothers, or at least 42 kg and 142 cm.
4. Ensure one iron tablet per week to adolescent girls to prevent anaemia.
5. Avoid dieting, missing meals and eating only fast food, soft drinks and confectioneries.
6. Avoid adolescent marriages and childbirth.
7. Avoid alcoholism, drug abuse and high-risk behaviour pattern that may lead to sexuality transmitted infections (STI) and reproductive tract infections (RTI).

Table 7.4 Immunization schedule (As approved by Indian Academy of Paediatrics)

Vaccine	**Age**
Tetanus Toxoid	Booster at 10 and 16 years
MMR Vaccine	1 dose if not given earlier
Rubella Vaccine	As part of MMR vaccine or 1 dose to girls at 12-13 years of age
Typhoid Vaccine	TA, Vi or Oral Typhoid vaccine every 3-5 years
Hepatitis-B Vaccine	3 Doses (0, 1 and 6 months) if not given earlier
Additional Vaccines	
Hepatitis-A Vaccine	2 doses (0 & 6 months) if not given earlier
Varicella Vaccine	1 dose below 13 years of age 2 doses at 1 month interval 13 years of age and above

(Each dose of immunization must be documented)

8. Strengthen nutrition and health education (NHE) using appropriate information, education and communication (IEC) techniques.

   Teenage day is celebrated on 1 st August and teenage week from 25th July to 1st August.

9. Ensure care and counselling in issues related to growth and development, emotional problems, sexuality, peer pressures, scholastic problems and career.

10. Impart family life education and parenting skills to adolescent boys and girls.

NHE is a million dollar investment that a country can accomplish for future security and productivity. This is true because the dietary habits of adolescents determine the adult size and health and also pave the foundation of the dietary habits of the future families.

7.4 Maternal Nutrition

The nine months of pregnancy represent the most intense period of growth and development humans ever experience. At no other time in life are the benefits of optimal nutritional status more obvious than during pregnancy.

Pregnancy and lactation are times of heightened nutritional vulnerability. However, the threat of malnutrition begins in the womb and continues throughout the life cycle (refer Section 3.3.4). Pregnant mother should be eating one extra meal for the baby (300 kcal and 15 g protein) and lactating mother should be eating one extra meal and a snack with plenty of fluids (500 kcal and 25 g protein).

Improving Nutrition through Life Cycle

A mother who was malnourished as a foetus, young child, or adolescent is more likely to enter pregnancy stunted and malnourished. Her compromised nutritional status affects the health and nutrition of her own children.

- Growth faltering earlier in life leaves women permanently at risk of obstetric complications and delivering low birth weight babies. Deficiencies of some micronutrients, such as folic acid and iodine, affect the foetus shortly after conception.
- By the time the pregnancy is detected, permanent damage is done. For these reasons, maternal malnutrition cannot be addressed during pregnancy alone.
- The periods before and between pregnancies provide an opportunity for women of reproductive age to prepare for pregnancy by consuming an adequate balanced diet, including supplements and fortified foods where available, and by achieving a desirable weight.
- Overweight and obesity at all ages, even in poor communities, present a difficult challenge for maternal and child health programs. Underweight and overweight often occur in the same communities and even the same households. Maternal overweight and obesity increase the risk of perinatal mortality, premature delivery, major birth defects, and maternal obstetric complications, including hypertension and gestational diabetes.
- Maternal and child health programs should alert women at all stages of the life cycle to the need to adjust diet and physical activity levels to achieve and

maintain a desirable weight for their own health as well as for better birth outcomes.

7.5 Geriatric Nutrition

Aging gracefully is an art. It needs lot of information, motivation and support, both physically and psychosocially.

With the dramatic decline in infectious disease over the last half century, more people around the world are living longer, and by 2050, around 30% of people in industrialised countries will be over 65. However, a simultaneous rise in chronic disease is increasing the risk of malnutrition. Up to 50% of residents in nursing homes and up to 70% of elderly patients in hospital are malnourished. The healthcare costs of malnutrition now exceed those of obesity. Malnutrition leads to decreased independence due to physical weakness and muscle wasting, which frequently leads to falls and fractures. Nearly half of all elderly patients with hip fractures are malnourished, while severely malnourished patients are more than three times more likely to suffer from infections.

Geriatric **nutrition** applies nutrition principles to delay effects of aging and disease, to aid in the management of the physical, psychological, and psychosocial changes commonly associated with growing old. "Elderly" was once defined as being age 65 or above, but the growing number of active and healthy older people has caused that definition to expand to

- "young old" (65 to 75),
- "old old" (75 to 85), and
- "oldest old" (85 and beyond).

The cornerstone of geriatric nutrition is a well-balanced diet. This provides optimal nutrition to help delay the leading causes of death: **heart** disease, **cancer,** and stroke. In addition, ongoing research indicates that dietary habits, such as restricting one's calorie intake and consuming antioxidants, may increase longevity.

What are the physiological changes that occur during aging?

Once the body reaches physiologic maturity, the rate of degenerative change exceeds the rate of cell regeneration. However, people age at different rates, so the elderly population is not a homogeneous group; there is great variability among individuals.

- Body composition changes as fat replaces muscle, in a process called sarcopenia. Research shows that **exercise,** particularly weight training, slows down this process. Because of the decrease in lean body mass, basal metabolic rate (BMR) declines about 5% per decade during adulthood. Total ca-



Fig. 7.9 An eldery man - The grace of aging

loric needs drop, and lowered protein reserves slow the body's ability to respond to injury or surgery. Body water decreases along with the decline in lean body mass.

Gastrointestinal (GI) changes include a reduction in digestion and absorption. Digestive hormones and enzymes decrease, the intestinal mucosa deteriorates, and the gastric emptying time increases. As a result, two conditions are more likely: pernicious anaemia and constipation. Pernicious anaemia may result because of hypochlorhydria, which decreases **vitamin B_p** absorption and affects approximately one-third of older Americans. Constipation, despite considerable laxative use among older people, may result from slower GI motility, inadequate fluid intake, or physical inactivity.

Musculoskeletal changes occur. A progressive drop in bone mass starts when people are in their 30s or 40s; this accelerates for women during **menopause,** making the skeleton more vulnerable to **fractures** or **osteoporosis.** Adequate intake of **calcium** and **vitamin D** helps to retain bone.

Geriatric nutrition must take into account sensory and oral changes. Decreases in all the senses, particularly in the **taste** buds that affect perception of salty and sweet tastes, may affect appetite. Xerostomia, lack of salivation,

affects more than 70% of the elderly. Also, denture wearers chew less efficiently than those with natural teeth.

- Other organ changes may occur. Insulin secretion is decreased, which can lead to carbohydrate intolerance, and renal function deteriorates in the 40s for some people.
- Cardiovascular changes may occur. Reduced sodium intakes become important, as **blood pressure** increases in women over age 80 (but, interestingly, it declines in older men). Serum cholesterol levels peak for men at age 60 but continue to rise in women until age 70.
- Immunocompetence decreases with age. The lower immune function means less ability to fight infections and malignancies. **Vitamin** E, **zinc,** and some other supplements may increase immune function.

What are the psychosocial changes that occur during aging?

These are the following:

- Depression, the most common cause of unexplained weight loss in older adults, occurs in approximately 15% of adults over age 65, with a much higher incidence in those living in extended-care facilities.
- Memory impairment caused by various types of **dementia, Alzheimer's disease,** or other neurological diseases rises dramatically, with half of all persons over age 85 affected. Weight loss and improper nutrition are potential problems.
- Alcohol abuse is often unreported, especially since approximately one-third of alcoholics age 65 years or older begin drinking later in life. Excessive alcohol intake (over 15% of total calories) increases morbidity and mortality, and leads to both physical and psychosocial problems.
- Social isolation becomes more common because of declining income, health problems, loss of spouse or friends, and assistance needs. All of these may affect appetite and possibly nutritional status.

Basic Energy and Nutrient Needs

Calorie requirements decrease with age, although individuals vary greatly depending on their activity level and health status. Diets that fall below 1,800 calories a day may be low in protein, calcium, **iron,** and **vitamins,** so should feature nutrient-dense foods.

Protein needs of healthy older adults are the same as for other adults, with 0.8 to 1 g of protein per kg of body weight recommended. Most older people without debilitating disease eat adequate protein, but those with infections or severe disease may become protein-malnourished due to increased protein requirements and poor appetites. Seniors do better eating more complex **carbohydrates,** which increase liber, vitamins, and **minerals,** and help with insulin sensitiv-

ity. Lactase-treated milk or fermented dairy products are suggested if lactose intolerance develops. Because caloric needs drop and heart disease is so prevalent, reducing total dietary fat and especially the amount of saturated **fats** is advised.

Mineral deficiencies are uncommon in older adults, and recommended amounts are the same or similar to those for younger adults. Iron-deficiency anaemia related to nutrition is rare, and more likely due to **blood** loss. Of the vitamins, vitamin D intakes are often lower than recommended, especially among homebound or institutionalized people who lack sun exposure (the most accessible source of vitamin D). The antioxidant vitamins, vitamin E, carotenoids, and **vitamin** C, continue to receive attention because of their potential to improve immune function and ward off disease. Requirements for **riboflavin,** vitamins B_6 and B_p, and zinc are increased in the elderly. However, needs for **vitamin** A decrease.

Aging and Dysphagia

The incidence of **dysphagia,** or difficulty in swallowing, increases with age. Dysphagia results from conditions such as stroke, Alzheimer's or **Parkinson's disease, multiple sclerosis,** or physiological changes such as loss of teeth or poorly fitting dentures. Inadequate dietary intake as a result of dysphagia can lead to weight loss, **dehydration,** and nutritional deficiencies. The American Dietetic Association has developed Level 1 through Level 4 dysphagia diets, which provide varying textures and liquids based on the severity of the condition.

Aging and Fluid Balance

Dehydration is the most common cause of fluid and electrolyte disturbances in older adults. Reduced thirst sensation and fluid intake, medications such as diuretics and **laxatives,** and increased fluid needs during illness contribute to dehydration. Adequate water-intake guidelines are 1 ml water/kcal energy consumed (for example, 1.8 L for an 1,800-calorie intake), or 25-30 ml/kg of weight for most individuals.

Aging and Skin Integrity

Skin breakdown is a common problem, particularly in bedridden or immunologically impaired people. The most common skin breakdown is the **pressure ulcer,** which occurs in 4% to 30% of hospitalized patients and 2% to 23% of residents of skilled-care **nursing homes.**

Pressure ulcers are graded or staged to classify the degree of tissue damage. Those with more serious Stage II to Stage IV ulcers have increased nutritional needs. Protein needs increase to 1-1.5 g protein/kg, caloric needs increase to 30-35 kcal/kg, and 25-35 cc fluid/kg is recommended.

Aging and Malnutrition

While most elderly people maintain adequate nutritional status, institutionalized and hospitalized older adults are at higher risk for malnutrition than individuals who are living independently. Cancer cachexia, the weak, emaciated condition resulting from cancer, accounts for about half of malnutrition cases in institutionalized adults.

Two common forms of malnutrition are protein-calorie malnutrition, in which the person appears ill-nourished; and protein malnutrition, in which an overweight person may have depleted protein stores. Nutrition support may involve higher protein and calorie amounts, nutritional supplements such as Ensure, or enteral tube feedings that provide nutrient solutions into the GI tract.

Assessment of Nutrition in the Elderly

The following are used to assess nutritional needs:

- A thorough medical history, **physical examination,** and dietary history can provide a general picture of the individual's nutritional status. Lab values also provide valuable information.
- Weight evaluation may be recommended. Normal weight status guidelines include a BMI of 21 to 27 (BMI = weight in pounds x 704.5/ht(in) squared) or Ideal Body Weight +/-10%. Guidelines for significant weight loss include 10% weight loss in six months, 5% in one month, or 2% in one week.
- Dehydration evaluation involves physical assessment (poor skin turgor, dark urine, flushed skin), and assessment of recent fluid and food intake. High laboratory levels of blood urea nitrogen (BUN), albumin, serum sodium and serum osmolality can indicate dehydration. Hyponatraemia is a very important parameter to be addressed, as most of them are on salt-restricted diet due to heart, kidney and high BP.

SECTION 8

Community Nutrition

"We are guilty of many errors and many faults, But, our worst crime is abandoning children, neglecting the fountain of life. Many of the things we need can wait, the child cannot. Right now is the time his bones are being formed, his blood is being made and his senses are being developed. To him, we cannot answer 'Tomorrow.' His name is 'Today'."

—Gabrieta Mistral (Chile)

8.1 NFHS Survey Reports and Summary

Malnutrition contributes to 60% of the 10 million deaths globally that occur every year among children under five years of age. Its contribution to child deaths is even higher during first six months of life, when mortality is highest. Breastfeeding can avoid this within one hour, and exclusive breastfeeding during first six months. A **National Nutrition Education Programme** (NNEP) **needs to be launched** to create a climate of nutritional awareness at different levels so that the invisible and silent emergency of malnutrition could be addressed effectively.

Nutrition Scenario

There has been a **conquest of nutritional deficiencies** - Florid nutritional deficiency syndromes like pellagra, beriberi, scurvy, kwashiorkor have disappeared; famines are no more; severe malnutrition among preschoolers has reduced appreciably; and nutritional status of adults has improved significantly. But still the high levels of malnutrition continue to influence morbidity and mortality rates in the country.

The **challenge** that still remains include

- High malnutrition levels particularly in women and children
- Undernutrition
- Micronutrient malnutrition

- Emerging diet-related diseases,
- High mortality rates—IMR, U5MR and MMR
- Inadequate Access to health care, immunization etc.

Undernutrition is defined as the outcome of insufficient food intake (hunger) and repeated infectious diseases. Undernutrition includes underweight (being underweight for one's age), stunted (being too short for one's age), wasted (being dangerously thin), and micronutrient malnutrition (being deficient in vitamins and minerals). As per NFHS-2 (1998-99) 47% of children under three years are **underweight,** 45.5% **stunted** and 15.5% **wasted.** India fares poorly even among the South East Asian countries, occupying the third place from the bottom with only Nepal and Bangladesh faring worse than India. The prevalence of **low birth weight** continues to be about 30% for last three decades. Low birth weight babies are more likely to die because of neonatal infections and under-nutrition. **Chronic energy deficiency** in adults is 39% in females and 37% in males (NNMB 2002).

Infant and young child feeding practices (IYCF) in India are far from optimal. According to NFHS-2, exclusive breastfeeding falls rapidly from 72% at one month to 20% at six months. Only about 16% babies in India start breastfeeding within one hour.

Other indicators are also disappointing. For instance, complementary feeding practices are very poor: only 33% of children aged between 6-9 months are given solid mushy foods (NFHS-2). Comparison with NFHS-1 highlights that we are able to stop the decline in breastfeeding. However, achieving the national goals will require massive action and resources. Results from a recent study from 49 districts (2003), with data for about 9000 mothers, also show a dismal picture. This is where malnutrition begins as *The Development Paradox.*

The paradox is that while India now in the front ranks of fast growing global economies, with a vibrant economic growth rate of around 7%, nearly 30% of the global burden of child deaths is borne by India. Economic growth is, at best, a slow and undependable way of eliminating child undemutrition. While income poverty in India is reduced to 26%, underweight prevalence in children under three years remains at 47% (NFHS-2 1998-99), reinforcing the argument that economic growth is a necessary, but not sufficient condition for improvements in young child survival, nutrition and development. Young babies are dependent on parents or other caregivers for their food intake.

Emerging Trends

The recently released NFHS-3 data (2005-2006) for 5 states is showing some positive improvements in nutritional status (underweight prevalence in children 0 to under 3 years) in Orissa, Chhattisgarh and Maharashtra, but is revealing stagnation of some key health and nutrition parameters in states like Gujarat and Punjab. There are discernible and largely consistent improvements in these three

states as related to infant and young child feeding, nutritional status and in infant mortality, (as also seen in SRS data). However, in the states for which data has been released, there is no improvement in nutritional anaemia in young children and this continues to be a major problem in women in the reproductive age group, especially adolescent girls and pregnant mothers. Vitamin A supplementation coverage also has shown a decline. Immunization coverage also remains a cause for concern.

Existing Nutrition and Health Interventions and Gaps

Malnutrition being a multi-faceted problem requires a multisectoral approach for its prevention and control. A number of direct and indirect nutrition interventions are being undertaken by different sectors of the government with a view to promote nutrition of the people. Some of the **direct nutrition interventions** are as under:

Ministry of Women and Child Development

- Integrated Child Development Services (ICDS) Scheme
- Nutrition Programme for Adolescent Girls (NPAG)
- Nutrition Advocacy and Awareness Generation Programmes of Food and Nutrition Board (FNB)
- Follow up action on National Nutrition Policy (1993)
- National Institute of Public Co-operation for Child Development (NIPCCD)

Ministry of Health and Family Welfare

- Iron and Folic Acid Supplementation of pregnant women
- Vitamin A Supplementation of children of 9-36 months age group.
- National Iodine Deficiency Disorders Control Programme

Department of Elementary Education and Literacy

- Mid Day Meal for primary school children

Some of the **indirect interventions** include:

Department of Agriculture and Cooperation

- Increased Food Production
- Horticultural Interventions

Food and Public Distribution

- Targetted Public Distribution System
- Antodaya Anna Yojana
- Annapurna Scheme

Rural and Urban Development

- Food for Work Programme

- Poverty Alleviation Programmes
- Safe Drinking Water and Sanitation
- National Rural Employment Guarantee Scheme

Ministry of Health

m National Rural Health Mission

- Integrated Management of Neonatal and Childhood Illnesses (IMNCI)
- Various Public Health Measures

Department of Elementary Education and Literacy

m Sarva Siksha Abhiyaan

- Adult Literacy Programme

Ministry of Women and Child Development

- Various women's welfare and support programmes.

The gigantic problem of malnutrition requires concerted efforts from different partners. There are innumerable **gaps** in the existing nutrition related interventions, some of which are as under:

- There is no national system of nutrition monitoring, mapping and surveillance. District level disaggregated data not available from NNMB and NFHS surveys.
- Nutritional concerns are not adequately reflected in the policies and programmes of the government.
- Intersectoral coordination mechanism, crucial for nutrition promotion, is inadequate.
- ICDS coverage is not universal. States are not able to allocate resources for supplementary nutrition.
- Tribal areas, food-scarce districts, chronically drought-prone rural and tribal hamlets have inadequate access to nutrition and health services.
- Nutrition education and awareness generation is weak. Except the programmes undertaken by the limited infrastructure of Food and Nutrition Board of this Department, it is not being undertaken by other nutrition and health interventions.
- Coverage of children under three years in vitamin 'A' prophylaxis programme is low.
- Iron and Folic Acid Supplementation for pre-school children, adolescent girls, pregnant women and lactating mothers is inadequate.
- Availability of iodized salt at household level had declined after the lifting of the ban on sale of non-iodized salt in October 2000.
- Micronutrient malnutrition is not being addressed in a comprehensive manner.

- Food Fortification programmes are negligible. Supplementation, dietary diversification and horticultural interventions are inadequate.
- Nutrition Programme for Adolescent Girls (NPAG) is only in 51 districts.
- Optimal infant and young child feeding practices (breast feeding and complementary feeding) need aggressive promotion.
- There are significant gaps found through a national assessment, in both policy and programmes on infant and young child feeding.

The vision of the National Nutrition Policy (1993) of achieving optimum nutrition for all, although is the ultimate goal to be achieved, all efforts need to be directed towards this aim if the country wants to accelerate its economic growth and development.

- **"Malnutrition Free India"** is the goal whose time has come and is the vision for National Nutrition Policy for the next decade. India's strong institutional and human resource base is capable of bringing about a transformation. The success will depend on the full involvement of all concerned sectors from Centre and State Governments, national institutions, community and social organisations, and women's groups in implementing the mandate of the National Nutrition Policy.
- With the constitution of the National Nutrition Mission under the chairmanship of the Hon'ble Prime Minister, there is an uncommon opportunity of mobilising all stakeholders towards achieving the goal of malnutrition-free India. The country has nation-wide Integrated Child Development Services scheme, Reproductive and Child Health Programme and Universalisation of Primary Education. All these development infrastructure need to be utilised to carry forward the task of meeting the goal of the National Nutrition Policy in the next decade.
- Good nutrition is the material basis for human resource development of a country or a community; nutrition is an issue of survival, health and development for current and succeeding generations. Children born underweight have impaired immune function and increased risk of diseases such as diabetes and heart disease in their later life.
- Malnourished children tend to have lower IQ and impaired cognitive ability, thus affecting their school performance and then the productivity in their later life. Such a vicious cycle of nutrition and development is not widely acknowledged and has very weak influence in policy making. **It has to be realised that the nutritional health in all age groups represents a national economic asset.**
- **For the first time the** X **Five Year Plan had set goals for infant and young child feeding indicators and reduction of undernutrition in children including micronutrient malnutrition.** Many of the X Five Year Plan Goals are yet to be realized. Keeping in view the mandate of the Millennium Development

Goals and the unmet goals of X Five Year Plan, the **following National Nutrition Goals are recommended for the XI Five Year Plan** to be met by 2012. The state-specific goals would need to be identified accordingly.

Reduce the prevalence of underweight in children under 5 years to 20%.
Eradicate the prevalence of severe undernutrition in children under five years.
First hour breastfeeding rates to increase to 80%.
Exclusive breastfeeding rates to increase to 90%.
Complementary feeding rate at six months to increase to 90%.
Reduce prevalence of anaemia in high-risk groups (infants, pre-school children, adolescent girls, pregnant and lactating women) to 25%.
Eliminate vitamin A deficiency in children under 5 years as a public health problem and reduce sub-clinical deficiency of vitamin A in children by 50%.
Reduce prevalence of Iodine Deficiency Disorders to less than 5%.

- **The achievement of the National Nutrition Goals for the XI F ive Year Plan would require a multi-pronged action on various issues. A number of policy decisions at the macro and micro level would be required at center and state levels to achieve the goals.**

Strategic Recommendations for the XI Five Year Plan

- **Articulating malnutrition as number one public health problem in the country.**
- **Greater emphasis on Nutrition Action by Health Sector at all levels**
- **Strengthening Nutrition in Medical, paramedical, indigenous and other systems of medicine like Ayurveda, Yunani, Siddha and homeo (AYUSH) and Agriculture**
- **Clinic/beds for severely malnourished children at PHC, CHC and district hospitals**
- **Establishing Nutrition Information System in the country**

A **five-pronged strategy** had been advised to accelerate the programmes to overcome micronutrient deficiency in the country. These related to (i) **Dietary Diversification** Awareness Creation concerning the Ministries of Health & Family Welfare, Women & Child Development and Information & Broadcasting. This is needed to be attempted through intensive IEC; (ii) **Nutrient Supplementation** concerning the Ministries of Health & Family Welfare, Women & Child Development and Department of School Education and Literacy. This could be achieved through biannual campaigns for administration of vitamin A to children between 6 months to 6 years, providing iron and folic acid supplements to children from 6 months to 2 years and to adolescent girls 10-19 years, administering iron tablets to all pregnant and lactating women and by emphasizing breastfeeding of infants up to 6 months under the NRHM Project Implementation Plans; (iii) **Food Fortifi-**

cation involving the Ministries and Department of Health, Food Processing Industries, Food & Public Distribution, Consumer Affairs, Finance, Panchayati Raj and State Governments. This would be achieved by providing the composition and quantity of fortificants to meet the micronutrient needs in different foods, by providing incentives to industry for production and identifying appropriate channels for distribution; (iv) **Horticulture Intervention** involving the Ministry of Agriculture for the supply of seeds, extension and storage support; and, (v) **Public Health Measures** involving the Ministries and Departments of Health & Family Welfare, Women & Child Development, Commerce, Rural Development and Urban Development. This would require streamlining procedures of procurement and supply, building institutional capacity in organizations for monitoring and mapping micronutrient deficiencies and provision of safe drinking water and sanitation.

To achieve the above goals, nutritional security needs to be prioritized during the XI Plan with the provision of earmarked funds. Estimated costs per day per beneficiary would be around 16 paise and with 50% cost sharing with the states, the total expenditure will be around Rs. 500 to 600 crores per month. The issues involved need a high degree of inter ministerial coordination necessitating a mission mode for achieving synergies for the best delivery of facilities. Panchayati Raj bodies would also need to be made partners in this endeavour. On these issues being decided. Planning Commission will be approached for funds for this mission.

8.2 National Nutrition Policy (NNP) and Programmes

CHILD WELFARE PROGRAMMES

The existing child welfare programmes include the various national nutritional supplementation programmes, the Universal Immunization Programme (UIP), the ICDS scheme and the Child Survival and Safe Motherhood (CSSM) Programme.

The Minimum Needs Programme, the 20 Point Programme, the Integrated Rural Development Programme, the Urban Basic Services for the Poor (UBSP), Adult Literacy Mission and all projects aimed at environmental sanitation, safe water supply and overall socioeconomic advancement contribute directly or indirectly to child survival and welfare.

The integrated management of childhood illness (the IMCI) and the integrated management of pregnancy and childbirth (IMPAC) are the new evolving strategies that integrate the vertical programmes.

1. Nutritional Supplementation Programmes in India

The Government of India has initiated several nutrition programmes in order

to prevent and control major nutritional disorders. These programmes have helped to bridge the calorie gap among children belonging to the low socio-economic status.

a) *Mid-Day Meal Programme:* This school feeding programme was initiated in 1962 in order to provide one-third of the daily requirement of calories and half the daily requirement of protein and other essential nutrients to the primary school children. The Co-operative for American Relief Everywhere (CARE), the Education Department and the Applied Nutrition Programme facilitated this school feeding programme initially. Currently it is run by the local administration.

b) *Supplementary Nutrition Programmes:* This special feeding programme for the pre-school children (6 months to 6 years) and pregnant and nursing mothers was started in 1970 for the weaker sections of the population. **The supplementary feeding supplies 300 kcal and 10 to 12 g protein per child per day. The mothers receive daily 500 kcal and 25 g protein.** This feeding is for 300 days in a year. The Urban Special Nutrition Programme (USNP) and ICDS are examples. The Ministry of Social Welfare is in charge of this programme. In places where Balawadis are there to give preprimary education, similar feeding is given by the Development Department.

c) *Specific Nutrient Supplementation:* This includes Iodine Deficiency Disorder (IDD) Control Programme, Vitamin A Prophylaxis Programme and Anaemia Control Programme. The supply of iodized salt in endemic areas, supply of five megadoses of vitamin A concentrate to children at an interval of six months and supply of iron-folic acid (IFA) tablets to expectant and nursing mothers as well as children are being implemented weekly; one IFA tablet to adolescent girls is yet another programme.

2. **The Universal Immunization Programme (UIP)**

This programme was initiated in India in 1985 and was dedicated to the memory of the late Prime Minister Mrs Indira Gandhi. It aims at 100 per cent immunization of all infants against six major vaccine-preventable killer diseases, namely, tuberculosis, diphtheria, pertussis, tetanus, poliomyelitis and measles before these children celebrate the first birthday. This programme has helped to improve the immunization status of children and to strengthen cold chain facilities.

3. **The Child Survival and Safe Motherhood (CSSM) Programme**

This was initiated in 1992 with preset goals by integrating all the available services and resources into a new package. It has child survival and safe motherhood components. The child survival components of CSSM include neonatal care, immunization, vitamin A deficiency control and prophylaxis, diarrhoea control and oral rehydration therapy, acute respiratory infection control and therapy, iron and folic acid supplementation, deworming and

others like limiting, timing and spacing of births. The impact of this programme is yet to be evaluated.

4. **The Integrated Child Development Services (ICDS) Scheme**

This is an excellent ongoing programme aimed at total development of the child. It is now renamed as **'Integrated Mother and Child Development (IMCD) Services'.** The ICDS programme was started in **1**975 with specific objectives. The main objectives of the programme are to improve the nutritional status of children (0-6 years), to reduce morbidity and mortality rates, to reduce school dropout rates through early stimulation programme for children (3-6 years old), to provide the environmental conditions necessary for mental, physical and social development of the child, to enhance mother's capacity to look after the health and nutritional needs of the child through nutrition and health education and to achieve effective coordination at the policy making and implementation level among Government departments to promote child development.

The services are delivered through a network of **Anganwadis.** Anganwadi is a community centre. The word literally means a courtyard. In project areas, one Anganwadi is established for every 1000 population.

The programme is supported by the Human Resource Development Department, Social Welfare Department and Health Department. The package of services available includes health check-up, immunization, referral service, supplementary nutrition, non-formal education for children (3-6 years), nutrition and health education for women, treatment of minor ailments and home visits.

The average calorie intake among one to five years old children was estimated to be 810 kcal per day as against the requirement of 1200 kcal. Food supplementation is planned to bridge this calorie gap. The food supplements were supplied initially by the CARE, the World Food programme and the State Government. The Central Government meets most of the other recurring expenditure. Now it is run by the local administration. Food supplementation is available for children and pregnant and lactating mothers. There is self-selection to some extent as better-off families do not enroll in the Anganwadi centers of ICDS. Food supplementation has been found to be effective in improving the nutritional status.

Growth monitoring is one of the major activities in the Anganwadi centre. The Anganwadi teacher is expected to weigh all the children regularly and record on the growth chart and thus make it visible to the mother. The Anganwadi teacher is also expected to report monthly the data concerning the percentage of children with normal nutritional status and those with various grades of PEM. However, *apart from the numbers, very little stress is given to which child moved up or down the nutrition ladder.*

Growth monitoring and growth charts form an integral part of the child survival programmes like **GOBIFFF (Growth monitoring, ORT, Breastfeeding, Immunization, Family Welfare, Female education, and Food supplementation).** These were expected to begin a revolution. The concept of under-five clinic was started by David Morley in Africa and growth chart was an integral part of the services. However, weighing is often done as a ritual and considered an end in itself rather than means to an end. The exact age of the child is often not known and hence plotting of weight at the growth chart becomes difficult. *Most of the weighing is done in the 3-6 years age group rather than in the younger age.* The problem of malnutrition and stunting are more prevalent in the younger children who are difficult to reach. They are not adequately covered in any programme. Growth monitoring has now become an expensive luxury than a necessity. Growth monitoring can help only if it is properly done and it is communicated to the mother. It can be of use only if it is accompanied by appropriate interventional strategies. In the ICDS, this is linked with nutritional supplementation. All the children are expected to get around 300 kcal per day and the severely malnourished are expected to get double the ration. But the services are underutilized in some areas and in some other areas, the supply is not sufficient to meet the demands. Nutrition and health education component is often neglected due to lack of time, lack of training and lack of suitable communication materials. The centres also lack storage facility leading to wastage of food materials.

Stimulation is mainly aimed at children above three years of age in the form of non-formal education. Lack of participation of the younger child is the main drawback in the ICDS programme. The crucial period in child development, i.e., below three years, is not made use of in imparting stimulation. These children are not sufficiently reached and are difficult to reach. There is nothing at the centre suitable for them. Often the food supplement is neither suitable for them nor available to them. Some centres arrange 'take home food' in addition to 'on the spot feedings'.

Community participation is always expected as the Anganwadi teacher herself is a member of the community. But often the community remains as a passive recipient. ICDS still remains a government programme rather than a community programme. Active efforts are now being made to integrate the programme of sanitation and safe water supply with ICDS to give more emphasis on family welfare services and to implement the programmes directed towards economic uplift especially among women with the help of the existing network. This programme has helped to improve immunization status and nutritional status of children. **Recently teenage clinics and adolescent care and counselling are also being integrated.** These clinics are run on Saturdays.

5. **Reproductive and Child Health Programme (RCH)**

As early as 1952, during the First Five Year Plan, India pioneered in the field of 'Reproductive and Child Health' by introducing a population control policy. Reduced fertility improves child survival and improved child survival reduces fertility. Target oriented approach was introduced in the Third Five Year Plan (1961-66) and the MTP Act was introduced in the Fourth Five Year Plan (1968-74). The National Health Policy was launched during the Sixth Five Year Plan (1980-85). The CSSM Programme was implemented in 1992. 'Target free approach' came into existence in demand-driven, high-quality programme. A target-free approach leading to a 'self-generated target' is the cornerstone of this programme.

a) **Definition:** RCH is defined as a state in which, people have the ability to reproduce and regulate their fertility; women are able to go through pregnancy and childbirth safely; the outcome of pregnancy is successful in terms of maternal and infant survival and well-being and the couples are able to have sexual relationship free of the fear of pregnancy and contracting diseases.

b) Components

i. **Child survival**
- Essential NB care
- Breastfeeding and complementary feeding
- * Immunization
- Vitamin A prophylaxis
- Integrated management of childhood ilnesses strategy (IMCI); like ARI, diarrhoea, VPDs, helminthiasis, anaemia, IDD etc.
- Referral services, setting up of FRUs
- Developmental surveillance and early intervention

ii. **Safe motherhood**
- Antenatal care including immunization and nutrition
- Essential obstetric care
- Emergency obstetric care including referral
- Postnatal care

iii. **Adolescent care and counselling**
- Family life education
- Teenage care and counselling
- Adolescent nutrition and anaemia prophylaxis
- Personal and menstrual hygiene
- Prevention of reproductive tract infections (RTI) and sexually transmitted infections (STI)

iv. **Others**
- Safe abortion
- Management of infertility

- Prevention and management of RTI and STIs
- Contraception—timing, limiting and spacing of births
- Screening and management of cancers of cervix and breast
- Information, education and communication **(IEC)**

v. **Community participation**

c) **Implementation:** The multipurpose health workers, supervisors and medical officers are given training and the subcentres are redesignated as RCH centres. The District Immunization Officer (DIO) is given charge of the programme and is given the designation of District RCH Officer.

6. **National Rural Health Mission (NRHM)**

Framework of implementation of the NRHM and the Indian Public Health Standards (IPHS) should clearly reflect nutrition inputs through action at village level, cluster of 4-5 villages and 30 villages, and at the Block level. Nutrition inputs can only be ensured through building of infant and young child feeding support centres/clinics at the cluster level and block level by skilled female workers. Accredited Social Health Activists (ASHA) at village level should actively engage in promotion of breastfeeding and provide support within one hour at birth to begin breastfeeding. The project implementation plans at district level under NRHM should include details on micronutrient malnutrition control programmes, namely Iron & Folic Acid Supplementation, Vitamin A Supplementation and National IDD Control programmes. The District Media Officers under NRHM need to be sensitised towards the problem of malnutrition including micronutrient malnutrition and equip to undertake nutrition orientation, awareness generation and IEC activities on various nutritional issues. At present the job responsibility of the District Media Officers do not include nutritional issues.

In Kerala, NRHM has taken up the issue of pain and palliative care among those with cancer and other chronic diseases. National Urban Health Mission with Urban Social Health Activists (USHA) is yet another programme to be put into popularity in urban areas.

7. **National Nutrition Monitoring Bureau** (NNMB)

NNMB undertakes diet and nutrition surveys periodically in 8-10 states in the country and projects the state level scenario for these states. NNMB continues to function in a project mode under ICMR since 1972 resulting in heavy turnover of staff, and poor staff strength. The National Nutrition Policy adopted by the Government in 1993 mandated that nutrition monitoring and surveillance system should be established in the country and that the National Institute of Nutrition entrusted with this responsibility and to be made accountable to MWCD for nutrition surveillance. In order to achieve this, it is imperative that NNMB is established in all states and UTs and is assigned the task of establishing a system of nutrition monitoring, mapping and surveillance. NNMB should also undertake district level food and nutrition surveys

including survey on prevalence of diet-related chronic diseases every five years in the country to enable area-specific planning and programming using disaggregated data.

- **Management Information System** (MIS) of various services under NRHM and ICDS also provides useful information on the coverage under various programmes. The MIS of various services need to be improved and utilised to serve as an important source of information for enabling nutrition monitoring, mapping and surveillance.
- **National Family Health Survey** (NFHS) undertaken by International Institute of Population Sciences are coordinated by the Ministry of Health & Family Welfare and are undertaken at an interval of 5-6 years. In fact this is the only source of nationwide data on under-nutrition among children under 5 years and anaemia among women and children. In fact, anaemia was added at NFHS-2 stage (1998-99) while NFHS currently in progress has taken a little more indicators on nutrition.
- **District Level Health Survey** (DLHS) covers all districts in a phased manner and is an important resource for projecting district level nutrition and health scenario in the country. Presently very few indicators on nutrition are included under the DLHS.

8. **Other National Programmes**

Many other programmes contribute directly or indirectly to child welfare.

a) *Minimum Needs Programme:* This was introduced during the fifth Five Year Plan (1974-1978). It includes rural health, rural water supply, rural electrification, adult education, nutrition, environmental health of slums and houses for the landless.

b) *20 Point Programme:* In addition to the Five Year Plans, the Government initiated the 20 Point Programme in 1975 and restructured it in 1986. The restructured programme constitutes the charter for the country's socio-economic development. Attack on rural poverty, provision of safe drinking water, health for all, two-child norm, expansion of education, housing, improvement of slum and protection of environment are the eight points in the programme related directly or indirectly to health.

c) *Integrated Rural Development Programme (IRDP):* It is a composite programme for rural development and economic uplift. This programme has helped in providing self employment.

d) *The Urban Basic Services Programme for the Poor (UBSP):* It is a composite programme for development and economic uplift of the poor people in the urban areas. Family units are established and the services are rendered with the help of **resident community volunteers (RCVs).** Family units will submit micro-plans for overall improvement and employment. The various micro-plans will be combined in the mini-plans and various mini-plans will be combined in the city-plans. Recently these

have been reorganised as the **'Kudumbasree' programme.**

e) *Applied Nutrition Programme (ANP):* It is a programme aimed at improvement in food production, distribution and nutrition education. This programme has created nutrition awareness and the habit of having kitchen garden in houses, institutions and schools.

f) *Adult Literacy Mission:* The adult literacy campaign has successfully implemented the first stage of adult education and also the second stage aimed at tribal and coastal population. The formation of family units and continuing education are planned in the third stage.

g) *India Population Project:* It is a programme aimed at population control and improved quality of life.

h) *Breastfeeding Promotion Network of India (BPNI):* Recognising the importance of infant and young child nutrition (IYCN) for promoting nutrition and health of the people, an exclusive institute for promoting IYCF would be required. There is no institute or NGO specialized in this area other than BPNI. The BPNI has a network in States and Districts with paediatricians working honorarily for the cause of IYCN.

BPNI with its national network needs to be adopted by the government to serve as an institute for promoting IYCN in the country.

9. **International Programmes**

WHO, United Nations International Children's Emergency Fund (UNICEF), World Bank, World Food Programme, Food and Agricultural Organization (FAO), United States Agency for International Development (USAID), CARE are some of the international agencies that initiate and support the child welfare programmes, especially in the developing countries including India. These agencies provide financial and technical support to implement various educational and nutritional programmes that help in child survival and child development.

THE NATIONAL NUTRITION POLICY

Government of India adopted the National Nutrition Policy in 1993 in view of widespread malnutrition and its effect on health and survival of population, especially young children. The policy advocates a comprehensive, integrated and intersectoral strategy for alleviating the multifaceted problem of malnutrition and achieving the optimal state of nutrition for the people.

Direct Interventions

Short-term Measures

1. Nutrition intervention for specially vulnerable groups
   - Expanding the safety net, particularly Integrated Child Development Services (ICDS) program
   - Appropriate behavioural changes among mothers

- Reaching the adolescent girls
- Ensuring better coverage of expectant women, for better health of women and reducing the incidence of low-birth weight babies

2. Fortification of essential foods with iron and iodine
3. Popularization of low cost nutritious food
4. Control of micronutrient deficiencies amongst vulnerable groups

Indirect Policy Instruments

Long-term Institutional and Structural Changes

1. Food security
2. Improvement of dietary pattern through production and demonstration
3. Policies for affecting income transfers
   Improving the purchasing power
   Streamlining the public distribution system
4. Land reforms
5. Health and family welfare
6. Basic health and nutrition knowledge
7. Prevention of food adulteration
8. Nutrition surveillance
9. Monitoring of nutrition program
10. Research into various aspects of nutrition
11. Equal remmuneration for men and women
12. Better communication strategies
13. Minimum wages administration
14. Community participation

A National Plan of Action on nutrition was developed in 1995 and the National Nutrition Goals for the year 2000 were identified as follows:

- Reduction in moderate and severe malnutrition among preschool children by half
- Reduction of incidence of low birth weight babies to less than 10 per cent
- Elimination of blindness due to vit A deficiency
- Reduction of iron deficiency anaemia among pregnant women to 25 percent
- Universal iodization of salt for reduction of iodine deficiency disorders to 10 per cent
- Giving due emphasis to geriatric nutrition
- Production of 250 million tonnes of food grains

THE INTEGRATED MANAGEMENT OF NEONATAL AND CHILDHOOD ILLNESS (IMNCI) STRATEGY

1. **Introduction**

The WHO/UNICEF algorithm for **Integrated Management of Neonatal Child-**

hood Illness (IMNCI) is specifically suited for developing countries like India due to the following reasons:

a. Co-existence of morbidities in the same child is a rule rather than an exception.
b. Various illnesses can be assessed with good sensitivity and specificity.
c. The IMNCI algorithm is diagnostically and therapeutically superior to the **vertical-disease specific algorithms'.**
d. The IMNCI algorithm has a provision for preventive services like immunization, breastfeeding counselling, complementary and supplementary feeding, etc.
f. There is an around 20% possibility of availing missed opportunities in preventive services.

We have generally shown slow enthusiasm in adopting foreign-made programmes. So, it is worth examining the background of the IMNCI strategy. This strategy is the product of dissatisfaction with the vertical-disease specific control programmes like those targetted at diarrhoea, acute respiratory infections, etc. The health worker and professionals often come across children whose symptoms have overlapping causes or for whom a single diagnosis is not appropriate. For example, cough and tachypnoea may be caused by pneumonia, as well as by anaemia or malaria. A very sick child may have pneumonia and meningitis or septicaemia. Misclassification of an illness is extremely common.

2. **Components**

The IMNCI strategy includes three components:

a. Improving the case evaluation and management skills of the health workers and professionals by a ‘guideline-driven programme of training’.
b. Providing essential drug supplies for effective management of childhood illnesses.
c. Optimizing the family and community practices in child health, especially care-seeking behaviour.

The IMNCI strategy can attract national and international investment into many child-oriented initiatives.

The IMNCI check list I and II for young infants of one week to 2 months of age and 2 months to 5 years (Tables 8.1 and 8.2) has been evaluated in various centres and the results have been compared with a ‘gold standard diagnosis' based on complete investigations and vertical-disease algorithms. The studies have shown that an integrated, rather than a vertical one, approach to diagnosis makes a sense because two-thirds of the children have two or more co-existent morbidities. The IMNCI algorithm covers more than 90% of the recorded illnesses and preventive services. It also suggests additional benefits like improved provider morale and quality care, rational drug usage and better health outcomes. It ensures thorough examination, adminis-

tration of the first dose of treatment onsite and better referral links. At present, many children who die do not reach a referral centre on time. The IMNCI strategy will improve maternal and community recognition of illness and an appropriate care seeking behaviour. But, there are certain challenges to the IMNCI strategy which need adaptations to suit our needs.

3. **Modifications/Challenges**
   a. As the diagnosis of pneumonia is based on a fast respiratory rate, there may be an overdiagnosis of pneumonia and an underdiagnosis of asthma and bronchiolitis. In settings where asthma is common and drugs are available to manage it, modified guidelines may be developed, for example, response to bronchodilator therapy.
   b. The IMCI strategy takes into account infants and children from 1 week to 5 years of age. Early neonatal period up to 1 week is not included. As infant mortality has fallen, the proportion of the Neonatal Mortality Rate (NMR) in relation to Infant Mortality Rate (IMR) has increased the NMR accounts for 65% or more of infant deaths. In places where 'Safe Motherhood and Mother Baby Package Programmes' are not in full swing, the IMNCI should include the early neonatal period also. Neonatal sepsis and other common morbidities should be included.
   c. The case definition of malaria in endemic areas based on unexplained fever alone leads to an overdiagnosis of malaria, which has serious consequences like antimicrobial resistance. Experts foresee a health calamity with widespread resistance to chloroquine and an increasing resistance to pyrimethamine-sulphadoxine (PSD). They recommend a combination therapy for malaria instead of a single drug therapy as in tuberculosis and AIDS. Artemisinin may be combined with chloroquine and PSD. Ablood film result may be incorporated into the algorithm at least in selected areas.
   d. In places where the prevalence of HIV is increasing, the IMNCI training and algorithm should incorporate HIV testing, preventive strategies like when to test, how to treat, how to prevent vertical mother to child transmission, etc.
   e. There should be awareness that there are other important, yet neglected areas of child health like environmental pollution, safe drinking water, accidents, mental health, emotional deprivation, childhood disabilities, child abuse and neglect, etc.

The general view is that it is better to allow it to start and not to piggy-back the programme and drown it. Costing and sustainability are the other aspects to be considered. Pre-service training in educational and other institutions will reduce the burden of training. In service training can be shared with partners like international agencies and other voluntary agencies.

Table 8.1 The IMNCI Checklist - I

MANAGEMENT OF THE SICK YOUNG INFANT AGE 1 WEEK UP TO 2 MONTHS

Name:_____ Age:_____Weight:_____ Kg. Temperature:_____ °C
Ask: What are the infants problems?_____ Initial visit?_____ Follow-up visit?_____

Assess	**Classify**
Check for possible bacterial infeciton ■ Has the infant had convulsions? ■ Count the breaths in one minute_____breaths per minute. ■ Repeat if elevated_____ Fast breathing? ■ Look for severe chest indrawing. ■ Look for nasal flaring. ■ Look and listen for grunting. ■ Look and feel for bulging fontanelle. ■ Look for pus draining from the ear. ■ Look at the umbilicus. Is it red or draining pus? ■ Does the redness extend to the skin? ■ Fever (temperature 37.5 °C or above feels hot) or low body temperature (below 35.5 °C feels cool) ■ Look for skin pustules. Are there many or severe pustules? ■ See if the young infant is lethargic or unconscious. ■ Look at young infant's movements. Less than normal?	

contd.

Does the young infant have diarrhoea?

■ For how long?_____ days.

■ Look at the young infant's general condition.
Is the infant:
Lethargic or unconscious?
Restless and Irritable?

■ Look for sunken eyes.

■ Pinch the skin of the abdomen. Does it go back:
Very slowly (longer than 2 seconds)?
Slowly?

Then check for feeding problem or low weight

■ Is there any difficulty in feeding? n Determine weight for age. Low_____Not low_____
Yes_____ No_____

■ Is the infant breast fed? Yes_____ No_____
If Yes, how many times in 24 hours?_____

• Does the infant usually receive any other foods or drinks? Yes_____No_____
If Yes, how often?

■ What do you use to feed the child?

contd.

If the infant has any difficulty in feeding; is feeding less than 8 times in 24 hours, is taking any other food or drinks, or is low weight for age AND has no indictions to refer urgently to hospital:

ASSESS BREAST FEEDING:

- Has the infant breast fed in the previous hour

If infant has not been fed in the previous hour, ask the mother to put her infant to the breast. Observe the breastfeeding for 4 minutes.

- Is the infant able to attach? To check attachment, look for:
  - Chin touching breast Yes_____ No_____
  - Mouth wide open Yes _____ No _____
  - Lower lip turned outward Yes_____ No_____
  - More areola above than below the mouth Yes_____ No_____
- No attachment at all _____ Not well attached _____ Good attachment_____
- Is the infant suckling effectively (that is, slow deep sucks, sometimes pausing)? Not suckling _____ Not suckling effectively _____ Suckling effectively
- Look for ulcers or white patches in the mouth (thrush).

Check the young infant's immunization status

BCG DPT 1 DPT 2

OPV 0 OPV 1 OPV 2

Return for next immunization on: Date:

Assess other problems:

Table 8.2 The IMNCI Checklist - II

MANAGEMENT OF THE SICK CHILD AGE 2 MONTHS UP TO 5 YEARS

Name:_____ _Age:„ _ Weight: _ **. Kg. Temperature:_____ °C**

Ask: What are the infants problems? _ Initial visit? .Follow-up visit?.

Assess		Classify
Check for general danger signs Not able to drink or breastfeed Vomits everything Convulsions	Lethargic or unconscious	General danger sign present? Yes____ No____ Remember to use danger sign when selecting classifications.

Does the child have cough or difficult breathing?

- **For how long?_____ days**
- **Count the breaths in one minute.**
  ____breaths per minute. Fast breathing?
- **Look for chest indrawing.**
- **Look and listen for stridor.**

contd.

Does the chiW have diarrhoea?

■ For how long?____ days
* Is there blood in the stool?

Look at the child's general condition.
Is the child:
Lethargic or unconscious?
Restless and irritable?
Look for sunken eyes.
Offer the child fluid. Is the child:
Not able to drink or drinking poorly?
Drinking eagerly, thirsty?
Pinch the skin of the abdomen. Does it go back:
Very slowly (longer than 2 seconds)?
Slowly?

Does the chitd have fever?

(by history/feels hot/temperature 37.5 °C or above) Yes _No_
Decide malaria risk: High___ Low_____

■ For how long?____days
■ If more than 7 days, has fever been present every day?
■ Has child had measles within the last 3 months?

Look or feel for stiff neck.
Look for runny nose.
Look for signs of Measles:
- Generalized rash and
- One of these: cough, runny nose, or red eyes.

contd.

If the child has measles now or within the last 3 months:

- Look for mouth ulcers.
  If yes, are they deep and extensive?
- Look for pus draining from the eye.
- Look for clouding of the cornea.

Does the child have an ear problem? Yes____ No ____

- Is there ear pain?
- Is there ear discharge?
  If yes, for how long?____ days

- Look for pus draining from the ear.
- Feel for tender swelling behind the ear.

Check for malnutrition and anaemia

- Look for visible severe wasting.
- Look for palmar pallor.
  Severe palmar pallor? Some palmar pallor?
- Look for oedema of both feet.
- Determine weight for age.
  Very low____ Not very low_____

contd.

Check the child's immunization status

BCG DPT 1 DPT2 DPT 3

OPVO OPV1 OPV 2 OPV 3 Measles

Return for next immunization on:

Date:

Assess the child's feeding if child has anaemia or very low weight or is less than 2 yr

- Do you breastfeed your child? Yes___ No____
  If yes, how many times in 24 yours?____times.
  Do you breastfeed during the night? Yes___No____
- Does the child take any other food or fluids? Yes___No_____
  If yes, what food or fluids?_____
  How many___ times per day? What do you use to feed the child?_____
  If very low weight for age: How large are servings?_____
  Does the child receive his own serving?____Who feeds the child and how?_____
- During this illness, has the child's feeding changed? Yes___ No____
  If yes, how?

Assess other problems:

4. The IMNCI check list
   The IMNCI check list I is used for the management of the sick young infant, 1 week to 2 months of age. The IMNCI check list II is used for the management of the sick child, 2 months to 5 years of age. These are given in tables 8.1 and 8.2.

8.3 Immunization

Immunization is the most important cost-effective strategy that has saved millions of children. There are various types of vaccines.

Table 8.3 Types of Vaccines

Type of antigen	Examples
Live bacteria, attenuated	BCG, Ty21a
Live virus, attenuated	OPV, MMR, Varicella
Inactivated bacteria	Pertussis, Whole cell killed typhoid
Inactivated virus	IPV, Rabies, HAV
Toxoid	DT, TT, Td
Capsular polysaccharide	Typhoid Vi, HIB, Meningococcal, Pneumococcal
Viral subunit	HbsAg
Bacterial Subunit	Acellular pertussis

Table 8.4 National Immunization Schedule (UIP)

Age	Vaccines
Birth	BCG, OPV_0, HBV*** (for institutional deliveries)
6 weeks	$DTP_{1(}$ OPV,, HBV, (& BCG, if not given deliveries
10 weeks	DTP_2, OPV_2, HBV_2
14 weeks	DTP_3, OPV_3, HBV_3
9 months	Measles
16-24 months	DTP, OPV
5-6 years	DT*
10 years	**"["I"****
16 years	TT

contd.

For pregnant women

Early in pregnancy	TTi or booster
One month after TTi	TT_2

*A second dose of DT vaccine should be given at an interval of one month if there is no clear history or documented evidence of previous immunization with DTPw

**A second dose of TT vaccine should be given at an interval of one month if there is no clear history or documented evidence of previous immunization with DTPw, DT or TT vaccines.

***HBV X 3 doses can be given along with DPT primary doses or as 0, 1, 6 month schedule in older children. HBV given at birth may be counted as zero dose

Table 8.5 IAP Immunization Time Table (Revised in 2008)

Age	Vaccine	Route
Birth	BCG OPVo Hepatits B_t	Intradermal Oral Intramuscular
6 weeks	DTPWj/DTPaj OPV/ + IPV 1 Hepatitis B_2 Hibj	IM Oral IM IM
10 weeks	$DTPw_2/DTPa_2$ OPV_2 + IPV 2 Hib_2	IM Oral IM
14 weeks	$DTPw_3/DTPa_3$ OPV_3 + IPV 3 Hepatitis $B_{3„}$ Hib_3	IM Oral IM IM
9 months	Measles	Subcutaneous
15-18 months	DTPwBj/DTPA Bj OPV_4 HIB Bj MMR1	IM Oral IM Subcutaneous
2 years	Typhoid #	IM

contd.

5 years	DTPw B_2/DTPa B_2 IM OPV5, MMR/ Oral
10 years	Td/Tdap#/TT, HPV^A IM
16 years	Td/Tdap#/TT IM

* OPV alone if IPV cannot be given
**3rd dose of Hepatitis B can be given at 6 months of age, HBV given at birth may also be considered as zero dose and 3 doses can be given along DTP 1,2 & 3
#Revaccination every 3 years
/N In females 3 doses at 0, 1 & 6 months
Pregnant Women 2 doses of TT

Table 8.6 Vaccines that can be given after discussion with parents

Age	Vaccine	Route
> 15 months of age	Varicella vaccine#	Subcutaneous
> 18 months of age	Hepatitis A vaccine^	Intramuscular
> 6 weeks of age	♦Rotavirus, Pneumococcal Conjugate Vaccine (PCV7)	IM

* < 13 years of age: 1 dose; > 13 years of age: 2 doses at 4-8 weeks interval
2 doses at 6-12 months interval
* Rotavirus vaccine - 2-3 doses as per brand at 4-8 week interval 6 week to 6 month old
PCV 7-3 primary doses at 6, 10 & 14 weeks followed by a booster at 15-18 months.

Newer & Other Vaccines

1. **Varicella (Chickenpox)**
   - It is a live attenuated Oka strain having 1000 plaque-forming units (PFU).
   - Can be used after 12 months, is usually given along with MMR at 15 months. It is also available as MMRV.
   - Up to 13 years, single dose SC/IM and after that 2 doses.
   - It can be used for post-exposure prophylaxis in 72 hr of contact.
   - Protect from light and use in 30 min of reconstitution. Breakthrough varicella is mild disease occurring in vaccinated children. Varicella zoster immunoglobulin (VZIg) can be given in high-risk cases on exposure (125 units/10 kg) and in neonates if mother gets varicella 5 days prior to and 2 days after delivery.

2. Hepatitis A Vaccine-HAV
   - It is formalin-inactivated and aluminium hydroxide adjuvanated, HM 175/ GBM strains cultured on HDC.
   - It is given after 18 months of age in doses at 6 months interval.
   - It can be used for post-exposure prophylaxis in 10 days of contact.
3. Pnuemococcal vaccines
   - 85 different serotypes of *Streptococcus pneumoniae* cause infections from 2 months of age onwards, like otitis, pneumonia, meningitis etc.
   - 1 and 5 account for one-third of diseases in India; others are 4,6,7,14 & 19.
   - The heptavalent conjugated pnuemococcal vaccine (PCV 7- Prevenar) contains 4, 6B, 9V, 14, 18C, 19F & 23 F, adjuvanated using diphtheria toxoid and aluminium phosphate.
   - It is given as 3 doses, 0.5 ml IM at 1 month interval + 1 booster at 18 months. 23 valent unconjugated polysaccharide vaccine (1,2, 3,4,6B, 7F, 8,9N, 9V, 10A, 12F, 14,15B, 17F, 18C, 19a, 19F,20,22F,33F) is also available for use > 2 yr of age for asplenia, sickle cell diseases, nephritic syndrome etc. It does not reduce otitis.
   - Route SC/ IM
4. Meningococcal vaccine
   - *Neisseria meningitides* accounts for 30-40% of meningitis < 15 yr of age.
   - 12 serotypes, especially A, B, C, Y, W135 cause infections. Unconjugated monovalent (C), bivalent (A & C) and tetravalent (A, C, Y, W135) can be used after 2 years of age.
   - Route SC/IM.
5. Japanese encephalitis - JE Vaccine
   - JE is currently given as a campaign to all 1-15-yr-old in endemic areas, single dose of Sa-14-14-2 live neuro attenuated JE vaccine. More than 95% protection using single dose.
   - Previously used inactivated mouse brain/hamster kidney cell derived vaccines are not in use currently.
6. Influenza vaccines
   - Orthomyxovirus A, B & C influenza viruses cause infections in pandemic proportions.
   - 'A' usually undertake mutations like antigenic shift and drift.
   - Each vaccine is useful only for 1 year. Trivalent vaccine of A (H_1N_1 & H^NJ and C is in use.
   - H_1N_1 swine flu and avian flu has resulted in recent pandemics.
7. Rabies vaccine
   - Anti-rabies vaccine (ARV) is used for primary prophylaxis in high-risk individuals (0, 7, 28 days IM) and for post-exposure prophylaxis as per the class of bite.

- Purified chick/duck embryo vaccine (PCEV & PDEV) and HDC/ purified Vero cell rabies vaccines (HDCV & PVRV) are in use currently instead of the nerve tissue vaccine.
- Rabies immunoglobulin (RIG), equine (40 U/kg) and human (20 U/kg) are available for Class II bites, half to be infiltered at the bite as such or diluted with equal volume N. saline to be used in multiple wounds and rest to be given as IM.
- Equine should be used only after test dose 0, 3, 7, 14. 28 and 90 is the original schedule of injection IM, to be discontinued if animal is alive after 10 days of the bite. Day 90 is optional, but be given in immunodeficient, extremes of ages etc.
- **Intradermal Rabies Vaccine (IDRV)**
- 0.1 ml ID on each arm on day 0, 3, 7 and 28 to a total of 0.8 ml is recommended for use in India - e.g., Rabipur & Verorab. Vaxirab (PDEV) and Rabivax (HDCV) are yet to be approved as IDRV.

Table 8.7 Post-exposure prophylaxis for bites

Category	Contact/Bite	Treatment
Class I	Licks on intact skin, touching/feeding animal	Observation if history is reliable
Class II	Minor scratches, abrasions and no bleeding, licks on broken skin	Initiate, may be discontinued if animal alive at 10 days/found rabies negative if euthanised
Class III	Multiple bites and bleeding wounds, contamination of mucosa with saliva	RIG + treatment as Class II

8. **Cholera vaccine**
   - Oral live vaccine is available for children 2 years and above
9. **Rotavirus vaccine**
   - Given < 6 months of age in endemic areas, there used to be a fear of causing intussusception.
   - Rotavirus vaccine - 2-3 doses as per brand at 4-8 week interval, 6 week to 6 month old
10. **Human Papilloma Virus (HPV) vaccine**
   - In females 3 doses at 0, 1 and 6 months, starting at 10 years of age
   - For primary prevention of HPV infection via intercourse and thereby carcinoma cervix

11. **Dengue fever vaccine & HIV vaccine** - under development & trial
12. **Yellow fever vaccine**
    - Live attenuated chick embryo vaccine SC for children above 9 months of age.

Cold Chain

It is the system of ensuring optimum temperature of vaccines from manufacturer to the beneficiary.

Table 8.8 Sensitivity of vaccines to freezing (T Vaccines should not be frozen)

Vaccines damaged by freezing	Vaccines that can be frozen without harm
DTP	OPV
DT	Measles/MMR
TT	BCG (before reconstitution)
Hepatitis B	
HIB	
Td	
Typhoid (whole cell killed vaccine)	
Hepatitis A	

Storage of Vaccines

All vaccines are safe at temperatures between 2-8°C for at least 6 months. If a freezer is available, it should be used for storage of OPC and Measles/MMR vaccines. The latter vaccines should be kept frozen at -20°C when stored for the long term - at this temperature these vaccines have a shelf-life of 2 years. Even these vaccines, however, can be kept at 2-8°C for shorter periods, e.g., 6-12 months for OPV and 18-24 months for measles.

DTP/DT/TT/Typhoid (T-series vaccines), HB and Hep A vaccines should never be frozen. The freezing point for adsorbed DTP vaccine is between -5 to 10°C. Freezing time depends on the number of doses in the vial and the temperature. It takes about 110 to 130 minutes at -10°C. The "Shake Test" can be used to determine if the vaccine has been frozen at any time: shake the vial so that the sediments, if any, are completely mixed; wait for 15 minutes; if the vaccine is not uniformly mixed or the sediments/flocculations are still found settled at the bottom, the vaccine is likely to have been frozen at some time. Such vials should be discarded.

At a temperature of 2-8°C, DTP/DT/TT/Hepatitis B/Hepatitis A/Varicella and HIB vaccine have a shelf-life of 24 months. Diluents should be stored at 2-8°C - these should not be used beyond 4 hours. Vaccines should be transported only in cold boxes or vaccine carriers - vacuum flasks should never be used for this purpose. It is recommended that vaccines should not be stored for more than 3 months at the district level and for more than 1 month at the level of primary health center. No vaccines should ever be stored at the sub-centers. Expiry dates of vials should be checked once a week. While storing vaccines follow **the "First-In-First-Out (FIFO)'' and "First to Expire First-Out (FEFO)"** rules.

During shipment and transportation, temperature and time sensitive monitor marks are used to check the cold chain. **Dial thermometers** are used to monitor the temperature in refrigerators/ice-lined refrigerators (ILRs) and are kept in every unit. Alcohol stem thermometers are much more sensitive and accurate than dial thermometers and can record temperatures from -50°C to +50°C. These can be used for deep freezers and ILRs. Temperature monitoring should be done twice a day in the case of ILRs and deep freezer and once a day in the case of walk-in coolers, where vaccines are stored in bulk for longer periods. A break in the cold chain is indicated if temperature rises above +8°C or falls below +2°C in the case of ILR and other refrigerators and above -18°C in the case of deep freezers.

The Vaccine Vial Monitor (VVM) is a time and temperature sensitive colorued label that provides an indication of the cumulative heat to which the vial has been exposed. VVMs were first introduced on OPV vials supplied to UNICEF and WHO in 1996. The VVM warns the end user when exposure to heat is likely to have degraded the vaccine beyond an acceptable level. It is used especially for temperature monitoring of OPV, which is the most thermolabile of all vaccines. If the VVM indicates proper storage of OPV in a given center, it can be presumed that other vaccines would also be potent. VVMs increase the flexibility in handing of vaccines in the field.

Interpretation of the colour change of VVM is as follows:

1. Inner square is lighter than outer circle: If the expiry date has not passed, vaccine can be used.
2. Inner square matches colour of outer circle or is darker than outer circle: Vaccine should be discarded.

The VVM provides information about the heat exposure of the vial over a period of time - the change in colour is gradual but irreversible. However, there may be other factors which can also affect the potency of vaccine (e.g., storage beyond the expiry date) and these may not be reflected in the VVM. Loss of potency depends upon the degree of temperature elevation as well as duration of exposure. Tetanus toxoid is the least heat-sensitive vaccine.

The manufacturer's instructions regarding shelf-life of a given vaccine must be rigorously followed. Measles vaccine loses viability quickly if kept at

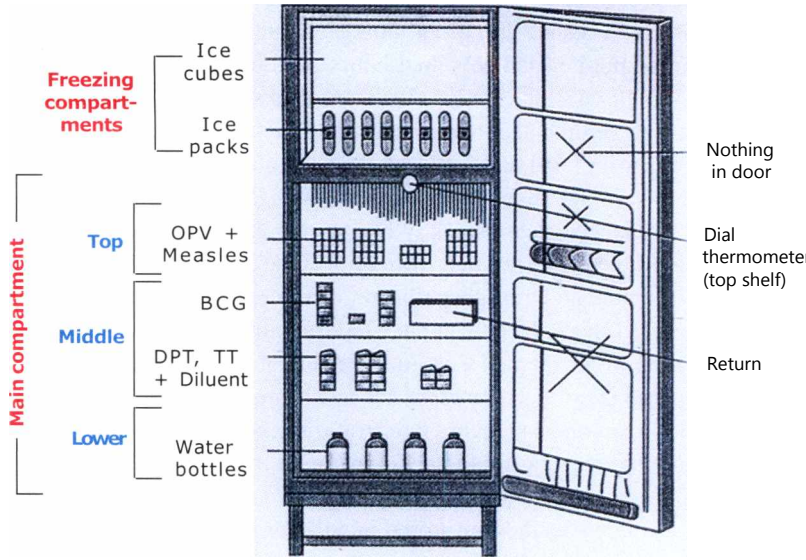


Fig. 8.1 Equipment: at health centre-refrigerator *(showing vaccines stored correctly)*

temperatures above 40°C. Reconstituted measles vaccine should be kept protected from heat and light during an immunization session and the left-over discarded after the session. OPV would lose viability if kept at 22-25°C for more than a day. Opened vials of OPV, however, may be used in subsequent sessions at a given health facility if it has been preserved at 2-8°C. OPV vials used in the field setting or an outreach facility or during a pulse immunization session must be discarded at the end of the day. Vaccine vials (single/multi-dose) should be gently shaken before use to ensure that the contents are clear and not granular or flaky. Vaccine vials should not be taken out to the field more than 3 times - after that these are best discarded irrespective of whether these have been opened or not.

8.4 Infestations and Infections

I. HELMINTHIASIS

The usual habitat of the helminths is the intestine, but some cause systemic invasion and some get encysted in the tissues when tiny larvae escape pulmonary filtration. Mechanical obstruction to the gut caused by worm balls and encysted tissue lesions may call for surgical intervention. The common helm-

inths are **nematodes and cestodes.** Most of them are **obligatory parasites** that need appropriate hosts for completion of life cycle, but *Strongyloides* are **facultative parasites** that can complete the life cycle in soil and they infect man if opportunity arises.

1. Nematodes

Ascariasis, ancylostomiasis, strongyloidiasis, enterobiasis, trichuriasis, visceral and cutaneous larva migrans, trichinosis and filariasis are the nematodal infections.

a) *Ascariasis (roundworm): Ascaris lumbricoides* is the most common of all helminths. Eggs laid by the mature female worm, about 2 lakhs per day, are transmitted by the faecal-oral route. Eggs may survive in soil for two years. The larvae that hatch out in the intestine pass through the pulmonary phase and finally settle down in the small intestine. During the pulmonary phase, it may cause **'Loeffler like syndrome'** characterised by cough, haemoptysis, eosinophilia and urticaria. In the intestine it may cause vague abdominal pain, worm mass obstruction, intestinal, pancreatic or biliary obstruction and worm migration through mouth, nose, ear via perforated ear drum, umbilicus via patent vitellointestinal duct or abdominal surgical wound via intestinal wounds. *Deworming prior to elective surgery is mandatory especially in children as fever and intake of certain drugs are known to produce worm migration.* **Worm mass obstruction is common in the ileo-caecal junction.** It is treated by IV fluids and Ryle's tube aspiration followed by anthelmintics like piperazine or pyrantel pamoate once the worm mass disappears. Rarely surgical resection may be needed if the mass does not disappear or if there is obstruction to blood flow. Round worms may cause vague periumbilical pain and may lead to up to 25% nutrient wastage.

b) *Ancylostomiasis (hookworm): Ancylostoma duodenale* and *Necator americanus* **suck up to 0.2 ml of blood per worm per day** and make the patients anaemic. Eggs laid by mature female worm, about 9,000-30,000/day hatch out in the soil. They penetrate the skin of the feet and pass through the pulmonary phase and finally settle down in the gut. **Ground itch** at the site of penetration, Loeffler-like syndrome, anaemia, hypoalbuminaemia and peptic ulcer-like abdominal pain are the manifestations. The worms may survive up to a decade in the gut and cause **chronic ulcer-type pain.** Larvae may remain dormant for a year or so and then may mature. Thus manifestations may occur even after leaving endemic areas. Hookworm larvae may move up to a height of 10 cm along grass and plants and may orally infect children who are in the habit of plucking and biting grass. Treatment is by pyrantel pamoate for 3 days, mebendazole for 3 days or albendazole. Bephenium hydroxy naphthoate and tetrachloroethylene were used earlier.

c) *Strongyloidiasis:* ***Strongyloides stercoralis*** larvae infect by penetrating the skin of the feet and pass through the pulmonary phase and finally settle in the gut. The mature female worms pass eggs that immediately hatch out into larvae that are detected in freshly passed stool. The eggs may rarely hatch out in the intestine before passage and may **autoinfect the same individual.** Dermatitis at the site of penetration, **'larva currens'** with moving edges in the perianal skin, Loeffler like syndrome, protein-losing enteropathy, diarrhoea alternating with constipation, malabsorption, segmental ileus and duodenitis are the manifestations. In immunosuppressed hosts, it may cause hyperinfection and fatal **disseminated strongyloidiasis** due to systemic invasion of larvae that hatch out in the gut. This is called autoinfection. Severe malnutrition and steroid therapy may also predispose to systemic invasion. Encephalopathy, Gram-negative sepsis and shock may occur along with disseminated strongyloidiasis. Transplacental transfer also has been reported. Thiabendazole or albendazole for three days is effective.

d) *Enterobiasis (pinworm):* ***Enterobius vermicularis*** eggs that are ingested hatch out in the gut and inhabit the caecum and appendix. The female worms migrate to perianal region especially in the night to lay eggs. The eggs may remain viable for 20 days. It can cause intense itching, sleeplessness, vulvitis, nocturnal enuresis, salpingitis etc. **Autoinfection** by finger-mouth transmission and **retroinfection** due to migration of larvae from perianal region back to caecum may occur. Family treatment and retreatment at 2-4 weeks interval, up to 3 courses, are often necessary. Mebendazole, pyrantel and piperazine for 7 days are found beneficial. Mebendazole, one tablet per week for 12 doses, is also found effective. It is the most successful parasite on earth which affects all types of people irrespective of their socioeconomic status all over the world.

e) *Trichuriasis (whipworm):* ***Trichuris trichiura*** eggs that are ingested hatch out in the gut and inhabit the colon. They suck blood up to 0.005 ml/worm/day and cause chronic bloody diarrhoea, **prolapse rectum** and anaemia. It comes in the differential diagnosis of intestinal polyposis and ulcerative colitis. Mebendazole and albendazole are effective. In resistant whipworm colitis, **mebendazole retention enema** is tried, 2 tablets per day in 100 ml N. saline for 3 days.

f) *Tissue nematodes:* These are the visceral and cutaneous larva migrans, trichinosis and filariasis.

   i) Cutaneous larva migrans is due to animal hookworm larvae like ***Ancylostoma braziliense, A. caninum*** etc., and other larvae like ***N. americanus, A. duodenale*** and ***S. stercoralis.*** It causes intense itching at the site of penetration like feet, leg, hand, buttock etc., and localize at the dermal-epidermal junction and move at a rate of 1-2 cm per day and produce serpiginous or bullous tracks up to 15-20 cm. They eventually resolve

after several weeks. Treatment includes local ethyl chloride spray, thiabendazole or albendazole suspension and oral thiabendazole or albendazole.

ii) Visceral larva migrans is caused by dog and cat parasites like ***Toxocara canis, T. catis*** and ***T. leonina.*** The eggs ingested by man hatch out in the gut and seed the liver, lung, brain, eye, kidney, heart etc., and evoke mass lesions and cause fever, hyper-eosinophilia. hepatosplenomegaly and lymphadenopathy. They are active up to 18 months and then resolve. Treatment is by low dose diethylcarbamazine for 3 weeks under cover of corticosteroids. Albendazole is also beneficial.

iii) Trichinosis is caused by ingested larvae of ***Trichinella spiralis*** through undercooked meat. The larvae that escape in the gut migrate to the striated muscles, eyes, heart, brain etc., and get encysted. They live for 5-10 years and act as **space-occupying lesions.** Treatment includes aspirin, corticosteroids, and thiabendazole, mebendazole or albendazole.

iv) Filariasis is caused by ***Wuchereria bancrofti*** and ***Brugia malayi.*** Filarial larvae are introduced into humans by mosquito bite. It produces lymphangitis, lymphadenitis, hydrocele, elephantiasis and tropical pulmonary eosinophilia. Treatment includes diethylcarbamazine for 3 weeks or single-dose ivermectin.

2. Cestodes

Cestodes include the various tapeworms. Adult worms inhabit the gut and interfere with host nutrition. Some of the larvae also cause dissemination and encystment in various organs.

a) *Taeniasis (tapeworm):* ***Taenia saginata*** (beef tapeworm), ***Taenia solium*** (pig tapeworm) and ***Diphyllobothrium latum*** (fish tapeworm) infect man through undercooked flesh. Larvae mature in the gut and adult worms measure up to 4-10 metres and live for many decades. Eggs that are passed out mature into infective larvae called cysticerci in the appropriate intermediate hosts. Passage of segments, vague abdominal pain, malnutrition and megaloblastic anaemia due to B_p deficiency produced by ***D. latum*** are the manifestations. Niclosamide and praziquantel are effective. A saline purge is advised after niclosamide to prevent cysticercosis.

b) *Cysticercosis:* It is encystment of larvae in brain, muscle etc., due to ingestion of eggs of ***T. solium.*** The encysted larvae may live up to 5 years and cause **space-occupying lesions.** In the brain, it is called neurocysticercosis. Dead cysticerci may calcify. Praziquantel and albendazole are effective. Surgical treatment may be needed in some. **Non-vegetarians who handle pigs may also get this.**

c) *Echinococcosis (hydatiddisease):* ***Echinococcus granulosus*** (dog tapeworm) eggs passed out by dogs when ingested by man hatch out into larvae and

seed the liver, lung, brain etc., and develop into **hydatid cysts.** These cause mass lesions. Rupture of the cysts may produce anaphylaxis. Injection of aqueous iodine or concentrated saline into the hydatid cyst will help to kill the embryos prior to surgery. Praziquantel x 15 days, albendazole x 28 days and mebendazole x 21 days are being tried. **Hydatid disease can affect even vegetarians through their contact with dogs.**

B. Diagnosis

a) *Stool microscopy:* The fertilised ova of roundworm have bile-stained coat and are elliptical in shape. The eggs of hookworm are non-bile stained; after one hour of preparation of slides the eggs may hatch out and larvae may be seen. These may be mistaken for the larvae of *Strongyloides stercoralis* which are seen in freshly passed motion. The eggs of pinworm are planoconvex and are also demonstrable on adhesive cellophane tapes pressed against perianal regions. Whip worm eggs are barrel shaped, brown, thick walled with knob-like ends. 10% of cases with negative stool test pass worms on treatment. Characterisation of passed out segments help in the diagnosis of tapeworms.

b) *Duodenal aspirate:* It is used to look for *Stercoralis* larvae and *Giardia.*

c) *Serology and skin tests:* These are useful in cysticercosis and hydatid cyst (Casoni test). Haemagglutination tests and ELISA are also useful in hydatid cyst and visceral larvae migrans.

d) *Imaging:* X-ray, ultrasound, CT and MRI scans are useful in space-occupying lesions of cysticercosis and hydatid cyst. Plain X-ray abdomen may show cigar-cut appearance in cases of worm mass obstruction.

e) *Tissue biopsy:* It is helpful in visceral larva migrans and trichinosis.

4. Anthelmintics

Anthelmintics are predominantly luminal, systemic or combined action drugs.

Luminal action drugs are selected when helminths in the gut are targeted. Systemic action drugs are selected when disseminated and encysted ones are targeted. Anthelmintics are vermifugal, vermicidal, ovicidal or larvicidal.

a) *Piperazine salts:* They cause **flaccid paralysis** of the roundworms and pinworms and lead to expulsion of the worms by peristalsis. They are readily absorbed and many produce ataxia, neurotoxicity etc. Dose is one ounce at bed time for two consecutive days. For children, the dose is 50-75 mg/kg x 2 days. **It is not safe in those with** CNS **disorders.**

b) *Pyrantel pamoate:* It causes **spastic paralysis** of the worms and leads to expulsion. They are less absorbed and are effective against roundworm, hookworm and pinworm. Dose is 600-800 mg as a single dose in adults. It may be repeated for 3 days in hookworm. For children, the dose is 11 mg/kg single dose. It is mutually antagonistic to piperazine. **It is not safe in liver disorders.**

c) *Levamisole:* It causes **muscle paralysis** and expulsion. It is effective against roundworm, hookworm and tropical eosinophilia. The dose is 150 mg single dose. In children, 50 mg single dose is given. **It may produce agranulocytosis. It should not be given with milk.** It is now used as an immunomodulator.

d) *Imidazoles (mebendazole, albendazole, thiabendazole etc.):* **Mebendazole** kills the parasite by inhibiting the **microtubular transport** system. It is broad spectrum and is effective against mixed infestations. **It may lead to worm migration.** Dose is 100 mg twice daily x 3 days for adults and children above 2 years. It is not usually given below 2 years. 400-600 mg thrice daily x 21 days is tried in hydatid disease. **Albendazole** blocks glucose uptake and kills the parasite. It is broad spectrum and is effective in multiple infestations. **It may also cause worm migration.** Dose is 400 mg single dose. It may be repeated for 3 days in strongyloidiasis and 400 mg twice daily x 28 days cycles at 2 week intervals x 3 cycles may be tried in hydatid disease. Albendazole is teratogenic. In children 1-2 years old, the dose is 200 mg single dose. **Thiabendazole** is not easily available. Dose is 25 mg/kg twice daily for 3-14 days. It may **produce Stevens-Johnson syndrome, neurotoxicity** and **cholestasis.**

e) *Niclosamide:* It inhibits anaerobic **phosphorylation** and is effective in tapeworms. Liberated ova of *T. solium* may cause cysticercosis and hence purging is needed after 2 hours. Dose is 1500 mg (3 tablets) single dose in adults. For children, the dose is 500 mg up to 2 years, 1000 mg up to 10 years and 1500 mg above 10 years.

f) *Praziquantel (cysticide/biltricide):* It **leads to loss of intracellular calcium and** dislodgement of the worms and larvae and subsequent phagocytosis of larvae. It is effective in tapeworms, cysticercosis, hydatid disease and schistosomiasis. Dose is 3000 mg in 3 divided doses x 15 days under cover of steroids in cysticercosis in adults. It may be repeated after 3 months. For children, the dose is 50 mg/kg/day in 3 divided doses x 15 days.

g) *Diethylcarbamazine:* It is a piperazine derivative. It is the drug for filariasis, pulmonary stage of larvae and eosinophilia. Dose is 3-6 mg/kg/day x 21 days. It may cause rebound eosinophilia.

h) *Bephenium hydroxy naphthoate (Alcopar):* It causes paralysis of worms and leads to expulsion. It is effective against hookworm. Dose is 2.5 g as single dose. It may be repeated after 1-2 days. In children half the dose is given. It is not readily available.

i) *Ivermectin (Mectizan):* **It causes spastic paralysis of worm** and is effective in filariasis, roundworm, pinworm, whipworm and strongyloidiasis. Dose is 0.2 mg/kg single dose. It may be repeated after 6-12 months.

5. Practical Guidelines in Treatment

Anthelmintic therapy is now integrated into primary health care and mebendazole

is made available nationwide through primary health centres. The limitations of mebendazole are multidosing, occasional roundworm migration and restarting of doses if interrupted. In children from low socioeconomic status previously untreated and having a heavy roundworm load, **a two-step approach** is better for fear of roundworm migration. **An initial roundworm treatment with pyrantel or piperazine and retreatment with mebendazole in view of mixed infestation are preferable.** Most worms mature in 3 months time and so retreatment quarterly is beneficial in children who are prone for reinfection. In others, half yearly treatment may be enough. As pinworms mature in *2-A* weeks, quarterly retreatment at 2-4 weeks interval and family treatment may be needed. The options for pinworm are mebendazole, albendazole, pyrantel, ivermectin or 7 days piperazine.

6. Prevention

Targeted mass therapy will decrease the reservoir of infection and interrupt the chain of transmission when coupled with sanitary measures. It is taken up in the Child Survival and Safe Motherhood (CSSM) programme. With the exception of *Strongyloides stercoralis,* helminths do not multiply in the same host, but have to come out and infect the same or a new host. **Periodic deworming of puppies, proper cooking of meat and slaughter hygiene are also to be ensured.**

II. SCARAB IASI S/DUNG BEETLE DISEASE

Passage of live dung beetles per rectum is called scarabiasis. Children belonging to low socioeconomic status are the usual victims.

1. **Agent**

   These beetles belong to the family Scarabaedae and subfamily Coprinae. More than one species have been noted in the same patient. *Onthophagus bifaciatus*, *(). unifaciatus, O. orientalis, O. damage, O. cervus, Caccobius*

   vulvanus etc., are the usual species. Several cases have been encountered and the first report from Kerala was in 1976 (Joseph A, Scarabiasis in Kerala. *Iud J of PH.* 1976,20:90-94).

2. **Route of entry**

   The usual habitat of the beetles is human or animal excreta present in open places. They are good fliers. Children who are unclothed, sleeping on the floor and those with faecal smell are usually affected. Due to the faecal smell and warmth, the beetles crawl up through the anus and are subsequently passed. There have been reports of passing 8-10 beetles per day. The current view is that the **beetles enter as adults** and they do not lay eggs or develop within the intestine. Eggs, larva, pupa and young adults have not been detected, but only mature adults are seen.

3. **Clinical features**

   One to five year old children are usually affected. It may be asymptomatic or

may manifest with loss of appetite, increased appetite, fever or diarrhoea. Passage of live adult beetles per rectum, several ones at a time on several occasions is the most common presentation. Parents often bring a few beetles.

4. **Treatment**

A warm soap and water enema is found beneficial in most cases to get rid of the beetles. But there may be some persistent cases. Some entomologists recommend the use of 2-5% gamma benzene hexachloride (BHC) along with the enema in such cases. BHC will not be absorbed in significant amounts. Care has to be taken to prevent reinfection.

III. TUBERCULOSIS

Tuberculosis is a chronic illness that is usually associated with malnutrition. It may manifest as an acute paediatric emergency in the form of TB meningitis, miliary TB etc. The sad tale about TB is that even 115 years after the discovery of TB bacilli in 1982 by Robert Koch and 35 years after **launching the National TB Control Programme (NTCP) in 1962,** it is still remaining unconquered. The prevalence of infectious TB cases is 4/1000 and this infectious pool is infecting the others, up to 40% of the population. Even though case finding is progressing, lack of unified and supervised treatment strategy and poor compliance are defeating the goal. Hence the intensive unified and supervised chemotherapy for TB was planned. It is being undertaken in selected districts since 1994. Chronic TB leads to malnutrition and malnutrition leads to flaring up of TB.

1. **Diagnosis**

In the revised strategy, TB is divided into pulmonary and extrapulmonary TB. Pulmonary TB is categorised into smear (AFB) positive and smear negative patients. Gastric aspirate for AFB is useful in children.

In children the diagnosis is based on epidemiological evidence of contact with TB, clinical features, Mantoux (Mx) positivity and X-ray evidences. X-ray diagnosis is often unreliable and hence a course of antibiotic should always be given and the patient should be reevaluated. Mantoux test also has only a limited value as a negative test does not rule out TB and a positive test more than 10 mm may be seen without active disease and in case of atypical mycobacteria. A false-negative test can occur in the incubation period, in severe forms of TB like miliary TB. in PEM and following immunosuppression, steroid therapy, viral infections etc. **A Mantoux-positive child less than 5 years of age with signs and symptoms should always be treated** and others should be observed for development of signs and symptoms. The signs and symptoms in extra-pulmonary TB depend on the site of involvement.

2. **Terminologies**

a) *New case:* A patient who has not taken anti-TB treatment (ATT) for more than one month before. Anybody who had ATT for more than one month is likely to have drug resistance.

x

b) *Relapse:* A patient declared cured by a physician is again found AFB postive.

c) *Smear positive failure case:* Remaining AFB positive after 5 months of ATT or after completing chemotherapy.

d) *Defaulter:* A patient who interrupted ATT for 2 months or more.

e) *Chronic case:* AFB positive after completing retreatment.

f) *Drug resistance:* Primary resistance is when drug-resistant bacilli infect a new person. Secondary resistance is when drug-resistant bacilli emerge in a patient due to inadequate chemotherapy with respect to regimen, dosage, duration or regularity.

3. **Drug and regimens**

Do not start ATT until a **firm diagnosis is** made and always document the reasons for starting ATT. In the unified revised strategy, the patients are divided into categories **I. II** and III (Table 8.9). **Short course chemotherapy (SCC)** is now accepted for adults. It may be effective in children also. In **directly observed therapy (DOT),** the health staff administer the drugs to the patients 3 times a week. When this is not feasible or acceptable, self administered therapy (SAT) by the patient is allowed. This is given daily as the compliance is uncertain. The commonly used drugs are INH (H), rifampicin (R), pyrazinamide (Z), ethambutol (E) and streptomycin (S).

When DOT or SCC is not feasible or tolerated, 9-12 months daily regimen may be given. **In TB meningitis (Cat I), steroids should be given for 6-8 weeks** to reduce exudates and blocking of CSF, and ATT may be given for 12-18 months. During pregnancy and lactation, ATT should be continued.

Streptomycin should be avoided as it can cross the placenta and affect hearing in the foetus. When the mother has TB, the baby should be put on chemoprophylaxis preferably with INH and rifampicin for 3 months or as long as mother is AFB positive. **Two drugs are preferred to prevent drug resistance.** Then the baby should be BCG vaccinated if Mx negative. If Mx positive, continue ATT as in a patient. Ethambutol is preferably avoided in children below 5 years and streptomycin may be substituted in Cat I instead of ethambutol. Baseline LFT values may be obtained in malnourished children and in those with hepatomegaly.

Rifampicin is the bactericidal drag active against persisters/dormant bacilli.

INH is the most powerful bactericidal drag, active against intracellular and extracellular bacilli.

Streptomycin is bactericidal drag active against extracellular rapidly multiplying bacilli.

Pyrazinamide is bactericidal drug active against slow-growing intracellular bacilli.

Others are bacteriostatic drugs.

Table 8.9 Antituberculosis therapy

Category	Criteria	Treatment	Dosage	Side effects
Category I	All sputum +ve pulmonary, All seriously ill	2 (HRZE)3 + 4 (HR)3	H-15 mg/kg If daily, 5 mg/ kg	Hepatitis, Peripheral neuritis, Aplastic anaemia, Psychosis, Hypersensitivity, Optic neuritis
Category II	All defaulters & retreatment cases	2 (HRZES)3 + 1 (HRZE)3 + 4 (HRE)3	R-10 mg/kg	Hepatitis, Flu like syndrome, Hypersensitivity
Category III	All AFB -ve, Not seriously ill extra pulmonary cases	2 (HRZ)3 + 4 (HR)	Z-50 mg/kg If daily 30 mg/ kg	Hepatitis, Arthralgia, Hyperuricaemia

E-30 mg/kg, if daily 15 mg/kg-SE: Optic Neuritis, color blindness
S- 15 mg/kg- SE: Ototoxicity, Nephrotoxicity, ataxia, vertigo

DOTS offers 90% cure rate as against 70% in self-administered daily therapy due to difference in compliance and completion rates.

BCG adenitis may be left alone if it is very small. A course of antibiotic is worth trying. When the BCG adenitis is big, more than 2.5 cm, caseated or requires drainage, a course of INH and rifampicin is given for 3-6 months. INH alone is not usually given to prevent drug resistance.

4. Evaluation

It is important to register and report all cases to the district TB coordinator. TB Identity Card (TIC) should be given to all patients and TB Treatment Card (TTC) should be maintained in the hospital or centre.

DOTS Plus (RNTCP Category IV)

It refers to DOTS prgramme that add components of MDR TB diagnosis, management and treatment under the Revised National TB Control Programme (RNTCP). Multi drug resistant TB (MDR-TB) is defined as resistance to INH and rifampicin with or without resistance to other drugs. It is thought to be up to 3% in new cases and 12-17% in retreatment cases. It is more commonly associated with HIV-AIDS.

MDR TB suspect is a TB patient who has failure after RNTCP Category I or III regimen or any RNTCP Category II patient sputum positive at the end of 4 months of treatment or later.

Any suspect should be referred to DTO within 2 weeks for diagnosis of MDR TB. DTO should arrange to send 2 early sputum samples to RNTCP-accredited Intermediate Research Lab (IRL) for culture and drug sensitivity, preferably in cetyl pyridium chloride (CPC) solution if expected delay is > 72 hr.

Cat IV uses 6 drugs—kanamycin, ofloxacin, ethionamide, pyrazinamide, ethambutol and cycloserine—for 6-9 months of intensive phase (IP) and 4 drugs—ofloxacin, ethionamide, ethambutol and cycloserine—during the continuation phase (CP) till 18 months. Para amino salicylic acid (PAS) may be substituted if any of the above drugs is not tolerated. All oral medicines are given on all 7 days and injections for 6 days, supervised by DOTS provider and on Sunday oral medicines are given non-supervised. Drug Kits are available for < 45 kg and > 45 kg weight. If the 4th month sputum is negative, IP is stopped after 6 months and CP started. If not, IP shall be extended. It sputum is positive after 7 months, further culture and drug sensitivity should be done.

Extremely drug resistant TB (XDR TB) is defined as resistance to INH and rifampicin plus any one of the chloroquinolone and any one of the injectable medicines. In XDR, daily DOTS is to be given for 2 years as per drug sensitivity.

8.5 Millennium Development Goals

The Millennium Declaration signed by world's leaders of 189 countries and the Millennium Development Goals (MDG) adopted by all United Nations Member States in 2000 have become a universal framework for development and a means for developing countries and their development partners to work together in pursuit of a shared future for all. This blueprint agreed by all the world countries have eight goals, which range from halving extreme poverty to halting the spread of HIV/AIDS and providing universal primary education to all by the target of 2015 and have galvanised the unprecedented efforts to meet the needs of the poorest. The Millennium Development Goals are interlinked and recognise the centrality of gender equality in the development agenda and set measurable time bound goals on commitments .The status report on Millennium goals for India published in 2005 evaluates the progress so far made from the base year 1990 and also highlights the strategies developed towards the attainment of the Goals in 2005.

India's tenth five year plan (2002-2007) has taken note of the Millennium Development Goals and included a number of targets, aiming higher than the ones accomplished in the MDG's and to be achieved during the plan period. The

unprecedented economic growth in India holds promise for the first MDG target to be met early. The monitorable targets for the tenth five year plan and beyond include the following: reduction of poverty ratio by 5 percentage points by 2007 and by 15 percentage points by 2012; providing gainful and high-quality employment at least to the addition to the labour force over the tenth plan period; all children in school by 2003; all children to complete 5 years of schooling by 2007; reduction in gender gaps in literacy and wage rates by at least 50 per cent by 2007; reduction in the decadal rate of population growth between 2001 and 2011 to 16.2 per cent; increase in literacy rates to 75 per cent within the tenth plan period (2002 to 2007); reduction of infant mortality rate (IMR) to45 per 1000 live births by 2007 and to 28 by 2012; reduction of maternal mortality rate (MMR) to 2 per 1000 live births by 2007 and to 1 by 2012; increase in forest and tree cover to 25 per cent by 2007 and 33 per cent by 2012; all villages to have sustained access to potable drinking water within the plan period; cleaning of all major polluted rivers by 2007 and other notified stretches by 2012 .The HIV/AIDS targets include 80% coverage of high risk groups through targeted interventions; 90% coverage of schools and colleges through education programmes; 80% awareness among the general population in rural areas; reducing transmission through blood to less than 1%; establishing of at least one voluntary counselling and testing centre (VCTC) in every district; scaling up of prevention of parent-to-child transmission (PPTCT) activities up to the district level; achieving zero level increase of HIV/AIDS prevalence by 2007. The malaria targets include Annual Blood Examination Rate (ABER) over 10 percent; Annual Parasite Incidence (API) 1.3 or less; 25% reduction in morbidity and mortality due to malaria by 2007 and 50% by 2010 (NHP 2002). Table 8.10 & 8.11 provides the values of the MDG indicators for available periods and also the progress towards achieving millennium goals in India.

It has been found by the Millennium Development Goals for India country report that some of the indicators could have been better presented in a manner differ-ent from the ones specified under the Millennium Development Goals. The non availability of some of the sufficiently reliable data was the reason for dropping some of the indicators such as proportion of population (PPP) below $1 per day, proportion of population below minimum level of dietary energy consumption, ratio of school attendance of orphans to school, attendance of non orphans aged 10-14 years, maternal mortality rate, proportion of population with access to secure tenure, unemployment rate of young people aged 15-24 years and proportion of population with access to affordable essential drugs on a sustainable basis.

India is moving in the direction of achieving all the goals much earlier than 2015 in spite of the vastness and complexities of our nation. It not only helps to attain human development and economic growth within the country, but also essential for reaching the Millennium Development Goals worldwide. While constant and high economic growth, accompanied by carefully designed and targeted

Table 8.10 India: Demographic changes from 2000 through 2016

Indicator	2000	2006	2011	2016
Population (in millions)	1028	1112	1192	1268
Life expectancy at birth (years)				
Males	63.8	65.8	67.3	68.8
Females	66.1	68.1	69.6	71.1
Crude birth rate (per 1000 population)	23.2	21.3	19.6	18
Crude death rate (per 1000 population)	7.5	7.3	7.2	7.1
Infant mortality rate (per 10000 live births)	61.3	54.3	49.2	44

Source: Populations for India and states 2001-2026, revised December 2006, Office of the Registar General and Census Commissioner, India

Table 8.11 Non communicable and communicable diseases in India

Disease	Total no. of cases*	Total no. of deaths*
Noncommunicable diseases		
IHD (1998)	25 million	119, 936
Stroke (1998	1 million	102, 620
Diabetes (1998)	28 million	21,000
Communicable diseases		
Tuberculosis (2001)	2.2 million	0.42 million
HIV/AIDS (2000)	3.86 million	Not available
Malaria (2001)	2 million	927 Deaths

Source: NCD in South-East Asia region-A profile WHO, New Delhi 2002 and National Institute of Health and Family Welfare. National Health Programmes on Noncommunicable Diseases. New Delhi, 2003 (*Ad hoc prevalence data)

geted pro-poor policies, are crucial for attaining the MDG's, effective decentralization, efficient delivery of services and respect for human rights, rule of law and account-ability have the potential to channel the MDG's into good governance. The interim order of the Supreme Court reported that the access of families below the poverty line (BPL) to grain at the set price at ration shops be improved and that individuals without means of support including older persons, widows and

disabled adults, be granted ration cards for free grain and state governments were also ordered to implement the 'mid-day meal scheme' in schools on a progressive basis. Significant progress in implementing this scheme has been reported. The ration shop system was introduced way back in 1970s.

There is plans for recasting and expanding the Integrated Child Development Services (ICDS) Scheme to make it more effective to attain the MDG targets. This is perhaps the largest scheme in the whole world with maximum number of beneficiaries, implemented in India from 1978.

The Self-Employed Women's Association (SEWA), a trade union registered in 1972 to represent low-income women, workers in India's vast informal sector, has regularly assisted its members, mostly women with marginal incomes from small farms or handicrafts, to spread the risk of income losses from draughts and cyclones through insurance and microfinance measures.

Now with the help of Panchayati Raj and local self government, the women employment schemes like 'Kudumbasree' and 'Mahilamandals' are in full swing contributing to economic progress in the families and societies and also in environmental health. States like Kerala and Tamil Nadu have set good examples in this respect. The progress in computer education and information technology (IT) is far enriching the economy, employment and social status.

It is essential that the next generation of reforms will need to focus on fostering a knowledge-based economy to maintain consistently high economic growth as average skill levels go on increasing. Inclusive public policies should focus on investments in health, education and infrastructure for future development, especially in the relatively backward areas and balanced development by region, communities and ethnic groups has to be emphasised. There is need for bridging the gap between the country's health policy and the health care needs of the poor in order to achieve the goals set for 2015. (see Appendix).

SECTION 9

Expanding Horizon in Nutrition

"Teenagers are not fed, they eat. For the first time in their lives, they assume responsibility for their own food intake. Social pressures thrust choices at them: to drink or not to drink, to smoke or not to smoke, to develop their bodies to meet sometimes, extreme ideals of slimness or athletic prowess."

—Hamilton & Whitney

9.1 Nutrition and Epigenetics

Epigenetics is a new stream of science which deals with the effect of environmental and other factors on our genetic phenotype. Epigenetics adds a whole new layer to genes beyond the DNA. It proposes a control system of 'switches' that turn genes on or off - and suggests that things people experience, like nutrition and stress, can control these switches and cause heritable effects in humans. The conventional view is that DNA carries all our heritable information and that nothing an individual does in their lifetime will be biologically passed to their children.

From the Greek prefix epi, which means "on" or "over", epigenetic information modulates gene expression without modifying actual DNA sequence. DNA methylation patterns are the longest-studied and best-understood epigenetic markers, although ethyl, acetyl, phosphoryl, and other modifications of histones, the protein spools around which DNA winds, are another important source of epigenetic regulation. The latter presumably influence gene expression by changing chromatin structure, making it either easier or more difficult for genes to be activated.

Only two percent of our DNA - via RNA - codes for proteins. Until very recently, the rest was considered "junk." the byproduct of millions of years of evolution. Now scientists are discovering that some of this junk DNA switches on RNA that may do the work of proteins and interact with other genetic material. Epigenetics delves deeper into our genome, involving "information stored in the proteins and chemicals that surround and slick to DNA."

The Three Main Types of Epigenetic Information

- Cytosine DNA methylation is a covalent modification of DNA, in which a methyl group is transferred from S-adenosylmethionine to the C-5 position of cytosine by a family of cytosine (DNA-5)-methyltransferases. DNA methylation occurs almost exclusively at CpG nucleotides and has an important contributing role in the regulation of gene expression and the silencing of repeat elements in the genome.
- Genomic imprinting is parent-of-origin-specific allele silencing, or relative silencing of one parental allele compared with the other parental allele. It is maintained, in part, by differentially methylated regions within or near imprinted genes, and it is normally reprogrammed in the germline.
- Histone modifications - including acetylation, methylation and phosphorylation - important in transcriptional regulation and many are stably maintained during cell division, although the mechanism for this epigenetic inheritance is not yet well understood. Proteins that mediate these modifications are often associated within the same complexes as those that regulate DNA methylation.

How do epigenetic modifications affect genes?

Genes carry the blueprints to make proteins in the cell. The DNA sequence of a gene is transcribed into RNA, which is then translated into the sequence of a protein. Every cell in the body has the same genetic information; what makes cells, tissues and organs different is that different sets of genes are turned on or expressed.

Starting from a zygote, an organism should successively activate most available genes during development in order to live. Thus, at adult age, all genes should be active. However, the simultaneous activity of all genes would produce an uncontrollable chaos of gene expression patterns not allowing coordinated cell- and organ-differentiation. Therefore, many genes need to be more or less permanently inactivated after they have done their job. Such a status can be triggered and maintained by an epigenetic tag. Because they change how genes can interact with the cell's transcribing machinery, epigenetic modifications, or "marks," generally turn genes on or off, allowing or preventing the gene from being used to make a protein. On the other hand, mutations and bigger changes in the DNA sequence (like insertions or deletions) change not only the sequence of the DNA and RNA, but may affect the sequence of the protein as well.

There are different kinds of epigenetic "marks," chemical additions to the genetic sequence. The addition of methyl groups to the DNA backbone is used on some genes to distinguish the gene copy inherited from the father and that inherited from the mother. In this situation, known as "imprinting", the marks both distinguish the gene copies and tell the cell which copy to use to make proteins.

Thus it appears that a simple environmental effect could switch genes on or off - and this change could be inherited. Our diet, behaviour, and environmental surroundings today could have a far greater impact than imagined on the health of our distant descendants.

The theory of epigenetics and the research that has been done so far makes one thing very clear - our diet, environment, toxins, our social environment and family bonds - all have the ability to modify our genetic code and its expression. This is not just a new understanding into our evolution and adaptation.

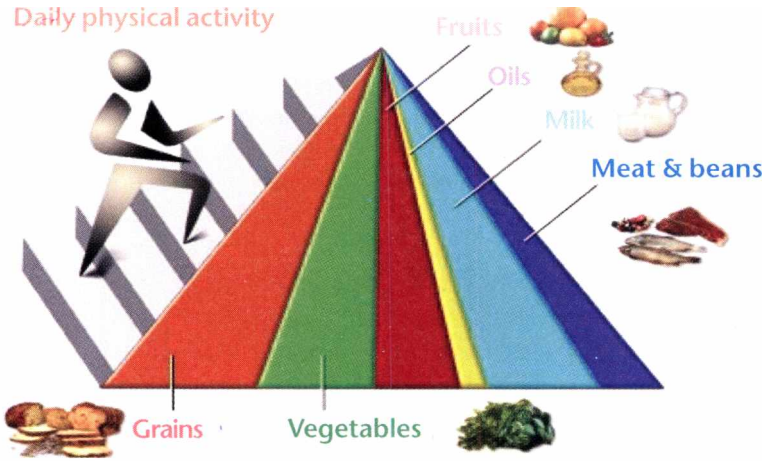


Fig. 9.1 Modified food pyramid

9.2 Sports Nutrition

Sports nutrition is a science that produces or provides and maintains the food or dietary ergogenic aids necessary for health, growth and physical performance. It deals with nutrients such as vitamins, minerals, supplements and organic substances such as carbohydrates, proteins and sugars in athletes of all sorts who want to make use of nutrition for their benefit (Fig. 9.1). **Sports nutrition** is the study and practice of nutrition and diet as it relates to athletic performance. Although an important part of many sports training regimens, it is most commonly considered only in strength sports like weight lifting and body building and endurance sports like cycling, running, and triathlon.

To many, sports nutrition means eating more proteins. Athletes shouldn't eat high-protein, low-carhohydrate diets because carbohydrates are the primary fuel for intense muscular effort. Keeping fit by healthy eating and good exercise is the best for best performance. Most experts agree that moderate exercise for 30 minutes every other day is all you need. This amount of exercise has been linked with improving cardiovascular function, lowering cholesterol and blood pressure, losing weight, and reducing stress. Intermittent exercise, several 10 minutes sessions a day, may be the best way to reduce the risk of heart disease.

Carbohydrate

Carbohydrates are one of the main dietary components. This category of foods includes sugars, starches, and fiber. Carbohydrates are classified as simple or complex. The classification depends on the chemical structure of the food, and how quickly the sugar is digested and absorbed. Simple carbohydrates have one (single) or two (double) sugars. Complex carbohydrates have three or more sug-

Examples of single sugars from foods include:

- Fructose (found in fruits)
- Galactose (found in milk products)

Double sugars include:

- Lactose (found in dairy)
- Maltose (found in certain vegetables and in beer)
- Sucrose (table sugar)

Honey is also a double sugar. But unlike table sugar, it contains a small amount of vitamins and minerals. (NOTE: Honey should not be given to children younger than 1 year old.)

Complex carbohydrates, often referred to as "starchy" foods, include:

- Legumes
- Starchy vegetables
- Whole-grain breads and cereals

Simple carbohydrates that contain vitamins and minerals occur naturally in:

- Fruits
- Milk and milk products
- Vegetables

Simple carbohydrates are also found in processed and refined sugars such as:

- Candy
- Regular (non-diet) carbonated beverages, such as soda
- Syrups (not including natural syrups such as maple)

- Table sugar

Refined sugars provide calories, but lack vitamins, minerals, and fiber. Such simple sugars are often called "empty calories" and can lead to weight gain. Also, many refined foods, such as white flour, sugar, and polished rice, lack B vitamins and other important nutrients unless they are marked "enriched." It is healthiest to get carbohydrates, vitamins, and other nutrients in as natural a form as possible—for example, from fruit instead of table sugar.

Getting too many carbohydrates can lead to an increase in total calories, causing obesity. Not getting enough carbohydrates can cause a lack of calories (malnutrition), or excessive intake of fats to make up the calories.

Carbohydrate is the predominant energy source for strength training. Stored as glycogen in the muscles, it is the fuel used to supply energy for short, intense bursts of power. The harder and longer you work out, the more glycogen your muscles require. Once these stores of glycogen are gone, your energy level will drop and you will run out of fuel to power muscle contractions. For this reason, athletes doing strength training exercise in the hopes of building lean muscle need to have adequate carbohydrates intake. Experts recommend at least 500 to 600 grams of carbohydrate per day to keep your muscle glycogen stores high. 55-60% of the energy can come from carbohydrate.

You can base your personal requirement on the following formula: 3.6 g carbohydrate x body wt (lb)= grams /day, or 8 g x body weight (kg) = g/ day

Protein

Proteins are complex organic compounds. The basic structure of protein is a chain of amino acids. Protein is the basic building material for muscle tissue and strength trainers need to consume more than the non-exercisers. However, most strength athletes still overestimate their protein needs. About 15% of the energy should come from protein. Every cell in the human body contains protein. It is a major part of the skin, muscles, organs, and glands. Protein is also found in all body fluids, except bile and urine.

You need protein in your diet to help your body repair cells and make new ones. Protein is also important for growth and development during childhood, adolescence, and pregnancy. Protein-containing foods are grouped as either complete or incomplete proteins.

Complete proteins contain all nine essential amino acids. Complete proteins are found in animal foods such as meat, fish, poultry, eggs, milk, and milk products such as yogurt and cheese. Soybeans are the only plant protein considered to be a complete protein.

Incomplete proteins lack one or more of the essential amino acids. Sources of incomplete protein include beans, peas, nuts, seeds, and grain. A small amount of incomplete protein is also found in vegetables.

Plant proteins can be combined to provide all of the essential amino acids and form a complete protein. Examples of combined, complete plant proteins are rice and beans, milk and wheat cereal, and corn and beans.

A diet high in meat can contribute to high cholesterol levels or other diseases such as gout. A high-protein diet may also put a strain on the kidneys.

Daily protein recommendations for strength athletes are about 0.6 to 0.8 grams per pound of body weight or 1.8 g/kg/day.

Fat

Fat is an essential nutrient. However, you require small amount of it to remain healthy. Less than 30% of your total daily calories should come from fat, especially unsaturated fat. Fats are organic compounds that are made up of carbon, hydrogen, and oxygen. They are a source of energy in foods. Fats belong to a group of substances called lipids, and come in liquid or solid form. All fats are combinations of saturated and unsaturated fatty acids, both mono and polyunsaturated.

Fat provides nine calories per gram, more than twice the number provided by carbohydrates or protein. Fat is essential for the proper functioning of the body. Fats provide essential fatty acids, which are not made by the body and must be obtained from food. The essential fatty acids are linoleic and linolenic acid. They are important for controlling inflammation, blood clotting, and brain development. Fat serves as the storage substance for the body's extra calories. It fills the fat cells (adipose tissue) that help insulate the body. Fats are also an important energy source. When the body has used up the calories from carbohydrates, which occurs after the first 20 minutes of exercise, it begins to depend on the calories from fat.

Healthy skin and hair are maintained by fat. Fat helps the body absorb and move the vitamins A. D. E. and K through the bloodstream.

Saturated Fats

These are the biggest dietary cause of high LDL levels ("'bad cholesterol"). When looking at a food label, pay very close attention to the percentage of saturated fat and avoid or limit any foods that are high. Saturated fat should be limited to 10% of calories. Saturated fats are found in animal products such as butter, cheese, whole milk, ice cream, cream, and fatty meats. They are also found in some vegetable oils—coconut, palm, and palm kernel oils. (Note: Most other vegetable oils contain unsaturated fat and are healthy.)

Unsaturated Fats

These are fats that help to lower blood cholesterol if used in place of saturated fats. However, unsaturated fats have a lot of calories, so you still need to limit them. Most (but not all!) liquid vegetable oils are unsaturated. (The exceptions

include coconut, palm, and palm kernel oils.) There are two types of unsaturated fats:

- Monounsaturated fats: Examples - olive and canola oils.
- Polyunsaturated fats: Examples - fish, safflower, sunflower, corn, and soybean oils.

Trans Fatty Acids

These fats form when vegetable oil hardens (a process called hydrogenation) and can raise LDL levels. They can also lower HDL levels ("good cholesterol"). Trans-fatty acids are found in fried foods, commercial baked goods (donuts, cookies, crackers), processed foods, and margarines.

Hydrogenated and Partially Hydrogenated Fats

This refers to oils that have become hardened (such as hard butter and margarine). Partially hydrogenated means the oils are only partly hardened. Foods made with hydrogenated oils should be avoided because they contain high levels of trans-fatty acids, which are linked to heart disease. (Look at the ingredients in the food label.)

Eating too much saturated fat is one of the major risk factors for heart disease. A diet high in saturated fat causes a soft, waxy substance called cholesterol to build up in the arteries. Too much fat also increases the risk of heart disease because of its high calorie content, which increases the chance of becoming obese (another risk factor for heart disease and some types of cancer). A large intake of polyunsaturated fat may increase the risk for some types of cancer. Reducing daily fat intake is not a guarantee against developing cancer or heart disease, but it does help reduce the risk factors.

Fiber

It is a substance found in plants. Dietary fiber—the kind you eat—is found in fruits, vegetables, and grains. It is an important part of a healthy diet. It is also named roughage/bulk of the diet. Dietary fiber adds bulk to your diet. Because it makes you feel full faster, it can be helpful in controlling weight. Fiber aids digestion, helps prevent constipation, and is sometimes used for the treatment of diverticulosis, diabetes, and heart disease.

There are two forms of fiber: soluble and insoluble. Soluble fiber attracts water and turns to gel during digestion. This slows digestion. Soluble fiber is found in oat bran, barley, nuts, seeds, beans, lentils, peas, and some fruits and vegetables. Soluble fiber has been scientifically proven to lower cholesterol, which can help prevent heart disease. Insoluble fiber is found in foods such as wheat bran, vegetables, and whole grains. It appears to speed the passage of foods through the stomach and intestines and adds bulk to the stool.

Eating a large amount of fiber in a short period of time can cause intestinal gas (flatulence), bloating, and abdominal cramps. This usually goes away once the natural bacteria in the digestive system get used to the increase in fiber in the diet. Adding fiber gradually to the diet, instead of all at one time, can help reduce gas or diarrhoea.

Too much fiber may interfere with the absorption of minerals such as iron, zinc, magnesium, and calcium. However, this effect usually does not cause too much concern because high-fiber foods are typically rich in minerals.

The average westerner eats 10-15 grams of fiber per day. The recommendation for older children, adolescents, and adults is 20-35 grams per day. Younger children will not be able to eat enough calories to achieve this, but it is a good idea to introduce whole grains, fresh fruits, and other high-fiber foods.

To ensure that you get enough fiber, eat a variety of foods, including:

- Cereals
- Dried beans and peas
- Fruits
- Vegetables
- Whole grains

Add fiber gradually over a period of a few weeks to avoid abdominal discomfort. Water aids the passage of fiber through the digestive system. Drink plenty of fluids (approximately 8 glasses of water or non-caloric fluid a day). Peeling can reduce the amount of fiber in fruits and vegetables. Eating fiber-containing food is beneficial, whether it is cooked or raw.

Water

Water is the most essential ingredient to a healthy life. Water has many important functions in the body including:

- Transportation of nutrients / elimination of waste products.
- Lubricating joints and tissues.
- Temperature regulation through sweating.
- Facilitating digestion.

Importance of Water During Exercise

Proper hydration is especially important during exercise. Adequate fluid intake for athletes is essential to comfort, performance and safety. The longer and more intensely you exercise, the more important it is to drink the right kind of fluids. In addition to the regular 2 liters or eight-ten glasses of water every day, you need to drink to replace fluids that are lost during exercise. To be confident that you are well hydrated before workouts, drink 2 cups of fluid 2 hours before exercise. During your workout, drink 4 to 8 ounces every 15 to 20 minutes. After exercise, replace any further fluid losses with 2 glasses of water. If you want to be precise,

you can weigh yourself before and after workouts. For each pound lost during exercise, you should be drink 2 glasses of fluid. (1 oz = 30 ml).

Sports Drinks

Energy bars and sports drinks may be helpful if exercise lasts longer than 1 hour. Carbohydrate supplements can be useful to get adequate carbohydrates into a busy day if you don't have time to eat a meal. Consuming a meal-replacement beverage just after muscle-building exercise is convenient, but you can do the same thing with a tuna sandwich, a banana or other food snack. You should try to consume some protein and carbohydrate after your workout in order to fuel muscle growth and replenish glycogen stores for your next workout.

Supplements

Most supplements that are supposed to help build muscle don't work. But some, such as creatine, fluid and electrolyte replacers, carbohydrate supplements, and liquid meal replacers may offer some benefits to strength training athletes. These are useful in these situations:

- Inadequate fluid intake
- Excessive sweating
- Failure to replace fluid losses during and after exercise
- Exercising in dry, hot weather
- Drinking only when thirsty

Hyponatraemia and Water Intoxication

Although rare, recreational exercisers are also at risk of drinking too much water and suffering from hyponatraemia and water intoxication. Clearly, drinking the right amount of the right fluids is critical for performance and safety while exercising.

Adequate Fluid Intake for Athletes

Because there is wide variability in sweat rates, losses and hydration levels of individuals, it is nearly impossible to provide specific recommendations or guidelines about the type or amount of fluids athletes should consume.

Finding the right amount of fluid to drink depends upon a variety of individual factors including the length and intensity of exercise and other individual differences. There are, however, two simple methods of estimating adequate hydration:

1. **Monitoring urine volume output and colour.** A large amount of light coloured, diluted urine probably means you are hydrated; dark coloured, concentrated urine probably means you are dehydrated.
2. **Weighing yourself before and after exercise.** Any weight lost is likely from fluid, so try to drink enough to replenish those losses. Any weight gain could mean you are drinking more than you need.

Things that Affect Fluid Loss in Athletes

- **High altitude.** Exercising at altitude increases your fluid losses and therefore increases your fluid needs.
- **Temperature.** Exercising in the heat increases your fluid losses through sweating, and exercise in the cold can impair you ability to recognize fluid losses and increase fluid lost through respiration. In both cases it is important to hydrate.
- **Sweating.** Some athletes sweat more than others. If you sweat a lot you are at greater risk for dehydration. Again, weigh yourself before and after exercise to judge sweat loss.
- **Exercise Duration and Intensity.** Exercising for hours (endurance sports) means you need to drink more and more frequently to avoid dehydration.

To find the correct balance of fluids for exercise, the American College of Sports Medicine suggests that "individuals should develop customized fluid replacement programs that prevent excessive (greater than 2 percent body weight reductions from baseline body weight) dehydration. The routine measurement of pre- and post-exercise body weights is useful for determining sweat rates and customized fluid replacement programs. Consumption of beverages containing electrolytes and carbohydrates can help sustain fluid-electrolyte balance and exercise performance."

According to the Institute of Medicine the need for carbohydrate and electrolytes replacement during exercise depends on exercise intensity, duration, weather and individual differences in sweat rates. Fluid replacement beverages might contain about 20-30 mEq sodium, 2-5 mEq of potassium and 5-10% carbohydrate. Sodium and potassium are to help replace sweat electrolyte losses, and sodium also helps to stimulate thirst. Carbohydrate provides energy for exercise over 60-90 minutes. This can also be provided through energy gels, bars, and other foods.

What about Sports Drinks?

Sports drinks can be helpful to athletes who are exercising at a high intensity for 60 minutes or more. Fluids supplying 60 to 100 calories per 1 glass help to supply the needed calories required for continuous performance. It's really not necessary to replace losses of sodium, potassium and other electrolytes during exercise since you're unlikely to deplete your body's stores of these minerals during normal training. If, however, you find yourself exercising in extreme conditions over 3 or 5 hours like marathon, Ironman or ultra marathon, you may likely want to add a complex sports drink with electrolytes.

General Guidelines for Fluid Needs During Exercise

While specific fluid recommendations aren't possible due to individual variabil-

ity, most athletes can use the following guidelines as a starting point, and modify their fluid needs accordingly.

Hydration Before Exercise

- Drink about 2 glasses (15-20 fl oz), 2-3 hours before exercise
- Drink 1 glass (8-10 fl oz), 10-15 min before exercise

Hydration During Exercise

m Drink 1 glass every 10-15 min during exercise

- If exercising longer than 90 minutes, drink 1 glass of a sports drink, with no more than 8 percent carbohydrate, every 15-30 minutes.

Hydration After Exercise

- Weigh yourself before and after exercise and replace fluid losses.
- Drink 2-3 glasses water for every 1 lb lost.
- Maintain 4:1 ratio of carbohydrate to protein in the supplements within the 2 hours after exercise to replenish glycogen stores.

Dehydration

Athletes need to stay hydrated for optimal performance. Studies have found that a loss of two or more percent of one's body weight due to sweating is linked to a drop in blood volume. When this occurs, the heart works harder to move blood through the bloodstream. This can also cause muscle cramps, dizziness and fatigue and even heat illness including:

- Heat exhaustion
- Heat stroke

Others

Stress should also be on other nutrients like sodium, calcium, iron etc.

Sodium

Sodium occurs naturally in most foods. The most common form of sodium is sodium chloride, which is table salt. Milk, beets, and celery also naturally contain sodium, as does drinking water, although the amount varies depending on the source. The body uses sodium to regulate blood pressure and blood volume.

Sodium is also added to various food products. Some of these added forms are monosodium glutamate, sodium nitrite, sodium saccharin, baking soda (sodium bicarbonate), and sodium benzoate. These are ingredients in condiments and seasonings such as Worcestershire sauce, soy sauce, onion salt, garlic salt, and bouillon cubes. Processed meats, such as bacon, sausage, and ham, and canned soups and vegetables are all examples of foods that contain added sodium. Fast foods are generally very high in sodium.

Too much sodium will contribute to high blood pressure in those who are sensitive to sodium. Most people with high blood pressure may be told to reduce their sodium intake. If you have high blood pressure, you should discuss this issue with your doctor. Sodium may lead to a serious build-up of fluid in people with congestive heart failure, cirrhosis, or kidney disease. Such people should be on a strict sodium-restricted diet, as prescribed by their doctor.

Dietary sodium is measured in milligrams (mg). Table salt is *40%* sodium; 1 teaspoon of table salt contains 2,300 mg of sodium. Healthy adults should limit sodium intake to 2,300 mg per day while individuals with high blood pressure should consume no more than 1.500 mg per day.

Iron

Iron deficiency is a common problem especially for women athletes. Studies have routinely found that athletes, especially female athletes, are often iron-deficient or anaemic. Iron is essential for athletic performance. One of its major functions is to carry oxygen to and carbon dioxide away from all the cells in your body. The brain also relies on oxygen transport and without enough iron you will find it hard to concentrate and feel tired and irritable. Iron is also needed to maintain a healthy immune system. If you don't have enough iron you may be prone to more frequent infections.

Athletes and Iron Deficiency

A combination of the following factors place athletes at risk of iron deficiency:

1. Inadequate supply of dietary iron. Athletes who avoid red meat have difficulty meeting the body's iron needs.
2. Increased demands for iron. Hard training stimulates an increase in red blood cell and blood vessel production, and increases the demand for iron. (Iron turnover is highest for endurance athletes training at high intensity.)
3. High iron loss. Blood loss through injury, or menstruation. In endurance athletes, 'foot strike' damage to red blood cells in the feet due to running on hard surfaces with poor quality shoes leads to iron loss. Finally, because iron is lost in sweat, heavy sweating leads to increased risk of deficiency.

The symptoms of iron deficiency include loss of endurance, chronic fatigue, high exercise heart rate, low power, frequent injury, recurring illness, and loss of interest in exercise and irritability. Other symptoms include poor appetite, and increased incidence and duration of colds and infections. Many of these symptoms are also common to overtraining, so misdiagnosis is common. The only sure way to diagnose a deficiency is a blood test to determine iron status. If you experience any of the symptoms above, and you are in one of the higher risk categories, you should visit your physician for lab work.

You can get iron in both animal and plant foods, but the iron in animal sources has an absorption rate of about 15 percent, compared to about 5 percent

for plants. So the more effective way to increase iron status is by eating animal products such as lean red meat, poultry or fish or liver. You can also increase the amount of iron in foods you eat by cooking with a cast iron skillet (especially if cooking acidic foods).

Iron absorption from any foods, whether plant or animal, is decreased if they are accompanied at meals by caffeine. Calcium and zinc also reduce the ability of the body to absorb iron. However, adding fruit (citrus fruit in particular), to meals enhances iron absorption. The best sources of iron in the diet include: Lean red meat, iron-fortified breakfast cereal, nuts and legumes (combine these with foods high in vitamin C).

All substances, whether medications or vitamins or supplements, carry potential risk of complications even if they are called "natural" and "herbal." While there is continued evidence indicating that some natural and herbal products have value, many also may cause a variety of side effects and interact negatively with prescription and other medications. In some cases these substances can interfere with, or reduce the effectiveness of, other medications.

Calcium

Athletes who are strict vegetarians should take a multivitamin to prevent deficiencies and a calcium supplement (1000 mg/day) to help prevent bone loss.

- Vitamin A and Vitamin D
  - No evidence of increased performance
  - May have toxic effects at high doses
- Vitamin E
  - No evidence of increased performance
  - Toxic effects are rare
- Vitamin C
  - Antioxidant effect may help decrease exercise-related muscle soreness
  - No effect on strength
  - Possible toxic effects at high doses
- Vitamin B_6
  - No evidence of increased performance
  - Toxic over 200 mg/day (nervous system side effects)
- Other antioxidants (Betacarotene, Bioflavinoids, Copper, Cysteine and Glutathione)
  - May help to protect against exercise-induced muscle damage
  - Study results are conflicting
  - Should not exceed 100% US RDA of antioxidants
- **Buyer beware!**
  - Some supplements have been found to contain up to 3000% of US RDA for vitamins and minerals.

Items to Improve Performance

If you are an endurance athlete, evidence suggests that eating some sugar (like energy bars, some types of candy bars, or sports drinks) 35 to 40 minutes before an event may provide energy (glucose) to your exercising muscles when your other energy stores have dropped to low levels. However, you should experiment with such strategies before competition because some people do not perform well after a blood glucose spike.

1. Caffeine and Performance

Caffeine acts as a stimulant on the central nervous system. It had been thought to boost endurance by stimulating a greater use of fat for energy, and thereby reserving glycogen in the muscles. Research, however, doesn't support that theory. When caffeine improves endurance, it does so by acting as a stimulant. Caffeine can have serious side effects for some people. Those who are very sensitive to its effects may experience nausea, muscle tremors, and headaches. Too much caffeine is a diuretic, and can result in dehydration, which decreases performance.

2. Creatine

- Chemical name: Creatine-Monohydrate
- Naturally available in meat and fish
- NCAA study found creatine supplements used by 12% of college athletes
- A subsequent survey of high school athletes showed similar usage rates
- There are many studies showing positive effects in healthy subjects published between July 1997 and November 2001

  Increased high intensity, intermittent exercise performance in squash players

  Increased cell hydration status and performance variables in Division I college football players more than training alone

  Augments repeated sprint cycle performance in hot environment without altering thermoregulatory responses

  Increases indices of high intensity exercise performance for both males and females

  Increased capacity of human muscle to perform work during alternating intensity contraction

  Ergogenic effect in elite ice hockey players

  Loading improves intermittent sprint capacity at end of endurance exercise to fatigue

  Adding creatine to glucose, taurine and electrolyte supplement promoted greater gains in fat and bone free mass, isotonic lifting volume and sprint performance during intense resistance and agility training

Helped to prolong time of maximal rate of power output could be maintained

There are some studies with no effect in healthy subjects published in November and December of 2001.

Did not positively influence isometric strength in untrained (sedentary) males.

Did not increase performance or training volume over placebo in rowers that performed a high intensity rowing and strength program.

No statistically significant difference in strength or fat-free mass gains after a resistance exercise training program compared with post exercise protein supplementation.

Supplementation in combination with high-intensity strength training increases strength during high intensity intermittent exercise 7 to 8% more than training alone. Supplementation probably increases performance in sports involving or requiring high intensity intermittent bursts of strength. Long-term effects (chronic use > 4 years) are not known and some experts have concern about potential for liver and kidney problems by long-term usage.

- **Dosing Loading:** 20 **to** 30 **g/day for one week, Maintenance:** 10 **to** 15 **g/day while training.**

3. *Hydroxy-Methylbutyrate - HMB*

- Metabolite of leucine (amino acid)
- Available naturally in catfish, citrus fruits and breast milk
- Some preliminary studies suggest that supplementation with HMB can suppress muscle protein breakdown.
- One placebo-controlled study in weightlifters reported slightly better strength increases and greater lean mass increases in the group taking HMB.
- No known adverse effects.
- Dosing: 1 g three times a day

4. *Ephedra-Herbal forms of the Stimulant Ephedrine*

m 80 confirmed deaths related to Ephedra use

- Experts suspect many more unconfirmed deaths

Adverse Effects

- High blood pressure (most common), palpitations and increased heart rate
- Seizure, thermoregulatory dysfunction
- Stroke, heart attack, sudden death

- Vasculitis,
- Allergic myocarditis (one case reported), acute hepatitis (one case report)
- Following the death of two professional athletes, FDA banned sale of Ephedra as a nutritional supplement. Since this time, manufacturers have started substituting other stimulants like orange extract.

5. Citrus Aurantium

n Orange extract; chemical structure very similar to ephedrine

6. Chromium Picolinate

No benefit demonstrated in studies. Adverse effects; stomach upset, anaemia, cognitive impairment, chromosome damage, interstitial nephritis.

7. L-Carnitine

No benefit demonstrated in studies. Adverse effect: significant muscle weakness

8. L-Tryptophan

No benefit demonstrated in studies. Adverse effect: eosinophilia-myalgia syndrome

9. Dehydroepiandrosterone (DHEA) and Androstenedione ("Andro")

- Chemicals that can be converted into testosterone in human biochemical pathways
- Naturally available in wild yams
- An early study done by a manufacturer of these products showed no significant increase in blood levels of testosterone. Study looked at lower doses of these supplements than are usually taken and did not measure ratio of testosterone to epitestosterone (T:E ratio).

DHEA - Does not seem to have much, if any, effect on fat-free body mass and strength

Androstenedione

m Causes a temporary increase in testosterone levels

- Has no effect on body's ability to make protein
- Does not seem to have any effect on strength
- No long term effect on blood testosterone levels
- Chronic use causes increase in estrogen levels by aromatization of androgen

Potential adverse effects

m May cause liver damage

- In females: Can cause male features in women; may increase risk of uterus cancer

- In males: Can cause female features in men; may increase risk of prostate cancer

Blood Doping

- You can be disqualified from participating in college sports if you test positive for a substance banned by the NCAA:
  - Whether or not you knew it was banned
  - Whether or not the product was mislabeled

Foods to Avoid Before Exercise

Any foods with a lot of fat can be very difficult and slow to digest and remain in the stomach a long time. They also will pull blood into the stomach to aid in digestion, which can cause cramping and discomfort. Meats, doughnuts, fries, potato chips, and candy bars should be avoided in a pre-exercise meal.

When To Eat

Exercising on a full stomach is not ideal. Food that remains in your stomach during an event may cause stomach upset, nausea, and cramping. To make sure you have enough energy, yet reduce stomach discomfort, you should allow a meal to fully digest before the start of the event. This generally takes 1 to 4 hours, depending upon what and how much you've eaten. Everyone is a bit different, and you should experiment prior to workouts to determine what works best for you.

If you have an early morning race or workout, it's best to get up early enough to eat your pre-exercise meal. If not, you should try to eat or drink something easily digestible about 20 to 30 minutes before the event. The closer you are to the time of your event, the less you should eat. You can have a liquid meal closer to your event than a solid meal because your stomach digests liquids faster.

The major source of fuel for active muscles is carbohydrate which gets stored in the muscles as glycogen in the days before exercise. This is one reason that the post-exercise meal is critical to recovery and being ready for the next exercise session.

These nutrients get converted to energy in the form of adenosine triphosphate or ATP. It is from the energy released by the breakdown of ATP that allows muscle cells to contract. However, each nutrient has unique properties that determine how it gets converted to ATP.

Carbohydrate is the main nutrient that fuels exercise of a moderate to high intensity, while fat can fuel low-intensity exercise for long periods of time. Proteins are generally used to maintain and repair body tissues, and are not normally used to power muscle activity.

Energy Pathways

Because the body cannot easily store ATP (and what is stored gets used up within a few seconds), it is necessary to continually create ATP during exercise. In general, the two major ways the body converts nutrients to energy are:

- Aerobic metabolism (with oxygen)
- Anaerobic metabolism (without oxygen)

These two pathways can be further divided. Most often it's a combination of energy systems that supply the fuel needed for exercise, with the intensity and duration of the exercise determining which method gets used when.

ATP-CP Anaerobic Energy Pathway

The ATP-CP energy pathway (sometimes called the phosphate system) supplies about 10 seconds worth of energy and is used for short bursts of exercise such as a 100 meter sprint. This pathway doesn't require any oxygen to create ATP. It first uses up any ATP stored in the muscle (about 2-3 seconds worth) and then it uses creatine phosphate (CP) to resynthesize ATP until the CP runs out (another 6-8 seconds). After the ATP and CP are used the body will move on to either aerobic or anaerobic metabolism (glycolysis) to continue to create ATP to fuel exercise.

Anaerobic Metabolism - Glycolysis

The anaerobic energy pathway, or glycolysis, creates ATP exclusively from carbohydrates, with lactic acid being a by-product. Anaerobic glycolysis provides energy by the (partial) breakdown of glucose without the need for oxygen. Anaerobic metabolism produces energy for short, high-intensity bursts of activity lasting no more than several minutes before the lactic acid build-up reaches a threshold known as the lactate threshold and muscle pain, burning and fatigue make it difficult to maintain such intensity.

Aerobic Metabolism

Aerobic metabolism fuels most of the energy needed for long duration activity. It uses oxygen to convert nutrients (carbohydrates, fats, and protein) to ATP. This system is a bit slower than the anaerobic systems because it relies on the circulatory system to transport oxygen to the working muscles before it creates ATP. Aerobic metabolism is used primarily during endurance exercise, which is generally less intense and can continue for long periods of time.

During exercise an athlete will move through these metabolic pathways. As exercise begins, ATP is produced via anaerobic metabolism. With an increase in breathing and heart rate, there is more oxygen available and aerobic metabolism begins and continues until the lactate threshold is reached. If this level is surpassed, the body can not deliver oxygen quickly enough to generate ATP and anaerobic metabolism kicks in again. Since this system is short-lived and lactic

acid levels rise, the intensity can not be sustained and the athlete will need to decrease intensity to remove lactic acid build-up.

Glycolysis

- 2 ATP
- 2 NADH (= 4 ATP; these are converted to ATP in the mitochondria during cellular respiration)

Formation of Acetyl-CoA

m 2 NADH (=6 ATP)

Krebs Cycle

. 6 NADH (= 18 ATP)
. 2 FADH, (= 4 ATP)
- 2 ATP

Total Yield

Glycolysis produces 2 ATP; aerobic respiration produces 34 more ATP

Pathway	Substrate-Level Phosphorylation	Oxidative Phosphorylation	Total ATP
Glycolysis	2 ATP	2 NADH = 4-6 ATP*	ర→ఐ *
CoA		2 NADH = 6 ATP	6
Krebs Cycle	2 ATP	6 NADH = 18 ATP 2 $FADH_2$ = 4 ATP	24
TOTAL	4 ATP	32 ATP	36-38

Fueling the Energy Systems

Nutrients get converted to ATP based upon the intensity and duration of activity, with carbohydrate as the main nutrient fueling exercise of a moderate to high intensity, and fat providing energy during exercise that occurs at a lower intensity. Fat is a great fuel for endurance events, but it is simply not adequate for high intensity exercise such as sprints or intervals. If exercising at a low intensity (or below 50 percent of max heart rate), you have enough stored fat to fuel activity for hours or even days as long as there is sufficient oxygen to allow fat metabolism to occur.

As exercise intensity increases, carbohydrate metabolism takes over. It is more efficient than fat metabolism, but has limited energy stores. This stored

carbohydrate (glycogen) can fuel about 2 hours of moderate to high level exercise. After that, glycogen depletion occurs (stored carbohydrates are used up) and if that fuel isn't replaced athletes may hit the wall or "bonk." An athlete can continue moderate to high intensity exercise for longer simply replenishing carbohydrate stores during exercise. This is why it is critical to eat easily digestible carbohydrates during moderate exercise that lasts more than a few hours. If you don't take in enough carbohydrates, you will be forced to reduce your intensity and tap back into fat metabolism to fuel activity.

As exercise intensity increases, carbohydrate metabolism efficiency drops off dramatically and anaerobic metabolism takes over. This is because your body can not take in and distribute oxygen quickly enough to use either fat or carbohydrate metabolism easily. In fact, carbohydrates can produce nearly 20 times more energy (in the form of ATP) per gram when metabolized in the presence of adequate oxygen than when generated in the oxygen-starved, anaerobic environment that occurs during intense efforts (sprinting).

Suggested Pre-Exercise Foods

Eating before exercise is something only the athlete can determine based upon experience, but some general guidelines include eating a solid meal 4 hours before exercise, a snack or a high carbohydrate energy drink 2 to 3 hours before exercise, and fluid replacement (sports drink) 1 hour before exercise.

1 Hour or Less Before Competition

m Fruit or vegetable juice such as orange, tomato, and/or

- Fresh fruit such as apples, watermelon, peaches, grapes, or oranges and/or
- Energy gels
- Up to 1 V_2 cups of a sports drink.

2 to 3 Hours Before Competition

m Fresh fruit

- Fruit or vegetable juices
- Bread, bagels
- Low-fat yogurt
- Sports drink

3 to 4 Hours Before Competition

a Fresh fruit

- Fruit or vegetable juices
- Bread, bagels
- Pasta with tomato sauce
- Baked potatoes
- Energy bar

- Cereal with low-fat milk
- Low-fat yogurt
- Toast/bread with limited peanut butter, lean meat, or low-fat cheese
- 30 oz of a sports drink

Healthy Eating

Start the day with 2 glasses of water.

Never miss the breakfast - the brain's food.

Derive 50-60% of total calories from carbohydrates, preferably from complex carbohydrates (starches) and natural sugars. Complex carbohydrates provide calories, vitamins, minerals, and fiber.

Foods that are high in processed, refined simple sugars provide calories, but very little nutrition. It is wise to limit these sugars.

To increase complex carbohydrates and healthy nutrients:

- Eat more fruits and vegetables.
- Eat more whole-grain rice, breads, and cereals.
- Eat more legumes (beans, lentils, and dried peas).

Here are recommended serving sizes for foods high in carbohydrates:

- Vegetables: 1 cup of raw vegetables, or 1/2 cup cooked vegetables, or 3/4 cup of vegetable juice
- Fruits: 1 medium-size fruit (such as 1 medium apple or 1 medium orange), 1/2 cup of a canned or chopped fruit, or 3/4 cup of fruit juice
- Breads and cereals: 1 slice of bread; 1 ounce or 2/3 cup of ready-to-eat cereal; 1/2 cup of cooked rice, pasta, or cereal; 1/2 cup of cooked dry beans, lentils, or dried peas
- Dairy: 1 cup of skim or low-fat milk

Sample 2,000 Calorie Menu with 50%-60% Calories from Carbohydrates

- **Breakfast**

  Cold cereal
  1 cup shredded wheat cereal
  1 tbsp raisins
  1 cup fat-free milk
  1 small banana
  1 slice whole-wheat toast
  1 tsp soft margarine
  1 tsp jelly

- Lunch

  Smoked turkey sandwich
  2 ounces whole-wheat pita bread

1/4 cup romaine lettuce
2 slices tomato
3 ounces sliced smoked turkey breast
1 tbsp mayo-type salad dressing
1 tsp yellow mustard
1/2 cup apple slices
1 cup tomato juice

- **Dinner**

Grilled top loin steak
5 ounces grilled top loin steak
3/4 cup mashed potatoes
2 tsp soft margarine
1/2 cup steamed carrots
1 tbsp honey
2 ounces whole-wheat dinner roll
1 tsp soft margarine
1 cup fat-free milk

- **Snacks**

1 cup low-fat fruited yogurt

General Tips

- Choose lean, protein-rich foods such as soy, fish, skinless chicken, very lean meat, and fat free or 1 % dairy products.
- Eat foods that are naturally low in fat such as whole grains, fruits, and vegetables.
- Get plenty of soluble fiber such as oats, bran, dry peas, beans, cereal, and rice.
- Limit fried foods, processed foods, and commercially prepared baked goods (donuts, cookies, crackers).
- Limit animal products such as egg yolks, cheeses, whole milk, cream, ice cream, and fatty meats (and large portions of meats).
- Look at food labels, especially the level of saturated fat. Avoid or limit foods high in saturated fat.
- Look on food labels for words like "hydrogenated" or "partially hydrogenated"—these foods are loaded with bad fats and should be avoided.
- Liquid vegetable oil, soft margarine, and trans fatty acid-free margarine are preferable to butter, stick margarine, or shortening.

Vegetarians are able to get adequate amounts of essential amino by eating a variety of plant proteins. Vegetarian athletes are at risk for being deficient in vitamins B_{12}, D, riboflavin, iron, zinc and calcium.

The following are the recommended serving sizes for protein:

- 2 to 3 ounces of cooked lean meat, poultry, or fish (a portion about the size of a deck of playing cards)

- 1/2 cup of cooked dried beans
- 1 egg, 2 tablespoons of peanut butter, or 1 ounce of cheese

Formula for Energy (kcal) and Fluid (ml) Calculation—Both Energy and Fluid Requirement Approximate

m Sedentary person

Weight (kg) x 25

- Moderately active person

  Weight (kg) x 30
- Active person (endurance athlete)

  Weight (kg) x 40
- Underweight person

  Weight (kg) x 45

(kg = lb/2.2)

1994 Dietary Supplement Health and Education Act

- Definition of a supplement: Any product that contains vitamins, minerals, amino acids, herbs, botanicals or a concentrate, metabolite, constituent, extract or combination of any of these ingredients.
- Removed dietary supplements from FDA regulation on the front end
- FDA must prove a supplement is dangerous before its sale can be prohibited

For information about how many servings are recommended, see the food guide pyramid (Fig. 9.2).



Fig. 9.2 Sports nutrition pyramid

9.3 The Concept of "Small But Healthy"

David Seckler, an economist, hypothesized that majority of the individuals who are categorized as malnourished may only be 'small, but healthy people'. They may be having optimum size with respect to their environment and genetic makeup. As early as 1959, Jelliffe had reported that those who are 'light in weight, short in stature with normal body proportions, and subcutaneous fat are healthy young children.'

The main anabolic hormones are:

- Insulin-like growth factors
- Insulin
- Growth hormone
- Modulated by steroids, thyroxine, sex hormones, receptors and their binding proteins.

Hypertrophic response by the fat cells makes you big circumferentially, whereas by the bones prior to epiphyseal fusion makes you grow taller.

Rapid growth is initiated by Insulin/IGF/GH axis which initiates cell division and inhibits apoptosis. Tall people have at least additional one trillion cells/ 10 cm increase in height. They are noted to be at high risk of developing skin cancers.

The risk factors for non-communicable diseases are:

- Higher birth weight
- Length and weight at one year
- Early catch-up growth
- Adiposity rebound
- Parameters attained at puberty

Recently three systematic reviews have described the consistent association between rapid infancy growth and subsequent obesity risk in childhood and later life. Recent studies have also described the very early development of insulin resistance in small for gestational age children who show catch-up growth, and this insulin resistance may, in turn, adversely affect body composition, growth and puberty. Long-term randomized trials of growth hormone therapy, however, remind us of the persisting short stature and significant adult height deficit in untreated children without early spontaneous catch-up. Even in modern societies with low rates of childhood infection and mortality, the small-for-gestational-age infant may face a dilemma over whether or not to catch up. Current nutritional strategies that promote catch-up growth should include some monitoring of weight-for-length and adiposity, and the concept of 'healthy catch-up growth' should be the goal of future research.

Growth potential is not something one can speed up and slow down. Those of us who suffer malnutrition in early life better remain thin, and short. Attempts to augment growth will result in undue fatness. Early catch up growth and adiposity rebound are examples of this.

SECTION 10

Child Development and Related Issues

"Your genes are the building blocks, your nutrition is the mortar & your environment is the architect that shape your destiny."

In spite of the constitutional provision of a right for optimum survival, many children are not getting the benefit of services like primary health care, adequate nutrition, proper environment, stimulation, education, recreation and tender loving parental care. Even though the precise mechanism is not clear, it has been established that growth, development and intelligence of such children are at risk. Structural and biochemical changes have been demonstrated in the growing brain due to malnutrition, in autopsy as well as in animal studies. Growth faltering, developmental retardation, intellectual impairments, reduced nerve conduction velocity and electrophysiological changes have been demonstrated in malnourished children. Such screening tests are of value only if they result in appropriate intervention strategies.

The expression of the endowed genetic potential for growth and intelligence is influenced by the interplay of nutritional, environmental and socioeconomic factors (Fig. 10.1). The role of malnutrition in reducing mental development is difficult to separate from the other associated retarding social and environmental factors (Fig. 10.2). Because multideprivation including malnutrition has been identified as the most important constraint in the total development of children, the mode of treatment has to be multidisciplinary including primary health care, nutritional supplementation, developmental stimulation, psychosocial support and socioeconomic advancement.

However, the existing child welfare programmes are not sufficient to meet all the needs and demands. It is known that the first two years of life include a period of rapid brain growth and development. **During the first 2 years, body growth becomes 20% of that of the adult and brain growth becomes 80 per cent and myelination becomes almost complete.** Hence any programme aimed at total development should start before the age of two years. Single point interventions like primary health care and nutritional supplementation have failed to deliver the

desired outcome. There are some reports that stimulation along with nutritional supplementation may be a better choice. A composite stimulation package including nutritional input, developmental stimulation, primary health care and psychosocial support is needed for optimum development and quality of survival.

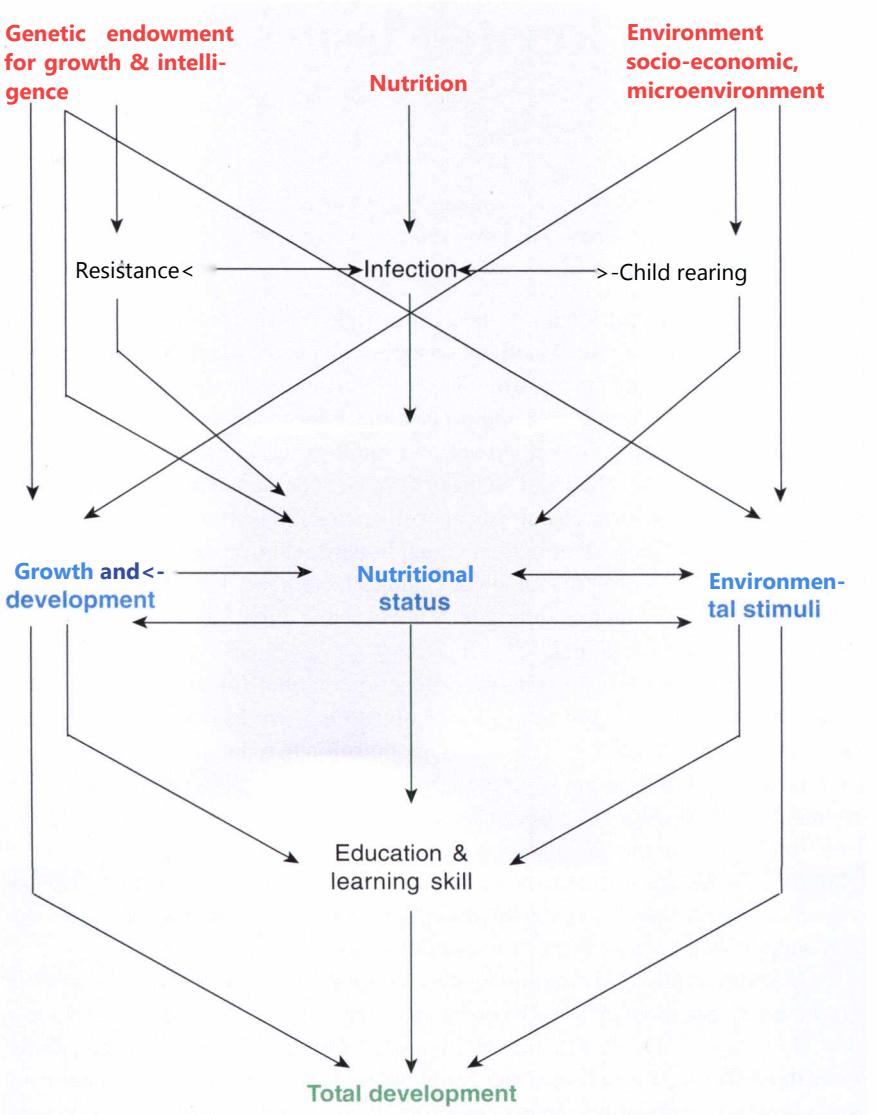


Fig. 10.1 Interplay of various host, nutritional and environmental factors in total child development

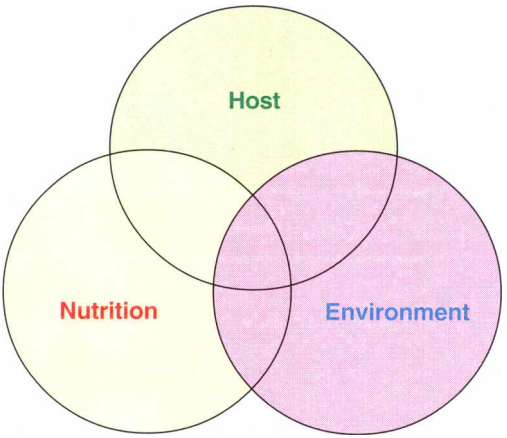


Fig. 10.2 Interaction between host, nutrition and environment

10.1 Normal Development

Development occurs in four areas, namely, gross motor, fine motor adaptive, language and personal-social. The developmental milestones are given in Table 10.1.

1. Gross Motor Development

This signifies the control of the body in a cephalocaudal order and development of locomotion. The neonatal reflexes have to disappear for the appearance of certain developmental milestones.

A young infant is first observed in the ***supine and prone position*** and ***is pulled to sitting position*** and is then examined in ***ventral suspension.*** In ventral suspension, the baby is placed in the prone position supported on both the palms of the examiner.

A newborn baby clears the nose in the prone position and lies with pelvis at a higher level. A newborn has a head lag on pulling to sitting position. In ventral suspension, the head and the limbs are flexed.

By three months, the head lag is minimal when pulled to sitting position. The baby can raise the chin in prone position and the pelvis is flat. On ventral suspension, the head and the hip are in line with the body.

By four months, the baby has head control when held up. In prone position, the baby can raise the head and shoulder and can stretch the arms. By 5 months,

Table 10.1 Developmental milestones of the child

AGE	MOTOR	ADAPTIVE	LANGUAGE	SOCIAL	VISION & HEARING
1 Month	Head lifts momentarily in the plane of body, ATNR predominates, complete head lag, momentarily holds chin off couch in prone position, sitting position -holds head up momentarily			Beginning to smile	Follows moving object, less than 90° turns his head to rattle
2 Months	Head in plane of body, head lag partial, sitting position - head bobs, plane of face at 45° by raising chin recurrently	Hands predominantly closed	Coos	Social smile +	Follows objects 180°
3 Months	Lifts head and chest, head above plane of body, moderate head control, bears weight on forearms	Reaches toward object and misses, hands open, no more grasp reflex, hand ure regard present, pulls at his dress	Says aah or naah, vocalizes with pleas-	Sustained social contact	Binocular vision develops by 3-6 months
4 Months	No head lag, head steady, enjoys sitting with full truncal support, when erect pushes with feet, ATNR none, holds head & chest off couch	Reaches and grasps object and brings to mouth, approaches object and overshoots, hands in midline & plavs with them, pulls his dress over the face, plays with rattle when kept in hand		Laughs out loud, excited at sight of food & breast	Turns head towards a sound at the same level at 3-4 months
5 Months	Full head control	Able to grasp objects deliberately, no more hand regard. crumples paper, plays with toys, bidexterous grasp	Smiles at self in the mirror	When he drops rattle looks to see where it has fallen	Turns head towards a sound below the level at 5-6 months

contd.

6 Months	Holds chest & abdomen off the couch, weight bearing on extended arms, rolls over from prone to supine	Grasps his feet & brings to mouth, holds bottle, if he has one cube in hand drops it if another is offered	Smiles & vocalizes at self in the mirror, monosyllabic babble	When he drops the rattle he tries to recover it, may protrude tongue as imitation, may show stranger anxiety, laughs when head is hidden in towel in peep-boo game, beginning to show likes & dislikes of food	
7 Months	Rolls over from supine to prone. sits briefly with support of pelvis, weight bearing present. object, bounces actively, weight bearing on one hand	Reaches out for large objects and grasps, transfers uses radial palm, ba rakes at pellet, if he has 1 cube in hand retains it even if another is offered, bangs object on table, takes all objects to mouth, feeds self with biscuit, palmar grasp	Polysyllabic sounds formed, savs da ma	Prefers mother, enjoys mother, pats image of self in mirror. resists if toy is pulled from hand, gastrocolic reflex weakens	Turns head towards a sound above the level at 7-9 months
8 Months	Sits alone, back straight, pulls to standing position, cruises	Grasps object with thumb & forefinger, picks up pellet with assisted pincer grasp, uncovers hidden toy, attempts to retrieve fallen toy, releases object grasped by another person	Savs mama or dada by combining syllables	Responds to sound of name, plavs peek a boo or pat a cake, waves bve-bve. reaches persistently to toys out of reach, responds to "no"	
9 Months	Stands holding on to furniture, may progress if in trying to crawl	Brings 2 cubes together as to compare the sizes &		Puts arm in front of face to prevent mother	

	backwards, sitting - can lean forward	bangs them on the table		from washing face
10 Months	Pulls self to standing position. pulls self to sitting position, crawls with abdomen on the couch	Lets go objects deliberately, picks up pellet neatly, coat sign	Can understand the meaning of some words	Pats a doll, can be placed on toilet seat
11 Months	Creeps - abdomen off ground, sitting - can lean sideways, walks with two hands held, sitting -can turn round to pick up object (pivots), walks sideways holding on to furniture	Will place object in examiner's hand but will not release it, rolls ball to examiner	Savs one word with-meanino	Lets go objects deliberately in order that they will be picked up, likes repetitive play
12 Months	Walks with one hand held, rises independently, bear walking	Unassisted pincer grasp. releases object to person on request, feeds with spoon with spilling	Few words besides mama or dada, 2-3 words with meaning	Plavs simple ball game, may kiss on request, mimicry
15 Months	Stand alone (13 months) Walks alone with broad base & high stepping gait; crawls upstairs, takes several steps sideward	Makes tower of 2 cubes: makes a line with crayon; inserts a pellet into a bottle; constantly throwing objects on the ground, takes off shoes, feeds with spoon without spilling, feeds self managing cup with slight spilling	Jargon: follows simple commands; may name an object	Indicate some needs by pointing; hugs parents; asks for objects by pointing
18 Months	Runs stiffly, sits on a small chair, walks upstairs with one hand held, walks normally, pulls hand as he walks, throws hall	Makes tower of 4 cubes, imitates scribbling, imitates vertical stroke, dumps pellet in the bottle, feeds	Average 10 words, names one or more parts of the body, points correctly to	Feeds self, tells when wet or soiled, clean & dry with occasional accident, carries out

contd.

	without falling	self managing cup without spilling, turns 2-3 pages at a time	1 picture, names 2 simple objects	orders, uses stick to reach toy, dry by day
21 Months	Walks backwards, picks up object without falling, walks upstairs with 2 feet per step	Tower of 6 cubes	Points correctly to 2 pictures, knows 4 parts of body, joins 2 words together, asks for food, drink & toilet	Obeys 3 simple orders
24 Months	Runs well, walks up & down stairs one step at a time 2 feet per step, opens doors, jumps, climbs on furniture	Tower of 7 cubes, circular scribbling, imitates horizontal stroke, turns paces 1 at a time, washes and dries hands	Puts 3 words together, talks incessantly, listens to stories with names 2 objects, tells a simple sentence	Handles spoon well, pictures, helps to undress, obeys 4 simple orders, dry at night, wears socks or shoes
30 Months	Goes upstairs with alternating feet, jumps with both feet, walks on tiptoe when asked	Tower of 9 cubes, makes vertical & horizontal strokes but generally not a cross, imitates circular stroke, forming closed figure, holds pencil in hand	Uses pronoun I. knows full name, names 3 objects, repeats 2 digits	Helps put things away, pretends to play, beginning to take interest in sex organs
36 Months	Rides tricycle, stands momentarily on single foot, goes upstairs with 1 foot per step & downstairs with 2 feet per step, jumps of bottom step	Tower of 10 cubes, copies a circle & imitates a cross (copies a cross by 3% years), beginning to draw spontaneously or on request	Counts three objects, repeats 3 numbers or a sentence of 6 syllables, constantly asking questions, knows some nursery rhymes, vocabulary = 250 words	Knows age & sex, parallel play present, washes hands, helps in dressing & does by self if helped with the buttons, postpone toilet movement

contd.

48 Months	Hops on 1 (oot. throws ball overhand, climbs well, goes downstairs with 1 foot per step	Copies a square, draws a man with 2-4 parts, names longer of the 2 lines, uses scissors, tripod grasp	Counts 4 numbers correctly, tells a story, obeys 4 commands, tells tall stories	Plays with children, role playing present, goes to toilet alone, right-left discrimination, imaginative play with a doll, can button clothes fully
60 Months	Skips	Names the heavier object, copies a triangle	Names 4 colours; counts to 10, distinguishes morning from afternoon, repeats 4 digits	Dresses and undresses; domestic role playing; asks questions regarding meaning of words
66 Months		Copies a diamond	Repeats 5 digits	Knows number of fingers, names week days, names 4 coins

Source: Nelson Textbook of Pediatrics; & Illingworth, The Normal Child

AMIEL TISON ANGLES

Age	Adductor angle	Popliteal	Dorsiflexion	Scarf sign
0-3	**40-80**	**80-100**	**60-70**	**Elbow does not cross the midline**
4-6	**70-110**	**90-120**	**60-70**	**Elbow crosses the midline**
7-9	**110-140**	**110-160**	**60-70**	**Elbow goes beyond the anterior axillary line**
10-12	**140-160**	**150-170**	**60-70**	**Elbow goes beyond the anterior axillary line**

CDC GRADING HEAD CONTROL (4 MONTHS)		SITTING (8 MONTHS)	STANDING (12 MONTHS)
I	Head erect & steady momentarily	Sits momentarily	Stand momentarily holding on to furniture
II	Dorsal suspension - lifts head along with body	Sit 30 seconds or more forward leaning	Take few steps while both hands supported
III	Prone position - elevates on arms, lifting chest	Sit with child's back straight	Can stand alone with legs wide apart
IV	Holds head steady while mother moves around	While sitting can turn around	Come to standing position with support of stool
V	Head balanced at all times	Raises to sitting position without support	Without support can take few steps

the baby bears weight on the forearm in prone position and can bear weight on standing. By 6 months, the baby bears weight on hands in prone position. Rolling over is noted around 5 months; but it is not a constant milestone. Some may skip this milestone. By 7 months, the baby can sit alone and lean forward and also bounce on standing.

By 8 months, the baby crawls on the abdomen. By 10 months, the baby creeps with the abdomen off the ground. By 10 months, most of the babies pull to standing posture and walk holding on to a piece of furniture. This is called *'cruising*

By 1 year, the baby can take a few steps. By 15 months, the baby can walk sideways and backwards. **The baby can run by 18 months.** By 24 months, the baby can climb stairs two feet per step and the baby can also kick a ball. **By 30 months, the baby can walk on tip toes.** By 3 years, the baby can climb stairs 1 foot per step and by 4 years the baby can climb downstairs one foot per step. **By 5 years, the child can skip.**

2. Fine Motor Adaptive Development

This includes eye coordination, hand and eyes coordination, hand and mouth coordination and hand-skills. The newborn can focus on objects. By 4 months, the baby can follow up to 90° and by 8 months, the baby can follow beyond the midline. By 4 months, the baby can hold objects with both hands and take them to the mouth. By 5 months, the baby can take the feet to the mouth.

By 6 months, the baby can transfer objects and can drink from a cup. By 9 months, the baby can hit two cubes together. **By 10 months, the baby has a** *'pineer grasp'* by approximating the thumb and the index finger and can uncover hidden things.

By 1 year, the baby can release objects on request and can make a **tower of two cubes.** By 15 months, the baby can feed self with a spoon without much spilling.

By 13 months, the baby can turn pages of a book, two to three pages at a time. By 2 years, the baby can turn page by page. By 2 years, the baby can scribble lines.

The baby can imitate **a circle by 3 years, a plus sign by 4 years, a multiplication sign by 5 years, a rectangle by** *4Vi* **years, a triangle by** *5Vi* **years and a diamond by 6 years.** The baby can make a tower of two cubes by **1** year and 3 cubes by 18 months and 6 cubes by 2 years. The baby can make postural adjustments for dressing by 1 year, can wear shoes by two years and can dress and undress by 3 years. An infant has *bidextrous approach* and uses both the hands with equal preference. Hand preference in an infant may point to hemiparesis. *Handedness evolves by two years of age.*

3. Language Development

By 1 month, the baby turns the head towards a sound and baby makes cooing sounds by 2 months. By 3 months, the baby can babble and by 4 months, the baby can laugh aloud.

By 7 months, the baby responds to his/her name. By 7-8 months, baby vocalises monosyllables and by 10 months combines monosyllables, waves bye-bye and understands spoken speech.

By 1 year, the baby speaks 2-3 words with meaning. By 18 months, one can speak 20 words and by 3 years about 250 words. The baby can make a sentence by two years and narrate recent events by 4 years.

4. Personal-social Development

The baby regards faces by 1 month, has social smile by 6 weeks and recognises the mother and caretakers by 3 months. The baby enjoys looking at the mirror by 6 months and imitates others by **1** year. **By 6 months, the baby has stranger anxiety.**

In the first 4 months, the baby has gastrocolic reflex and frequent passage of stool. By 10 months, the baby can be made to sit on the toilet seat and by 18 months, can walk to the toilet. By 2 years, most of the babies are toilet trainable. By 6 months, the baby can drink from a cup and by 15 months the baby can self feed with a cup and a spoon with little spilling. The baby is generally dry by day by 18 months and dry by night by 3 years. By 3 years, the baby has bowel control. By 3 years, the baby also has gender identity. By 2 years of age. baby refers to self as 'I'.

10.2 Developmental Assessment

Growth and development are sometimes used interchangeably. But **growth** implies *increase in size of organs and body* and *development implies differentiation and maturation of function.* The former indicates quantitative growth and the latter indicates qualitative growth. Development is influenced by the physical, emotional and social environment. In early childhood, cognitive growth and development are difficult to differentiate from neurologic and behavioural maturation. In later childhood, it can be measured by communicative skills and cognitive abilities.

The development of each child is unique and the pattern of development may be profoundly different for each child within the broad limits of 'normality'.

1. Factors Affecting Child Development

Various factors determining the development of the child are detailed below.

a) *Genetic factors:* Even though genetic factors are thought to be the final limits of biologic potential, they are intimately interwoven with the environment.

b) *Physical factors:* Prenatal as well as postnatal physical insults affect growth and development.

c) *Nutritional factors:* Nutritional factors influence growth and development. Chronic malnutrition causes stunting of physical growth. Prenatal and early postnatal malnutrition affects development and reduces the ability of the individual to adapt to the environment.

d) *Emotional factors:* Emotional factors like position of the child in the family, the child rearing practices in the family and community etc., affect growth and development.

e) *Sociocultural factors:* Sociocultural factors either limit or expand the range of behaviour of children. The schedule for acquisition of skills, such as sitting, walking etc., which were earlier thought to be the result of maturation alone are now found to be influenced by the conventional expectations. Socioeconomic factors are also reflected in the nutritional status of the child.

2. Interaction between Various Factors and Child Development

There is an interplay between genetic, nutritional and environmental factors that influence growth, development and intelligence. However, the contribution of each of them is difficult to separate and evaluate. The effect of malnutrition in reducing the intellectual achievement is difficult to separate from other associated retarding social and environmental factors. Genetic endowment, nutrition and environment are the importtant determinants of the overall development.

Some claim that genetic factors account for 80 per cent of development and only 20 per cent is accounted by the other factors. But some others are of the opinion that this is an underestimate of the other factors. *The child's ultimate intelligence is the result of the interaction between host, nutrition and environment.*

Psychosocial factors have been shown to influence growth and intelligence in various studies. Good nutrition in the pre-school years will enable the child to make better use of the available environmental stimulation in order to achieve the endowed genetic potential for growth and intelligence. Various studies have shown better growth and development in children from high socioeconomic status and reduced growth and development in children from low socioeconomic status. It has been concluded that children from privileged communities show the expression of the genetic potential similar to that of the western standards like the NCHS standards. The NCHS standards are the highest reference standards ever set.

The sequelae of deprivation alone are found to be similar to that of PEM. The retarding influences are multifactorial and that may be the reason why single

point interventions like nutritional supplementation and primary health care have failed to deliver the desired outcome. A multidisciplinary approach that includes social and psychomotor stimulation has been suggested as a better choice. As growth and development go hand in hand, nutrition and stimulation are the two components to be combined for total development of children.

3. Assessment Tools

Maturity, behaviour and mental functions can be evaluated by the assessment of development and intelligence. The **neurodevelopmental status** of children should be assessed in order to understand the deviation, impairment or retardation and to plan appropriate recommendations and interventions. Observation based on casual examination should be interpreted with caution because a child who is irritable, hungry, sleepy or ill, does not perform at his or her expected level. A future examination may be needed in such children. For infants born **prematurely,** the developmental level may be compared to **'corrected chronological age'** during the first two years of life, i.e., obtained by reducing the period of prematurity from the chronological age.

The role of developmental assessment is to understand whether the child is progressing as per norms set by large majority of children of the same age group. However, it is **not a predictor of future IQ** and any deviation from normal should be brought to the notice of the parents in a reassuring way. The developmental tests mainly measure maturity and behaviour in four functional areas, namely, gross motor, fine motor adaptive, language and personal-social. The four functional areas are closely related and overlapping. But in defective development, they show some dissociation. A child may be advanced in one area and retarded in the other. Thus each function must be evaluated separately. A battery of developmental tests are available.

a) *The Denver Developmental Screening Test (DDST)'.* The DDST was originally designed as a screening test (Frankenburg, 1967). It is now being increasingly used as a tool for routine developmental assessment (Glascoe, 1992). The Denver developmental reference chart is suitable for quick assessment of all the four areas of development in children up to 6 years of age (Frankenburg, 1981). This will take 10-25 minutes only.

b) *Gesell Developmental Schedule*: This measures the four functional areas of development in children up to five years of age. It will take 30—40 minutes. It is more concerned with the diagnosis and evaluation of abnormalities than the attainment of various milestones (Gesell et al.. 1962).

c) *Bayley Scale of Infant Development (BSID) :* This scale provides the motor scale, the mental scale and the infant behaviour record in children up to 30 months of age. An overall developmental index is based upon the combined scores on the motor and mental scales. It takes approximately 30-60 minutes (Bayley, 1969).

d) *Baroda Developmental Screening Test:* This is a screening test based on BSID, Baroda norms (Phatak et al., 1984). The Baroda norms were standardized on Indian children. It evaluates motor and mental development and infant behaviour. An abbreviated BSID is also available for follow up of high risk neonates (Phatak, 1990).

e) *Trivandrum Developmental Screening Chart (TDSC):* Based on 17 selected items from BSID Baroda norms, the test was designed in the Child Development Centre, SAT Hospital, Trivandrum, for children up to 24 months of age. This simple tool can be administered and interpreted by any person with minimal training. It takes 5 to 7 minutes only (Nair et al., 1991).

f) *Developmental Screening Test (DST):* This is a simple scale that can be administered up to the age of 15 years. It was standardized on Indian children (Bharathraj, 1983).

g) *Brazelton Neonatal Behavioural Assessment Scale.* This scale is based on the observation of the baby and the response to 20 primitive reflexes (Brazelton, 1981).

h) *Developmental Observation Card (DOC):* This was designed in the Child Development Centre of SAT Hospital, Trivandrum, based on the observation that large majority of developmental delays can be identified using four key milestones, namely, social smile, head holding, sitting alone and standing alone and taking a few steps with or without support that generally appear not later than 2,4, 8 and 12 months respectively (Nair, 1992).

However, these developmental tests have very low predictive value regarding future IQ and have several limitations. The cross-cultural use of these scales is also not often ideal. Indian children have motor skills ahead of others. But the language and personal-social skills are often behind. Developmental assessment furnishes information on the stage of development and gives the parents a chance to perceive in what stage the child is and the degree of retardation if there is any. But the selection of the test is very important.

4. Developmental Quotient (DQ)

DQ is computed using the following formula:

$$\frac{\text{Developmental age}}{\text{Chronological age}} \times 100$$

10.3 Assessment of Intelligence

This is generally done in children above three years of age by a trained clinical

psychologist. Intelligence tests measure several brain functions including auditory memory, visual-spatial capability and receptive and expressive language. **The Intelligence Quotient (IQ) is computed using the following formula:**

$$\frac{\text{Mental age}}{\text{Chronological age}} \times 100$$

The calculated IQ is an average of various mental functions measured. Alfred Binet of Stanford University, who designed the first IQ test did not intend it as a measure of cognitive ability, but as a simple tool to predict school performance. IQ may not reflect the optimal cerebral function or potential of the individual. It is insensitive to the adverse effects of sociocultural and environmental factors that affect the potential. Adaptations to the environment, social skills etc., also need to be assessed. As age advances, most children make satisfactory adaptation in the community, both vocationally and socially. In spite of all the limitations, IQ remains the major diagnostic criterion used to deal with the retarded children. The levels of intellectual retardation according to IQ are given in Table 10.2.

Table 10.2 Various levels of retardation according to IQ

Level of retardation	IQ	Remarks
Borderline/Average	70-85	Vulnerable to educational problems
Mild/Educable	50-70	Often need special classes
Moderate/Trainable	35-50	Trainable in workshop setting
Severe	20-35	Trainable for self-care skills
Profound	Below 20	Need custodian care

Even the retarded children should be given stimulation and educational services either in regular classes with extra resources or in special classes. **'Mainstreaming'** is the term used for including the retarded children in the regular classes. Education does not mean studying academic subjects alone, but also means learning self-care activities and social skills. The various available intelligence tests are detailed below:

a) *Stanford-Binet Intelligence Scale*: This takes into account verbal ability, perceptual skills, short-term memory, and hand and eye coordination. It takes 45-60 minutes (Terman et al,, 1937).

b) *Binet-Kamat Test:* This is an Indian adaptation of the Stanford-Binet scale (Kamat, 1967). This is also available in Hindi (Kulshrestha, 1971).

c) *Wechsler Intelligence Scale for Children (WISC):* It has a verbal scale and a performance scale. It can be administered in the 5 to 15 years age group and will take 45-60 minutes (Wechsler, 1949).

d) *Malin Intelligence Scale for Indian Children*: This is an Indian adaptation of WISC. Since the items are mostly influenced by formal schooling system, it may not give the real capabilities in non-school going children (Malin, 1969).

e) *Goodenough's Draw-A-Man Test for Indian Children*: This can be administered in children of 3 to 13 years of age group. The child is asked to draw a man and he/she receives 1 point for each of the items present in the drawing. For each 4 points, 1 year is added to the basal age of 3 (Phatak, 1987).

I) *Tests of Cognitive Functions :* Based on Piaget's theory of intellectual development, cognitive functions can be measured in four major stages of development, namely, **sensorimotor stage** (0-2 years), **preoperational stage** (2-7 years), **concrete operational stage** (7-11 years) and formal operational stage (above 11 years). The cognitive development of Indian children has been shown to follow a similar pattern. However, this is not an estimate of IQ. Intelligence tests when culturally appropriate, measure major cognitive and mental abilities. The Indian modification of WISC, Stanford-Binet test, Draw-a-Man test etc., are suitable for Indian children. Nutritional status, socio-economic factors, and maternal IQ etc., have been found to influence growth, development and intelligence. Malnutrition and other retarding environmental variables also adversely influence the cognitive development and school performance.

These IQ ranges are based on the American Association on Mental Deficiency (AAMD) terminology and classification and the American Psychiatric Society Diagnostic and Statistical Manual III (DSM III). According to the WHO International Classification of Diseases (ICD), borderline retardation is not included.

10.4 Health Care Delivery Systems

The interventions aimed at total development of children include nutritional inputs, medical inputs and developmental stimulation. Adequate nutrition, housing, basic sanitation, healthy lifestyle and protection against environmental hazards and communicable disease come under health care delivery systems. Health services are designed to meet the health needs of the people through the use of available knowledge and resources through a health care network and to reach every one by 2000 AD, i.e.. **Health for All** (the Alma Ata Declaration).

Health care services are organized at three levels, primary, secondary and tertiary. Each level represents different types of care.

1. Primary Health Care

This is the first level of contact of the people with the national health system and is also termed essential health care. At this level, most of the health problems can be dealt with and resolved and so this is the most effective care available to one and all. In India, it is provided through the network of primary health centers, sub centers, public health staff, community volunteers, trained traditional dais etc. This health team bridges the cultural and communication gap between the rural people and the organized health sector. It functions at the grass root level.

a) *Definition of primary health care.* The primary health care was regarded as synonymous with basic health services, first contact care and services provided by generalists. It was redefined in the Alma Ata conference in 1978. **Primary health care** is defined as *'essential health care made universally accessible to individuals and acceptable to them through their full participation and at a cost the community and the country can afford'* (WHO, 1978). The concept of primary health care has been accepted by all countries as the key to the attainment of Health for All by 2000 AD. In the 30th World Health Assembly. **Health for All** has been defined as **attainment of a level of health that will enable every individual to lead a socially and economically productive life. Health is not merely absence of disease, but complete physical, mental, spiritual and social well-being.**

b) *Components of primary health care.* The eight components of primary health care are the following:
   i) Maternal and child health services including family planning.
   ii) Education regarding health problems and their prevention.
   iii) Drugs that are essential.
   iv) Immunization against major killer diseases.
   v) Control of communicable and common diseases and injuries.
   vi) Indigenous or endemic diseases control.
   vii) Nutrition including promotion of supply and intake.
   viii) Environmental sanitation including safe water.

Considering the first letters of the eight components, primary health care can be equated to MEDICINE. Primary health care measures like health promotive, preventive and curative services act as very good intervention strategies. Services like immunization, care during illness, referral services, medical check up etc., are available to all the children.

2. Secondary Health Care

This is the first referral level and is also termed intermediate health care. The district hospitals and community health centers are included in this. They are

called **first referral units (FRUs).** These centres have better infrastructural facilities and skills.

3. Tertiary Care Level

This is the final referral level and is also termed **specialized health care.** The medical colleges, national institutes, regional centers and other apex institutions are included in this. These centres have well-equipped critical care facilites and expertise.

Community participation is recognized as a major attribute in the approach to the health care system. The shift from medical care to health care and from urban population to rural population are the other attributes in the modern health care system.

10.5 Nutritional Inputs for Intervention

Nutritional supplementation, nutritional therapy, nutrition education and nutritional rehabilitation are the most important nutritional inputs.

1. Nutritional Supplementation/Supplementary Feeding

It is given to children under five years of age through various feeding centers and to school children through the school feeding programmes. On an average, supplements that contain around 300 kcal and 10-12 g protein per child is given 300 days a year. The average calorie gap in preschoolers has been assessed to be around 400 kcal. Both governmental and non-governmental agencies are involved in the various feeding programmes.

2. Nutritional Therapy

It is given to children with PEM. Treatment of PEM involves **resuscitation** of the child from-life threatening medical emergencies, **restoration of** nutritional status and **rehabilitation.** Resuscitation is undertaken in a hospital setting. Restoration and rehabilitation are either centre based or home based. Restoration of weight for height is achieved by nutritional therapy. Rehabilitation is achieved by continuing nutritional supplementation and primary health care. In nutritional therapy, therapeutic diet is given by administering 150 to 200 kcal and 3 to 4 g of protein per kg per day to the malnourished child. (Also refer section on PEM in section 6).

3. Specific Nutrient Supplementation

Specific nutrients like oil, vitamins and minerals are being supplemented,

a) *Oil supplementation:* It is a very good method to overcome the calorie gap. **Coconut oil** is found to be most effective compared to other oils. This is due

to the better absorption of **medium chain triglycerides** present in coconut oil. Moreover, it has a low omega-6 to omega-3 fatty acid ratio. High omega-6 content has been reported to produce a lot of free radicals that cause angiotoxicity, impaired immune function, impaired glucose tolerance, increased platelet aggregation, albuminuria etc., especially in diabetic patients. **The desired ratio of omega-6 to omega-3 fatty acid is less than 5:1.** In sunflower and safflower oils, the ratio is almost 150:1 and omega-3 fatty acids are deficient.

b) *Vitamin supplementation:* Vitamin A prophylaxis by administering 5 doses of two lakhs units of vitamin A concentrate every 6 months to infants and young children has been found to be effective in preventing blindness as well as morbidity and mortality. **Vitamin** A **prophylaxis was first started in Indonesia.** Vitamin A supplementation leads to boosting up of immunity. It stabilizes the lysosomal membrane and augments immune responses and potentiates interleukin-I response. Vitamin A deficiency leads to binding of the bacteria to the respiratory epithelial cells. Vitamin A deficiency has been found to predispose to acute respiratory infection and diarrhoea. Vitamin A supplementation during measles has been shown to reduce morbidity and mortality. Other vitamin supplementations are usually undertaken when there is associated deficiency and malnutrition.

c) *Mineral supplementation:* Iron supplementation is undertaken when there is anaemia. Iron and folic acid tablets are given for anaemia prophylaxis to underfive children and pregnant and lactating mothers. Zinc supplementation has been found to be effective in those with PEM and in LBW babies. Iodine fortified salt (IFS), and iodine and iron double fortified salt (DFS) are yet other interventions.

d) *Micronutrient initiative:* The micronutrients that are currently supplemented are vitamin A, iron, folic acid, iodine etc. Zinc is supplemented in some target groups like **LBW.** malnutrition and diarrhoeal disease. **Spirulina** is a promising biological supplement for most of the micronutrients and antioxidants.

4. Nutrition Education

This is undertaken through discussion groups and mother's meetings. Main emphasis is given to breastfeeding, weaning, nutritional requirements of children, diet during illness and also methods to achieve adequate nutrition in the presence of various constraints like low income, lack of time and fuel, food fads, dietary habits of the family etc. Combined **nutrition and health education (NHE)** is another approach.

5. Nutritional Rehabilitation

The concept of nutritional rehabilitation was first introduced by Bengoa in 1967. **To get well and to keep well** are the basic ideas of **rehabilitation.** It can be centre

based or home based. Home-based rehabilitation utilizing the existing child welfare programmes is most successful.

A combined approach involving both nutrition education and nutritional supplementation and education has been found to be better than supplementation alone. A weight gain of 10 to 15 g/kg/day, or at least 500 g per week which is five times more than that of a normal child for the same height and ten times more than that of a normal child of the same age, is expected during nutritional therapy. Satisfactory weight gain has been found in various nutritional intervention trials. In one to three months, the child is expected to attain weight for height which is considered clinical recovery. Studies from Guatemala, Mexico and United States have also shown that nutritional supplementation produced a small but statistically significant effect on the different aspects of cognitive development.

10.6 Developmental Stimulation

Early intervention by stimulation is available for infants at risk for developmental delay. Early stimulation has shown immediate positive effect. *The objectives are to stimulate the child through the normal developmental channel and to prevent developmental delay.* As growth and development go hand in hand, nutrition and stimulation need to be clubbed. Children with severe malnutrition functionally isolate themselves and fail to interact with the environment. The psychosocial as well as physical development of children are to be taken care of. Stimulation has been found to be effective in malnourished children. A home-based stimulation is more cost effective. Age-appropriate tasks are best suited. In children below two years, search for hidden objects, verbal and gestural imitation and scribbling and in children above two years, the concept of size and shape, interaction with mother and environment, action songs, finger and toe games, sorting and matching of objects and pictures etc., are found to be most suitable.

Studies from Spain, Beirut, Chile, Columbia, Jamaica and Peru have shown various stimulation models to be effective. The stimulated group was provided with a colourful environment, plenty of toys, warm nursing care, sensori-motor stimulation, psychological stimulation, physical and emotional contacts, affective stimulation, structured play, physical exercise, home education and frequent home visits. However, the knowledge regarding stimulation of malnourished children is not sufficient. The site of intervention, the curriculum, the duration and the cost effectiveness need further clarification.

An account of a **composite stimulation model** developed in the Nutrition follow-up clinic of the Department of Paediatrics, SAT Hospital, Medical College, Trivandrum, is given below.

Planning and Execution of Comprehensive Rehabilitation

The Composite Stimulation Package (STIM)

This includes the following attributes:

A. Medical
- Primary health care
- Immunization
- Treatment of intercurrent infections
- In-patient services
- Periodic deworming

B. Nutritional
- Dietary evaluation
- Nutritional assessment and monitoring
- Nutritional supplementation
- Specific nutrient supplementation
- Nutrition education

C. Stimulation
- Developmental evaluation
- Developmental information
- Individualized tasks for catch-up
- Play therapy
- Motor co-ordination tasks
- Training activities of daily living

D. Psychosocial
- Social interview
- Psychosocial counselling
- Decision making
- Child rearing skills

A. Medical

1. *Primary health care:* Essential services that are universally available to all children, through the primary health centre network should be made available. These are the services included in the National and State programmes, namely, health promotive, preventive and curative services, medical check-up and necessary investigations.
2. *Immunization:* The children should be administered all the immunizations appropriate for the age.
3. *Treatment of intercurrent infections:* All intercurrent illnesses are to be managed by early diagnosis and appropriate treatment of infections like diarrhoea, acute respiratory infections, scabies etc.
4. *In-patient services:* Those who require hospitalization should be referred for admission and given specialized care.

5. *Periodic deworming:* Deworming should be done initially and repeated quarterly or half yearly or earlier if indicated. In infants, initial deworming should be done around 1 year of age.

B. Nutritional

1. *Dietary evaluation:* This can be done by a 24-hour dietary recall method and considering the calorie and protein intake and also by gathering information regarding breastfeeding and weaning practices and reasons for not breastfeeding if not breast fed etc.
2. *Nutritional assessment and monitoring:* This includes assessment of nutritional status by clinical signs and anthropometry, growth monitoring by periodic measurements and conveying of the same to the mother.
3. *Nutritional supplementation:* This includes food supplementation from the ICDS and Nutrition Clinic in addition to the food from the child's home. Breastfeeding should be encouraged in all. Breastfeeding is advised to be continued till two years of age.
   a) ***Food supplementation from ICDS:*** All the children are to be referred to the nearest ICDS Anganwadi from where each child is expected to get around 300 kcal and 10 g protein/day and the severely malnourished child is expected to get double the share. Also ensure **family pot feeding** at home.
   b) ***Food supplementation from Nutrition Clinic:*** In the model study, during each visit to the Nutrition Clinic, each child received a palatable ready-to-mix protein-energy mix **(SAT Mix) 1** kg per month during the first 3 months and then *Vi* kg per month throughout the study. This was given in installments so that during each visit the child received a ration. SAT Mix was prepared by adding preroasted and powdered rice, wheat and black gram to powdered sugar in the ratio 1:1:1:2 (100 g = 380 kcal and 8 g protein). Each serving was advised to be **about 6 rounded teaspoons** (30 g) **of SAT Mix made into a semisolid paste using hot water in order to supply 125 Cal and 2.5 g protein.**

      The severely malnourished children should be advised 200 Cal/kg/day and 4 g/kg of protein/day. As this approximates the Recommended Dietary Allowances (RDA) for the age (ICMR) the mother or the worker can be tuned to give the RDA for the age in 6-8 feeds/day and to mobilize the extra calories either from home or the Anganwadi.
4. *Specific nutrient supplementation:* This includes oil, vitamins and minerals supplementation.
   a) ***Oil Supplementation***: All the mothers should be advised to give extra oil to their children. Coconut oil that has the advantage of medium chain triglyceride and an optimum omega-6-omega-3 fatty acid ratio of < 5:1. One teaspoonful three times a day can be given either along with semi-

solids and hot rice or as such care should be taken to prevent aspiration of oil.

b) *Vitamin A prophylaxis:* All the children should be given 2 ml of vitamin A concentrate containing 2 lakhs of vitamin A every six months. Those with vitamin A deficiency should be given extra doses. (Also refer Section 4.1)

c) *Iron and folic acid supplementation:* Tablets containing 20 mg elemental iron and 100 mg folic acid can be given daily during the first three months for anaemia prophylaxis. Those with anaemia should be given 6 mg/kg elemental iron/day or two tablets per day for three months.

d) *Others:* Children are also given a multivitamin syrup which includes minerals like zinc, 1 teaspoon twice daily during the first three months.

5. *Nutrition education:* This is achieved by focus group discussions arranged for mothers. The discussion includes the following:

a) Demonstration of food items as models and charts.

b) Points to be remembered while cooking, storing and buying food.

c) RDA for the various age groups and physiological conditions like pregnancy, lactation etc., with the help of nutrition information charts like the ten commandments (refer Section 1.3).

d) Role of protective foods like vitamins and minerals and information regarding available food supplementation programmes.

e) Cooking demonstration and advice to use locally available and cheap food items.

f) Role of proper and prompt weaning or complementary feeding.

g) Advantages of breastfeeding and the need for exclusive breastfeeding during the first four to six months of age and also the need for continuing breastfeeding during the first two years of life.

The most important principle of the communication is the *development of a 'we' rather than a 'they' relationship between the health team and the parents.* The second principle adopted is the essence of the proverb regarding the methods of learning: *"What I hear, I forget; what I see, I remember; what I do, I know."*

C. Stimulation

a) *Developmental Evaluation:* Developmental evaluation may be done using the Denver Developmental Reference Chart (Frankenburg, 1981). The four functional areas, namely, gross motor, fine motor adaptive, language and personal-social are assessed separately.

b) *Developmental information:* The information obtained by developmental assessment about the stage of development of the child is utilized to give the parents a chance to perceive in what stage the child is and the degree of retardation, if there is any. A child who is advanced in one area of development is often found to be retarded in the other. Developmental information

can be imparted by displaying simplified developmental information charts (Table 10.3 and 10.4) and through discussions regarding the developmental age of a child in comparison to the chronological age.

Table 10.3 Simplified Developmental Information Chart-1 (SDIC-1)

Milestone	Age
Social smile	½ months
Cooing sounds	2 months
Head holding, voluntary grasp	4 months
Reach out, play with rattle and transfer objects	6 months
Pivot on abdomen/able to sit & bang 2 cubes	8 months
Pincer grasp, vocalize amma, acha, tata	10 months
Able to stand/make steps, pick up pellets, speaks 1-3 words, able to feed solids and uses cup and spoon	12 months

Table 10.4 Simplified Developmental Information Chart-II (SDIC-II)

Age (months)	Gross motor	Fine motor	Personal-social	Language
1-3	Head holding while prone	Coordinate head & eyes, follow object	Social smile	Cooing sounds
3-6	Head control	Grasp objects & bring to mouth	Enjoy mirror	Vocalize vowels
6-9	Self sitting	Hand transfer	Play peek-a-boo	Imitate speech
9-12	Self standing, walking with support	Pincer grasp, drink from a cup	Wave bye-bye, specific	Dada-mama
12-18	Self walking	Scribble, tower of 2-3 cubes	Use spoon, house work	3-6 words imitate

contd.

18-24	Run, walk up steps	Imitate vertical lines, tower of 4-6 cubes	Remove and put on cloth	Point to body parts/ pictures
24-36	Broad jump, Copy O, tower pedal tricycle of 8-9 cubes		Play tag, group play	Name picture, give name
36-48	Balance on 1 foot	Copy +, draw a man	Dress & brush Name with no help & colours gender identity	
48-60	Hops, walk up steps with alternate feet	Copy, cut with scissors, throw & ball over head play	Locate persons places, group &	Give name age, tell a story

c) *Individualized developmental tasks for catch-up:* Individualized approach is accomplished with respect to the four functional areas by teaching the mother to do the tasks at home. The objectives are to stimulate the child through the normal developmental channel and to prevent developmental delay. Improvement in one functional area helps the child to improve functions in other areas also.

The method adopted is to teach the mother, make her do, compliment the mother and make her repeatedly do during follow up and thus make her a developmental therapist. Individualized developmental tasks appropriate for the maturity age can be taught with the help of a chart (Table 10.5). The Trivandrum Child Development Centre (CDC) model is found effective. The special sense organs are also to be stimulated.

i) ***Stimulation of vision:*** The visual sense is stimulated by decorating the rooms and surroundings and using mobile and brightly coloured objects.

ii) ***Stimulation of hearing:*** The auditory sense is stimulated by talking, singing and by using music boxes, music sandals, music brushes etc.

iii) ***Stimulation of tactile sense:*** The tactile sensation is stimulated by handling, positioning, stroking etc.

iv) ***Stimulation of vestibular system:*** The position sense is stimulated by picking up, rocking etc.

d) *Play Therapy.* Interaction with the mother and surroundings, playing with toys alone and later in groups are included in play therapy. Various colourful toys, musical toys and pull toys are offered. Mother is instructed to make toys from articles available in and around the house and bring them during follow-up visits and also see what other mothers have made and exchange toys whenever possible.

Play therapy places emphasis on symbolic understanding and stimulation of child development through play. *It is defined as the use of a natural activity with a young child to help the child consolidate the level of development that has been reached and encourage the child to move on to the next level.* Playing with the mother or a familiar teacher will give the true picture of the child's abilities. The therapist should work on the child's attention, communication, comprehension, expressive speech, auditory and visual skills etc., in addition to gross motor and fine motor skills. Most mothers are undertaking this play therapy, but they may want some more new ideas. They also want it to be structured and the results to be evaluated. They should get help and guidance from the therapist. Play therapy is planned according to the age of the child or the developmental stage of the child. In the first year the baby is very distractible and has a very short attention span. In the second year, the child can concentrate for a short time on an activity he likes. This is called rigid attention. He cannot tolerate any interruption by another child or an adult without losing interest. In the third year, his attention is single channelled.

Table 10.5 Individualized developmental tasks for infants 0-12 months of age

0-3 months

1. Carry the baby with head and body fully supported.
2. Put a bright coloured mobile toy above the bed enabling the baby to find it easily.
3. Put the baby in different places at different times and change the position of the baby frequently; on the back, on the tummy, on the sides etc.
4. Encourage the baby to hear sounds, look at faces and objects. Use musical rattles, coloured balls etc.
5. Talk and sing to the baby.

3-6 months

1. Put the baby on the ground and put toys around; allow the baby to move freely.
2. Carry the baby straight with head supported and when sufficient head control is there, carry straight with head unsupported.
3. Put the baby on the tummy and using a rattle moved up & down encourage the baby to practise lifting the head and shoulders.
4. Help and encourage the baby to roll over showing a toy from either side.

contd.

5. Rub a toy across the palm of the baby and encourage to grasp & then encourage to squeeze a toy that makes noise when squeezed.

6-9 months

1. Make the baby sit up and reach out for toys and grasp them.
2. Give the baby one toy and allow to play for a few minutes and then offer another toy to the same hand and encourage the baby to transfer the first toy to the opposite hand instead of dropping it and then give the second toy and allow to play with both the toys.
3. Make the baby stand up on the mother's knees and allow to bounce gently up and down and to support own weight.
4. Allow the baby to enjoy the mirror and vocalize.
5. Call the baby by name.

9-12 months

1. Encourage the baby to crawl by moving a pull toy in front.
2. Now since the baby has some understanding and responds to simple instructions, name and teach body parts, pictures, toys, animals etc.
3. Give containers to fill and empty; allow to poke with fingers into a toy with holes and encourage pincer grasp.
4. Make the baby imitate waving bye-bye, shaking head, clapping hands, uncovering hidden toys etc.
5. Stimulate the baby to pull to standing by keeping a favourite toy on the cot and allow the baby to cruise by moving the toy along the cot.

12-24 months

1. Encourage self walking, walking up steps, running etc.
2. Pile up blocks and make the baby playfully knock them down and help to rebuild again.
3. Give a crayon and a book and allow to scribble and also to turn pages.
4. Teach parts of the body, common objects, animals, birds, numbers etc., and also activities of daily living (ADL) like bathing, dressing, brushing, eating, drinking etc.
5. Allow the baby to kick and throw the ball.

Source: Modified from the Trivandrum CDC model

He can now tolerate some instructions without losing interest. In the fourth year, the child is able to control his own attention. He seeks help and enjoys taking turns with other children. In the fifth year, the child can be taught in a group as in a classroom. Noisy distraction should be avoided.

Generally, two types of play therapy are undertaken. **Directive play therapy** is the type where the therapist is responsible for guidance and interpretation. In **non-directive play therapy,** the therapist leaves the responsibility to the child and closely observes the child. Children are given the opportunity to play out feelings of tension, anger, confusion etc. Symbolic plays like doll plays are most useful. Play materials may include dolls, doll family, doll house with furniture, animals, household items like cups, plates, telephone etc., pictures, crayons, cubes and cars. Girls generally enjoy imitating the mothers, nursing the dolls etc., and boys enjoy driving cars etc. Building towers and bridges are constructional tasks. Matching shapes help in visual perception and concept formation.

A problem child needs more attention and appreciation. A hyperac- tive child, a miserable unhappy child, a severely malnourished child, a handicapped child etc., belong to this group. They tend to have poorer attention span and are easily distractible.

e) *Motor Coordination Tasks:* This includes hand and feet exercises in order to encourage hand and feet skills (Table 10.6).

Table 10.6 Motor Coordination Tasks (MCT)

Age group (months)	Task I	Task II	Task III
6-12	Bang 2 cubes	Pat with palm	Pincer grasp
12-18	Scribble from cup	Use spoon, drink with spoon	Feed doll
18-24	Kick ball	Throw ball	Brush teeth
> 24 Phase I	Hand clapping while counting up to 3	Hand floor tapping while counting up to 3	Imitate vertical lines while counting up to 3
Phase II	Alternate hand clapping while counting up to 4	Thumb to finger tapping while counting up to 4	Copy 0 while counting up to 4
Phase III	Successive thumb to finger tapping while counting up to 4	Feet floor tapping while counting up to 4	Copy + while counting up to 4

f) *Training activities of daily living:* Training of activities like brushing the teeth, self dressing, self feeding especially with spoon and cup are also included in the intervention.

When the mother is not available or compliant, an elder sibling or any other family member may be instructed as in David Morley's Child to Child Programme.

D. Psychosocial

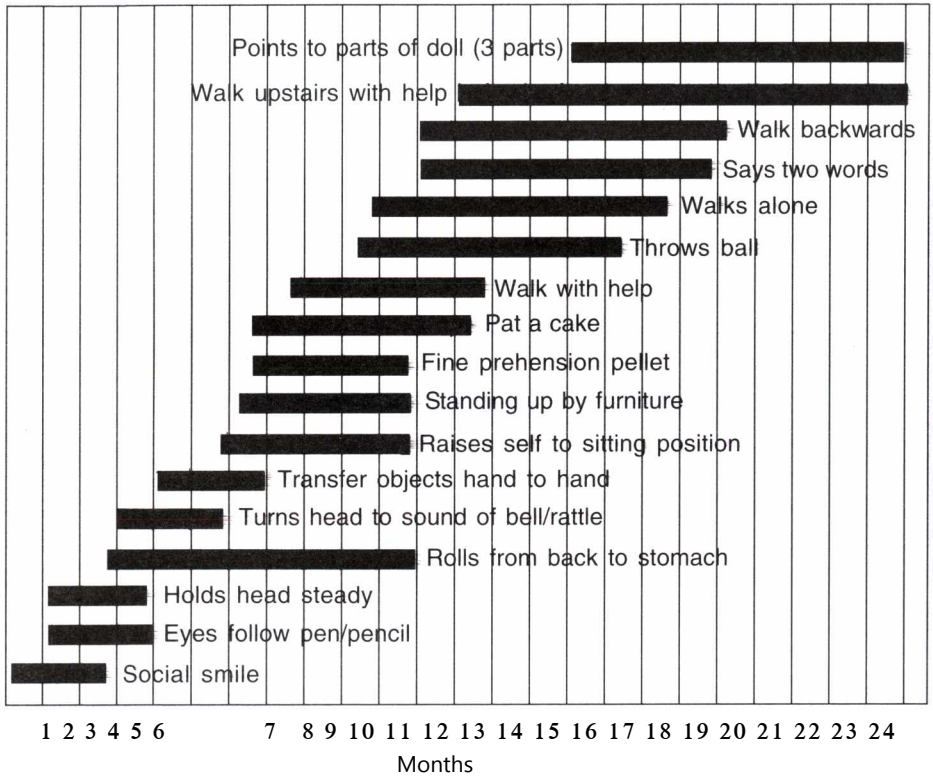
1. *Social interview:* Interview with the parents especially the mother is done in order to understand the standard of living, the pitfalls in child rearing practices, child neglect, family problems, parental disharmony etc.
2. *Psycho-social counselling:* Appropriate counselling is given to the parents with the help of the social scientist and clinical psychologist in the Department.
3. *Decision making:* Support is given for decision making regarding family size, earnings, employment etc.
4. *Child rearing skills or parenting skills:* Skills in child rearing, good attitude towards the child and care during illness are taught to the mother and the family.

Make sure that the baby has vision and hearing. There is physiological variation in walking, speech, tooth eruption etc., and some children may skip certain milestones like rolling over and progress to the next.

Nutritional Management (NUT) vs Developmental Stimulation Package (STIM)

A before and after comparison study utilizing nutritional supplementation and comprehensive stimulation package with both nutritional and developmental inputs done in malnourished children showed that both the interventions produced a positive impact on growth and development. The stimulation package was rated superior. However, the catch-up noted at the end of the **interventions** fell short of that of the well-nourished controls from high socioeconomic status. The malnourished children were noted to have delay in all the four areas of development, namely, gross motor, fine motor adaptive, language and personal-social development. The delay in gross motor development was most marked among the malnourished and the catch-up in gross motor milestones was marked after nutritional supplementation. Among those with poor environment and poor maternal child rearing practices, delay in other spheres of development like personal-social and language was also marked. In those with brain involvement, there will be **global delay** in all areas of development.

It was observed that developmental stimulation and psychosocial support to the mother led to catch up in these spheres as well. Thus it was clear that *all children with poor environment and those with malnutrition need develop-*



Trivandrum developmental screening chart (TDSC) for assessing the development of children less than two years

A vertical line is drawn, or a pencil is kept vertically, at the level of the age of the child (in months) being tested. If the child fails to achieve any item that falls short on the left side of the vertical line, the child is considered to have a developmental delay

Fig. 10.3 TDSC chart

mental surveillance and stimulation in addition to food supplementation. This can easily be integrated with the existing ICDS programme. Children above the age of three years who sit in the Anganwadis are getting stimulation by way of non-formal education. The younger children who are also the beneficiaries of ICDS should also get the benefit of developmental surveillance and stimulation. This can be achieved by disseminating the knowledge about normal milestones and by picking up developmental delay for early intervention. Developmental information should be included in the discussions during mother's meetings in the Anganwadis and should be popularised using simplified Developmental Observation Charts. This is very essential for ensuring quality of survival. The concept of **Child Development Referral Units (CDRUs)** in all the districts is meant to provide training in this respect and also early detection and early intervention to those who need special care.

Projects and Proposals

NUTRITION FOLLOW-UP CLINICS, NUTRITION REHABILITATION CENTRES & COMMUNITY EXTENSION SERVICES

Nutrition Rehabilitation Centres are established in various places to provide nutritional supplementation, nutrition education and follow-up to malnourished children. This concept was first introduced by **Bengoa** in 1967. Some of the centres have outpatient and inpatient facilities, kitchens, metabolic labs, kitchen garden etc. Most of the major hospitals for children have at least Nutrition Follow-up Clinics. The six strategies for comprehensive care of children includes Nutrition, Immunization, Medical check-up and medical care. Family health. Education and Stimulation (NIMFES). These are depicted in Fig. 4.13.

1. Services

The Department of Paediatrics in SAT Hospital, Medical College, Trivandrum, is running a Nutrition Follow-up Clinic. Once a week new cases are registered and old cases are followed up. Malnourished children are referred to the clinic from the OPD. in-patient wards, peripheral hospitals and from the ICDS network. They are given the benefit of nutritional supplementation, nutrition education, dietary evaluation, growth monitoring, immunization, medical check-up and care during illness, family health advice including planned maternity and child rearing skills, health education and developmental stimulation **(NIMFES).** This is in accordance to GOBIFFF (Growth monitoring, ORT, Breast feeding, Immunization, Food supplementation, Female education and Family Health).Those with developmental delay are subjected to evaluation using the Denver Developmental Screening Chart and are given individualized stimulation. For nutritional rehabilitation, a precooked, ready to mix cereal, pulse, sugar mixture (SAT Mix), coconut oil, vitamin and mineral supplements and family pot feeding are utilized. For developmental stimulation, individualized developmental tasks, play therapy, motor coordination tasks and activities of daily living (ADL) are resorted to (refer Sections 11.4—11.6)

Psychosocial support given to the mother is the key to the success of such clinics.

2. Distribution of SAT Mix

SAT Mix is a research and development (R&D) product of the clinic. It is now made available in 100 g packings at a very low cost through the SAT Hospital Health Education Society (SATHHES).

3. Extension Services

The clinic also extends services to peripheral health clinics in the Trivandrum Corporation and undertakes training of postgraduate students and research. Several nutrition related research projects have been undertaken in the clinic.

RESEARCH PROJECTS

A brief account of the projects undertaken is given below. Studies were done in under-five children and were compared with appropriate controls,

a) **Completed Projects**

1) **Growth faltering and developmental delay in children with PEM:** In addition to physical retardation like wasting and stunting, they also had gross developmental delay. This observation paved the way to a series of further research in the field to develop models for comprehensive rehabilitation.

2) **Serum biochemical profile in children with PEM:** Serum protein, albumin, phospholipids and enzymes like LDH, gamma GT were found low. Serum protein electrophoresis showed that alpha-1 globulins which contain acute phase reactants were increased, alpha-2 and beta globulins that contain carrier proteins were reduced and gamma globulins were generally increased. Serum total lipids and cholesterol were variable.

3i) **Biochemical predictors of mortality in PEM:** In addition to low S. albumin level, low S. phospholipid and gammaglobulin levels were observed as predictors of mortality in children with PEM.

4) **CSF biochemical profile in children with PEM:** CSF sugar, protein, total lipids were normal, but CSF cholesterol, phospholipids and LDH were significantly lower in those with PEM. The low lipid fractions in CSF may be a reflection of the low brain lipids or may be due to maximum conservation of lipids by the growing brain.

5) **Nerve conduction velocity in children with PEM:** Nerve conduction velocity done in motor nerves showed a definite delay in those with PEM.

6) **Hepatic and renal function studies in children with PEM:** Hepatic enzymes and blood urea were low normal except in those with dehydration, renal shut down, hepatitis and ascending cholangitis. Hepatic and renal dysfunction indicated poor prognosis.

7) **Vitamin status in children with PEM:** Vitamin A and E levels were demonstrated low. Clinically many of them had riboflavin deficiency.
8) **The role of a home-made protein calorie mix in prevention and management of protracted diarrhoea:** The mix was found to be effective, well tolerated and well accepted. This was later named as SAT Mix.
9) **Nutritional assessment of children below two years in a deprived community:** The prevalence of PEM, stunting and wasting were alarmingly high in the coastal area of Poonthura close to Trivandrum city. Up to 10% were found have grade III and IV PEM according to IAP classification as against the state average of around 2%. This study was an eye opener and led to several interventions in the area.
10) **Influence of sociocultural factors and microenvironment on nutritional status of under-five children in coastal Kerala:** The children were noted to have very poor physical, biological and psychosocial environment. Maternal attitudes, skills and support to the mothers were far from optimum. They also had poor nutritional status.
11) **Morbidity and mortality pattern of under-fives in coastal Kerala:** The morbidity due to ARI. diarrhoea, skin infections, chronic ear infections, caries tooth and VPDs like TB were quite high, but the mortality was not high.
12) **Growth and development of under-five children in a deprived community:** This study evaluated the impact of the interventions in a deprived area after a period of three years and the results were gratifying. The immunization coverage was nearly 90%. But malnutrition was still a problem, but majority of them were low birth weight babies. This points to the need for concentrating on the girl child, on the adolescent child and the antenatal mother.
13) **A comparative study of brainstem auditory evoked potentials (BAEP) in PEM before and after nutritional supplementation:** The BAEP latencies and interpeak latencies were found lower which subsequently improved to the normal range after rehabilitation. This is due to progressive myelination. This points to the need for early intervention in the growing stage of the brain.
14) **The effect of protein-energy malnutrition and environmental deprivation on child development:** Environmental factors, namely, socioeconomic status, maternal education, standards of sanitation, microenvironment as assessed by maternal attitude and supportive system to the mother were found to correlate with nutritional status and child development. Even the well nourished from low socioeconomic status had lesser developmental and intelligence scores than those from high socioeconomic status. There is an urgent need to improve the environment, especially microenvironment, for optimum growth and development.

15) **The role of developmental stimulation in nutritional rehabilitation—a hospital study:** Developmental stimulation imparted along with nutritional supplementation was found superior than nutritional supplementation alone. It was identified as a necessity in children with PEM and those with poor environment.

Measures are to be taken to integrate it with the existing child welfare programmes like the ICDS programme.

16) **The role of developmental stimulation in nutritional rehabilitation—a community study:** The hospital model for comprehensive stimulation was worked out in the coastal community at Poonthura and was found highly beneficial.

17) **A comparative study of serum elemental profile with special reference to calcium, magnesium, zinc and copper in children with PEM before and after nutritional supplementation:** Except calcium, the other elements were much lower and even after nutritional supplementation zinc and copper continued to be low indicating the need for trace element supplementation. These levels were estimated using atomic absorption spectrophotometer in the Division of Cellular and Molecular Cardiology, SCTIMST, Trivandrum.

18) **Feeding practices among children with PEM:** Majority of the children were noted to take a very unbalanced diet. They were taking milk and cereal daily, but did not take any pulses (legumes), vegetables including greens, fresh fruits etc. This leads to micronutrient deficiencies in addition to calorie gap. Phosphate in milk and oxalates and phytates in cereals reduce the absorption of minerals like calcium, iron etc. Micronutrient deficiency is thought to be one of the major causes for stunting and increased susceptibility to infection.

19) *Serum micronutrient profile in children with diarrhoea:* The level of serum elements, namely, magenesium, copper and zinc were found to be low in children with diarrhoea. The level of calcium was within normal limits. The level of zinc was markedly depressed in those with persistent diarrhoea and malnutrition. These children are likely to benefit by micronutrient supplementation to replenish their eroded stores.

20) **The outcome of zinc supplementation is nutritional rehabilitation:** In a randomized control study on malnourished children, it was found that, even though catch up in weight was comparable in the zinc-supplemented and the zinc-unsupplemented groups, the catch up in height was significantly more in the zinc-supplemented group. Micronutrient supplementation, especially zinc, may be benefical in those with PEM and stunting.

21)**Reevalidation of Mid Arm Circumference (MAC) and Body Mass Index (BMI) among children:** The present cut-off values of MAC were found not to tally with the grade of PEM as assessed by weight-for-age criterion.

It was found that values above 14 corresponded with normal nutritional status and less than 12 corresponded with severe PEM and values between 12-14 corresponded with mild to moderate PEM. Similarly, the present cut-off value of > 18.5 to denote nomal nutritional status was found inappropriate. Far too many nomal children and growing adolescents were categorzied as chronic energy deficient (CED) by this. But, BMI > 15 was found to be appropriate in the growing age and BMI < 13 was found to denote severe CED. These figures are in accordance with the Rao and Singh index.

22) Status of micronutrients in malnutrition before and after rehabilitation. ***Indian Pediatrics***, 2000, 37: 912-913. (Zn & Cu low even after rehabilitation)

23) Outcome of nutritional rehabilitation with and without zinc supplementation. ***Indian Pediatrics,*** 2000,37: 650-655. (Zn supplementation for 3 mo resulted in normal Zn level and better weight & height gain at the end of 6 mo)

24) Auxologic, Biochemical and Clinical (ABC) Profile of LBW Babies - 2 year prospective study. ***J Tropical Pediatr***, Oxford University Press, London, 2007, 53: 374-382 9Low micronutrients among preterm LBW and term LBW in comparison to normal weight babies with inadequate catch up growth emphasizing the need for special packages)

25) Umbilical Cord blood nutrients in LBW babies in relation to birth weight and gestational age. ***Indian J Med Res.*** 2008, 128: 128-133 (lower levels in preterm LBW than term LBW and lower calcium and iron levels even in normal weight babies)

26) Mid Arm Circumference and Body Mass Index- The two Auxological Parameters in Neonates. ***J Tropical Pediatrics,*** Oxford University, 2006. (MAC of 9 cm is noted in normal newborn babies and < 8 cm indicates LBW. The BMI of newborns babies is 13.)

b) Ongoing Projects

- Community Study on Anemia among rural Adolescent girls
- Community Study on Pscho-Social aspects among Rural Adolescent girls
- Folate for primary prevention of Congenital Heart Disease along with Neural tube defect. Clustering of folate related polymorphism (MTHFR gene) in mother's of children with CHD
- CD 34 stem cell reaping from Umbilical cord blood (UCB)

c) **The Future**

The potential application of information stemming from the Human Genome Project (HGP) has to new branches of research like Nutrigenomics, Nutrigenetics, Proteomics, Peptidomics, Metabolomics and so on. The func-

tional interaction of food with genome at molecular, cellular and systemic level is called Nutrigenomics, while Nutrigenetics referes to genetically determined differences in how individuals react to specific food. Proteome is the collection of all the proteins in an organism. Metabolomics refers to the metabolic profiling in an organism.

RECOGNITIONS

The various projects undertaken in the clinic have been presented and published and the author has won a number of recognitions.

a) Dr. CO Karunakaran Award for best research paper 1982, 1984.
b) Indian Academy of Paediatrics (IAP), Kerala Chapter, Award for the best research paper, 1991
c) Cochin Paediatric Society Award for the best research paper from teaching institutions, IAP South Zone Conference, 1992
d) Nutrition Society of India, National Senior Award in Community Nutrition, 1994
e) International Ambulatory Paediatric Research Award, Virginia, 1994
f) IAP South Zone Conference Award for best research paper from teaching institutions, 1996
g) Dr TN Krishnan Award for outstanding contribution in the field of Health and Family Welfare, 1996
h) Nana Mini Screen Awards for best TV program.

IMPORTANT PUBLICATIONS

1. Elizabeth KE, Sathy N, The role of developmental stimulation in Nutritional Rehabilitation.***Indian Pediatrics***, 1997,34:681-695
2. Elizabeth KE, Management of Protein Energy Malnutrition, ***IAP Journal of Practical Pediatrics,*** 1998,6:339-344
3. Elizabeth KE, ***Nutrition & Child Development***, 2006, 4th edition. Paras Medical Publisher, Hyderabad
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RAP 15 (2005): Chapters—Sex determination, Genetic Counselling
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Appendices

Appendix 1: Socio-economic status according to updated Kuppuswami's Scale (2007)

Item	Score
Education	
1. Professional degree/Hons., MA and above	7
2. BA, BSc degree	6
3. Intermediate/Post high school certificate	5
4. High school certificate	4
5. Middle school completion	3
6. Primary school/literate	2
7. Illiterate	1
Occupation	
1. Profession	10
2. Semi profession	6
3. Clerical, shop/farm owner	5
4. Skilled worker	4
5. Semi-skilled worker	3
6. Unskilled worker	2
7. Unemployed	1
Income*	
1. Rs. 19575 and above	12
2. Rs. 9788-19574	10
3. Rs. 7323-9787	6
4. Rs. 4894-7322	4
5. Rs. 2936-4893	3
6. Rs. 980-2935	2
7. Rs. < 979	1

Total score	Socio-economic status scale
26-29	Class I (upper)
16-25	Class II (upper middle)
11-15	Class III (lower middle)
5-10	Class IV (upper lower)
< 5	Class V (lower)

Source: Updated Kuppuswami SESS (IJP 2007)
**Monthly income*

Appendix 2: Standards of sanitation according to Briscoe's scale (1978)

Behaviour	Points		
	3	2	1
1. Water:			
Drinking	Tube well/tap	Ring well	Pond
Washing	Tube well/tap	Ring well	Pond
Bathing	Tube well/tap	Ring well	Pond
2. Defecation children < 5 yrs	Latrine/disposed off	Open within the compound	Anywhere
3. Hand washing by mother before handling/ eating food	Yes	Occasional	No
4. Hand washing by mother after defecation	Yes with soap	Yes with sand/ ash	Yes with water
5. Appearance of mother's hands and cloth	Clean	One clean	Unclean
6. Drinking water storage	Direct use	Clean, covered	Unclean, uncovered
7. Water for washing	Direct use	Clean, covered	Unclean, uncovered

Points	Behaviour
18-21	Good
13-17	Fair
7-12	Poor

Source: Briscoe, 1978

Appendix 3: Micro-Environment Scoring Scale, Elizabeth (1994)

		Score		
		3	2	1
A.	Maternal attitude*			
	1. Do you yell at your child when you are angry?	No	Occasionally yes	Yes
	2. Do you punish your child when you are angry?	No	Occasionally yes	Yes
	3. Do you hit your child when you are angry?	No	Occasionally yes	Yes
	4. Do you ignore your child when he/she asks for help?	No	Occasionally yes	Yes
B.	Supportive system to the mother**			
	1. Do your in-laws help in household work and child rearing?	Yes	Occasionally yes	No
	2. Do you have someone to turn to during any difficulty?	Yes	Occasionally yes	No
	3. Do your friends/neighbours look after your child in your absence?	Yes	Occasionally yes	No
	4. Are you happy about your life/fate?	Yes	Occasionally yes	No

Score	Micro-environment
19-24	Good
15-18	Fair
8-14	Poor

Source: Elizabeth, 1994
**Items selected from the Handbook of Parental Acceptance and Rejection (Rohner, 1980)*
***Items selected from the Supportive System Scale (Vazir, 1983)*

Appendix 4: Infant Milk Substitutes Act (1992)

The Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act, 1992, is in force since August 1, 1994. It extends to the whole of India. Certain highlights of the Act are as follows:

1. *Advertisements:* No advertisement of infant milk substitutes or feeding bottles

2. *Incentives:* No incentives of any kind for the purpose of promoting the use or sale of infant milk substitutes or feeding bottles. No free samples to mothers.

3. *Donations:* No donation or subsidised supplies of infant milk substitutes or feeding bottles to any person except to an orphanage.

4. *Healthcare system:* No display of placards or posters for the purpose of promoting the use or sale of infant milk substitutes, feeding bottles or infant foods (weaning foods).

5. Information to health workers should be scientific and factual.

6. *Health worker:* No inducement to health worker for promoting the use of infant milk substitutes, feeding bottles or infant foods.

7. Any direct or indirect expenditure incurred on a health worker by the manufacturer, distributor or retailer of infant milk substitutes, feeding bottles or infant foods should be disclosed (both by the giver as well as the health worker) to the institution or organization to which such health worker is attached.

8. *Labels:* Containers of infant milk substitutes or infant foods should indicate in a clear, easily, readable manner, the words "Important Notice'' in capital letters in English and Hindi (use of any local language in addition to these can be made). The labeling must include certain details outlined in the Act.

9. No container or label of infant milk substitutes shall
   a) Have pictures of an infant or a woman or picture of phrases designed to increase the saleability of the product,
   b) Use of the word "humanized" or "maternalized" or any other similar word.

10. *Educational material:* Every educational material, whether audio or visual dealing with prenatal or postnatal care or with infant feeding shall include clear information relating to:

    a) Benefits of breastfeeding
    b) Preparation and continuance of breastfeeding, and
    c) The hazards, financial and social, of improper use of infant milk substitutes and feeding bottles.

    The details included in the labels and educational material should adhere to the guidelines outlined in the Act of 1992 and the relevant rules of 1993.

11. *Company sales persons:* No company sales representative shall perform any function which relates to educating a pregnant woman or mother of an infant or prenatal or postnatal care of the infant.

12. No linking of remuneration to sales persons or employees with the volume of sales of infant milk substitutes, feeding bottles or infant foods.

13. *Standards:* Infant milk substitutes, feeding bottles or infant foods shall conform to the standards as per the Act.

14. *Penalty, etc.* Details of power of entry and search, confiscation of products within the scope of the Act, fine, imprisonment, appeal etc.

The Infant Milk Substitute, Feeding Bottles and Infant Foods (Production, Supply and Distribution) Act, 1992 prohibits

- Distribution of free samples to mothers
- Advertising to the public
- Promotion in health care facilities
- Distribution of gifts or samples to health workers
- Promotion of words and pictures that idealize bottle feeding
- Advice to mothers by company sales staff
- Financial assistance to health organizations or associations of doctors to organise conferences, seminars, etc.
- Incentives to sale personnel or retailers on the basis of volume of sales

Breastfeeding Promotion Network of India (BPNI) has contributed a great deal in training and ensuring that there is no infringement of the Act.

Appendix 5: Demographic indicators and vital statistics (Source: The state of the world's children, UNICEF, 2009)

1. Basic indicators

Basic Indicators	Toal population (1000s)	Under-5 mortality rate		Infant mortality rate (under 1)		Total population (thousands)	Annual no. of births (th-ousands)	Annual no. of under-5 deaths (th-ousands)	GNI per capita (US$)	Life expec-tancy at birth	Total adult ilteracy rate (%)	Net primary school enro-lment/atten-dance (%)	% Share of household income 1995-2005*	
													Lowest	Highest
	2007	'09	'07	'90	'07	2007	2007	2007	2007	2007	2007	2007	40%	20%
India	1169016	117	72	83	54	1169016	27119	1953	950	64	66	98	19	45
South Asia	1567187	125	78	89	59	1567187	37986	2985	889	64	63	80	19	46
World	6655406	93	68	54	47	6655406	135770	9216	7952	68	81	85	19	42

2. **Nutrition**

			% of children ('00-2007) who are			% of under-fives ('00-07) suffering from				
Nutrition	Under-5 mortality rank (2007)	% of infants with low birth weight (2007)	Exclusive breastfed (< 6 months)	Breastfed with complementary food (6-9 months)	Still breast feeding (20-23 months)	Underweight Moderate & severe	Wasting moderate & severe	Stunting moderate & severe	Vit. A supplementation coverage rate (6-59 months) 2007	% of households consuming iodized salt 2007
India	**49**	**28**	**46K**	**57**	**77**	**43**	**19**	**38**	**33w**	**51**
South Asia		**27**	**44**	**53**	**75**	**45**	**18**	**38**	**50**	**51**
World		**14**	**38**	**55**	**50**	**25**	**11**	**28**	**62e**	**68**

Note: **'K' refers to exclusive breastfeeding for less than 4 months, w - target less than up to 59 months of age, e - excluding China**

3. Health

Health	Under-5 mortality rank	% of population using improved drinking water sources (2006)			% of population using adequate sanitation facilities (2006)			% of routine EPI vaccines financed by govt (2007)	% immunized (2007) 1-year-old children					% NB protected against tetanus (2007)
		Total	Urban	Rural	Total	Urban	Rural	Total	TB	DPT3	Polio3	Measles	Hep 83	AN-TT
India	**49**	**89**	**96**	**86**	**28**	**52**	**18**	**100**	**85**	**81**	**62**	**67**	**6**	**86**
South Asia		**87**	**94**	**84**	**33**	**57**	**23**	**83**	**87**	**84**	**69**	**71**	**29**	**85**
World		**87**	**96**	**78**	**62**	**79**	**45**	**71**	**89**	**90**	**82**	**82**	**65**	**81**

4. Education

Educa-tion	Under-5 mortality	Youth literacy rate (15-24 yr) rank				Primary school enrolment ratio				Net primary school atten-dance (%)		Secondary school enrol-ment ratio	
		1990		2007		00-07 (gross)		00-07 (net)		2000-2007		00-07 (gross)	
		Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
India	**49**	**62**	**34**	**87**	**77**	**114**	**109**	**90**	**87**	**85**	**81**	**59**	**49**
South Asia		**60**	**32**	**84**	**74**	**111**	**104**	**88**	**83**	**81**	**77**	**54**	**45**
World		**82**	**69**	**90**	**85**	**108**	**103**	**90**	**87**	**80**	**77**	**67**	**63**

5. **Demographic indicators**

Demo-graphic indica-tors	Under-5 mortality rank	Population (thousands) 2007		Population annual growth rate (%)		Crude death rate		Crude birth rate		Life expectancy		Total fertility rate	%of population urbanized	Average annual growth rate of urban population (%)	
		Under 18	Under 5	'70-'90	'90-'01	1970	2007	1970	2007	1970	2007	2007	2007	'70-'90	'90-2007
India	**49**	**446646**	**126808**	**2.1**	**1.8**	**17**	**8**	**40**	**23**	**49**	**64**	**2.8**	**29**	**3.5**	**2.8**
South Asia		**614747**	**175250**	**2.2**	**2.0**	**17**	**8**	**41**	**24**	**48**	**64**	**3.0**	**29**	**3.6**	**3.0**
World		**221356**	**629106**	**1.8**	**1.5**	**13**	**9**	**32**	**20**	**56**	**68**	**2.6**	**49**	**2.4**	**2.3**

6. Economic indicators

Economic indicators	GNI per capita (US$)	GDP per capita average annual growth rate (%>		Average annual rate of Inflation %	% of popul-ation below $1.25 a day	% of central government expenditure allocated to: (1997-2006)			ODA inflow in millions US$	ODA inflow as a % of recipient GNI	Debt service as a exports of goods and services	
	2007	■60-'90	90-2007	'90-'07	2005	Health	Education	Defence	2006	2006	'90	2006
India	**950**	**1.6**	**4.5**	**6**	**4.2**	**14**	**2**	**4**	**1379**	**0**	**25**	**7**
South Asia	**889**	**1.7**	**4.1**	**6**	**40**	**14**	**2**	**5**	**9191**	**1**	**21**	**7**
World	**7952**	**3.1**	**2.4**	**8**		**11**	**14**	**6**	**81045**	**0**	**18**	**11**

7. **Women's status**

Women	Life expectancy: females as a % of males	Adult literacy rate: females as a % of males	Gross 6 nrolment ratios: fe males as a % n males		Contra-ceptive prevalence (%)	Antenatal care coverage 4 times (%)	Skilled atten-dant at delivery (%)	Maternal mortality ratio adjusted
			Primary	Secondary				
	2007	2007	2007	2007	2007	2007	2007	2000-2007
India	**105**	**71**	**96**	**85**	**56**	**37**	**47**	**450**
South Asia	**104**	**71**	**94**	**85**	**68**	**34**	**41**	**500**
World	**106**	**88**	**97**	**98**	**62**	**47**	**62**	**400**

Appendix 6: Indian Recipes

1. ***Ragi malt***

   Clean the grain, wash it well and soak in double volume water for about 16 hours. Then scrub the grain in water, remove the seed coat by washing and soak for another 3-4 hours. Then drain off the water and spread the grain on a damp cloth and cover it with another damp cloth and keep it for a day. The grain will germinate. This is called malting. The next day, the ragi will have a small white sprout. The sprout should not be allowed to grow long; if it grows, the taste will be altered. Then dry the grain in sun by spreading on a tray and then roast it till 70°C in an iron pan till the malt flavour comes. Then powder it into a fine flour and sieve it. Store in airtight container. Germination enhances digestibility, vitamin content and reduces the bulk on cooking. It is rich in amylase and is also called Amylase Rich Food (ARF). ARF powder can be added to other recipes towards the end of cooking to reduce the bulk on cooking and to increase digestibility.

2. ***Ragi porridge***

   Ragi malt 25 g
   Jaggery 20 g

   Add enough water for cooking and little coconut milk or cow's milk at the end. Ragi porridge can also be prepared for each day by overnight soaking and grinding of the grain. After grinding, sieve through a fine cloth and discard the seed coat. The excess water in the flour can be drained and cooked then and there or can be sun dried and used on subsequent days.

3. ***Ragi Cake/Ragi Adai***

   Ragi flour 50 g
   Jaggery 20 g
   Coconut grating 20 g
   Salt to taste

   Water sufficient quantity. Make a paste of the flour and prepare small balls. Spread the paste on plantain leaf or other suitable leaf. Mix the jaggery and coconut grating and add 1-2 spoons to the middle and fold the leaf and cook in steam. The jaggery and coconut grating can also be mixed with the flour and can be spread on a greased iron pan or tawa to make the adai.

4. ***Panjiri***

   Wheat atta 50 g

Jaggery/sugar 25 g
Oil/ghee 2 teaspoon

Roast wheat atta in oil till it turns light brown. Add the jaggery solution and mix well.

5. *Rice-Milk-Conjee*
Rice 25 g
Jaggery 20 g
Cow's milk/coconut milk q.s
Water q.s

Boil the water, add rice and when the rice is half cooked, add the jaggery solution and cook well. Add milk towards the end.

6. *Wheat and green gram laddoos*
Wheat flour 50 g
Green gram/dal 50 g
Sugar 100 g
Oil 2 tablespoon

Roast the flour and dal separately till light brown. Grind the green gram dal to a fine powder. Mix them together and fry for 10 minutes with oil. Remove from fire, add powdered sugar and prepare them into small balls. Any cereals-pulse combination or any dal or cereal alone can be prepared similarly.

7. *Wheat dalia porridge*
Wheat dalia 25 g
Sugar 20 g
Coconut/cow's milk q.s
Water q.s

Roast wheat dalia and add to boiling water and cook until soft. Add sugar and milk towards the end. Variation: Combination of wheat dalia and moong dal can be prepared into a similiar porridge.

8. *Kheer*
Suji/rice 25 g
Sugar 20 g
Milk q.s

Boil milk, add the grain and cook until soft and then add sugar.

9. *Khichri*
Wheat dalia/rice 50 g
Dal 25 g
Spinach 25 g
Ghee/oil 2 teaspoon
Salt to taste

Water q.s

Other green leafy vegetables can also be used. Cook cereal and dal together and mash well. Boil leafy vegetable, mash and strain. Add the leafy vegetable puree, salt and heated ghee/oil to the rice dal mixture and stir well.

10. ***Sweet khichri***

Sugar or jaggery to the above to make it a sweet khichri.

11. ***Groundnut and gingelly laddoos***

Groundnut 25 g
Jaggery 25 g
Gingelly seeds 10 g

Roast groundnut and powder it and add to gingelly and pound with jaggery and make it into small balls.

12. ***Banana powder***

Banana powder 25 g
Sugar 20 g
Water q.s

Remove the banana coat the cut into thin slices. Sun dry the pieces and then grind into a powder and store it in airtight container. Cook the banana powder with enough water to a semi-solid and add sugar.

13. ***Banana groundnut mix***

Ripe banana 1
Jaggery 100 g
Milk q.s
Groundnut 10 g

Mash the ripe banana; roast and grind the groundnuts. Mix together with jaggery and little milk into a smooth paste.

14. ***Precooked ready to mix cereal-pulse (SAT Mix)***

Rice 1/2 kg, Wheat 1/2 kg
Black gram 1/2 kg, Sugar 1 kg

Rice, wheat and black gram are cleaned and roasted separately in equal proportion. Then these items are mixed together and powdered into fine powder. Sugar is also powdered separate and mixed together. The mix can be kept in airtight container and enough quantity can be taken and mixed into a semisolid with enough hot water or hot milk.

Appendix 7: Various types of ORS and ReSoMal

Item	WHO ORS	Hypoosmolar ORS	IAP Hypoosmolar ORS
NaCI	3.5 g	2.6 g	1.75 g
KCI	1.5 g	1.5 g	1.5 g
Na citrate	2.9 g	2.9 g	2.9 g
Glucose	20 g	13.5 g	15 g
Na	90 mEq/L	75 mEq/L	60 mEq/L
K	20 mEq/L	20 mEq/L	20 mEq/L
Cl	80 mEq/L	65 mEq/L	50 mEq/L
Citrate	10 mEq/L	10 mEq/L	10 mEq/L
Glucose	111 mEq/L	75 mEq/L	84 mEq/L
Osmolarity	311 mOsm/L	245 mOsm/L	224 mOsm/L

Composition of stool (mEq/L)

	Na K	a	HCO_3
Cholera Child	101 27	92	32
Adult	140 13	104	44
Non Cholera	56 25	55	14

ReSoMal

ORS	**1 pakt**	**Na (mEq/L)**	**45**
H_2O	**2 L**	**K (mEq/L)**	**40**
Sugar	**50 g**	**Cl (mEq/L)**	**70**
Electrolyte		**Citrate (mEq/L)**	**7**
mineral		**Mg (mEq/L)**	**3**
mixture (EMM)	**40 ml**	**Zn (mEq/L)**	**0.3**
Composition		**Cu (mEq/L)**	**0.045**
KCI	**22.4 g**	**Osmolarity**	**300**
Tripot, citrate	**81 g**	**(mOsm/L)**	
MgCI	**76 g**		
Cu sulphate	**1.4 g**		
Zn acetate	**8.2 g**		
Water	**2.5 L**		

ReSoMal is rehydration solution for the malnourished. Mineral mixture has to be prepared as per the composition given and added to ORS.

Appendix 8: Terms used for infant feeding

Exclusive breastfeeding means giving a baby no other food or drink including water, in addition to breastfeeding (except medicine and vitamin or mineral drops; expressed breast milk is also permitted).

Predominant breastfeeding means breastfeeding a baby, but also giving small amounts of water or water-based drinks.

Full breastfeeding means breastfeeding either exclusively or predominantly.

Bottle feeding means feeding a baby from a bottle, whatever is in the bottle, including expressed breast milk.

Artificial feeding means feeding a baby on artificial feeds and not breastfeeding at all.

Partial breastfeeding means giving a baby some breastfeeds and some artifical feeds, either milk or cereal or other food.

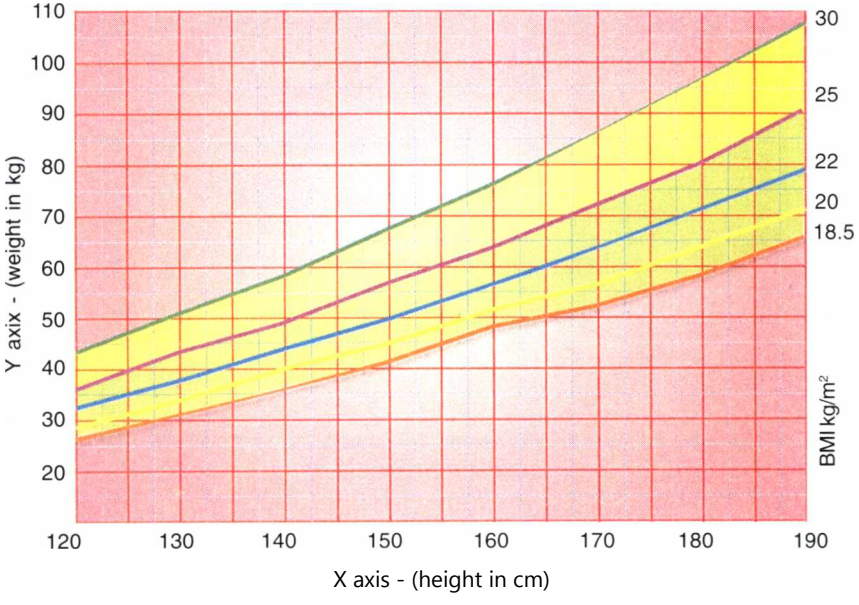
Complementary feeding means giving a baby other food in addition to breastfeeding when it is appropriate, after the age of 4-6 months.

Appendix 9: ELIZ Solution for potassium and magnesium supplementation in malnutrition

Compsition

	(Qty)		(mEq/L)
Glucose	50 g	Na	70
NaCI	2 g	K	40
15% KCI	20 cc	Cl	70
7.5% $NaHC0_3$	20 cc	HCO_3	20
MgCI 6 H_20	30 g	Mg	3
Water	1 L	Glucose	278

Appendix 10: ELIZ Health Path for Adults (EHPA)



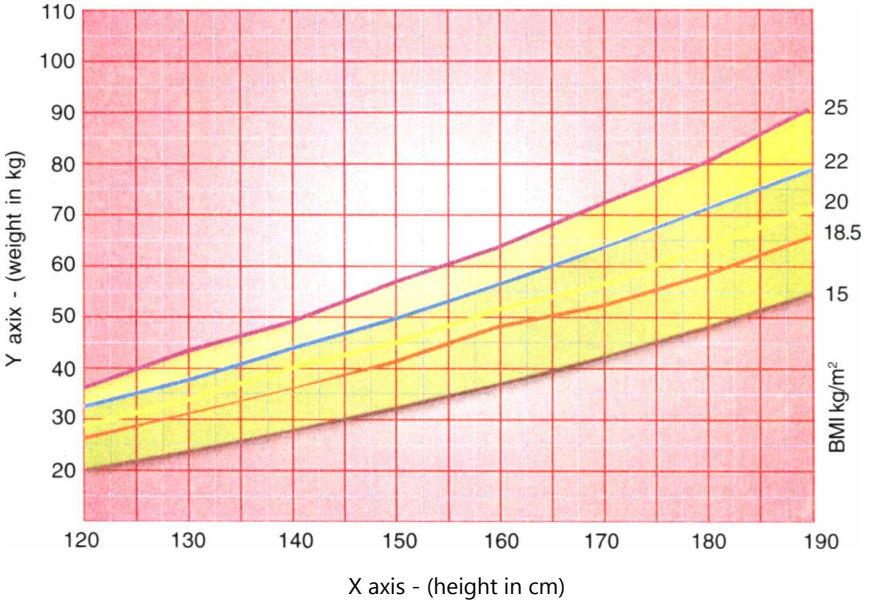
Plot the height on X-axis and weight on Y-axis. Mark the meeting point and project the point along or parallel to the dotted line and directly read the BMI from right margin.

Ex: If the height is 150 cm and weight is 50 kg, the BMI is 22
If the height is 155 cm and weight is 55 kg, the BMI is between 22 and 25

Readings in shaded area are in normal range.
Readings < 18.5 indicate underweight
Readings > 25 indicate overweight
Readings > 30 indicate obesity

Elizabeth KE., Three-in-one Weight, Height and BMI Charts for children and adults. *Journal of Tropical Pediatrics,* Oxford University Press, London, 2003, 49:224-227.

Appendix 11: The ELIZ Health Path for Adolescent Children (EHPAC)

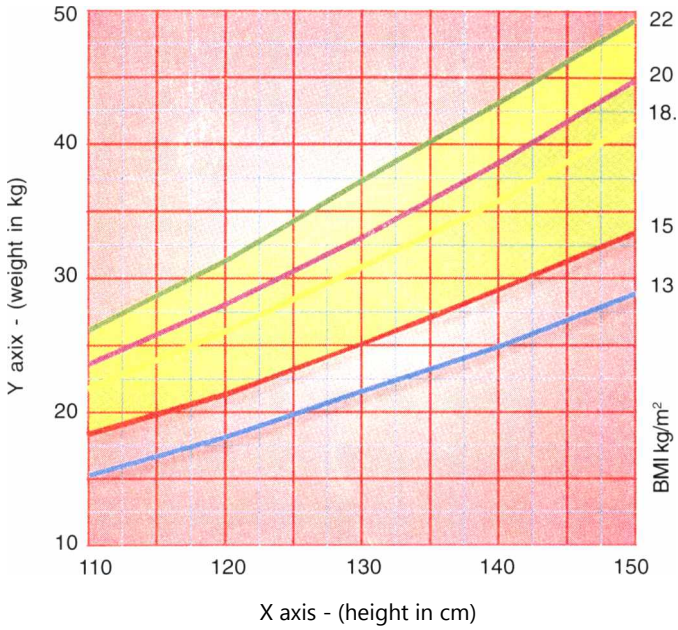


Plot the height on X-axis and weight on Y-axis. Mark the meeting point and project the point along or parallel to the dotted line and directly read the BMI from right margin.

Ex: If the height is 140 cm and weight is 40 kg, the BMI is 20
If the height is 155 cm and weight is 50 kg, the BMI is between 20 and 22

Readings in shaded area are in normal range.
Readings < 15 indicate underweight
Readings > 22 indicate overweight
Readings > 25 indicate obesity

Appendix 12: The ELIZ Health Path for Older Children (EHPOC)

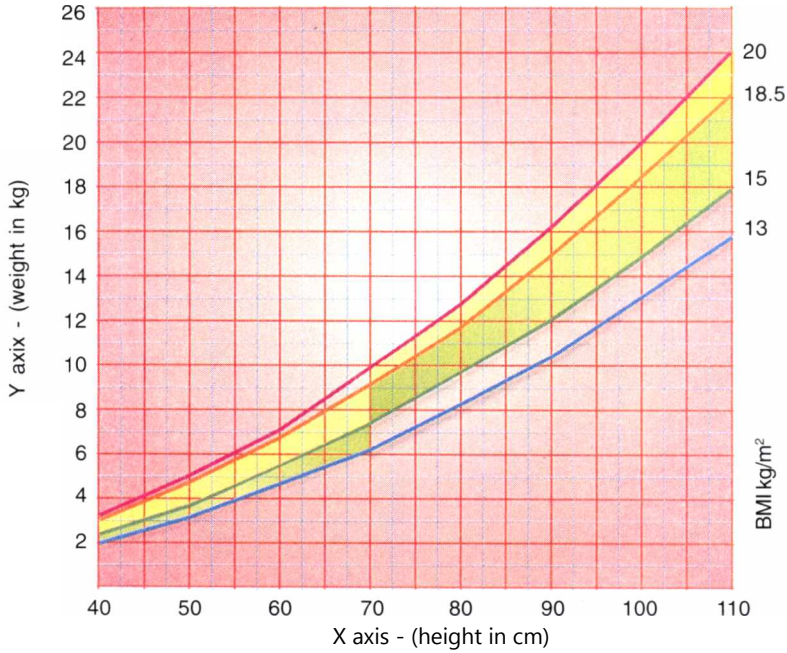


Plot the height on X-axis and weight on Y-axis. Mark the meeting point and project the point along or parallel to the dotted line and directly read the BMI from right margin.

Ex: If the height is 130 cm and weight is 25 kg, the BMI is 15
If the height is 140 cm and weight is 35 kg, the BMI is between 15 and 18.5

Readings in shaded area are in normal range.
Readings < 15 indicate underweight
Readings > 20 indicate overweight
Readings > 22 indicate obesity

Appendix 13: The ELIZ Health Path for Under-Five Children (EHPUC)



Plot the height on X-axis and weight on Y-axis. Mark the meeting point and project the point along or parallel to the dotted line and directly read the BMI from right margin.

Ex: For infants (0-1 years) between 40-70 cm
If the height is 55 cm and weight is 4 kg, the BMI is 13
If the height is 60 cm and weight is 5 kg, the BMI is between 13 and 15

Readings in shaded area are in normal range.
Readings < 13 indicate underweight
Readings > 15 indicate overweight
Readings > 18.5 indicate obesity

Ex: For children (1-5 years) between 70-110 cm
If the height is 90 cm and weight is 12 kg, the BMI is 15
If the height is 95 cm and weight is 14 kg, the BMI is between 15 and 18.5

Readings in shaded area are in normal range.
Readings < 15 indicate underweight
Readings > 18.5 indicate overweight
Readings > 20 indicate obesity

Appendix 14: Comparison of Growth - Weight & Hieght

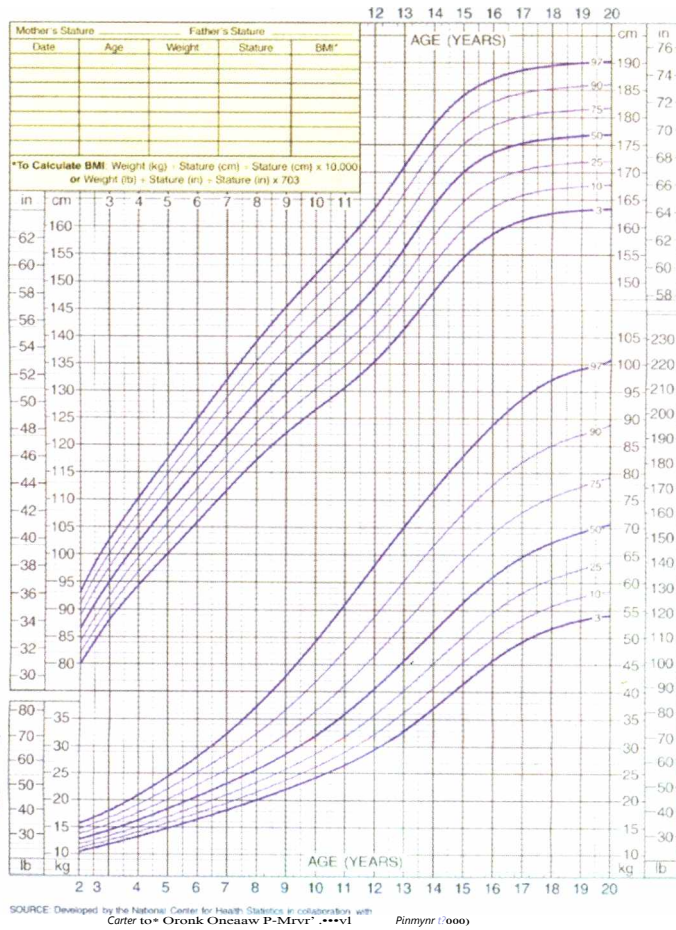
Comparison of Growth - Weight & Hieght

Age	ICMR (1990)		CDC (2000)		WHO (2006)	
	kg	cm	kg	cm	kg	cm
Birth	3.2	49.9	3.3	50.5	3.5	50
1 yr	9.5	75.1	10.2	76.1	9.5	72.1
2 yrs	11.8	84.5	12.3	85.6	11.25	85.1
3 yrs	14.1	93.9	14.6	94.9	13.8	94.0
4 yrs	16.0	101.6	16.7	102.9	15.75	101.5
5 yrs	17.7	108.4	18.7	109.9	18.3	108.0
6 yrs	19.5	114.6	20.7	116.1	20.5	114.0
7 yrs	21.8	120.6	22.9	121.7	23.0	119.5
8 yrs	24.8	126.4	15.5	127.0	25.5	124.5
9 yrs	28.5	132.2	28.1	132.2	27.5	130
10 yrs	32.5	138.	31.4	137.5	30.0	136.0

Appendix 15A: Boys (2 to 20 yr) Statu re-for-age and Weight-for-age percentiles (CDC)

2 to 20 years: Boys **name**----------

Stature-for-age and Weight-for -age percentiles **record#**

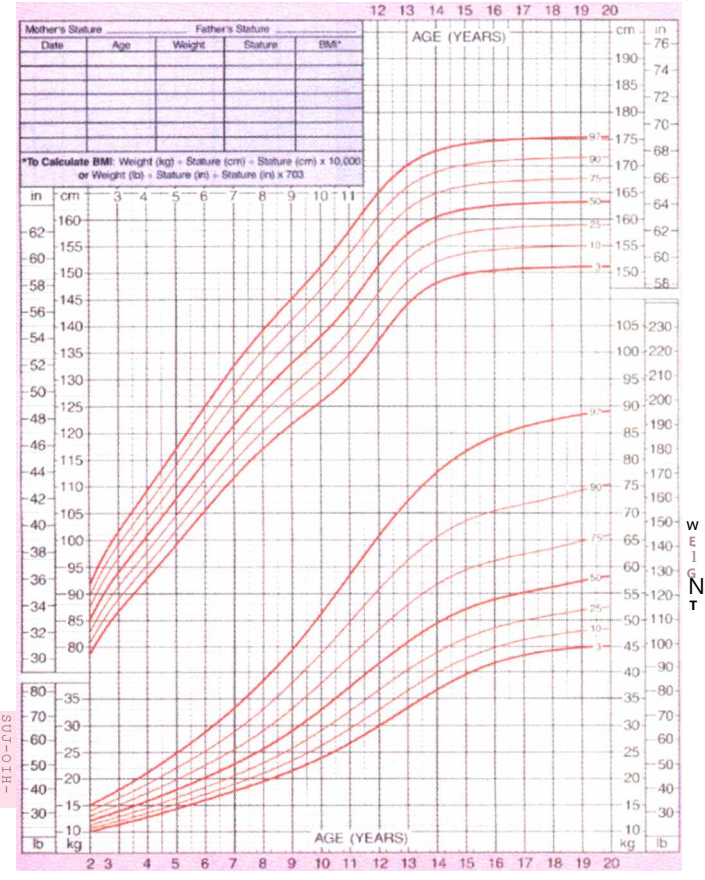


SOURCE: Developed by the National Center for Health Statistics in collaboration with

Carter to* Oronk Oneaaw P-Mrvr' .•••vl *Pinmynr* (©000)

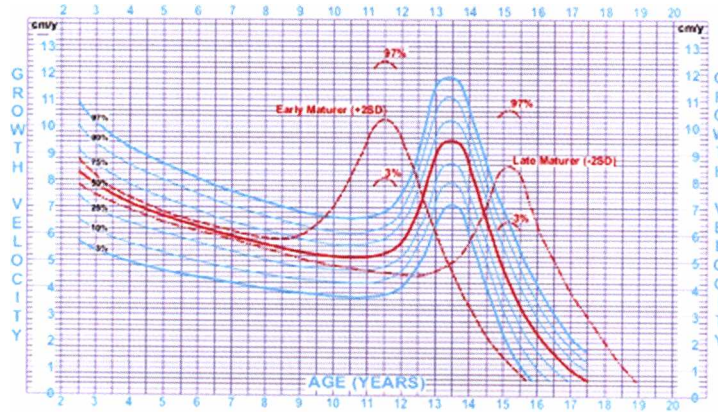
Appendix 15B: Girls (2 to 20 yr) Stature-for-age and Weight-for-age percentiles (CDC)

2 to 20 years: Giris
Stature -for -age and Weight-for-age percentiles

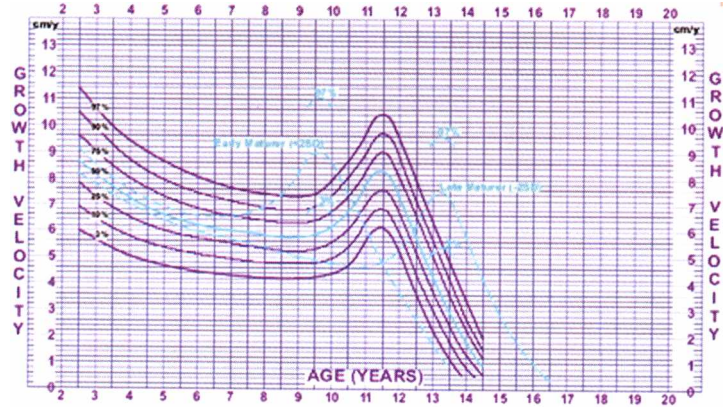


Appendix 16: Growth Velocity Curves

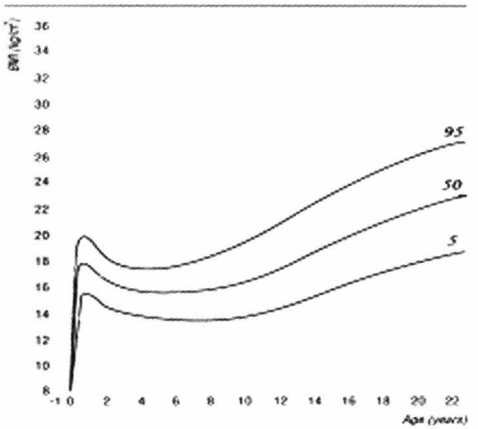
Growth Velocity - Boys



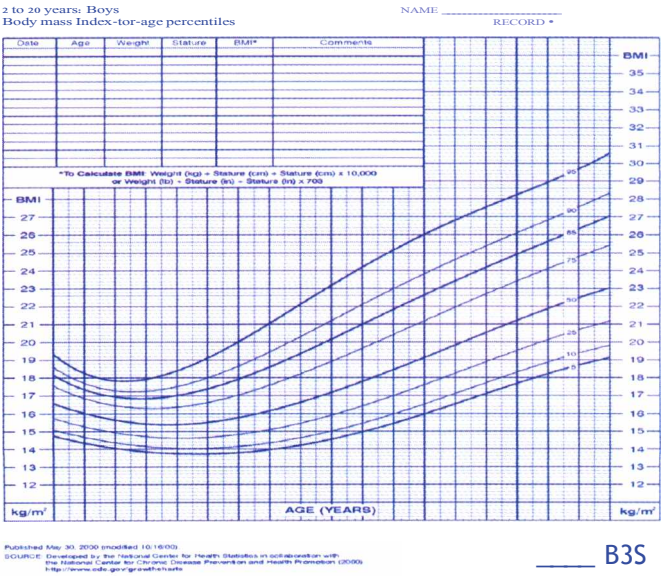
Growth Velocity -Girls

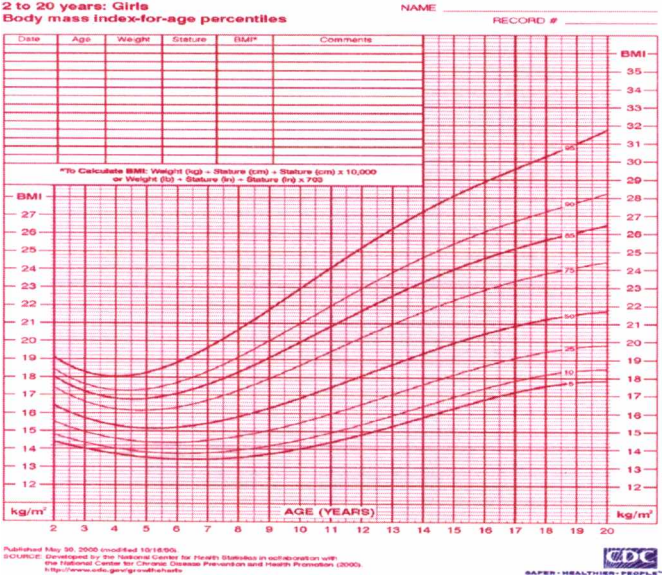


Appendix 17: BMI Centiles

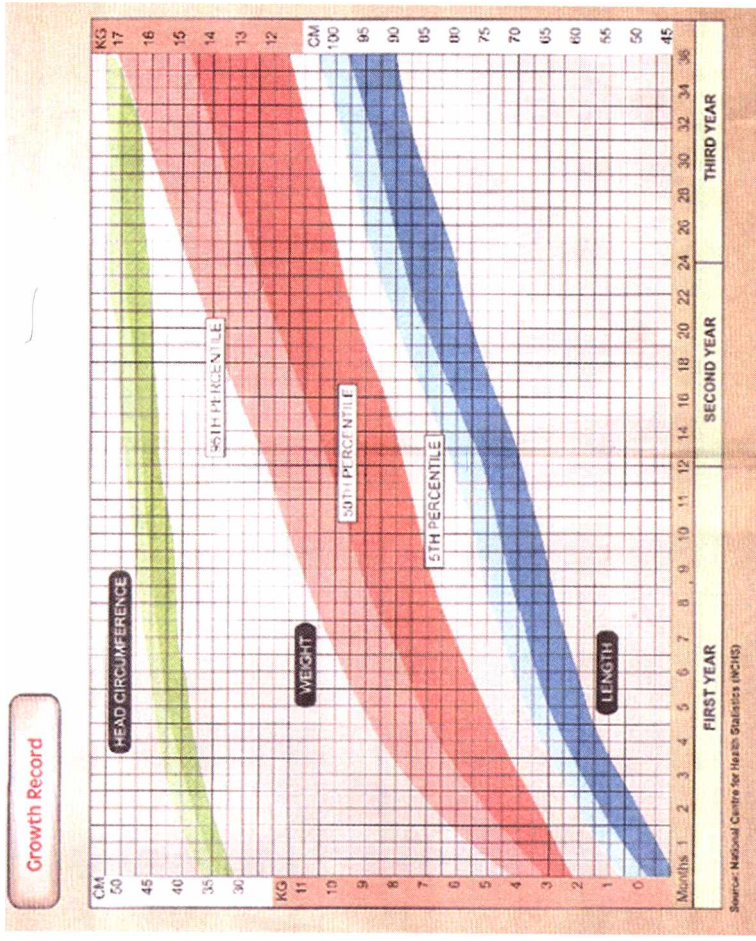


Schematic representation of body mass index (BMI) centiles for obesity, median and under weight in a sample population




2 to 20 years: Girls
Body mass index-for-age percentiles
NAME
RECORD #
Date
Age
Weight
Stature
BMI*
Comments
*To Calculate BMI: Weight (kg) ÷ Stature (cm) ÷ Stature (cm) x 10,000
or Weight (lb) ÷ Stature (in) ÷ Stature (in) x 703
BMI
kg/m²
AGE (YEARS)
95
90
85
75
50
25
10
5
Published May 30, 2000 (modified 10/16/00).
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
http://www.cdc.gov/growthcharts
CDC
SAFER · HEALTHIER · PEOPLE™

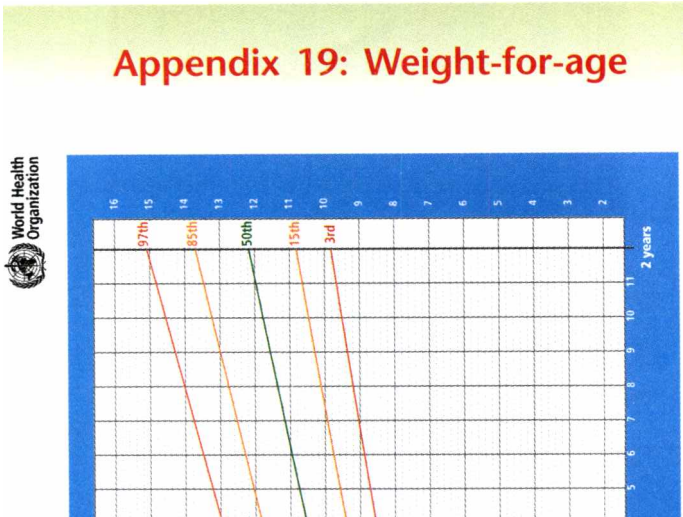
Appendix 18: Growth Record (NCHS)



® Weight-for-age BOYS

Birth to 2 years (percentiles)

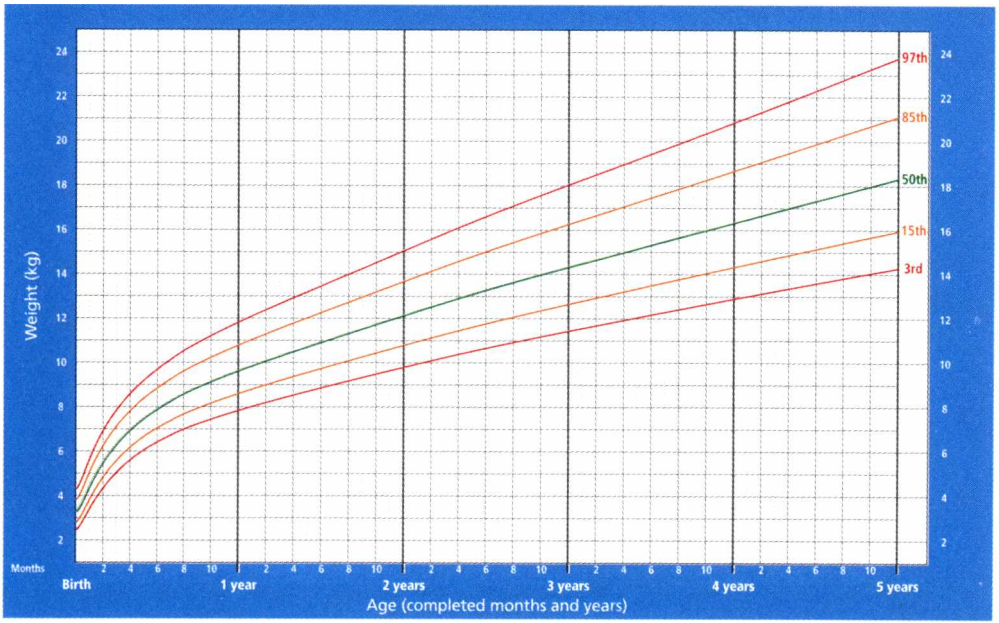
Appendix 19: Weight-for-age



© Weight-for-age BOYS

Birth to 5 years (percentiles)

^ World Health
* Organization



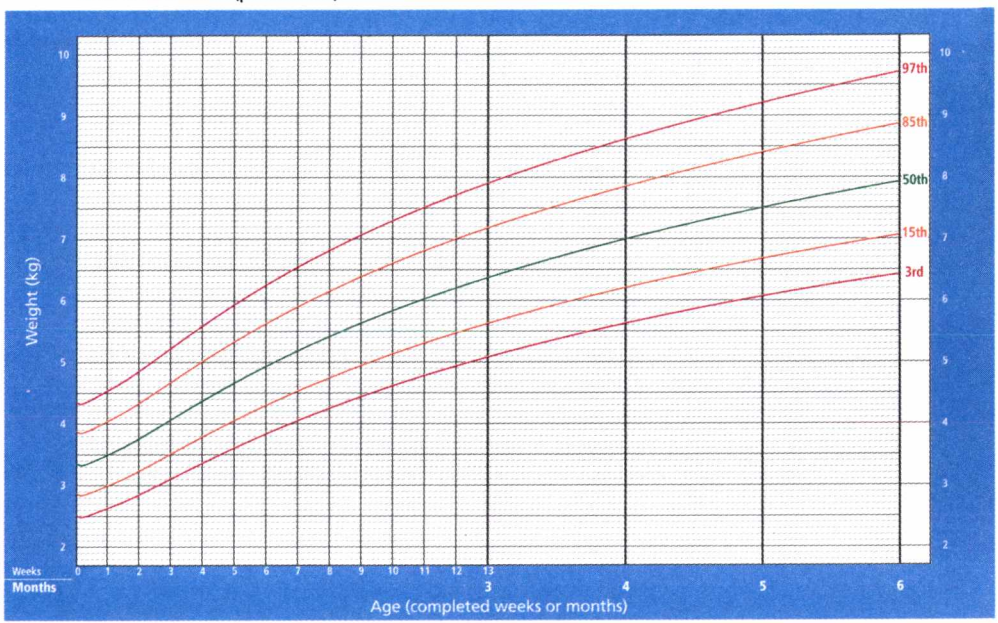
WHO Child Growth Standards

C

Weight-for-age BOYS

World Health Organization

Birth to 6 months (percentiles)

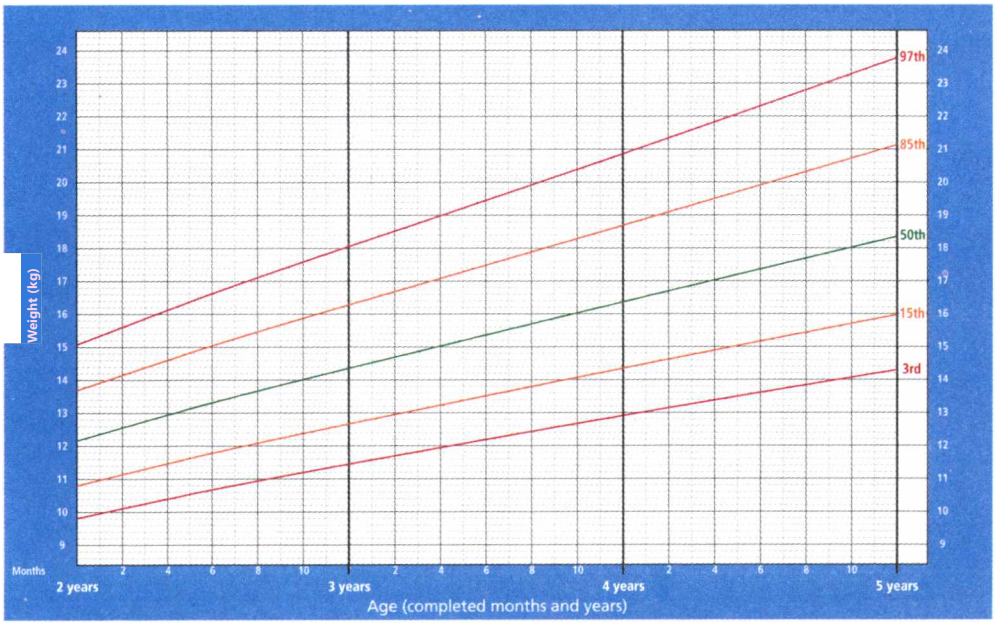


WHO Child Growth Standards

O Weight-for-age BOYS

World Health Organization

2 to 5 years (percentiles)

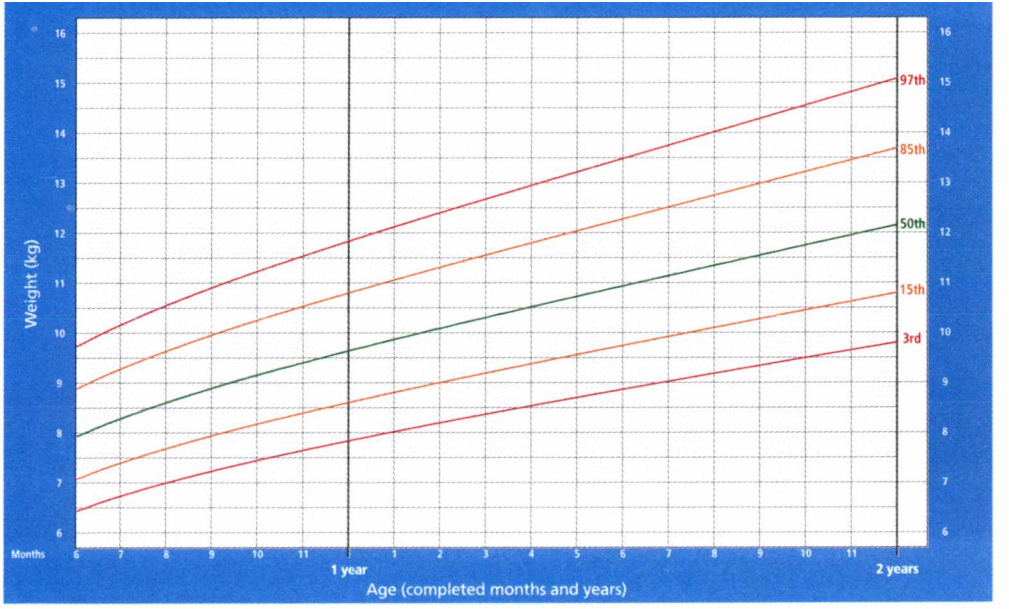


WHO Child Growth Standards

Weight-for-age BOYS

6 months to 2 years (percentiles)

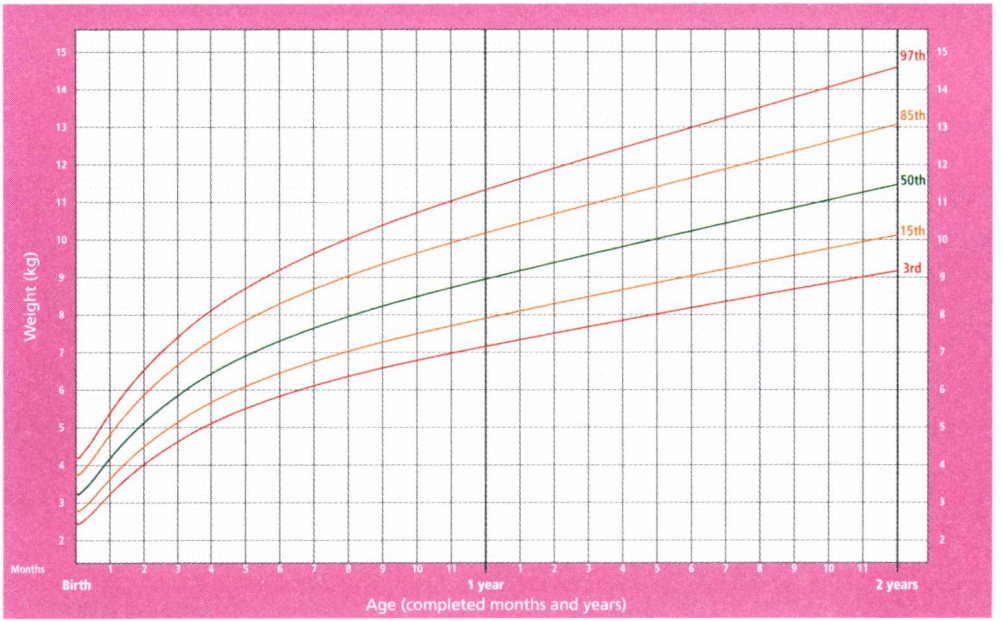
WHO Organization /jtaworMHearn



Q Weight-for-age GIRLS

Birth to 2 years (percentiles)

World Health Organization

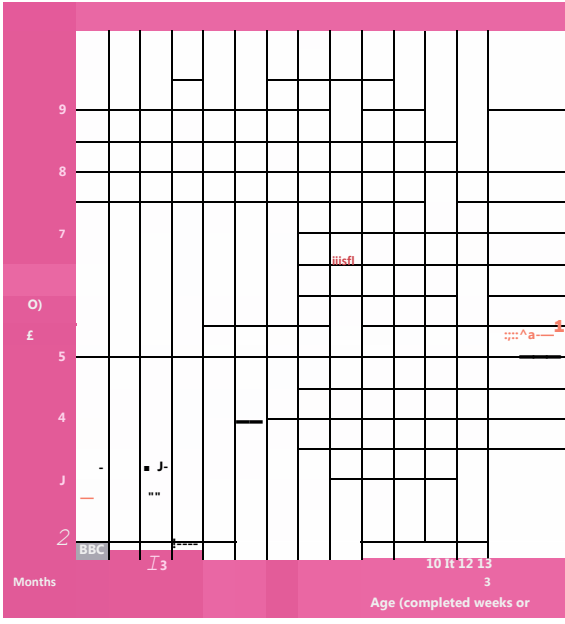


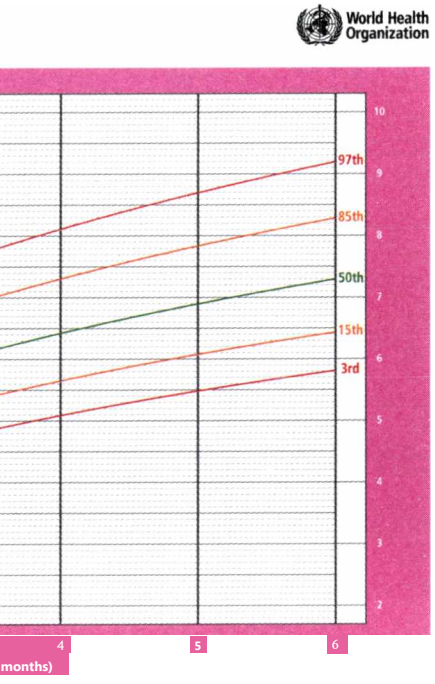
WHO Child Growth Standards

©

Weight-for-age GIRLS

Birth to 6 months (percentiles)



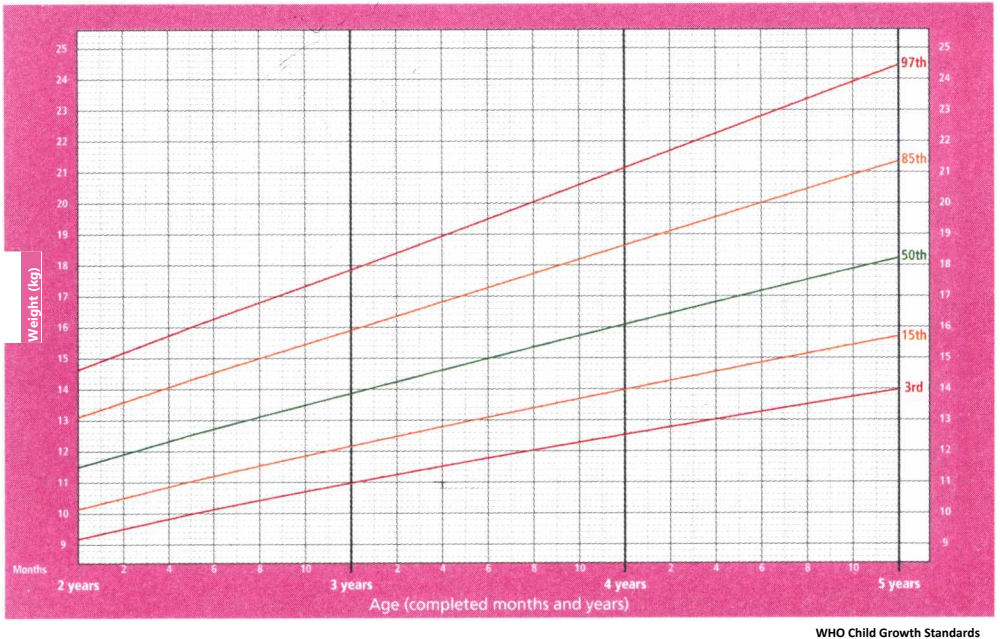


WHO Child Growth Standards

© Weight-for-age GIRLS

World Health Organization

2 to 5 years (percentiles)

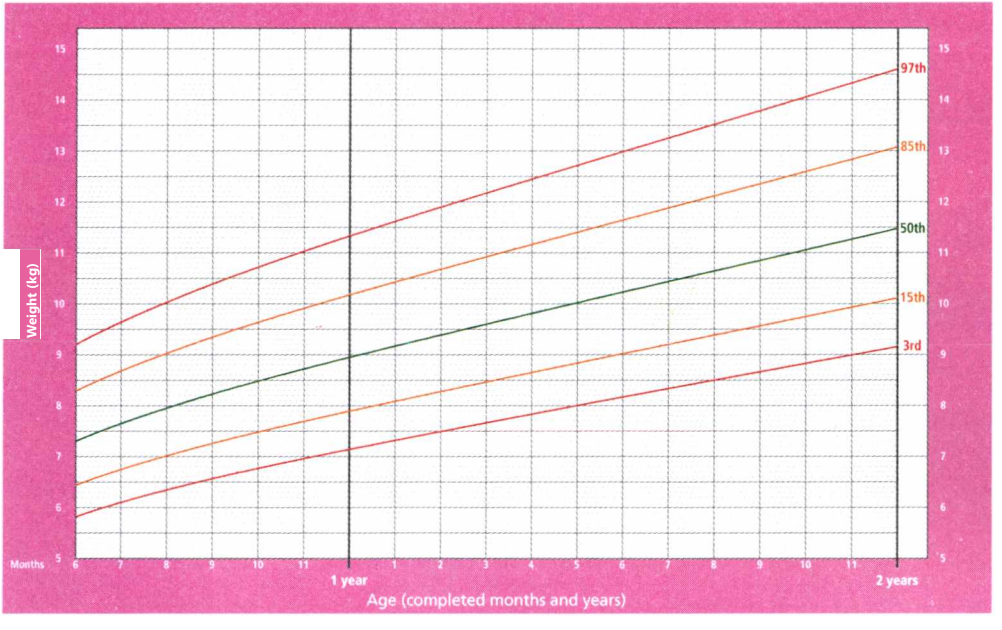


WHO Child Growth Standards

Weight-for-age GIRLS

6 months to 2 years (percentiles)

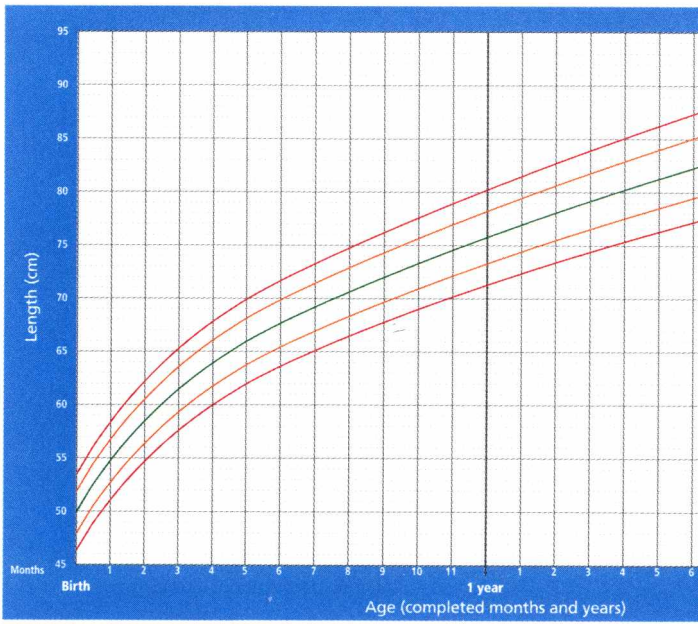
World Health Organization



WHO Child Growth Standards

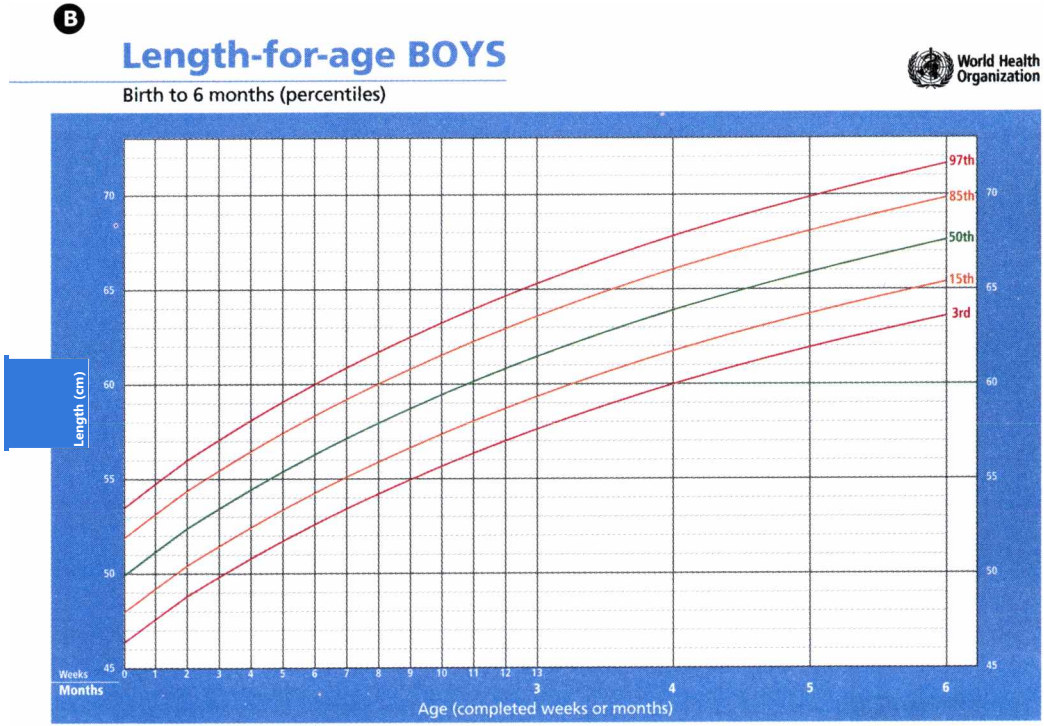
© Length-for-age BOYS

Birth to 2 years (percentiles)



Appendix 20: Length-for-age

WHO Child Growth Standards

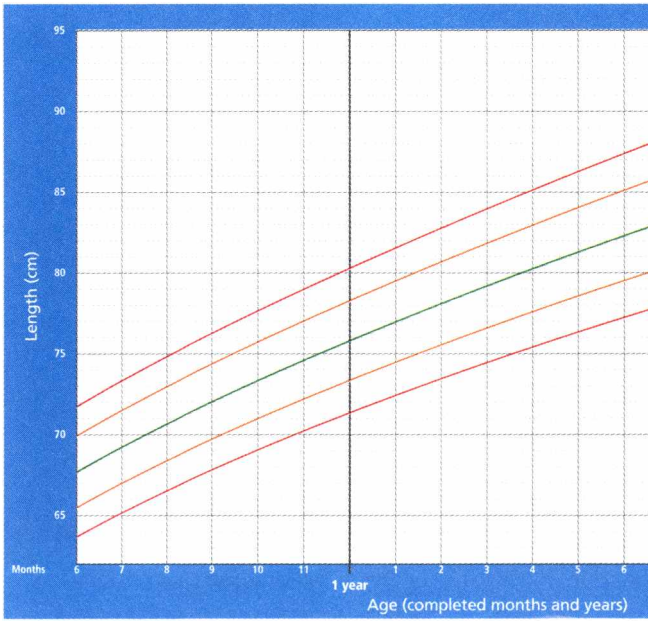


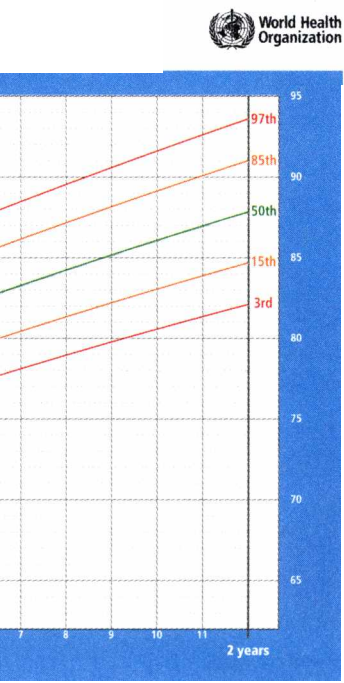
WHO Child Growth Standards

©

Length-for-age BOYS

6 months to 2 years (percentiles)



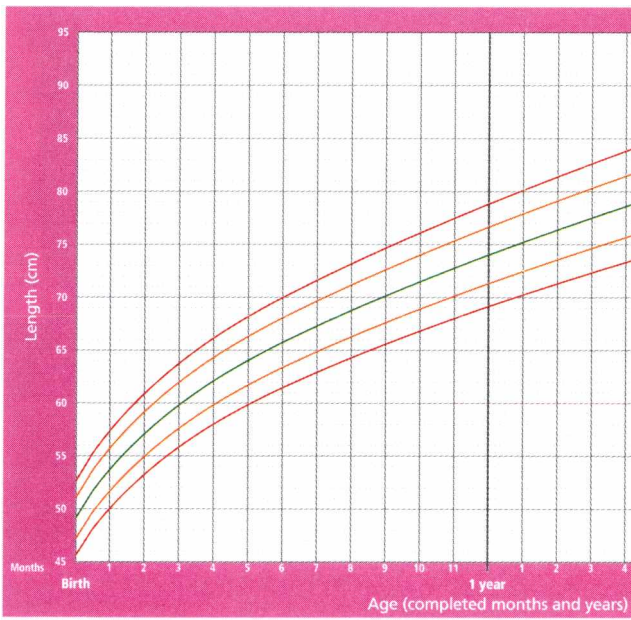


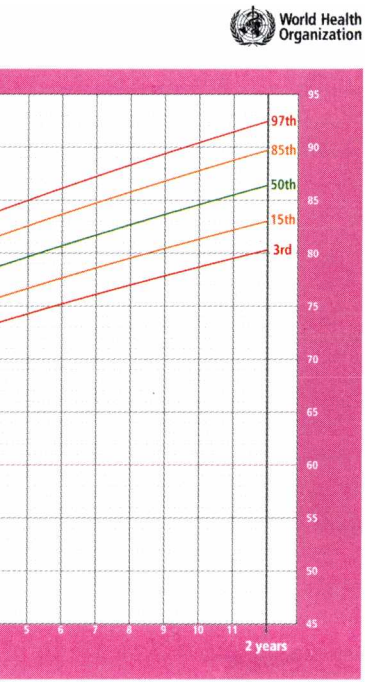
WHO Child Growth Standards

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Length-for-age GIRLS

Birth to 2 years (percentiles)



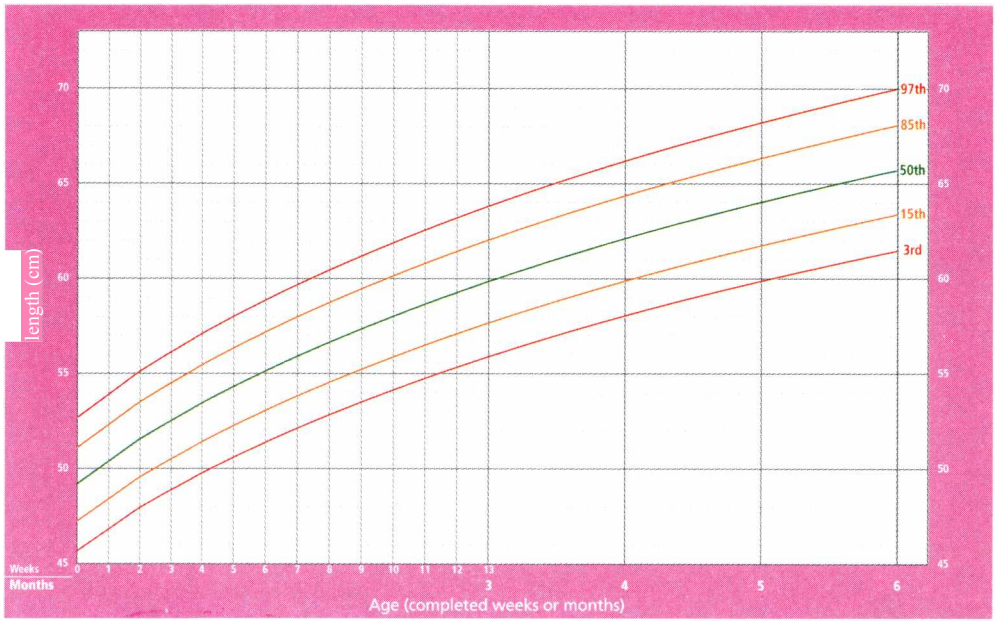


WHO Child Growth Standards

° Length-for-age GIRLS

Birth to 6 months (percentiles)

World Health Organization

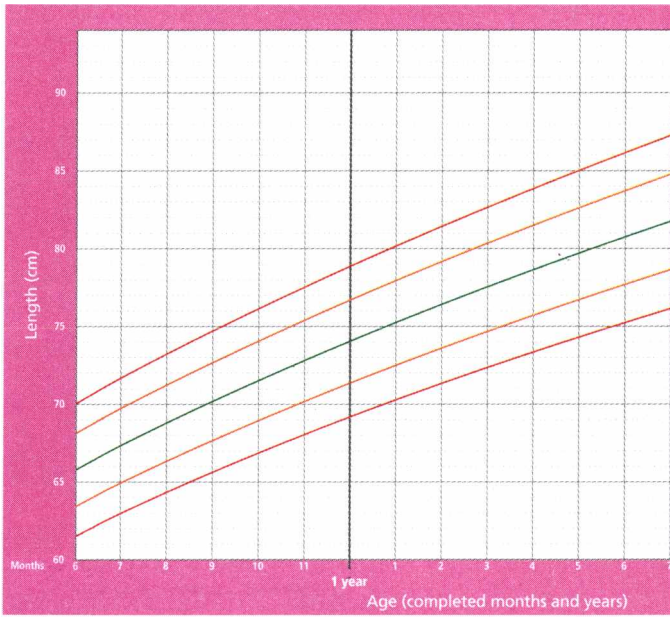


WHO Child Growth Standards

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Length-for-age GIRLS

6 months to 2 years (percentiles)

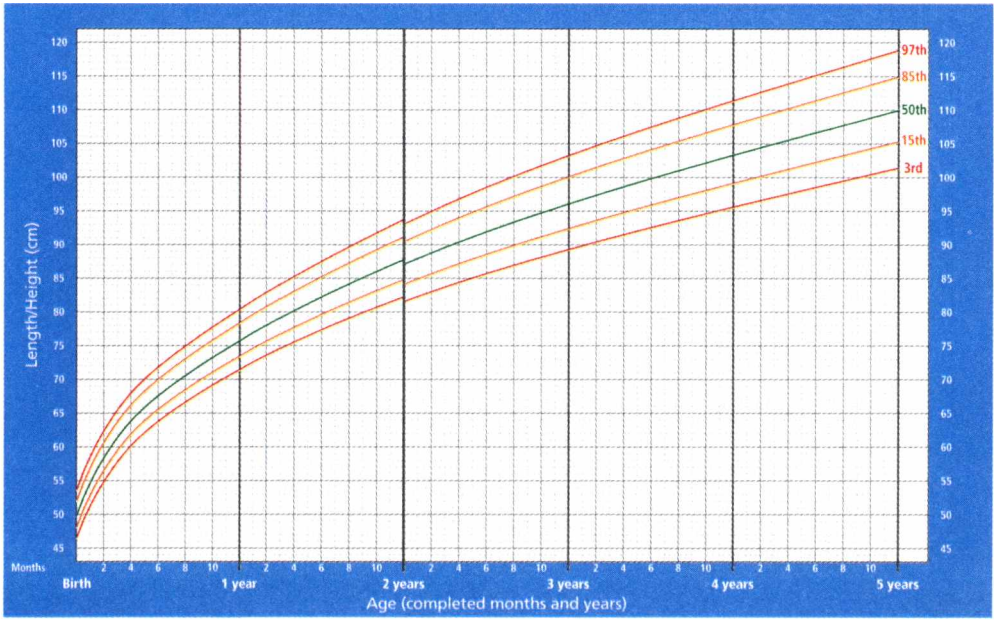


WHO Child Growth Standards

Length/height-for-age BOYS

Birth to 5 years (percentiles)

| World Health
* Organization



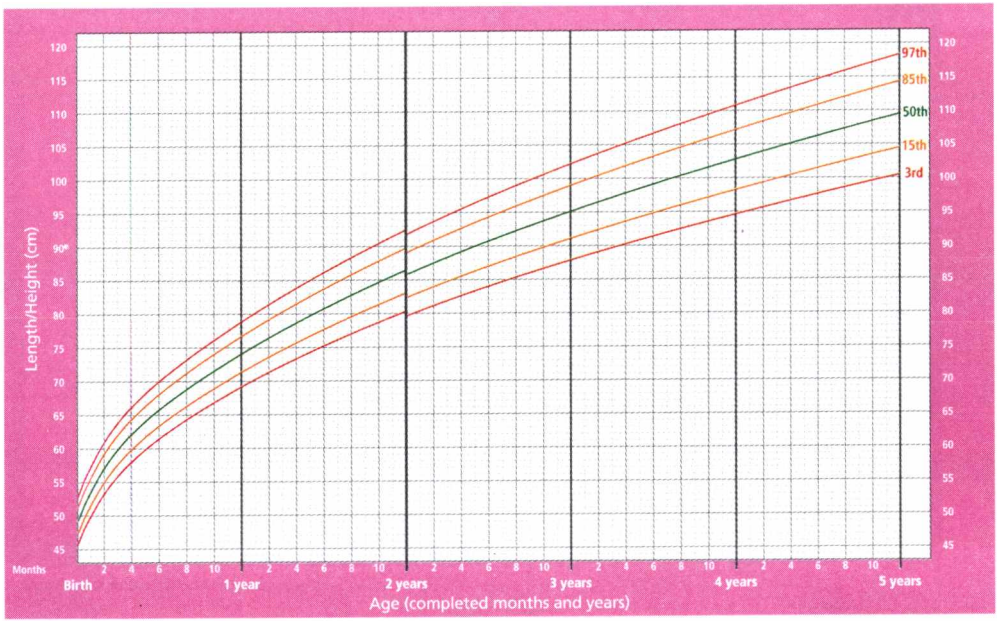
WHO Child Growth Standards

Appendix 21: Length/Height-for-age

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^WoHdHealth

Birth to 5 years (percentiles)



WHO Child Growth Standards

2 to 5 years (percentiles)

Appendix 22: Height-for-age

/J&l World Health
yjzWj? Organization

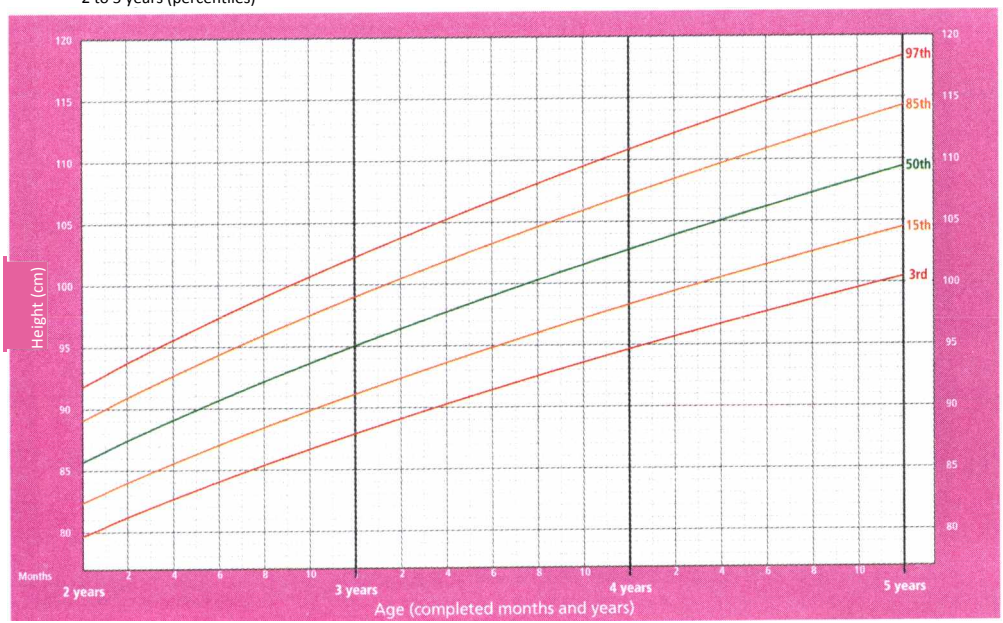
WHO Child Growth Standards

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Height-for-age GIRLS

World Health Organization

2 to 5 years (percentiles)

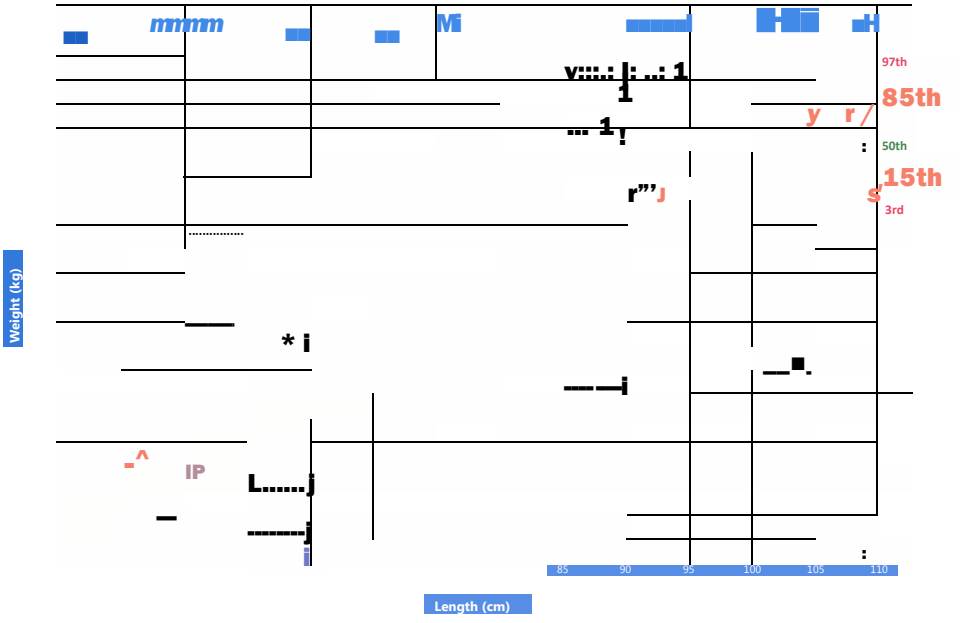


WHO Child Growth Standards

Weight-for-length BOYS

Birth to 2 years (percentiles)

World Health Organization



WHO Child Growth Standards

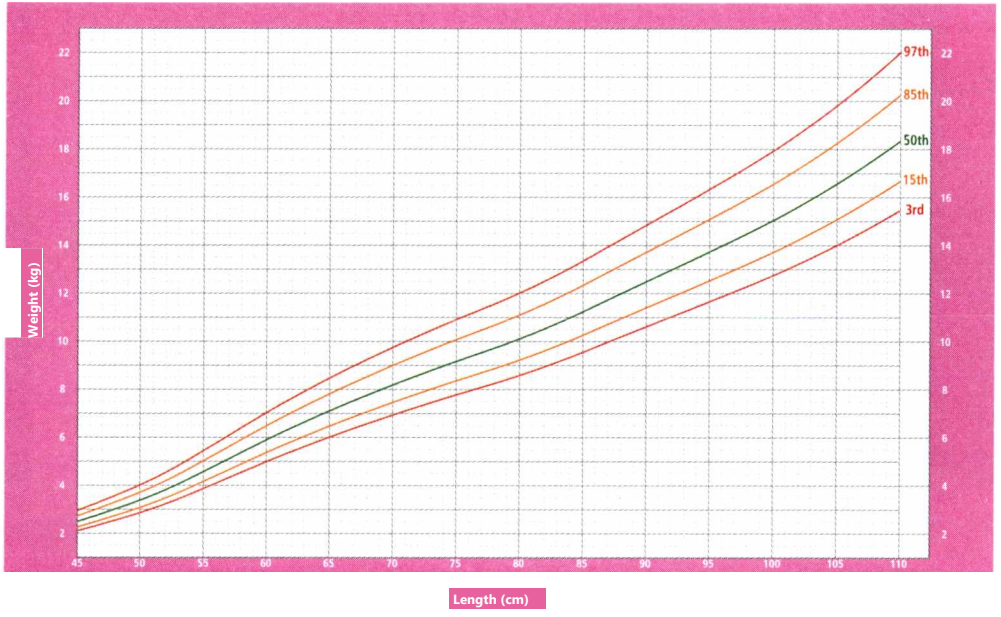
Appendix 23: Weight-for-Length

B

Weight-for-length GIRLS

World Health Organization

Birth to 2 years (percentiles)



WHO Child Growth Standards

A

Weight-for-height BOYS

2 to 5 years (percentiles)

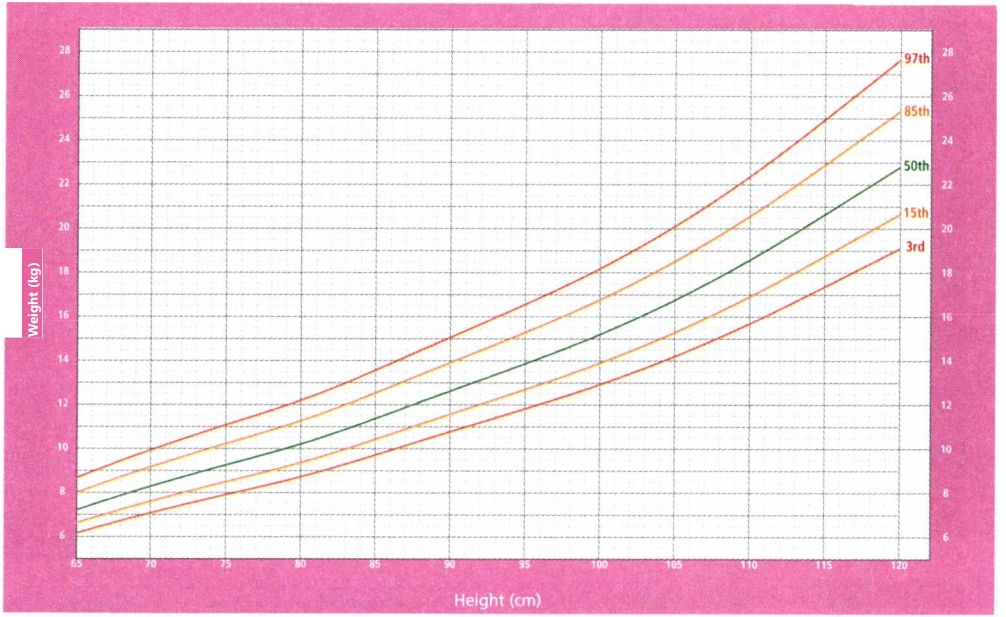
Appendix 24: Weight-for-Height

ft World Health
f Organization

WHO Child Growth Standards

World Health Organization

2 to 5 years (percentiles)



WHO Child Growth Standards

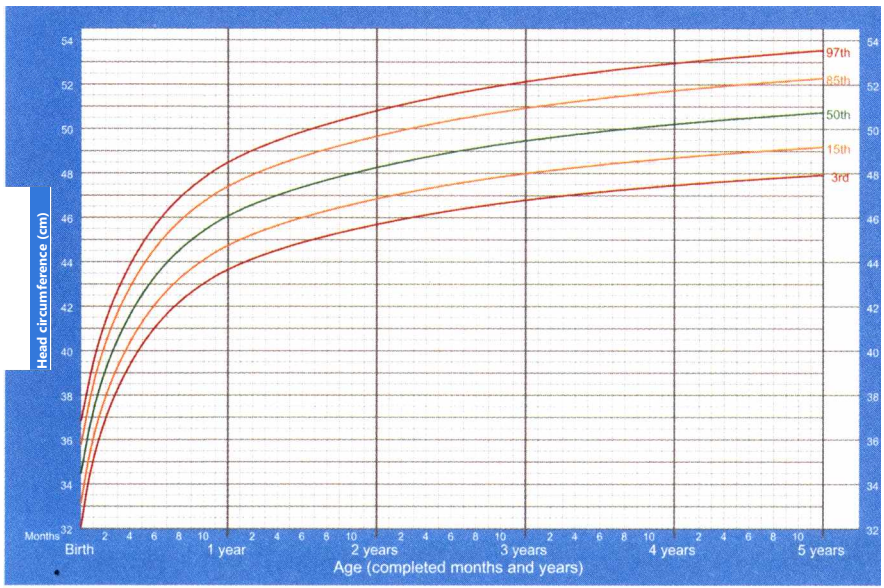
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Appendix 25: Head circumference-for-age

Head circumference-for-age BOYS

Birth to 5 years (percentiles)

World Health Organization

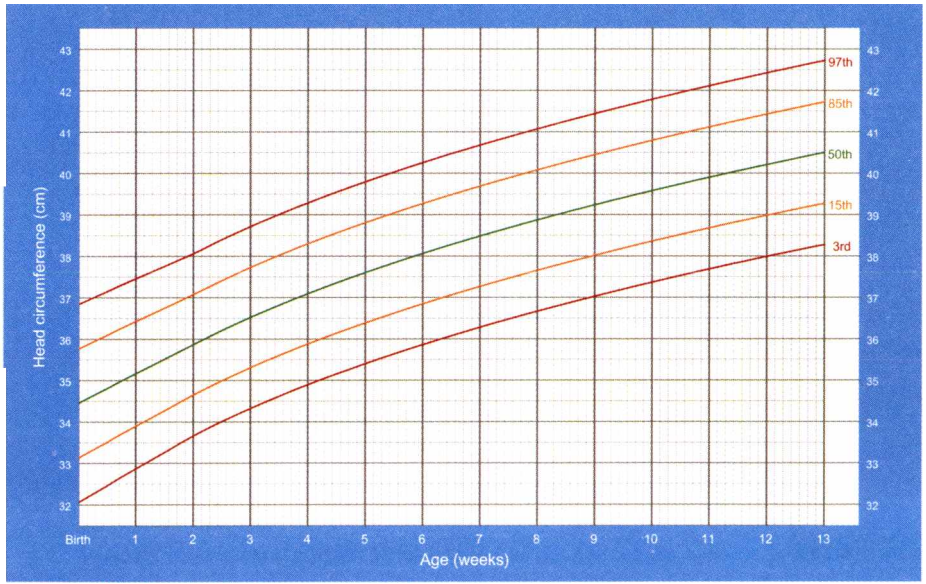


WHO Child Growth Standards

Head circumference-for-age BOYS

World Health Organization

Birth to 13 weeks (percentiles)



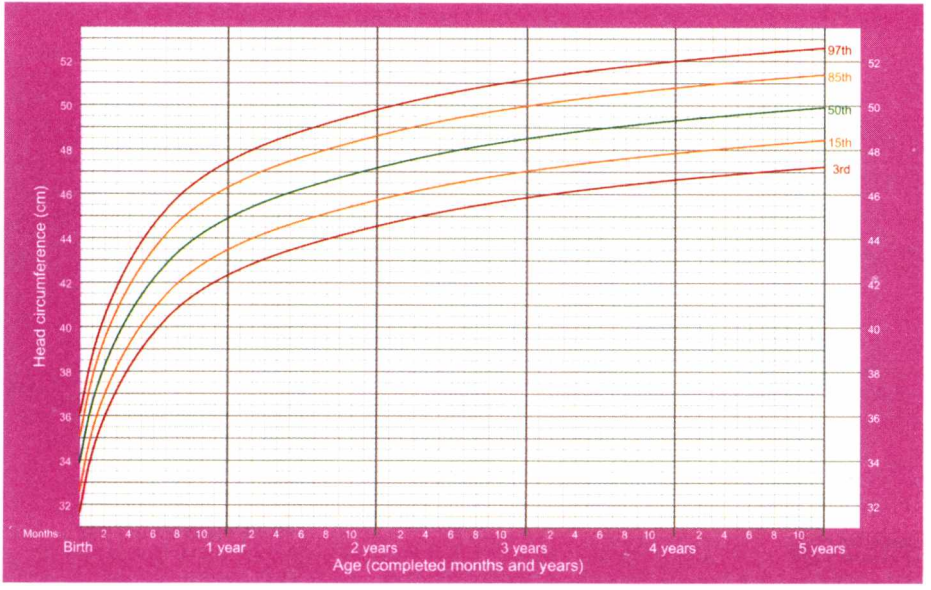
WHO Child Growth Standards

C

Head circumference-for-age GIRLS

World Health Organization

Birth to 5 years (percentiles)

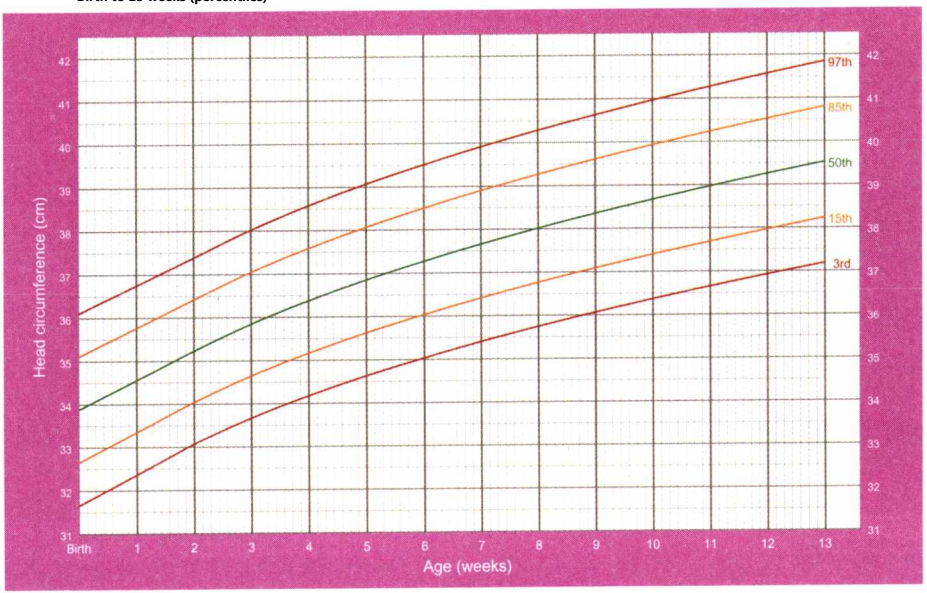


WHO Child Growth Standards

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Head circumference-for-age GIRLS

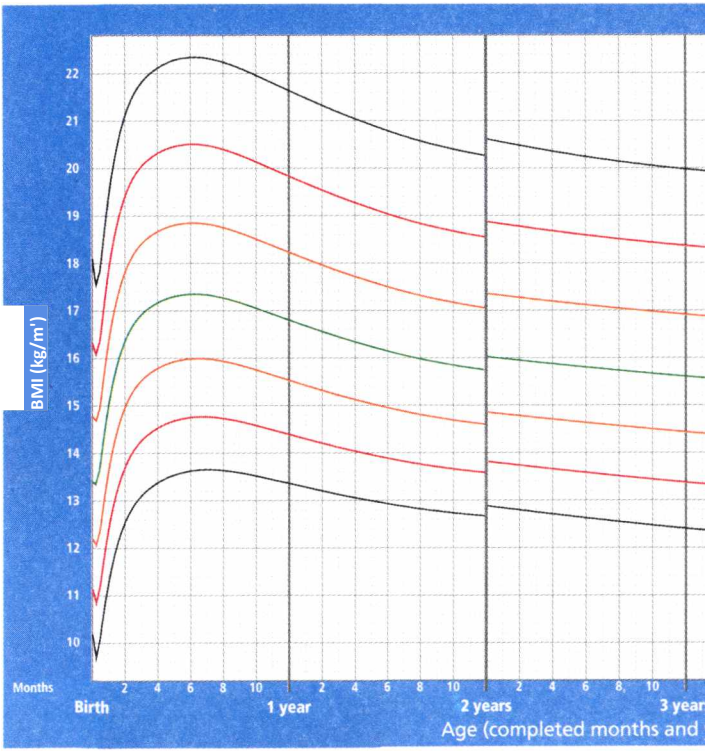
Birth to 13 weeks (percentiles)



WHO Child Growth Standards

BMI-for-age BOYS

Birth to 5 years (z-scores)



Appendix 26: BMI-for-Age

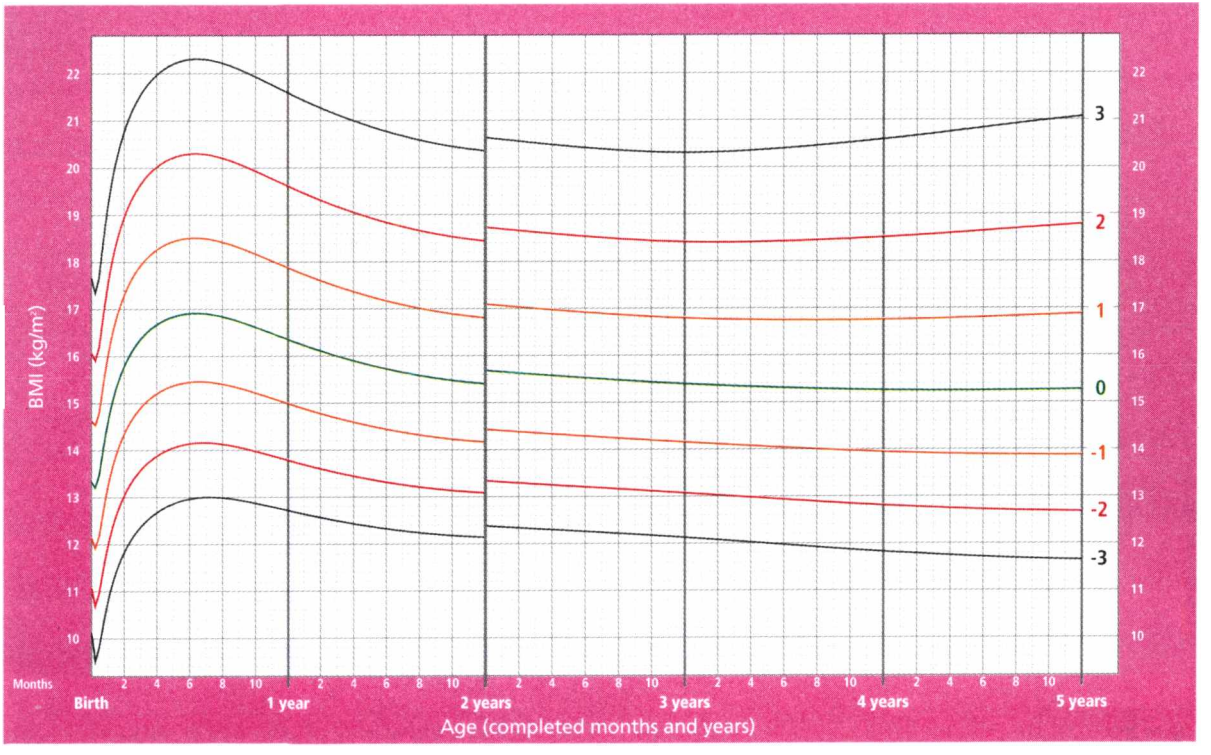
World Health Organization

WHO Child Growth Standards

BMI-for-age GIRLS

World Health Organization

Birth to 5 years (z-scores)



WHO Child Growth Standards

Appendix 27: BMI Cut-off Values

BMI Cut offs as per Classification of Body Mass Index (BMI) for Adults

Classification	BMI (kg/m²)	Risk of co-morbidities
Underweight	< 18.5	Low (but risk of other clinical problems increased)
Normal range	18.5-24.9	Average
Overweight	> 25.0	
Pre-obese	25.0-29.9	Increased
Obese class I	30.0-34.9	Moderate
Obese class II	35.0-39.9	Severe
Obese class III	>40.0	Very severe

International Obesity Task Force (IOTF) cut-off Points of BMI for the Diagnosis of Overweight and Obesity in Children Defined to Pass through BMI of 25 and 30 kg/m² at Age 18

Age (years)	BMI 25 kg/m²		BMI 30 kg/m²	
	Males	Females	Males	Females
2	18.41	18.02	20.09	19.81
2.5	18.13	17.76	19.80	19.55
3	17.89	17.56	19.57	19.36
3.5	17.69	17.40	19.39	19.23
4	17.55	17.28	19.29	19.15
4.5	17.47	17.19	19.26	19.12
5	17.42	17.15	19.30	19.17
5.5	17.45	17.20	19.47	19.34
6	17.55	17.34	19.78	19.65
6.5	17.71	17.53	20.23	20.08

contd.

Age (years)	BMI 25 kg/m²		BMI 30 kg/m²	
	Males	Females	Males	Females
7	17.92	17.75	20.63	20.51
7.5	18.16	18.03	21.09	21.01
8	18.44	18.35	21.60	21.57
8.5	18.76	18.69	22.17	22.18
9	19.10	19.07	22.77	22.81
9.5	19.46	19.45	23.39	23.46
10	19.84	19.86	24.00	24.11
10.5	20.20	20.29	24.57	24.77
11	20.55	20.74	25.10	25.42
11.5	20.89	21.20	25.58	26.05
12	21.22	21.68	26.02	26.67
12.5	21.56	22.14	26.43	27.24
13	21.91	22.58	26.84	27.76
13.5	22.27	22.98	27.25	28.20
14	22.62	23.34	27.63	28.57
14.5	22.96	23.66	27.98	28.87
15	23.29	23.94	28.30	29.11
15.5	23.60	24.17	28.60	29.29
16	23.90	24.37	28.88	29.43
16.5	24.19	24.54	29.14	29.56
17	24.46	24.70	29.41	29.69
17.5	24.73	24.85	29.70	29.84
18	25	25	30	30

Values of the MDG indicators for available periods in

No	Indicator	Year	Value
1	Proportion of population below 1990 poverty Iine(%)		37.5
2	Undernourished people as % 1990 of total population		62.2
3	Proportion of under nourished children	1990	54.8
4	Literacy rate of 15- 24 year olds	1990	64.3
5	Ratio of girls to boys in primary education	1990-91	0.71
6	Ratio of girls to boys in secondary education	1990-91	0.49
7	Under 5 mortality rate (per 1000 live birth)	1988-92	125
8	Infant mortality rate (per 1000 live birth)	1990	80

Appendix 28: Millennium Development Goals

India

Year	Value	MDG target value
1999-2000	26.1	18.75
1999-2000	53	31.10
1998	47	24.7
2001	73.7	100
2000-01	0.78	1
2000-01	0.63	1
1998-2002	98	41
2003	60	27

9	Maternal mortality rate (per 100,000 live births)	1991	**437**
10	Population with sustainable access to an improved water source(rural)(%)	1991	55.54
11	Population with sustainable access to an improved water source(urban)(%)	1991	81.38
12	Population with access to sanitation(urban)(%)	1991	47
13	Population with access to sanitation(rural)(%)	1991	9.48
14	Deaths due to Malaria per 100,000	1994	0.13
15	Deaths due to TB perl00,000	1999	56
16	Deaths due to HIV/AIDS	2000	**471**

1998	407	109
2005	90	80.5
2001	82.22	94
2001	63	72
2005	32.76	72
2004	0.09	-
2003	33	-
2004	1114	

Appendix 29: Nutrition Websites

Today we find mushrooming of nutrition websites us they see the field of nutrition as a fertile land. But it is always difficult to assess which website offers the perfect information. Here is the list of websites which could be used to access information on nutrition and public health.

Food and Agriculture Organization (FAO)—**www.fao.org**

International Life Sciences Institute (ILSI)—**www.ilsi.org**

International Nutrition Foundation (INF)—**www.inffoundation.org**

The Micronutrient Initiative—**www.micronutrient.org**

Nestle Nutrition Institute (NNI)—**www.nestlenutrition-institute.org**

Public Health Foundation of India—**www.phfi.org**

United Nations Children's Fund (UNICEF)—**www.unicef.org**

U.S. Food and Drug Administration—**www.fda.gov**

World Health Organization (WHO)—**www.who.int/en**

American Diabetes Association—**www.diabetes.org**

American Heart Association—**www.americanheart.org**

American Society for Nutritional Sciences—**www.faseb.org/asns/intro.html**

CDC: National Center for Chronic Disease—**www.cdc.gov/nccdphp**

C FT R I. Mysore—**www.mylibnet.org.in/cftri/cftri.html**

FDA Search—**www.fda.gov/search/index.html**

Indian Council of Medical Research—**www.icmr.nic.in**

India- ICDS—**http://wcd.nic.in/childdet.htm#icdsg**

Indian Dietetic Association—**www.idaindia.com**

Indian Medical Journals (INDMED)—**www.indmed.nic.in/jrnl.html**

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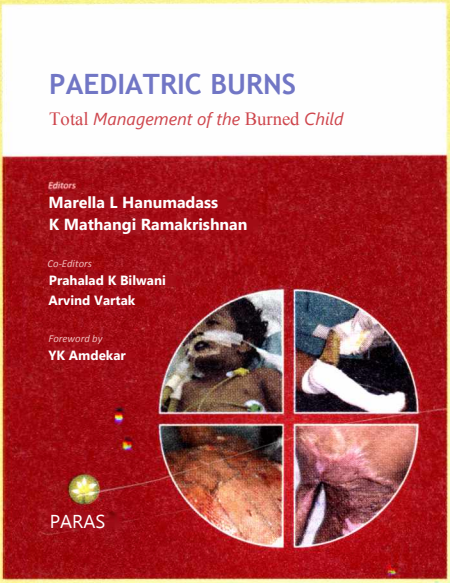
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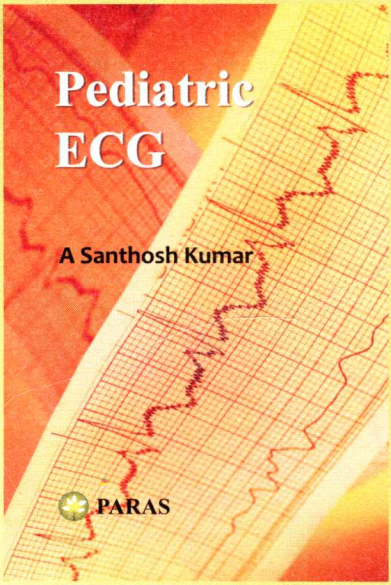
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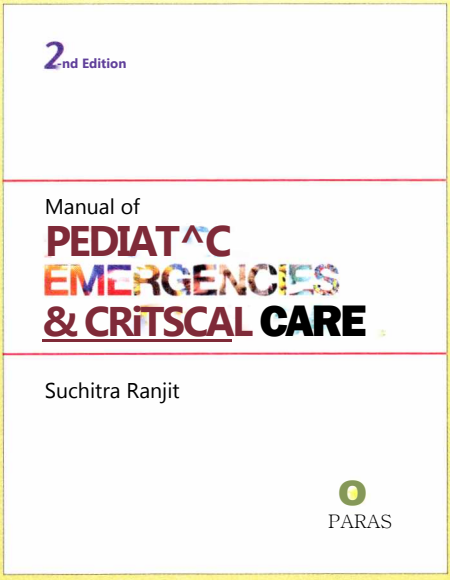
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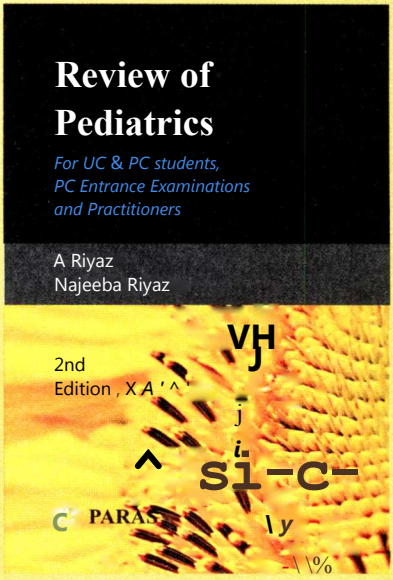
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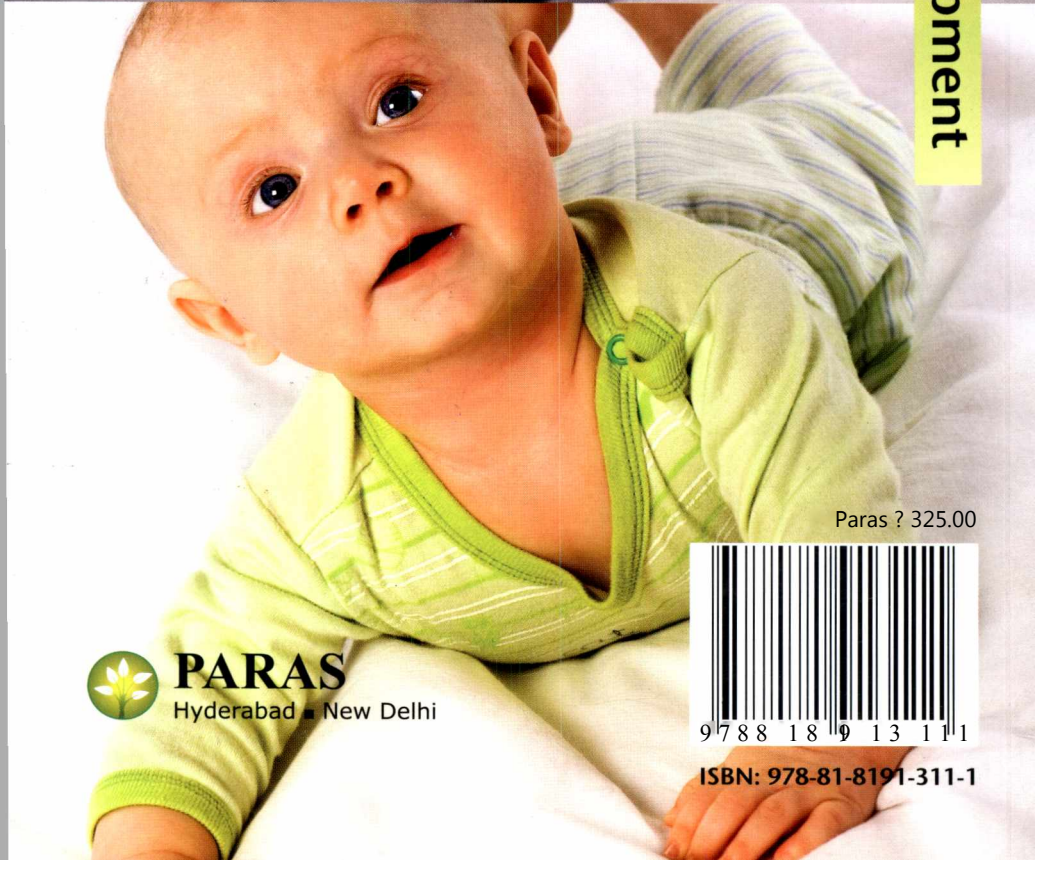
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